2009 Honda Civic SI (FG2)

Complete Build Guide

Chassis: FG2 (8th Gen Coupe)

Engine: K20Z3 -- 2.0L i-VTEC, 197 HP / 139 lb-ft (stock)

Transmission: 6-Speed Manual

Mileage: ~170,000 miles

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2009 Honda Civic SI Coupe -- My Build Log

Chassis: FG2 (8th Gen Coupe) **Engine:** K20Z3 -- 2.0L i-VTEC, 197 HP / 139 lb-ft (stock) **Transmission:** 6-Speed Manual **Color:** Blue **Mileage:** ~170,000 miles

What I'm Going For

Daily-drivable with a sport mode toggle. Quiet when I need it, loud when I want it. I'm building a full bolt-on NA K20Z3 targeting **220-240 WHP on pump gas, 230-250 WHP on E85** -- headers, intake manifold, throttle body, exhaust, pulleys, flex fuel, and a Hondata tune to tie it all together. I'm doing a valved 3" exhaust with a QTP cutout so I can switch between a quiet daily tune and an aggressive straight-piped sport tune. Flex fuel with real-time ethanol content sensing lets me run pump gas or E85 seamlessly. And eventually, I'm putting a LattePanda mini PC in the car for permanent Hondata management and exhaust valve control.

The rule that overrides everything: engine longevity comes before power. Every tune ships as two calibrations (Daily + Sport), and Sport means "squeeze the engine for everything it's got WITHIN the reliability envelope" -- not "push it to the edge." See 10-Hondata-FlashPro/tuning-workflow-and-maps.md for the tuning rules I'll live by.

Modification Status

Installed

#	Mod	Part	Actual Cost	Status
1	Clutch & Flywheel	Exedy Stage 1 Organic (08806) + Exedy Lightweight Flywheel (HF02)	$960.43*	Installed
2	Cold Air Intake	K&N Typhoon 69-1014TS	$504.17	Installed
3	Short Shifter + Bushings	Acuity 3-Way Adjustable + Trans-Side Bushings	$622.20	Installed
4	Tires	Continental ExtremeContact DWS06 Plus 215/45ZR17 91W XL	$715.00	Installed

\$960.43 includes clutch, flywheel, and strut bar bundled together (incl. tax/shipping)*

Purchased -- Awaiting Install

#	Mod	Part	Actual Cost	Status	Notes
5	Engine Mounts	Hybrid Racing 70A	$657.15	Awaiting install	
6	Brakes	Hawk HPS Pads + R1 Concepts Slotted Rotors	$532.14	Front installed 2026-04-20 -- rear pending	Full 4-corner upgrade
7	Strut Bar	Megan Racing Race Spec Front Upper (Polished)	(in #1)	Deferred until coilovers -- BC Racing camber plates will change strut-top geometry anyway	
8	Hondata FlashPro	Hondata FlashPro	$859.59	Awaiting first flash	ECU tuning, datalogging, gauges

Total spent to date: $4,850.68 (all prices include tax, shipping, and insurance)

Planned -- Not Yet Purchased

#	Mod	Details	Notes
8	Valved Exhaust	DIY 3" QTP cutout bypass (keep OEM mufflers)	3" dump, step-down to 2.5" for muffled path
9	Headers	Skunk2 Alpha 4-2-1 (412-05-1930) + Berk HFC	Best available, stepped primaries, 10-18 WHP
10	Intake Manifold	Skunk2 Ultra Street + bored stock TB (66mm)	Larger plenum, shorter runners, DBW compatible
11	Pulleys & Harmonic Balancer	ATI Super Damper + NST accessory pulleys	Proper damping + reduced parasitic loss
12	Flex Fuel & Fuel System	Flex fuel sensor + Bosch EV14 550cc injectors + Acuity rail + DW300C pump	E85 capable, 5-15 WHP on NA over pump gas
13	Clutch Hydraulics	Hybrid Racing HYB-CMC-01-20 complete kit + rear main seal + throwout bearing	170k-mile reliability upgrade + "free labor while trans is out" bonus items
14	Suspension	BC Racing BR coilovers + full bushing refresh + sway bars + camber kit	Complete suspension overhaul -- coilovers, bushings, geometry correction
15	Ignition Refresh	NGK plugs + OEM Honda coils (30520-RWC-A01) + VTC actuator (14310-RBC-003 updated part)	New. Critical before any tune. Fixes cold-start rattle.
16	Cooling System	Koyorad HH060063 aluminum radiator + OEM thermostat + hoses + water pump	New. Plastic end tank failure insurance + headroom for E85 heat. Must precede headers.
17	Wideband AFR	AEM X-Series 30-0300 wideband UEGO	New. Required before any dyno tune. Integrates with FlashPro analog input.
18	Timing Chain Maintenance	Honda OEM chain 14401-PNA-004 + tensioner 14510-PRB-A01 + guides + VTC actuator (bundled with Ignition Refresh)	New. Phase 0 maintenance at 170k. Decision rule: 13.5mm tensioner rod = go/no-go.
19	Cylinder Head (Drop-In)	4 Piston Pro 156v2 All Motor Package + ARP 208-4701 studs + Cometic C4561-030 + Skunk2 Tuner Stage 2 cams + Skunk2 Pro cam gears	New. "Go all out" decision. ~$4,130 parts + tune. Bundle with headers, IM, pulleys, VTC install.
20	Built Motor Bottom End	Future phase appendix -- Wiseco + BC625+ + ACL Race + PRB-A01 pump	New. Not current purchase. Reference for future 400-500 WHP E85 or ultra-high-RPM NA.
21	Purple Cosmetics	Acuity 1927-PPL oil cap + K-Tuned DP2-K20-PRP dipstick + Acuity ESCO-T6 knob + Dress Up Bolts HON-027-TI + NST22015K-Purple alt/idler + Samco purple hoses + powder-coat batch	New. Aesthetic layer. Tier-1 package ~$966; full purple theme ~$1,836.
--	In-Car PC	LattePanda 3 Delta permanent install	Controls FlashPro + exhaust valve

How This Repo Is Organized

Each mod has its own folder with some subset of:

- **overview.md** -- What the mod is, why I chose it, part numbers, status
- **install-guide.md** -- Step-by-step install instructions
- **brainstorm.md** -- Research, alternatives I considered, pros/cons, things I ruled out
- **purchasing.md** -- Parts list with part numbers and prices

Simple bolt-ons might only have an overview. Complex mods (valved exhaust, LattePanda, headers) have all four.

Special Folders

- 10-Hondata-FlashPro/Temporary-Setup/ -- Getting started with a laptop before the permanent install
- 10-Hondata-FlashPro/Permanent-LattePanda-Install/ -- Full guide for the in-car PC build
- 10-Hondata-FlashPro/tuning-workflow-and-maps.md -- Daily/Sport map rules, reliability-first ceilings, and how I work with Claude on tune changes
- 12-Valved-Exhaust/ -- My DIY valved exhaust design, parts, and planning
- docs/ -- Cross-cutting reference documents

Docs

- docs/mod-order-and-maintenance.md -- Chronological mod order + 170k-mile maintenance plan
- docs/vehicle-specs.md -- Stock specs for the FG2
- docs/install-references.md -- Guides, videos, and community threads I've collected
- docs/torque-specs.md -- **New.** K20Z3 / FG2 torque reference for every fastener I'll touch
- docs/maintenance-parts-catalog.md -- **New.** OE + aftermarket part numbers in one place
- docs/baseline-logs.md -- **New.** A/B datalog protocol for every mod

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Quick Reference

Spec	Value
Engine Code	K20Z3
ECU Code	PRB (37820-RRB-A58 or similar)
Displacement	1998cc (2.0L)
Compression	11.0:1
Stock HP / TQ	197 HP @ 7800 RPM / 139 lb-ft @ 6200 RPM
Redline	8000 RPM (fuel cut at 8200 RPM)
Stock Exhaust Diameter	2.36" (60mm) from cat back
Fuel	Premium 91+ octane required
OBD2 Port	Under dash, left of steering column
Interior Fuse Box	Driver side lower dash, behind removable panel
Fuse Type (Interior)	Mini blade

Build Principles I'm Living By

1. **Reliability over power, always.** At 170k I'd rather leave 10 WHP on the table than risk a bearing.
2. **Every tune = Daily + Sport maps.** Quiet daily mode, aggressive-but-safe sport mode.
3. **Cross-reference every decision.** If Mod X changes the situation for Mod Y, the Y doc says so explicitly.
4. **Phase 0 maintenance comes first.** A sick engine can't be tuned well no matter what parts go on.
5. **No lean WOT fuel, ever, on any tune.** Pump gas = 12.0 AFR minimum; E85 = lambda 0.82 minimum.
6. **Dyno verification for every timing change.** No armchair advances.

Contributing / AI Assistant Reference

See CLAUDE.md for the AI assistant reference file containing full context about this build, conventions, and instructions for AI tools assisting with this project.

Last updated: 2026-04-20

2009 Honda Civic SI Coupe (FG2) -- Vehicle Specifications

Stock specs for my car. I put this together as a quick reference since I'm constantly looking up bolt sizes, torque specs, and dimensions when planning mods.

General

Spec	Value
Year	2009
Make / Model	Honda Civic SI
Trim	SI (top performance trim)
Body Style	Coupe (2-door)
Generation	8th Gen (2006-2011)
Chassis Code	FG2
Production	East Liberty Auto Plant, Ohio

Engine

Spec	Value
Engine Code	K20Z3
Type	2.0L DOHC i-VTEC Inline-4
Displacement	1998cc
Bore x Stroke	86mm x 86mm
Compression Ratio	11.0:1
Horsepower	197 HP @ 7800 RPM
Torque	139 lb-ft @ 6200 RPM
Redline	8000 RPM (fuel cut at 8200 RPM)
VTEC Crossover	~5800 RPM (stock)
Fuel System	Sequential multi-port fuel injection
Fuel Requirement	Premium unleaded 91+ octane
Engine Management	Keihin PRB ECU
Emissions	ULEV-2

Transmission

Spec	Value
Type	6-speed manual
Transmission Code	K20Z3 close-ratio 6MT
Final Drive	4.764:1
LSD	Helical limited-slip differential (front)

Brakes (Stock)

Spec	Front	Rear
Type	Ventilated disc	Solid disc
Diameter	11.8" (300mm)	10.2" (260mm)
Caliper	Single-piston sliding	Single-piston sliding

Suspension (Stock)

Spec	Front	Rear
Type	MacPherson strut	Multi-link
Anti-roll bar	Yes	Yes

Dimensions

Spec	Value
Wheelbase	104.3"
Length	175.4"
Width	68.9"
Height	54.7"
Curb Weight	~2,877 lbs
Weight Distribution	~61/39 F/R

Electrical

Spec	Value
Battery	12V, Group 51R
Alternator	~130A
OBD2 Port	Under dash, left of steering column
Interior Fuse Box	Driver side lower dash, mini blade fuses
Engine Bay Fuse Box	Near battery, standard blade fuses

Exhaust (Stock)

Spec	Value
Header	Cast iron log-style manifold
Catalytic Converter	Close-coupled to header
Pipe Diameter (cat-back)	2.36" (60mm)
Resonator	Yes (mid-pipe)

Muffler	Single oval canister, rear
Tailpipe Exit	Single, driver side

Chronological Mod Order & 170k-Mile Maintenance Plan

2009 Honda Civic SI (FG2) -- K20Z3, ~170,000 miles, 6MT

This is my game plan -- the order I'm tackling every mod and the maintenance I need to knock out first. **Maintenance intervals here are based on community experience (8thcivic.com, Reddit r/CivicSi, K-series forums) -- NOT Honda's factory intervals**, which are way too conservative for a 17-year-old, 170k-mile sport compact that's actually been driven.

Phase 0: Critical Maintenance (Do First -- Before Any More Mods)

I need to handle these before adding any more power or stress to the drivetrain. At 170k miles, some of these are honestly overdue.

#	Item	Why	Community Interval	Est. Cost (DIY)	Est. Cost (Shop)
1	Timing chain inspection	K20Z3 uses a timing chain, not a belt -- no replacement interval. BUT chain stretch is real at 170k. A stretched chain causes retarded cam timing, VTEC engagement issues, and eventually jumps teeth. I already know this needs checking.	Inspect at 150k+. Replace if stretch measured or chain noise present.	$20 (inspection) / $300-500 (replacement parts)	$100 (inspection) / $800-1,200 (replacement)
2	Valve adjustment	K20Z3 has solid lifters (no hydraulic lash adjusters). Valves go tight over time -- tight valves burn. Community consensus: check every 30k-60k on a driven SI. At 170k, if never done or unknown history, this is urgent.	Every 30,000-60,000 miles	$10 (feeler gauges)	$150-250
3	Spark plugs	NGK ILZKR7B-11S (stock iridium, pre-gapped). Even iridium plugs degrade by 100k. At 170k, if they're original, they're well past done. Misfires, poor fuel economy, lost power.	Every 60,000-80,000 miles (community). Honda says 100k -- too long.	$40 (set of 4)	$80-120
4	Coolant flush + thermostat	OEM Honda Type 2 coolant (blue). At 170k, the coolant has been through thousands of heat cycles. Thermostat can stick partially open (overcooling) or closed (overheating). I'm replacing both proactively.	Coolant every 50,000-60,000 miles. Thermostat at 100k+ or with coolant flush.	$30 coolant + $15 thermostat	$100-150
5	Transmission fluid change	Honda MTF or Amsoil Synchromesh. The 8th gen SI 6MT synchros are sensitive to fluid quality. Notchy 2nd gear is often a fluid issue, not a synchro issue.	Every 30,000-40,000 miles	$25 (3 quarts Honda MTF)	$80-120
6	Brake fluid flush	DOT 4 (Motul RBF 600 or ATE TYP 200). Brake fluid is hygroscopic -- absorbs water, boiling point drops, internal corrosion increases. If never flushed, the fluid is original from 2009.	Every 2 years or 30,000 miles	$15	$80-100
7	Clutch fluid flush	Same as brake fluid but separate reservoir on the FG2. Especially important before the CMC/CSC upgrade.	Every 2 years or 30,000 miles	$10	$50-80
8	Serpentine belt	Visual inspection -- cracking, glazing, squealing. At 170k, I'm likely on the 2nd or 3rd belt. Replace if there are any signs of wear.	Every 60,000-80,000 miles or at first sign of wear	$25	$80-120
9	Water pump	K20Z3 water pump is driven by the timing chain. I'll inspect it during the timing chain inspection. Weeping from the weep hole = replace immediately.	Inspect at timing chain service. Replace proactively at 150k+.	$50 (pump)	Included in timing chain labor

10	Motor oil + filter	Honda recommends 0W-20, but community consensus for a 170k-mile K20Z3 that's driven hard: 5W-30 full synthetic (Mobil 1, Pennzoil Platinum, or Castrol Edge). Thicker oil compensates for worn bearing clearances.	Every 5,000-6,000 miles with synthetic. Honda says 7,500 -- too long for an SI driven spiritedly.	$35	$60-80
11	PCV valve	17130-RTA-000 or equivalent. Cheap, easy, often neglected. A failed PCV causes oil consumption, rough idle, and boost leaks if ever turbo'd.	Every 60,000-80,000 miles	$10	$30-50
12	VTC actuator (REVISED part 14310-RBC-003)	#1 K20Z3 gripe on 8thcivic -- the cold-start rattle. Original actuator's spring weakens; new part has stronger spring and better locking pin. Universal replacement at 170k. Free labor if doing timing chain inspection. Critical before any dyno tune.	At timing chain service or 150k+	$280 (DIY)	$280 + $150-300 labor if standalone
13	Valve cover gasket + plug tube seals	OEM 12030-RRA-A01 kit. Universal 170k oil leaker. Oil in plug wells will foul new spark plugs in weeks. Cheap fix.	150k+	$40	$120-180
14	Oil filter housing gasket/o-ring	OEM 15301-PT0-003. Weeps at 170k. $8 fix while oil is drained for oil change.	150k+	$8	$50

Timing Chain Deep Dive

See ../05-Timing-Chain-Maintenance/overview.md for the complete parts list, step-by-step procedure, torque specs, and common mistakes.

The K20Z3 timing chain is rated for the life of the engine, but "life of the engine" assumes Honda's idea of a normal lifecycle (~200k miles of grandma driving). On a 170k-mile SI that's been driven like an SI:

- **Decision rule (from Honda service manual):** measure tensioner rod protrusion. If >= **13.5 mm**, chain is stretched -- replace chain + guides + tensioner as a set. Under 13.5 mm, serviceable.
- **Chain noise:** A rattling or slapping noise at cold start (first 5-10 seconds) that goes away once oil pressure builds is the classic symptom of a stretched chain with a maxed-out tensioner.
- **Why bundle with VTC actuator:** VTC actuator 14310-RBC-003 (Phase 0.5 Ignition Refresh) lives behind the timing cover. Labor is identical -- do them together.
- **If chain is OK but I'm in there anyway:** replace the tensioner (~$65) as cheap insurance, reuse chain and guides.
- **Budget:** ~$445 DIY parts (chain set + oil pump chain + all gaskets), ~$200 additional if bundling VTC actuator.

Phase 0.5: Ignition Refresh -- Between Maintenance and Mods

After the fluids/belt/PCV maintenance but BEFORE any dyno tune session, the ignition system gets its own focused refresh. See 06-Ignition-Refresh/overview.md for full detail.

#	Item	Why	Approx Cost
1	NGK ILZKR7B-11S plugs x4 (pump gas)	New iridium plugs before any tune work -- clean spark baseline	~$40
2	Honda OEM coils 30520-RWC-A01 x4	Tired coils misfire on E85 tunes and get blamed on tuning	~$160-200

3	Honda VTC actuator 14310-RBC-003 (updated part)	Fixes cold-start rattle. Free labor during timing chain inspection. Must be fresh before any tune.	~$280
	Subtotal		~$480-520

Cross-reference: Do VTC actuator DURING timing chain inspection (Phase 0). Plugs + coils can wait until just before Phase 1E (FlashPro first flash).

Phase 1: Immediate Installs (Already Purchased or Priority Purchase, No Dependencies)

These mods are already bought (or need to be bought first) and have no dependencies. Install order within this phase doesn't really matter -- they're all standalone.

Order	Mod	Time (DIY)	Standalone?	Notes
1A	Clutch hydraulics (CMC/CSC)	3-5 hrs	Yes	Not yet purchased -- priority buy. I need to do this before anything else. 170k miles = ticking time bomb. Improves pedal feel immediately. Also do: rear main seal (91214-RDA-A01), throwout + pilot bearings, fresh MTF while trans is out. See 07-Clutch-Hydraulics/overview.md cross-reference section.
1B	Engine mounts (Hybrid Racing 70A)	3-4 hrs	Yes	Reduces wheel hop, improves shift feel. NVH increase is noticeable but livable.
1C	Brakes (Hawk HPS + R1 Concepts rotors)	2-3 hrs	Yes	Full 4-corner upgrade. I need to bed the pads properly (follow Hawk's bedding procedure). Want this done before I have more power to stop.
1D	Strut bar (Megan Racing)	30 min	DEFERRED -- don't diagnose now	BC Racing BR coilovers (Phase 5) replace the strut-top geometry entirely. Whatever fits or doesn't fit today will be different once coilovers are in. Revisit after suspension overhaul. See 18-Strut-Bar/overview.md.
1E	Hondata FlashPro -- first flash	1-2 hrs	Yes	Register unit, flash base map, verify no CELs. I'm not tuning yet -- just getting it online. This enables datalogging for all future work. See 10-Hondata-FlashPro/tuning-workflow-and-maps.md for the tuning rules.
1F	Cooling system refresh	2-3 hrs	Yes	New phase addition. Koyorad aluminum radiator + fresh OEM thermostat + hoses. Must precede headers and E85. See 09-Cooling-System/overview.md.
1G	Wideband AFR install	1-2 hrs + bung welding	Tied to header install	AEM X-Series 30-0300 wideband. Bung welds into Skunk2 Alpha header collector during Phase 2B. See 11-Wideband-AFR/overview.md.

My planned order within Phase 1: 1A -> 1F -> 1C -> 1B -> 1E -> 1G (at header install) -> 1D (at coilover install)

- **1A Clutch hydraulics first** -- reliability risk at this mileage + RMS job shares labor
- **1F Cooling refresh** -- must be done before headers; not time-critical but same weekend as 1A works fine since I'm already under the car
- **1C Brakes** -- more power needs more stopping; independent of other jobs
- **1B Engine mounts** -- quality-of-life; no dependencies

- **1E FlashPro first flash** -- AFTER ignition refresh (Phase 0.5) is done; clean baseline for datalogging
- **1G Wideband** -- physically ties into the 2B header install; bung gets welded when the system is off the car
- **1D Strut bar** -- defer entirely until after coilovers (Phase 5)

Phase 2: Exhaust System (Must Be Done Together)

The valved exhaust and headers are interrelated. My plan is to install the exhaust cutout first, tune it, then add headers and retune.

Order	Mod	Time (DIY)	Standalone?	Notes
2A	Valved exhaust (QTP QTEC30 cutout)	4-6 hrs (shop recommended for welding)	Semi-standalone	Install cutout + dump pipe + reducer. Get both tunes working (valve open/closed). I can do this without headers using the stock manifold.
2B	Headers (Skunk2 Alpha + Berk HFC)	6-10 hrs DIY / 3-5 hrs shop	Must be done with 2A complete	PB Blaster the manifold studs 3-5 days before. I'll probably have a shop do this ($300-500 labor) -- seized stud risk at 170k is real.
2C	Hondata retune (dyno)	2-3 hrs on dyno	Must be done after 2A+2B	Tune both calibrations (daily quiet + sport loud) with headers installed. Verify AFR in both valve positions. Budget $300-500 for dyno tune.

Important: Do NOT install headers without planning the exhaust cutout first. Headers change the entire exhaust flow, and the cutout position needs to account for header length + HFC placement. I'm installing the cutout first (2A), validating it works, then adding headers (2B) and retuning everything (2C).

Stud prep is critical: I'm starting PB Blaster on the exhaust manifold studs at least 3-5 days before the header install. Heat cycle the car + spray daily. At 170k miles, those studs WILL be seized. I'll have new M10x1.25 stainless studs and copper-plated flange nuts ready.

Phase 3: Intake System (Must Be Done Together, After Exhaust)

The intake manifold and throttle body work together and need a retune after install.

Order	Mod	Time (DIY)	Standalone?	Notes
3A	Intake manifold (Skunk2 Ultra Street)	3-5 hrs	Paired with 3B	I could technically install this alone, but doing it with the TB at the same time saves a retune.
3B	Bored throttle body (stock DBW -> 66mm)	N/A (send out for boring)	Paired with 3A	I'll send the stock TB to Drag Cartel or HeelToe Auto for boring to 66mm. Lead time: 1-2 weeks. I need to order this BEFORE buying the manifold so both arrive together.
3C	Hondata retune (dyno)	2-3 hrs on dyno	Must be done after 3A+3B	Retune for the new manifold + TB. This is the "full bolt-on" tune. Budget $300-500.

Why after exhaust: The intake manifold tune depends on exhaust flow. If I tune the manifold first and then change headers, I'd need to retune again -- wasted dyno time and money. Headers -> intake manifold -> single comprehensive tune is more efficient.

Note: My K&N Typhoon cold air intake (already installed) stays in the system. The flow path becomes: K&N filter -> intake pipe -> bored TB -> Skunk2 Ultra Street manifold -> K20Z3.

Phase 4: Rotating Assembly (Standalone, After Full Bolt-On Tune)

Order	Mod	Time (DIY)	Standalone?	Notes
4A	Pulleys + Harmonic Balancer (ATI Super Damper + NST kit)	3-5 hrs	Yes	Standalone install. Requires crank bolt removal (impact gun needed, 250+ ft-lbs). Doesn't require a retune -- it's a mechanical change, not an airflow change.

Why after the tune: The pulley system doesn't affect tuning, but it does affect accessory drive belt routing. I'm installing it after the full bolt-on tune so I'm not juggling multiple changes at once. The ATI Super Damper is especially important at 170k miles -- the stock damper's elastomer is likely degraded by now.

Phase 5: Flex Fuel System (Final Power Adder)

Order	Mod	Time (DIY)	Standalone?	Notes
5A	Fuel pump (DeatschWerks DW300C)	1-2 hrs	Can be done early (standalone)	Drop-in replacement. I can install this anytime, even before injectors -- the stock ECU handles it fine.
5B	Fuel rail + injectors (Acuity + ID1050x)	2-3 hrs	Paired with 5C	Install rail, injectors, and ethanol sensor at the same time.
5C	Ethanol content sensor + fuel lines	1-2 hrs	Paired with 5B	Inline sensor before fuel rail. -6 AN PTFE braided lines.
5D	Hondata flex fuel tune (dyno)	3-4 hrs on dyno	Must be done after 5A+5B+5C	Flex fuel calibration: gasoline map + E85 map with real-time interpolation. This is the most complex tune in the build. Budget $400-600.

Why last: E85 is the biggest power adder (5-15 WHP) but also the most complex. I want the full bolt-on tune dialed in on pump gas first, THEN add flex fuel as the final layer. That way the tuner has a known-good base to build the flex fuel calibration on top of.

Clutch warning: Full bolt-on + E85 targets 230-250 WHP / 165-180 lb-ft. My Exedy 08806 Stage 1 is rated to ~280-300 WHP / ~200-210 lb-ft. I'm within the margin, but I'll be monitoring for clutch slip under hard acceleration in 3rd-6th gear after the E85 tune.

Phase 6: In-Car PC (Quality of Life -- Anytime After FlashPro)

Order	Mod	Time (DIY)	Standalone?	Notes
6A	LattePanda 3 Delta permanent install	6-10 hrs (spread over a weekend)	Yes	Can be done anytime after the FlashPro is registered and working. The PC replaces my laptop for Hondata management and adds exhaust valve control via Arduino GPIO.

Note: The LattePanda is a quality-of-life upgrade, not a performance mod. It doesn't need to happen in any specific order relative to power mods. That said, having it installed before the flex fuel tune would make the dyno session easier -- permanent Hondata connection, real-time monitoring, no laptop balancing act.

Master Timeline (Recommended Execution Order)

```
PHASE 0 -- MAINTENANCE (do immediately, before any mods)
  |-- Timing chain inspection + replacement if stretched
  |     (05-Timing-Chain-Maintenance/overview.md -- full parts + procedure)
  |-- Valve adjustment
  |-- Coolant flush (Honda Type 2 base)
  |-- Transmission fluid (Honda MTF)
  |-- Brake fluid flush (DOT 4)
  |-- Clutch fluid flush (DOT 4)
  |-- Serpentine belt (inspect/replace)
  |-- PCV valve
  |-- Water pump (inspect during timing chain; replace if weeping)
  |-- Valve cover gasket + plug tube seals (12030-PNC-000)
  \-- Oil filter housing O-ring (15301-PT0-003)

PHASE 0.5 -- IGNITION REFRESH (before any dyno tune)
  |-- NGK ILZKR7B-11S plugs x4
  |-- Honda OEM coils 30520-RWC-A01 x4
  \-- VTC actuator 14310-RBC-003 (bundle w/ timing chain -- free labor)
    See: 06-Ignition-Refresh/overview.md

PHASE 1 -- ALREADY PURCHASED / PRIORITY BUYS (install in order 1A -> 1G)
  |-- 1A: Clutch hydraulics (Hybrid Racing HYB-CMC-01-20 + RMS + throwout + pilot)
  |-- 1B: Engine mounts (Hybrid Racing 70A)
  |-- 1C: Brakes (Hawk HPS + R1 Concepts -- 4 corners)
  |-- 1D: Strut bar (DEFERRED until coilovers -- Phase 5)
  |-- 1E: FlashPro first flash (register + base map, no tuning yet)
  |-- 1F: Cooling system refresh (Koyorad + OEM thermostat/pump/hoses)
  \-- 1G: Wideband AFR install (AEM X-Series 30-0300 -- bung welds during 2B)

PHASE 2 -- EXHAUST SYSTEM
  |-- 2A: Valved exhaust (QTP QTEC30 + 3" dump)
  |-- 2B: Headers (Skunk2 Alpha + Berk HFC)        <- PB Blaster studs 3-5 days before!
  \-- 2C: Dyno tune #1 (both calibrations)

PHASE 3 -- INTAKE SYSTEM
  |-- 3A: Intake manifold (Skunk2 Ultra Street)
  |-- 3B: Bored throttle body (stock DBW -> 66mm)    <- Send out 1-2 weeks before
  \-- 3C: Dyno tune #2 (full bolt-on tune)

PHASE 4 -- ROTATING ASSEMBLY
  \-- 4A: ATI Super Damper + NST pulleys             <- No retune needed

PHASE 4.5 -- CYLINDER HEAD (aftermarket drop-in)     <- NEW phase
  |-- 4.5A: 4P Pro 156v2 All Motor Package + ARP 208-4701
  |-- 4.5B: Skunk2 Tuner Stage 2 cams + Skunk2 Pro cam gears
  |-- 4.5C: Install bundles with Phase 2 + 3 + VTC actuator (single teardown)
  \-- 4.5D: Dyno retune MANDATORY after head swap  <- ~$500-800
  See: 16-Cylinder-Head/overview.md

PHASE 5 -- SUSPENSION (can be done anytime, no tune dependency)
  |-- 5A: BC Racing BR coilovers + rear camber arms
  |-- 5B: Full bushing refresh (Energy Suspension + Hardrace)
  |-- 5C: Sway bars (Progress front + rear) + end links
  |-- 5D: Ball joints + tie rod ends + roll center adjusters
  \-- 5E: Professional 4-corner performance alignment  <- MANDATORY

PHASE 6 -- FLEX FUEL
```

```
  |-- 6A: Fuel pump (DeatschWerks DW300C 9-307-1008) <- Can be done earlier
  |-- 6B: Acuity 1913-PPL rail (satin purple) + Bosch EV14 550cc injectors (0280158117)
  |-- 6C: Continental 13577429 ethanol sensor + Innovate MTX-D converter + PTFE fuel lines
  \-- 6D: Dyno tune #3 (flex fuel calibration)       <- Most complex tune
  See: 19-Flex-Fuel-and-Fuel-System/acuity-fuel-rail-and-flex-fuel-build.md

PHASE 7 -- IN-CAR PC (anytime after FlashPro)
  \-- 7A: LattePanda 3 Delta permanent install

PHASE X -- FUTURE / ASPIRATIONAL (not a current purchase plan)
  |-- Built Motor Bottom End -- Wiseco + BC625+ + ACL Race + PRB-A01 oil pump
  |                            Only if target moves past ~270 WHP NA or going forced induction
  \-- See: 20-Built-Motor-Bottom-End/overview.md

AESTHETIC LAYER -- PURPLE COSMETICS (anytime)
  |-- Acuity 1927-PPL oil cap + K-Tuned DP2-K20-PRP dipstick + Acuity ESCO-T6 shift knob
  |-- Dress Up Bolts HON-027-TI engine bay kit + Skunk2 649-05-0120-PPL valve cover hardware
  |-- NST22015K-Purple alt + idler pulleys (? NOT crank -- conflicts with ATI Super Damper)
  |-- Samco Sport purple silicone hoses (custom order, 4-6 wk lead)
  |-- Radium 20-0270-00 coolant reservoir (custom purple anodize)
  |-- Powder-coat batch: valve cover + strut bar + fluid caps (BMC, PS, washer)
  |-- G2 G2165 purple caliper paint (do with rear brake install, cover all 4 corners)
  \-- See: 21-Purple-Cosmetics/overview.md, 04-Brakes/purple-calipers.md, 09-Cooling-System/overview.md
```

Dyno Tune Summary

I'll need **3 dyno sessions** total (or 2 if I combine exhaust + intake in one session -- but that's risky because troubleshooting gets harder if something doesn't tune well):

Session	After Installing	What's Tuned	Est. Cost
Tune #1	Exhaust (cutout + headers + HFC)	Both calibrations (daily quiet + sport loud), AFR, timing with headers	$300-500
Tune #2	Intake manifold + bored TB	Full bolt-on pump gas tune -- the main event	$300-500
Tune #3	Flex fuel system (injectors + ethanol sensor)	Flex fuel calibration (gas + E85 maps, real-time interpolation)	$400-600
Total tuning budget			$1,000-1,600

Tip: I need to find a tuner experienced with K-series Hondata BEFORE starting Phase 2. The Skunk2 Alpha + Ultra Street combo is the most common full bolt-on setup -- any good K-series tuner will have extensive experience with it. I'm going to book the dyno session before ordering parts so I'm not sitting around with untuned mods on the car.

Money Already Spent (Actual Costs -- incl. tax, shipping, insurance)

Item	Actual Paid
Exedy Stage 1 clutch + flywheel + Megan Racing strut bar (bundle)	$960.43

K&N Typhoon 69-1014TS intake	$504.17
Acuity short shifter + cable bushings	$622.20
Hybrid Racing 70A engine mounts	$657.15
R1 Concepts slotted rotors (WCPN2-59004)	$309.69
Hawk HPS brake pads (front + rear)	$222.45
Hondata FlashPro	$859.59
Continental ExtremeContact DWS06 Plus 215/45ZR17 91W XL (15572680000) -- purchased 2026-06-16	$715.00
Total Spent to Date	$4,850.68

Budget Summary (All Remaining Mods + Maintenance)

Maintenance (Phase 0)

Item	Cost (Parts)
Timing chain inspection/replacement	$20-500
Valve adjustment	$10 DIY
Spark plugs	$40
Coolant + thermostat	$45
Transmission fluid	$25
Brake + clutch fluid	$25
Serpentine belt	$25
PCV valve	$10
Oil change	$35
Phase 0 Total (parts)	$235-715

Modifications (Remaining -- Not Yet Purchased)

Phase	Mod	Est. Parts Cost
1A	Clutch hydraulics	~$326 (Hybrid Racing kit + Motul RBF 600)
2A	Valved exhaust	~$485 + $150-250 labor
2B	Headers + HFC	~$1,000-1,200 + $300-500 labor
2C	Dyno tune #1	$300-500
3A+3B	Intake manifold + bored TB	~$725
3C	Dyno tune #2	$300-500
4A	ATI Super Damper + NST pulleys	~$640-760
5A	Coilovers (BC Racing BR)	~$1,195
5B	Bushing kit + trailing arm bushings	~$428
5C	Sway bars (Progress front + rear)	~$637

5D	Ball joints + tie rods + roll center adj	~$390
5E	Performance alignment	~$150
5	Rear camber arms (Skunk2) + camber bolts	~$363
6A-6C	Flex fuel system	~$1,075
6D	Dyno tune #3	$400-600
7A	LattePanda PC	~$400-500
Remaining Mods Total		~$8,864-11,160

Grand Total Summary

Category	Amount
Already Spent	$4,850.68
Maintenance (Phase 0, remaining)	$235-715
Modifications (remaining)	$8,864-11,160
Remaining to Spend	~$9,099-11,875
Full Build Total (spent + remaining)	~$13,950-16,726

That assumes I'm doing DIY labor for most installs except headers (shop recommended), exhaust welding (muffler shop), and possibly suspension bushing pressing. Paying a shop for everything would add $2,500-5,000 in labor.

Mods That CANNOT Be Done Standalone

These have hard dependencies -- I should never install them without completing the prerequisites:

Mod	Requires First	Why
Headers (2B)	Valved exhaust cutout working (2A)	Cutout position depends on header/HFC length. Install order matters.
Hondata tune #1 (2C)	Exhaust complete (2A+2B)	Tune must account for header + cutout flow characteristics
Intake manifold (3A)	Full exhaust installed + tuned (Phase 2)	Changing intake after exhaust tune wastes a dyno session
Bored TB (3B)	Install with manifold (3A)	Single retune for both saves money
Hondata tune #2 (3C)	Manifold + TB installed (3A+3B)	Full bolt-on pump gas tune
Flex fuel tune (6D)	All fuel system components (6A+6B+6C) + full bolt-on tune (3C)	Flex fuel calibration is layered on top of the full bolt-on tune

Mods That CAN Be Done Standalone (Any Order)

Mod	Notes
Clutch hydraulics (CMC/CSC)	No dependencies. Do early.

Engine mounts (Hybrid Racing 70A)	No dependencies.
Brakes (Hawk HPS + R1 Concepts)	No dependencies.
Strut bar (Megan Racing)	No dependencies (once clearance resolved).
FlashPro first flash	No dependencies (just needs a laptop).
ATI Super Damper + pulleys	No dependencies. No retune needed.
Fuel pump (DW300C)	Can be done anytime -- stock ECU handles it fine.
Suspension (coilovers + bushings + sway bars)	No tune dependency. Can be done anytime. Do all at once -- car is already apart.
LattePanda PC	Only needs FlashPro registered first.

Last updated: 2026-04-16

Maintenance Parts Catalog -- OE + Aftermarket Part Numbers

> Every part number I need for 170k-mile Phase 0 maintenance and ongoing upkeep, in one place. OE numbers I buy from Honda dealers or majestichonda.com; aftermarket from AutoZone / RockAuto / Amazon. Prices are approximate -- I verify before ordering.

This catalog is the single source of truth for "what do I order." When something needs replacing, I check here first.

Engine -- Fluids

Fluid	Spec	OE Part	Aftermarket Equivalent	Capacity	Change Interval
Motor oil	5W-30 full synthetic (community), 0W-20 (Honda spec)	N/A (Honda doesn't brand oil anymore)	Mobil 1 5W-30 EP, Pennzoil Platinum 5W-30, HPL No-VII 5W-30	4.4 qt (w/ filter)	5,000 mi / 6 months
Oil filter	K-series short	15400-PLM-A02	Mobil 1 M1-110A, Purolator PL14610	1	Every oil change
Coolant	Honda Type 2 (blue, pre-mix)	OL999-9011	NONE -- use OEM only	~1.3 gal	50-60k mi
Transmission fluid	Honda MTF-3	08798-9031	NONE -- stock Honda MTF only, synchros hate GL-4/5	2.2 qt	30-40k mi
Brake fluid	DOT 4 LV	N/A	Motul RBF 600 (boiling point 594F dry / 421F wet)	~1 qt (bleed)	2 yr / 30k mi
Clutch fluid	DOT 4 LV	Same as brake	Motul RBF 600 (share bottle with brakes)	~0.5 qt	2 yr / 30k mi
Power steering	Honda PSF-S	08206-9002	NONE -- Honda PSF-S only	~0.8 qt	60k mi

Aftermarket oil preference: Mobil 1 EP 5W-30 or HPL No-VII 5W-30. I'm running 5W-30 over Honda's 0W-20 spec because of 170k-mile bearing clearances -- the slightly heavier oil keeps pressure up.

On E85: oil change interval drops to **3,000-4,000 mi**. Ethanol combustion byproducts contaminate oil faster. Non-negotiable maintenance tax for running ethanol.

Ignition

Part	OE P/N	Qty	Price	Notes
Spark plug (pump gas)	NGK ILZKR7B-11S	4	~$10/ea	Pre-gapped 0.043". Do NOT re-gap iridium.
Spark plug (E85 flex)	NGK ILFR7H	4	~$12/ea	One heat range colder for E85. Swap at flex-fuel tune session.
Ignition coil	30520-RWC-A01	4	~$40-50/ea OEM	Buy OEM Honda from dealer. Fakes on eBay are common.
VTC actuator (updated)	14310-RBC-003	1	~$280	Superseded revision with stronger spring. Cold-start rattle fix.
Cam chain tensioner	14510-RAA-A02	1	~$75	Replace with timing chain if stretch detected.

Cooling System

Part	OE P/N	Aftermarket	Price	Notes
Radiator (OEM)	19010-RRB-A51	Koyorad HH060063	OEM $180 / Koyo $250	Koyo is the move -- same money, aluminum end tanks, no cracking.
Thermostat	19301-RNA-305	Stant or Motorad 180F	~$25	Keep stock spec. Tune fan tables in Hondata instead of going low-temp.
Thermostat gasket/O-ring	19305-PR3-004	--	~$5	Fresh gasket every time.
Upper radiator hose	19501-RRB-A00	Gates 22918	~$30	OEM fits right.
Lower radiator hose	19502-RRB-A00	Gates 22919	~$30	OEM fits right.
Heater inlet hose	79721-SNA-A00	--	~$20	
Heater outlet hose	79725-SNA-A00	--	~$20	
Water pump	19200-RAA-A01	Aisin WPH-802	~$80-100	Replace proactively at timing chain service.
Water pump gasket	19222-PNA-003	--	~$8	Mandatory fresh.
Radiator cap	19045-PH7-003	NGK/Motorad 1.1 bar	~$15	

Fuel System

Part	OE P/N	Aftermarket	Price	Notes
Fuel filter (in-tank)	17048-SNA-A32	Hitachi / Beck Arnley equivalent	~$60	Replace before first E85 tank -- ethanol dislodges deposits.
Fuel pump assembly (OEM)	17040-SNA-A62	DW200 (9-201-1001) or DW300C (9-307-1008)	OEM $250 / DW300C $130	DW300C for E85.
Fuel pressure regulator	(in fuel pump assembly)	N/A	--	Returnless system. No separate FPR on FG2. Stock is fine for NA.
Fuel injectors (stock)	16450-RRA-A01	Bosch EV14 550cc or ID725	Stock ~$60/ea	Stock 310cc is too small for E85. See 19-Flex-Fuel-and-Fuel-System.
Fuel rail (stock)	16620-RBC-003	Acuity 1913-K20	--	Keep stock until flex-fuel conversion.

Drivetrain / Clutch Hydraulics

Part	OE P/N	Aftermarket	Price	Notes
CMC (stock)	46920-SVA-A03	Hybrid Racing HYB-CMC-01-20 (full kit)	OEM $120 / HR kit $290	Kit = CMC + CSC + braided line. Do the full kit.
CSC (stock)	46930-SVA-A02	(included in HR kit)	--	
Clutch pedal bushing	46991-S5A-004	Hybrid Racing solid	~$25	Often dry-rotted at 170k. Cheap fix.
Clutch line (rubber OEM)	46961-SNW-A51	(included in HR kit -- stainless braided)	--	Replace with braided. Stock swells.
Rear main seal	91214-RDA-A01	--	~$20	Replace while trans is out for clutch job.

Front crank seal	91212-PNC-003	--	~$15	Replace with pulley upgrade.
Flywheel bolt set	90001-PAA-A01	ARP 208-2802	OEM $20 / ARP $60	

Engine -- Seals / Gaskets

Part	OE P/N	Price	Notes
Valve cover gasket kit (includes plug tube seals)	12030-RRA-A01	~$40	Universal 170k leaker. Oil in plug wells = time.
Oil filter housing gasket/o-ring	15301-PT0-003	~$8	Weeps at 170k. Cheap while oil is drained.
Exhaust manifold gasket (header-to-head)	18115-RRB-A01	~$20 OEM	OEM only. Aftermarket gaskets blow.
Cat-to-manifold donut gasket (EP3)	18229-S6D-T31	~$10	Stock fitment for 8th gen SI too.
Cat-to-B-pipe donut gasket	18393-SR3-A01	~$10	
Oil drain plug crush washer	94109-14000	$1	Every oil change.

Suspension -- Wear Items (170k Replace Baseline)

Part	OE P/N	Aftermarket	Price	Notes
Front outer tie rod (passenger)	53560-SNA-A01	Moog ES800373	~$30	
Front outer tie rod (driver)	53540-SNA-A01	Moog ES800374	~$30	
Front inner tie rod	--	Moog EV800246 (pair)	~$60	
Front lower ball joint	51220-SNA-A03	Moog K500149	~$40	Or K-Tuned RCA if lowering 1.5"+
Front sway bar end links	51320-SNA-A02	Moog K80768 / K80769	~$35 each	Whiteline KLC139 adjustable if lowering 1.5"+
Rear shock top mount	52675-SNA-A03	--	~$25/ea	Replace during coilover install.
Rear upper control arm bushings	--	Energy Suspension 16.18114 kit	(in kit)	
Rear trailing arm bushings	--	Hardrace 6926	~$150 pair	Known 170k failure point.
Wheel bearing (front)	44300-SNA-951	Koyo or Timken equivalent	~$80/ea	Replace only if humming.

Motor Mounts

Part	OE P/N	Aftermarket	Price	Notes
Rear mount	50820-SNA-A82	Hybrid Racing 70A (in kit)	OEM $150 / HR kit $657	Kit replaces all 3.
Transmission mount	50850-SNA-A82	(in HR kit)	--	
Side mount	50820-SNA-A01	(in HR kit)	--	
Hasport 62A inserts (budget alt)	N/A	HSP-MO-CV-SI	~$110	Inserts only; keeps OEM rubber structure. Quieter than full mounts.

Exhaust Hardware

Part	OE P/N	Aftermarket	Price	Notes
Exhaust manifold studs (8x)	90201-PRB-A00 (M10x1.25 x 45mm)	ARP 208-4301 kit	OEM ~$15/set / ARP ~$90	ARP for header install -- worth it.
Exhaust manifold flange nuts (copper-plated)	90212-SA5-003	ARP 200-8321	~$15	Copper-plated resists seizing.
Header-to-cat spring bolt assembly	19103-RDJ-A01	--	~$15	Fresh springs + gaskets at install.
O2 sensor (upstream, front)	36531-RRB-A01	Denso 234-9049	~$100	Replace if check engine light issue.
O2 sensor (rear, downstream)	36532-RRA-A01	Denso 234-4133	~$80	Rear O2 gets repurposed for flex fuel signal -- may not need OEM spare.

Filters

Part	OE P/N	Aftermarket	Price	Notes
Engine air filter (stock box)	17220-RRB-A00	K&N 33-2342 (same form factor)	OEM $25 / K&N $50	N/A for me -- running K&N Typhoon.
Cabin air filter	80292-SVA-A01	Fram CF10133	~$15	Every 15k.

Accessory Belts

Part	OE P/N	Aftermarket	Price	Notes
Serpentine belt	56992-RNA-A04	Gates K070663	~$25	Inspect every 60-80k. Visual cracks = replace.

Small Wear Items

Part	OE P/N	Aftermarket	Price	Notes
PCV valve	17130-RTA-000	Beck Arnley 045-0316	~$15	60-80k interval.
Battery (Group 51R)	N/A (Honda doesn't sell)	Odyssey PC925LMJT (AGM), Optima Yellow, or Interstate MT-51R	~$120-220	AGM preferred for LattePanda constant draw.

What To Buy As Spares

These I keep on the shelf so I'm never stranded:

- **2x oil filter (15400-PLM-A02)** + matching oil (I get a case when on sale)
- **1x set of spark plugs** (the pump gas set)
- **1x air filter element** for the K&N (clean + re-oil kit)
- **1x PCV valve**
- **1x set of drain plug crush washers**
- **Assorted M10x1.25 copper-plated flange nuts** (exhaust)

Ordering Preferences

Vendor	Best For	Notes
Honda dealer (VIN quote)	Genuine OEM where it matters (VTC actuator, gaskets, seals, head fasteners)	Order by VIN to get the revised part.
majestichonda.com	OEM online, discounted	Same dealer parts at 20-30% off MSRP.
RockAuto	Aftermarket (Moog, Gates, NGK)	Consistently the cheapest. Slower shipping.
AutoZone / O'Reilly	Fluids, same-day needs	Borrow-a-tool program is clutch for pullers.
Amazon	K&N, Hawk, Hybrid Racing, Acuity	Prime shipping worth it for weekend jobs.
HeelToe Auto	Skunk2, Hardrace, harder-to-find K parts	Honda specialist. Excellent customer service.
K Series Parts	Specialty K-series bits	Best catalog for Hondata/FlashPro accessories.
Summit Racing	ATI, AEM, QTP	Usually cheapest on performance parts.

Last updated: 2026-04-18

Torque Specs -- K20Z3 / FG2 Quick Reference

> I got tired of hunting forums at 11pm for torque values. This is my cached reference for every fastener I'm likely to touch during this build. Specs pulled from the Honda factory service manual (FSM) and cross-checked with 8thcivic.com community threads.

Quick sanity check rule: if I'm not sure and can't find the spec, don't guess on anything load-bearing. OEM specs only. Aftermarket hardware follows aftermarket hardware spec (e.g., ARP), not the part it replaces.

Wheels

Fastener	Torque	Notes
Lug nuts	80 ft-lb	Star pattern, final pass. Re-torque after 50 miles on new wheels.

Engine -- Accessories and External

Fastener	Torque	Notes
Spark plug	13 ft-lb (156 in-lb)	Gentle. Never muscle iridium plugs. Apply anti-seize sparingly on threads only.
Ignition coil hold-down bolt	108 in-lb (9 ft-lb)	
Oil drain plug	29 ft-lb	New crush washer every time.
Oil filter	Hand tight + 3/4 turn	Honda A02 filter. Pre-lube gasket with fresh oil.
Oil filter housing bolt (center stud)	18 ft-lb	
Valve cover bolts	89 in-lb (7.5 ft-lb)	Star pattern from center out, light blue Loctite on threads.
Spark plug tube seal bolts	Same as valve cover	Replace seals if oil in plug wells.
Thermostat housing bolts	9 ft-lb	New gasket mandatory.
Water pump bolts	9 ft-lb	Fresh gasket. Align carefully.
Coolant bleed screw	Hand tight	Plastic -- do not over-torque.

Engine -- Serpentine / Accessories

Fastener	Torque	Notes
Crankshaft pulley bolt	181 ft-lb	Bolt is torque-to-yield on some years -- check FSM. Use impact or long breaker bar. Anti-seize on threads.
Alternator pulley nut	80 ft-lb	Hold alternator shaft. Loctite medium strength.
Alternator mounting bolts	33 ft-lb	
Idler pulley bolt	29 ft-lb	
A/C compressor mounting bolts	17 ft-lb	Do not loosen without recovering refrigerant first.

Exhaust

Fastener	Torque	Notes
Exhaust manifold studs to head	23 ft-lb (stud-in)	Anti-seize on threads going in, copper anti-seize on exhaust side. ARP studs follow ARP spec if using ARP kit.
Exhaust manifold flange nuts (to head)	23 ft-lb	Torque in 3 passes, star pattern.
Header-to-cat spring bolts	16 ft-lb	Do not overtighten -- springs compress to set pressure.
Cat-to-B-pipe flange bolts	25 ft-lb	New donut gasket every disassembly.
Resonator/muffler flange bolts	25 ft-lb	
O2 sensor	33 ft-lb	Anti-seize on threads. Do not get sensor element contaminated.
Wideband O2 sensor (AEM)	33 ft-lb	Same as stock O2.
QTP QTEC30 cutout flange bolts	25 ft-lb	Treat like any 3" exhaust flange.

Intake / Fuel

Fastener	Torque	Notes
Intake manifold bolts (to head)	17 ft-lb	Star pattern from center. Fresh gasket. Replace if warped.
Throttle body bolts	17 ft-lb	Fresh O-ring.
Fuel rail bolts (to IM)	108 in-lb (9 ft-lb)	Gentle -- crushing aluminum.
Fuel line banjo bolt	25 ft-lb	New crush washers both sides.
Fuel pump mounting bolts (tank assembly)	39 in-lb	Very gentle -- plastic tank threads.
PCV valve	Hand tight + 1/8 turn	Plastic threads.

Clutch / Transmission

Fastener	Torque	Notes
Flywheel bolts	87 ft-lb	Loctite 242 (blue) mandatory. Torque-to-yield on some Exedy kits -- check Exedy instructions.
Pressure plate bolts	19 ft-lb	Star pattern in 3 passes. Alignment tool for disc.
Clutch master cylinder mounting nuts	10 ft-lb	Firewall side.
Clutch slave cylinder bolts	16 ft-lb	At bellhousing.
Clutch line banjo bolt (at CSC)	25 ft-lb	New crush washers.
Transmission drain plug	29 ft-lb	New crush washer.
Transmission fill plug	33 ft-lb	
Axle nut	181 ft-lb	Front wheel bearing preload critical. Re-torque after 100 miles.
Bellhousing bolts	47 ft-lb	

Suspension -- Front

Fastener	Torque	Notes
Strut-to-knuckle bolts	83 ft-lb	Replace if removed -- stretch bolts.
Strut top mount nuts (to body)	29 ft-lb	
Coilover lock rings	Hand tight + spanner	Follow coilover manufacturer spec (BC Racing BR uses their spanner wrenches).
Lower ball joint pinch bolt	40 ft-lb	Replace if removed.
Lower control arm front bolt	54 ft-lb	
Lower control arm rear bolt (to subframe)	76 ft-lb	
Tie rod end castle nut	40 ft-lb + cotter pin	Turn up to next slot if needed, never back off.
Sway bar end link nuts	29 ft-lb	
Sway bar bracket bolts (to subframe)	29 ft-lb	
Subframe-to-body bolts	65 ft-lb	If subframe was lowered for header access, torque all 4 to spec.

Suspension -- Rear

Fastener	Torque	Notes
Shock lower bolt	40 ft-lb	
Shock top nut (to body)	22 ft-lb	
Upper control arm bolts	47 ft-lb	
Lower control arm bolts	47 ft-lb	
Trailing arm bolts (both ends)	54 ft-lb	
Rear camber arm lock nuts (Skunk2)	35 ft-lb	Follow Skunk2 spec.
Sway bar end link nuts	29 ft-lb	
Sway bar bracket bolts	29 ft-lb	

Brakes

Fastener	Torque	Notes
Caliper bracket bolts (front, 19mm)	80 ft-lb	
Caliper bracket bolts (rear, 12mm)	54 ft-lb	
Caliper slide pin bolts (12mm)	25 ft-lb	Fresh silicone grease on pins.
Rotor retaining screw (small cross-head)	7 ft-lb	Tiny -- do not over-torque. Anti-seize to prevent future seizing.
Brake line banjo bolt (at caliper)	25 ft-lb	New crush washers.
Bleed screw	Hand tight + 1/8 turn	Do not over-torque; will strip.

Engine Mounts

Fastener	Torque	Notes
Rear mount through-bolt	47 ft-lb	
Side mount (passenger) through-bolts	47 ft-lb	
Transmission mount	47 ft-lb	
Hybrid Racing 70A mount hardware	Follow Hybrid Racing spec	Ships with instructions. Generally 40-50 ft-lb.

Interior / Chassis

Fastener	Torque	Notes
Seat belt anchor bolts	32 ft-lb	Safety-critical. Never use heat.
Front seat mounting bolts (4x)	29 ft-lb	
Shift knob thread	Hand tight	
Shift base bolts	17 ft-lb	
Center console fasteners	Hand tight	Plastic clips mostly.

Head / Valvetrain (Reference -- I'm not opening this)

Fastener	Torque	Notes
Head bolts	Multi-step angle (FSM)	Replace bolts every time. Follow exact FSM sequence.
Cam caps	108 in-lb (9 ft-lb)	Final torque only -- sequence matters.
VTC actuator bolt	59 ft-lb	Loctite 262 (red). Hold cam with wrench on flats.
Valve adjuster lock nut	14 ft-lb	Gentle -- brass nut deforms if overtightened.

Fastener Replacement Rules (One-Time Use)

These bolts I replace every disassembly, no exceptions:

- Head bolts
- Flywheel bolts (if Exedy instructions say so)
- Strut-to-knuckle bolts (stretch bolts)
- Lower ball joint pinch bolt
- Axle nuts
- All cotter pins
- All crush washers (banjo bolts, drain plugs)
- All exhaust donut gaskets
- Valve cover gasket (if removed)

Thread Locker Reference

Strength	Color	Use
Low (222)	Purple	Small fasteners, easy removal
Medium (242)	Blue	General -- flywheel, pulley bolts, suspension
High (262/271)	Red	Permanent -- cam gear, VTC actuator. Heat required to remove.
Anti-seize	Copper/silver	Exhaust threads, spark plug threads, rotor screws

Default to medium (blue 242) unless the spec calls for something else. Never use red where future disassembly is needed.

Tools I Need for This Spec List

- 1/2" drive torque wrench, 30-250 ft-lb range (CDI, Tekton, or Snap-On)
- 3/8" drive torque wrench, 10-80 ft-lb range
- 1/4" drive torque wrench, 20-200 in-lb range (for the valve cover and small bolts)
- Calibrated within the last year (I can send it out for recal every ~2 years)
- A harmonic balancer puller (for the crank pulley work)
- An angle gauge for torque-to-yield head bolts (I'm not opening the head, but it's in the kit if I ever do)

Last updated: 2026-04-18

Baseline Datalogs -- A/B Tracking for Every Mod

> The point: before and after every tune change or hardware mod, I run the same standardized set of datalogs. Apples-to-apples. This way I can see what each change actually did instead of guessing from the seat of my pants.

Rule: Log BEFORE I change anything. No exceptions. Once a mod is on, I can't go back in time.

Why I'm Doing This

I'm about to spend a lot of money on a lot of different mods that all affect each other. Without baseline datalogs at each phase:

- I can't tell which mod actually moved the needle
- I can't catch a mistuned calibration early
- I have no way to argue with a dyno tuner if results come back lower than expected
- If something goes wrong later (a sensor goes bad, a misfire starts), I have no "healthy" reference

FlashPro's datalogger captures everything I need. It's free. The only cost is ~15 minutes per log session. No excuse to skip this.

Standard Log Set -- Every Time

Four logs, run in sequence, same day, same route:

Log 1 -- Cold Start / Idle Quality

- **Duration:** 3-5 min from key-on
- **Conditions:** Engine COLD (sat overnight)
- **Procedure:**

1. Key to ON, FlashPro connected, logger running
2. Start engine, let idle for 2 minutes without touching pedals
3. Blip throttle 3x to 3000 RPM, return to idle
4. Rev to 5000 RPM and hold 5 seconds

- **Looking for:** Cold start fuel trim, VTC actuator behavior (phase lock time), coolant temp rise rate, idle stability, misfire counts

Log 2 -- Cruise / Steady State

- **Duration:** 10-15 min
- **Conditions:** Engine fully warmed (coolant 185F+, IAT stable)
- **Procedure:**

1. 35 mph, 4th gear, level ground, 3 minutes
2. 55 mph, 5th gear, level ground, 3 minutes
3. 70 mph, 6th gear, level ground, 3 minutes
4. Light throttle transitions between speeds

- **Looking for:** Closed-loop fuel trims (STFT / LTFT), knock counts at cruise, timing behavior, MPG reference

Log 3 -- Part-Throttle Pulls

- **Duration:** 5-10 min
- **Conditions:** Fully warm
- **Procedure:**

1. 3rd gear, 2000 to 5000 RPM, 50% throttle pull x 3
2. 3rd gear, 2000 to 5000 RPM, 75% throttle pull x 3
3. Let cool ~30 seconds between pulls

- **Looking for:** Partial-load fuel tables, timing ramp, VTEC engagement behavior (below crossover)

Log 4 -- WOT Pulls

- **Duration:** 5-10 min
- **Conditions:** Fully warm, clear road, safe conditions
- **Procedure:**

1. 3rd gear, 2500 RPM to redline, 100% throttle x 3
2. 4th gear, 2500 RPM to redline, 100% throttle x 2 (if space allows)
3. Let cool fully between pulls

- **Looking for:** WOT AFR hitting target, knock counts (MUST be 0), peak timing, peak RPM, injector duty cycle

Total time: ~45 minutes per log session. I do this quarterly in stock form, and before/after every tune or hardware change.

What Parameters I Log

FlashPro lets me configure which parameters are captured. My standard log profile:

Category	Parameters	Why
Engine basics	RPM, Load (g/rev), TPS, MAP, VSS, gear	Context for every other reading
Fuel	STFT, LTFT, AFR (narrowband + wideband via analog input), injector duty cycle, injector pulse width	Tune health
Timing	Ignition advance (actual), timing correction, knock count	SAFETY -- knock count is the #1 watchdog
Cam / VTEC	VTC target, VTC actual, VTEC solenoid state, VTC error	Actuator health
Thermal	ECT (coolant), IAT (intake air), oil temp (if wired)	Heat soak detection
Electrical	Battery voltage, MIL status	System health
Flex fuel	Ethanol content %, fuel temp	Once flex fuel sensor is wired
Wideband	External AFR signal (via analog input)	Tuning ground truth

---

Baseline Milestones I'll Capture

Each milestone gets a full 4-log set + date-stamped filename. I'll keep logs in docs/logs/ (I'll create the folder once

I have my first session).

#	Milestone	What's Changed	Goal
0	Today -- stock ECU, current mods (intake + clutch + shifter + tires)	None (baseline)	My reference "before" state. Captures K&N intake on stock calibration.
1	Post-maintenance (Phase 0 done)	Plugs, VTC actuator, coils, fluids, thermostat, radiator	Clean healthy-engine baseline before any tune changes.
2	Post-FlashPro first flash (base map, no mods)	Community base map for stock + intake	Tune-only delta vs stock ECU.
3	Post-mounts + brakes + hydraulics + FlashPro	All Phase 1 mods installed	Ready-for-tuning baseline.
4	Post-exhaust (cutout + headers + HFC)	Phase 2 complete	Exhaust flow improvements.
5	Post-dyno tune #1 (both daily + sport)	Dual calibrations dialed in	Verify reliability-first rule. Knock counts = 0.
6	Post-intake mfld + bored TB	Phase 3 complete	Full bolt-on hardware.
7	Post-dyno tune #2 (daily + sport, pump gas)	FBO pump gas maps dialed	Pump gas final.
8	Post-ATI damper + NST pulleys	Phase 4	Rotating assembly.
9	Post-flex fuel install (hardware only)	Phase 6 hardware, running pump	Still on pump, new injectors/pump/sensor.
10	Post-dyno tune #3 (flex fuel final)	Daily + sport, gas + E85 interpolated	Final state.

Log File Naming Convention

YYYY-MM-DD_milestone_NN_log-type.fpdl

Examples:

- 2026-04-18_milestone-00_baseline_log-1-cold-idle.fpdl
- 2026-04-18_milestone-00_baseline_log-4-wot-pulls.fpdl

This way chronological sorting works automatically, and I can tell at a glance what each file represents.

Red Flags I'm Watching For

If ANY of these show up in a log, I pull over immediately and investigate before continuing:

Symptom in log	What It Means	Action
Knock count > 0 under sustained load	Engine is knocking	Pull timing immediately. Do not drive hard until resolved.
Knock correction > -2 deg	ECU is fighting knock	Pull over. Check fuel quality, sensors, timing tables.
STFT/LTFT > ±10%	Fuel trim running extreme	Check injectors, fuel pressure, O2 sensor, MAF (if applicable)
AFR leaner than 12.0 WOT (pump gas) or lambda > 0.82 (E85)	DANGEROUS lean condition	STOP. Pull over. Do not drive until fixed. Increase fuel.
VTC error > 3 deg sustained	VTC actuator failing	Replace actuator -- engine is not running commanded cam timing.
Injector duty cycle > 85% at any RPM	Injectors near limit	Upsize injectors or verify fuel pressure.

ECT > 220F sustained	Cooling system inadequate	Stop, let cool. Check coolant level, thermostat, radiator condition.
IAT > ambient + 50F at speed	Heat soak -- intake cooking	Confirm heat shield function; a normal hot-side intake runs IAT elevated but not extreme.
Battery voltage < 13.2V with alternator on	Charging system weak	Check alternator, belt, battery.

Sharing Logs with Claude / Tuner

FlashPro exports logs to CSV via FlashProManager -> File -> Export Datalog. That's the format I use for analysis.

For the Claude workflow (see feedback_tuning_priorities.md in my memory system if you're reading this through me): paste the CSV content directly, or attach the file. I'll analyze specific cells, flag outliers, and propose table edits. I cannot flash the ECU -- I can only recommend what to change.

For my tuner: .fpdl binary files directly. He loads them in his FlashProManager and sees everything native.

What a "Healthy" Baseline Looks Like on a Stock 8th Gen SI

Rough reference values so I know when something's off:

Parameter	Healthy Range (stock + K&N intake)
Idle RPM	740-780
Idle AFR (warm, closed loop)	14.5-14.9
Cruise AFR (45 mph, level, 5th gear)	14.5-14.9
WOT AFR (stock tune)	11.5-12.2 (Honda is intentionally rich)
Knock count at WOT (stock)	0 (should never see any)
ECT at cruise	185-195F
IAT at cruise (65F ambient)	85-110F (K&N runs hotter than stock airbox)
LTFT cruise	±5% (stock + intake -- intake tune will bring closer to 0)
Peak ignition advance, WOT	25-32 deg at 5000-6500 RPM
VTEC engagement	5800 RPM (stock), tunable to 4500-5500
Battery voltage, alternator charging	13.8-14.4

Anything outside these ranges in my current "stock tune + intake" state is a flag to investigate before I flash anything.

Created: 2026-04-18

Intake -- K&N Typhoon (Short Ram)

Already on the car

Part Information

Field	Value
Part	K&N Typhoon Short Ram Intake
Part Number	69-1014TS
Fitment	Honda Civic SI 2.0L (2006-2011)
Filter Type	K&N high-flow cotton gauze
Finish	Polished aluminum intake tube
Actual Cost (incl. tax/shipping)	$504.17

Note: K&N markets this as a "Cold Air Intake," but it's technically a **short ram intake** -- the filter mounts in the engine bay (hot side) near the exhaust manifold, not in a cold air location behind the bumper/fender. It does use the stock fresh air duct to feed cooler outside air to the filter area, though.

Why I Went With This

I replaced the stock airbox with the K&N Typhoon mainly because the K20 sounds absolutely incredible with an open intake. Beyond the sound, it gives me:

- Slightly improved airflow at high RPM
- A modest 3-8 HP gain (varies by dyno and conditions)
- Better throttle response feel

The stock airbox is honestly pretty well-designed on these cars, but I wanted the sound and the small gains. Worth it for how much more alive the car feels up top.

Installation Notes

- **Placement:** Hot side of the engine bay. The filter sits near the exhaust manifold side, so it gets exposed to engine heat.
- **Heat Shield:** The kit includes a partial heat shield, but it doesn't fully isolate the filter from engine bay temps.
- **Fresh Air Duct:** I kept the stock fresh air duct (routes air from behind the bumper/fender toward the filter area). It helps feed cooler outside air to the filter despite the hot-side placement.
- **Intake Air Temps:** Real-world IATs run higher than a true cold-side setup. The fresh air duct helps, but on hot days or in traffic, IATs will climb. I'll keep an eye on this through Hondata datalogging once I get it set up.

Tuning Impact

This is the only installed mod that affects the ECU tune.

The changed airflow characteristics (different tube diameter, less restriction, different sensor placement) mean the stock ECU calibration isn't perfectly matched anymore. The car runs fine on the stock tune -- closed-loop O2 correction handles the difference -- but once I flash a Hondata base map calibrated for an intake, it'll:

- Correct fuel tables for the intake's airflow characteristics
- Optimize timing for the slightly different charge characteristics
- Potentially adjust VTEC engagement for the modified intake flow

There's a base map for "stock + intake" on the 8th gen SI available from Hondata's downloads, which is what I'll start with.

Cold Air Intake Install Guide -- K&N Typhoon 69-1014TS

Status: ALREADY INSTALLED -- this is my reference doc for re-install, reverse-engineering, and future reference if I pull the intake for any reason (base-map flashing, MAF cleaning, sensor swap, shop inspection)

Reality Check

This is the easiest mod on the whole build. The actual wrenching is maybe 45 minutes of real work for someone who's never opened the hood. I'm writing this guide anyway because:

1. I'll probably pull the intake at some point to clean or re-oil the K&N filter, or to swap back to the stock airbox for a dealer visit or emissions concern. Re-install is less obvious the second time around when the novelty's worn off.
2. If I ever re-flash the ECU for a fresh Hondata base map, I want to know exactly what I installed so the tune matches reality.
3. K&N's own instructions are fine but thin. Honda-specific tips (the hidden plenum clip, the MAF sensor orientation, the fresh-air duct trick) come from 8thcivic.com threads, not the kit.

Difficulty: Beginner (2/10 on wrenching scale) **Time Budget:** 45-90 minutes. Budget 90 if it's the first time. **Shop-worthy?** No. Not even close. Pay a shop $100+ for this and you're paying them to not let you learn your own engine bay. The only fasteners I touch are hose clamps, 10mm bolts, and one sensor plug. If any of those scare me, I shouldn't be building the car at all.

Reliability-first framing on this mod: The ONLY thing that can hurt the engine here is (a) over-oiling the K&N filter and contaminating the MAF sensor, or (b) missing a vacuum/PCV hose connection that causes a lean code. Both are preventable with patience. Power gain is 3-8 HP and is the secondary reason I did this -- primary was the sound.

Prerequisites and Cross-References

Things That Must Be Done First

Do First	Why
Baseline datalog on stock airbox	Without a before/after, I can't tell if the new intake + tune actually improved anything. See docs/baseline-logs.md.
Confirm stock MAF sensor is clean and functional	A dirty or drifting MAF on the stock airbox is a pre-existing problem that'll haunt me on the new intake. Spray it with CRC MAF cleaner (NOTHING ELSE) before I swap.
Have the Hondata base map (stock + intake) downloaded	I don't need to flash it on install day, but I want it on hand. Car runs on stock calibration after intake -- closed-loop O2 correction handles the fueling shift -- but it's not optimized until I flash.

Things to Do at the Same Time (if the car is already apart)

Also Do	Why
Inspect PCV valve and hose	It's right there. 170k miles, cheap to replace (Honda PCV is ~$15). Cracked PCV hose causes lean codes that'll get blamed on the new intake.
Inspect and clean the throttle body	Throttle plate is exposed while intake tube is off. Quick wipe with throttle body cleaner + rag. Do NOT spray aggressively -- DBW actuator is behind the plate.
Inspect the fresh-air duct (behind the driver-side headlight, routes to airbox area)	This is the duct I'll reuse to feed cooler air toward the K&N filter. If it's cracked from 170k miles of heat cycling, I want to know now.

Mods That Change or Obsolete This Install Later

- **14-Intake-Manifold/** -- When I install the Skunk2 Ultra Street IM with a bored throttle body, the intake tube's throttle-body-side coupling may need a different adapter. K&N's coupler is sized for the stock 62mm TB; bored 66mm TB needs a different reducer or the next size up silicone coupler. Plan ahead: don't discard the K&N packaging.
- **13-Headers/** -- Headers radiate significantly more heat under the hood. Once headers go in, the hot-side placement of this intake becomes a bigger concern. I may add a better heat shield (DEI reflective wrap around the filter) or eventually switch to a true cold-side relocation kit. For now, stock fresh-air duct + partial heat shield is the compromise.
- **10-Hondata-FlashPro/tuning-workflow-and-maps.md** -- This is the one installed mod that affects the ECU tune. Base map flashes after this is on the car and baseline-logged.

Tools Required

Critical:

- 10mm socket + short extension, 1/4" drive ratchet (all airbox bolts are 10mm)
- Flathead screwdriver (hose-clamp duty -- the K&N kit uses worm-gear clamps, stock airbox uses spring clamps)
- Spring-clamp pliers (for stock MAF-to-airbox spring clamps; a stubborn-grip channel-lock works in a pinch)
- Long-nose pliers (for the PCV and IAT sensor connectors; the clips are tiny)
- Torque wrench, 3/8" drive, 10-80 ft-lb range (only for the intake-to-throttle-body clamp area, honestly -- most of this install is hand-tight territory)

Useful:

- CRC MAF Sensor Cleaner (red can, specifically marked for MAF use -- **NOT** carb cleaner, **NOT** brake cleaner, **NOT** throttle body cleaner)
- Clean microfiber rag
- Shop light or headlamp
- Zip ties (K&N kit comes with a few; I wanted more for routing the intake temp sensor wire neatly)
- Sharpie -- for marking clamp positions and hose orientations before removal
- Phone camera -- take "before" photos of every connector, every hose, every clip orientation. This is free insurance and I use it on every install.

Parts List

These are the real part numbers from overview.md. I'm not fabricating anything.

#	Item	Source	Part Number
1	K&N Typhoon Short Ram Intake kit	K&N direct / Summit / Amazon	69-1014TS
2	(Included in kit) K&N high-flow cotton gauze filter	K&N	Included
3	(Included in kit) Polished aluminum intake tube	K&N	Included
4	(Included in kit) Silicone couplers + worm-gear clamps	K&N	Included
5	(Included in kit) Partial heat shield	K&N	Included
6	(Included in kit) MAF sensor adapter / grommet (if needed)	K&N	Included

Total kit cost as purchased: $504.17 (including tax/shipping -- see overview.md).

Things I wanted on hand but are NOT in the K&N kit:

- CRC MAF Sensor Cleaner -- ~$8, AutoZone/O'Reilly
- Fresh PCV valve (if the old one looks tired) -- Honda OEM, ~$15
- Extra worm-gear clamps (two spare, just in case) -- ~$5 total

Step-by-Step

Phase A: Prep (10-15 min)

1. Park on level ground. Engine stone cold -- ideally overnight cool. Hot engine = burned hands on the intake tube and heat shield.
2. Disconnect the battery negative terminal (10mm). Intake swap doesn't strictly require this, but I'm touching the MAF sensor connector and I don't want a momentary short to fault-code the ECU.
3. Pop the hood all the way up, prop with the stock prop rod.
4. Unbox the K&N kit on a clean surface. Inventory every piece against the K&N instruction sheet. Missing a coupler or clamp and finding out mid-install is the worst way to learn that lesson.
5. Pre-install the MAF grommet into the new intake tube if the kit uses one (check K&N instructions -- the 69-1014TS uses a specific grommet orientation; the MAF element must face the airflow correctly).

Phase B: Remove Stock Airbox (15-20 min)

6. Locate the MAF sensor on the stock intake tube, between the airbox and the throttle body. There's a 5-pin electrical connector on top.
7. Unclip the MAF connector (squeeze the tab, pull straight off). Tuck the connector out of the way so it doesn't flop around and catch on something.
8. Locate the IAT (intake air temperature) sensor -- on the 8th gen SI, this is integrated into the MAF sensor on most trims, but double-check. If there's a separate connector, unplug it the same way.
9. Loosen the hose clamp at the throttle-body side of the stock intake tube (10mm or flathead, depending on clamp style). Slide the clamp back but don't fully remove -- it'll be easier to reuse as reference later.
10. Disconnect the PCV hose from the stock intake tube (squeeze, pull). Honda's PCV hose has a little clip; be gentle or it cracks.
11. Disconnect any other small vacuum lines on the stock tube -- typically there's one breather line. Sharpie a

mark on each hose and its port so I can verify orientation on reassembly.

12. Remove the three 10mm bolts holding the stock airbox to the inner fender / engine bay mount points.

13. Lift the stock airbox + intake tube assembly up and out. It'll come out as one piece. Watch for the fresh-air intake snorkel -- it slots into a duct at the front of the airbox and typically stays behind in the car (which is what I want).

14. Set the stock airbox in a box and label it "STOCK AIRBOX -- FG2 SI K20Z3 -- 2026-04." I'm keeping this forever. Emissions/dealer/resale/"what if" insurance.

Phase C: Install K&N Typhoon (20-30 min)

15. Verify the throttle body is clean. If it's gunky (170k miles, probably is), wipe the throttle plate perimeter gently with a rag lightly dampened with throttle body cleaner. DO NOT push the plate open with force -- DBW actuator is sensitive.

16. Install the K&N rubber coupler onto the throttle body inlet. Slide it on fully until it seats against the throttle body flange. Install a worm-gear clamp on the throttle-body side but leave it loose for now.

17. Install the second coupler clamp on the intake-tube side of the coupler -- also loose for now.

18. **Install the MAF sensor into the new K&N tube** in the correct orientation. K&N's instructions show an airflow arrow on the sensor -- it MUST point toward the throttle body, not toward the filter. This is the single most common mistake on a Honda intake swap and it throws an immediate CEL plus wildly wrong fueling. Check, double-check, take a photo before I tighten.

19. Torque the MAF sensor mounting screws to hand-tight + a gentle snug. These are plastic-into-plastic threads on the K&N tube and over-torque strips them. 25 in-lb range -- just firm.

20. Thread the new K&N intake tube into the coupler at the throttle body. Wiggle until fully seated.

21. Position the heat shield against the inner fender per K&N's diagram. It bolts to two of the same mount points the stock airbox used. Thread the bolts but do NOT torque yet -- the shield position needs to clear the filter with a small gap (5-10mm) so vibration doesn't rub through the filter cage over time.

22. Install the K&N filter onto the filter-side end of the intake tube. Slide fully on. Install the worm-gear clamp, tighten snug (not crushing -- I'm compressing rubber against aluminum, over-tight just bulges the rubber).

23. Verify the filter clears the heat shield, the inner fender, and any nearby hoses/wires by at least ~10mm. Reposition the shield if needed, then torque the shield bolts to snug (~9 ft-lb -- these are into captive nuts in the chassis, not load-bearing).

24. Route the **stock fresh-air duct** from behind the driver-side headlight to point toward the filter area. The duct terminates in an open end -- I'm not connecting it to anything, just aiming it at the filter so cooler outside air gets fed into the filter's vicinity. This is what makes the Typhoon a "short ram with fresh-air assist" instead of a pure engine-bay snorkel.

25. Reconnect the PCV hose to the K&N tube's PCV port. Double-check orientation against my sharpie mark from step 11.

26. Reconnect any secondary vacuum/breather lines to their respective ports. Every one of them, no exceptions. A missed line = lean code, rough idle, and me cursing at the dash.

27. Reconnect the MAF sensor electrical connector. It should click positively into place. Tug it lightly to verify latch.

28. Tighten both coupler worm-gear clamps at the throttle body. Snug but not crushing. The rubber should look evenly compressed, not ballooned out the sides.

Phase D: Reassemble and Sanity Check (10-15 min)

29. Walk around the engine bay and visually confirm: no loose hoses, no dangling connectors, no tools left behind, no rags forgotten. Twice.
30. Reconnect the battery negative terminal. 10mm, snug but not bottomed out into the post.
31. Close the hood, but don't latch it fully yet -- I'll want it back up for the first-start inspection.

First Start and Verification

32. Key to ON (NOT START) for 5-10 seconds. Listen for the fuel pump priming. No warning lights beyond the normal bulb check? Good. If I get a CEL immediately, it's the MAF connector not seated, the MAF oriented backward, or a vacuum line I missed.
33. Start the engine. Idle should settle within 5-10 seconds to around 800-900 RPM. A slight 30-second elevated idle (~1100 RPM) is normal for the cold-start strategy.
34. Pop the hood back up while the car idles. Listen carefully near the throttle body coupler and the MAF area. A hissing / whistling sound = vacuum leak. A thin puff of white "smoke" near the filter on first start = normal factory-oil burnoff from the filter. Goes away within a minute.
35. Feel around every hose junction -- no sucking air, no warm exhaust smell (shouldn't be any near the intake, but confirms nothing's obviously cross-connected wrong).
36. Let the engine come to full operating temperature (~190degF). Watch the IAT reading if I have an OBD-II reader plugged in -- it should climb from ambient to somewhere around 20-30degF above ambient at idle. If it's pegging at 150degF+ at idle with a cold engine, the sensor is in a weird spot or reading wrong. (Normal: IAT will climb significantly once I shut the engine off and heat soaks; this is the hot-side placement penalty.)
37. Short drive around the block, 10-15 minutes of mixed driving. No WOT yet. Listen for the new intake note -- distinctly different from stock, a deeper growl above 4500 RPM, borderline glorious above 6000.
38. Return home, pop the hood, re-check every clamp. Vibration may have loosened one. Snug as needed.
39. **After 50 miles:** re-torque / re-snug the throttle-body couplers. Rubber settles.
40. **After 100 miles:** final re-check. Should be stable from here on.

Post-Install Notes

MAF Recalibration and Fuel Trims

- **The car runs on the stock calibration after this swap.** The ECU uses the MAF signal in closed-loop operation, so it corrects fueling across most of the operating range. Expect **long-term fuel trims (LTFT)** to shift by 3-7% in the first 50-100 miles as the ECU re-learns. This is normal. If LTFT goes above +10% or below -10%, there's a real leak or a MAF issue to diagnose.
- **Short-term fuel trims (STFT)** will look bouncy for the first drive or two. They'll settle.
- **I will flash a Hondata base map ("stock + intake" community map for 8th gen SI) once I have FlashPro set up.** This is NOT urgent -- the car runs fine on stock calibration -- but it's the first tune I'll do. See 10-Hondata-FlashPro/tuning-workflow-and-maps.md.
- **Do NOT clear fuel trim adaptation (disconnect battery + throttle relearn)** unless I'm specifically told to by a tune workflow. Let the ECU adapt naturally.

Heat Soak Behavior to Expect

- **IAT will climb significantly at idle / in traffic / after shutdown.** This is the hot-side placement penalty -- the

K&N Typhoon's filter is in the engine bay, not behind the bumper. On a hot summer day in traffic, IAT can hit 140-160degF. The ECU pulls timing as IAT rises, so peak power suffers in heat-soak conditions. This is the tradeoff I accepted for the sound, the simpler install, and the lower cost vs. a true cold-side relocation.

- **Moving cruise recovers fast.** At highway speed, fresh air flowing past the filter drops IAT to 20-30degF above ambient within a couple minutes.
- **I'll monitor IAT via Hondata datalogs** once FlashPro is set up. If hot-day IAT is genuinely hurting performance, I'll revisit (DEI heat wrap around the filter, a proper cold-side relocation kit, or eventually a different intake design).

K&N Filter Oil Maintenance

- **DO NOT clean and re-oil the K&N filter until at least 50,000 miles** unless it's visibly caked. The factory oil is correct from the box. Early cleaning is how people over-oil and contaminate their MAF.
- **When it IS time to clean:** use K&N's Recharger kit (cleaner + oil). Follow K&N's instructions exactly -- spray oil application, not squirt. Light coat. Let it wick in for 20-30 minutes before reinstalling. Over-oiled filter + spinning engine = oil mist on the MAF element = drifting MAF readings = fueling weirdness.
- **Warning signs of MAF contamination:** hesitation off idle, LTFT drifting consistently negative (car thinks it's getting more air than it is), stumble on transition from closed to open throttle. Cure: pull MAF, spray CRC MAF cleaner (red can, MAF-specific), let dry fully, reinstall. Do not wipe the MAF element with anything. Do not use any other cleaner.

Common Mistakes

1. **MAF oriented backward.** Airflow arrow must point toward the throttle body. Immediate CEL and wildly wrong fueling if reversed. Check twice.
2. **Over-oiling the K&N filter at first service.** Factory oil is correct. First cleaning should not be for 50k+ miles and should be K&N's light spray application only.
3. **Missing a vacuum/PCV line on reassembly.** Sharpie every hose before removal. A forgotten breather line causes a lean code and rough idle, and I'll spend an hour chasing it.
4. **Over-torquing the MAF sensor screws into the plastic K&N tube.** Strips the threads. Hand-snug only.
5. **Crushing the silicone couplers with over-tight worm-gear clamps.** Rubber should look evenly compressed, not ballooned. Snug, not crushing.
6. **Skipping the fresh-air duct.** The stock fresh-air duct routes from the front of the car to the airbox area. It's easy to leave it dangling after removing the airbox. Aim it at the K&N filter -- this is half the reason the Typhoon works as well as it does despite hot-side placement.
7. **Flashing an aggressive intake tune on day one.** Even the community "stock + intake" base map should go on after a few days of baseline driving and datalogging. Reliability first.
8. **Assuming the stock MAF is fine.** At 170k miles, the stock MAF might be drifting. A quick CRC MAF spray on the stock sensor before it goes into the new tube is cheap insurance.
9. **Forgetting to reconnect the battery before trying to start.** Did this once. Felt stupid. It happens.

See Also

- overview.md -- parts, part numbers, why I picked the Typhoon despite the hot-side placement

- ../docs/torque-specs.md -- general fastener specs (no intake-specific torques are critical, but referenced for anything adjacent like throttle body bolts at 17 ft-lb)
- ../docs/baseline-logs.md -- the A/B datalog protocol I'm using to confirm this intake actually did what K&N says
- ../10-Hondata-FlashPro/tuning-workflow-and-maps.md -- the base map that makes this intake actually worth the gains
- ../13-Headers/install-guide.md -- future header install will significantly increase bay temps; may force a heat-management revisit on this intake
- ../14-Intake-Manifold/ -- future IM swap with bored TB may require a different K&N coupler
- ../06-Ignition-Refresh/ -- fresh plugs and coils must happen before any tune, including the post-intake base map

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Last updated: 2026-04-20

Short Shifter & Bushings -- Acuity Instruments

Already on the car

Part Information

Short Shifter

Field	Value
Part	Acuity Instruments 3-Way Adjustable Short Shifter
Fitment	8th Gen Honda Civic SI (2006-2011) 6-speed
Adjustment	3-way adjustable throw reduction

Shifter Cable Bushings

Field	Value
Part	Acuity Instruments Shifter Cable Bushings
Location	Transmission side (where the cables attach to the transmission)
Material	Stiffer than OEM rubber

Cost

Field	Value
Short shifter (base price)	$459.00
Cable bushings (base price)	$79.00
Actual Cost (both, incl. tax/shipping)	$622.20

Why I Went With These

The stock SI shifter has a long, kind of vague throw. I hated how imprecise it felt, especially trying to bang through gears quickly. The Acuity 3-way adjustable short shifter massively reduces shift throw, and the trans-side cable bushings replace the soft OEM rubber with stiffer units that eliminate the slop in the linkage.

Together, they made a night-and-day difference. The shift feel is so much tighter and more direct now. Combined with the fresh Exedy clutch and the Hybrid Racing 70A engine mounts I've got waiting to go in, this is going to be a complete drivetrain feel overhaul.

Notes

- The Acuity shifter is adjustable -- I've been experimenting with the three settings to find my preferred throw length
- Shorter throw = more effort per shift (leverage trade-off), but it feels way sportier
- The trans-side bushings are honestly the single most impactful improvement for shift precision -- if I could only pick one mod for shift feel, it'd be these

- Acuity also makes shifter-side (cabin-side) bushings, but the trans-side ones address the biggest source of slop
- No tuning impact -- purely mechanical

Short Shifter + Bushings Install Guide -- Acuity 3-Way + Cable Bushings

Status: ALREADY INSTALLED -- reference and re-install notes

Both parts are on the car. This guide exists so (a) if I pull the shifter to tweak throw or swap bushings I'm not re-learning from scratch, and (b) when I'm eventually in there for a clutch-hydraulics or trans job, the trans-side bushing procedure is already captured. See overview.md for parts and rationale.

Reality Check

Honestly, one of the easier wins on the whole build. The shifter swaps in under the center console -- bolts and a circlip. The trans-side bushings are a bit fussier because you're working by feel from below, but nothing here is intimidating. No fluids, no torque-critical engine fasteners, no harness work.

Metric	Reality
Difficulty	Easy-intermediate
Time (first attempt)	2-3 hours
Time (re-install or throw adjust only)	30-45 min
Worth DIY?	Yes. Shop labor is $150-250; you can do this in an afternoon.
Special tools	None exotic -- standard hand tools + snap-ring pliers
Reversible?	Fully. Keep OEM shifter + bushings bagged.

The one real bite: the trans-side cable end clips are small, under tension, and you're working by feel. Drop one onto the subframe crossmember and you'll waste 20 minutes with a magnet. Have spares.

Prerequisites / Cross-references

Do first

Do First	Why
Transmission in NEUTRAL, e-brake set	If the trans is in gear the cables are loaded and the trans-side ends fight you
Car fully cool	The selector lever area sits near exhaust
Acuity's install PDF open on a tablet	Throw-position diagrams I'm not reproducing here

"Free labor" opportunities -- do this while already under there

Reliability-first logic: if the car is apart for a bigger job, do the adjacent small jobs at the same time.

If I'm doing...	...also do
07-Clutch-Hydraulics (CMC/CSC -- any time the trans is out or clutch line is off)	Inspect/replace trans-side cable bushings. Already under there.

Clutch + flywheel replacement (done Phase 0)	Cables should have been inspected then; add an inspection task if I skipped it
Engine mounts (../04-Engine-Mounts -- Hybrid Racing 70A, pending)	70A mounts transmit more shift vibration. Stiffer shifter + bushings will feel sharper and noisier after that goes in. Expected, not a defect.

No conflicts. Thousands of 8th-gen SIs have this exact combo.

Tools Required

Critical:

- 10mm socket + ratchet (center-console fasteners)
- 12mm and 14mm sockets (shifter base + trans-side cable bracket)
- Needle-nose pliers (C-clips on pivot and cable ends)
- Snap-ring / circlip pliers, internal + external (HF $15 kit is fine)
- 1/4" drive torque wrench, 5-25 ft-lb (shifter base bolts -- 17 ft-lb per torque-specs.md)
- Plastic trim pry tool (blue plastic set; metal will mar the console)
- Headlamp (trans-side work is by feel)

Also useful:

- Jack + 2 jack stands or ramps (for trans-side access)
- Small parts magnet (for the clips I will drop)
- Zip-lock bags + Sharpie (console has multiple screw lengths -- bag each group)
- Silicone grease, NOT lithium (petroleum greases swell poly bushings)

No lift, no transmission jack, no specialty Honda tool.

Parts List

From overview.md.

#	Item	Source	Notes
1	Acuity Instruments 3-Way Adjustable Performance Shifter	Acuity direct / Hybrid Racing	8th Gen Civic SI 2006-2011 6MT
2	Acuity Instruments Shifter Cable Bushings (trans-side)	Acuity direct / Hybrid Racing	Trans-side set specifically; shifter-side is a separate kit I'm not running
3	Silicone grease, small tube	AutoZone / Permatex 22058	For bushing install; NO copper anti-seize or lithium
4	Spare OEM cable clips (optional)	Honda dealer	Cheap; exactly the clip I will drop

Combined cost: $622.20 (tax + shipping included, per overview.md).

No fluids, no gaskets, no consumables beyond the grease.

Step-by-Step

Phase A: Center Console Removal (20-30 min)

1. Trans in NEUTRAL, e-brake up, key OUT. Disconnect battery (-) if paranoid about SRS light -- this is nowhere near an SRS module, but it's 2 minutes of insurance.
2. Unscrew shift knob counter-clockwise (hand tight from factory).
3. Pry up the shift boot / bezel trim with the plastic tool. Spring clips all around, no screws. Work in from one corner.
4. Pull the ashtray / small storage tray; remove the 2 screws inside.
5. Lift the armrest, remove 2 screws in the cupholder well.
6. Pry up the rear console trim, remove the 2 screws underneath.
7. Slide console back and up. Watch for:
   - 12V outlet wiring (driver side) -- unclip at the back, don't pull on the wire
   - Shift boot surround trim -- separate piece on some trims
8. Set console aside on a towel. **Bag and label each screw set.**

Phase B: Shifter Base Access (5 min)

9. Shifter base is now exposed: metal bracket, stalk, two cables running forward to the trans, rubber dust boots.
10. **Take a photo.** Cable routing and clip orientation matter.
11. Peel back the cable boots at the shifter-side ends.

Phase C: Disconnect Cables from OEM Shifter (10 min)

12. Each cable terminates in a ball socket that clips onto a shifter-base pin via a small C-clip / circlip.
13. Remove retaining clips with needle-nose or circlip pliers (style depends on year -- check Acuity's PDF).
14. Pop each cable off its pin; let them hang -- they're captured at the firewall.
15. Clips into a labeled bag immediately.

Phase D: Swap the Shifter Base (15-20 min)

16. Remove the 4 base bolts (12mm or 14mm depending on production run). Factory Loctite, so they'll be stubborn -- breaker bar to crack, ratchet to spin.
17. Lift OEM shifter out. Heavier than expected.
18. Scrape gasket residue / dirt from floor-pan mating surface, wipe clean.
19. Drop in the Acuity shifter. Same 4-bolt pattern.
20. Start all 4 bolts hand-tight before torquing any -- lets the bracket self-align.
21. Torque to **17 ft-lb** in a cross pattern (docs/torque-specs.md "Shift base bolts"). Do NOT overtighten; the floor pan is thin steel and will dimple.

Phase E: Throw-Position Selection (5 min)

22. The 3-Way has three pivot positions:
    - **Position 1 (longest throw, closest to OEM):** ~20% reduction
    - **Position 2 (middle):** ~30-35% reduction
    - **Position 3 (shortest):** ~40% reduction
23. **Start at Position 2.** See "Tips on throw-position choice" below. Do NOT jump to Position 3 day one.
24. Remove the C-clip, slide the pivot pin, reposition the lever, re-insert pin, re-install clip.
25. Verify the clip is fully seated. If it pops off while driving, the shifter comes apart in my hand.

Phase F: Reconnect Cables (10 min)

26. Pop each cable end onto its Acuity pin.
27. Install retaining clips (new from Acuity if supplied).
28. **Tug-test each cable end.** A cable that pops off while driving = no shifting. Catastrophic.
29. Re-seat cable boots.

Phase G: Trans-Side Bushing Replacement (30-45 min)

30. Lift front, 2 jack stands on subframe.
31. Locate the shift cable bracket on the upper-front face of the trans (behind the intake manifold area). Two cables run to a selector lever on top of the trans and a second selector behind it.
32. Each cable end has:
    - An outer-cable retainer clip (housing to bracket)
    - An inner-cable ball clip (ball-end into the trans selector)
    - An OEM rubber bushing between ball and selector -- **this is the replacement target**
33. Work one cable at a time. Don't disconnect both or orientation gets scrambled.
34. Remove the outer clip first (needle-nose). Pocket it -- do NOT set it on the subframe.
35. Pop the inner ball clip. The cable end lifts off.
36. Pry out the old OEM bushing. On a 170k car it'll be deformed or torn. Wipe the selector hole clean.
37. **Thin film of silicone grease** on the new Acuity bushing -- too much attracts grit.
38. Press the new bushing into the selector hole. Firm thumb pressure. If I'm hammering, stop and check orientation.
39. Seat the cable ball back into the bushing.
40. Inner clip, then outer clip. Tug-test.
41. **Repeat for the second cable.**

These are trans-side bushings only -- the bracket isolators stay OEM. Acuity's shifter-side bushing kit is separate and I didn't buy it.

Phase H: Reinstall (15-20 min)

42. Double-check no tools or clips in the shifter tunnel.
43. Reverse Phase A: console in, 12V outlet reconnected, console screws hand-tight + snug (NOT torqued -- plastic threads).
44. Bezel / boot back on -- align tabs, press firmly all the way around.
45. Shift knob hand tight.
46. Reconnect battery (-) if disconnected.

Phase I: Calibration Test -- engine OFF, car stationary (5 min)

47. Key off, work the shifter through every gear slowly. Feel for binding, graunch, or a gate that won't find home.
48. **Every gear finds its gate cleanly, including reverse.** Reverse lockout (press-down) must still work.
49. If a gear feels blocked or the lever sits wrong at neutral, STOP. Cable end not seated, or cables crossed at the trans selector.
50. Engine running, clutch in: step 1->6 and back. Smooth, no fight.

First Drive Checklist

Before leaving the driveway:

- [ ] All 6 gears + reverse engage without force
- [ ] Reverse lockout still requires pressing down
- [ ] Shifter self-centers in neutral -- no drift
- [ ] No new rattles / buzzes from the console at idle
- [ ] No tugging / resistance as the engine rocks on its mounts at idle (would indicate a cable routed too tight)

First 5 miles:

- [ ] 1->2 shift clean (this is the most common "notchy after short-shifter" surface)
- [ ] 3->2 downshift clean (reveals bushing pre-load issues)
- [ ] No cable rub -- squeaky / rubbery noise on shift = boot or cable hitting bodywork or heat shield
- [ ] Park, get under the car, confirm trans-side bracket secure and no clips walked loose

After 50 miles:

- [ ] Re-torque shifter base bolts to 17 ft-lb. Standard practice for anything heat-cycled under load.
- [ ] Re-inspect trans-side clips. Any clip off its groove = replace.

Common Mistakes

Cable pre-load on the trans side

No torque spec on the bushings; they press in. But the outer-cable housing clip can get clocked "one slot over" from where OEM sat, leaving the cable under permanent tension. Symptoms:

- Shifter resting off-center at rest
- One gear easy, the other end of the same gate needs effort (e.g., 1st easy, 2nd fights)
- Accelerated bushing wear

Fix: reposition the outer-cable clip so the cable is relaxed with the shifter in neutral.

Over-torquing the shifter base bolts

17 ft-lb. Not 30, not "snug with my elbow." The floor pan is thin; dimple it and the bracket no longer sits flat, which throws off pivot geometry.

Picking the shortest throw on day one

Position 3 is genuinely short. Too short for a 170k-mile trans on most drivers. More below.

Losing the cable clips

Guaranteed if I'm not careful. 4-6 spare OEM clips on hand. Magnetic parts tray under the work area.

Grease errors

Dry install = squeak within 100 miles. Over-greased = grease collects grit, which abrades the bushing. Thin film, silicone only.

Skipping the photo before disconnecting cables

Crossed cables or a cable pinched against a heat shield edge will chafe through in months. Photo before, compare after.

Tips on Throw-Position Choice

This is where reliability-first actually matters -- wrong throw can accelerate synchro wear.

Position 1 -- Least aggressive (~20%)

- Still noticeably shorter than OEM but retains mechanical sympathy.
- Best pick if the trans is high mileage (mine is) and synchros feel tired in 2nd or 3rd.
- Lowest mis-shift risk, lowest chance of a missed 3->2 under brakes.
- **The reliability pick if I'm ever unsure.**

Position 2 -- Middle (~30-35%)

- Sweet spot for a street car that occasionally sees canyon / track.
- Where I've been running it.
- More effort per shift than Position 1 but the feel improvement is big.
- Low mis-shift risk IF technique is clean (fully declutched, deliberate gate motion).

Position 3 -- Max (~40%)

- Genuinely aggressive lever ratio. Any sloppy technique shows up as grinding or a refused gate.
- Honda's synchros are soft brass. Too-short-throw + notchy gearbox = accelerated synchro wear at minimum, bent synchro keys if I rush a 1->2 cold.
- Tempting for bragging rights. **Not worth it** on a 170k trans with no rebuild planned.
- Reserved for post-transmission-rebuild with upgraded synchros.

Rule I'm giving myself

Start at Position 2. If after 500 miles I'm missing gates or crunching, drop to Position 1. Do NOT escalate to Position 3 unless the trans has been freshened.

Shorter throw also means more lever effort (less mechanical advantage). If shift effort is climbing enough that I'm muscling the lever at speed, drop back a position -- that's leverage, not a trans problem.

See Also

- overview.md -- parts and why
- ../01-Clutch-and-Flywheel/ -- shifter feel is downstream of clutch feel
- ../07-Clutch-Hydraulics/ -- the logical "do this at the same time" future job
- ../04-Engine-Mounts/ -- 70A mounts will make this setup feel tighter and noisier once installed
- ../docs/torque-specs.md -- shift base bolts (17 ft-lb)
- ../docs/mod-order-and-maintenance.md -- where this sits in the master order

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Last updated: 2026-04-20

Clutch & Flywheel -- Exedy Stage 1 Organic + Exedy Lightweight Flywheel

Already on the car

Part Information

Clutch

Field	Value
Part	Exedy Organic Clutch Kit Honda Civic SI (2006-2011) Stage 1
Part Number	08806
Purchased From	shop.redline360.com
Type	Stage 1 organic disc with upgraded pressure plate
Rating	~280-300 WHP / ~200-210 lb-ft TQ reliable limit
Actual Cost (with flywheel + strut bar, incl. tax/shipping)	$960.43 combined

Flywheel

Field	Value
Part	Exedy Lightweight Flywheel
Material	Chromoly steel
Weight	Significantly lighter than stock flywheel (~8-9 lbs vs ~14-15 lbs stock)
Part Number	HF02
Purchased From	shop.redline360.com
Fitment	Honda Civic SI (2006-2011) K20Z3

Exedy 08806 vs OEM Replacement (KHC10)

Spec	KHC10 (OEM Replacement)	08806 (Stage 1 Organic)
Disc material	OEM organic	Upgraded organic
Pressure plate	Stock clamping force	Higher clamping force (upgraded)
Pedal feel	Identical to stock	Slightly firmer, still very streetable
Power limit	~230 WHP / ~170 lb-ft	~280-300 WHP / ~200-210 lb-ft
Daily drivability	OEM feel	Excellent -- organic disc is smooth, no chatter

I went with the 08806 over a basic OEM replacement and I'm glad I did. It gives me way more headroom above where I'll be with bolt-ons (215-225 WHP) while still being totally smooth for daily driving. No chatter, no drama -- just a better clutch.

Why I Went With These

At ~170k miles, my stock clutch was toast and needed to go anyway. Rather than just throwing in another OEM replacement, I figured I'd spend a little more and get something with real headroom. The Stage 1 has a beefier pressure plate so it'll hold up to whatever power I make with the bolt-on setup, and it's still completely streetable.

The lightweight flywheel was honestly a no-brainer while I was already in there. Swapping from the heavy stock single-mass unit to the lighter chromoly one made a noticeable difference:

- **Faster revs:** Less rotational mass means the engine spins up and down quicker. VTEC engagement feels sharper, rev-matching is way easier.
- **Better throttle response:** Less inertia to fight -- the engine just responds more immediately.
- **Quicker shifts:** RPM drops and rises faster between gears, and it makes the Acuity short shifter feel even more alive.
- **Slightly rougher idle:** The heavier stock flywheel smooths out combustion pulses. The lighter one lets more vibration through at idle. Totally normal, and honestly pretty mild with the Exedy unit.

Power Headroom

Power Level	Torque (est.)	Clutch Status
Stock (197 HP / 139 lb-ft)	139 lb-ft	Well within limits
Bolt-ons + tune (210-225 WHP)	155-165 lb-ft	Comfortable. Plenty of margin.
~250 WHP	~180 lb-ft	Still fine. No issues.
~280 WHP	~200 lb-ft	Upper edge of reliable daily use.
300+ WHP	210+ lb-ft	Slipping likely. Upgrade needed.

With all my planned bolt-ons (intake + headers + exhaust + tune = ~215-225 WHP), I've got ~60-75 WHP of headroom. I won't need to touch this clutch again unless I go forced induction at serious boost levels.

Notes

- The lightweight flywheel makes clutch engagement feel slightly more abrupt -- less rotational inertia to buffer things. Totally normal, and I got used to it within a day or two.
- Combined with my Acuity short shifter, Acuity trans-side bushings, and the Hybrid Racing 70A mounts I'm planning to install, the whole drivetrain feel is going to be completely transformed.
- The lighter flywheel technically makes low-RPM drivability slightly worse (less momentum to keep things spinning through lugging situations). I just avoid lugging it in high gear at low RPM.
- The Exedy HF02 flywheel is designed for the K20Z3 crank, so any future clutch upgrade will bolt right to it.

Clutch & Flywheel Install Guide -- Exedy Stage 1 (08806) + LW Flywheel (HF02)

Status: ALREADY INSTALLED -- reference doc

I did this job myself at ~170k miles. The original clutch was smoked and I wasn't paying a shop $1,200+ to drop the trans when I had a weekend and a buddy. This doc is my record of what I did and the playbook if I ever go back in there.

If you are about to do this: read the "While You're In There" section twice. The single biggest mistake on this job is pulling the trans and not refreshing every wear item while the driveline is open. You do not want to do this job twice.

Reality Check

This is a **transmission-out** job. That's the whole story. The actual clutch-and-flywheel swap is maybe 90 minutes of real wrenching -- the other 7+ hours is getting access and reassembling.

Difficulty for a first-timer: high. Not because any single step is technical, but because the job is long, parts are heavy and awkward, and there are a dozen small things (dowel pins, pilot bearing, input shaft alignment, bellhousing bolt lengths) that will bite you if you skip them.

Time budget:

- **DIY first-timer:** 10-12 hours. Full weekend. Nothing else that day.
- **DIY experienced:** 6-8 hours.
- **Shop labor:** 6-8 billed hours, ~$800-1,200.

Reliability-first rule for this job: every accessible seal, bearing, and bushing behind the bellhousing gets refreshed now. Every one I skip is a future trans-drop I'm signing up for. That's the heart of the "while you're in there" philosophy and it applies here harder than anywhere else on the car.

Prerequisites / Cross-References

Dropping the transmission is the single most-enabling disassembly on this car. Several unrelated jobs become effectively free labor the moment the trans is on the floor. **If any of the following are on the to-do list, do them now:**

Also Do	Why it's free labor now	Cross-ref
Clutch master + slave cylinder + braided line (Hybrid Racing HYB-CMC-01-20)	CSC is on the bellhousing. Doing it separately later = pulling the trans twice.	07-Clutch-Hydraulics/
Rear main seal (91214-RDA-A01)	Crank is exposed the moment the flywheel comes off. $20. 170k RMS weeps within a year guaranteed.	--
Pilot bearing (91006-RWC-003)	Lives in the crank center. Accessible only with flywheel off. ~$15.	--
Throwout bearing (22810-RWC-003)	Lives on the input shaft. Trans-out only. ~$40.	--

Fresh Honda MTF-3 (08798-9031, 2.2 qt)	Trans is drained for removal anyway.	--
Trans-side shifter cable bushings (Acuity)	If not already done.	02-Short-Shifter-and-Bushings/
Engine/trans mounts (Hybrid Racing 70A)	Trans mount has to come out anyway.	08-Engine-Mounts/

I did the first five on my install. Shifter bushings were already done on a previous session. I deferred the 70A mounts and regret it -- pulling the trans mount with stock hardware installed is the same effort as pulling it to install the Hybrid Racing piece.

While You're In There -- Read This Section Twice

The entire point of this job is to not do this job twice. Trans-out labor is what's expensive. The parts you touch while it's out are cheap.

Do now, every time, no arguments:

1. **Clutch hydraulics (CMC + CSC + braided line).** This is the single biggest parallel job. At 170k the stock CMC/CSC is on borrowed time -- it's a known 8th gen failure point. The Hybrid Racing HYB-CMC-01-20 is a complete drop-in kit and the CSC lives on the bellhousing. Doing this clutch-and-flywheel install without the hydraulics upgrade is leaving a trans-out job on the calendar for next year. **This doc and 07-Clutch-Hydraulics/ are meant to be read as a pair. I consider them one single job.**
2. **Rear main seal.** Pop old, press new, done. 15 minutes.
3. **Pilot bearing.** Slide-hammer puller or the grease-packing trick. Pre-lube the new one.
4. **Throwout bearing.** Verify the Exedy 08806 kit ships one; if not, OEM is $40.
5. **Fresh MTF-3.** 2.2 qt Honda MTF-3, no substitutes -- aftermarket GL-4 makes the synchros hate you.

Don't skip. Any of them. Ever.

Tools Required

Critical:

- **Flywheel holder tool.** K-Tuned or similar. Improvisable with a ratchet strap + bellhousing bolt but the real tool is $30.
- **Clutch alignment tool** (ships with Exedy 08806).
- **Transmission jack** or floor jack + wide wood cradle + helper.
- **Impact gun** (1/2"). Axle nuts are 181 ft-lb and 170k old.
- **Torque wrenches:** 1/2" (30-250 ft-lb) and 3/8" (10-80 ft-lb), both in-lb for small stuff.
- **14mm, 17mm, 19mm sockets** + wrenches, shallow and deep.
- **32mm axle nut socket.**
- **Pilot bearing puller** (or grease-packing method).
- **4-ton jack stands** on subframe pinch welds. Never under a jack alone.
- **2x new ARP-rated or OEM flywheel bolts** -- flywheel bolts are one-time-use per Exedy.

Also useful: creeper, headlamp, drip pan, magnetic tray, Loctite 242 (blue), 2+ cans brake cleaner, sharpie.

Parts List

#	Item	Part Number	Source
1	Exedy Stage 1 Organic Clutch Kit (disc + PP + TOB + alignment tool)	08806	shop.redline360.com
2	Exedy Lightweight Chromoly Flywheel	HF02	shop.redline360.com
3	OEM Rear Main Seal	91214-RDA-A01	Honda dealer
4	OEM Pilot Bearing	91006-RWC-003	Honda dealer
5	OEM Throwout Bearing (verify kit)	22810-RWC-003	Honda dealer
6	Honda MTF-3, 2.2 qt	08798-9031	Honda dealer
7	Loctite 242 (blue)	--	AutoZone

Paired install: Hybrid Racing HYB-CMC-01-20 clutch hydraulics kit -- see 07-Clutch-Hydraulics/.

Step-by-Step

Phase A: Disconnect and Lift (30-45 min)

1. Disconnect battery negative.
2. Loosen front lug nuts while car is on the ground.
3. Raise front, set on 4-ton jack stands on subframe pinch welds. Chock rear.
4. Remove both front wheels.
5. Remove engine bay splash shield.

Phase B: Axle Removal (30-45 min)

6. Remove 32mm axle nuts, both sides. Impact gun.
7. Remove lower ball joint pinch bolts (these are stretch bolts -- replace them).
8. Pop ball joints out of knuckles with pickle fork or separator.
9. Walk axle stubs out of hubs.
10. At the trans, pry axles out of the diff. Fluid drips -- drip pan ready. Plug the trans holes with rag or caps.
11. Set axles aside.

Phase C: Top-Side Disconnects (20 min)

12. Disconnect shifter cables from trans selectors (two cable ends, clips).
13. Disconnect reverse lockout switch, speed sensor, backup light switch, any other harness plugs.
14. Unbolt ground strap from bellhousing.
15. Remove starter: two wires + two mounting bolts.

Phase D: Support Engine + Drop Trans Mount (15-20 min)

16. Support the engine from above (support bar) or from below (floor jack + wood block under pan). **The engine cannot drop when the trans mount comes out.**
17. Position trans jack under the transmission, snug up.
18. Unbolt trans mount from chassis and from trans bracket. Remove mount.

Phase E: Unbolt Bellhousing (20-30 min)

19. Remove bellhousing bolts from under the car (6-7 of them, various lengths -- **phone picture which length goes in which hole**). Some are buried up top; long extension from below.
20. Unbolt CSC line bracket / clutch fork cover.
21. Leave two bellhousing bolts finger-tight until you've confirmed nothing else is attached.

Phase F: Lower the Trans (30-45 min)

22. Final walkaround: harness clips, ground straps, anything still bolted. **Anything.**
23. Remove last two bellhousing bolts.
24. Pull trans straight back off the input shaft dowels. **Do not let it hang on the input shaft.** If it won't release, wiggle side-to-side -- it's hung on dowel pins, not the input shaft.
25. Lower the jack, walk the trans out from under.
26. Beer checkpoint.

Phase G: Pressure Plate + Flywheel Off (20 min)

27. Install flywheel holder.
28. Unbolt pressure plate in a star pattern, a little at a time -- the diaphragm spring is under tension and uneven loosening warps the cover. 6 bolts.
29. Pull PP off, catch falling disc.
30. **Inspect input shaft splines on the trans.** Sharp edges + light film of grease = good. Rounded or pitted = trans needs a second look before reinstall.
31. Unbolt flywheel. 8 bolts, star pattern, staged. **One-time-use per Exedy.**
32. Lift flywheel off (heavy).

Phase H: Rear Main Seal + Pilot Bearing (30-45 min)

33. **Rear main seal:** pry old seal out carefully -- do not gouge the aluminum housing or crank sealing surface. Plastic trim tool or seal puller.
34. Brake cleaner + lint-free rag.
35. Press new seal (91214-RDA-A01) in square and flush. A large socket matching the seal OD, tapped gently with a dead-blow, does the job.
36. **Pilot bearing:** slide-hammer puller, or pack the cavity behind the bearing with wheel bearing grease, insert a snug wooden dowel, hammer -- hydraulic pressure ejects the bearing. Messy but works.
37. Brake cleaner in the bore.
38. Press new bearing in with a socket that contacts the outer race only (never the inner -- you'll brinell it). Drive until it bottoms.
39. Dab of fresh high-temp grease on the inner race.

Phase I: Flywheel Install (20 min)

40. Clean crank flange mating face with brake cleaner. No oil, no fingerprints, no old Loctite crumbs.
41. Place HF02 flywheel on the crank. **Note the dowel pin** -- the flywheel only goes on one way. If it doesn't sit flat, the dowel is missed. Do not force.
42. Drop of Loctite 242 on each new flywheel bolt thread. Do not drown.
43. Thread all 8 by hand. Snug in star pattern.
44. Install flywheel holder.

45. Torque in 3 passes, star pattern: 30 ft-lb -> 60 ft-lb -> **87 ft-lb final** per docs/torque-specs.md. Verify against Exedy instructions -- some kits spec torque-to-yield (angle gauge on final pass).

Phase J: Disc + Pressure Plate (15 min)

46. Identify which side of the disc faces the flywheel. Stamped marking ("FLYWHEEL SIDE" or similar). **Installing the disc backwards is the classic first-time mistake and means dropping the trans again.** Double-check.
47. Hold the disc against the flywheel, insert the Exedy alignment tool through the disc hub into the pilot bearing -- centers the disc.
48. Place pressure plate over disc, aligning dowel pins with flywheel holes.
49. Thread all 6 PP bolts by hand.
50. Torque in 3 passes, star pattern: 8 ft-lb -> 14 ft-lb -> **19 ft-lb final**. **Do not over-torque** -- these are small bolts and over-torquing warps the cover permanently.
51. Pull alignment tool. Disc should wiggle slightly between flywheel and PP.

Phase K: Throwout Bearing + Fork (10 min)

52. Inspect clutch fork pivot ball -- replace if sloppy.
53. Light dab of high-temp grease on pivot ball and TOB fingers. **Light.** Excess grease slings onto the disc and causes chatter forever.
54. Install new TOB on the fork or input shaft collar (kit-dependent -- the 08806 ships with a sleeve-style TOB).
55. Clip fork retainer so TOB doesn't walk off during reinstall.

Phase L: Trans Reinstall (45-60 min -- reverse of removal)

56. **If doing the CMC/CSC upgrade, install the new CSC on the bellhousing now** -- trivially easier with the trans out. See 07-Clutch-Hydraulics/.
57. Roll trans under car, raise to mating height.
58. Line up input shaft to disc hub **by eye** -- the alignment tool work means the disc is centered, so the input shaft slides home with gentle pressure.
59. **Do not hammer the trans onto the engine.** If it won't mate, the input shaft is misaligned or dowel pins aren't seated -- pull back and check.
60. Install all bellhousing bolts finger-tight, then torque to **47 ft-lb** cross-pattern.
61. Reinstall trans mount, torque to spec.
62. Remove engine support.

Phase M: Reconnect Everything (30-45 min)

63. Reinstall starter (wires + torque).
64. Reconnect all harness plugs.
65. Reconnect shifter cables to trans selectors. Verify each clips in solidly.
66. Reinstall axles, trans-side first (push until circlip clicks into diff -- tug to confirm). Reinstall hub-side, new lower ball joint pinch bolts to spec.
67. Torque axle nuts to **181 ft-lb.** Re-torque after 100 miles.
68. Refill trans with 2.2 qt fresh MTF-3. Fill to dripping out of the fill plug.
69. Splash shield back on.
70. Wheels on, lugs **80 ft-lb** star pattern.
71. Lower car.

72. **Bleed clutch hydraulics** -- full procedure in 07-Clutch-Hydraulics/. Do not skip.
73. Reconnect battery.
74. Key on, do NOT start -- verify clutch pedal feel. Pump 10-15 times, re-check. Mushy = bleed more.

First Start / First Drive

75. Start car. Idle 2 minutes. Listen at the bellhousing -- light TOB whir for the first few hours is normal; grinding or screeching is not.
76. Foot on brake, clutch in, shift through every gear: 1, 2, 3, 4, 5, 6, R. Each should engage cleanly.
77. Clutch out slowly in neutral -- pedal returns smoothly.
78. Empty parking lot test drive. Engage at low RPM. **Engagement point mid-pedal, clean progressive bite.**

First-drive checklist

- [ ] Engagement point mid-pedal, clean bite
- [ ] No chatter on launch
- [ ] No shudder at partial engagement
- [ ] No slip at light throttle in any gear
- [ ] Reverse engages without grinding
- [ ] No fluid dripping from bellhousing
- [ ] No MIL

Do not WOT for 500 miles. See bed-in below.

Bed-In Procedure (Per Exedy)

An organic disc needs heat cycles to mate properly to the flywheel and PP surfaces. Skip bed-in and you glaze the disc -- it'll slip or chatter forever.

Exedy's spec for the 08806:

- **First 500 miles: no WOT, no launches, no aggressive engagements.** Drive it like it's your grandmother's car.
- **Vary load.** Mix stop-and-go with highway. Multiple full heat cycles (warm up, many engagements, cool down) are better than one long highway cruise.
- **No holding the clutch at the engagement point.** Partial engagement burns the disc.
- **No slipping the clutch on hills.** Either fully engaged or handbrake.
- **After 500 miles:** gradually increase load. First short WOT at ~750 miles. Full-send after 1,000 miles.

I logged ~700 miles of mixed commuting before my first proper pull. No slip, no chatter, engagement still crisp. Bed-in worked.

Common Mistakes

1. **Disc backwards.** Disc is asymmetric. Backwards = clutch won't release = drop the trans again. **Check the stamped side marking.**
2. **Missing the flywheel dowel pin.** If the flywheel doesn't sit dead flat, the dowel is missed. Forcing bolts

cracks the flywheel or the crank flange.

3. **Forgetting to prime the pilot bearing.** No grease = dry chirp after 50 miles.
4. **Cross-threading flywheel or PP bolts.** Always hand-start every bolt. Ratchet only after all are finger-tight.
5. **Over-torquing the pressure plate.** 19 ft-lb final. Over-torque warps the cover and causes permanent chatter. Calibrated torque wrench, not feel.
6. **Skipping the alignment tool.** Eyeballing disc centering doesn't work -- trans won't mate, or it will mate and the clutch won't release.
7. **Dropping the trans onto the input shaft during removal or install.** Bends input shaft, kills pilot bearing. Keep the trans supported at bellhousing height.
8. **Reusing flywheel bolts.** Torque-to-yield = one-time use. Reused can fail catastrophically.
9. **Skipping the bleed before first start.** Air in system = pedal to floor = stuck in gear = dangerous.
10. **Cleaning flywheel face with WD-40 or oil.** Brake cleaner only. Oily film = immediate slip.

See Also / Cross-References

- overview.md -- why I chose the 08806 + HF02 combo, power headroom, daily-driver feel.
- **../07-Clutch-Hydraulics/ -- do this install at the same time. Single session. The CSC lives on the bellhousing and doing hydraulics separately means pulling the trans twice. Most important cross-reference in this document.**
- ../02-Short-Shifter-and-Bushings/ -- if trans-side bushings aren't already done, do them while the trans is accessible.
- ../08-Engine-Mounts/ -- trans mount comes out anyway; best opportunity for the Hybrid Racing 70A kit.
- ../docs/torque-specs.md -- every torque value used here, FSM-cross-referenced.
- ../CLAUDE.md -- overall build context.

Last updated: 2026-04-20

Brakes -- Hawk HPS Pads + R1 Concepts Slotted Rotors

Status

- **Front rotors + front pads:** ? **INSTALLED 2026-04-20**
- **Rear rotors + rear pads:** ? Still in the box -- installation pending
- **Bed-in procedure:** ? Pending (do after rears are in -- single combined bed-in on all four corners)

Part Information

Rotors

Field	Value
Part	R1 Concepts Slotted Black Disc Brake Rotor Set
Part Number	WCPN2-59004
Type	Slotted, black e-coated
Coverage	Full set (front and rear)
Actual Cost (incl. tax/shipping)	$309.69

Brake Pads -- Front

Field	Value
Part	Hawk HPS Street/Sport
Part Number	HB361F.622
Position	Front
Compound	HPS (High Performance Street)
Actual Cost (front + rear, incl. tax/shipping)	$222.45

Brake Pads -- Rear

Field	Value
Part	Hawk HPS Street/Sport
Part Number	HB145F.570
Position	Rear
Compound	HPS (High Performance Street)

Why I'm Doing This

Full 4-corner brake upgrade. At 170k miles, the stock rotors and pads need to be replaced anyway, so I figured I'd upgrade while I'm at it rather than just throw on another set of OEM pads.

- **Slotted rotors:** Better heat dissipation, help evacuate gas and debris from the pad surface, and maintain consistent bite as temperatures climb

- **Hawk HPS pads:** Higher friction coefficient than stock, improved initial bite, better fade resistance during spirited driving, and still totally daily-drivable (low dust, low noise)

For what I paid, getting both rotors and pads upgraded all four corners felt like a solid deal.

Notes

- Hawk HPS is a street-performance compound -- not a track pad. It's designed for daily driving with improved performance, not sustained track abuse. Perfect for what I need.
- The slotted rotors will make a slight hissing sound when cold, especially first thing in the morning. Totally normal, goes away once the pads warm up.
- **I need to do the bed-in procedure after installing.** Hawk has a specific pad bed-in process (a series of moderate stops from speed to allow the pad material to transfer to the rotor surface evenly). Skipping bed-in causes uneven pad deposit, vibration, and reduced performance. Not skipping that.
- No tuning impact -- purely mechanical

Purple Calipers -- Phased Plan

Owner's Decision

- **NOW (Option A):** purple caliper paint on stock calipers when rear pads/rotors are installed. Keeps the 3-day-old fresh front brake job intact.
- **FUTURE (Option C):** plan a full Wilwood / StopTech / Brembo big-brake kit for a later phase (~260+ WHP or ~40-50k miles from now). **Do not buy now.** Documented here so the decision is already researched when the time comes.
- **NOT DOING (Option B):** full Wilwood front kit right now -- rejected. Would throw away the Hawk HPS + R1 Concepts rotors installed 2026-04-20.

Part 1 -- NOW: G2 Purple Caliper Paint Kit

What to buy

Item	Part #	Source	Price
G2 High Temp Epoxy Paint -- Purple	G2165	Amazon / Walmart / G2USA direct	~$55-75
Prep supplies (brake cleaner 2x, red Scotchbrite, painter's tape)	--	Any auto parts	~$25
Nitrile gloves (hardener is nasty)	--	Hardware store	~$12
One kit covers all 4 corners.			
Total			~$95-120

G2165 is the official G2 Purple kit: 3.5 oz epoxy base + 0.5 oz hardener = 4 oz mixed, 980degF rating (overkill for Hawk HPS street use), 2-hour wheel-on cure, 24-hour drive cure.

Why paint instead of covers

Factor	G2 Paint (winner)	MGP Covers (rejected)
Cost	$95-120	$400-500 (custom PPG purple + Si logo)
Lead time	Buy today	12 business days custom color
Slotted rotor visible	YES (my R1 Concepts stays visible)	No -- cover hides caliper body
Install timing	Do during rear brake install	Separate labor day
Long-term durability	Epoxy won't flake if prepped correctly	Clips can rattle or pop off at speed
Reversibility	None (acceptable -- calipers are at 170k)	Fully reversible

Alternatives considered and rejected

- **Foliatec Deep Violet FT-2179** ($50-80, 572degF rating): great color, 3-part kit with cleaner + hardener. Slightly lower heat rating than G2 but still plenty for Hawk HPS street use. Brush application only. Buy this if G2 is out of stock.
- **Dupli-Color Plum Purple DE1640** ($10-15): NOT marketed as caliper-specific. High Temp Coating intended for engines. Used on calipers commonly but chips where pads contact bracket. Recoat every 2-3 years.

- **Plasti-Dip purple**: peels within 30 days on brake calipers. Don't.
- **VHT Caliper Paint**: no purple SKU in their SP line. Black/red/blue/gold/yellow/silver only.

In-car paint procedure (rotors + pads stay on)

1. Work one corner at a time. Car stays anchored on the other three.
2. Park flat, chock opposite wheels.
3. Loosen lugs 1/2 turn while wheel is still on ground.
4. Jack up, support on jack stand. **Never trust the jack alone.**
5. Remove wheel.
6. **Mask everything not going purple:**
   - Rotor face (both sides) -- slide a disc of scrap cardboard between pad and rotor, or mask the edge with tape
   - Brake pads and backing plates
   - Brake hose
   - Dust shield (optional -- if you want it clean)
   - Slider boots (heat-sensitive rubber -- do NOT paint)
7. **Clean caliper thoroughly** with brake cleaner. Spray -> drip -> spray again. Keep cleaner OFF the pad friction material.
8. **Wire-brush loose rust** on visible surfaces. Don't worry about behind the caliper -- invisible through the wheel anyway.
9. **Scuff with red Scotchbrite.** Kill the gloss, give the paint a key.
10. **Final brake-cleaner wipe.** Spotless. No oil, no fingerprints.
11. **Mix G2 kit** per instructions (hardener into paint bottle, cap, shake 2 minutes).
12. **Apply in thin even coats.** 2-3 coats minimum. 15-20 min tack time between coats.
13. **Don't drip onto the rotor surface.** If you do, wipe immediately with brake cleaner on a rag.
14. **Wait 2 hours** before reinstalling wheel.
15. **Reinstall wheel**, torque lugs to **80 ft-lbs** (FG2 spec). Move to next corner.
16. **First drive:** keep it gentle for 50 miles -- no hard stops. Let the epoxy finish fully cross-link under normal heat cycling.
17. **Don't power-wash** the calipers for 7 days.

Ideal timing -- paint when installing rear brakes

The rear pads (HB145F.570) + rotors (part of WCPN2-59004 set) are still in the box. Do them together:

1. Unbox rear pads + rotors
2. Jack up all 4 corners over a weekend, supported properly
3. Install rear rotors + pads
4. **While all 4 calipers are exposed**, paint them G2 purple
5. Let cure overnight
6. Wheels back on, bed in all four corners together with Hawk's factory bed-in procedure

Single weekend, one paint mix, zero wasted labor.

Part 2 -- FUTURE: Big-Brake Kit (Deferred)

Trigger points to revisit this section:

- Power crosses ~260 WHP at the wheels (post-FBO + E85)
- 40-50k miles from now when Hawk HPS + R1 Concepts wear out
- First track day where HPS fade is a problem
- Decision to go to 18" wheels

Wilwood Forged Narrow Superlite 6R -- front (RECOMMENDED)

Field	Value
Part #	140-11978
Price	~$1,400-1,700 base; +$250-400 for custom purple anodize
Caliper	6-piston Forged Narrow Superlite 6R
Rotor	13"+ (multiple options)
Wheel clearance	15" minimum; stock FG2 SI 17" clears (verify with Wilwood wheel template before purchase)
Color	Black/Red standard. Custom PPG purple via direct Wilwood order (~4-8 week lead, +$250-400)
Pad options	Wilwood Polymatrix BP-10 / BP-20 (street), wide aftermarket
ABS compat	Retains factory ABS
Brake lines	Stainless included (most kits)

Wilwood D154 -- rear (matches front)

Field	Value
Part #	140-11979-R (standard), 140-11979-DR (drilled rotors)
Price	~$964
Caliper	D154 2-piston
Compat	Retains OEM parking brake on FG2
Color	Black/Red standard. Custom purple anodize available.

Full Wilwood front + rear, purple anodize, installed: ~$3,000-3,300.

Alternative: StopTech ST-40 front

Field	Value
Part # base	83.434.4300
Suffix	.51 Black, .71 Red, .81 Yellow, .61 Silver
Price	~$1,827-2,175
Rotor	328x28mm 2-piece slotted AeroRotor
Colors	Red/Black/Silver/Blue/Yellow/Gold standard. No purple option from StopTech. Requires custom post-purchase paint.
17" wheel clearance	Yes (17" minimum)

Alternative: Brembo GT front (for FG2)

Field	Value
Part #s	1A1.6022A1 (drilled), 1A2.6022A1 / 1A2.6022A3 (slotted)
Price	~$2,500-3,200
Rotor	328x28mm 2-piece
Colors	Red/Black/Yellow/Silver standard. No purple from Brembo. Custom paint +$300.
17" wheel clearance	Yes
Rear kit	Brembo does NOT list a dedicated FG2 rear -- use Wilwood D154 or keep OEM

Skip list

- **AP Racing Essex Radi-CAL** -- no FG2 kit; custom bracket build only ($4,500-6,500). Track-only. Not for this build unless it becomes a dedicated track car.
- **TSX caliper OEM swap** -- cheap ($500 all-in) mid-step upgrade but not a big-brake kit. Worth noting as an option if budget is the constraint.

What gets thrown away at BBK time

- Hawk HPS pads front + rear (~$222 sunk; maybe $50-100 resale if low wear)
- R1 Concepts slotted rotors (~$310 sunk; maybe $50-100 resale if straight)
- Stock calipers (core value only, ~$20)
- Stock rubber brake lines (trash -- 17 years old at 170k, they need replacing anyway)
- G2 purple paint on calipers (gone with the calipers)

Effective sunk cost if BBK happens in Year 2-4: ~$380-430. **This is exactly why the recommendation is G2 paint ($95) instead of MGP custom covers ($500) for Option A** -- minimize throwaway.

Phased approach recommendation

- **Years 0-2, FBO + tune, <230 WHP:** Hawk HPS + R1 Concepts + G2 purple. Total brake investment: ~$620.
- **Years 2-4, post-full-BO + E85:** reassess. If Hawk HPS shows fade on canyon runs (IR-thermometer the rotor after a spirited descent), upgrade to Hawk HP+ or HPS5.0 pads ($200) before committing to a BBK.
- **Year 4-5 or at 260+ WHP or first track day:** full Wilwood FNSL 6R + D154 rear, custom purple anodize (~$3,000-3,300 all-in).
- **Track-only future:** AP Racing Radi-CAL only if the car becomes a dedicated track build.

Sources

Paint & covers

- G2 Purple G2165 -- Amazon
- G2USA instructions
- Foliatec FT-2179 -- A4H-Tech
- MGP Caliper Covers -- 2009 Civic fitment
- MGP customizer (custom PPG purple)

Big-brake kits (for future reference)

- Wilwood 140-11978 front FNSL 6R -- K-Series Parts
- Wilwood 140-11979-R rear D154 -- Speedway Motors
- Wilwood Honda Civic disc brake index
- StopTech ST-40 83.434.4300 Civic SI -- Import Image Racing
- Brembo GT 1A2.6022A1 -- Evasive Motorsports

Cross-References

- 04-Brakes/overview.md -- Hawk HPS + R1 Concepts (current install)
- 21-Purple-Cosmetics/overview.md -- overall purple theme
- docs/torque-specs.md -- 80 ft-lb lug spec for FG2

Last updated: 2026-04-20

Brake Install Guide -- Hawk HPS + R1 Concepts Slotted + G2 Purple Calipers

Status: Fronts installed 2026-04-20. Rears + G2 purple caliper paint PENDING -- planned as a combined weekend session so the paint and rear R&R share labor.

Reality Check Before I Start

Brakes are the most-commonly-DIY'd job on any car, and that's exactly why it's easy to get complacent. **This is the mod with the highest safety stake on the whole build.** A caliper bracket bolt that wasn't torqued, or a rushed bed-in that crystallizes the Hawk HPS friction layer, costs me a rotor or an intersection.

Time budget:

- Fronts: 2-3 hours DIY for both sides (actual 2026-04-20 session: ~2.5 hours)
- Rears: 2-3 hours DIY for both sides -- slightly slower because the rear pistons screw in
- Paint session (on top of rear install): add 1.5-2 hours of active work plus 24-hour air-cure before driving

Difficulty: Intermediate. Nothing here is rocket science, but it is unforgiving of skipped steps. No shortcuts. Rushed bed-in = warped rotors + crystallized pad surface = buy the set twice. Reliability-first: I would rather drive home on an unpainted caliper than an untorqued one.

Prerequisites / Cross-References

Things that need to be done FIRST or AT THE SAME TIME

Do First / Also Do	Why
New DOT 4 brake fluid flush simultaneous with rear install	Fluid is overdue at 170k. Rotating the rear pistons pushes old fluid back up the lines -- flushing at the end turns a contamination event into a maintenance win. Shares DOT 4 stock with ../07-Clutch-Hydraulics/.
Torque wrench, calibrated	Non-negotiable. Caliper bracket bolts are the fastener that turns a weekend mod into a crash. ../docs/torque-specs.md has every value.
Bed-in procedure after all four corners are done	Non-negotiable. Hawk's bed-in is what makes HPS compound deposit correctly on the R1 rotor. Skip it and I get judder + a hot-spot pattern I can't polish out.
Alignment re-check after rear install	Any hub play or loose suspension found during rotor R&R means a check is prudent. See ../17-Suspension/.

What this mod does NOT need

- No tuning impact. Pure mechanical change.
- No electrical. No sensors. No ECU concerns.
- No cross-impact on the rest of the build until the BBK question re-opens at 260+ WHP (see Defer Plan below).

Tools Required

Critical:

- 1/2" drive torque wrench, 30-150 ft-lb range (caliper bracket and lugs)
- 3/8" drive torque wrench, 10-40 ft-lb range (slide pins)
- C-clamp OR dedicated caliper piston compression tool (**fronts only -- rears need the cube tool, see below**)
- **Rear caliper piston rotation tool** -- 4-way cube or dedicated rear caliper windback tool. FG2 rear pistons **screw in, not compress.** Trying to C-clamp them will tear the piston boot.
- 14mm socket (caliper slide pin bolts)
- 17mm socket (front caliper bracket bolts -- double-check on my car, some FG2s use 19mm)
- 19mm socket (front caliper bracket bolts on most FG2 SI fronts; 12mm on rear brackets per torque-specs)
- Lug nut socket (21mm for OEM, check aftermarket)
- Breaker bar (18"+) for lugs that have been on a while
- Slide-pin silicone grease (NOT copper anti-seize -- silicone is what the rubber boots tolerate)
- Pad abutment / shim grease (Permatex Ceramic Extreme or equivalent)
- Wire brush
- Brake cleaner (2-3 cans -- I always underestimate this)
- Jack + 4-ton jack stands (never trust the jack alone)
- Wheel chocks
- Nitrile gloves
- Shop rags

Useful:

- Bungee cord or wire to hang the caliper off the strut (never hang a caliper by the flex hose)
- Rubber mallet or dead-blow hammer (for rotors that don't want to come off)
- Sharpie (mark left vs right pads so they go back in the same position if ever removed)
- Penetrating oil for the 7 ft-lb rotor retaining screw (tiny cross-head on the rotor face -- strips easily)

Paint session adds (from purple-calipers.md):

- G2 G2165 Purple High Temp Epoxy kit (3.5 oz base + 0.5 oz hardener, one kit does all 4 corners)
- Red Scotchbrite pads
- Painter's tape
- Cardboard scrap (rotor face masking)
- Extra nitrile gloves (G2 hardener is nasty)

Parts List

#	Item	Part #	Position	Source	Cost
1	R1 Concepts Slotted Black Disc Brake Rotor Set (full set, F+R)	WCPN2-59004	All 4	R1 Concepts	$309.69
2	Hawk HPS Street/Sport pads, front	HB361F.622	Front	Hawk distributor	(part of $222.45)
3	Hawk HPS Street/Sport pads, rear	HB145F.570	Rear	Hawk distributor	(part of $222.45)

4	G2 High Temp Epoxy Paint -- Purple	G2165	All 4 calipers	Amazon / G2USA	~$55-75
5	DOT 4 brake fluid (1 qt minimum for flush)	Motul RBF 600 or ATE Type 200	Reservoir	Any	~$18-25
6	Fresh crush washers (banjo fittings, if disturbed)	OEM	Lines	Dealer	~$5

See overview.md for rotor + pad rationale and purple-calipers.md for the G2 decision.

Step-by-Step

This install runs in three phases. Phase A is already done and documented below so I have a reference for when I do the rears. Phases B and C happen together.

Phase A: Front Install (COMPLETED 2026-04-20 -- documented for reference)

Done passenger-side then driver-side. Recording here so the rear session has a known-good procedure to mirror.

1. **Park flat, chock opposite wheels** (rears, chocked diagonally).
2. **Loosen front lug nuts 1/2 turn** while wheel is still on the ground.
3. **Jack up front corner, set on jack stand** on the pinch weld. Never trust the jack alone.
4. **Remove wheel.**
5. **Retract the front caliper piston BEFORE touching the bracket.** Loosen the brake fluid reservoir cap. Place an old pad or block of wood between the piston and rotor, compress slowly with the C-clamp. Watch the reservoir rise -- wipe any overflow immediately, brake fluid eats paint. (Skipping this step leaves pads jammed against the rotor and the caliper won't come off cleanly.)
6. **Remove the two caliper slide pin bolts** (14mm). Back them out fully.
7. **Slide the caliper body off the bracket.** Hang it on the coil spring with bungee cord or wire. **Never let it dangle from the brake hose** -- tears internal reinforcement that fails later at speed.
8. **Remove the old pads from the bracket.** Note orientation (inner vs outer, wear indicator tang direction).
9. **Remove the two caliper bracket bolts** (19mm on the fronts). Breaker bar -- torqued to 80 ft-lb for 170k miles.
10. **Remove the bracket.**
11. **Remove the old rotor.** Small cross-head retaining screw on the rotor face -- usually seized; penetrant + hand impact driver. If rotor is stuck on the hub (it will be), **tap the rotor hat evenly around the circumference with a dead-blow.** Never strike the rotor face.
12. **Wire-brush the hub face** down to bare metal. Scale here causes rotor runout.
13. **Thin layer of anti-seize on the hub face only.** Not on the rotor face, not anywhere near the pads. Purely prevents future seizing.
14. **Install the new R1 Concepts WCPN2-59004 rotor.** Slots point in direction of wheel rotation at outer edge -- R1 ships them marked "left" and "right." Thread the rotor retaining screw back in, torque **7 ft-lb** (anti-seize on threads -- it's tiny, do not over-muscle).
15. **Reinstall caliper bracket. Torque 80 ft-lb** per ../docs/torque-specs.md.
16. **Install Hawk HPS HB361F.622 pads into the bracket.** Wear indicator tang on inboard pad, facing front of car. Shims come pre-attached -- do not peel. Thin coat of pad slide grease on the abutment ears only, NEVER on

the friction face.

17. **Slide the caliper body back over the new pads.** Should seat cleanly if piston is fully retracted. If it doesn't, retract more -- forcing it bends the slide pins.
18. **Install caliper slide pin bolts** with fresh silicone grease on the pin shafts. **Torque 25 ft-lb.**
19. **Reinstall wheel.** Hand-thread all 5 lugs, lower until wheel bites the ground, torque star pattern to **80 ft-lb.**
20. **Repeat for driver side.**
21. **Pump the brake pedal 10-15 times with engine off** before driving. First presses will be soft as pistons travel out. Do not skip -- first press in the driveway otherwise goes floor-to-firewall.
22. **Check brake fluid reservoir**, top up to MAX with DOT 4 if needed.

Outcome 2026-04-20: Firm pedal after 12 pumps. No leaks at flex hose or bracket. Low-speed drive confirmed no pull, no grinding. Bed-in deferred until rears are in.

Phase B: Rear Install (PENDING -- planned procedure)

Same basic job as fronts with two critical differences: parking brake integration, and the rear pistons **screw in, they do not compress.** Wrong tool = torn piston boot + destroyed parking brake mechanism.

1. **Parking brake OFF.** Piston can't rotate against e-brake engagement. Car in gear on jack stands.
2. **Chock front wheels.**
3. **Loosen rear lug nuts 1/2 turn** on the ground.
4. **Jack up rear corner, set on jack stand.**
5. **Remove wheel.**
6. **Check e-brake cable tension at the caliper arm.** If tight, loosen the e-brake adjuster at the handle to free slack. Note current adjustment position to restore later.
7. **Disconnect e-brake cable from caliper arm** (small clip retainer).
8. **Remove caliper slide pin bolts** (14mm).
9. **Slide caliper off bracket, hang on wire** -- not on the hose, not on the e-brake cable.
10. **Rotate the rear piston INWARD using the cube tool.** The FG2 rear piston has two notches on its face; the parking brake mechanism screws it in/out. Turn clockwise (viewed from piston face) with gentle inward pressure until fully seated. **C-clamping a rear piston destroys the internal screw mechanism -- #1 rear-brake mistake.**
11. **Remove old pads.**
12. **Remove rear caliper bracket bolts** (12mm). Breaker bar.
13. **Remove bracket.**
14. **Remove old rotor.** Retaining screw, tap hat evenly if seized.
15. **Wire-brush hub face. Anti-seize on hub face only.**
16. **Install new R1 Concepts rear rotor** (from WCPN2-59004 set). Same directional check. Retaining screw **7 ft-lb**, anti-seize on threads.
17. **Reinstall caliper bracket. Torque 54 ft-lb** per ../docs/torque-specs.md.
18. **Install Hawk HPS HB145F.570 rear pads.** Shims pre-attached. Slide grease on abutment ears only.
19. **Slide caliper body over pads -- align piston face notches with the raised bumps on the inner pad backing plate.** If they don't line up, nudge piston with cube tool, retry. This keeps pad orientation correct as piston rotates under e-brake use.
20. **Torque slide pin bolts to 25 ft-lb.**

21. **Reconnect e-brake cable to caliper arm** (clip retainer).
22. **Reinstall wheel, torque lugs to 80 ft-lb star pattern.**
23. **Repeat for driver side.**
24. **Re-adjust e-brake** at the handle to restore original tension. Car should hold on moderate incline at 4-5 clicks.
25. **Pump the pedal 10-15 times** to extend rear pistons back out against new pads.
26. **Low-speed road test (5 mph parking lot crawl) BEFORE bedding in.** Apply parking brake -- does it hold? Pull firmly -- does it grab evenly without pulling? If e-brake doesn't hold, diagnose before higher-speed driving; usually means piston didn't fully seat back out or cable isn't reconnected. Do NOT bed in until e-brake confirms good.

Phase C: Purple Caliper Paint -- G2 G2165 (during rear install)

Fronts aren't painted yet, so this session paints **all four calipers at once.** The rears are already exposed from Phase B. Fronts get pulled a second time -- extra labor, but the alternative is mismatched cure or a third paint session.

Procedure integrates G2 instructions from purple-calipers.md. One corner at a time. Car anchored on the other three jack stands.

1. **After Phase B is reassembled and confirmed (through step 26)**, come back and paint.
2. **Per corner: remove wheel, remove caliper fully** (slide pins out, lifted off bracket). **Support on wire.** No weight on the flex hose.
3. **Wire-brush loose rust** on visible caliper faces. Not worrying about what's hidden behind the caliper.
4. **Scuff all paintable surfaces with red Scotchbrite.** Kill the factory gloss so the epoxy has a key to grip.
5. **Brake cleaner -- spray, drip, spray again.** Spotless. No oil, no fingerprints, no brake dust. Highest-leverage step for paint durability. Keep cleaner off the pad friction faces.
6. **Mask everything not going purple:**
   - Rotor face (cardboard disc between pad and rotor, or tape the edge)
   - Brake pads and backing plates
   - Brake flex hose
   - Bleeder screw (keep paintable-free for future flushes)
   - Slide pin boots (heat-sensitive rubber -- do not paint)
   - Dust shield (optional)
7. **Mix G2 kit per instructions.** Hardener bottle into the paint bottle, cap, shake 2 minutes. Pot life ~2 hours once mixed.
8. **Apply 2-3 thin coats with ~30 minutes flash between coats.** Thin is the word -- thick coats run and pool. Thin coats self-level and cure harder.
9. **No drips on rotor surface.** If it happens, wipe immediately with brake cleaner on a rag.
10. **24-hour air cure before driving.** G2 is a two-part epoxy that cross-links at room temp over 24 hours, then finishes curing under brake heat.
11. **During air cure, reinstall caliper** (slide pins torqued 25 ft-lb -- paint is on the body, not pin bores, no torque impact). Wheel on with lugs finger-tight so the corner is safe overnight.
12. **Next day: torque lugs to 80 ft-lb** before driving.
13. **Heat-cure happens during bed-in.** Intentional: Hawk bed-in cycles calipers from ambient through

~500degF repeatedly, which is exactly how G2 epoxy finishes its full cross-link. G2165 is rated to 980degF -- zero risk of bed-in damaging the paint. Natural cure cycle, which is why I paint the weekend of the install rather than a separate day.

On doing fronts retroactively later: considered and rejected. (a) Front pads are already bedded -- re-pulling fronts breaks the seated pad-to-rotor transfer, wasteful. (b) Separate paint day = separate full mask-and-prep cycle. (c) Rears would cure before fronts, mismatch visible until fronts caught up. All four together is the right call.

Hawk HPS Bed-In Procedure (from Hawk's published spec)

Do not skip. Do not abbreviate. Rushed bed-in = crystallized pad surface (glazed, loss of bite) + uneven pad deposit on the rotor (braking vibration I can't polish out -- buy rotors again).

Find an empty stretch of road where I can do this without stopping completely at either end.

1. **6-10 moderate stops from 40 mph down to 5 mph.** Firm but not panic-brake. Do not come to a full stop -- roll through and accelerate back to 40. Heats pads into operating range and deposits an even friction layer without letting the pad sit stationary on a hot rotor (which imprints).
2. **Cool down: drive 5-10 min at steady cruise** with minimal brake use. Airflow sheds the heat.
3. **2-3 harder stops from 60 mph down to 10 mph.** Closer to ABS-threshold firm, but not triggering ABS. Do not fully stop. Roll through, accelerate again.
4. **Drive 10 minutes at cruise to cool fully.** Do NOT park on hot pads -- that's exactly how a pad shape gets imprinted into the rotor.
5. **Normal driving from here.** Full performance available after ~200 miles of mixed driving as the transfer layer matures.

If I feel pulsing or vibration during bed-in: stop, cool completely, resume gently. First sign of uneven deposit -- continuing to hammer makes it worse.

First Drive Checklist

After pump-up pedal test and bed-in, on the first real drive:

- **No pull** to either side under normal braking
- **No pedal fade** over repeated stops -- pedal height and firmness consistent
- **No grinding, scraping, or rhythmic ticking** (slotted rotors will make a faint cold hiss -- that's normal, not grinding)
- **No burning smell persisting past 10 minutes** (slight smell during bed-in is expected; persistent = something is dragging)
- **Even pad deposit on rotor face when cool** -- uniform dark transfer. Blue patches = overheated spot; bare silver rings = uneven deposit, may need to redo bed-in
- **E-brake holds on moderate incline at 4-6 clicks** (rears)

If any fail, diagnose before highway-speed driving.

Common Mistakes (the ones I'm specifically avoiding)

1. **Compressing rear pistons with a C-clamp instead of rotating with the cube tool.** Destroys the parking brake screw mechanism -- replacement caliper territory.
2. **Forgetting to retract the piston before removing the bracket.** Old pads stay jammed against the rotor; swinging the caliper off rips the slide pin boots.
3. **Greasing the pad friction face.** Contaminated pads go in the trash. Grease on abutment ears and slide pins only.
4. **Not torquing lugs in star pattern.** Warps the rotor over the next 200 miles as hub face stress settles unevenly. Star pattern, 80 ft-lb final, every time.
5. **Rushing or skipping bed-in.** Glazed pads + uneven deposit = buying the set again in 5,000 miles. 20 minutes is cheap insurance on the whole mod.
6. **Hanging the caliper from the brake hose.** Invisible internal damage surfaces later as a hose failure at speed. Always wire or bungee.
7. **Painting before cleaning properly.** G2 won't bond to factory gloss + brake dust + 170k of road grime. Prep is 70% of a durable paint job.
8. **Torquing lugs with wheel hanging free.** Always bring wheel to the ground (or just barely loaded) so the hub isn't spinning against the wrench.

Defer Plan -- Big Brake Kit

Per purple-calipers.md, the Wilwood FNSL 6R front (140-11978) + D154 rear (140-11979-R) BBK is **deferred, not cancelled.** Trigger points to revisit:

- Power crosses ~260 WHP (post-FBO + E85 -- see ../13-Headers/, ../14-Intake-Manifold/, ../19-Flex-Fuel-and-Fuel-System/)
- 40-50k miles from now when Hawk HPS + R1 Concepts wear out
- First track day where HPS shows fade (IR-thermometer the rotor; upgrade pad compound first before BBK)
- Move to 18" wheels

The G2 purple paint is intentionally low-investment (~$95) to minimize sunk cost when the BBK happens. Custom purple anodize on the Wilwoods is the endgame; this is the interim that looks intentional without pre-committing $3,000. Throwaway if BBK lands Year 2-4: ~$380-430. Known number, accepted.

See Also / Cross-References

- overview.md -- parts selected, rationale for Hawk HPS + R1 Concepts over OEM refresh
- purple-calipers.md -- full G2 G2165 research, Wilwood BBK deferral plan, MGP cover rejection
- ../docs/torque-specs.md -- caliper bracket (F: 80 ft-lb, R: 54 ft-lb), slide pins (25 ft-lb), rotor screw (7 ft-lb), lug nuts (80 ft-lb), all cross-checked against Honda FSM
- ../07-Clutch-Hydraulics/ -- DOT 4 flush shares fluid stock with the clutch hydraulic refresh, plan them together
- ../17-Suspension/ -- alignment re-check after brake work is prudent, and the coilover + camber arm install is the right session to verify hub health at the same time
- ../21-Purple-Cosmetics/overview.md -- the purple calipers are the cornerstone of the overall purple cosmetic theme

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Last updated: 2026-04-20

Timing Chain Maintenance -- K20Z3 Preventive Service

Status: Planned -- Phase 0 Maintenance (do before any performance mods)

The Decision Rule

Honda does not specify a mileage interval for the K20Z3 timing chain. The factory service manual defines one objective measurement: **tensioner rod protrusion length**.

Rod extension	Action
>= 13.5 mm	Chain is stretched -- replace chain + guides + tensioner.
10-13 mm	On the edge. At 170k miles, reliability-first says replace the whole set now while in there.
< 10 mm	Chain is healthy. Replace the tensioner only ($75) as insurance, reuse chain and guides.

My plan: pull the cover (because I'm doing the VTC actuator anyway as Phase 6 Ignition Refresh). Measure. Decide on the spot.

Why It Makes Sense Now

The VTC actuator (Phase 6 Ignition Refresh) lives behind the timing chain cover. Every labor hour spent accessing the VTC is the same labor hour to inspect the chain. Marginal parts cost for the full chain service while the cover is already off: **~$200 in chain + guides + tensioner**. Marginal labor: under an hour. It would be a mistake not to bundle these.

No Symptoms -- But Going In Anyway

Cold start clean. No rattle. No retard DTCs. No misfires. That's a good sign, not a guarantee. Stretched chains usually announce themselves with cold rattle before catastrophic failure, and the 13.5 mm tensioner measurement catches the rest. At 170k with **plans for a tune**, a drop-in head eventually, and E85 on the horizon, I need a known-clean mechanical baseline. Hidden chain stretch would skew every cam timing measurement during dyno tuning.

Parts List

Core Timing Parts (OEM Honda -- do not cheap out here)

Part	OEM P/N	Qty	Est. Price	Notes
Timing chain (cam)	14401-PNA-004	1	$55-75	IWIS-made for Honda, 86 links, with 3 colored alignment links
Cam chain auto-tensioner	14510-PRB-A01	1	$55-65	The "13.5 mm rod" part. Always new with a new chain.
Cam chain tensioner arm (pivoting guide)	14520-PNA-003	1	$35-45	Plastic wears
Cam chain guide (fixed)	14530-PNA-003	1	$25-35	Plastic wears
Oil pump chain	13441-PNA-004	1	$30-40	Lives behind the cam chain -- easy replace while in there
Oil pump chain tensioner	13450-RAA-A02	1	$25-35	Cheap insurance

Oil pump chain guide	13460-PNC-004	1	$25-35	Plastic wears
VTC actuator (REVISED updated part)	14310-RBC-003	1	$250-300	Covered under Phase 6 Ignition Refresh budget, not double-counted here
Crank pulley bolt (torque-to-yield, ONE-TIME-USE)	90017-RWC-A01	1	$10-15	Do not reuse the old one.
Front crank main seal	91214-RZY-A01	1	$15-20	Press into cover before installing cover

Gaskets & Seals (all new, every time)

Part	P/N	Price	Notes
Valve cover gasket kit (incl. plug tube seals + grommets)	12030-PNC-000	$25-40	Always the KIT, not bare gasket alone
Water pump gasket (if pulling WP at same job)	19222-PNA-003	$8-10	If bundling with cooling system job
Oil pan gasket (only if dropping pan -- optional)	11251-RAA-A02	$15	Skip if you don't drop the pan

Consumables

Item	P/N	Use
Hondabond HT (high-temp RTV)	08718-0004	Timing cover sealing corners
Moly 60 assembly grease	08798-9010	VTC actuator vanes, cam nose, crank nose
Engine oil (4.4 qt 5W-30 full synthetic)	--	Fresh on first start
Oil filter	15400-PLM-A02	Always new
Coolant	Honda Type 2 OL999-9011	Top off if WP is swapped
Dielectric grease	--	Coil boots on reinstall

Optional Upgrades -- My Verdict

Upgrade	Verdict
BillETomic / Blackworks / K-Tuned billet tensioners	Skip. Designed for race/boost. On a 240 WHP street build, OEM 14510-PRB-A01 is correct. Billet can over-tension and accelerate guide wear.
TODA Heavy Duty tensioner	Skip. Race-tier only.
Adjustable cam gears	Skip unless dyno-tuning serious big cams later. VTC actuator already varies intake cam 50deg.
IWIS aftermarket chain vs OEM Honda	Same chain. IWIS is Honda's OEM supplier. Buy OEM packaging.
Aftermarket "performance" guides	Skip. Stock guides are fine.

Tools Required

Specialty Tools

Tool	Purpose	Cost
Honda K-series cam alignment tool (Lisle 37950)	Locks both cams at TDC. NON-NEGOTIABLE -- without it the cams spring-rotate under valve pressure and hit valves.	Lisle 37950 ~$35-45

Crank pulley holder tool	Holds crank for bolt breakaway. Impact wrench is alternative.	AutoZone loan-a-tool or Evergreen HCPH1 ~$35
Large breaker bar (24"+)	Crank bolt is 181 ft-lb factory and can be 300+ after 170k heat cycles	Own
3/8" torque wrench, in-lb range (0-250 in-lb)	Valve cover + timing cover bolts are 87 in-lb and 106 in-lb. Big torque wrench can't measure this low accurately.	Harbor Freight Icon or similar ~$60-120
Torque angle gauge	Crank bolt is 37 ft-lb + 90deg angle	~$15-25
Magnetic pickup / flexible grabber	Dropped bolts WILL happen	HF ~$10
Paint marker (white/silver)	Backup chain-to-gear marks	Any

Standard Tools (should already have)

- Metric sockets 10/12/14/17/19mm (shallow + deep)
- 1/4", 3/8", 1/2" ratchets + extensions + swivel joint
- Metric combination wrenches, especially 10/12/14/17mm
- Allen/hex bits (some M6 hex-head cover bolts)
- **Plastic or brass gasket scraper (NOT steel)** -- steel gouges aluminum = permanent leak
- Shop light, rags, brake cleaner
- Engine oil drain pan
- Floor jack + stands + chocks
- Small block of wood (support engine via oil pan)
- Zip ties (hold chain against cam gear during swap)

Step-by-Step Procedure

See install-guide.md for the full walkthrough. Summary:

Phase A -- Prep (30-45 min): level ground, battery negative off, 4-24 hr cooldown, remove K&N intake, engine cover, drain coolant (if WP swap), remove alternator + A/C compressor + PS pump bracket.

Phase B -- Support engine + remove passenger mount (20 min): floor jack + wood block under oil pan, take up weight, remove upper passenger mount + engine bracket.

Phase C -- Valve cover off (30 min): unplug coils, unbolt cover in spiral-outward pattern, inspect cams.

Phase D -- Set TDC cyl 1 (15 min): rotate crank clockwise via 19mm on crank bolt, align pulley mark to pointer, verify both cam gear triangle marks point up parallel to head surface, mark chain-to-gear with paint marker.

Phase E -- Crank pulley off (20-40 min): impact or breaker bar, remove bolt (discard -- one-time-use), slide pulley off.

Phase F -- Timing chain cover off (45-60 min): 13 bolts of varying lengths -- document positions, remove 3 oil-pan-face bolts from below, score Hondabond, tap off, clean both mating surfaces spotless.

Phase G -- MEASURE TENSIONER ROD (10 min, the go/no-go decision).

Phase H -- Remove tensioner, guides, chain (30 min): **INSTALL CAM LOCK TOOL FIRST** (Lisle 37950). Compress tensioner rod with retainer pin, unbolt tensioner, remove arm + fixed guide, slip chain off cam gears,

slide off crank gear.

Phase I -- VTC actuator swap (30 min): hold intake cam with wrench, loosen actuator bolt, slide old off, apply Moly 60, install 14310-RBC-003, torque to **83 ft-lb**. No impact wrench.

Phase J -- Install new chain (30-60 min): route on crank gear first (single colored link to crank mark), route up to cams (two colored links align to cam gear marks -- 9 links apart), install fixed guide, install chain arm, install tensioner (pin still in), PULL PIN last, hand-rotate crank 2 full turns, verify cam marks return to TDC.

Phase K -- Oil pump chain (20 min, since you're there): same conceptual procedure, behind the cam chain.

Phase L -- Reassemble timing cover (45-60 min): press new crank seal into cover, clean mating surfaces to bare aluminum, apply Hondabond HT 2-3mm bead + extra at corners, install cover within 5 min (skin-over time), torque bolts crisscross from center outward to **8.7 ft-lb**.

Phase M -- Reinstall crank pulley with NEW bolt (15 min): Moly 60 on crank nose, install damper, NEW 90017-RWC-A01 bolt, torque sequence: **131 ft-lb -> loosen fully -> 37 ft-lb -> +90deg angle**.

Phase N -- Valve cover, engine mount, accessories (45 min): new 12030-PNC-000 gasket kit, Hondabond HT dabs at cam cap corners (4 locations), valve cover torque **7.2 ft-lb** in spiral-out-from-center pattern. **DO NOT OVER-TORQUE** -- strips aluminum head.

Phase O -- First start (30 min): fuel pump prime 3x, start within 3 sec expected, idle 3-5 min watching for leaks/CEL/rattle, short drive 5-10 min low-RPM only.

Phase P -- 100-mile re-torque (10 min): re-torque valve cover to **7.2 ft-lb**. Visual-check timing cover for weeps. Scan OBD2 pending codes.

Torque Specs

Fastener	Torque	Notes
Valve cover bolts	7.2 ft-lb (9.8 N·m / 87 in-lb)	Spiral outward from center. Re-torque at 100 mi.
Timing chain cover bolts (M6)	8.7 ft-lb (12 N·m)	All positions. Hondabond HT is primary seal.
Cam chain tensioner bolts	8.7 ft-lb (12 N·m)	Pin still in until bolted down
Cam chain guides (fixed + arm pivot)	8.7 ft-lb (12 N·m)	
Oil pump chain guide/tensioner bolts	8.7 ft-lb (12 N·m)	
VTC actuator center bolt	83 ft-lb (113 N·m)	Hold cam with wrench. No impact. Dry.
Exhaust cam sprocket bolt (if touched)	51 ft-lb (69 N·m)	
Cam pulse plate bolts (if touched)	29 ft-lb (39 N·m)	
Crank pulley bolt (NEW 90017-RWC-A01)	131 -> loosen fully -> 37 ft-lb -> +90deg	Torque-to-yield. One-time-use.
Passenger engine mount bracket to cover	~33 ft-lb (44 N·m)	Confirm against kit
Passenger engine mount main bolts	~47 ft-lb (64 N·m)	Hybrid Racing 70A -- confirm kit docs
Alternator mounting bolts	33 ft-lb (45 N·m)	
A/C compressor mounting bolts	17 ft-lb (23 N·m)	If touched
Oil drain plug	29 ft-lb (39 N·m)	

Cross-reference docs/torque-specs.md for build-wide reference.

Common Mistakes

Mistake	Consequence	Prevention
Rotating crank with chain off and no cam tool	Valve-to-piston or valve-to-valve contact, bent valves	Install Lisle 37950 BEFORE removing chain. Never rotate without it.
Chain installed off by one tooth	Low power, DTCs, possible valve damage over time	Verify cam marks align with each other AND crank TDC after 2 full crank rotations
Reused crank pulley bolt	Stretches past yield, loosens, eats crank gear, catastrophic	Buy 90017-RWC-A01 new with the kit
Reused valve cover gasket	Oil weep day one	Always new. Kit is $25.
Skipped spark plug tube seals	Oil pools in plug wells, shorts coils	Kit 12030-PNC-000 includes them
Old Hondabond / mating surface not cleaned	Persistent oil leak	Plastic scraper + brake cleaner until bare aluminum
Too much Hondabond	Squeezes internal, clogs oil galleries	2-3 mm bead, not a finger's width
Too little Hondabond at corners	Leak at pan/cover/head junctions	Extra dab (4-5 mm) at gasket-meets-gasket junctions
VTC actuator installed with impact wrench	Internal damage, phantom VTC DTCs later	Hand wrench only. Torque to 83 ft-lb dry.
Moly 60 skipped on cam nose / VTC vanes	Dry start first fire, accelerated wear	Light coat. K20 standard.
Valve cover over-torqued	Stripped M6 head threads	7.2 ft-lb. Use proper in-lb torque wrench.
Skipped 100-mile re-torque	Valve cover weeps within a month	Calendar reminder. 10 minutes.
Tensioner pin pulled too early	Chain into wrong position, cover won't seat	Pin stays in until ALL guides + tensioner installed, chain routed, everything bolted.

Cost Estimate

DIY Parts

Category	Cost
Chain set (chain + tensioner + arm + fixed guide)	~$170
Oil pump chain set	~$95
VTC actuator	$280 (budgeted under Phase 6 Ignition Refresh)
Crank bolt + crank seal	~$30
Gaskets + Hondabond + Moly 60 + misc	~$75
Oil + filter + coolant	~$75
Parts only (chain job)	~$445
Parts w/ VTC actuator included	~$725

Specialty Tool Spend (one-time)

- Lisle 37950 cam lock: $40
- In-lb torque wrench (if not owned): $60-120
- Angle gauge: $20
- Crank pulley holder (if not impact): $35
- **Tools subtotal:** ~$155-215

Shop Labor (for comparison)

- Independent shop: 6-8 hours x $90-130/hr = $540-1,040
- Honda dealer: 6-8 hours x $150-200/hr = $900-1,600
- **Shop total (parts + labor): $1,200-2,000**

DIY Grand Total

- Parts only: $445 (chain) / $725 (+ VTC actuator bundled)
- With tools: $600-940 first time
- **Time:** 1.5 full weekend days for first-timer, 6-8 hours for experienced

Symptoms Dictionary (If Something Goes Wrong After Install)

Symptom	Probable cause	Fix
Rattle at first start, disappears after 2-3 sec	Tensioner taking up slack on cold oil -- normal for a new chain	Monitor. If it persists past a week, pull cover and re-check.
Persistent rattle, warm engine	Tensioner retainer pin never pulled, OR chain one tooth off	Pull cover. Fix alignment.
P0341 (cam position sensor A circuit) / P0344	Cam timing off by one tooth, OR cam sensor connector not fully seated	Re-verify cam timing. Check sensor connectors.
P0011 / P0014 (VTC phase error)	VTC actuator installed at wrong orientation, OR old -002 actuator installed by mistake	Pull cover, verify -003 actuator, verify orientation matches factory marks
Rough idle, no DTCs	Valve cover gasket leaking onto coils (plug tube seals)	Re-torque valve cover. If leak persists, re-do plug tube seals.
Rough idle + P03xx misfire	One tooth off on timing	Pull cover, recheck
Oil weeping from timing cover seam	Hondabond gap or too little at corner	Clean thoroughly, re-seal
Oil weeping from valve cover seam	Gasket pinched or under-torqued	Re-torque to 7.2 ft-lb. If persists, replace gasket again.
Oil in spark plug wells	Plug tube seals not seated / not installed	Pull valve cover, re-seat or replace seals
No crank / no start	Cam and crank out of time by more than one tooth -- ECU refuses to fire	Pull cover, verify timing. Do NOT crank repeatedly -- risk of valve contact.
Start + immediate loud clatter	Chain jumped, valve contact happening	Shut off immediately. Pull cover. Leakdown test may be needed.
Coolant in oil (milky dipstick)	If WP was touched, gasket failure	Pull WP, replace gasket. Drain + refill oil.

Cross-References

- 06-Ignition-Refresh/overview.md -- VTC actuator install (14310-RBC-003) bundles here. Free labor.
- 09-Cooling-System/overview.md -- Water pump (19200-RBC-013) bundles here if coolant is drained.
- 15-Pulleys-and-Harmonic-Balancer/overview.md -- ATI Super Damper install uses same crank-bolt yield cycle. Bundle if possible.
- 08-Engine-Mounts/overview.md -- Hybrid Racing 70A passenger mount is removed for access. Install while in there if not already in.
- 10-Hondata-FlashPro/overview.md -- First FlashPro flash only after chain + VTC actuator is done.
- docs/torque-specs.md -- build-wide torque reference.

Reference Resources

- K-POWERED -- K-Series Timing Chain Inspection & Replacement -- Best overview with 13.5mm rod measurement photos
- 8thCivic -- DIY Timing Chain Removal -- FG2-specific thread with 3-bolt oil-pan-flange gotcha
- 8thCivic -- K20Z3 Teardown -- full teardown photos
- How to Replace K20 Timing Chain YouTube
- How To Set Cam Timing Honda K20/K24 -- watch 3x before starting
- 8thCivic -- All Torque Specs Thread
- K20a.org -- How long do timing chains last -- 13.5mm rule community discussion
- K20a.org -- Crank pulley bolt torque -- yield torque procedure
- K Series Parts -- Camshaft Install Guide

Last updated: 2026-04-20

Ignition Refresh -- Spark Plugs + Coils + VTC Actuator

Status: Planned -- Do BEFORE any dyno tune

Why This Gets Its Own Folder Now

I originally lumped spark plugs into Phase 0 maintenance, but after digging deeper into 8thcivic.com and k20a.org threads for 170k-mile K20Z3s, it became obvious that ignition is a whole system that deserves its own treatment -- especially before I flash anything with Hondata. Three things go together here:

1. **Spark plugs** -- almost certainly cooked at 170k
2. **Ignition coils** -- coil-on-plug (COP) units that get tired and start dropping cylinders under load, especially on E85 where more fuel needs more spark energy
3. **VTC actuator** -- the #1 K20Z3 gripe on the forums (that cold-start rattle everyone hears)

Doing these together, before a dyno session, eliminates the single biggest source of "tune won't dial in / mystery misfires / ragged idle" headaches that get blamed on the tuner.

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1. Spark Plugs

Stock Spec

Field	Value
OEM plug	NGK ILZKR7B-11S
Type	Iridium, laser-welded, pre-gapped
Gap	0.043" (1.1mm) -- pre-gapped from factory, do NOT re-gap
Heat range	7 (NGK scale)
Change interval (Honda)	100,000 mi
Change interval (community)	60,000-80,000 mi on a driven SI

At 170k, if these are originals they're cooked. Even if they were replaced once around 85k, they're on borrowed time. Symptoms I'd be looking for: slightly rough idle, MPG drop, misfire codes, hesitation off VTEC.

What I'm Going With

For pump gas (now through full bolt-on tune): NGK ILZKR7B-11S OEM replacement. 4 plugs, ~$40.

For E85 / flex fuel tune (later): NGK ILFR7H (or ILKAR7L11) -- **one heat range colder**. E85 burns cooler than gas but produces more combustion volume; a colder plug resists pre-ignition under the higher effective octane timing I'll be running. Community consensus on k20a.org for E85 K-series builds.

Do NOT re-gap iridium plugs. The fine iridium electrode is fragile. If the plug isn't at spec gap out of the box, it's defective -- send it back.

Buy Links

Plug	P/N	Use	Price	Link
NGK ILZKR7B-11S	4912	Pump gas tune	~$10/ea	Any auto parts store / Amazon
NGK ILFR7H	6102	E85 flex fuel tune	~$12/ea	Amazon / RockAuto

I'll verify current pricing before buying.

2. Ignition Coils (Coil-on-Plug)

A Note First

The K20Z3 is **coil-on-plug (COP)** -- there are no spark plug wires. One coil sits directly on each plug. People confuse this on forums constantly. No wires. Don't let anyone sell me wires.

Stock Spec

Field	Value
OEM coil	Denso / Honda
Part number	30520-RWC-A01
Qty	4 (one per cylinder)
Community-reported lifespan	100-150k on a driven SI

Why OEM, Not "Performance" Coils

I looked at the Accel, MSD, and Granatelli "hi-output" coils. Verdict: marketing. The K20Z3 OEM coil is already adequate for 300+ WHP on pump gas. The aftermarket "upgrade" coils add zero real spark energy at normal combustion pressures -- they just cost 3x more and sometimes ship with worse build quality than OEM Honda units.

The actual problem at 170k is that my stock coils are tired, not that stock coils are underpowered. Fresh OEM Honda coils (30520-RWC-A01) solve the problem completely.

Symptoms of Tired Coils on a K20Z3

- Random misfire codes (P0301-P0304) especially at high RPM / high load
- "Stutter" under WOT above 6000 RPM
- Rough cold-start idle that smooths out after 30 seconds
- Slight MPG drop

All of these get worse on E85 because E85 needs more spark energy to ignite (higher latent heat of vaporization). If I tune for E85 on tired coils, I'll chase phantom knock that's actually misfire.

Buy

Part	P/N	Qty	Approx Price	Notes
Honda OEM Ignition Coil	30520-RWC-A01	4	~$40-50/ea	OEM Denso. Buy from a Honda dealer or majestichonda.com for the real deal -- eBay has a lot of fakes.

3. VTC Actuator -- The Big One

What It Is

The Variable Timing Control (VTC) actuator is the cam phaser that sits on the front of the intake camshaft. It uses oil pressure to advance/retard intake cam timing through the rev range. This is part of what makes the K20Z3 feel like it has an expanding torque band -- the VTC works in combination with VTEC (which is a different system -- VTEC switches between low- and high-lift cam lobes, VTC continuously varies intake cam phase).

The Problem

The original VTC actuator has a **known weak spring** that loses preload over time. When the engine is cold and oil pressure is low (first 1-3 seconds of startup), the actuator rattles against its stops before oil pressure locks it in place. **That cold-start rattle everyone on 8thcivic complains about -- that's it.**

Left alone long enough, the actuator can fail catastrophically (lose phase lock under load, jumping cam timing) which bends valves. Not common but it happens.

Revised Honda Part

Honda revised the part silently. The new actuator has a stronger internal spring and better locking pin geometry.

Part	P/N (superseded)	P/N (current)	Price
VTC Actuator	14310-RBC-002 (old)	14310-RBC-003 (revised)	~$280 OEM

Buy the 003 part. Not 002. Any dealer ordering by VIN will ship the 003; the 002 has been end-of-life'd but aftermarket sellers still list it.

When To Do It

Ideal: during timing chain inspection. The actuator is the first thing you pull when the timing cover comes off. Labor is effectively free if I'm already in there.

Also ideal: before the full bolt-on tune. If I tune on a weak VTC actuator, the tuner will chase phantom cam timing issues. Fresh actuator = clean baseline.

Absolutely required: before E85 tune. E85 runs more timing advance; if the VTC can't hold phase precisely, I'll get unpredictable tune behavior.

Symptom Check (How I'll Know Mine Is Bad)

1. **Cold-start rattle** for 1-3 seconds after first morning start, goes away once oil pressure builds. That's the diagnostic.
2. **Worst case:** persistent rattle even warm. Replace immediately, don't drive.
3. **FlashPro datalog:** cam timing actual vs. commanded will show oscillation or lag at idle-to-2500 RPM transitions.

Cost Summary

Item	Price
NGK ILZKR7B-11S x4 (pump gas)	~$40
Honda OEM coils 30520-RWC-A01 x4	~$160-200
Honda VTC actuator 14310-RBC-003	~$280
Later: NGK ILFR7H x4 (E85 swap)	~$50
Subtotal (do-it-all-now)	~$480-520

Labor if someone else does it: add ~$150-300 for the VTC actuator (timing cover off). Plugs + coils are a 30-minute DIY job.

Where This Sits in the Mod Order

Phase 0.5 -- between Phase 0 (fluids/maintenance) and Phase 1 (already-purchased installs). Specifically:

- BEFORE FlashPro first flash -> clean baseline for datalogging
- BEFORE any dyno session -> no phantom misfire/timing issues
- WITH timing chain inspection -> VTC labor is free

See Also

- docs/mod-order-and-maintenance.md -- where this slots in the master plan
- 10-Hondata-FlashPro/ -- the tune this enables

Last updated: 2026-04-18

Ignition Refresh Install Guide -- Plugs + Coils + VTC Actuator

Status: Pending (Phase 0.5 -- before any tune)

This is the clean-baseline job. Plugs, coils, and VTC actuator all get replaced before the FlashPro ever gets plugged in. At 170k miles, any one of the three being tired will lie to a tuner -- phantom misfires, drifting cam angle, inconsistent AFR -- and I refuse to chase ghosts on the dyno. Fix the hardware first, then tune.

Reality Check

Two jobs under one folder, with an enormous difficulty gap.

Phase A -- Plugs + Coils. 45 minutes, easy. A 10mm socket, a 5/8" plug socket, and a torque wrench. Zero scary moments unless I cross-thread an iridium plug.

Phase B -- VTC Actuator. 4-6 hours standalone, much harder. Pull crank pulley (one-time-use yield bolt), pull timing cover, lock cams with a Lisle 37950, release chain tension, break the actuator center bolt with a controlled breaker-bar pull (no impact), fit the revised 14310-RBC-003, rebuild with fresh gaskets and a new crank bolt. Install wrong -- bad orientation, cams not locked, reused crank bolt -- and I bend valves.

The single most important decision on this page: do not schedule the VTC actuator standalone. Bundle it with the timing chain service (see ../05-Timing-Chain-Maintenance/). The cover is already coming off for chain inspection; the actuator is the first thing behind it. Labor-wise, the actuator goes from "4-6 hours, scary" to "+30 minutes while I'm in there." Doing VTC standalone means 80% of a timing chain job without doing the chain -- insane at 170k.

Rule: if the timing cover comes off, actuator and chain set both get serviced.

Prerequisites and Cross-References

Hard rule -- do this BEFORE any dyno session or FlashPro flash

Phase 0.5 sits between basic maintenance and the first tune.

- No FlashPro flash until Phase A (plugs + coils) is complete.
- No FlashPro flash until the cold-start rattle is resolved -- Phase B done, or confirmed-not-needed via clean cold-start datalog.

Bundle with timing chain (strongest recommendation here)

See ../05-Timing-Chain-Maintenance/overview.md. The 14310-RBC-003 actuator is shared labor with the chain service -- chain inspection + actuator replacement share the same 4-6 hours of access work. Every downstream decision -- FlashPro flash, dyno, headers, E85 -- assumes this cover has been opened and both jobs done.

Related files

- ../05-Timing-Chain-Maintenance/overview.md -- bundle target, labor sharing
- ../05-Timing-Chain-Maintenance/install-guide.md -- master procedure for cover removal, TDC setup, cam lock install, reassembly
- ../10-Hondata-FlashPro/tuning-workflow-and-maps.md -- the tune this unblocks
- ../docs/torque-specs.md -- every fastener in this guide cross-referenced
- overview.md -- parts picks, reasoning, forum research

Tools Required

Phase A (plugs + coils)

Tool	Purpose
10mm socket, 1/4" drive	Coil hold-down bolts
5/8" spark plug socket (thin-wall, rubber insert)	Iridium plugs
6" and 10" extensions, 3/8" drive	Plug well access
3/8" ratchet	General
Torque wrench, in-lb range (0-250 in-lb)	Coil bolts at 108 in-lb (9 ft-lb)
Torque wrench, ft-lb range	Spark plugs at 13 ft-lb
Copper anti-seize (sparingly)	Plug threads
Dielectric grease	Coil boot interiors
Shop vac or compressed air	Plug-well debris

Phase B (VTC actuator)

Everything in ../05-Timing-Chain-Maintenance/install-guide.md, plus:

Tool	Purpose	Cost
Lisle 37950 K-series cam lock	Locks both cams at TDC. Without it the cams spring-rotate and hit valves. Non-negotiable.	~$40
19mm socket + 24"+ breaker bar	Crank pulley bolt (131 ft-lb factory, often 200+ after 170k)	Own
Crank pulley holder (or impact)	Holds crank while breaking bolt	~$35 / loaner
Torque wrench to 131 ft-lb+	Crank bolt step 1 and VTC actuator at 83 ft-lb	Own
Torque angle gauge	Crank bolt: 37 ft-lb + 90deg	~$20
3/8" in-lb torque wrench	Timing cover at 8.7 ft-lb	Own
Plastic or brass gasket scraper	Cover mating surfaces -- NO STEEL on aluminum	~$10

Consumables specific to Phase B

Part	P/N	Notes
Valve cover gasket kit	12030-PNC-000	Kit -- includes plug tube seals + grommets
Timing cover sealant	Hondabond HT 08718-0004	RTV, not a paper gasket
Front crank seal	91214-RZY-A01	Press into cover before cover goes back on

Crank pulley bolt (TTY, one-time-use)	90017-RWC-A01	Old bolt is scrap the moment it comes off
Moly 60 assembly grease	08798-9010	VTC vanes, cam nose, crank nose

Parts List

Per overview.md:

#	Part	P/N	Qty	Notes
1	NGK iridium spark plug	ILZKR7B-11S (NGK 4912)	4	Pre-gapped at 0.043" (1.1mm). Do NOT re-gap.
2	Honda OEM ignition coil	30520-RWC-A01	4	OEM Denso. Honda dealer or majestichonda.com -- eBay has fakes.
3	Honda VTC actuator (REVISED)	14310-RBC-003	1	The 003 part only. The old 002 is what I'm removing.
4	Crank pulley bolt (TTY, one-time-use)	90017-RWC-A01	1	New every time the pulley comes off.
5	Front crank main seal	91214-RZY-A01	1	Press into cover on reassembly.
6	Valve cover gasket kit	12030-PNC-000	1	Kit, not bare gasket.
7	Hondabond HT	08718-0004	1 tube	Timing cover sealing surfaces.

Step-by-Step -- Phase A: Plugs and Coils (45 min)

Standalone, bolt-on. Any time before the FlashPro flash.

A1. Warm the engine, then cool

Start, bring to operating temp (thermostat open, fan cycles once). Shut off. Cool **at least 20 minutes.** The brief warm cycle expands plug threads in the aluminum head slightly, helping break the seal without galling. Head should be cool enough not to burn my hand.

A2. Disconnect battery negative

10mm. Electrical safety plus ECU short-term fuel trim reset so post-install datalogs start from zero.

A3. Pull coil #1

- Unplug the harness connector (press tab, pull gently -- 170k connector tabs are brittle).
- Remove the single 10mm hold-down bolt.
- Pull the coil straight up with a gentle twist. If it fights, rotate 1/4 turn both ways to break the boot seal, then lift.

A4. Vacuum the plug well

Shop vac or compressed air at the top of the well. Debris falling into an open combustion chamber scores cylinder walls. 10 seconds of cleaning prevents hours of regret.

A5. Break the plug loose

Socket + extension, steady pull. **If resistant, do not force.** Small spray of penetrant, wait 15 minutes, retry. Stripped threads in an aluminum head = pulling the head for a Heli-Coil. Patience is free. Once loose, unscrew by hand.

A6. Inspect the old plug (free diagnostic)

- **Light tan ceramic, mild wear:** healthy.
- **Black sooty carbon:** rich -- leaky injector, failing O2, or tired coil misfiring.
- **White/bleached ceramic:** lean or hot -- potential pre-ignition.
- **Oily fouling:** valve stem or plug tube seals leaking. Tube seals in kit 12030-PNC-000.
- **Damaged/missing electrode:** detonation. **Stop.** Diagnose before continuing. Do not tune.

Photograph each plug in cylinder order (1 on left) for the record.

A7. Prep the new plug

Confirm ILZKR7B-11S on box, electrode intact, ceramic clean. **Gap is pre-set at 0.043". Do NOT re-gap iridium** -- the fine electrode is fragile. If gap looks wrong, the plug is defective -- return it.

A8. Anti-seize (sparingly)

NGK says skip it on plated threads; on a fresh engine I'd agree. On a 170k engine where previous plugs fought coming out, I apply a thin film of copper anti-seize on threads only -- not the electrode, not the first thread from the tip. "Barely visible sheen," not "coated." Over-anti-seize over-torques the plug at a given wrench reading and cracks iridium.

A9. Hand-thread and torque

Hand-start, all the way down, until seated. **If it binds, back it out and restart.** Cross-threaded iridium in aluminum = head-off repair. Torque wrench to **13 ft-lb (156 in-lb).** Firm click. Do not "give it a little extra."

A10. Coil back in

- Small dab of dielectric grease inside the coil boot. Keeps moisture out, helps the boot release next service.
- Coil straight down, feel it seat. Reinstall 10mm bolt.
- **Torque to 108 in-lb (9 ft-lb)** with the in-lb wrench. M6 strips easily.
- Reconnect harness, listen for the click.

A11. Repeat for 2, 3, 4

One cylinder at a time. Never all four out simultaneously -- if I'm interrupted, I know exactly where I stopped.

A12. Reconnect battery, verify start

Battery (-) back on. Key to ON 5 seconds (fuel prime), then start. Idle smooths within 10 seconds as the ECU relearns trims. Any misfire, rough idle, or CEL = a coil connector isn't fully seated. Pull and re-seat.

Step-by-Step -- Phase B: VTC Actuator (4-6 hrs standalone, or +30 min bundled with timing chain)

The master procedure for timing cover removal, TDC verification, and cam lock install lives in ../05-Timing-Chain-Maintenance/install-guide.md. I am not duplicating it. This section covers actuator-specific steps, assuming the cover is off and the cams are locked with a Lisle 37950.

If doing the actuator standalone, follow ../05-Timing-Chain-Maintenance/install-guide.md Phases A-G to get the

cover off, come back here, then return to that document's Phases L-P for reassembly.

B1. Crank pulley removal (see 05 guide)

Loosen crank pulley bolt (131 ft-lb factory, expect more after 170k). Breaker bar + holder tool or impact. Bolt is **dis carded** -- TTY one-time-use. Slide pulley off.

B2. Timing cover removal (see 05 guide)

13 bolts of varying lengths. Three are on the oil-pan face and easy to miss. Score the old Hondabond, tap the cover off gently, clean both mating surfaces with a plastic scraper + brake cleaner to bare aluminum. No steel scrapers.

B3. TDC, cam marks, cam lock -- prerequisite state

- Crank at TDC cylinder 1 (pulley mark to pointer).
- Both cam gear marks (triangles/dots) point straight up, parallel to head surface.
- **Install Lisle 37950 now.** Cams must be mechanically locked before any torque touches the actuator bolt. Without the lock, cams spring-rotate under valve spring pressure and valves hit pistons.

B4. Release chain tension

Locate the cam chain auto-tensioner on the passenger side. Retract the piston per its design (small lever/pin; see 05 guide Phase H). Insert the retainer pin. Chain is now slack enough to work.

B5. Break the VTC actuator center bolt

- Reinstall torque is **83 ft-lb.** Coming out after 170k it may be tighter.
- **NO IMPACT WRENCH.** Actuator internals (vane, spring) are shock-sensitive, especially on a failing unit. Controlled pull only.
- Cam is held by the Lisle 37950. Breaker bar on the bolt, steady pull, break loose. Spin off by hand.
- Remove the old actuator -- it slides off the intake cam nose. Keep for comparison (002 and 003 look nearly identical externally, but 003 is what I'm installing).

B6. Chain off old actuator, clean cam nose

Slip chain off the actuator sprocket. **Zip-tie chain to the head** so it doesn't fall into the oil pan. Wipe cam nose, inspect for scoring/galling. Expected: light wear. If chewed, the old actuator was failing a while and may have contaminated oil -- early oil change at break-in.

B7. Install the new 14310-RBC-003

- **Verify part number on the box.** 14310-RBC-003. Not 002. Not an aftermarket knockoff.
- Light coat of Moly 60 on the cam nose and the actuator's internal vane surface.
- Slide the actuator onto the cam nose. **Orientation matters.** The actuator has an alignment dot/arrow that must match the cam reference feature. **Wrong orientation = bent valves the instant the chain is loaded.** See Honda service manual or ../05-Timing-Chain-Maintenance/install-guide.md Phase I for exact mark locations.
- Route chain over the actuator sprocket, keeping colored alignment links on the cam marks if reusing the chain.

B8. Torque actuator center bolt to 83 ft-lb

- Hand-thread fully.
- Torque wrench to **83 ft-lb (113 N·m).** One steady click. Cam held by Lisle 37950 throughout -- the lock absorbs torque reaction, not the chain or valvetrain.
- **No impact. Dry threads** (unless Honda FSM page that ships with the actuator specifies Moly 60 on threads -- some years do).

B9. Release chain tensioner, verify slack

Pull the tensioner retainer pin. Tensioner extends and pulls chain slack. Visually inspect: taut on tension side, free on slack side. Cam marks still aligned. Crank still at TDC.

B10. Release cam lock, rotate crank 2 full turns, re-verify

- Remove Lisle 37950.
- Using a 19mm on the crank bolt hole (or snug the pulley temporarily for grip), **rotate crank BY HAND, clockwise only, two full turns (720deg).**
- Return to TDC cylinder 1. Verify cam marks still aligned with each other AND the crank TDC mark.
- **If marks don't line up, STOP.** Chain is one tooth off. Re-lock cams, release tension, correct, torque again, repeat. Do not start the engine until right.

B11. Timing cover back on with fresh Hondabond

- Press new **91214-RZY-A01** front crank seal into the cover (lip inboard, clean press -- see 05 guide).
- 2-3mm bead of Hondabond HT around the mating surface. Extra dab at corners (cover-to-head-to-block junctions).
- Cover on within 5 minutes (skin-over time).
- **Torque bolts to 8.7 ft-lb (12 N·m)** crisscross center-outward. In-lb torque wrench -- the big ft-lb one can't measure this low accurately.

B12. Crank pulley reinstall -- NEW yield bolt

- Moly 60 on crank nose.
- Slide pulley on.
- **NEW 90017-RWC-A01 bolt.** Old one is trash.
- Sequence per Honda FSM: **131 ft-lb, then loosen fully, then 37 ft-lb, then +90deg angle.** The breakaway-then-retorque cycle is how TTY bolts reach correct preload.

B13. Reassemble accessories, coolant (if drained), oil

Follow ../05-Timing-Chain-Maintenance/install-guide.md Phases L-N for full reassembly order. Valve cover with new 12030-PNC-000 kit, torque 7.2 ft-lb spiral-out, Hondabond HT dabs at the 4 cam cap corners. Fresh oil + filter. Coolant top-off if the cooling system was opened.

First Start Checklist

1. **Key to ON, listen for fuel pump prime (2-3 sec buzz).** Cycle 3 times (ON, pump stops, OFF, ON).
2. **Start.** Should catch within 3 seconds. If cranking past 5-6 seconds, stop -- something is wrong, likely cam timing or a disconnected sensor.
3. **Listen at idle for:**

- **No cold-start rattle** in the first 1-3 seconds. If I hear the familiar K20Z3 rattle, the new actuator isn't working -- wrong orientation, wrong part (002 vs 003), or install fault. Shut down, pull the cover.
- **Smooth idle within 10-20 seconds.** Rough idle usually means a loose coil connector or ECU relearning trims -- give it a minute.
- **No CELs.** P0341/P0344 = cam position. P0011/P0014 = VTC phase error. Any of these = pull cover, verify timing.

4. **Idle 3-5 minutes.** Watch for leaks -- valve cover, timing cover, coolant if water pump was touched.
5. **Short drive, 5-10 min, below 4000 RPM.** Verify it drives. No WOT, no VTEC engagement.

Dyno-Ready Verification (Before FlashPro Tune #1)

Before the first flash, I pull a baseline datalog on the factory calibration with fresh ignition hardware. Three things must be true:

1. **Short-term fuel trim (STFT) within ±8%** at idle and 2500 RPM cruise. Larger = vacuum leak or fuel/air issue, fix before tuning.
2. **Misfire counters zero** over a 10-minute mixed-RPM log (Mode 6 via FlashPro datalog viewer). Any cylinder misfiring = bad coil connector or unseated plug.
3. **Cam angle actual tracks commanded** within ~2deg, no oscillation at idle-to-2500 transitions. VTC health check -- weak actuator shows lag/oscillation, healthy -003 tracks cleanly.

All three clean = dyno-ready baseline. ../10-Hondata-FlashPro/tuning-workflow-and-maps.md is the next file to open.

Common Mistakes

Mistake	Consequence	Prevention
Cross-threading iridium into aluminum head	Stripped threads = Heli-Coil or head removal	Hand-thread fully before any wrench touches it
Re-gapping an iridium plug	Broken electrode = defective plug	Pre-gapped from factory. Don't touch the electrode.
Over-torquing the plug	Cracked ceramic, broken tip in cylinder	13 ft-lb. Torque wrench only.
Over-torquing coil hold-down bolt	Stripped M6 in head	108 in-lb (9 ft-lb). In-lb wrench.
Over-torquing VTC center bolt	Actuator damage, phantom VTC DTCs	83 ft-lb. Hand wrench. Cam held.
Reusing the crank pulley bolt	Stretches past yield, loosens, eats crank gear	Always new 90017-RWC-A01. Old is scrap.
Not locking cams during actuator install	Cams spring-rotate, valves hit pistons, bent valves	Lisle 37950 before the actuator bolt gets touched. Non-negotiable.
Wrong actuator orientation	Cam phase impossible, bent valves on first start	Verify alignment marks per FSM. Slow-rotate 2 turns by hand before starting.
Installing superseded 002 instead of -003	Cold-start rattle doesn't go away; job wasted	Verify part number on the box. Order by VIN from Honda dealer.
Impact wrench on the actuator bolt	Damages internal spring and vane	Breaker bar only. Cam lock absorbs torque.

No dielectric grease in coil boots	Water intrusion, arcing, stuck boots next service	Small dab inside every boot.

See Also

- overview.md -- part selection, forum research, cost summary
- ../05-Timing-Chain-Maintenance/overview.md -- **bundle target.** Chain service shares 100% of the actuator's access labor.
- ../05-Timing-Chain-Maintenance/install-guide.md -- master procedure for cover removal, TDC verification, cam lock install, reassembly
- ../10-Hondata-FlashPro/overview.md -- first flash only after Phase A and Phase B complete and the dyno-ready checklist passes
- ../10-Hondata-FlashPro/tuning-workflow-and-maps.md -- daily/sport map rules, datalog workflow
- ../docs/torque-specs.md -- build-wide torque reference
- ../docs/maintenance-parts-catalog.md -- full OEM part number catalog

Last updated: 2026-04-20

Clutch Hydraulics Upgrade -- 2009 Honda Civic SI (FG2 / K20Z3)

Vehicle: 2009 Civic SI Coupe (FG2), K20Z3, ~170,000 miles **Date:** 2026-04-16 **Goal:** Preventative upgrade of clutch master cylinder (CMC) and clutch slave cylinder (CSC) before they leave me stranded

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My Pick: Hybrid Racing Complete CMC Upgrade Kit

I'm going with the best available option -- the Hybrid Racing drop-in kit. It includes everything.

Field	Value
Product	Hybrid Racing Clutch Master Upgrade for Accord/Civic/RSX/TSX
Part Number	HYB-CMC-01-20
Price	$289.99
Buy Link	hybrid-racing.com
Install Guide	guides.hybrid-racing.com/8)
Fitment	2006-2011 Honda Civic SI (FG2 coupe / FA5 sedan) -- LHD only

What's Included in the Kit

- (1) Upgraded clutch master cylinder with clevis
- (1) Upgraded slave cylinder
- (1) Stainless steel braided clutch line with fittings
- (1) Firewall seal
- (1) PTFE grease packet
- (1) Hardware pack and mounting clips

Everything I need in one box -- CMC, CSC, and braided line all included. No adapter bracket needed. No extra parts to source. That's why this is the move.

> **Note:** The link I originally found (hybrid-racing.com/products/8th-gen-clutch-master-cylinder-upgrade) goes to the **HYB-CMC-01-02** ($199.99), which is the **K-swap version** for older Civics/Integras (1992-2001). It does NOT fit a stock 8th gen SI. The correct product for my car is the **HYB-CMC-01-20** linked above. Glad I caught that before ordering.

Additional Components Needed

The Hybrid Racing kit includes almost everything, but I still need clutch fluid and a bleeder tool.

Clutch Fluid -- Motul RBF 600 DOT 4

Field	Value
Product	Motul RBF 600 Factory Line DOT 4 Racing Brake Fluid

Size	500mL (0.5L) -- one bottle is enough, two recommended
Dry Boiling Point	594F (312C)
Wet Boiling Point	421F (216C)
Buy Links	Amazon -- ~$18/bottle
	Walmart -- ~$17/bottle
	BimmerWorld -- $17.99/bottle

Why Motul RBF 600: Highest dry boiling point of any readily available DOT 4 fluid. Amber color makes it easy to see when it's dirty during future flushes. Honestly the enthusiast standard for Honda clutch and brake systems.

Alternative: ATE TYP 200 Gold -- Dry 536F / Wet 396F, slightly cheaper. You can alternate gold and blue versions at each flush to visually confirm old fluid is fully purged. Clever system.

Bleeder Tool (Optional But Recommended)

Option	Product	Price	Notes
Best	Motive Products Power Bleeder	~$50-70	One-person operation, pressurizes reservoir
Good	Mityvac hand vacuum pump	~$30-40	One-person, pulls fluid from bleeder valve
Free	Two-person manual bleeding	$0	One pumps pedal, one opens/closes bleeder

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Complete Shopping List

#	Item	Product	Price	Link
1	Complete CMC/CSC/Line Kit	Hybrid Racing HYB-CMC-01-20	$289.99	Buy -- Hybrid Racing
2	Clutch Fluid	Motul RBF 600 DOT 4 (500mL x2)	~$36	Buy -- Amazon
3	Clutch pedal bushing (OEM)	Honda 46991-S5A-004	~$8	Honda dealer -- dry-rotted at 170k, 30-sec install while system is apart
4	Motive Power Bleeder (one-time, reusable for brakes too)	Motive 0100	~$60	Amazon
	Subtotal (CMC upgrade alone)		~$393	

Cross-Reference: "Do This At The Same Time" Opportunities

I'm pulling the transmission to access the CSC. While it's out, several other jobs become effectively free labor. **I'm doing all of these in one session** -- no way I'm pulling the transmission twice.

Add-on	OEM P/N	Price	Why It's Free Labor Now
Rear main seal	91214-RDA-A01	~$20	Trans is out, crank is exposed. 170k RMS will weep inside a year otherwise. Do it now.
Throwout bearing (OEM)	22810-RWC-003	~$40	Accessible only with trans out. Cheap insurance.
Pilot bearing	91006-RWC-003	~$15	Same logic.
Transmission inspection	--	$0	Visual check for input shaft seal weeping, bellhousing cracks, synchro rings.

Fresh Honda MTF-3	08798-9031, 2.2 qt	~$25	Refill with fresh fluid -- catches any residual clutch debris.

Total with add-ons: ~$493. About $100 more than the core kit, and it saves me from pulling the transmission again for avoidable reasons. The reliability-first rule says: if it's accessible cheaply now, do it now.

That's it. The Hybrid Racing kit provides the main CMC/CSC/line/fittings/grease, plus the bonus items cover every other 170k-mile wear point that's accessible only during this job.

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1. Why the 8th Gen SI CMC/CSC Fails

The Problem

The 8th gen Civic SI (2006-2011, FA5 sedan / FG2 coupe) is notorious in the Honda community for clutch hydraulic failures. Every 8th gen owner knows about this, and at 170k miles I'm basically on borrowed time. The two primary failure modes:

1. **Soft / spongy clutch pedal** -- The pedal gradually loses firmness. Engagement point drifts toward the floor. Eventually the clutch won't fully disengage, making shifting difficult or impossible.
2. **Internal seal bypass (no visible external leak)** -- The most frustrating failure mode. Fluid bypasses the internal seals inside the CMC under pressure, so the pedal sinks to the floor during use (especially in stop-and-go traffic or aggressive driving) but recovers after sitting. The reservoir stays full, there's no puddle on the ground, and it looks perfectly fine from the outside.
3. **External leaking** -- Less common initially but happens as seals deteriorate further. Fluid weeps from the CMC pushrod seal (visible on the firewall inside the cabin near the pedal) or from the CSC at the transmission bellhousing.

Typical Failure Mileage

- **60,000-100,000 miles** is the most commonly reported window for the first CMC failure on 8th gen SIs.
- Some owners report issues as early as 40,000-50,000 miles, particularly in hot climates or with aggressive driving.
- By 120,000-150,000 miles, the vast majority of 8th gen SIs have had at least one CMC replacement if still on the original unit.
- The CSC tends to last somewhat longer than the CMC but often fails within 20,000-40,000 miles of the CMC going.
- At 170k miles, if my originals are still in there, they're on extremely borrowed time (or they've already been replaced at some point -- worth checking service records).

Root Causes

1. **Inferior internal seal material** -- The 8th gen SI CMC uses rubber cup seals that swell, harden, and crack over time. Honda used a lower-grade rubber compound compared to what went in the RSX, S2000, and even the base model Civic.
2. **Heat soak** -- The CMC is mounted on the firewall directly behind the engine. The K20Z3 runs hot, and the CMC gets significant heat soak -- especially in traffic. Heat accelerates seal degradation.
3. **Fluid contamination / moisture absorption** -- DOT 3/DOT 4 brake fluid is hygroscopic (absorbs moisture from the air). The 8th gen clutch fluid is rarely flushed during routine maintenance. Moisture-contaminated fluid

lowers the boiling point and attacks rubber seals from the inside.

4. **Bore surface finish** -- The stock CMC has an aluminum bore with a relatively basic surface finish. As the seals degrade, they can score the bore, which further accelerates leakage past the seals.

5. **Small bore diameter** -- The 8th gen SI CMC bore is small (roughly 5/8" or 15.87mm), which means it operates at higher hydraulic pressures for a given pedal force. More stress on the seals.

CMC vs CSC -- How to Tell Which One Is Failing

Symptom	CMC (Master)	CSC (Slave)
Pedal slowly sinks to floor when held	Yes (classic sign)	No
Pedal is soft/spongy but stays in position	Could be either (air in system)	Could be either
Fluid on driver's side carpet or firewall	Yes -- leaking past pushrod seal	No
Fluid weeping near transmission bellhousing	No	Yes
Reservoir level dropping with no visible leak	CMC internal bypass (most common)	Possible CSC internal leak
Clutch won't disengage after sitting in traffic	Yes -- heat soak causing seal bypass	Less likely
Pedal feels fine cold, gets soft when hot	Yes -- classic heat-related CMC failure	Less likely
Hard pedal / pedal won't return	Rare CMC issue (sticking piston)	More commonly a CSC issue or throwout bearing

The "parking lot test": Pump the clutch pedal 15-20 times rapidly. If the engagement point drops noticeably and the pedal feels mushier, the CMC is bypassing internally.

TSBs and Recalls

- **No official recall** was issued for the CMC/CSC on the 8th gen SI.
- **TSB 07-006** (early 2007) addressed clutch judder and chatter on 2006-2007 SI models but was related to the clutch disc, not hydraulics.
- Honda did quietly revise the CMC part number during the 8th gen production run (see OEM part numbers below), which tells me they knew about the issue, but they never issued a formal TSB specifically for CMC/CSC failure.
- The Honda community widely regards this as a "known but unacknowledged" design weakness. Par for the course.

2. Stock CMC/CSC Specifications

OEM Part Numbers

Clutch Master Cylinder (CMC):

- **46920-SVA-A01** -- Original part number (2006-2008 models, early production)
- **46920-SVA-A02** -- Revised part number (later production and replacement)
- **46920-SVA-A03** -- Further revised (current replacement part from Honda)
- Approximate dealer price: $80-$120

Clutch Slave Cylinder (CSC):

- **46930-SVA-A01** -- Original

- **46930-SVA-A02** -- Revised
- Approximate dealer price: $50-$80

Note: The revisions are primarily to the internal seals. Physical dimensions and mounting stayed the same across all revisions.

Why the Stock Units Are Inferior

Compared to the RSX (DC5), S2000 (AP1/AP2), and even the 7th gen Accord V6 6MT:

1. **Cheaper seal compounds** -- Cost-cutting during the 8th gen SI's development cycle. The SI was Honda's entry-level sport compact; the RSX Type S and S2000 were premium products with better component sourcing.
2. **No seal revision from Honda until failures were widespread** -- The initial design went to production without adequate thermal testing for seal life.
3. **Less robust bore finish** -- The RSX CMC has a better honed bore surface that causes less seal wear.
4. **Packaging / heat exposure** -- The 8th gen engine bay is tighter than the RSX (EP3/DC5 platform). More heat reaches the CMC.

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3. Why the Hybrid Racing Kit Is the Best Option

What I Get with HYB-CMC-01-20

The Hybrid Racing Clutch Master Upgrade ($289.99) is a **complete, drop-in solution**:

- **Upgraded CMC** -- OE genuine JDM master cylinder with better seals, better bore finish, and proven longevity
- **Upgraded CSC** -- Included in the kit, no need to source separately
- **Stainless steel braided clutch line** -- Replaces the stock rubber line that swells under pressure. PTFE inner liner doesn't expand = 100% of pedal force reaches the CSC
- **All hardware** -- Banjo fittings, crush washers, firewall seal, mounting clips, PTFE grease
- **Direct bolt-on** -- No adapter bracket fabrication, no drilling, no guesswork

Community Reception

- **8thcivic.com** forum: Hundreds of positive install reports. Considered the gold standard upgrade.
- **Reddit r/CivicSi**: Frequently recommended as "the first thing you should do" on an 8th gen SI.
- **Long-term reliability:** Owners report 80,000-100,000+ miles with zero issues after the upgrade.
- **Complaints are rare.** The most common negative feedback is occasional fitment fiddling during install.

Why I Ruled Out the Cheaper Options

Option	Price	Why Not
OEM replacement (46920-SVA-A03 + 46930-SVA-A02)	~$130-200	Same fundamental design weakness. I'd be doing this again in 60k-80k miles. Not interested in repeating the job.
DIY RSX CMC swap + separate parts	~$200-380	Need to source CMC, CSC, adapter bracket, and line separately. More research, more fitment risk, no single-source warranty.
HYB-CMC-01-02 ($199.99)	$199.99	Wrong product -- this is the K-swap version for 1992-2001 Civics/Integras. Does NOT fit a stock 8th gen SI.

The Hybrid Racing HYB-CMC-01-20 costs ~$90-110 more than cobbling together RSX parts myself, but I get a

complete tested kit with a single part number, install guide, and warranty. For a "do it once and forget it" upgrade at 170k miles, this is honestly the right move.

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4. Installation Overview

Difficulty Level

Moderate DIY -- 6/10 difficulty.

If I can do brake pads and rotors, I can do this. The hardest part is bleeding the system properly. The actual physical swap is straightforward.

Time Estimate

- **First-timer (me):** 3-5 hours (including bleeding)
- **Experienced:** 1.5-2.5 hours
- **Shop labor:** Typically quoted at 2-3 hours

Install Guide

Hybrid Racing provides a dedicated install guide for the 06-11 Civic SI: **guides.hybrid-racing.com -- CMC Upgrade (06-11 Civic Si)/8)**

Key Steps (Summary)

Does the dash need to come apart? No. The CMC pushrod-to-pedal connection is accessible under the dash on the driver's side. I might want to remove the lower kick panel (a few clips) for better access.

CMC Removal:

1. Disconnect clutch line from CMC (banjo bolt). Have rags ready.
2. Siphon old fluid from reservoir.
3. From inside the cabin, disconnect CMC pushrod from clutch pedal (clevis pin with cotter pin).
4. Remove two mounting bolts on firewall.
5. Pull CMC out through engine bay.

CMC Installation:

1. Bench bleed the new CMC before installing (critical -- see below).
2. Insert new CMC, attach pushrod to pedal.
3. Bolt to firewall, connect clutch line.

CSC Swap:

1. CSC is bolted to outside of transmission bellhousing (accessible from under the car).
2. Disconnect line, remove two bolts, swap CSC.
3. Torque to ~16 ft-lbs.

Clutch Line:

1. Remove entire OEM line (CMC to CSC).
2. Route new SS braided line, secure with zip ties or P-clamps away from exhaust.

Bench Bleeding (Critical Step)

1. Clamp new CMC in vise with soft jaws.
2. Fill reservoir with Motul RBF 600.
3. Use bench bleed kit (short hoses looping output back to reservoir).
4. Slowly push piston in and release -- 20-30 times until no bubbles.
5. Cap output port, install immediately.

Bleeding the Installed System

1. Fill CMC reservoir to MAX with fresh Motul RBF 600.
2. Bleed at CSC bleeder valve (furthest point from CMC).
3. **Pressure bleeder method:** Attach to reservoir, pressurize to 10-15 PSI, open bleeder, let fluid flow until clear with zero bubbles.
4. **Manual method:** Helper pumps pedal 3-4 times, holds down. Open bleeder to release fluid/air. Close bleeder. Release pedal. Repeat.
5. Pump clutch pedal 20-30 times -- should be firm with consistent engagement point.
6. If soft, repeat bleeding. Air in the system is the #1 post-install problem.
7. Check all connections for leaks. Top off reservoir.

Tips From My Research

- **Do NOT let the reservoir run dry during bleeding.** Air in the CMC = start over.
- **Brake fluid destroys paint.** Keep a spray bottle of water handy for immediate rinse.
- **New crush washers every time** you disconnect a banjo bolt.
- **The clevis pin under the dash is fiddly.** Small flashlight + magnetic pickup tool helps.
- **Second bleed after 1 week of driving** is good practice -- lets micro-air work out.

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5. Ongoing Maintenance

Clutch Fluid Flush Interval

- **Honda's recommendation:** No specific interval (this is part of the problem with the stock system failing).
- **Community recommendation: Every 2 years or 30,000 miles**, whichever comes first.
- **Track use / aggressive driving:** Every year or 15,000 miles.
- Clutch fluid has its own reservoir (separate from brakes on the 8th gen SI), so flushing is quick and cheap.

Post-Upgrade Checklist

- [ ] Flush clutch fluid every 2 years / 30,000 miles with fresh Motul RBF 600
- [ ] Visually inspect CMC reservoir level at every oil change
- [ ] Check for leaks at CMC, CSC, and all line fittings annually
- [ ] If the pedal ever feels different, bleed the system immediately

Is This a "Do It Once and Forget It" Upgrade?

Yes, with basic maintenance. The Hybrid Racing kit components are fundamentally better than stock. Owners routinely report 100,000+ miles with zero issues. Flush the fluid on schedule and I should never have to think about clutch hydraulics again for the life of this car. The stainless braided line is a lifetime part.

Last updated: 2026-04-16

Clutch Hydraulics Install Guide -- Hybrid Racing HYB-CMC-01-20 + RMS + Throwout + Pilot

Status: Not purchased yet -- priority buy

The Hybrid Racing HYB-CMC-01-20 kit isn't in my garage yet, but it's next in the shopping queue. At 170k miles on the original CMC and slave, the 8th gen SI clutch hydraulic system is living on borrowed time. This is a reliability buy. I don't want to be stranded at a red light because the master cylinder decided today was the day its seals gave up.

Reality Check Before I Start

There are two very different jobs hiding inside "clutch hydraulics." I need to be honest about which one I'm doing before buying parts and clearing a weekend.

Path A -- CMC, line, slave only (no transmission removal)

- **Time:** 3-5 hours
- **Difficulty:** Easy intermediate. If I can bleed brakes, I can do this.
- **What I touch:** Firewall CMC, stainless braided line, bellhousing slave. That's it.
- **What stays put:** Transmission, clutch, flywheel, RMS, throwout, pilot.
- **When:** My clutch and flywheel are healthy and not planned for replacement.

Path B -- Full bundle with transmission out

- **Time:** 8-12 hours realistic for me (5-7 for an experienced tech)
- **Difficulty:** Hard. Pulling the K20Z3 trans is a weekend commitment.
- **What I touch:** Everything in Path A plus RMS, throwout + fork, pilot bearing, plus the 03-Clutch-and-Flywheel work if the clutch is being replaced.
- **When:** The clutch is being replaced anyway. Pulling the trans exposes every wear item that's otherwise inaccessible. Doing RMS/pilot/throwout separately later means pulling the trans again, which I am not doing if I can avoid it.

My situation: The Exedy Stage 1 (08806) + HF02 is already installed per 03-Clutch-and-Flywheel/overview.md. The transmission is not coming back out for routine reasons. **Path A is what I'm actually doing on this job.** Path B is documented for completeness in case the clutch ever fails early or I pull the trans for another reason.

Decision tree

```
Is the clutch/flywheel healthy and already installed?
|
|-- YES -> Path A (CMC + line + slave only)
|         3-5 hours, no trans out
|
\-- NO / also being replaced -> Path B (full bundle)
          8-12 hours, trans out, bundle with 03-Clutch-and-Flywheel
          Add RMS, pilot, throwout while trans is out
```

Prerequisites and Cross-references

Document	Why
overview.md	Source of truth for parts and rationale
03-Clutch-and-Flywheel/overview.md	Only relevant for Path B -- the clutch that would be swapped in the same session
Hybrid Racing official guide/8)	Manufacturer's published torque and routing specifics
docs/torque-specs.md	CMC mounting nuts, slave bolts, line banjo

Cross-reference interactions

- **03-Clutch-and-Flywheel** -- already installed. **If it weren't, I'd bundle this entire job with that one.** Transmission removal is the expensive step; everything accessible with the trans out should happen simultaneously.
- **02-Short-Shifter-and-Bushings** -- no hydraulic interaction. Separate subsystem.
- **DOT 4 fluid** -- Motul RBF 600 is shared across brakes and clutch.

Decision rule

- **Clutch OK, not planned for replacement:** Path A standalone.
- **Clutch needs work, or planned within next 30k miles:** Wait, bundle as Path B. Pulling the trans twice is a bad trade against waiting a few months.
- **CMC actively failing right now (pedal to floor, no disengagement):** Path A immediately.

Tools Required

Critical

- **Flare nut / line wrenches** -- 10mm and 12mm. A regular open-end wrench will round the flare fittings.
- **10mm, 12mm, 14mm sockets + wrenches**
- **Clear vinyl bleed hose (3/16" ID) + catch bottle**
- **Motive Power Bleeder 0100** -- $50-70, one-person bleeding, also works for brakes
- **Floor jack + 4-ton jack stands**
- **Cotter pin pliers / pick set**
- **Magnetic pickup tool + headlamp** -- the under-dash clevis pin loves to disappear
- **Turkey baster or fluid syringe** (dedicated to brake fluid, not kitchen use)
- **Torque wrench, 3/8" drive, 10-80 ft-lb**

Fluids and consumables

- **Motul RBF 600 DOT 4** -- minimum 500mL, I'd buy 1L
- **Brake cleaner + spray bottle of water** -- DOT 4 eats paint on contact
- **Clean shop rags** -- more than I think I need
- **New copper crush washers** -- every banjo I disconnect gets fresh washers, no exceptions
- **Blue painter's tape** -- mask the firewall around the CMC before starting

Path B additional tools

- Seal driver sized for the rear main seal. **Do not hammer directly on the seal.**
- Pilot bearing puller (slide-hammer style) or the bread-packing trick
- Pilot bearing installation driver
- Transmission jack (O'Reilly/AutoZone rental)
- Everything else 03-Clutch-and-Flywheel calls for

Parts List

#	Item	Part Number	Source
1	Hybrid Racing CMC kit (CMC, slave, SS braided line, firewall seal, PTFE grease, hardware)	HYB-CMC-01-20	hybrid-racing.com
2	Motul RBF 600 DOT 4 (500mL x2)	--	Amazon / BimmerWorld
3	Clutch pedal bushing, OEM	46991-S5A-004	Honda dealer
4	Motive Power Bleeder	0100	Amazon

Path B additional parts

#	Item	Part Number
5	Rear main seal, OEM	91214-RDA-A01
6	Throwout bearing, OEM Koyo	22810-RWC-003
7	Pilot bearing	91006-RWC-003
8	Honda MTF-3 (2.2 qt)	08798-9031
9	Fresh clutch kit (if replacing)	See 03-Clutch-and-Flywheel

Step-by-Step -- Path A: CMC / Line / Slave Only

Phase A1: Drain the reservoir (20 min)

1. Park on level ground, wheels chocked. Pop the hood.
2. Mask the firewall, strut tower, and fender around the clutch reservoir with blue painter's tape. DOT 4 eats paint and I will spill some.
3. Locate the clutch reservoir -- smaller of the two on the driver-side firewall, separate from the brake reservoir on the 8th gen SI.
4. Suction out as much old fluid as possible with a dedicated turkey baster. It will be brown or amber -- confirmation I'm doing this job for the right reason.
5. Cap the reservoir loosely.
6. Lift the front of the car onto 4-ton jack stands on subframe pinch points. Never work under a jack alone.

Phase A2: Disconnect CMC under the dash (30-45 min)

7. Sit in the driver's seat, head and shoulders in the footwell. Headlamp > flashlight.
8. Locate the clutch pedal pushrod where it comes through the firewall and terminates at a clevis/clevis pin with a cotter pin into the pedal arm.

9. Remove the cotter pin. Save only if clean and unbent.
10. Slide the clevis pin out. The pushrod drops free. Magnetic pickup tool catches the pin when it tries to escape.
11. In the engine bay, locate the 2 CMC mounting nuts (on engine-bay side, threading onto studs through the firewall). These torque to **10 ft-lb** on reassembly per docs/torque-specs.md.
12. Unclip any hard line retainers from the CMC body.

Phase A3: Remove the old line (20-30 min)

13. Rags under the CMC and slave.
14. At the CMC end, use a flare nut wrench to crack the line fitting loose. Support the CMC with my other hand.
15. Thread the fitting off by hand.
16. At the slave (under the car, at the bellhousing), same procedure -- flare nut wrench, crack loose, unthread by hand.
17. If either end is a banjo instead of a flare, use a 12mm socket on the banjo bolt and **throw away the old crush washers**.
18. Remove the old line. Note the routing so I can replicate it with the new braided line.

Phase A4: Install the new HR CMC (30-45 min)

19. With line off and pushrod disconnected, pull the old CMC off the firewall studs. Rag underneath for the last fluid dribbles.
20. Inspect and replace the firewall grommet/seal with the fresh one from the HR kit.
21. **Bench bleed the new CMC before installing.** This single step decides whether my final bleed is smooth or an afternoon of cursing. Clamp the CMC in a soft-jaw vise. Fill the reservoir. Using the included bench bleed tubing (short hoses looped back into the reservoir), slowly stroke the piston 20-30 times until no more bubbles come out. Cap the outlet port immediately.
22. Install the new CMC onto the firewall studs. Thread the 2 mounting nuts hand-tight.
23. Torque both CMC mounting nuts to **10 ft-lb**.
24. Reattach the pushrod to the pedal with the clevis pin. **Install a NEW cotter pin** and bend the legs apart. This is the one fastener whose failure puts me on the side of the road -- no skipping.

Phase A5: Install the new stainless braided line (30 min)

25. Route the new HR line along the same path as the old one.
26. **Keep it well clear of the exhaust manifold/header.** Stainless braid transmits heat into the PTFE liner. 1+ inch of clearance, more with the Skunk2 Alpha per 13-Headers.
27. Thread the upper fitting into the CMC by hand first, then snug with a flare nut wrench.
28. Thread the lower fitting into the slave (or banjo if that's what the HR kit uses). New crush washers on any banjo.
29. Torque banjo fittings to **25 ft-lb**. Flare fittings per HR's printed spec -- typically hand-tight plus 1/6 to 1/4 turn with a flare nut wrench.
30. Secure the line at the original clip points; zip ties if clips don't match the braid profile.

Phase A6: Install the new slave cylinder (20 min)

31. The HYB-CMC-01-20 kit includes a slave. I'm installing it -- no reason not to with 170k on the old one.
32. From under the car, remove the 2 slave mounting bolts at the bellhousing.

33. Pull the slave off. Note whether the pushrod stays in the bellhousing or comes with the slave; match on reassembly.
34. Install the new slave. Torque mounting bolts to **16 ft-lb**.

Phase A7: Fill and bleed (45-60 min -- see bleed section below)

Phase A8: Verify pedal feel + disengagement test (15 min)

35. Lower the car off jack stands.
36. Key ON, engine OFF. Pump the pedal 10 times. Should feel progressively firmer with a clean, consistent engagement point. Spongy or sinking = air still in the system, redo the bleed.
37. Start the engine. Clutch in, shift into 1st. Release slowly -- car should want to move. Back to neutral. Reverse. Repeat.
38. Short test drive, 5 minutes, residential. Work through every gear. Confirm clean disengagement, no grinding, no dragging.
39. Back home, check every fitting for weeping. Top off if the level dropped.

Step-by-Step -- Path B: Full Bundle with Transmission Out

Documented for completeness. My clutch is already fresh so I'm not doing Path B now. If the clutch ever fails early or I swap in a higher-capacity unit, this is the sequence.

Step 1: Pull the transmission per 03-Clutch-and-Flywheel

Everything up through "transmission separated and on the floor" is covered there. Don't duplicate it here.

Step 2: Rear main seal (trans out, flywheel off)

- Remove the flywheel per the 03 guide. The RMS is now visible around the crank snout at the back of the block.
- Extract the old seal with a seal pick. **Do not gouge the block or crank.**
- Clean the seal bore with brake cleaner.
- Install the new **91214-RDA-A01** using a proper seal driver matched to the seal OD. **Do not hammer directly on the seal.** Hammering distorts the lip and guarantees a leak inside a year. Driver seats the seal flush with the block face.
- Light coat of fresh oil on the seal lip before installation.

Step 3: Pilot bearing

- Pull the old pilot bearing with a slide-hammer puller or the bread-packing trick (pack the cavity with white bread, insert a dowel, tap -- hydraulic pressure pushes the bearing out).
- Clean the bore.
- Install the new **91006-RWC-003** with a driver that contacts the outer race only. Drive squarely until fully seated.

Step 4: Throwout bearing and fork

- Remove old throwout bearing and fork from the transmission.
- Inspect the fork pivot for wear. Replace if worn.

- Install new OEM Koyo **22810-RWC-003**. Light PTFE grease on the fork contact points (HR kit includes a packet; otherwise Honda moly paste).

Step 5: CMC, line, slave per Path A

Everything in Path A Phases 1-6 still applies. With the trans out, slave swap is even easier from above.

Step 6: Reassemble per 03-Clutch-and-Flywheel

Trans back on, bellhousing bolts to **47 ft-lb**, refill with fresh Honda MTF-3, everything per docs/torque-specs.md.

Step 7: Bleed per procedure below

Same bleed either path.

Bleed Procedure (do not rush this)

Air in the clutch hydraulics is the #1 post-install failure mode. Spongy pedal, sinking pedal, fine-cold-soft-hot pedal -- almost always trapped air, not a bad part. I do this right the first time.

Step 1: Gravity bleed (15 min, zero effort)

1. Fill the CMC reservoir to MAX with fresh Motul RBF 600.
2. Clear vinyl hose on the slave bleeder nipple. Other end into a catch bottle with an inch of fresh fluid in it (submerged tip prevents air siphoning back in).
3. Crack the bleeder 1/2 turn. Leave it.
4. Gravity works for 15 minutes. Keep the reservoir topped off -- if it runs dry, I start over.
5. Close the bleeder.

Step 2: Pedal bleed (20+ cycles)

Two-person manual:

1. Helper in the driver's seat.
2. Helper pumps the pedal slowly 3-4 times, holds to the floor.
3. I crack the bleeder. Fluid and air bubbles exit through the clear hose.
4. I close the bleeder.
5. Helper releases the pedal slowly.
6. Repeat 20+ cycles. Top off every 5 cycles -- NEVER run dry.
7. When I see zero bubbles for 3 cycles in a row, I'm done.

One-person Motive power bleeder:

1. Attach Motive cap to the reservoir (verify adapter fits 8th gen SI).
2. Pump to 10-15 PSI. Do not exceed 15.
3. Crack the slave bleeder. Fluid flows continuously.
4. Close when clear with zero bubbles. Top off Motive reservoir as needed.

Step 3: Final verification

1. Watch the bleed hose on the last cycle -- any bubble activity at all, keep going.
2. Pump the pedal 20 times with engine OFF. Firm, consistent, engagement point in the upper third of travel.

3. Pedal sinks when held or feels mushy? Back to Step 2.

Alternative: vacuum bleeder (Mityvac)

Pulls fluid from the bleeder directly. Faster than gravity + pedal alone, but power bleeder is still superior for keeping positive pressure on the system while fluid flows.

First Drive Checklist

- [ ] Pedal firm from fully released to fully depressed, no mushy spot
- [ ] Engagement point consistent, upper third of pedal travel
- [ ] Clean disengagement at idle -- 1st and reverse without grinding
- [ ] No pedal fade after 10+ minutes of stop-and-go (the classic 8th gen symptom I'm curing)
- [ ] No weeping at CMC, line fittings, or slave
- [ ] Reservoir level stable after a 20-30 minute drive
- [ ] Pedal returns fully with no stickiness (pedal bushing 46991-S5A-004 swap helps here)
- [ ] Path B only: no slip under WOT load after the 500-mile break-in per 03-Clutch-and-Flywheel

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Common Mistakes I'm Watching For

1. **DOT 4 on paint.** Strips clearcoat on contact. If I spill, flush with water immediately, then wipe with a damp rag. Painter's tape up front catches the worst.
2. **Cross-threading a flare fitting.** Start by hand, never with a wrench. Back off if it doesn't thread smoothly -- cross-threading destroys the aluminum port.
3. **Air left in the system.** Spongy pedal = air = redo the bleed. Don't convince myself "firm enough" is firm. It gets worse with heat cycles.
4. **Skipping the bench bleed.** Getting a dry CMC air-free in-car is miserable and often incomplete.
5. **Forgetting a new cotter pin on the pedal clevis.** Old cotter is bent or corroded. New pin, legs spread. No skipping.
6. **Not torquing the CMC mounting nuts.** 10 ft-lb is gentle but specific. Loose = pedal feel drift + firewall seal weep.
7. **Hammering the RMS in (Path B).** Fragile seal lip. Use a proper driver. Hammered = weeps oil onto the clutch within 10k miles and I'm pulling the trans again.
8. **Letting the reservoir run dry during bleeding.** One lapse and the CMC sucks air. Top off every 5 cycles.
9. **Reusing old crush washers.** Work-hardened copper doesn't seal the second time. $0.50 beats a weeping banjo.
10. **Routing the braided line near the exhaust.** Heat shortens PTFE liner life. 1+ inch clearance, more with headers.

See Also

- overview.md -- part selection rationale and OEM part numbers
- 03-Clutch-and-Flywheel -- the "bundle!" cross-reference; if clutch ever needs service, everything in Path B

happens in the same session

- 02-Short-Shifter-and-Bushings -- different subsystem, but relevant if I'm already under the dash
- docs/torque-specs.md -- CMC 10 ft-lb, slave 16 ft-lb, clutch line banjo 25 ft-lb, bellhousing 47 ft-lb (Path B)
- Hybrid Racing CMC install guide (06-11 Civic Si)/8) -- manufacturer's published procedure

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Last updated: 2026-04-20

Engine Mounts -- Hybrid Racing 70A

Purchased, sitting in the box

Part Information

Field	Value
Part	Hybrid Racing Engine Mount Kit
Durometer	70A (polyurethane)
Material	Polyurethane inserts in billet aluminum housings
Fitment	8th Gen Honda Civic SI (2006-2011)
Note	These mounts may be manufactured by Hasport and sold/rebranded through Hybrid Racing
Actual Cost (incl. tax/shipping)	$657.15

Why I'm Doing This

At 170k miles, my stock rubber mounts are almost certainly degraded and soft. Stiffer mounts will reduce engine movement under load, which means:

- Better shift feel (less engine rock during gear changes)
- Improved throttle response (less drivetrain slop)
- Less wheel hop under hard launches
- Ties everything together with the Acuity short shifter and bushings for an overall tighter drivetrain feel

This is the last piece of the drivetrain feel puzzle. Once these go in alongside the clutch, flywheel, short shifter, and bushings that are already on the car, it should feel like a completely different machine.

Trade-Offs

70A is on the **streetable end** of performance mounts:

Durometer	Character
Stock rubber (~40-50A)	Smooth, quiet, isolates all vibration
70A (this kit)	Noticeable vibration at idle, moderate NVH increase, significant improvement in drivetrain feel
85-90A	Harsh at idle, very noticeable cabin vibration, track-oriented
Solid (100A+)	Race car. Unbearable for street use.

What I'm expecting: A noticeable buzz/vibration at idle and low RPM, especially through the steering wheel, shift knob, and pedals. At driving RPMs it should disappear. Most 8th gen SI owners running 70A mounts say it's a worthwhile trade-off for the improved driving feel, and I'm fine with a little vibration at idle.

Notes

- No tuning impact -- purely mechanical
- Install requires supporting the engine from above (engine support bar) or below (jack under oil pan with wood

block) while swapping mounts

- Typical install time: 2-4 hours

Engine Mounts Install Guide -- Hybrid Racing 70A

Status: Purchased, install pending

The full kit is in the box -- passenger, driver (transmission), rear, and front torque/lower mount. $657.15 spent. Last piece of the drivetrain-feel puzzle, going in alongside the already-installed Exedy Stage 1 clutch, Exedy lightweight flywheel, Acuity short shifter, and Acuity trans-side cable bushings.

Reality Check Before I Start

Intermediate-level, 3-4 hour job if nothing goes sideways. Every mount is 2-4 bolts and a swap -- the difficulty is what's around the mounts:

- Engine has to be supported the entire time a mount is out. Drop it and I'm buying an oil pan at best, a pickup tube at worst.
- Harnesses, coolant hoses, and the brake line route near the passenger and rear mounts. Pinch one and I've added a new problem.
- Access is tight up top. I'll be removing the K&N intake tube (and probably the air box) to see the passenger mount (-> 01-Cold-Air-Intake/).

Where people get hurt: rushing the engine support. Jack under the oil pan without a wide wood block dents the pan (stamped steel on the K20Z3). Engine support bar over the shock towers is safer and leaves both hands free.

Why 70A and not stiffer: reliability-first decision. 88A+ polyurethane mounts on the FG2 are known to crack the stamped OEM mount *brackets* (not the mounts -- the brackets they bolt into) over time. The 8th gen passenger mount bracket is a particular weak point. 70A gives me the vast majority of the drivetrain-tightening benefit without transmitting enough load into the brackets to tear them. Stiffer is not better past this point on this chassis.

Prerequisites / Cross-references

Accept before install

- **NVH will go up, permanently.** Idle buzz through the shift knob, pedals, and steering wheel will be noticeably stronger. Cold-start vibration is the most obvious change for the first 30 seconds each morning. At driving RPM it mostly disappears. This is the documented 70A tradeoff -- livable, but real.
- **Clunk on throttle on/off transitions is normal on polyurethane mounts.** It's the small amount of drivetrain rotation the stiffer bushings allow snapping to the loaded position. Fades over break-in. Loud metallic banging is different -- stop and inspect.

"Do This With" -- shared-labor overlap

Also doing	Overlap
Clutch & Flywheel (-> 03)	Trans mount comes out for bellhousing access -- free mount swap. Mine is already done, but for anyone reading before their clutch job: do the mounts during the clutch job.
Clutch Hydraulics (-> 07)	Doesn't lift the engine, but passenger-side access overlaps with brake/clutch line routing up top. If I'm already in there, I might as well knock out the passenger mount.

Ignition Refresh (-> 06)	Coil pack wiring routes past the passenger mount. Plugs/coils are a 15-minute add-on while the area is open.

Rule of thumb: if I'm lifting the engine for any reason -- clutch, oil pan, timing chain, rear main seal -- do the mounts at the same time. Supporting the engine is 80% of the effort.

Tools Required

Critical:

- **Engine support bar** (Harbor Freight item 61238 or similar, ~$80) OR a floor jack with a wide piece of 2x6 or 2x8 hardwood to spread the load across the oil pan. Support bar is strongly preferred.
- 14mm, 17mm, 19mm sockets (6-point, not 12-point -- these bolts will be tight)
- Matching wrenches in the same sizes (for the nut-on-the-back-side fasteners)
- Breaker bar, 18" minimum -- rear mount bolts are torqued to 47 ft-lb from factory and haven't been out since 2009 in my car
- **Torque wrench, 10-80 ft-lb range** (3/8" drive preferred for the mount bolts)
- Needle-nose pliers -- for the small harness clips around the mount brackets
- Pry bar, 12" -- for nudging the engine laterally to line up new mount bolt holes
- Floor jack + 4-ton jack stands (for the car itself, separate from engine support)
- Penetrating oil (PB Blaster) -- mount bolts are weather-exposed for 17 years

Also useful:

- Impact gun (makes initial bolt removal much faster, but do final install with torque wrench)
- Magnetic parts tray -- I will lose a bolt down into the belly pan otherwise
- Droplight / headlamp
- Sharpie to mark bolt locations and engine position before anything comes out

Parts List

#	Item	Part / Source	Status
1	Hybrid Racing Engine Mount Kit (full set -- passenger, driver/trans, rear, front torque)	Hybrid Racing, 70A polyurethane in billet aluminum housings	Purchased, $657.15
2	OEM mount hardware (reusing existing bolts unless damaged)	Honda dealer backup if threads are FUBAR	--
3	Medium-strength threadlocker (Loctite 242 blue)	AutoZone / Amazon	On hand
4	Dielectric grease (for any harness clips I unclip)	AutoZone	On hand

Note: HR mounts for the 8th gen SI are widely believed to be Hasport-manufactured and rebranded. If the sheet in the box says "Hasport" anywhere, that's not a wrong-kit problem.

Step-by-Step

Phase A: Lift and Access (20-30 min)

1. Let the car sit cold. Park on level ground. Chock the rear wheels.
2. Disconnect the battery (-) terminal. Mounts are not electrical, but the passenger mount area is right next to the alternator B+ cable and I don't need that cable arcing against a wrench.
3. Lift the front of the car, set on 4-ton jack stands on the subframe pinch welds. Both sides. Never work under a jack alone.
4. Drop the belly pan (plastic undertray) -- several 10mm bolts and plastic pushpins. Set aside.
5. Spray penetrant on every mount bolt I can see, top and bottom. Let sit while I set up the engine support.

Phase B: Engine Support (15-20 min)

6. Position the engine support bar across the shock towers, over the engine. Feet resting on the flat shock tower tops (use a towel or thin rubber under each foot to protect the paint).
7. Attach the support chain/hook to the engine lift point on the cylinder head (the cast eyelet on the back of the head, driver side, is factory-intended for this).
8. Take up slack in the chain. I want the engine supported but **not lifted** yet. The suspension should still be carrying the weight through the subframe.
9. Alternative if no support bar: position a floor jack under the oil pan with a 2x8 board across the pan to spread load over the widest area possible. **Do not jack directly on the oil pan metal -- it will dent.** The pan is stamped steel and the pickup tube is right above it.

Phase C: Passenger-Side (Upper) Mount FIRST (45-60 min)

Most accessible mount, and with the support bar at tension the engine can't move while I work.

10. Remove the K&N intake tube between air box and throttle body (3-4 hose clamps, MAP vacuum line). See 01-Cold-Air-Intake/ for hose routing on reassembly.
11. If the air box blocks mount bolts (it will, on the forward corner), pop it out. 2 bolts + MAF connector.
12. **Take a phone photo** of mount orientation, bracket orientation, and bolt positions before anything comes out.
13. Remove 2-3 bolts holding the mount bracket to the **engine** (17mm).
14. Remove 3 bolts holding the mount to the **chassis** (14mm).
15. Lift the mount + bracket out as one assembly.
16. **Lift the engine slightly** with the support bar -- just 1/4" to 1/2" -- to clear the new mount's bolt holes. No more than necessary; I don't want to stress the other mounts or axles.
17. Position the new HR mount. **Orientation matters** -- look for the alignment pin or the arrow cast into the housing. Mounting backward loads the urethane at the wrong angle and causes premature failure.
18. Install all bolts hand-tight first, then torque per the HR sheet. **Typical spec is 40-54 ft-lb depending on the bolt.** Factory reference is 47 ft-lb on the side mount through-bolts (docs/torque-specs.md). Blue Loctite on chassis-side bolt threads.
19. Release support bar tension and let the engine settle. Should sit square, no sag. Re-tension for the next mount.

Phase D: Rear Mount (Engine Block to Subframe) (30-45 min)

This one takes the most torque reaction load in 1st and 2nd gear -- biggest contributor to wheel hop on a soft mount.

20. Under the car, locate the rear mount. 3 bolts -- 2 into the subframe cradle, 1 through-bolt to the engine bracket.
21. **Support bar under tension before pulling the through-bolt.** Don't let the engine shift rearward.
22. Remove the 2 subframe bolts first. Through-bolt last.
23. Pull the old mount down and out.
24. Install the new HR rear mount -- asymmetric, find the alignment feature and install in the factory-intended direction.
25. Torque through-bolt and subframe bolts per HR spec (47 ft-lb factory reference, but **HR sheet is the authority on HR parts**). Blue Loctite.

Phase E: Driver-Side (Transmission) Mount (45-60 min)

26. Top of the transmission, bolted to the chassis frame rail. Access from above.
27. **Watch the harness routing.** Trans harness is zip-tied to the mount bracket in a couple places on the FG2 -- cut and save those positions, I'll replace on reassembly.
28. Remove bolts holding bracket to trans case (17mm).
29. Remove bolts holding mount to frame rail (typically 3x 14mm).
30. Lift mount + bracket out as one assembly.
31. Install new HR trans mount. Mind orientation, torque to spec, blue Loctite.
32. Re-route harness with new zip ties. **Do not pinch the harness between bracket and frame rail** -- chafes through over time and throws random electrical codes.

Phase F: Front Lower / Torque Mount (30-45 min)

The "dogbone" mount runs from the front of the engine down to the subframe. Controls engine rotation under throttle.

33. Locate the torque mount: small rectangular dogbone, 2 bolts.
34. **Watch for exhaust proximity.** The downpipe and stock manifold pass very close -- if the exhaust is still hot, wait.
35. Remove the 2 bolts. Tilt the mount to clear the subframe.
36. Install the new HR torque mount. **Most orientation-critical of the four** -- the urethane bushings at each end are loaded differently under forward vs reverse torque. Factory-intended direction only.
37. Torque to HR spec (typically 40-47 ft-lb). Blue Loctite.

Phase G: Final Checks and Reassembly (20-30 min)

38. **Torque every single bolt** per the HR instruction sheet. Mount bolts take shock loads -- under-torqued bolts walk loose.
39. **Gently test engine lift** with the support bar after each mount and once after all four. Engine should settle square, no twist or sag. If sagging, back out and check mount orientation.
40. Release support bar fully. Engine rides on the new mounts.
41. Visual sweep:
    - No pinched harness (especially driver-side trans mount area)
    - No hose touching exhaust or moving parts
    - Intake tube fully seated, hose clamps tight
    - No dropped hardware sitting on the subframe

42. Reinstall air box, intake tube, belly pan.
43. Lower car off jack stands. Reconnect battery (-).

Post-Install Break-In

Polyurethane mounts settle over the first several heat cycles. Urethane compresses slightly under the engine's steady weight, and the torque on the bolts relaxes as the mount cures into its final position.

- **First 500 miles:** drive normally. No repeated hard launches. Let the mounts settle.
- **At ~500 miles:** re-torque every mount bolt. I will find 1-2 that have backed off slightly. Non-optional.
- **At ~1000 miles:** final re-torque pass. Stable after this.

Skipping the re-torque is how people end up with a mount bolt backing out at 10k miles and a half-loose mount they don't know about.

First Drive Checklist

- **Idle vibration** -- expected, moderate. Buzz through the shift knob and pedals is normal. Violent shaking is not -- stop and inspect (wrong orientation, loose bolt, mount not seated).
- **Shift feel** -- tighter immediately. Less engine rock during gear changes. With the short shifter and bushings, shifts should feel surgical.
- **Clunk on throttle on/off transitions** -- **normal on 70A.** Fades over break-in. Loud metallic banging is NOT normal.
- **Wheel hop under hard launches** -- dramatically reduced. Biggest functional win of the upgrade.
- **Cold start** -- noticeably more vibration the first 15-30 seconds. Smooths as engine warms.
- **Axles** -- listen for new clicking or binding. Stiffer mounts can reveal a tired axle at 170k by letting more load through.

Common Mistakes

1. **Mount orientation wrong.** Urethane mounts are asymmetric and directional. Backward loads the insert at the wrong angle and it fails premature. Alignment pin or arrow on every mount before bolting.
2. **Forgetting the 500-mile re-torque.** Bolts loosen as urethane seats. Re-torque or accept eventual walk-out.
3. **Pinching the trans harness** between driver-side mount bracket and frame rail. Chafes through and throws random electrical codes.
4. **Engine on the bare oil pan without a wood block.** Dents the pan. Worst case dents the pickup tube bracket inside and starves the engine of oil under hard cornering. Wide 2x8 minimum, support bar preferred.
5. **Skipping the phone photo before disassembly.** I promise I'll remember bracket orientation. I won't.
6. **Over-torquing.** HR spec is 40-54 ft-lb depending on bolt. Torquing to 80+ "to make sure" strips the stamped engine-side brackets. Brackets are the weak point, not the bolts.

Expected NVH Changes -- Honest Version

Gaining:

- Significantly less wheel hop on hard launches
- Tighter shift feel (compounds with the short shifter and bushings)
- Sharper throttle response (less drivetrain slop eating inputs)
- Engine and drivetrain move as one unit under load

Giving up:

- **More cabin buzz at idle** through the shift knob, steering wheel, pedals. Noticeable at every light.
- **More cold-start vibration** the first 15-30 seconds.
- **Slight highway drone** around 2500-3000 RPM in 6th. Drive around it by staying off that exact RPM.
- **Clunk on throttle transitions** during break-in.

Net: tradeoff is worth it. Every FG2 owner running 70A in the research threads says the same thing -- livable, and the drivetrain-feel gain is real.

See Also

- overview.md -- parts, cost, rationale
- ../03-Clutch-and-Flywheel/ -- if the clutch hadn't been done yet, the trans mount would come out for free during that job
- ../07-Clutch-Hydraulics/ -- overlapping access on the passenger side
- ../01-Cold-Air-Intake/ -- air box and intake tube routing (removed for passenger mount access)
- ../06-Ignition-Refresh/ -- coil/harness routing near the passenger mount area
- ../docs/torque-specs.md -- mount torque references and the 70A hardware spec note

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Last updated: 2026-04-20

Cooling System Refresh -- Radiator, Thermostat, Hoses

Status: Planned -- Do BEFORE headers and BEFORE E85 tune

Why This Gets Its Own Folder

At 170,000 miles, my radiator is a ticking time bomb. The FG2 OEM radiator has plastic end tanks that crack around 150-180k -- the passenger-side end tank is the usual first failure. Symptom is steam from the front of the engine bay and coolant on the driveway, usually on a hot summer day when I least want to deal with it.

On top of that, full bolt-ons + E85 + 240 WHP will dump meaningfully more heat into the cooling system than the car was designed for. More combustion -> more heat -> higher coolant temps -> potential heat soak during a summer dyno session or a hard canyon run.

I'm proactively replacing the whole cooling side of the engine while I'm doing everything else. No surprises, better thermal headroom.

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1. Radiator

Why I'm Upgrading, Not Just Replacing

If I was staying stock I'd just buy another OEM plastic-tank radiator and drive on. But at my power target, the all-aluminum aftermarket radiator does two things that actually matter:

1. **More heat rejection** -- roughly 15-20% larger core volume, full-aluminum end tanks, better fin density
2. **No plastic end tanks to crack** -- the #1 failure point on the OEM unit is simply gone

The gain isn't just peace of mind. E85 dumps noticeably more heat into coolant than pump gas (higher fuel volume, more combustion mass). A hot dyno session on pump gas with the stock radiator would probably be fine. On E85 full bolt-on, margins get thinner.

Options

Radiator	Core	End Tanks	Fitment	Approx Price	Notes
Koyorad HH060063	All-aluminum, 36mm core	Aluminum	Direct fit FG2/FA5	~$250	My pick. Proven on 8thcivic. No modification needed.
Mishimoto MMRAD-CIV-06SI	All-aluminum, 42mm core	Aluminum	Direct fit	~$370	Thicker core. Overkill for NA. Lifetime warranty.
OEM Honda 19010-RRB-A51	Plastic/aluminum	Plastic	Direct fit	~$180	Same design that failed once already. Don't buy this twice.
PWR C-4418	All-aluminum	Aluminum	Fit requires trimming	~$450	Race-grade. Overkill for street.

Decision: Koyorad HH060063. Sweet spot for a street/autox FG2 at 240 WHP on E85 -- enough extra capacity to matter, no fitment drama, and $120 cheaper than the Mishimoto for essentially the same functional outcome at my power level. The Mishimoto's thicker core is nice but genuinely unnecessary for NA.

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2. Thermostat

Stock vs Low-Temp -- Why I'm Going Stock

A lot of guys default to a 160F low-temp thermostat with every cooling upgrade. **I'm not doing that.** Here's why:

- FlashPro has full control of the cooling fan tables. I can tune my fan on-point to 185-195F and keep coolant temps right where I want them through software, without hacking the thermostat spec.
- The K20Z3 wants to be at 190-200F for optimal combustion efficiency. Running it cold hurts fuel economy, increases wear, and causes more blow-by (PCV makes more work).
- Honda's 180F-opening stock thermostat is correct for the engine.

I'll replace with the OEM part and tune the fan tables in Hondata to manage heat.

Part	P/N	Price
Honda OEM Thermostat	19301-RAF-004 (K20Z3 Civic Si/TSX -- confirmed. The 19301-RNA-305 part is for the R18 non-Si Civic, do NOT order that)	~$30
Honda OEM Thermostat O-ring	19305-PR3-004	~$6

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3. Hoses

OEM rubber hoses at 170k are almost certainly leaching plasticizer, soft, and overdue. Doing them now while the system is drained means no extra coolant loss and no extra labor.

Hose	Honda P/N	Price
Upper radiator hose	19501-RRB-A00	~$30
Lower radiator hose	19502-RRB-A00	~$30
Heater hoses (pair)	79721-SNA-A00 / 79725-SNA-A00	~$40

Silicone hose upgrade? Functionally unnecessary for NA -- OEM rubber at fresh condition is fine and cheaper. But the build has a purple aesthetic theme, and silicone is the only path to a purple coolant system. **If going purple: see the Purple Cooling System section below.**

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4. Water Pump

The water pump is driven by the timing chain -- inspect when the timing chain comes off, replace proactively if I'm already in there. This is already covered in my timing chain inspection section.

Part	P/N	Price	Notes
Honda OEM Water Pump	19200-RBC-013 (K20Z3 Civic Si -- confirmed. 19200-RAA-A01 is a different family)	~$90	Replace proactively at 170k timing chain service
Honda OEM Water Pump Gasket	19222-PNA-003	~$10	Fresh gasket mandatory

Aftermarket option	Aisin WPH-802 (OEM-quality)	~$65	Acceptable substitute if Honda pump isn't available

5. Coolant

Type

Honda Type 2 coolant (blue). Premix, do not dilute further. Using anything else voids the water pump and head gasket longevity guarantees (the additive package is specific).

- **Honda OL999-9011 (Type 2 premix, 1 gallon)** -- ~$25/gallon, buy 2 gallons (system holds ~1.3 gallons but I need extra for bleeding/topping)

Do NOT use:

- Prestone "universal" -- wrong additive package
- Green ethylene glycol -- completely incompatible
- "Water Wetter" only -- not a standalone coolant, and I don't need it at NA power levels

Bleeding Procedure

The K20Z3 has an air bleed screw on the upper radiator hose neck. Critical for proper fill:

1. Fill radiator slowly with engine cold, bleed screw open
2. Close cap, run engine with heater full hot, bleed screw cracked open
3. Burp/tilt car as needed until no air comes out
4. Top off expansion tank to MAX cold line

Skip this and I get an air pocket at the thermostat -> overheating with a "good" temp sensor reading because the sensor is in a dry spot. Common 8th gen SI fail.

Cost Summary

Item	Price
Koyorad HH060063 radiator	~$250
Honda OEM thermostat + gasket	~$30
Honda OEM upper + lower radiator hose	~$60
Honda OEM heater hoses	~$40
Honda OEM water pump (with timing chain service)	~$80
Honda Type 2 coolant (2 gallons)	~$50
Bleed/fill consumables	~$10
Total (DIY)	~$520

Labor if I have a shop do the radiator alone: add $150-200. Water pump labor is effectively free during timing chain service.

When To Do This

Must happen BEFORE:

- Headers (I want fresh cooling headroom when I add combustion heat from the larger exhaust flow)
- E85 dyno tune (see above -- E85 dumps more heat)

Can happen alongside:

- Timing chain service (for water pump)
- Any other phase where car is up on stands

Cannot be skipped:

- If I see ANY weeping from the plastic end tank
- If coolant temps creep above 205F in summer traffic on the stock setup
- If there's a dye stain on the radiator fins (diagnostic of a seeping end tank)

Purple Cooling System (Optional Aesthetic Layer)

The cooling system carries the purple theme more visibly than anywhere else in the engine bay because the hoses are long and prominent. This section is **all optional** -- everything above covers the reliability-correct baseline.

6.1 Purple silicone hose kit -- Samco Sport (custom color order)

Samco Sport is the only reputable brand that offers purple as a factory color for silicone hose kits. HPS, Mishimoto, Hybrid Racing, and K-Tuned all ship black/blue/red only for the FG2 kit.

Item	Brand / Route	Price	Lead Time
Samco Civic SI 06-11 (FD2/FG2) silicone kit in purple	Order direct at samcosport.com or through Andy's Autosport	$280-$340	4-6 weeks
Alt (faster): Mishimoto MMHOSE-CIV-06BL in blue (reads purple-family in certain light)	Mishimoto direct / Amazon	~$115	1-2 days

Samco kits are 4-ply reinforced, 350degF+, highly reviewed across K-series forums for fit and longevity. The lead time is the only annoyance.

6.2 Radium coolant reservoir (custom purple anodize)

Radium Engineering 20-0270-00 universal billet coolant tank with custom purple anodize:

- Base price: $294 black; add ~$20-50 for custom anodize
- 750 mL capacity, -6AN fittings, sight tube, 32mm pressure cap receiver
- Fitment: universal -- requires bracket fabrication or creative mounting on FG2 (multiple builds use passenger strut tower)
- **Contact Radium direct** (503-244-0990) for custom purple -- not published on their website

6.3 Radiator cap -- no purple factory option

No reputable brand ships a purple-anodized cap rated for K20 system pressure (1.1 bar / 16 psi minimum). Options:

1. **Koyorad SK-C13** ($30, 1.3 bar, correct for the HH060063's Koyo-neck) powder-coated purple by a local shop for ~$20-40. Functionally correct, visually on-theme.
2. Keep Koyorad SK-C13 in factory red finish. Red reads well against purple silicone hoses in classic JDM aesthetic.

?? The Koyo SK-C13 is a **deep-plunger** cap that fits Koyo necks only -- it does NOT fit OEM Honda radiator necks. Keep a spare Mishimoto MMRC-13-SM ($25) on hand for any emergency OEM-radiator run.

6.4 T-bolt hose clamps -- stainless only

Silicone hoses require T-bolt clamps (worm-gear fails on silicone). **No reputable brand ships purple-anodized T-bolts** -- Vibrant, HPS, Mishimoto, Hybrid Racing are all stainless. Low-cost Hymee/Amazon anodized purple T-bolts have documented corrosion issues on coolant systems.

Recommended: Hybrid Racing HYB-TBC-00-10 ($45) -- stainless, correct sizes (4x 37-42mm + 2x 31-36mm) for the Civic SI FG2 hose kit.

6.5 Coolant -- the iconic purple question

Option	Purple?	Compatibility	Recommendation
Honda Type 2 OL999-9011 (blue-green)	No	Factory-correct for Honda aluminum	Primary coolant. Non-negotiable for long-term reliability.
Royal Purple Purple Ice 01600 (12oz additive)	Slight tint, ~5-10degF real-world summer temp drop	Compatible with Honda Type 2	Add. Low risk, fits theme.
Engine Ice HPC1001 (purple, propylene glycol based)	Yes -- true purple	Aluminum-safe, but different inhibitor chemistry from Honda Type 2	Avoid for full system. Don't mix with Honda Type 2. Only use if fully flushing to Engine Ice, and even then most K20 builders revert after photos.

Decision: Honda Type 2 ($50 for 2 gallons) + Royal Purple Purple Ice ($12) as the conditioner. Reliability first; purple tint as a bonus.

6.6 Skip list (not worth the purple chase)

- **Mishimoto fan shroud kit (MMFS-CIV-06SI)** -- street benefit is minimal, $215, no purple. Skip.
- **K-Tuned billet thermostat housing** -- adds failure points, $135, no purple for K20Z3. Skip.
- **Low-temp thermostat** -- fights ECU, no real benefit with FlashPro fan tuning. Skip (already covered above).
- **Powder-coating the Koyorad radiator core** -- reduces cooling efficiency (fin powder acts as insulator). End-tank-only powder is OK (~$75-125) but skip the full radiator.

6.7 Purple theme cost summary

Configuration	Cost
Critical path (reliability, OEM parts)	~$660
+ Samco purple hoses + Radium reservoir + T-bolt kit	+$665 = ~$1,325

+ Purple Ice + powder coat cap	+$40 = ~$1,365

Complete Cost Summary (reliability baseline + optional purple)

Configuration	Total
Reliability baseline (OEM Honda parts, no purple)	~$520-670
Reliability baseline + Samco purple silicone kit	~$900
Reliability baseline + Samco + Radium reservoir	~$1,250
Full purple theme (Samco + Radium + Purple Ice + Koyo cap powder)	~$1,400-1,600

See Also

- 21-Purple-Cosmetics/overview.md -- overall purple theme plan
- 05-Timing-Chain-Maintenance/overview.md -- water pump replaced at same job
- docs/mod-order-and-maintenance.md -- master plan sequencing
- docs/fluids-and-intervals.md -- coolant interval and spec

Last updated: 2026-04-20

Cooling System Install Guide -- Koyorad + OEM Pump/Thermostat + Purple Hoses

Status: Pending -- Phase 1F; must precede headers and E85

The FG2's OEM plastic-tank radiator is at the back half of its life at 170k, and thermal load goes UP when I add headers and E85. Doing it now, in my garage, beats a 95-degree breakdown three states from home.

Coolant bleeding is the #1 DIY-caused failure on the K20Z3. Rushed bleed = air pocket at the thermostat = sensor reads fine while a hot spot warps the head. That's a $2,000 mistake that takes an extra 20 minutes to avoid.

Reality Check

- **Radiator + thermostat + hoses (no water pump):** 2-3 hours wrench time, plus drive/cool cycles for the bleed.
- **Water pump added:** +2-3 hours standalone. But the water pump is timing-chain-driven -- **don't do it standalone.** Bundle with timing chain service in 05-Timing-Chain-Maintenance/. Incremental labor there is ~30 minutes because the cover is already off.
- **Purple cosmetic add-on:** no additional wrench time, but Samco kit has a **4-6 week lead** and Radium custom anodize needs a phone call (503-244-0990). Order parts before picking up a tool.

Full Saturday first time through. Fresh coolant, slow bleed, two drive cycles, check next morning, check again in a week.

Prerequisites / Cross-references

Do First	Why
Order Samco purple hose kit if going purple	4-6 week lead -- call Samco direct or Andy's Autosport
Order Radium 20-0270-00 w/ purple anodize	Call Radium at 503-244-0990; not on website
Have all OEM part numbers in hand	No substituting. See Parts List below.
Decide on water pump timing	Standalone = don't bother. Bundle with 05-Timing-Chain-Maintenance

Must Happen Before	Why
13-Headers/install-guide.md	Skunk2 Alpha dumps serious heat into the bay. I want cooling headroom in place before I add heat.
E85 tune (see 19-Flex-Fuel-and-Fuel-System/)	E85 needs more fuel mass per combustion event -> more heat rejected to coolant.
Any summer dyno session	Dyno cells get hot and still air doesn't cool a failing radiator.

Can Happen At The Same Time	Why
05-Timing-Chain-Maintenance/ water pump replacement	Timing chain comes off anyway; water pump becomes a 30-min add
21-Purple-Cosmetics/ Samco + Radium + Koyorad cap powder	This IS the purple cooling execution phase

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Tools Required

Critical:

- 2-gallon drain pan, clean, labeled "COOLANT"
- 10mm, 12mm, 14mm sockets on a 3/8" ratchet
- Extension set (3", 6", and a wobble end for the block drain)
- Long-handled flathead screwdriver for the bleed screw on the upper rad hose neck
- Hose clamp pliers (OEM spring clamps are a pain without them)
- Torque wrench, 3/8" drive, 10-80 ft-lb range
- Coolant funnel kit with rad neck adapters (Lisle 24680 or similar -- this is what makes the bleed not-awful)
- Spill mats or cardboard under the car -- Honda Type 2 stains concrete
- 1 gallon of distilled water (for 50/50 mix if using concentrate; none needed if using premix)
- Jack + 4-ton stands -- I need the car up to reach the lower hose and the block drain
- Chassis-safe jack pad for the subframe pinch weld

Nice to have:

- Inspection mirror + headlamp for the bleed screw (it's tucked behind the upper hose)
- Small pick for teasing old gasket material off the thermostat housing
- Shop rags (a lot -- coolant gets everywhere no matter what)
- Latex or nitrile gloves -- Honda Type 2 is not friendly to skin

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Parts List (verified against overview.md)

#	Item	Part Number	Notes
1	Koyorad aluminum radiator	HH060063	All-aluminum, direct fit FG2/FA5
2	Honda OEM thermostat (K20Z3)	19301-RAF-004	NOT 19301-RNA-305 -- that's R18 non-Si
3	Honda OEM thermostat O-ring	19305-PR3-004	Fresh every install. No reuse.
4	Honda OEM water pump (K20Z3)	19200-RBC-013	NOT 19200-RAA-A01 -- different family
5	Honda OEM water pump gasket	19222-PNA-003	If doing the pump
6a	Samco purple silicone hose kit	Samco Civic SI 06-11 custom purple	4-6 week lead; Samco direct or Andy's
6b	OR Honda OEM upper hose	19501-RRB-A00	Non-purple reliability baseline
6c	OR Honda OEM lower hose	19502-RRB-A00	Non-purple reliability baseline
6d	OR Honda OEM heater hoses (pair)	79721-SNA-A00 + 79725-SNA-A00	Non-purple reliability baseline
7	Radium coolant reservoir	20-0270-00 w/ custom purple anodize	Call 503-244-0990 direct
8	Hybrid Racing stainless T-bolt clamps	HYB-TBC-00-10	4x 37-42mm + 2x 31-36mm
9	Primary coolant	Honda Type 2 OL999-9011	Buy 2 gallons premix
10	Coolant additive	Royal Purple Purple Ice 01600	Additive only, not a coolant
11	Koyorad rad cap	SK-C13	Koyorad necks only -- will NOT fit OEM
12	Backup rad cap	Mishimoto MMRC-13-SM	Emergency OEM-neck cap; keep one in the trunk

Part number discipline: Honda cross-year K20 parts look similar. No substituting 19301-RNA-305 (R18) or

19200-RAA-A01 (wrong pump family) because "the dealer had it in stock." Wrong P/N = wrong temperature, wrong flow, wrong fit.

Step-by-Step

Phase A: Drain the System

1. **Engine MUST be cold.** Overnight ideal. Warm coolant under 16 psi will burn me and give false bleed readings. Non-negotiable.
2. Lift the front onto jack stands on the subframe. Chock the rear wheels.
3. Drain pan under the radiator.
4. Remove radiator cap slowly. Any hiss = not fully cold; wait.
5. Open the drain petcock (plastic thumbscrew, driver-side lower corner). Drains ~1 gallon from the radiator.
6. Move pan under the block drain -- small hex bolt on the drive-belt (passenger) side, near the oil filter housing. Another ~0.3 gallon comes out of the block.
7. Reinstall block drain with fresh washer, hand tight + 1/8 turn (soft bolt).
8. Close the radiator petcock.

Skipping the block drain keeps 30% of the old coolant in the system and dilutes the fresh fill.

Phase B: Radiator Swap (if no water pump this session)

9. Disconnect both fan shroud electrical connectors (rad fan + condenser fan), push-tab release.
10. Remove 4 fan shroud bolts (10mm). Shroud transfers to the Koyorad -- don't trash it.
11. Loosen upper hose clamp at the radiator. Twist off gently (cracked plastic neck = new problem).
12. Loosen lower hose clamp at the radiator.
13. Remove the 2 upper radiator mount brackets (10mm, 1 per side).
14. Lift the radiator straight up and out.
15. **Dry-fit the fan shroud to the Koyorad HH060063 before bolting.** Koyorad publishes direct fitment; I verify anyway. Shroud tabs should line up. If not, confirm I have HH060063 (not the 42mm Mishimoto).
16. Transfer shroud to Koyorad. Re-bolt.
17. Lower Koyorad into the lower rubber mounts. Bolt upper brackets down.
18. Reconnect fan shroud harnesses.
19. Slide upper and lower hoses onto Koyorad necks. **Use NEW Hybrid Racing HYB-TBC-00-10 T-bolt clamps.** Don't reuse OEM spring clamps on silicone -- they cold-flow through and leak.
20. Torque T-bolts to HR spec (~35-45 in-lb). Silicone deforms easily; if the hose "gives," back off 1/4 turn.

Phase C: Water Pump (IF bundled with timing chain service -- see 05-Timing-Chain-Maintenance/)

If I'm doing this standalone, stop and reschedule for timing chain service. Water pump solo is 2-3 hours of timing cover work for a $90 part. Bundled, it's 30 minutes.

21. Front cover should already be off per 05-Timing-Chain-Maintenance/install-guide.md. If not, go execute that phase first.
22. Remove the 3-4 water pump bolts (10mm). Pump pulls off with the gasket.
23. Clean the block mating surface with a plastic razor. No metal -- aluminum gouges become leak paths.

24. Brake cleaner + lint-free cloth. Let flash off.
25. Dry-fit Honda **19200-RBC-013** with new gasket 19222-PNA-003. Honda spec is no RTV; I'm following it. (Thin film of Permatex Ultra Grey optional -- OEM gasket dry is warranty-correct.)
26. All bolts hand-tight in cross pattern, then torque.
27. Torque water pump bolts to **9 ft-lb** (per docs/torque-specs.md), star pattern, 2 passes.
28. Return to 05-Timing-Chain-Maintenance/ to reinstall chain, tensioner, cover.

Phase D: Thermostat Swap

29. Thermostat housing is on the block, driver side, near the lower rad hose. 2 bolts (10mm).
30. Disconnect lower rad hose from the housing. Residual coolant drips -- rag ready.
31. Remove the 2 housing bolts. Lift off.
32. Pull old thermostat. Note orientation: **jiggle valve UP** (12 o'clock). The jiggle valve is the small spring-loaded pin that lets air escape when closed. Wrong orientation = chronic air pocket.
33. Scrape old O-ring residue from the housing groove. Brake cleaner on mating surface.
34. Install NEW **19301-RAF-004** thermostat into the block recess. Jiggle valve UP.
35. Install NEW **19305-PR3-004** O-ring. Do NOT reuse -- it's 170k miles compressed.
36. Reinstall housing, both bolts hand-tight first, then torque.
37. Torque housing bolts to **9 ft-lb** (per docs/torque-specs.md).
38. Reconnect lower rad hose with NEW T-bolt clamp.

Phase E: Hose Installation (if doing Samco purple set)

39. Lay out Samco set on cardboard -- upper, lower, heater pair, any small bypass hoses. Verify kit matches Samco's FG2 06-11 list.
40. Upper hose first -- radiator neck, then engine side. Clock so Samco logo faces forward.
41. Lower hose next -- routes past alternator. **Confirm >=1" clearance from the alternator case.**
42. Heater hoses -- run to firewall. Route away from header area; leave slack for post-header re-route.
43. Install HYB-TBC-00-10 T-bolts. Torque to spec -- **don't crank down on silicone.** Overclamping = permanent deformation and pressure-cycle weeping.
44. Confirm no hose contacts alternator case, exhaust manifold, AC compressor, or any sharp edge.

Phase F: Reservoir Swap (Radium 20-0270-00 with purple anodize)

45. Drain the stock plastic reservoir -- unclip, invert over drain pan.
46. Disconnect overflow hose at the reservoir.
47. Remove stock reservoir bracket from chassis. Save hardware.
48. Mount Radium 20-0270-00 per supplied instructions. Most FG2 builds mount on the passenger-side strut tower; verify no conflict with 18-Strut-Bar/.
49. Install -6AN fittings per Radium's diagram. Thread sealant per their spec.
50. Route overflow line from rad cap neck to Radium inlet. Trim to length; zip-tie away from hot surfaces.
51. Torque Radium mount hardware to supplier spec.

Phase G: Fill + Bleed -- THE CRITICAL PROCEDURE

This is where a lazy DIY becomes a head job. Read it twice.

52. Walk the system -- confirm all clamps torqued, hoses seated, drains closed.
53. Coolant: **Honda Type 2 OL999-9011 premix -- do NOT dilute.** If I bought concentrate by mistake, mix

50/50 with distilled (not tap) water.

54. Add Royal Purple Purple Ice **01600** per bottle (~12oz for this system). **Purple Ice is an additive, not a coolant replacement.** Goes IN ADDITION to Honda Type 2.

55. Install Lisle funnel kit on the radiator neck -- the key to filling without air lock.

56. Pour coolant slowly into the funnel.

57. **Open the bleed screw** on the upper hose junction (small plastic screw on the metal fitting near the thermostat). Close when coolant streams without bubbles.

58. Keep topping the funnel. When it stops draining, system is approximately full cold -- air pockets remain.

59. **Start the engine, cap OFF** (Lisle funnel sealed to the neck). Idle.

60. Heater to FULL HOT, fan low -- opens the heater core loop and pulls air from there.

61. Rev to ~2000 RPM, hold 30 sec, release to idle. Pumps coolant, breaks air pockets loose.

62. Repeat every minute. Coolant drops suddenly when the thermostat opens (upper hose gets hot). Top off immediately.

63. Continue until no bubbles in the funnel, upper hose fully hot, heater blows HOT, level stable.

64. Remove funnel carefully (it's full). Install the Koyorad SK-C13 cap (Koyorad cap on Koyorad neck only -- Mishimoto MMRC-13-SM stays in the trunk for OEM-neck emergencies).

65. Top off Radium reservoir to MAX COLD.

66. Drive gently 10 min -- no WOT, no sustained high RPM. Full heat cycle.

67. Park. Cool COMPLETELY -- 4 hours minimum, overnight better.

68. Re-check Radium reservoir. It will have dropped as remaining air purged. Top up to MAX COLD.

Bleed Verification

- **+1 day:** Check reservoir, top off, look for drips.
- **+1 week:** Check reservoir. Cold engine, pop the cap, verify the radiator neck is full.
- **+1 month:** Check again. Still dropping = slow air-pull or leak; investigate.

Stable at MAX COLD after a week = properly purged.

First Drive Checklist

- Temp gauge reaches normal (dead center / slightly below) in ~5 min and stays there.
- Heater blows HOT with temp knob full hot. Cold = air pocket in heater core, back to bleed.
- No coolant smell, cabin or engine bay.
- Radium reservoir stable (slow drop in first week = residual air purge, OK).
- No drips after drive + overnight sit.
- Rad fan cycles on and off correctly.
- No steam from the engine bay -- ever. Steam = hot spot or leak, shut down immediately.

Common Mistakes (Don't)

1. **Wrong thermostat P/N.** 19301-RNA-305 is R18 non-Si -- NOT K20Z3. Verify **19301-RAF-004** on the box.
2. **Wrong water pump P/N.** 19200-RAA-A01 is a different K-family. Verify **19200-RBC-013**.
3. **Koyorad SK-C13 cap on OEM neck.** SK-C13 is a deep-plunger cap for Koyorad necks only. On an OEM

neck it won't seal -- lose system pressure, boil below normal temp. That's what the Mishimoto MMRC-13-SM backup is for.

4. **Mixing Engine Ice with Honda Type 2.** Different inhibitor chemistry. Reacts, drops out of solution, clogs the heater core. Full flush first if switching.
5. **Skipping the block drain.** Keep 30% old coolant, dilute the fresh fill. 5-minute step.
6. **Skipping or rushing the bleed.** Air pocket at the thermostat = sensor reads fine while a hot spot warps the head. Most common DIY-caused K20 death. Spend the full hour.
7. **Powder-coating the Koyorad core.** Fin powder = insulator = 10-20% less heat rejection. **End tanks only** for powder coat.
8. **Over-torquing T-bolts on silicone.** Silicone cold-flows -- crank down, clamp loosens at the first heat cycle, weeps. Spec only.
9. **Reusing the thermostat O-ring.** 170k miles compressed. $6 new.
10. **Tap water.** Minerals coat the water pump impeller and head jackets. Distilled only.

Purple Cosmetics Integration Note

The cooling system is the most visible purple layer because hoses run half the bay. Two lead-time gotchas:

1. **Samco Sport purple silicone kit:** 4-6 week lead, custom color run. Samco direct or Andy's Autosport. **Don't start this phase with OEM hoses planning to "swap to Samco later"** -- filling the system twice is wasted coolant and an extra bleed.
2. **Radium 20-0270-00 custom purple anodize:** not on Radium's website. Call **503-244-0990**. ~$20-50 surcharge over black, 2-3 week lead.

Plan both orders at least 6 weeks before install day. For the rest of the purple theme, see 21-Purple-Cosmetics/overview.md.

See Also

- overview.md -- parts decision rationale, including the full purple cooling section (source of truth)
- ../05-Timing-Chain-Maintenance/ -- water pump labor bundles here; do not replace water pump standalone
- ../13-Headers/install-guide.md -- must happen AFTER this phase (heat dependency)
- ../19-Flex-Fuel-and-Fuel-System/ -- E85 tune must happen AFTER this phase (heat dependency)
- ../21-Purple-Cosmetics/overview.md -- Samco + Radium + rad cap powder-coat under the broader purple theme
- ../docs/torque-specs.md -- thermostat housing (9 ft-lb), water pump (9 ft-lb), coolant bleed screw (hand tight)
- ../docs/fluids-and-intervals.md -- coolant spec and replacement interval

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Last updated: 2026-04-20

Hondata FlashPro -- ECU Tuning

Purchased, haven't used it yet

Part Information

Field	Value
Part	Hondata FlashPro
Type	Original FlashPro (NOT FlashPro 2 -- no Bluetooth)
Software	FlashProManager (Windows only, x86)
Connection	USB Mini-B to OBD2 (built-in OBD2 connector)
Pairing	Locks to one ECU at a time
Calibration Files	.fpc format
Datalog Files	.fpdl format
Actual Cost (incl. tax/shipping)	$859.59

What It Does

The FlashPro reflashes the stock Honda ECU, and it opens up pretty much everything:

- Custom fuel maps, ignition timing, VTEC engagement point
- Rev limiter and speed limiter adjustment
- Launch control, flat-foot shifting
- Real-time datalogging (AFR, knock, temps, fuel trims, etc.)
- Real-time gauge display via FlashProManager
- Multiple calibration storage and switching

This is probably the single most impactful mod on the list. Everything else is supporting hardware -- this is what actually wakes the car up.

My Setup Plan

Phase 1: Temporary Laptop Setup

- Use any Windows laptop with the FlashPro
- Flash a community base map for "stock + intake" on the 8th gen SI
- Learn the software, datalog, monitor AFR and knock
- Add a 7" external screen for live gauges while driving
- See: Temporary-Setup/

Phase 2: Permanent In-Car PC

- LattePanda 3 Delta mini PC permanently installed in the car
- 7" touchscreen on the dash (same screen from Phase 1)
- Mausberry Circuits car power supply for clean power and auto-shutdown
- Wired to the FG2 fuse box (constant 12V + ignition-switched 12V)

- Arduino GPIO controls exhaust valve relay (future)
- See: Permanent-LattePanda-Install/

Phase 3: Dual Calibrations

- **Daily tune:** Conservative, valve closed, quiet
- **Sport tune:** Aggressive, valve open, loud
- One-tap switching via touchscreen app (Python/tkinter script)

Tuning Notes

- The K&N Typhoon 69-1014TS is the only mod that affects the tune right now
- I'm starting with the Hondata community base map -- conservative and safe
- I'm NOT adding ignition timing or leaning out fuel tables without a dyno tune
- A professional dyno tune ($300-500) is the plan once I get headers + exhaust on
- Always running 93 octane with any performance calibration
- Always saving the stock calibration as a backup before touching anything

Hondata FlashPro Install Guide -- First-Flash Procedure

Status: Purchased, first flash pending

This doc covers ONLY the first flash: unboxing the FlashPro, registering it, reading and saving my stock calibration, flashing a conservative community base map matched to my current mod state (K&N intake only), verifying no CELs, and capturing my first baseline datalog.

It does NOT cover:

- The permanent LattePanda in-car install -- see Permanent-LattePanda-Install/full-guide.md
- Actual tuning, table edits, or dyno work -- see tuning-workflow-and-maps.md
- Daily/Sport dual-map creation -- that's post-exhaust, covered in the tuning workflow doc

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Reality Check Before I Start

Time budget: 1-2 hours. ~30 min prep, ~20-30 min connect/read/backup, 2-3 min actual write, the rest for verification and the first datalog drive.

Technical difficulty: LOW. Almost no wrenching -- plug a cable into OBD2, click buttons in a Windows app, watch a progress bar.

Cognitive stakes: HIGH. A mid-write interruption is recoverable (the ECU keeps the old calibration and FPM offers Resume). But flashing on a weak battery can damage the FlashPro-to-ECU pairing and cost me a $200+ Hondata recovery fee. And flashing without a stock backup first is a choice I can't undo -- once a new calibration is over the factory one, the only way back is the .fpc I saved.

Rule #1, no exceptions: Stock backup before any write. That saved file is the factory reset button. Without it, every subsequent decision becomes a bet on "will this work?" instead of "I can always roll back."

Rule #2: Stable 12V during the write. Trickle charger on, accessories off, key held steady.

First-time ECU tuner. I take this slow, read every FPM dialog twice, and if anything feels off I stop and ask before clicking Yes.

Prerequisites -- Don't Skip

Cross-references -- things that must be done FIRST

Do First	Why
Phase 0.5 Ignition Refresh complete	Fresh plugs, healthy OEM coils, and the revised 14310-RBC-003 VTC actuator. If baseline ECU state is already throwing P0011/P0014 or misfires, my first datalog will be noise and I can't tell if a new calibration made anything better or worse.
K&N Typhoon intake installed	Already done. This is the one mod driving the need for a base map over factory.
FlashPro unboxed, cable verified in bag	USB Mini-B specifically. Don't lose it -- original FlashPro uses Mini-B, not Micro or C.

Hondata account created at hondata.com	Registration binds the unit to my account. Do this BEFORE I'm sitting in the driveway with the cable in my hand.
93-octane tank, at least half full	Stable fuel pressure matters during idle verification, and any performance calibration wants premium.

Vehicle + Environment Prerequisites

- **Stable 12V battery.** 12.6V resting, 13.8V+ with alternator charging. If mine is marginal, I replace it before flashing. A weak battery under ignition-on load for 10+ minutes will sag, and a sagging battery mid-write is the scenario Hondata specifically warns against.
- **Trickle charger / battery tender connected during the flash.** Hondata recommends this explicitly. Clamp to battery terminals (NOT the jump-start points under the hood -- those have a fuse that can isolate), set to maintenance mode.
- **Windows x86 laptop or desktop.** FlashProManager (FPM) is Windows-only, x86 only. No Mac, no Linux, no ARM tablet. My Windows laptop is what I'll use for Phase 1 (this first flash) before the LattePanda takes over permanently.
- **Admin rights on the laptop.** FPM installer + FTDI driver both need admin.
- **Antivirus disabled for the flash session.** Windows Defender occasionally quarantines the FlashPro USB-serial driver or flags FPM. Disable for the duration, re-enable after.
- **USB Mini-B cable** -- ships in the box. Don't substitute a random drawer cable unless mine is actually damaged; cheap unshielded cables have been blamed for comms dropouts on the 8thcivic.com forum.
- **Key ON, engine OFF during the write.** Do not touch the key, start the engine, or disturb the ignition while the progress bar is moving. Period.

Don't start this if

- Battery is old or weak and I haven't put the tender on it for 2+ hours
- My laptop's battery is <50% and no charger is plugged in
- I'm rushed -- can't give this an uninterrupted 90 minutes
- There's a pending Windows Update that wants to reboot
- Any existing CEL is lit (diagnose and clear first -- a stored P0011 or P0300 will poison my first datalog)

Tools Required

Item	Notes
Windows x86 laptop	Dedicated for Phase 1. Will be replaced by LattePanda in Phase 2.
FlashProManager (FPM) software	Latest version from hondata.com/downloads. Free, requires account.
FTDI VCP driver	From ftdichip.com/drivers/vcp-drivers/. FPM installer usually includes it, but a fresh install from FTDI is the troubleshooting-proof path.
USB Mini-B cable	From the FlashPro box.
OBD2 cable	BUILT INTO the FlashPro unit -- the opposite end of the housing from the USB port. No separate cable needed.

Trickle charger / battery tender	OEM-style or CTEK. 2A maintenance mode is plenty.
Multimeter (optional but wise)	Verify battery voltage before flashing rather than trusting the dash gauge.
Notepad	Physical or digital. Write down serial numbers, dialog text, any error codes. I don't want to be searching my browser history later.

Parts List

#	Item	Source	Status
1	Hondata FlashPro (original, not FP2)	Hondata retailer	Owned
2	USB Mini-B cable	In FlashPro box	Owned

That's it. Everything else (laptop, charger) is existing gear.

Step-by-Step -- Phase A through Phase H

Phase A: Pre-Flash Setup (Laptop Prep) -- 20-30 min, done the day before

Done on the couch, not in the car. Getting software and drivers right ahead of time means I'm not troubleshooting Windows in the driveway.

1. **Create a Hondata account** at hondata.com. Use a real email -- registration mail goes there.
2. **Download FlashProManager (FPM)** from hondata.com/downloads. Install with admin rights; accept FTDI driver prompts; reboot if asked.
3. **Launch FPM once, no FlashPro connected.** Confirm it opens cleanly, then close.
4. **Plug the FlashPro into a direct USB-A port** (NOT a hub, dock, or USB-C adapter -- FTDI serial comms are latency-sensitive).
5. **Open Device Manager -> Ports (COM & LPT).** I should see "USB Serial Port (COMx)". Yellow warning or under "Other Devices" = reinstall the FTDI VCP driver from ftdichip.com/drivers/vcp-drivers/.
6. **Disable Windows Update auto-restart, sleep timers, and antivirus** for the session. Revert immediately after.
7. **Register the FlashPro.** In FPM, open the registration dialog, enter the unit serial (from the housing sticker) and my Hondata credentials. This binds the unit to my account. No flashing works until registration is done, and re-registering it onto a different ECU later is a $200 fee.
8. **Verify FPM shows FlashPro as connected** via View -> FlashPro Status -- unit serial and firmware version visible.
9. **Let FPM update FlashPro firmware if prompted.** Do this with the unit on USB only (NOT plugged into OBD2), on laptop AC power.

Close FPM, unplug the FlashPro, set it aside with the cable.

Phase B: Vehicle Prep -- 10 min

Car parked in the garage, cool, overnight if possible. Cold sensors give the cleanest first datalog.

10. **Park nose-in on a flat surface.** Parking brake on.
11. **Connect trickle charger to the battery** per its manual. Verify 12.6-13.8V maintenance reading. Leave connected the entire session.
12. **Hood up** for battery access and engine-bay eyeballing on first start.
13. **All accessories OFF.** Radio, climate, headlights, cabin lights. Every amp the accessories pull is an amp the ECU doesn't get.
14. **Verify battery voltage** via multimeter or dash gauge at key-ON: 12.6V+ resting.
15. **Set up driver's-seat workstation.** Laptop on passenger seat or lap desk, charger plugged in, water/coffee ready. I'll be sitting here 45-60 min.
16. **Confirm no dash CELs lit** at key-ON. Any existing stored code gets written down now before I touch anything.

Phase C: Initial Connection + ECU Read -- 10-15 min

17. **Plug FlashPro's OBD2 end into the OBD2 port** (under dash, driver's side, left of steering column on FG2). Push firmly -- connector clicks when fully seated. A loose OBD2 seat is the #1 cause of mid-session comms dropout.
18. **Plug USB Mini-B into FlashPro, USB-A into the laptop's direct port** (same port I tested in Phase A).
19. **Key to ON (position II), engine OFF.** Dash wakes up, ECU powered. Do NOT crank.
20. **Verify FlashPro LEDs.** Power LED solid (USB enumerated); OBD/ECU link LED solid within a few seconds (K-line handshake established).
21. **Launch FPM.** Auto-detects the FlashPro -- confirm via View -> FlashPro Status.
22. **Run Tools -> Read from ECU.** Pulls the factory PRB calibration off the ECU into FPM as a saveable file.
23. **WAIT.** The read is slower than a write -- 2-5 minutes is realistic. DO NOT interrupt, touch the key, or wiggle the cable. Watch the progress bar.
24. **FPM displays ECU type.** Should read PRB (any sub-rev like PRB-A00 or PRB-A58 is fine). Anything else = STOP; wrong car or comms issue.

Phase D: STOCK BACKUP -- 5 min, critical

The single most important step in this document.

25. **File -> Save As** in FPM.
26. **Name it** STOCK_PRB_2026-04-20_pre-any-flash.fpc. YYYY-MM-DD for clean sort order.
27. **Save to local disk** under my build folder (C:\Users\extra\OneDrive\Desktop\Civic-SI-Build\10-Hondata-FlashPro\calibrations\). OneDrive syncs to cloud automatically -- but I don't trust that alone.
28. **Copy the file to a second location** -- external USB or a fresh cloud upload. Two independent copies.
29. **Verify file size non-zero** (typical .fpc is a few hundred KB).
30. **Close FPM, reopen FPM, open the saved .fpc** to confirm it loads and shows tables. An unreadable backup is no backup.
31. **Write the filename and save location in my build notes.**

This file is now my factory reset button. If anything downstream goes wrong, flashing this file returns the car to factory ECU state.

Phase E: Base Map Selection -- 10 min

Current mod state is stock everything + K&N Typhoon intake, so I want an "intake-only, 91/93 pump gas" 8th-gen SI community base map. Hondata ships a library of these with FPM, plus more on 8thcivic.com.

32. **Open the base map library** (File -> Open, typically Documents\Hondata\FlashProManager\Calibrations\).
33. **Pick the closest match to my hardware.** Intake-only, pump gas, no other mods. DO NOT pick anything assuming exhaust, headers, intake manifold, or E85 -- mismatched maps run lean or pig-rich and CEL or worse.
34. **Review the base map in FPM before flashing:**
    - **AFR at WOT:** commanded >= 12.0. Leaner on pump gas = red flag.
    - **Rev limiter:** <= 8200 RPM (stock). A 9000 RPM rev limit is not a first-flash map.
    - **Ignition timing at WOT:** not more aggressive than stock.
    - **VTEC engagement:** stock-ish (~5400-5800 RPM).
    - **Knock correction:** enabled, not disabled.
35. **Save a working copy** as BASE_intake-only_<date>_working.fpc. Never flash the original base file -- keep the source pristine.

If no community base map is a perfect match, I'd rather flash the stock backup right back and wait for a tuner recommendation than flash something mismatched.

Phase F: THE FLASH -- 5 min, no touching anything

36. **Verify battery voltage still 12.6V+.** Below 12.3V = STOP, fix, restart.
37. **Close all other Windows programs.** FPM only. No browser, no Slack, no background updaters.
38. **In FPM: Tools -> Flash Calibration** (or Program ECU, version-dependent). Select the working copy of my base map.
39. **Read the confirmation dialog.** Should show calibration name, ECU type (PRB), and not-interrupting warning. Accept.
40. **Progress bar starts.** Write takes ~20-45 seconds.
41. **DO NOT TOUCH ANYTHING.** Not laptop, key, OBD2 connector, or USB. Hands in lap.
42. **Wait for "Program Complete" dialog** -- flash is done when it appears.
43. **Do NOT immediately power cycle the key.** Let the ECU settle a few seconds, dismiss the dialog, then touch the key.

If the flash fails mid-write: DO NOT restart the flash on my own. The ECU keeps its previous (stock) calibration. FPM detects incomplete flashes on next connection and offers Resume. If Resume fails or anything feels wrong, call Hondata support BEFORE further action.

Phase G: First-Start Verification -- 10-15 min

44. **Cycle key OFF, wait 10 sec, back to ON.** Listen for 2-second fuel pump prime from the rear. No prime = stop, something's wrong.
45. **Start the engine.** Should fire cleanly on first crank. Long crank or rough start = yellow flag.
46. **Idle 3-5 min, hands off throttle.** Watch dash for CEL.
47. **If a CEL lights:** shut off, connect FPM, pull the code, do not drive. P0301-P0304 = likely mismatched base map; flash stock backup and regroup. P0420 = base-map rear-O2 settings; not dangerous but worth understanding. P0011/P0014 = base-map issue OR pre-existing VTC actuator issue that Phase 0.5 should have fixed.
48. **No CEL = good.** Verify basics:

- Idle RPM stable 700-800
- ECT climbing normally toward 190-200F
- No exhaust smoke (brief cold-start vapor is normal)
- No unusual ticking, pinging, rattle

49. **Short test drive, 5-10 min mixed conditions.** Neighborhood streets, gentle acceleration. NO WOT, NO VTEC pulls. Verifying the car behaves normally, not tuning.

Phase H: First Datalog -- 20-40 min

First real log with the FlashPro. Goal is a clean baseline I can compare future drives against, not a performance session.

50. **Open FPM -> Datalogger view.**
51. **Configure datalog parameters.** Explicitly include: RPM, VSS, TPS, APP, MAP, IAT, ECT, Commanded AFR, STFT, LTFT, Primary O2, Rear O2, Knock Count, Knock Retard, Commanded Ignition Advance, Actual Ignition Advance, Battery Voltage, VTEC state, VTC target, VTC actual, Injector Duty %. I can't recover data I didn't log.
52. **Datalog -> Start.**
53. **Drive 15-30 min of mixed conditions:**
    - 2-3 min idle in driveway before pulling out (cold-start behavior)
    - Low-speed neighborhood (25-35 mph, light throttle)
    - Highway cruise (60-70 mph, steady throttle) for 5+ min to stabilize trims
    - Light-to-moderate partial throttle, 2500-4500 RPM
    - **NEVER WOT.** Not without a wideband, not before Tune #1 on the dyno.
    - **DO NOT hit VTEC deliberately.** Keep under 5000 RPM.
54. **Back home, engine idling: Datalog -> Stop.**
55. **File -> Save Datalog As** -> baseline_intake-only_<YYYY-MM-DD>.fpdl. Double-save (local + cloud + external).
56. **File -> Export -> CSV** for a human-readable copy I can paste into Claude.
57. **Quick review in FPM:**
    - Knock Count = 0 across entire log. Any non-zero = investigate.
    - STFT within +/- 8% during closed-loop cruise.
    - LTFT still learning -- will drift for the first few drives.
    - ECT 190-210F at cruise.
    - Commanded vs actual ignition advance track together. Large gap = knock correction active.
    - No limp mode, no CEL, no stored codes.

Clean across the board = verified success. Subsequent work (headers, exhaust, dyno tunes) lives in other docs.

First-Start Checklist

- [ ] Stock calibration saved locally AND to a second location
- [ ] Stock .fpc file opened and verified readable
- [ ] Base map reviewed for safe AFR, rev limit, timing, VTEC
- [ ] Battery 12.6V+ with tender connected

- [ ] Accessories off, hood up, key to ON only (not crank)
- [ ] Flash completed with "Program Complete" dialog
- [ ] Engine starts cleanly, idles stable 3-5 min
- [ ] No CEL after 10 minutes of idle + gentle driving
- [ ] First datalog captured, exported to .fpdl + .csv
- [ ] Knock count = 0 across the entire first log
- [ ] STFT within +/- 8% during closed-loop cruise

Troubleshooting Shortlist

Symptom	First Check
FPM won't see the FlashPro	FTDI driver in Device Manager -> reinstall from ftdichip.com if missing or yellow-flagged. See troubleshooting-and-faq.md for the full walkthrough.
FPM sees FlashPro but can't read ECU	Key at ON (not ACC, not OFF). OBD2 connector fully seated. Battery above 12.5V. No other scantool sharing the port.
Flash fails partway through	STOP. Do not start another flash. Reconnect; FPM will detect incomplete flash and offer Resume. If Resume fails or anything feels off, call Hondata support before further action.
CEL after flash	Read the code in FPM, compare against the code table in troubleshooting-and-faq.md. For anything I don't understand: flash the stock backup back, regroup.
Car won't start after flash	Most likely a base-map mismatch. Flash the stock .fpc backup to return to factory. Verify start. Re-evaluate which base map I picked.
Rough idle after flash	Give the ECU 5-10 minutes of runtime to adapt trims. If persistent, base-map mismatch -- revert.
USB cable feels loose	Try the other known-good USB Mini-B cable before assuming the FlashPro port is damaged. Never flash on a wobbly connection.

Full troubleshooting coverage lives in troubleshooting-and-faq.md.

Common Mistakes I Am Specifically Watching For

1. **Skipping the stock backup** because "the base map is fine." No. Stock backup every time, before every write. Rule #1.
2. **Flashing on a weak battery.** The one scenario that damages unit-to-ECU pairing and lands me in Hondata's paid recovery. Tender on, 12.6V+, verified.
3. **USB hub or dock** instead of a direct laptop USB-A port. FTDI serial comms are latency-sensitive. Direct only.
4. **Flashing a FlashPro 2 calibration onto my original FlashPro.** Feature flags differ. Only use .fpc files confirmed compatible with FlashPro original + 06-11 Civic SI PRB.
5. **Aggressive base map** (headers/exhaust/E85) on an intake-only car. Intake-only community map or nothing.
6. **Interrupting a flash to "check something."** Nothing checks better than letting the write finish.
7. **Antivirus still running** -- Windows Defender can quarantine a driver mid-session. Disable explicitly before Phase F.
8. **Losing the USB Mini-B cable** that came in the box. FlashPro original uses Mini-B specifically.

9. **Not writing down the stock filename and backup locations.** Three months from now, finding my stock backup should be a 10-second task.
10. **First-flash + first-datalog in a rushed 30-min window.** This deserves an unhurried afternoon.

Dual-Map Pre-Flight

This guide covers ONLY the first flash of a single, conservative community base map. Daily/Sport dual-map setup is NOT part of the first flash:

- Daily and Sport maps are the product of Tune #1 on a dyno, after headers + HFC + exhaust cutout are installed.
- Full dual-map rules (AFR targets, timing ceilings, VTEC crossover, rev limiter strategy) live in tuning-workflow-and-maps.md.
- Switching calibrations on original FlashPro requires a fresh flash each time (~20-45 sec). That's a LattePanda-install concern, not a today concern.

If I catch myself thinking "let me just try a Sport map right now" -- no. Not without a wideband, a dyno, a tuner, or knock monitoring during real WOT pulls. First flash is the conservative community base map, full stop.

See Also

- overview.md -- FlashPro hardware basics, cost, and my setup plan
- hardware-software-deep-reference.md -- full engineering detail on the unit and FPM software
- tuning-workflow-and-maps.md -- Daily/Sport map rules, reliability-first ceilings, how tune sessions are planned
- troubleshooting-and-faq.md -- full code/problem reference
- ../06-Ignition-Refresh/ -- required prerequisite (plugs + coils + VTC actuator)
- ../11-Wideband-AFR/ -- required before any dyno tune; bung gets welded during header install
- ../19-Flex-Fuel-and-Fuel-System/ -- flex fuel sensor wiring is a separate install, future phase
- Permanent-LattePanda-Install/full-guide.md -- the in-car PC install this laptop workflow bridges toward
- ../docs/baseline-logs.md -- A/B datalog protocol I should follow for every mod milestone after this first log

Last updated: 2026-04-20

FlashPro Hardware + FlashProManager Software -- Deep Reference

> Everything I know about the original Hondata FlashPro unit and FlashProManager (FPM) software, consolidated. This complements the basic overview.md with the engineering-level detail I'll actually need during install, setup, and tuning.

Verification note: The live forums and Hondata docs aren't accessible from this research session, so items flagged **[VERIFY]** are claims from my accumulated K-series/FlashPro knowledge (through Jan 2026) that I want to confirm against Hondata's current support KB or the quick-start card in my box before I bet anything important on them. Everything else is high-confidence.

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1. Hardware -- What I Own vs What's Current

FlashPro Original (mine)

Field	Value
PC connection	USB Mini-B
ECU connection	Built-in male OBD-II connector on the opposite end of the housing
Bluetooth	None
Wi-Fi	None
Analog inputs	2 external inputs, each 0-5V DC single-ended
Calibration storage	One active calibration at a time (the one currently flashed to the ECU)
Datalog storage	Internal flash for a limited-length log, plus unlimited via FPM streaming to PC
LEDs	2-3 status LEDs on top of unit ([VERIFY] exact count -- Hondata quick-start card has definitive diagram)
Side button	Used for datalog start/stop on some revisions ([VERIFY] my specific unit's behavior from the box doc)
Internal chipset	FTDI USB-to-serial bridge -- shows up in Windows Device Manager as "USB Serial Port (COMx)"

FlashPro 2 (current retail, NOT mine)

Difference	Impact for Me
Adds Bluetooth	Enables Hondata Mobile iOS/Android app -- my unit doesn't have this
Different USB connector ([VERIFY] USB-C vs Micro-B)	Cable incompatibility if I borrow a FlashPro 2 unit
More analog inputs (I believe 4)	For my build, 2 is sufficient (wideband + flex fuel sensor)
Newer ECU support (Type R FL5, 11th gen Si)	Not relevant to my K20Z3

Bottom line: for the FG2 K20Z3, the original FlashPro is **fully supported** and gives me everything I need. FlashPro 2 doesn't add any function I'd use on my car.

ECU Compatibility -- PRB Variants

The K20Z3 in the 06-11 Civic SI ships with PRB-family ECUs. Hondata's FlashPro supports **any PRB ECU for a**

06-11 Civic SI (coupe FG2 or sedan FA5). Sub-revisions (PRB-A00, -A10, -A58, etc.) all work -- Hondata doesn't discriminate by sub-variant for the Civic SI application.

What does NOT work:

- Non-SI Civic ECUs (SDA for automatic, RNA for base)
- ECUs from JDM FD2 (Japanese-market Civic Type R -- different ECU family)
- Any ECU not originally an 06-11 Civic SI PRB

LEDs and Status Indicators

My best understanding of what the LEDs mean (**[VERIFY with box quick-start]**):

LED	State	Meaning
Power/USB	Solid	USB is connected, unit is enumerated on the PC
Power/USB	Off	Not connected or driver issue
OBD/ECU link	Solid	K-line/ISO communication established with the ECU
OBD/ECU link	Blinking	Flash or datalog operation in progress
Activity	Blinking	Writing internal datalog
Activity	Solid	Actively flashing ECU calibration

Recovery from Failed Flash

If a flash gets interrupted (car shuts off, USB unplugged, etc.), the ECU is NOT bricked. FlashProManager detects an incomplete flash on the next connection and offers to resume. **Recovery is software-driven**, not via physical button combinations on the unit.

The one thing that CAN permanently damage the pairing: trying to flash with a battery below ~11V. Always flash with engine running or with a battery charger attached maintaining voltage.

Driver Issues (the #1 Support Issue)

Windows 10/11 sometimes replaces the FTDI VCP driver with a generic Microsoft serial driver after Windows Update. Symptoms:

- FPM fails to see the FlashPro unit
- Unit shows in Device Manager but under different category
- Flash attempts timeout

Fix: reinstall the FTDI VCP driver from ftdichip.com/drivers/vcp-drivers/. This is the top item in any FlashPro troubleshooting guide.

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2. FlashProManager (FPM) -- Software Deep Dive

Top-Level Views

View	Function	Keyboard / Menu
Calibrator	Edit tunable tables and settings	File -> Open Calibration, then Edit menu

Parameter Monitor	Live gauges from ECU over K-line	View -> Parameter Monitor
Datalogger	Start/stop recording, browse past logs	View -> Datalogger
FlashPro Status	Unit info, firmware version, current paired ECU	View -> FlashPro Status
Flash Calibration	Upload a calibration to the ECU	Tools -> Flash Calibration (or equivalent)
Read from ECU	Download the currently-flashed calibration from ECU	Tools -> Read from ECU

The Calibrator -- Organization

Detailed breakdown in calibrator-tables-reference.md. Top-level groups I recall:

- Fuel (main fuel, low-cam vs high-cam, cranking, warmup, after-start, flex fuel tables)
- Ignition (main timing, low-cam vs high-cam, cranking, per-gear trims where available)
- VTEC (engagement RPM, throttle threshold, hysteresis, oil pressure gating)
- Cam Angle / VTC (target phase, low-cam vs high-cam, min ECT)
- Idle (base RPM, A/C idle-up, electrical-load idle-up, IACV calibration)
- Rev Limits (main fuel cut, per-gear, valet)
- Speed Limiter
- Launch Control / Flat-Foot / Two-Step
- Shift Cut / No-Lift-Shift
- Torque Management
- Fans (low-speed / high-speed ON/OFF temps, A/C override)
- A/C (WOT cut, high-RPM cut)
- O2 / Fuel Trims (closed-loop target, trim authority limits, rear O2 enable)
- EVAP (purge valve duty, gating conditions)
- Cranking / Startup
- Analog Inputs / Gauges (wideband voltage->AFR curve, flex fuel freq->% curve, scaling for optional oil pressure / boost / etc.)
- DTC / CEL options (rear O2 defeat, IMRC defeat for Ultra Street, EVAP DTCs)
- Misc / Options

Parameter Monitor -- Live PIDs

The K-line runs at ~10-20 Hz for a typical parameter set. Parameters I recall being available (non-exhaustive, **[VERIFY against current FPM]**):

Core engine: RPM, VSS, TPS %, APP % (accelerator pedal), MAP, MAF, IAT, ECT, Calculated Load, Ignition Advance (commanded), Ignition Advance (actual, post-knock-retard), Knock Count, Knock Retard amount, STFT, LTFT, Commanded AFR, Primary O2 narrowband voltage, Rear O2 narrowband voltage, Injector Duty %, Injector Pulse Width (ms), Battery Voltage.

VTEC/VTC: VTEC solenoid state (on/off), VTC target (deg), VTC actual (deg).

Control signals: IACV duty %, Purge Valve duty %, A/C request, Fan request (low/high), Commanded Throttle, Actual Throttle, Clutch switch, Brake switch, Neutral switch, Closed-loop flag, Open-loop flag, Gear (calculated).

User analog inputs: Analog 1 (raw voltage + configurable scaled value -- e.g., wideband AFR), Analog 2 (same -- e.g., flex fuel %).

Datalogger Details

- **File format:** .fpdl -- **binary, proprietary, not publicly documented.** Cannot be parsed by external tools directly.
- **Sample rate:** ~20 Hz typical per parameter (**[VERIFY in FPM settings]**).
- **File size:** 10-minute mixed-drive log with ~30 parameters typically produces ~500KB-2MB.
- **Export to CSV: File -> Export -> CSV** (or Datalog menu -> Export -- exact location varies by FPM version). Produces a human-readable CSV with a header row of parameter names + units and one row per sample.
- **What's logged by default:** Core engine parameters (RPM, TPS, ECT, IAT, MAP, STFT/LTFT, Knock Count, Knock Retard, Commanded/Actual Timing, VSS).
- **What I must manually enable:** Analog inputs (wideband, flex fuel), VTC actual/target, injector duty, per-cell fuel trim, detailed cam phase.

Before my first dyno session, I open Datalog -> Parameters and check every box I might want data on. Logging more doesn't cost anything; missing data is expensive.

Analog Inputs -- How They Work

My original FlashPro has 2 analog inputs, each 0-5V DC single-ended referenced to FlashPro ground. Typical use:

Analog 1 -> AEM X-Series Wideband (30-0300)

- Wire the AEM's "Analog Output 1" wire to FlashPro's analog 1 input
- Common scaling: 0V = 10.0 AFR, 5V = 20.0 AFR (AEM X-Series gasoline default)
- Configure in FPM: Analog Inputs -> Analog 1 -> Wideband O2 preset (AEM 30-0300)

Analog 2 -> Flex Fuel Sensor

- The GM/Continental 13577429 sensor outputs a **frequency** signal (50 Hz = 0% ethanol, 150 Hz = 100%), NOT a voltage.
- **[VERIFY -- load-bearing for my flex fuel build]:** whether the original FlashPro's analog 2 can directly read frequency, or if it requires:
    - Wiring to the rear O2 signal input pin at the ECU (bypassing the analog input entirely), OR
    - An external frequency-to-voltage converter before feeding analog 2
- FlashPro 2 reads frequency natively. I'm less sure about the original unit. **This is the single most important thing to verify before I commit to flex fuel wiring.**

Cross-reference: Both 19-Flex-Fuel-and-Fuel-System/overview.md and 11-Wideband-AFR/overview.md already flag this wiring question. My tuner's answer (or Hondata's flex-fuel setup KB article) is what I trust.

Calibration Files (.fpc)

- **Binary proprietary format.** Cannot be edited outside FPM.
- **Storage:** I can save as many .fpc files as I want on my PC. FPM opens one at a time for editing.
- **Unit-side storage:** The FlashPro holds only the **currently-flashed** calibration. To switch tunes (daily ? sport), I open FPM and flash the other .fpc.
- **Read from ECU:** Tools -> Read from ECU downloads whatever is currently flashed to the ECU into a new .fpc file. This is how I'd back up my stock calibration before flashing anything else.
- **Diff two calibrations:** FPM has a "compare" feature (**[VERIFY name]**) that shows which cells differ between

two .fpc files. Essential for understanding what a tuner changed.

Multi-Calibration Switching (Daily ? Sport)

The original FlashPro cannot switch calibrations without a PC flash. This is a meaningful limitation for my two-map goal.

Options:

1. **Laptop in car + flash on-demand:** inconvenient but works. Each flash is ~20-45 seconds (**[VERIFY -- I initially said 5-6 min but that's more likely the full calibration READ time, not the WRITE]**).
2. **LattePanda in car + permanent FPM install:** what I'm actually building toward. One-tap calibration switching via a Python script that calls FPM's flash operation.
3. **FlashPro 2 upgrade:** supports on-unit calibration switching natively via the mobile app. Would cost ~$700 to upgrade from original. Not worth it for me.

Flash Time Reality Check

[VERIFY actual times by stopwatch on my first flash]:

- Calibration write (flash to ECU): ~20-45 seconds
- Calibration read (download from ECU): ~2-5 minutes (slower)
- Firmware update (FPM -> FlashPro unit itself): ~1-3 minutes

The "5-6 minutes" I saw in forum posts is likely a worst-case or older-FPM-version read operation. Real-world flash is fast.

Export Capabilities

- Calibration -> CSV: No direct export of every table to CSV. FPM has a printable/exportable **calibration report** (HTML or PDF, depending on version -- **[VERIFY]**) that lists all table values. Useful for sharing with a tuner who doesn't have FPM handy.
- Datalog -> CSV: Yes. **File -> Export -> CSV** (or equivalent).
- Calibration compare -> CSV: Possible via the compare feature if it exists.

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3. What's NOT Possible (Important for Setting Expectations)

These things I cannot do with the original FlashPro, and no amount of clever wiring or software will change that:

1. **No public API.** Hondata does not publish an SDK, REST endpoint, or documented IPC interface. Third-party tools cannot read .fpdl or write .fpc files in a supported way.
2. **No Bluetooth.** Mobile app support is FlashPro 2 only.
3. **No on-unit calibration switching.** Always requires FPM + a flash.
4. **No .fpdl parsing outside FPM.** CSV export is the only way to get data into Python / Excel / another tool.
5. **No command-line flash operations (that I'm aware of). [VERIFY]** whether FPM has a documented CLI -- last I checked, it didn't.
6. **No odometer adjustment.** K-Pro (a different Hondata product for older Hondas) has more ECU manipulation capability than FlashPro. FlashPro is calibration-focused only.

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4. Ownership Housekeeping

ECU Re-Registration

If I ever sell the car or swap ECUs, Hondata charges a fee to re-pair my FlashPro to a new ECU. **Commonly cited as $200** (**[VERIFY current price at hondata.com/support]**). One-time per transfer.

Counterfeits

Counterfeit FlashPro units have been sold on eBay and Facebook Marketplace. Signs of a fake:

- Price significantly below new retail
- Seller won't provide serial number before purchase
- Missing Hondata hologram sticker
- FPM won't enumerate the unit despite driver install

I bought mine from a legit retailer. Anyone buying used: verify serial with Hondata support before paying.

Firmware Updates

FPM auto-updates FlashPro firmware when a new version is available. I do firmware updates with:

- Laptop plugged into stable power (not battery-only)
- FlashPro plugged into USB (not into OBD2)
- No interruption (don't unplug during update)

A failed firmware update is recoverable -- FPM detects it on next connection and retries.

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5. Integration Paths for My Build

Phase 1 Workflow (Laptop-Based)

1. Windows laptop on passenger seat during drives
2. FlashPro plugged into OBD2 + laptop USB
3. FPM running with datalogger armed
4. 7" external screen on dash showing live parameter monitor gauges (via FPM's display-only mode)
5. Post-drive: CSV export for analysis, compare calibrations, plan changes

Phase 2 Workflow (LattePanda 3 Delta In-Car)

Covered in detail at Permanent-LattePanda-Install/full-guide.md. High-level:

1. LattePanda permanently mounted with auto-boot on ACC
2. FPM auto-launches on Windows login
3. FlashPro permanently wired via USB Mini-B inside the cabin (no connector exposed)
4. 7" touchscreen on dash = FPM display
5. Arduino GPIO on the LattePanda drives the QTP exhaust valve relay
6. **[ASPIRATIONAL]** Python script watches the datalog folder, auto-exports .fpdl -> .csv, uploads to cloud for Claude to analyze remotely (**[VERIFY if FPM has a CLI or if this requires UI automation -- see agent research notes above]**)

Phase 3 Workflow (Future -- Claude Integration)

Covered in tuning-workflow-and-maps.md. I paste exported CSV datalogs to Claude for analysis, get specific table-cell recommendations back, apply them in FPM manually, flash, verify.

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6. What I'd Do Differently In Hindsight

If I were buying a FlashPro today from scratch:

- **For a daily-driven street car that never leaves the country:** the original FlashPro does everything I need. Stick with it.
- **For track days where I want mobile datalogging:** FlashPro 2 for the Bluetooth -> mobile app.
- **For anyone doing only quick tuning without a PC ever in the car:** FlashPro 2.

But my build is primarily a PC-based workflow (the whole LattePanda plan), so the original FlashPro was honestly the right buy.

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7. Action Items Before My First Flash

Everything I need to verify / do before I press Flash for the first time:

- [ ] Confirm FlashPro enumerates in FPM (driver installed correctly)
- [ ] Confirm FlashPro firmware is current (FPM will prompt)
- [ ] **Read stock calibration from ECU** -> save as STOCK_ORIGINAL_20260419.fpc (the factory reset file)
- [ ] Register the FlashPro with Hondata (ties unit serial to my ECU serial)
- [ ] Pick starting calibration -- Hondata community base map for "stock + intake" ([VERIFY correct base map name])
- [ ] Verify battery voltage above 12.5V before flashing (ideally engine running)
- [ ] After flash: verify no CELs, idle reads sane, no knock in first 10 minutes of driving
- [ ] Set up datalog parameters to include everything relevant, not just defaults
- [ ] **Verify analog input wiring works** (wideband reads sensibly at idle) before trusting any datalog

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See Also

- overview.md -- hardware overview + setup-path summary
- tuning-workflow-and-maps.md -- Daily/Sport map rules, reliability-first ceilings
- calibrator-tables-reference.md -- every tunable table with safe ranges
- datalog-analysis-guide.md -- how to read the logs I'll record
- advanced-features.md -- launch, flat-foot, per-gear, shift lights, torque management
- troubleshooting-and-faq.md -- CEL codes, common issues, Windows driver fixes
- tuner-directory.md -- shops I'm considering for the three dyno sessions
- Temporary-Setup/getting-started.md -- laptop-based first-flash walkthrough
- Permanent-LattePanda-Install/full-guide.md -- in-car PC build

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Last updated: 2026-04-19

FlashPro Calibrator Tables -- My Reference

> Every tuning table in FlashProManager, what it controls, typical values, and safe-range guidance for my K20Z3 at 170k miles. This is the file I open when a tuner says "bump cell 4500-LOAD8 by 0.5 degrees" and I need to know which menu to look in.

Verification note: I pulled this together from my accumulated research + Hondata documentation + community knowledge. Before trusting specific numerical ranges on a dyno session, I'm cross-checking with my tuner and against the current Hondata help docs (FlashProManager -> Help menu -> Help Contents, or hondata.com/flashpro-civic-si-2006-2011-us). Items I flagged with **[VERIFY]** below are things I'll double-check before I apply them.

How FlashProManager Is Organized

FlashProManager is a single Windows app with a handful of top-level views:

View	What It Shows	When I Use It
Calibrator	Every tunable table + setting in the current calibration	Anytime I want to see or change what the car is calibrated to do
Parameter Monitor	Live data stream from the ECU via OBD2	Troubleshooting, verifying behavior after a change, gauge-style dashboard
Datalogger	Start/stop recording, review past logs	Gathering data for tuning, A/B comparisons, diagnosis
FlashPro Status	Unit status, firmware, calibration currently loaded, VIN pairing	Sanity check before/after a flash
Help	Built-in reference for every calibrator parameter	First place to look when a table name isn't obvious

The Calibrator is organized by function. What follows is my cheat sheet for what lives where and what each one actually controls.

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1. Fuel Tables

Low Cam Fuel Table (Pre-VTEC)

Controls the air-fuel ratio target when the engine is running on the low-lift cam profile (below VTEC engagement). Axes are RPM vs engine load (g/rev or MAP depending on calibration).

What I tune here:

- AFR targets at cruise (closed-loop -- usually stays at 14.7 stoich regardless)
- AFR targets at mid-throttle (should trend slightly richer, 13.8-14.2)
- AFR targets at WOT below VTEC (12.5-13.0 typical for pump gas)

Safe range for my NA 170k K20Z3 on pump 93:

Condition	AFR Target
Idle / closed-loop	14.6-14.8 (let ECU do its thing)

Cruise	14.5-14.9 (closed-loop zone)
Part-throttle	13.5-14.2
Low-cam WOT	12.5-13.0 (Sport) / 12.8-13.2 (Daily)

Never lean past 12.0 AFR at WOT on pump gas. Never.

High Cam Fuel Table (VTEC Engaged)

Same axes, but for the high-lift cam profile. This is the table that actually matters for WOT power -- once VTEC engages, this is where the engine lives at high RPM.

Safe range for my NA 170k K20Z3 on pump 93:

Condition	AFR Target
High-cam light load (rare -- usually you only hit high cam at WOT)	13.0-13.5
High-cam WOT	11.8-12.2 (Sport) / 12.0-12.4 (Daily)

Honda's stock high-cam WOT AFR is intentionally rich (~11.5 in places) for emissions and reliability. A well-tuned Hondata calibration will lean this up slightly for more power while staying in the reliability envelope.

Flex Fuel Fuel Tables (E85 Endpoint)

When flex fuel is enabled, FlashPro stores a **second set of fuel tables** for the E85 endpoint. The ECU interpolates between gasoline and E85 tables based on the sensor's ethanol reading.

Safe range for E85 WOT on my engine:

Condition	Lambda	AFR (E85 stoich = 9.765)
Low-cam WOT	0.82-0.85	8.00-8.30
High-cam WOT (Sport)	0.78-0.80	7.62-7.81
High-cam WOT (Daily)	0.80-0.82	7.81-8.00

Richer lambda on E85 is safer because E85 cools the charge more aggressively, and the extra fuel volume buffers against lean spikes during transients.

Cranking Fuel

Short table that governs how much fuel gets injected during engine cranking.

When I touch this:

- After switching to E85 for the first time -- E85 needs significantly richer cranking fuel, especially in cold weather
- If cold starts get rough after hardware changes

Don't touch unless: I have a cold-start problem I'm diagnosing. Stock cranking fuel is well-dialed for gasoline.

After-Start Enrichment

Rich-mixture period immediately after starting, tapers off over seconds. Helps smooth cold idle.

Warm-Up Enrichment

Extra fuel based on coolant temp during warmup. Tapers to 0 once ECT hits target.

[VERIFY] The exact axes (ECT thresholds and duration) vary by calibration version -- I'll confirm with tuner.

---

2. Ignition Tables

Low Cam Ignition Table

Timing advance (degrees BTDC) at each RPM x load cell during low-cam operation.

Safe conservative peak values on my pump 93 build:

RPM	Load	Peak Timing (pump 93)
2000-3000	Light (cruise)	30-35 deg
2000-3000	WOT	20-25 deg
4000-5000	WOT (low cam)	22-28 deg
5500-6000	Just before VTEC	24-28 deg

High Cam Ignition Table

Same, for VTEC-engaged operation. This is where most of the power-vs-knock fight happens.

Safe conservative peak values on pump 93:

RPM	Load	Peak Timing (Sport)
5500-6000	WOT	26-29 deg
6000-7000	WOT	28-32 deg
7000-8000	WOT	26-30 deg (backing off near redline is normal)

Rule I won't break: If knock count in the datalog is > 0 at any cell for sustained operation, I pull timing at that cell. No "maybe it's sensor noise" excuses. Every knock event matters on a 170k-mile engine.

Flex Fuel Ignition Tables (E85 Endpoint)

E85 tolerates more timing. Typical gains on a well-tuned NA K20Z3:

- Low cam: +1 to +3 degrees across the band
- High cam: +2 to +5 degrees, concentrated in the 5000-7000 RPM range where knock margin on pump gas is tightest

[VERIFY] I'll let my tuner dial these. Arm-chair advancing timing on E85 "because it can take more" is the single most common way people hurt NA K-series engines.

Knock Retard

Not directly a table I tune, but a setting that controls how aggressively the ECU pulls timing when it detects knock. Stock is conservative. I leave this alone -- the whole point is that knock correction is my safety net.

---

3. VTEC Settings

VTEC Engagement RPM (Low to High Cam)

Single RPM value where the ECU activates the VTEC solenoid to switch from low-lift to high-lift cam profile.

Stock: ~5800 RPM **Safe range for tuning:** 4500-6000 RPM **My plan:**

- **Daily map:** 5400-5600 RPM (close to stock -- smoother VTEC transition, preserves fuel economy)
- **Sport map:** 4800-5200 RPM post-full-bolt-on (earlier VTEC = more area under the curve in hard driving)

Hard floor I won't go below: 4500 RPM. Below this, the cam profile doesn't make sense for the airflow -- you get a weird lumpy off-VTEC power band and nothing gained.

VTEC Engagement Throttle

Some calibrations gate VTEC on throttle position too (so light-throttle cruise at 5500 RPM doesn't wake up the high cam). I leave this at stock unless specifically addressing a drivability issue.

VTEC Disengagement RPM / Hysteresis

The RPM at which VTEC drops back to low cam. Typically 300-500 RPM below engagement to prevent flutter.

---

4. VTC (Cam Phase) Tables

Variable Timing Control phases the intake cam by 0-50 degrees (crank degrees). This is continuously variable based on RPM x load.

VTC Target Table

What phase the ECU asks for at each RPM x load cell. Honda's stock cam phase map is already highly optimized -- there's not much room for gains here on an NA build.

When to touch this:

- After intake manifold swap (Skunk2 Ultra Street changes breathing dynamics -- tuner may recommend 2-5 degrees of advance at specific cells)
- After headers -- similar small adjustments possible

When to NOT touch this:

- On day 1 with stock hardware -- Honda engineered this; I don't know better yet

VTC Minimum / Maximum

Limits on how far the ECU is allowed to phase the cam. **I don't change these.** Going past Honda's engineering limits risks valve-to-piston clearance issues.

---

5. Idle Settings

Target Idle RPM

Stock: 740-780 warm **Tunable:** 650-1000 RPM **My plan:** leave stock for Daily. Sport map may bump to 800-850 if engine mounts are causing idle roughness at lower RPM.

IACV (Idle Air Control Valve) Settings

Controls how the ECU manages airflow at idle. After intake manifold swap, this sometimes needs recalibration -- Skunk2 Ultra Street has different idle airflow characteristics than stock RRC.

Idle Spark Advance

Typically stock or slightly bumped for idle stability after aftermarket mods.

---

6. Rev Limits

Fuel Cut RPM (Hard Limiter)

Stock: 8200 RPM **Tunable:** up to ~8800 RPM on stock internals (don't) **My plan:**

- **Daily:** 8200 RPM (stock)
- **Sport:** 8300-8400 RPM (100-200 RPM over stock max)

Hard ceiling I won't cross: 8400 RPM on stock internals at 170k miles. Community consensus is that K20Z3 bottom ends survive 8500+ in built/fresh engines, but mine is neither built nor fresh. Reliability-first.

Per-Gear Rev Limits

FlashPro lets me set different fuel-cut RPMs per gear. Useful for:

- Lower limit in 1st gear (prevents over-rev on a 1-2 upshift)
- Higher limit in 5-6 (not really needed on NA K20Z3)

[VERIFY] Per-gear rev limit is commonly in the "advanced" or "misc" section -- exact menu location varies by FPM version.

---

7. Launch Control

Launch Control RPM

RPM target when driver is at WOT with car stationary. Clutch-engagement-sensitive.

My plan (Sport map only):

- 4500 RPM launch limit
- No 2-step aggression -- just a rev limiter at 4500 when stationary
- Disable in Daily map

Not recommended for me until: engine mounts are in, clutch hydraulics are upgraded. Launching a worn clutch at 4500 RPM = new clutch next week.

Flat Foot Shift (No-Lift-Shift)

Allows the driver to keep WOT through the upshift -- ECU cuts spark briefly during clutch depression to prevent

over-rev. Nice feature but hard on the drivetrain.

My plan: disabled. My 170k-mile transmission doesn't need the abuse, and the time saved on a shift is negligible for street driving.

---

8. Torque Management

Torque Management Disable

Honda's stock ECU pulls timing during 1-2 and 2-3 upshifts to smooth the shift. Community consensus is this feels like a dead spot and is worth disabling for a sporty feel.

My plan: disable in Sport map. Leave stock in Daily for smoother commuting.

Ignition Retard per Gear

Some calibrations allow per-gear timing maps. Not commonly used on NA builds -- more of a turbo tuning feature.

---

9. Fan and Cooling Tables

Primary Fan ON Temperature

Stock: ~205F **My plan:** 195F on Daily, 190F on Sport. With the Koyorad radiator (09-Cooling-System/) and 5W-30 synthetic, tighter fan control keeps ECT stable under sustained load.

Primary Fan OFF Temperature

Typically 5-10F below ON temp to prevent hunting.

Secondary Fan (A/C) Triggers

Controls the second fan for A/C operation. Leave stock unless diagnosing A/C-related overheating.

Cross-reference: the cooling system refresh feeds directly into these tables. New radiator + tuner-dialed fan maps = peace of mind on a hot dyno day.

---

10. Sensor / Input Settings

O2 Sensor Settings

- **Primary (upstream) O2:** tune uses it for closed-loop fuel correction
- **Secondary (downstream) O2:** normally monitors cat efficiency; often **repurposed for flex fuel signal input on FG2** (VERIFY the exact pin with tuner)

MAP Sensor Calibration

If I add a larger MAP sensor (unusual on NA), it's recalibrated here. Stock MAP is fine for my build.

Wideband O2 (Analog Input)

FlashPro has configurable analog inputs that can read a 0-5V wideband O2 signal. This is where the **AEM X-Series 30-0300** (11-Wideband-AFR/) gets wired. Configuration:

- Set analog input type: "Wideband O2 (AEM 30-0300)" (or Innovate, PLX, etc. -- tool has presets)
- Calibration: verify idle reads ~14.7 AFR when the engine is warm

Ethanol Content Sensor Input

Second analog input (or repurposed rear O2 signal) for the flex fuel sensor. **[VERIFY exact pin for FG2]** -- forum consensus points to the rear O2 signal wire, but I want Hondata's official wiring doc or my tuner's sign-off before I splice.

---

11. DTCs / CELs (Diagnostic Trouble Codes)

Rear O2 CEL Defeat

If I run a test pipe or delete the cat downstream of the OEM position, the rear O2 throws P0420 (catalyst efficiency). FlashPro can suppress this code.

I don't need this -- I'm running the Berk HFC which keeps cat efficiency within spec.

EVAP / Canister CEL Defeat

If the EVAP system is removed (race-only), FlashPro suppresses the related codes. **I don't need this.**

Intake Manifold Runner Control CEL

Stock RBC/RRC intake manifolds have a runner control flap (IMRC). Skunk2 Ultra Street deletes this. The calibrator has a setting to disable the CEL that would otherwise throw after the IM swap.

Injector Calibration (Slopes / Offsets / Dead Times)

When I swap to Bosch EV14 550cc injectors, I enter their characterization data here:

- Injector flow rate (cc/min at base fuel pressure)
- Dead time at various battery voltages
- Slope correction factors

[VERIFY] Most injector manufacturers publish Hondata-ready characterization sheets. I'll ask Bosch or check DeatschWerks/ID documentation for whichever injector I end up with.

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12. What I Leave Alone (Mostly)

Settings I don't touch as a novice -- they're either stock-calibrated well, or only matter for forced induction / built engines:

- Knock sensor settings (stock threshold is what I want catching real knock)
- Barometric pressure compensation
- Intake air temp compensation (stock is good)
- Coolant temp fuel/timing compensation

- Startup crank timing
- Closed-loop gain / PID tuning
- Misfire detection thresholds

My tuner may touch some of these during a session. I don't DIY any of them.

Priority of Changes -- What to Tune in What Order

When a session starts, this is the order of operations:

1. **Base tables match hardware** (correct injector size, MAP, VTC range)
2. **Closed-loop fuel trims clean up** (cruise AFR hits 14.7 consistently, LTFT ±3%)
3. **Low-cam WOT AFR on target** (verify with wideband)
4. **High-cam WOT AFR on target** (most important -- this is peak power territory)
5. **Ignition timing -- low cam, WOT** (verify via dyno pulls, knock = 0)
6. **Ignition timing -- high cam, WOT** (same)
7. **VTEC crossover adjustment** (based on torque curve from dyno)
8. **VTC cam phase tweaks** (small gains, last to touch)
9. **Idle / drivability polish**
10. **Flex fuel E85 tables** (entire process repeats for E85 endpoint)

See Also

- overview.md -- FlashPro hardware + setup paths
- tuning-workflow-and-maps.md -- Daily/Sport map rules, safety ceilings
- datalog-analysis-guide.md -- how to read the logs that feed these tables
- advanced-features.md -- launch, flat-foot, per-gear, shift lights
- ../19-Flex-Fuel-and-Fuel-System/ -- flex fuel hardware that feeds the E85 tables
- ../11-Wideband-AFR/ -- wideband that validates every AFR change

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Last updated: 2026-04-19

Datalog Analysis Guide -- Reading FlashPro Logs Like a Tuner

> How I interpret a datalog after a drive. What to look for first, what patterns mean, what's a red flag vs normal noise. This is the file I open after every session so I'm not wasting tuner time on obvious stuff.

Reliability-first reminder: every analysis here assumes the engine is more valuable than any number on a dyno sheet. If something looks wrong, I pull over, investigate, and fix before driving hard again.

What I Log -- Standard Parameter Set

These are the parameters I configure FlashProManager to log on every session. Sample rate: highest available (usually ~20 Hz / 50ms intervals).

Engine Basics

- RPM
- Engine Load (g/rev)
- MAP (kPa)
- TPS (%)
- VSS (vehicle speed)
- Gear (derived from VSS x RPM ratio)

Fuel

- Short-Term Fuel Trim (STFT, %)
- Long-Term Fuel Trim (LTFT, %)
- Lambda (from stock narrowband O2)
- **Wideband AFR (via analog input, from AEM X-Series)**
- Injector duty cycle (%)
- Injector pulse width (ms)
- Target AFR (what the calibration is asking for)

Ignition

- Ignition advance (actual, degrees)
- Ignition advance correction (negative = knock retard)
- **Knock count (the watchdog)**
- Knock volts (if logged at fast rate -- high-frequency knock sensor data)

Cam / VTEC

- VTC target (degrees advance commanded)
- VTC actual (degrees advance achieved)
- VTC error (target ? actual)
- VTEC solenoid state (on/off, boolean)

Thermal

- ECT (coolant temperature, degF)
- IAT (intake air temperature, degF)
- Oil temp (if wired -- optional analog input)

Electrical

- Battery voltage
- MIL (Malfunction Indicator Light) status

Flex Fuel (when installed)

- Ethanol content (%)
- Fuel temperature (degF)

The Analysis Order -- What to Check First

When I open a datalog, I go through it in a specific order. Each step either passes (move on) or flags something that needs investigation.

1. Knock Count (Safety First -- Non-Negotiable)

First thing I look at. Every time.

Filter the log for any row where knock count > 0. Then look at what the engine was doing at that moment.

Pattern	Interpretation	Action
0 knock events across entire log	Healthy, proceed to next check	--
Isolated spike during rough road / pothole	Chassis noise false positive	Document + ignore, but flag if it persists
Recurring events in specific RPM x load cell	Real knock	Pull timing at that cell, investigate fuel quality / sensor health
Cluster during WOT pull at high RPM	Aggressive timing	Pull 1-2 deg at affected cells, retest
Event during shift	Transient lean, usually torque-management-related	Tune shift cut smoothing, not timing

Hard rule: I never ship a tune with known recurring knock. Period.

2. Fuel Trims (STFT and LTFT)

Short-Term Fuel Trim: what the ECU is actively adding/subtracting RIGHT NOW to hit target AFR. Long-Term Fuel Trim: long-running average bias (what the ECU has learned it needs to add/subtract consistently).

Healthy ranges at cruise (closed-loop):

STFT	LTFT	Meaning
±3%	±3%	Tune is dialed in for cruise
±5%	±5%	Minor issue -- dirty MAF, slight vacuum leak, injectors slightly off
±10%	±10%	Real problem -- vacuum leak, failing sensor, wrong injector data
0 (pegged)	±15%+	Closed-loop running out of authority -- ECU can't correct, something is significantly wrong

Pattern analysis:

Pattern	Likely Cause
LTFT consistently -5 to -10% (running rich)	MAF reading high, fuel pressure too high, injector dead time wrong
LTFT consistently +5 to +10% (running lean)	Vacuum leak, dirty MAF, injector clog, weak fuel pump
LTFT shifts from +5 to -5 between logs	Could be ambient temp / altitude -- normal within limits
LTFT drifts over a single log from 0 to +10%	Fuel pump pressure drop under heat, failing sensor

3. AFR at WOT (Wideband Ground Truth)

The stock narrowband O2 pegs rich outside the stoich range -- I can't use it for WOT tuning. The **AEM X-Series wideband** signal coming in on the FlashPro analog input is my real data.

Comparing logged wideband AFR to calibration target AFR:

Delta	Meaning
Actual = Target ± 0.1	Tune is locked in
Actual > Target + 0.3 (leaner than commanded)	Fuel delivery issue -- pump, injectors, pressure
Actual < Target ? 0.3 (richer than commanded)	Injectors slow to close (dead time wrong), leaky injector, wrong injector constant
Actual sawtooths around target	Noisy sensor, wiring issue, or wideband sensor aging

Red flag at WOT on pump gas: AFR > 12.5. That's leaner than any reasonable tune would target. Pull over.

Red flag at WOT on E85: lambda > 0.85. Same deal.

4. Ignition Advance vs Target

FlashPro logs both "commanded" timing (what the calibration table says) and "actual" timing (after corrections -- knock retard, IAT compensation, ECT compensation).

Healthy: actual = commanded within ±0.5 deg across most of the log.

Red flags:

Pattern	Meaning
Actual consistently 2-3 deg below commanded at high RPM WOT	Knock correction active -- usually accompanied by knock count > 0
Actual drops sharply during a shift	Torque management active (if not disabled)
Actual varies wildly cell-to-cell with no knock	IAT or ECT compensation triggering -- check if engine is overheating or intake getting heat-soaked

5. VTC Error (Actuator Health Check)

VTC error = VTC target ? VTC actual

Error	Meaning
±1 deg sustained	Healthy VTC actuator
±2-3 deg transient during fast transitions	Normal -- actuator takes time to hit target

±3+ deg sustained	Failing VTC actuator -- this is the cold-start-rattle actuator problem. See 06-Ignition-Refresh/.
Error oscillates at idle	VTC actuator spring is weak -- replace before further tuning
Error ~0 at idle, grows at high RPM	Oil pressure issue (worn pump, wrong oil viscosity)

If I see sustained VTC error > 3 deg anywhere in my log, I stop tuning and replace the actuator (OEM 14310-RBC-003, revised part). Tuning on a failing VTC is a waste of money because the engine isn't doing what the tune asks.

6. Injector Duty Cycle

Healthy at WOT peak:

- Stock 310cc injectors at stock power: 50-60% duty
- Stock 310cc injectors on NA full-bolt-on: 75-85% duty (starting to run out of headroom)
- Bosch EV14 550cc on full-bolt-on pump gas: 45-55% duty (comfortable)
- Bosch EV14 550cc on full-bolt-on E85: 60-75% duty (comfortable)

Red flag: sustained duty cycle > 85%. Injectors are maxed out. Fuel delivery is about to fall off a cliff and cause a lean spike. Either:

- Upsize injectors immediately, OR
- Verify fuel pressure is within spec, OR
- Reduce target AFR richness until hardware can be upgraded (temporary band-aid)

Never try to compensate by raising fuel pressure -- that's an E85-world fuel line failure in the making.

7. Thermal Behavior

Healthy ECT:

- Cruise on highway: 185-195degF
- Cruise in traffic (no A/C): 195-205degF
- WOT pulls: brief rise to 200-210degF, recovers in seconds
- Sustained WOT (dyno pull, canyon run): should not exceed 215degF with fresh Koyorad radiator

Red flag: ECT > 220degF -- stop driving hard, let it cool, check coolant system.

Healthy IAT (my K&N Typhoon is hot-side):

- Cruise at speed: ambient + 15-25degF
- Traffic at idle: ambient + 30-50degF (hot-side intake absorbs heat soak -- this is the trade-off I accepted)
- After WOT: drops within 10-20 seconds back to baseline as airflow recovers

Red flag: sustained IAT > 150degF -- intake has insufficient cooling, heat shield may be inadequate.

8. Battery Voltage

Should stay at 13.8-14.4V with the alternator charging. If I see it drop below 13.0V during engine running:

- Alternator failing (common at 170k)
- Loose battery terminal or bad ground
- Overloaded electrical system (rare on NA -- becomes relevant with LattePanda installed)

9. Flex Fuel Specific (Once Installed)

Ethanol content sensor readings:

- Should be stable (±2%) between shifts in fuel source
- Pump gas fills read E0-E15 (ethanol in regular gas)
- "E85" at the pump actually reads E60-E83 depending on season/station
- Winter blend E85 often reads E51-E65

Red flag: sensor reads 0% or 100% constantly -- sensor failure or wiring issue. ECU should fall back to a safe "assume E0" map, but I'll verify by checking fueling behavior.

Common Datalog Patterns -- What They Mean

Pattern: "Lean spike at 3500 RPM on a light-cruise pull"

Common 8th gen SI complaint. Usually caused by:

- Incorrect VTC phase at that cell (transitional region)
- Torque management timing pull creating airflow change the fuel didn't anticipate
- Stock lean-cruise calibration designed for emissions

Fix: tuner adjusts low-cam fuel table at 3500 RPM x medium load. Not a DIY fix.

Pattern: "Rough idle after Skunk2 Ultra Street install"

Ultra Street has different plenum volume than stock RBC. IACV calibration is off.

Fix: tuner recalibrates IACV tables. Some trial and error.

Pattern: "Engine stutters at exactly 4800 RPM"

Two common causes:

- VTEC engagement hitch (cam transition timing slightly off)
- Knock retard kicking in repeatedly at that cell

Fix: check knock count at 4800 RPM first. If knock-driven, pull timing at that cell. If VTEC-transition-driven, adjust VTEC engagement RPM or hysteresis.

Pattern: "WOT power falls off flat after 7000 RPM"

On a supposedly FBO car, this is usually:

- Injectors hitting duty cycle ceiling
- Fuel pump pressure dropping under load
- Stock intake restriction (if manifold isn't yet upgraded)
- Stock exhaust restriction (if exhaust isn't yet upgraded)
- Cam phase not optimized for top-end

Fix: datalog injector duty and fuel pressure. Hardware gap or tune gap.

Pattern: "STFT/LTFT clean at cruise but lean at part-throttle acceleration"

Tip-in enrichment (transient fuel) is under-calibrated.

Fix: tuner dials in tip-in fuel table.

Pattern: "Knock count spikes at 6200 RPM on a hot day only"

IAT compensation insufficient -- timing not being pulled enough when intake temps climb.

Fix: IAT compensation table gets more aggressive at high IAT, OR I improve intake cooling (heat shield, true cold-air duct).

What I Share With Claude / My Tuner

For Claude (CSV export workflow)

From FlashProManager: **File -> Export Datalog -> CSV**.

The CSV contains one row per sample (~20 Hz), one column per logged parameter. Typical file: 10-minute drive = ~12,000 rows x ~30 columns = ~500KB plain text. Pasteable into chat in chunks, or attachable directly.

When I paste a log to Claude:

- Include the calibration name I was running
- Include what mods were on the car at time of log
- Specify what question I'm asking ("why did I see knock at WOT in 4th?" not "look at this log")
- Include a datalog range of interest if the log is long (e.g., "rows 8500-9200 is the WOT pull")

For my tuner

The tuner wants the raw .fpdl file -- he loads it in his own FlashProManager and sees everything native. I email / upload it. He sends back a new calibration .fpc file.

Red-Flag Quick Reference (One-Glance Checklist)

After any drive, I scan for these:

- [ ] Knock count = 0 everywhere (no exceptions)
- [ ] Knock retard correction stays above -1 deg (no active pulling)
- [ ] LTFT within ±5% at cruise
- [ ] Wideband AFR at WOT within 0.2 of target
- [ ] Injector duty < 85% at peak
- [ ] ECT < 215degF sustained
- [ ] VTC error < 2 deg sustained
- [ ] Battery voltage > 13.2V running
- [ ] No MIL codes thrown

Any NO above = investigate before the next hard pull.

See Also

- tuning-workflow-and-maps.md -- safety ceilings these analyses enforce
- calibrator-tables-reference.md -- the tables I'd edit based on these findings

- ../docs/baseline-logs.md -- the log session protocol (pre/post every mod)
- troubleshooting-and-faq.md -- CEL codes and common issues

Last updated: 2026-04-19

FlashPro Advanced Features -- Launch, Flat-Foot, Per-Gear, Torque Management

> The features that make FlashPro more than "just a fuel-and-timing reflash." What each does, when I'd use it, and the reliability-first settings for my 170k-mile K20Z3.

Verification note: Exact FPM menu names, parameter availability, and default values for the PRB calibration (06-11 Civic SI) change slightly between FPM software versions. Items flagged **[VERIFY]** I'll confirm against my current FPM version or with my tuner before enabling.

---

1. Launch Control (2-Step)

What It Does

Holds engine RPM at a user-set value while the car is stationary with clutch in. Releases immediately on clutch pedal release. Activation is gated on clutch switch, VSS (speed), and sometimes brake.

Typical Settings

Parameter	Value (my plan)
Activation: Clutch switch	Required (pressed)
Activation: VSS below	5 mph
Activation: Min TPS	90%+ (only engaged at WOT)
Activation: Min ECT	150degF (no cold-engine launches)
Launch RPM -- Daily map	Disabled entirely
Launch RPM -- Sport map	4500 RPM (conservative -- protects clutch/axles at 170k)

Why 4500 RPM, Not 5500

The K20Z3 makes meaningful torque below VTEC, so I don't need to pre-load VTEC to get off the line. Launching at 5500+ RPM with my stock engine mounts (before Hybrid Racing 70A mounts are in) and 170k-mile axles = grenading something on the drivetrain.

- Stock clutch (new Exedy Stage 1): rated ~280 WHP, so shock-loading at 5500 RPM is pushing it
- 170k-mile axles: inner CV joints are a known wear point; sudden high-torque engagement accelerates failure
- Stock or fresh engine mounts: high-RPM launches with soft mounts transfer more torque to the drivetrain than to the wheels

After my Phase 5 suspension work and axle refresh (if I ever do them), I could bump launch to 5000-5200. Not before.

How the RPM Is Held

The ECU uses one of two mechanisms:

- **Ignition cut:** ECU skips spark events to limit RPM. Preferred for street use -- softer, quieter.

- **Fuel cut:** ECU skips fuel injection. Harsher, louder, potentially lean transients.

My setting: Ignition cut. Fuel cut is what race cars use for the flamethrower effect; daily-drivable isn't worth the lean risk.

Ignition Retard During Launch

Some calibrations retard timing while launch control is active to reduce power to the wheels (so RPM stays at target without all the exhaust drama). **Stock retard: around -5 to -10 degrees during launch.** Leave alone.

When I Won't Use Launch Control

- Before engine mounts are upgraded
- Before clutch hydraulics are upgraded
- On public roads (outside tracked spaces, per the reliability-first rule)
- In the rain
- With a cold engine (below 150degF ECT)

---

2. Flat-Foot Shifting (No-Lift-Shift)

What It Does

Allows the driver to keep the throttle pinned during the upshift. The ECU detects the clutch depression and briefly cuts ignition (or fuel, or both) to prevent over-rev while the transmission is between gears. Throttle stays 100%, shift is cleaner.

The Trade-Off

Pros:

- Slightly faster upshifts (0.05-0.1 sec)
- No throttle lift = no intake vacuum spike = cleaner AFR during shift
- Flashier, more "race-y" feel

Cons:

- Shock-loads the drivetrain harder than a normal lift-shift
- Accelerates clutch wear
- At high RPM, the shock can damage transmission synchros on a 170k-mile gearbox
- Shift cut isn't free -- every cut is a fuel/spark interruption the engine doesn't love

My Decision

Disabled on both maps. My 170k-mile transmission doesn't need the abuse, and 0.1 seconds is meaningless on the street. Might revisit after a full clutch + trans refresh. Until then, traditional clutch-in/throttle-lift/shift/clutch-out is fine.

If I Ever Enable It

Parameter	Conservative setting
Activation	Clutch pressed + TPS > 80%

RPM threshold	> 5000 RPM (don't cut at low RPM)
Cut duration	100-200 ms max
Cut type	Ignition cut (not fuel cut)
Gear range	1->2, 2->3, 3->4 only (above 4th, normal shifts are fine)

---

3. Per-Gear Rev Limits

What It Does

FlashPro lets me set different fuel-cut RPMs per gear. Useful for protecting the drivetrain from over-rev scenarios.

What I'd Set

Gear	Rev Limit (Sport map)	Why
1st	8200 RPM	Prevent massive over-rev on a missed 1->2 shift
2nd	8400 RPM	Same, slightly more headroom
3rd	8400 RPM	Comfortable peak-power range
4th	8400 RPM	Same
5th	8400 RPM	Same
6th	8200 RPM	Highway cruise gear -- no reason to push past stock

Daily map: all gears at stock 8200 RPM.

The Hard Ceiling

I won't go above 8400 RPM on any gear on stock internals at 170k miles. Full stop. **[VERIFY]** the exact per-gear parameter location in FPM -- it may be under "Rev Limits" or "Advanced" depending on FPM version.

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4. Torque Management Disable

What Honda's Stock Tune Does

During upshifts (especially 1->2 and 2->3), Honda's stock calibration pulls ignition timing by 5-15 degrees for 100-200 ms to "cushion" the shift. The result: a slight hesitation in the pull, often perceived as a dead spot around VTEC engagement during aggressive shifts.

What Disabling Does

Flattens the torque-reduction tables so timing stays commanded throughout the shift. The pull feels uninterrupted.

Community Consensus

Most Hondata tuners on K20 builds flatten these tables as part of a Sport tune. It's a well-understood modification with no reliability cost as long as:

- The clutch is healthy
- Rev limits are respected
- The shift isn't so violent it exceeds transmission torque capacity

My Plan

- **Daily map:** leave stock. Smoother commuting.
- **Sport map:** tuner flattens the torque management tables during Tune #1. Cleaner VTEC pulls through shifts. **I don't flatten these myself.** They interact with ignition timing authority in subtle ways, and a dyno-verified flattening is safer than armchair editing.

---

5. Shift Light / Shift Indicator

The Reality

The 8th gen Civic SI does NOT have a factory shift light on the gauge cluster. The tachometer needle is the "shift indicator" -- you watch the VTEC crossover and redline yourself.

What FPM Offers

FPM has a **software shift-light display** in the parameter monitor gauge view. A configurable RPM threshold triggers a visual alert (color change, flash, etc.) on the laptop/LattePanda screen.

[VERIFY] whether the original FlashPro has a **hardware digital output pin** that could drive an external LED shift light. I don't recall one. FlashPro 2 has more I/O. If I want a physical shift LED, I'd need an Arduino on the LattePanda reading RPM from the FPM datalog stream and driving the LED.

My Plan

- **Software shift light in FPM gauge view:** set to 7800 RPM (200 RPM before my 8200 rev limit, gives me a visual cue to upshift).
- **Physical LED:** possible future project via Arduino, not a priority.

---

6. Rev Hang / Throttle Response

What Rev Hang Is

When I lift off the throttle, the stock calibration holds the throttle plate slightly open (for emissions purposes -- cleaner combustion during deceleration) before fully closing. Result: engine revs "hang" at the lift-off point for 0.3-0.8 seconds before decaying, making rev-matching on downshifts awkward.

The Fix

FPM has a setting to reduce or eliminate rev hang by closing the throttle plate more aggressively on lift-off. **[VERIFY exact parameter name]** -- commonly called something like "Lift-off throttle plate control" or "Deceleration throttle position."

My Plan

Daily map: minor reduction in rev hang (better driveability, still clean emissions). **Sport map:** aggressive reduction. Sharper throttle response, easier rev-matching.

Dyno tuner handles this during Session #1 or #2.

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7. Gear Detection Calibration

What It Does

The ECU calculates which gear you're in based on VSS ÷ RPM ratio. Per-gear features (rev limits, flat-foot, etc.) depend on this working correctly.

When It Needs Recalibration

- After changing tire size (e.g., going from 215/45R17 to 235/40R17 shifts ratios by ~2-3%)
- After changing final drive (axle ratio swap -- not planned for my build)
- If the ECU misidentifies a gear (FPM parameter monitor shows wrong gear under load)

How to Calibrate

FPM has a **gear-learning mode** where I drive in each gear at steady RPM and the ECU records the ratio. Alternative: manually enter known VSS/RPM values per gear in the calibration table.

My Plan

I'll let the ECU auto-learn after the first flash. Check gear detection in the parameter monitor during a test drive (drive in each gear, verify FPM displays the correct gear number). Recalibrate if I ever change tire sizes.

---

8. Speed Limiter Removal

What Stock Does

The 8th gen SI has an electronic top-speed limiter (**[VERIFY]** approx 110-125 mph). Engages via fuel cut when VSS exceeds the threshold.

FPM Setting

"Remove Speed Limiter" checkbox or adjusting the limiter to 180+ mph (effectively removed). Common setting for any performance tune.

My Plan

Both maps: speed limiter off. I don't plan to drive at those speeds on public roads, but I don't want a fuel cut surprising me on a track day or open-road event.

---

9. A/C Cut at WOT

What It Does

On WOT with A/C on, the ECU disengages the A/C compressor clutch. Recovers 3-8 HP that would otherwise be absorbed by compressor drag.

Stock Behavior

Already enabled on the stock calibration -- Honda does this by default. **Leave alone.**

[VERIFY] whether the stock threshold is adjustable (some calibrations let me change the TPS% that triggers A/C cut).

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10. Cooling Fan Tables

What It Does

Controls when the primary and secondary (A/C) cooling fans turn on and off based on ECT.

Stock Settings

- Primary fan ON: ~205degF
- Primary fan OFF: ~195degF
- Secondary fan: triggers with A/C or with high ECT

My Plan (with Koyorad radiator + Phase 0 coolant)

- **Daily map:** Fan ON 195degF, Fan OFF 185degF
- **Sport map:** Fan ON 190degF, Fan OFF 180degF

Tighter fan control keeps ECT stable during sustained load (canyon runs, dyno pulls). The Koyorad has extra heat rejection capacity; running the fan earlier uses that capacity proactively. See 09-Cooling-System/.

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11. CEL Defeat Flags

Rear O2 Sensor CEL (P0420 Cat Efficiency)

When I'd need this: If I ran a catless test pipe (not in my plan -- I have the Berk HFC).

My plan: Not needed.

IMRC (Intake Manifold Runner Control) CEL

When I'd need this: After installing the Skunk2 Ultra Street intake manifold, which deletes the stock IMRC flap. Without this defeat flag, I get a persistent P2004/P2005 code.

My plan: Tuner enables this flag during Tune #2 (post-intake manifold install). See 14-Intake-Manifold/.

EVAP System CELs

When I'd need this: Only if I physically removed the EVAP canister (race-only). Not in my plan.

My plan: Not needed.

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12. Closed-Loop Authority Limits

What They Are

The ECU's short-term and long-term fuel trims operate within a limited authority range -- they can add/subtract fuel up to ±25% from the calibration's target, but clamp beyond that.

When They Matter

If my injectors, MAF, or fuel pressure is significantly off, trims can peg at the limit. FPM has tunable limits:

- **STFT limit:** typically ±25%
- **LTFT limit:** typically ±20%

I don't touch these unless diagnosing a specific trim-pegging issue. Widening the limits masks the underlying problem.

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13. Features I Won't Touch

These exist but are either well-calibrated stock or too specialized for my build:

- **Boost control tables** (N/A on NA)
- **Wastegate duty** (N/A on NA)
- **Intercooler sprayer** (N/A on NA)
- **Map switching via clutch-pedal tap** (FlashPro 2 feature, not available on original unit)
- **TPS calibration** (only adjusted after a throttle body replacement or cleaning)
- **CAM sensor learning** (auto-learned by ECU; don't manually adjust)
- **Immobilizer settings** (never touch -- I'll brick the car)

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14. Summary -- My Advanced Feature Matrix

Feature	Daily Map	Sport Map
Launch control	Disabled	4500 RPM (post-mount upgrade)
Flat-foot shift	Disabled	Disabled (reliability call)
Per-gear rev limit	Stock 8200	8400 in 2-5, 8200 in 1 and 6
Torque management	Stock	Flattened (tuner)
Shift light	FPM gauge at 7800	FPM gauge at 7800
Rev hang reduction	Mild	Aggressive
Speed limiter	Removed	Removed
A/C WOT cut	Stock (enabled)	Stock (enabled)
Fan ON temp	195degF	190degF
Rear O2 defeat	No (have HFC)	No (have HFC)
IMRC defeat	Yes (after IM swap)	Yes (after IM swap)

See Also

- overview.md -- FlashPro hardware overview
- tuning-workflow-and-maps.md -- reliability-first rules these features operate under
- calibrator-tables-reference.md -- tables these features interact with
- datalog-analysis-guide.md -- verifying behavior after enabling features
- troubleshooting-and-faq.md -- issues related to these features

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Last updated: 2026-04-19

FlashPro Troubleshooting + FAQ

> When something goes wrong. When I see a code I don't recognize. When FPM won't connect. When a flash fails. My first-stop diagnostic guide.

Verification note: Claims marked **[VERIFY]** are from my accumulated knowledge (Jan 2026 cutoff) that I want to confirm against current Hondata KB articles or forum threads before acting. Everything else is high-confidence.

Connection and Software Issues

"FlashProManager doesn't see my FlashPro unit"

Most common cause: Windows replaced the FTDI driver during a Windows Update.

Fix:

1. Open Device Manager
2. Look under "Ports (COM & LPT)" for the FlashPro -- should appear as "USB Serial Port (COMx)"
3. If it's under "Other Devices" or showing a yellow warning icon, the driver needs fixing
4. Download the latest FTDI VCP driver from ftdichip.com/drivers/vcp-drivers/
5. Install, reboot, re-try FPM

"FlashPro enumerates but FPM says 'No communication with ECU'"

Check in order:

1. Ignition is ON (key in position II) -- not just ACC, and not off
2. OBD2 port is clean (no bent pins, no dirt)
3. FlashPro OBD2 connector is fully seated (push hard, it should click)
4. Battery voltage above 12.5V (charger or engine running is safest)
5. No other scantool is sharing the OBD2 port
6. Try another cable / USB port

"Flash failed mid-write"

Not a brick. The ECU holds its previous calibration. Symptoms: FlashPro shows error, FPM reports flash incomplete.

Recovery:

1. Reconnect FlashPro to OBD2 + USB
2. Launch FPM
3. FPM detects incomplete flash, offers to resume
4. Click Resume -- it picks up where it left off
5. Verify with Tools -> Read from ECU afterward

Prevention:

- Flash with engine running or battery charger
- Don't unplug anything during the flash
- Don't start/stop the engine during flash

- 20-45 seconds of patience

"FPM crashes when I open a .fpc file"

Causes:

- .fpc was created with a newer FPM version than I'm running -- update FPM
- File is corrupted -- re-download or re-request from tuner
- FPM installation is corrupted -- reinstall

"Firmware update failed"

Same recovery as flash failure -- reconnect, FPM detects and retries. If the unit seems completely unresponsive after multiple retries, contact Hondata support.

CEL / DTC Codes -- K20Z3 Specific

Codes That Matter vs Codes I'd Tune Out

Never tune out without understanding why the code is triggering. A CEL is information. Suppressing it without fixing the cause hides real problems.

Common Codes After Modifications

Code	Meaning	Likely Cause on Modded Car	Action
P0420	Catalyst efficiency below threshold	Aftermarket HFC flows faster than stock cat; rear O2 sees less cleanup	Verify HFC is actually functional (Berk HFC should pass). If test pipe/catless, rear O2 defeat flag in FPM
P0171	System too lean (Bank 1)	Vacuum leak, dirty MAF, weak fuel pump, wrong injector data	Don't tune out -- fix the fuel delivery issue
P0172	System too rich (Bank 1)	Stuck injector, fuel pressure too high, bad MAF	Don't tune out -- find the cause
P0300-P0304	Misfire (random or per-cylinder)	Bad coil, fouled plug, low compression, wrong timing	Diagnose, don't suppress
P0325	Knock sensor circuit	Sensor failure, wiring damage	Replace sensor; this code disables knock correction (dangerous)
P2004 / P2005	IMRC stuck open	Skunk2 Ultra Street deletes IMRC	Enable IMRC defeat flag in FPM
P0335	Crankshaft position sensor	Sensor failure, wiring, flywheel reluctor damage	Replace sensor; car won't run well
P0340	Camshaft position sensor	Sensor or wiring	Replace sensor
P0011 / P0014	VTC actuator / camshaft timing	Weak VTC actuator -- this is the 14310-RBC-003 revised part 06-Ignition-Refresh/ fixes	Replace actuator
P0128	Coolant temp below thermostat regulating temp	Stuck-open thermostat	Replace thermostat
P0116	Coolant temp sensor circuit range	Sensor failure	Replace sensor

P0135 / P0141	O2 sensor heater circuit	Sensor or wiring	Replace sensor
P0446	EVAP vent control circuit	Vent valve or wiring	Fix or (if race) defeat flag
P0505	Idle air control malfunction	IACV stuck or wiring	Clean IACV, replace if needed

Codes I Can Tune Out (Because the Hardware Change Justifies It)

- **P0420** after installing Berk HFC: only if rear O2 continues complaining despite functional HFC. **[VERIFY]** Berk HFC passes stock rear O2 threshold.
- **P2004/P2005** after Skunk2 Ultra Street install: IMRC is physically deleted, rear O2 flag should suppress.

Codes I Will NEVER Tune Out

- P0300-P0304 (misfire -- always a real issue)
- P0325 (knock sensor -- critical safety system)
- P0335/P0340 (crank/cam position -- car won't run safely)
- P0011/P0014 (VTC -- engine damage risk)
- P0171/P0172 (fuel trim extremes -- real problem upstream)

Rule: a CEL that doesn't have a known cause from a modification I did recently is a real problem. Investigate before suppressing.

Tuning / Datalog Issues

"I see knock count > 0 but I don't feel anything"

Knock you can't feel can still hurt the engine over time. On a 170k-mile motor, even isolated knock events are too many.

Process:

1. Open the datalog, filter for knock count > 0 rows
2. Note the RPM x load cells where it occurred
3. If it's a specific cell pattern: **pull 1-2 degrees timing at those cells** via FPM, re-flash, retest
4. If it's random/scattered: likely chassis noise (bad road, impact from pothole) -- document and monitor
5. If it's repeatable: real knock. Root cause: fuel quality, sensor health, timing too aggressive, insufficient cooling, or hardware problem

"Wideband reads 'error' or 'invalid'"

Possible causes:

- Sensor warmup incomplete (takes 30-60 seconds on cold start)
- Wiring issue (signal or power)
- Sensor is dying (2-3 year lifespan on E85 usage)
- AEM X-Series controller needs free-air calibration (annual, per AEM's instructions)

"Idle AFR is 12.0 instead of 14.7"

On a warm engine in closed loop, AFR should be right at 14.7.

Check:

- Is the engine fully warmed? (ECT > 180degF)
- Is FPM showing closed-loop flag = ON?
- Is the narrowband O2 sensor responding (voltage swings 0.1-0.9V)?
- Is there a vacuum leak (idle AFR can't lean out despite ECU commanding)?

"Car stumbles at 3500 RPM under light cruise"

Common 8th gen SI complaint. Usually a fuel table transition issue, sometimes VTEC transition hitch.

Process:

1. Datalog a cruise through 3500 RPM, gentle throttle
2. Look at AFR at the stumble point
3. Look at commanded vs actual timing
4. Look at VTC target vs actual
5. Send datalog to tuner -- usually a low-cam fuel table adjustment at that cell

"Car won't start after flashing a new calibration"

Check:

- Did the flash complete successfully? (Read back from ECU, compare to what I flashed)
- Are there any CELs?
- Is the base map correct for my hardware? (e.g., flashed an E85 map while running pump gas = won't run)
- Is the battery healthy?

Recovery:

- Flash the stock calibration back (STOCK_ORIGINAL_YYYYMMDD.fpc)
- Verify car starts normally on stock
- Then re-evaluate the new calibration

"Rough cold start after switching to E85"

Expected behavior. E85 has higher latent heat of vaporization and doesn't ignite as readily in cold weather.

Fix:

- Richer cranking fuel table (tuner adjustment)
- Longer after-start enrichment
- Winter blend E85 (lower ethanol percent, better cold start)
- Don't run straight E85 below 20degF ambient

Hardware Issues

"FlashPro USB connector is loose / wobbly"

USB Mini-B has a reputation for wearing out. If the connector is loose:

- Try a different Mini-B cable (cables wear too)

- If the connector on the FlashPro unit itself is worn, contact Hondata support for repair/replacement
- Long-term fix: permanent install (LattePanda build) where the cable isn't plugged/unplugged

"FlashPro shows a red/orange status LED"

[VERIFY] exact LED meanings from box quick-start card. Typical meanings:

- Red = error during operation
- Orange = waiting for user action (flash confirm, etc.)
- Green/white = normal operation

Calibration Issues

"My calibration makes less power than stock"

Possible causes:

- Wrong base map for my hardware
- Fuel tables too rich
- Timing too retarded
- VTEC engagement too high for mods

Fix: back to stock calibration, then load a known-good community base map for my specific mod combo, then dyno-verify.

"Fuel trims are pegged at +15% (running lean)"

ECU can't add enough fuel to hit target AFR.

Root causes:

- Vacuum leak (smoke-test intake)
- Dirty / failing MAF
- Clogged fuel filter
- Weak fuel pump
- Wrong injector data in calibration
- Injector clog (send for cleaning)

"Fuel trims are pegged at -15% (running rich)"

ECU can't subtract enough fuel.

Root causes:

- Leaky injector (stuck open, dripping)
- Fuel pressure too high
- MAF reading high
- Wrong injector scalar in calibration

FAQ -- Stuff I Googled More Than Once

"Can I flash multiple calibrations to the FlashPro unit at once?"

No. The original FlashPro stores one active calibration at a time. To switch daily ? sport, I open FPM and flash the other .fpc. Takes ~20-45 seconds per flash.

FlashPro 2 supports on-unit calibration switching via the mobile app. My original FlashPro does not.

"How often should I re-flash my tune?"

Never, unless:

- I changed hardware (adds, removes, upgrades)
- My tuner sends a revision
- I'm switching between daily and sport (frequent)

A calibration doesn't "wear out."

"Does Hondata charge per flash?"

No. I flash unlimited times after purchasing the FlashPro. The only ongoing cost is if I transfer the unit to a different ECU -- **$200 re-registration fee [VERIFY current price]**.

"Can I switch Daily ? Sport calibrations while driving?"

No. Flashing requires ignition ON + engine OFF (ideally) or engine idling. I can't flash with the car rolling.

Workaround for quick switching: park, clutch in, engine idling, open FPM on my phone (FlashPro 2) or laptop, flash, done. ~45 seconds.

"Can I read the VIN from my ECU?"

[VERIFY] -- FPM may display VIN during the read-ECU operation. The FlashPro pairs to ECU serial, not VIN, so this doesn't affect functionality.

"Do I need a tuner for my first flash?"

No. First flash can be a Hondata community base map matched to my hardware. I register the FlashPro, read my stock calibration for backup, then flash the base map. Takes an hour max.

I DO need a tuner for: any WOT calibration change, timing advance, VTEC engagement changes, flex fuel calibration.

"Can I datalog without FPM running?"

[VERIFY] -- the FlashPro has internal flash for short logs triggered by the unit's side button. These auto-save to internal memory and can be downloaded later when plugged into a PC. Useful for a "track day" recording without a laptop in the car.

"Will Hondata void my engine warranty?"

The 2009 Civic is well out of Honda's original warranty (3 year/36,000 mi basic). Not a concern for me at 170k miles.

For someone still in warranty: flashing the ECU and having the dealer detect it could void powertrain coverage. Reading the stock calibration and flashing it back before dealer visits is the common workaround (ECU looks

stock to Honda's tools).

"Does FlashPro work with my aftermarket gauge cluster?"

FlashPro talks to the ECU. Gauge clusters are separate. My OEM cluster will continue to work normally.

If I ever install an aftermarket digital dash (e.g., Haltech IC-7), it connects to the ECU via CAN bus and works in parallel with FlashPro -- no conflict.

"How much of my build requires a tuner vs DIY in FPM?"

DIY in FPM	Requires Tuner
First flash of community base map	Any WOT calibration change
Loading saved calibrations	Timing advance
Datalog setup + CSV export	VTEC engagement changes
Parameter monitor / gauge display	Flex fuel calibration
CEL defeat flags (IMRC after IM swap)	Closed-loop trim adjustments
Rev limiter adjustment (within spec)	Anything that touches the fuel or timing tables
Fan table adjustment	Idle recalibration after IM swap
Gear detection calibration	Cam phase (VTC) optimization
Speed limiter removal	Injector characterization data for new injectors

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Emergency Protocols

"Something sounds wrong -- what do I do?"

1. Pull over safely
2. Turn car off, don't start again
3. Connect FlashPro + laptop, read DTCs
4. Flash stock calibration back as a precaution (rules out tune issues)
5. Check oil, coolant, listen for mechanical noises
6. If anything obvious (no oil, overheating, leaking fluid): tow, don't drive
7. If nothing obvious and car starts/runs normally on stock: drive gently home, diagnose properly

"Car won't start at all after I flashed"

1. Try to start 3 times, 5 seconds crank each time, 10 seconds rest between
2. If no start: flash stock calibration back via FPM
3. If stock calibration also won't start: check for CELs, check fuel delivery, check spark
4. If flashing fails: contact Hondata support

"I think I damaged my engine from tuning"

1. Stop driving the car immediately
2. Pull a complete datalog from the most recent drive
3. Send logs + tune to my tuner

4. Do NOT attempt to "tune your way out" of a suspected mechanical issue
5. If the engine needs inspection, open it -- find-it-before-it-finds-you saves money

The reliability-first rule exists specifically to prevent this scenario.

Where to Get Help

Hondata-Official

- FPM Help menu -> Help Contents (built-in reference)
- hondata.com/support (**[VERIFY current URL]**)
- Hondata support ticket system (email)
- Hondata official forum (**[VERIFY URL]** -- historically at forum.hondata.com)

Community

- 8thcivic.com FlashPro subforum -- best 8th-gen-specific help
- r/CivicSi -- Reddit, casual
- k20a.org -- deeper K-series technical discussions
- BrenTuning YouTube -- datalog walkthroughs

My Tuner

- First call for anything tune-related
- Don't waste his time with driver issues or basic troubleshooting -- use the resources above first

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See Also

- overview.md -- FlashPro hardware basics
- hardware-software-deep-reference.md -- full hardware/software reference
- tuning-workflow-and-maps.md -- reliability-first rules
- datalog-analysis-guide.md -- how to read logs for diagnostics
- calibrator-tables-reference.md -- which tables to edit for which issue
- tuner-directory.md -- who to call

Last updated: 2026-04-19

Tuner Directory -- Who I'm Considering for My Three Dyno Sessions

> Shops I'd trust for a K20Z3 FlashPro tune on a 170k-mile FG2, plus what to look for, what to avoid, and how to vet any shop I didn't list. This is a starting point -- **I re-verify every shop's current status (website active, owner still there, reputation intact) before booking.**

Verification note: URLs, pricing, and shop statuses below are from my accumulated K-series community knowledge through Jan 2026. Shops open, close, and change owners regularly. Every entry should be treated as a **last-known-good starting point, not a live-verified 2026 directory.** Before sending money to anyone, I check the current 8thcivic.com dyno-results thread and search their recent forum mentions.

Tuner Selection Criteria -- What Matters

Non-Negotiable

1. **Returns my stock calibration backup on request.** If they won't save and hand back my OEM .fpc, walk away.
2. **Uses datalogs during the tune.** Not just dyno sheets -- actual parameter-level logs. This is the work product.
3. **Runs multiple baseline pulls before any changes.** If their first move is "let me advance some timing," find another shop.
4. **Documents what they changed.** I want a calibration-diff I can review.
5. **Has public dyno results on 8thcivic/Honda-Tech/YouTube.** Verifiable track record, not Instagram marketing.

Strongly Preferred

1. **Specializes in K-series / 8th gen Si specifically.** Shops that tune 50 platforms and barely see SIs will miss K20Z3-specific quirks.
2. **Writes detailed tune notes.** Not "dyno'd at 215 whp, see you next time."
3. **Honors the reliability-first rule.** Tuner agrees before the session that my 170k-mile motor gets safe AFR/timing, not max-possible AFR/timing.
4. **Hondata FlashPro native** -- they tune Hondata every week, not once a year.

Red Flags (Walk Away)

- Quotes a peak WHP guarantee before seeing the car
- Won't show me a recent datalog from another customer
- Pushes upsells (cams, turbo kit) before even doing the base tune
- Reviews on 8thcivic or r/CivicSi include the word "melted"
- No mention of knock count or AFR safety margins on their site
- Very cheap relative to others (~$200 when everyone else is $500)
- Won't tune my specific hardware combo ("we don't really work with Skunk2 alphas")

By Region -- Known Shops

West Coast (CA, OR, WA, NV)

Church Automotive Testing -- Wilmington, CA

[VERIFY current status and URL]

- **Owner:** Shawn Church
- **Dyno:** Dynapack hub dyno (highly repeatable, reads slightly lower than Dynojet but more consistent)
- **Specialization:** K-series, RSX, 8th gen Si, Honda/Acura in general. A K-series icon.
- **Reputation:** Meticulous, safety-first. Known to refuse to push a car past his own limits even when a customer asks. Reliability-first tuner by philosophy -- matches mine.
- **Typical cost per session:** $400-600 (**[VERIFY current pricing]**)
- **Booking lead time:** 4-8 weeks historically
- **Why consider:** If I'm anywhere near LA, this is probably my top pick.
- **Why skip:** Distance if I'm not on the West Coast.

SpeedFactory Racing -- Kent, WA

- **Dyno:** Dynojet
- **Specialization:** K-series builds, B-series, K-swaps. FG2 FlashPro work is routine.
- **Reputation:** Big YouTube presence, well-regarded nationally.
- **Typical cost:** $500-700 per session
- **Why consider:** PNW-accessible, lots of public FBO K20Z3 dyno content.

Others to verify in region

- **HARDmotion** (Atlanta-based but travels to West Coast events occasionally -- verify current presence)
- **Hasport Performance** (Arizona -- legendary for mounts/swaps, refers out for tuning)
- **4Piston Racing** (Indiana-based but remote-tunes West Coast clients)

Rocky Mountain / Southwest (AZ, CO, UT, TX)

Reckless Motorsports -- Phoenix, AZ [VERIFY]

- **Dyno:** Dynojet
- **Specialization:** K-series regulars
- **Typical cost:** $450-600

Texas (Dallas/Fort Worth / Houston / Austin)

Texas K-series scene is healthy but shops rotate frequently. **Search the current 8thcivic.com Texas Meetups thread or r/CivicSi for current go-to recommendations.** Historical shops have included:

- **K-Tuned / MD Automotive** (verify status)
- **Dezod Motorsports** (historical CA/TX presence -- verify 2026 status)

Midwest (IL, MI, OH, IN, MO, WI)

4Piston Racing -- Danville, IN

- **Specialization:** K-series engine builder (legit machine shop + tuning). Routinely posts K20Z3 FBO and built-engine dyno pulls.

- **Dyno:** In-house Dynojet
- **Reputation:** Top-tier, huge YouTube following, real engineering depth.
- **Typical cost:** $500-700 per session
- **Why consider:** If I'm willing to road-trip from anywhere midwest, this is the destination pick. Also does full engine builds if I ever go that route.

King Motorsports Unlimited -- Sussex, WI

- **Specialization:** Mugen North America distributor, K-series royalty. Deep OEM + aftermarket knowledge.
- **Dyno:** Dynojet
- **Typical cost:** Premium ($600+)
- **Booking:** Often booked far out
- **Why consider:** Absolutely top-shelf if budget allows and I want the name.
- **Why skip:** Pricing and availability vs competitors.

Other Midwest

- **PRL Motorsports** (Ohio -- more 10th gen/turbo focused, but has FlashPro capability)

Southeast (FL, GA, NC, TN)

HARDmotion -- Atlanta, GA

- **Tuner:** Anthony
- **Dyno:** Mustang dyno (reads slightly lower than Dynojet, usable tuning reference)
- **Specialization:** K-series including 8th gen Si. Huge YouTube presence covering K20Z3 tunes specifically.
- **Typical cost:** $450-600 per session
- **Why consider:** Top Southeast pick, public dyno results shop. Their videos are some of the best K20Z3 tuning content on the internet.

BrenTuning -- Massachusetts (covered under Northeast below)

Remote-tune specialist, works for Southeast clients routinely.

Northeast (NY, NJ, MA, PA, CT, VA, MD)

BrenTuning -- Tewksbury / Peabody area, MA [VERIFY]

- **Specialization:** National remote-tune kingpin for FlashPro. THE name in mail-in FlashPro tuning.
- **In-shop tuning:** Yes, Dynojet
- **Remote tuning:** Signature service. Workflow: I email/upload datalogs + car info + current calibration -> they send revised .fpc -> I flash, drive, log, send back -> iterate.
- **Typical cost (in-shop):** $450-650 per session
- **Typical cost (remote):** ~$350 first calibration, ~$100 revisions **[VERIFY current pricing]**
- **Reputation:** Hundreds of successful K20Z3 tunes, strong 8thcivic forum footprint, known methodology.
- **Why consider:** For remote tuning, effectively the community default. Strong recommendation for Tunes #1 and #2 on a budget.
- **Why be cautious:** Remote tuning requires discipline on my end (accurate datalogs, honest reporting of symptoms). Less hand-holding than an in-shop session.

PTuning -- Maryland / DC area [VERIFY status]

- Historical K-series reputation in the Mid-Atlantic. Verify current operating status before booking.

Others to verify

- **Circuit Motorsports** (NJ)
- **Evolved Injection / Delicious Tuning east coast partners** (occasional K work)

Remote-Only / Mail-In Specialists

BrenTuning (covered above) -- the default remote option.

Boutique / Instagram operators

Several small operators advertise remote FlashPro work on social media. **Vet heavily before sending money:**

- Search 8thcivic for their name -- look for recent customer reviews
- Ask for 3+ references (customer phone numbers/emails)
- Ask to see a sample datalog from their most recent tune
- Pay via credit card, not Zelle/Venmo (chargeback protection)

What Each Session Should Cost

Tune #1: Post-Exhaust (Headers + HFC + Cutout)

- Dual calibration (Daily quiet + Sport with cutout open)
- Pump gas only
- **Expected:** $400-600 in-shop, $300-450 remote
- **Session length:** 2-4 hours on dyno
- **What gets tuned:** fuel tables (low-cam + high-cam), ignition timing, VTEC engagement, rev hang, torque management flattening for Sport map

Tune #2: Post-Intake Manifold (Full Bolt-On Pump Gas)

- Dual calibration (Daily + Sport)
- Still pump gas
- **Expected:** $400-600 in-shop, $300-450 remote
- **Session length:** 2-4 hours
- **What gets tuned:** everything from Tune #1 refined, plus IACV recalibration (new IM), IMRC defeat flag, potentially VTC phase adjustments for Ultra Street manifold

Tune #3: Flex Fuel (Gas + E85 Dual-Fuel Interpolation)

- Dual calibration (Daily + Sport) x dual fuel (gas + E85 endpoints) = **4 effective tunes**
- **Expected:** $500-800 in-shop, $400-600 remote
- **Session length:** 4-6+ hours (flex fuel is the long day)
- **What gets tuned:** E85 endpoint fuel + timing tables, sensor calibration verification, cold-start enrichment, interpolation verification with gradual E-content changes

Combined Pricing?

Some shops offer a package across all three sessions for a ~10-15% discount. **I'm not interested** -- separation of hardware changes means clean attribution of gains. Combined sessions also make troubleshooting harder if something regresses between changes.

Dyno Type -- What to Expect

Dyno	Character	Typical K20Z3 FBO Reading
Dynojet (inertia roller)	Reads highest, most common for K-series bench-racing comparisons	210-225 WHP / 150-160 WTQ
Mustang (eddy-current load)	Reads 5-12 WHP lower than Dynojet on same car	200-215 WHP / 145-155 WTQ
Dynapack (hub dyno)	Best repeatability, slightly lower readings than Dynojet	200-215 WHP / 145-155 WTQ

Honest expectation for a well-tuned FBO K20Z3 in 2026: 210-225 WHP on a Dynojet. Anything above 240 WHP pump gas on stock internals is suspect or built.

E85 flex fuel adds 10-18 WHP typically. **Above 250 WHP pump, below 265 WHP E85 is the realistic envelope for NA FBO.**

My Three-Session Plan

Given my geography (**[NOT YET DECIDED -- TBD based on where I'm actually located]**) and my reliability-first rule:

- **Tune #1 (post-exhaust):** BrenTuning remote -- low risk session, BrenTuning has done this combo hundreds of times
- **Tune #2 (post-intake manifold):** BrenTuning remote again -- consistency with Tune #1's methodology
- **Tune #3 (flex fuel):** In-shop session, ideally Church Automotive or 4Piston or HARDmotion depending on geography. Flex fuel is complex enough that eyes-on-the-car is worth the travel.

Total tuning budget: $1,300-1,800 across all three sessions.

Questions to Ask Every Shop Before Booking

1. "Can you show me a recent datalog from a 06-11 Civic SI K20Z3 FBO tune?" (Verifies they actually tune these)
2. "Do you save and return the stock calibration to the customer?" (Non-negotiable)
3. "What's your approach when you see knock counts in a datalog?" (Answer should include 'pull timing immediately')
4. "What AFR target do you run for pump gas WOT on a 170k motor?" (Expect 11.8-12.2, not 12.5+)
5. "Do you offer separate daily + sport calibrations?" (Essential for my build)
6. "How do you handle flex fuel tuning?" (For Tune #3 specifically)
7. "Can I see your shop and dyno before booking?" (Weed out garage hack operations)
8. "What's your lead time?" (Tells me how busy and reputable they are)
9. "What's included in the price? Revisions? Follow-up datalog review?" (Clarifies scope)
10. "Can you refuse to push the car past safe limits even if I ask?" (The right answer is yes)

See Also

- overview.md -- FlashPro hardware basics

- tuning-workflow-and-maps.md -- the reliability-first rules any tuner I hire must agree to
- datalog-analysis-guide.md -- what I want the tuner to be looking at in their datalogs
- calibrator-tables-reference.md -- what tables the tuner will touch
- ../docs/baseline-logs.md -- my pre-session datalog protocol

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Last updated: 2026-04-19

Tuning Workflow, Daily/Sport Maps, and the Reliability-First Rule

The One Rule That Overrides Everything

> **Engine life and reliability come before power. Every time. No exceptions.**

I'd rather leave 10 WHP on the table than bend a valve or spin a bearing on a 170k-mile K20Z3. This rule applies to every calibration decision, every parts choice that feeds into tuning, and every "should I bump this one more degree?" moment on a dyno. Safety first, power second.

If a number on a dyno sheet costs me the engine, that number doesn't count.

Every Tune = Two Maps

Whenever I do a tune (base map flash, post-exhaust retune, post-intake-manifold retune, flex fuel), the output is **two calibrations, not one:**

Daily Map

- **Valve:** Cutout closed (quiet)
- **Timing:** Conservative -- 2-3 deg under max safe
- **AFR at WOT:** 12.0-12.2 on pump gas, lambda 0.80 on E85
- **VTEC crossover:** Stock-adjacent (~5400-5800 RPM)
- **Rev limiter:** Stock (8200 RPM fuel cut)
- **Launch control:** Off
- **Idle:** Smooth, OEM-feeling
- **Cold start:** Polished, no rough minute-one
- **Goal:** Drives like a new SI from the dealer. Quiet. Predictable. Nothing on the edge.

Sport Map

- **Valve:** Cutout open (loud)
- **Timing:** Max safe -- always verified with knock datalog at 0 counts AND wideband AFR at target
- **AFR at WOT:** 11.8-12.0 on pump gas, lambda 0.78-0.80 on E85 (slightly richer = cooler, safer)
- **VTEC crossover:** Lowered to 4800-5200 RPM (only after the exhaust flows enough to justify it)
- **Rev limiter:** +100 to +200 RPM over stock (8300-8400), not more
- **Launch control:** Optional, conservative (4500 RPM, no 2-step aggression)
- **Idle:** Tolerable. Slight lope allowed from lower VTEC crossover, nothing more.
- **Goal:** Squeezes the engine for everything it's got, WITHIN THE RELIABILITY ENVELOPE. Not "let's see what happens." Not "maybe push it one more click." Verified safe, every parameter.

What "Sport = max" actually means:

Sport is NOT "remove safety margins." It IS "stop leaving free power on the table." Free power lives in:

- Timing advance that the engine can take without knock, proven by datalog
- Fuel enrichment that's cooler and safer than what a stock conservative tune runs

- VTEC crossover dropped to where the exhaust/intake combo actually supports it
- Fan tables dialed tight so heat soak doesn't force defensive retune

Every one of these is a gain measured against a conservative baseline, not against the engine's destruction point.

The Three Tune Sessions Plan

Session	After Installing	Both Maps Tune	Budget
Tune #1	Exhaust (cutout + headers + HFC)	Daily + Sport on pump gas, no intake mfld yet	$300-500
Tune #2	Intake manifold + bored TB	Daily + Sport on pump gas -- FULL BOLT-ON pump final	$300-500
Tune #3	Flex fuel system (sensor + injectors + pump)	Daily + Sport with gas?E85 real-time interpolation	$400-600

Never combine sessions. Each tune session isolates one set of hardware changes so I can attribute gains (or regressions) cleanly. Combining intake + flex fuel + exhaust retune in one dyno day = four hours of the tuner guessing which mod caused what.

Cross-reference: Each session maps to a datalog milestone in docs/baseline-logs.md. Pre-session logs + post-session logs + dyno graph = complete paper trail.

Pre-Session Checklist (Runs Before Every Dyno Day)

Hardware Health

Any NO answer below = reschedule. Tuning a sick engine is throwing money away.

- [ ] Spark plugs fresh (< 5,000 mi on current set)
- [ ] Ignition coils healthy (no misfire codes in 500 mi)
- [ ] VTC actuator replaced or confirmed silent at cold start (see 06-Ignition-Refresh/)
- [ ] Valve adjustment in spec (within last 60k mi)
- [ ] Timing chain inspected (measured via FlashPro cam phase datalog, < 2-3 deg retardation)
- [ ] Coolant system fresh (see 09-Cooling-System/)
- [ ] Oil fresh, correct viscosity (5W-30 for my 170k motor)
- [ ] Fuel filter clean (replaced post-E85 introduction if applicable)
- [ ] Battery healthy (13.8+V at idle with alternator charging)
- [ ] Wideband O2 sensor installed and reading sensibly at idle (see 11-Wideband-AFR/)

Software / Calibration

- [ ] **Stock calibration saved as backup** (the factory reset button)
- [ ] Previous dyno's final calibration saved to a known filename (for rollback)
- [ ] FlashProManager on current version
- [ ] All FlashPro firmware updates applied (test flash at home before dyno day)
- [ ] My log set from the last baseline session is reviewed -- no ignored knock events, no drifting trims

Session Plan

- [ ] Tuner knows both calibrations are needed (daily + sport)

- [ ] Tuner has the datalog files from previous session
- [ ] Tank is HALF FULL of the target fuel (not empty, not full -- stable fuel pressure matters)
- [ ] If running E85: verified station I filled at, and flex fuel sensor is reporting sensible %

The Safety Ceilings I Won't Cross

These are bright lines. Below any of them, no tune ships.

Parameter	Safe Ceiling	Reliability Margin
Knock count at WOT	0 (strictly zero)	Ever. If I see any count, timing gets pulled immediately.
Pump gas WOT AFR	Never leaner than 12.0	Cool-running richness is free reliability.
E85 WOT lambda	Never leaner than 0.82	E85 forgives rich, punishes lean.
Peak cylinder temp proxy (IAT + ignition advance)	Stay well under OEM map's hot cells	Datalog ECT + IAT + timing. Retreat at 220F ECT.
Injector duty cycle	Never above 85% sustained	If I hit 85%, upsize injectors -- never compensate with raised fuel pressure.
Rev limiter	+200 RPM over stock, max	The factory 8200 is already conservative by design. Moving it further is pure risk.
VTEC engagement RPM	Never below 4500	Below this, the cam profile doesn't make sense for the airflow.
Fuel pressure at WOT	Within 2 PSI of base spec	Pressure drop = lean spike. Hard fail condition.
Coolant temp sustained	Target <= 200F at cruise, <= 210F at WOT	Above this, fan tables need work, not tune changes.

If any of these get crossed mid-session, the session stops. Fix the cause, come back another day.

How Each Mod Affects The Tune

Mod	Affects Fuel?	Affects Timing?	Affects VTEC?	Retune Required?
K&N intake (installed)	Yes -- more air -> leaner baseline	Slight -- can advance a hair	Slight -- crossover could drop	Base map only, no dyno
Clutch + flywheel (installed)	No	No	No	None -- pure mechanical
Shifter + bushings (installed)	No	No	No	None
Engine mounts	No	No	No	None (maybe idle AFR hair off until ECU adapts -- stock tune self-corrects)
Brakes	No	No	No	None
Strut bar	No	No	No	None
Headers + HFC	YES -- major	YES -- significant	YES -- crossover drop opportunity	Dyno required
Valved exhaust cutout	Yes -- AFR shifts valve open vs closed	Minor	No	Dyno -- tune both valve states

Intake manifold + bored TB	YES -- major	YES -- significant	YES -- crossover drop opportunity	Dyno required
Pulleys + damper	No	No	No	None (mechanical -- slight ECU idle relearn)
Flex fuel system	YES -- total recalibration	YES -- interpolated	Optional	Dyno required, long session
Clutch hydraulics	No	No	No	None
Suspension / brakes	No	No	No	None
LattePanda install	No	No	No	None -- just monitoring

Tune-required mods get grouped into dyno sessions. Tune-unaffected mods get installed whenever -- they don't drive the calendar.

What "Sport Map" Looks Like Per Phase

As the hardware progresses, the Sport map gets more aggressive -- but always within the reliability ceilings above.

Phase	Hardware	Sport Map Character
Now (intake only)	K&N, no cat/exhaust mods	Community base map, no sport/daily split yet -- just "don't break anything"
Post-exhaust	Cutout + headers + HFC	First real Sport map: +1-2 deg timing, AFR 11.9, VTEC at 5200
Post-intake mfld	FBO complete	Refined Sport: +2-3 deg over stock, VTEC at 4900-5000, rev to 8300
Post-flex fuel	Gas + E85 dual	E85 Sport: lambda 0.79, +4-5 deg over stock, VTEC at 4800, rev to 8400

None of these exceed the safety ceilings. Each phase has more to work with because the hardware lets the engine breathe better -- not because I'm pushing it harder into the red.

Working With Claude On Tune Changes (Eventually)

Right now I can't directly give Claude live ECU access -- FlashProManager is Windows-only and hardware-paired. What I can do:

The CSV Export Workflow

1. Drive with FlashPro datalogging on.
2. Back at my PC, open FlashProManager -> File -> Export Datalog -> CSV.
3. Paste the CSV content into a chat with Claude (or attach the file).
4. Claude analyzes -- flags outliers, identifies unsafe cells, proposes specific table edits.
5. I open the corresponding fuel/timing/cam table in FlashProManager.
6. I type the proposed values into specific cells (or reject them if the reasoning is off).
7. Save calibration, flash, verify.
8. Log again, repeat.

What Claude CAN Do

- Read my CSV datalog and flag knock events, lean AFR, trim drift, over-temp cells

- Compare two sessions (before/after) and quantify what changed
- Propose conservative timing/fuel adjustments with specific cell addresses
- Spot patterns I'd miss (e.g., a specific load cell that's consistently lean-biased)
- Talk me through which cells a given mod should affect most

What Claude CANNOT Do

- Flash my ECU (no hardware access)
- Read my .fpc or .fpdl binary files directly (I need to export to CSV/text)
- Replace a professional tuner for dyno-verified timing advances
- Know what my engine "sounds like" or respond to physical feedback
- Make safety calls that override the ceilings above -- even at my explicit request

What I Share With Claude Per Session

- Datalog CSV (or excerpt of relevant cells)
- Screenshot of the fuel, timing, VTEC, or cam phase table I'm considering editing
- My current calibration name and any notes from my tuner
- What I'm trying to accomplish (e.g., "smooth out the transition from closed to open loop at 3500 RPM")

The Future: LattePanda as Live Tune Companion

Once the LattePanda 3 Delta is installed, the workflow gets tighter. The LattePanda runs FlashProManager natively -- so it can be logging continuously, and the wideband O2 signal is coming in on an analog input that the same machine reads. A lightweight Python script can:

- Auto-export datalogs after each drive to a dedicated folder
- Compute simple alerts ("today you had 3 knock events above 6500 RPM in cells X, Y, Z -- investigate")
- Sync logs to cloud storage so I can paste them to Claude from any device

But this is a Phase 7+ project. The core workflow (drive -> export CSV -> analyze -> edit -> flash) works fine on a laptop at home today.

See Also

- overview.md -- the FlashPro hardware + my temporary/permanent setup paths
- Temporary-Setup/getting-started.md -- getting online with a laptop first
- Permanent-LattePanda-Install/full-guide.md -- in-car PC install
- ../docs/baseline-logs.md -- my datalog protocol
- ../11-Wideband-AFR/ -- the wideband this workflow relies on

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Last updated: 2026-04-18

FlashProManager 4.6.6.0 -- Master Reference

> **Single source of truth for everything Hondata FlashPro + FlashProManager.** This file is built from Hondata's own official help content (pasted section-by-section from the live help site), cross-referenced with community sources and my own FlashPro install evidence. Future Claude instances should load this before any Hondata conversation.

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0. Navigation Index (Mirrors the Hondata Help TOC)

Jumping-off points organized exactly like the FlashPro Help tree:

Basics

- 1. What's New -- Version Timeline
- 2. FlashPro Features
- 3. Hardware Specifications
- 4. System Requirements
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- 6.98 Update Boot Firmware
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- 6.100 Compatibility Matrix for Civic Si '06
- 6.101 Per-Sensor Definitions

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- 9. Plugins and Scripting (Lua)
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- 11. Vehicle-Specific Notes: Civic Si 2006-2011 (MY CAR)
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Practical Quick-References

- PRB Calibration Window Tree (what I actually see in FPM)
- Pin Allocation Decision for My Build

PRB Calibration Window Tree (What I Actually See in FPM)

From screenshot of FPM's Calibration window with my PRB K20Z3 calibration loaded. This is the navigable tree in the software -- each ? expands to reveal parameters and tables. Maps 1:1 to sections 6.64-6.84 below.

```
? Calibration
|- ? Fuel
|  |- ? WOT lambda adjustment low
|  |- ? WOT lambda adjustment high
|  |- ? Cranking fuel
|  |- ? Cylinder trim
|  |- ?? Air temperature compensation
|  |- ?? Coolant temperature
|  |- ? Active fuel tuning low
|  |- ? Active fuel tuning high
|  \- ?? Active Tuning (enable/disable + settings)
|- ? Ignition
|  |- ? Ignition low
|  |- ? Ignition high
|  |- ? Cylinder trim
|  |- ?? Air temperature compensation
|  \- ?? Coolant temperature
|- ? VTEC
|  (no sub-tables -- parameters only: rpm window, pressure window, conditions)
|- ?? VTC
|  |- ? Cam angle low (low-cam VTC targets, 3D by cam angle)
|  |- ? Cam angle high (high-cam VTC targets)
|  |- ? Cam angle index low (RPM/load breakpoints for low-cam table)
|  \- ? Cam angle index high (RPM/load breakpoints for high-cam table)
|- ?  Closed Loop
|  \- ? Lambda (target lambda + related)
|- ? Knock Control
|  |- ? Knock ignition limit low
|  |- ? Knock ignition limit high
|  |- ? Knock sensitivity low
|  |- ? Knock sensitivity high
```

```
|  |- ? Knock retard low
|  \- ? Knock retard high
|- ? Rev Limits
|  (parameters only: overall, launch, speed, boost cut, full throttle shift)
|- ? Sensors
|  \- ? AFM (air flow meter calibration table)
|- ? Boost Control
|  \- ? Boost control duty lookup (not usable on NA)
|- ? Idle
|  (idle speed tables + parameters)
|- ?? Throttle
|  \- ? TPlate Normal (target throttle plate table)
|- ?? Misc
|  (VSA, EPS, cruise control, fan temps, etc.)
|- ? Traction Control
|  (if Hondata TC module connected -- hardware + software)
\- ? Ethanol
   \- ? Ethanol ignition compensation
   (+ Ethanol percentage table, fuel multiplier, boost limit)
```

Key observations:

- Fuel and Ignition are **mirrored** -- same sub-structure for both (low, high, cylinder trim, IAT comp, ECT comp). Learning one teaches the other.
- VTC has **4 tables** (cam angle low, cam angle high, and their separate index tables). The index tables are where I do VTEC crossover optimization per Section 6.30.
- Knock Control has **6 tables** (3 parameters x low/high cam). Deep tuner territory.
- **WOT lambda adjustment has two copies** (low/high). Critical since Hondata ceiling of 12.50:1 applies to BOTH.
- **Active fuel tuning low AND high** -- ECU-side auto-tuning works on BOTH cam profiles.
- **Ethanol ignition compensation** lives here -- my flex fuel interaction point.

Pin Allocation Decision for My Build

Based on official Hondata ECU Connectors topic + Wideband + Ethanol Parameters docs:

My 2006-2011 Civic Si (PRB ECU) pinout for aftermarket inputs:

Pin	Stock Function	Repurposable For	My Plan
A23	ELD (Electrical Load Detector)	Wideband input	WIDEBAND (AEM X-Series 30-0300) <- my wideband goes here
A33	ECT2 (second coolant temp sensor)	Wideband OR Flex fuel input	FLEX FUEL (Innovate ECF-1 controller output) <- flex fuel goes here
B2	EGR output	Boost control output	Unused (NA) -- future turbo prep only
B29	EGRL (EGR lift input)	Wideband input	Unused -- redundant for my needs
C27	SO2 (Rear O2 sensor input)	NOT RECOMMENDED for wideband	Keep stock function (with Berk HFC, rear O2 is valid)

Why this allocation:

- **A23 / ELD for wideband:** Hondata explicitly lists as good wideband input. ELD must be disabled in misc params + factory pin removed from ECU connector.
- **A33 / ECT2 for flex fuel:** Ethanol Parameters topic confirms ECT2 is the K-series flex fuel input path. Must disable ECT2 in misc + remove factory pin.
- **C27 / SO2 left stock:** Section 6.8 says SO2 is NOT recommended for wideband (ECU voltage limiting). I have the Berk HFC so rear O2 stays functional and verifies catalyst efficiency.
- **B2 unused on NA:** saves a pin if I ever go turbo.

Wiring reality: this means I need to:

1. Remove the ECT2 pin from the ECU connector -> solder flex fuel controller signal to that wire (or new wire going to ECT2 pin location)
2. Remove the ELD pin from the ECU connector -> solder AEM wideband analog output signal to that wire
3. Disable ELD AND ECT2 in FPM Misc parameters before flashing
4. Configure the wideband input calibration (0V=10 AFR, 5V=20 AFR for AEM X-Series)
5. Configure the ethanol percentage table (Innovate ECF-1: 0.00V=0%, 5.00V=100%)

Critical: disable BEFORE flashing, flash with ELD/ECT2 disabled, THEN do the wiring. Otherwise ECU throws DTCs during boot.

My install:

- **FlashPro hardware:** FlashProOG (original, not Mini, not FlashPro 2)
- **FlashProManager version:** 4.6.6.0 (installed from FlashProManagerV4-6-6-Full.exe in C:\Users\extra\Downloads\)
- **FlashProManager exe:** C:\Program Files (x86)\FlashPro\FlashProManager.exe (+ 64-bit variant)
- **Help file (local):** C:\Program Files (x86)\FlashPro\FlashPro.chm
- **Config file:** C:\Users\extra\AppData\Local\Hondata\FlashPro\FlashPro.ini
- **Calibration directory:** C:\Users\extra\OneDrive\Documents\FlashPro Calibrations\
- **Internal FlashPro unit version (per ini):** FlashPro=461004

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1. What's New -- Version Timeline

Source: https://www.hondata.com/help/flashpro/index.html?whatsnewtopic.htm (fetched 2026-04-19)

Version	Released	Changes
4.6.6	2026-04-02	Firmware updated to v278. Added Spoon Sports 2026 N One 5K7-J11. CANFlex support for 2016-2021 Civic. Extended 2022+ Civic to 64A-X110 Mexico variant.
4.6.5	2026-02-26	Firmware v277. Enhanced 2012-2015 Civic Si boost target/duty params. Fixed ethanol boost limit on FK8/FL5/DE5.
4.6.4	2026-02-13	Firmware v276. Expanded US FL5 engine swap compatibility.
4.6.3	2026-01-15	Firmware v275. FK8 support for 5BF-S13 Singapore ECU.
4.6.2	2025-11-26	FL5 compatibility with 66V-Q04 binary. Fixed cranking on 2014 Civic Si Mexico.
4.6.1	2025-11-21	Fixed no-boost on 2020 Civic MT Japan.
4.6.0	2025-10-30	Firmware v274. Broadened 2019+ RDX support.

Key observation for my K20Z3 FG2 build: the 4.6.x series has been entirely focused on newer platforms (FK8,

FL5, 2012+ Civic Si, RDX). **The PRB calibration for 06-11 Civic SI has not changed in any 4.6.x release.** This means my calibration files and community base maps from older FPM versions remain valid on 4.6.6.0 -- no migration concerns, no lost features.

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2. FlashPro Features (Hardware + Software)

Source: FlashProManager help, "Features" topic (pasted by user)

Hardware Features

- Programmable ECU interface
- Connects via OBDII diagnostic port
- **No ECU modification necessary** (exception: Bosch ECUs)
- Fast ECU programming time
- **20 hours on board datalogging memory** (unit stores logs independently of PC)
- **Fast datalogging -- 4x faster than OBDII protocols** (uses CAN, not K-line)
- Check and clear diagnostic codes (DTCs)

Software Features (FlashProManager)

- Custom laptop gauges (on-screen display while connected)
- Readiness codes and smog check status
- Cam angle aware table editing
- **? Dual calibration support** -- **THIS IS SIGNIFICANT**

The Dual Calibration Question (Open)

The earlier reference docs I wrote stated that the original FlashPro stores **one** calibration at a time. Hondata's own help says **"Dual calibration support."** This contradicts what I had written. What "dual calibration" means in FPM context needs verification:

- **Possibility A:** The FlashPro unit holds both a primary AND secondary calibration, and the driver switches between them via some input (a physical button tap, a dashboard toggle wired to an input). If true, this enables **Daily ? Sport switching without a PC flash** -- a game-changer for my build.
- **Possibility B:** "Dual calibration" means FPM on PC can manage two calibrations (which is trivially true -- you can save unlimited .fpcal files on disk).
- **Possibility C:** "Dual calibration" means per-ECU slots (primary + secondary ROM slots in the ECU itself that the FlashPro can flash independently).

[VERIFY] with a Hondata official docs page specifically about dual calibrations, or via the "Dual Cal" menu in FPM.

If the answer is Possibility A, my entire Daily/Sport workflow changes -- no LattePanda required for on-demand switching. If it's C, the LattePanda still matters.

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3. Hardware Specifications

Source: FlashProManager help, "Specifications" topic (pasted by user)

Physical -- FlashProOG (what I have)

Spec	Value
Weight	210g / 7.4oz
Storage temperature range	-20 to 85 degC / 0 to 185 degF
Operational temperature range	0 to 70 degC / 30 to 160 degF
OBDII cable length	2m / 6'

Physical -- FlashProMini (smaller variant, not mine)

Spec	Value
Weight	85g
Storage temperature range	-20 to 85 degC / 0 to 185 degF
Operational temperature range	0 to 70 degC / 30 to 160 degF

Electrical (both variants)

Spec	Value
Supply voltage	9-16V
Current consumption (active)	< 100 mA
Current consumption (standby)	< 10 mA

Communication Protocols

- **ISO 15765 CAN** -- note this is CAN bus, not K-line. Explains the "4x faster than OBDII" claim.

Practical Implications

- Operational temp ceiling of 70degC / 160degF = careful about engine-bay placement. The cabin is fine. Under the dash is fine. Not in the engine bay.
- 100mA active current draw is trivial for the car's 12V system. Doesn't load the alternator.
- 10mA standby is low enough that leaving it plugged into OBDII indefinitely won't drain the battery in a reasonable timeframe (weeks), but I'll unplug when parking for extended periods.

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4. System Requirements

Source: FlashProManager help, "Requirements" topic (pasted by user)

FlashPro Requirements

- Laptop or Desktop
- **Windows 10 or 11 (32 or 64 bit)** -- official support
- USB Port (1.1 or 2.0 or 3.0)
- **Internet connection** (for registration, updates, some features)

Power Management (laptop users)

- **Recommended: disable automatic standby** in Power Options. Standby during a flash = potentially bricked calibration.
- **Not recommended: allow hibernation.** Same reason.

Implication for LattePanda Install

- LattePanda 3 Delta runs Windows 11 x64 natively -- **confirmed compatible.**
- USB 3.0 port on the Panda (for the FlashPro unit) -- confirmed.
- I need to disable Windows sleep/hibernation in the LattePanda's Power Options. Critical for reliability.

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5. Help Site Structure (as discovered)

Live Hondata help root: https://www.hondata.com/help/flashpro/

Known working topic URLs (confirmed via WebFetch):

- https://www.hondata.com/help/flashpro/index.html?whatsnewtopic.htm -- What's New / Changelog
- https://www.hondata.com/help/flashpro/whatsnewtopic.htm -- Same content, direct topic URL
- https://www.hondata.com/help/flashpro/introduction.htm -- Introduction overview (confirmed via fetch -- includes getting started, calibration creation, reflashing, datalogging)

Known NOT working (returned generic Hondata homepage):

- https://www.hondata.com/help/flashpro/tuningguidetopic.htm
- https://www.hondata.com/help/flashpro/calibrationtopic.htm
- https://www.hondata.com/help/flashpro/datalogtopic.htm
- https://www.hondata.com/help/flashpro/parameterstopic.htm
- https://www.hondata.com/help/flashpro/tocindex.htm
- https://www.hondata.com/help/flashpro/helpindex.htm

This means URL pattern is **not predictable by topic name guessing**. Hondata's CHM-to-web export uses specific filenames that don't match the topic titles in obvious ways. The user feeding content directly (copy-paste from the help page) is the most reliable way to build this reference.

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6. Topics Captured So Far

6.1 Features -- CAPTURED (see Section 2)

6.2 Specifications -- CAPTURED (see Section 3)

6.3 Requirements -- CAPTURED (see Section 4)

6.4 What's New / Changelog -- CAPTURED (see Section 1)

6.5 Introduction -- CAPTURED

The Introduction topic confirms workflow sequence:

1. Install software, check for updates, connect USB drivers, complete owner info, register with Hondata
2. Locate OBDII port, insert device, ignition ON (engine off), lock to vehicle

3. Open Calibration window, select ECU type, choose starting calibration matched to mods, save
4. Ignition on / engine off, Upload to flash
5. Datalog with F9, save from menu, open graph windows with templates
6. Refer to Tuning Guide for actual tuning

Known Hondata tutorial videos (require internet): FlashPro Introduction, Getting Started, Race Calibration, Datalogging and Air/Fuel ratios.

6.6 The FlashPro (Hardware Unit) -- CAPTURED

Physical variants:

Variant	Connection to car	USB connector	Bluetooth	Colors available
FlashProOG (mine)	OBDII via 2m cable	USB Type B	Yes, if after December 2013	Race (grey), Dealer (black), CARB (blue)
FlashProMini	Directly onto OBDII connector	USB Type C	Yes	Race (grey) only

Both are functionally identical -- differences are only size, shape, appearance, connectors. Same software, same features, same ECU support.

Indicator lights (on the unit itself):

LED	Behavior	Meaning
Power	On	FlashPro has power (from USB or OBDII diagnostic connector)
	Off	Disconnected or in power saving mode
Program	On / steady flashing	RE-FLASHING ECU -- do NOT unplug or switch off ignition
Datalog	Steady flashing	Datalogging from ECU to onboard memory, or indicating USB activity
USB	On	FlashPro connected and USB drivers installed correctly
	Off	USB not connected OR drivers not installed correctly

Physical buttons:

Button	Function
Program	Initiates ECU re-programming (flash). Short press primary calibration.
Datalog	Starts/stops datalogging from vehicle

Dual calibration button combo (from "Uploading a Calibration" topic):

- **Primary calibration:** press Program for at least 1 second
- **Secondary calibration:** hold Datalog button down, THEN press Program for at least 1 second, release Program before releasing Datalog

This is on the FlashPro unit itself -- no PC connection needed to switch.

6.7 Installation Warnings -- CAPTURED (CRITICAL)

Hondata's warranty will NOT cover damage from:

- Low-impedance injectors

- Non-resistor spark plugs
- Incorrect engine ground location

Injector impedance:

- Honda ECU designed for **high-impedance ("saturated") injectors**
- Low-impedance ("peak and hold") injectors will damage ECU and/or wiring
- Low-Z can be used with a resistor box, but high-Z strongly preferred
- **For my build:** verify Bosch EV14 550cc are high-impedance (they are -- standard saturated EV14 platform). ID1050x are also high-impedance. Safe either way but confirm part-specific spec.

Spark plugs:

- **Do NOT use non-resistor plugs** -- causes electrical interference that can affect ECU
- **My NGK ILZKR7B-11S and ILFR7H are resistor plugs** (the "R" in ILFR indicates resistor type) -- compatible

Engine Ground (G101) -- extremely important:

- Ground from wiring harness to engine at G101 must make good electrical contact with the cylinder head
- If G101 ground is poor, return path for sensors/ignition/VTEC can flow through OBDII -> FlashPro -> laptop -> laptop charger -> inverter -> vehicle body
- **This can damage ECU and/or FlashPro**
- **Recommended ground location:** stud on cylinder head (best electrical connection)
- **Acceptable alternative:** top bolt holding radiator water output to cylinder head
- **Stock location on intake manifold:** usually works but inferior to cylinder head

For my 170k K20Z3: I need to inspect G101 condition during the next engine work. A corroded G101 ground at this mileage is plausible. Cleaning or relocating to the cylinder head stud is cheap insurance.

Software versions:

- **Do NOT roll back or backdate FPM versions**
- FlashPro firmware gets updated by newer FPM -- older FPM trying to access newer-firmware FlashPro may malfunction

Reflashed ECUs:

- If ECU has been reflashed by anyone other than Hondata (uprev, Ktuner, etc.), **do NOT use FlashPro until ECU is reflashed back to stock by the original company**
- Most aftermarket reflashers corrupt ECU flash memory in a way that prevents future reflashing
- **My ECU:** confirmed as stock (Factory Flash backup = original OEM calibration). Safe.

6.8 Wideband Lambda Interface -- CAPTURED (CRITICAL for Phase 6 + 11-Wideband-AFR)

This contradicts my earlier reference docs that said "Analog Input 1" of the FlashPro unit. The wideband actually wires into unused **ECU** input pins (not FlashPro analog inputs). The ECU's K-series pinout has several available inputs:

Available ECU inputs for wideband (with caveats):

ECU Input	Viability	Notes
SO2 (secondary O2 / rear O2)	NOT RECOMMENDED	ECU internally limits voltage; accurate wideband reading not possible. Also conflicts with flex fuel sensor if using this pin.
EGRL (EGR lift)	OK	Requires adding an ECU pin and wire
ECT2 (second water temp sensor)	OK	Second ECT must be disabled; factory pin must be removed from ECU connector
ELD (electrical load detector)	OK	ELD must be disabled; factory pin must be removed from ECU connector

See: ECU Connectors topic (not yet captured) for pin locations.

Grounding -- critical to avoid ground loops:

- Wideband meter MUST use same ground as ECU for BOTH power AND analog output
- **For K-series: G101 on the intake manifold is the correct ground point**
- Do NOT ground the ECU or wideband to any other points
- Do NOT share wideband ground wire with any other devices

Settings in FPM:

- Wideband input settings under: **Closed loop parameters**
- Select which analog input pin (from the 4 options above)
- Voltage offset can be configured -- **but if you need a voltage offset you have a ground problem that should be fixed, not compensated**
- Voltage-to-lambda translation: typically 0V = 10.0 AFR, 5V = 20.0 AFR (linear). If wideband has more than 2 voltage/lambda pairs, it may be non-linear and cannot be used.

Wideband-specific output specs (from Hondata help):

Brand	Output spec
PLX	0V = 10:1, 5V = 20:1 (standard)
AEM (later models)	0V = 10:1, 5V = 20:1 (earlier AEMs had different outputs -- check manual)
Innovate	Program to "standard" 0V=10:1, 5V=20:1 via their programming software/cable

For my AEM X-Series 30-0300: It's the "later AEM" spec = 0V=10:1, 5V=20:1. No reprogramming needed.

Implications for 11-Wideband-AFR/overview.md:

- That doc said "FlashPro analog input 1" -- **wrong**. Needs updating.
- The ECU pin to use depends on what I'm doing with flex fuel. ECT2 or ELD are the best candidates for my build. EGRL requires adding a pin/wire.
- Both wideband AND flex fuel sensor need ECU input pins. I need to plan which pin each gets.

6.9 Calibrations (New Cal Workflow) -- CAPTURED

About:

- A calibration contains all settings and table values that can be changed in the ECU
- Normal extension: .fpcal (CONFIRMED -- was wrong in earlier docs that said .fpc)

New calibration workflow:

1. File -> New Calibration
2. Verify correct vehicle model and type. If FlashPro plugged in, defaults to correct; otherwise uses last-used vehicle
3. Double-click calibration name to select and close window
4. Calibrations marked:
   - **Stock equivalent** = same as what car shipped with
   - **Tuned** = calibration is tuned for best performance from that part combination
5. For some vehicles, a **New Calibration Wizard** is available -- select mod combo + fuel octane -> wizard picks best calibration

For me: When creating a new calibration for my car, I'd pick one that matches my current installed mods. Starting point likely "stock equivalent + K&N Typhoon intake" since that's my current hardware state.

6.10 Modifications (Reusable Calibration Snippets) -- CAPTURED

What a Modification is:

- A **subset** of a calibration -- reusable across compatible calibrations
- Example: a custom throttle mapping table that can be imported into any calibration without touching the rest of the cal
- Installed/removed via Modifications window

How to use:

- File menu -> Modifications, or toolbar button
- Click checkbox or double-click modification to install/remove
- Warning: a mod may change several parameters at once, so installing one may cause another to show as "installed" too. Some mods can appear installed even on a stock cal. This is normal.

Creating a mod (to share or archive my own):

- In Modifications window: Export button
- Set Category, Author, Revision, Description, Notes
- Select the specific Parameters (P) and Tables (T) to include
- Export creates the mod file
- Update button: refreshes an existing mod file
- **Storage location:** C:\Program Files (x86)\FlashPro (Modifications directory)
- Windows won't allow saving directly there from an app -- must save to different dir then copy via Explorer

Usefulness for my build:

- If a tuner gives me a "modification file" for (e.g.) flex fuel tables, I can layer it onto my base calibration
- If I make useful changes (e.g., my custom fan tables), I can export them as a mod for future reuse
- Don't need to keep re-doing the same changes across every new base cal

---

6.11 Quick Modifications (In-Window Shortcuts) -- CAPTURED

- Quick Modifications are mods accessible directly from the Calibration window

- Example: drop-down for different intake AFM calibrations
- When regular mods are added/removed, the Quick Modification drop-down updates to match
- If an underlying table is edited, the drop-down either switches to a different modification or shows blank (if no mod matches)

Practical: a fast way to swap between "stock airbox" / "short ram" / "cold air intake" calibrations without hunting through Modifications window.

6.12 Programming Your ECU -- CAPTURED

Key stats (correcting my earlier estimate of 20-45 seconds):

- **Actual flash time: approximately 90 seconds**
- **Do NOT interrupt the flash** -- unplugging cable or cancelling upload puts ECU in recovery mode
- Interrupted flash = ECU won't run engine, but CAN still be re-programmed with FlashPro (see 6.15 Recovery Mode)

For my workflow: 90 seconds is still fast enough that Daily ? Sport switching via re-flash is viable (if I didn't want to use the on-unit dual cal feature). But dual calibration + button switching is faster still.

6.13 Opening a Calibration -- CAPTURED

- Before programming the ECU, open a calibration (New, or from file)
- Calibrations from older FPM versions OR non-calibration files: FPM shows error
- Implication: if a tuner sends me an old .fpcal, FPM may refuse to load it until they resend from a current version

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6.14 Uploading a Calibration -- CAPTURED (GAME CHANGER)

This is the dual calibration mechanism. It's not PC-dependent at runtime.

Uploading from PC to ECU (standard workflow):

1. Load calibration in FlashProManager
2. Plug FlashPro into PC via USB
3. Plug FlashPro into OBDII, switch on ignition
4. Select Upload from Online menu

Uploading from FlashPro to ECU (using stored calibration, NO PC needed):

1. Plug FlashPro into OBDII, switch on ignition
2. **Primary calibration:** press Program button for at least 1 second
3. **Secondary calibration:** hold Datalog button down, press Program for at least 1 second, release Program before Datalog

Storing calibrations on the FlashPro unit:

1. Load calibration in FPM
2. Plug FlashPro into PC via USB
3. Open FlashPro window -> Calibrations tab -> Upload

4. Choose primary or secondary slot

Prerequisites:

- FlashPro must be locked to your vehicle
- FlashPro must be registered with Hondata

? THE DAILY/SPORT WORKFLOW I NOW WANT

1. Build a Daily calibration, upload to **primary slot** on FlashPro
2. Build a Sport calibration, upload to **secondary slot** on FlashPro
3. In the car:
   - Want daily quiet? Ignition on, press Program 1 sec, wait ~90 sec, start car.
   - Want sport loud? Ignition on, hold Datalog + press Program 1 sec, wait ~90 sec, start car.
4. **No PC required for the switch.** Previously I assumed I'd need the LattePanda permanently connected.

Now: the LattePanda is still valuable for datalogging + exhaust valve control, but **calibration switching works without it**.

This changes the game for Phase 7 planning. The LattePanda becomes more about monitoring + logging + valve relay, less about "flash button replacement."

6.15 Downloading a Calibration -- CAPTURED

- The last uploaded calibration is stored on the FlashPro
- Download via: Online menu -> Download
- Once downloaded, the cal can be edited, saved, or re-uploaded

Useful when:

- I don't have the .fpcal file locally but want to pull back what's on the unit
- Someone else has flashed the car and I need to know what's running

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6.16 ECU Recovery Mode -- CAPTURED

What triggers it: interrupted flash (power loss, cable unplug, cancel).

What it is: ECU won't run engine, but CAN still be re-programmed.

How to recognize: FPM status bar shows exclamation icon.

Recovery procedure:

1. Upload a calibration to the ECU (can be the same as before)
2. **Switch ignition OFF then back ON** (important step!)
3. Verify ECU icon changes to indicate normal mode

Implication: Failed flash is NOT a brick. Always recoverable. Reduces flash-day anxiety significantly.

6.17 Maintenance Minder (Service Light Reset) -- CAPTURED

When switching between AFM and race calibrations, the service maintenance reminder can illuminate the dash wrench light and display "service A" or "service B" with negative miles.

Reset procedure:

1. Ignition ON, do NOT start engine
2. Hold **SEL/RESET** button for 10 seconds until display starts flashing
3. Release SEL/RESET
4. Hold it down AGAIN for 6 seconds -- display clears

For me: Worth knowing before/after any race calibration flash.

6.18 Live Tuning -- CAPTURED (MAJOR FEATURE)

What it is: Real-time table updates in the ECU while the engine is running. Changes take effect immediately -- no re-flash needed.

Recommended: Only use Live Tuning while actively tuning the vehicle. **Disable for regular driving.**

How it works:

- FPM alters the ECU so it copies table data from flash memory to RAM on reset
- Table edits in FPM get sent to ECU RAM live
- ECU tables update in fractions of a second

Limitations:

- FlashPro + laptop MUST be connected to vehicle for changes to take effect
- Not all calibration tables support live tuning (depends on how ECU accesses them internally)
- Only a subset of live-tuning tables can be active simultaneously (RAM limit)
  - Non-VTC engines (e.g., S2000): all live-tuning tables can be used at once
  - **VTC engines (my K20Z3):** 2-3 "major" tables (fuel low cam, fuel high cam, ignition low cam) + many supporting tables (injector latency, etc.)
- **If ignition cycles off/on:** live tuning tables reset to "last calibration upload" values -- NOT to stock. FPM detects key-off-key-on if connected and re-uploads live changes.

Workflow:

1. Select tables to live-tune (via table window OR Online menu -> Select Live Tuning Tables)
2. Upload calibration to ECU
3. Edit/tune tables in real-time
4. When done: de-select live tuning, upload calibration again to bake in changes

Icons shown next to tables that support Live Tuning:

- Live Tuning not enabled
- Live Tuning enabled, pending upload
- Live Tuning active

Practical for my dyno sessions: this is how tuners iterate fast -- make a cell change, see it immediately on the dyno run, no re-flash. Saves hours.

6.19 Instant Live Tuning -- CAPTURED

What it is: Live Tuning without the initial upload step.

Difference from regular Live Tuning: you don't need to upload a calibration before starting -- edits apply directly.

Same limitations apply:

- Needs FlashPro + laptop connected
- Not all tables supported
- RAM limit on simultaneous tables
- Ignition cycle resets live values (but FPM re-uploads on key-off-key-on if connected)

Workflow:

1. Click the live-tuning icon next to a parameter/table, OR use Select Live Tuning Tables window
2. Edit -- takes effect immediately
3. When done: de-select live tuning, upload calibration to bake changes permanent

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6.20 Tuning Your Vehicle (Overall Process for VTC Engines) -- CAPTURED

Dyno tuning principle: Maintain consistency between runs. Start each run with same coolant temp + intake air temp. Maintain same time interval between runs.

Tuning a VTC engine (my K20Z3 IS a VTC engine -- this procedure applies directly):

The K20Z3 has 5 copies of each major table, one per cam angle. Standard sets:

- **Angles: 0, 10, 20, 30, 40** OR **0, 15, 30, 40, 50 degrees**

Step-by-step procedure (Hondata official):

1. **Start with low cam.** Force ECU into low cam by setting VTEC point HIGH. Only dyno to above expected VTEC point (6500 RPM typically enough). **Do NOT run over 7000 RPM on low-speed cam** -- lost motion assemblies can float, damaging spring retainers. Big warning for me.
2. **Lock cam to a single angle.** Find what angles my calibration uses. Starting at first cam angle (e.g., 0deg), set the low-speed cam angle table to that value.
3. **Tune fuel for this cam angle** until measured AFR is acceptable.
4. **Tune ignition for this cam angle.** Default ignition settings are usually close to optimum.
5. **Copy fuel and ignition tables** from the tuned cam angle to the next -- next cam angle starts tuned.
6. **Lock to next cam angle, repeat.** All 5 angles.
7. **Once all cam angles tuned for fuel + ignition**, tune VTC to generate composite cam angle table.
8. **Switch to high cam.** Set VTEC point LOW. Start each dyno run just past VTEC point, continue to redline. Repeat steps 2-7 for high-speed cam.
9. **Tune the VTEC point**, then the VTEC crossover VTC.
10. **(MAP calibrations only)** Tune part-throttle fuel.

Advanced tips from Hondata:

- **More productive to first tune the cam angle where engine makes most power** (usually 30deg). Tune 30deg fuel+ignition, copy to tables above AND below, then tune remaining tables. Faster than sequential.
- **VTEC smoothing:** Advance VTC cam angle just BEFORE VTEC point so the jump to high cam needs less VTC rotation. See 6.30 VTEC Crossover Tuning below.
- **For fuel tables, edit by selecting ROWS.** Ensures part-throttle fuel is adjusted based on full-load lambda

readings -- less part-throttle tuning needed later.

- **Forced induction:** start by tuning on minimum boost. Wastegate or bypass held open lets you tune atmospheric columns first, then boost columns. **Never disconnect intercooler pipes -- overspeed risk.**

Implication for my build: the Sport tune build procedure is exactly this 10-step process executed on a dyno. The tuner handles it. My job is to provide a clean engine, wideband data, and a healthy fuel system.

6.21 Fuel Tuning (MAF vs MAP Methods) -- CAPTURED

Two calculation methods:

Method	Based On	Has VE Tables?
Mass flow (AFM / MAF)	Air flow meter reading	No -- simple mass ratio
Speed/density (MAP)	Manifold pressure + volumetric efficiency tables	Yes

For tuning, a lambda meter is required.

Civic Si stock oxygen sensor:

- **Semi-wideband type** (not narrowband)
- **Range: 10.7:1 to 20:1** (that's a usable range)
- **Reads progressively richer as exhaust gas temperatures increase**
- **Useful for:** partial throttle and naturally aspirated tuning (my K20Z3 at my target power levels)
- **For forced induction:** aftermarket wideband recommended

For my NA build: I can partly rely on the stock o2 for baseline tuning, but **AEM X-Series wideband is still the right call** for dyno-session ground truth and E85 verification where richer mixtures matter. Also: AF.Corr (Section 6.35 below) is Hondata's software-corrected channel that partially fixes the rich-reading bias.

6.22 Tuning Mass Flow (AFM) Fuel -- CAPTURED

?? Hondata's own warning (matches my reliability-first rule):

> "A lean air/fuel condition will damage the engine. Make sure that you monitor the air/fuel ratio at all times, and abort any dyno run if the air/fuel ratio becomes too lean."

Mass flow tuning basics:

- AFM measures mass of air into engine
- Fuel = ratio of fuel to air (no VE tables)
- **Critical: AFM must read incoming air accurately** -- aftermarket intakes usually alter AFM readings
- AFM auto-compensates for headers, exhaust, cams, but NOT for intake standing waves / reversion (common with race headers + large cams)

Low load fuel (part throttle / closed loop):

- ECU uses AFM reading directly
- No VE adjustment
- If closed-loop fuel isn't at stoich, the AFM is reading wrong -- usually from aftermarket intake
- AFM can be re-calibrated (Tuning AFM Flow -- Section 6.23)

High load fuel (WOT / open loop):

- ECU uses additional enrichment table ON TOP OF AFM-based calculation
- Table name: **WOT compensation table**

? CRITICAL WARNING FROM HONDATA:

> "Values in the WOT compensation tables should only be changed within certain limits. The ECU will not switch into open loop correctly if the WOT compensation table values are raised so that less fuel would be required in open loop than closed loop. Generally values should not be increased over the stock values to prevent this from occurring. **As a guide, do not set the WOT air/fuel ratio above 12.50:1.**"

This directly matches the reliability-first rule I wrote into tuning-workflow-and-maps.md. Hondata's own official ceiling is 12.50:1 AFR at WOT. My doc says >=12.0 -- I can tighten to 12.2-12.5 and still be within Hondata's sanctioned range. Never lean past 12.50 per Hondata. Never.

6.23 Tuning AFM Flow (Calibration Procedure) -- CAPTURED

Prerequisites:

- Stock injectors
- Stock AFM flow values
- Stock WOT compensation values

Procedure:

1. Warm car to normal operating temp
2. Datalog 5-10 minutes of varied driving
3. In FPM -> XY Graph (from Advanced Graphs):
   - X-axis: AFM.v
   - Y-axis: S.TRIM
   - Check 'Show mean', check 'Closed loop', uncheck 'Open loop'
4. Result: graph of AFM reading vs fuel trim
5. Identify trends -- alter AFM voltage-to-flow calibration accordingly
6. Example: ECU trims 2-3% from 1.7V to 2.8V? Increase flow numbers on AFM Flow table by 2.5% in that range
7. **Only make 2-3 changes per iteration**
8. Save, upload, re-test until fuel trims are within 1%

Example: "AFM is reading approx 15% low, first tuning change would be to increase AFM flow numbers by 15%, across whole table, then re-test."

For my build: My K&N Typhoon intake definitely shifts AFM readings. If I'm on an AFM-based calibration, this is the process the tuner will run. If I'm on a MAP-based calibration (see 6.24 and 6.31), the AFM is irrelevant.

6.24 Tuning Speed/Density (MAP) Fuel -- CAPTURED

?? Same lean warning as mass flow.

Method:

- MAP sensor measures manifold pressure
- **Volumetric efficiency (VE) lookup tables** indexed by engine speed -> mass of air in cylinder
- Additional compensations: coolant temp, battery voltage, intake air temp
- Tune by editing VE tables

Fuel table structure:

- Typical indices: RPM (bottom axis) x Intake manifold pressure (side axis)
- 3D shape (because K20Z3 has 5 cam angle slices)
- Interpolation between cells

Fuel table used for BOTH:

- Part throttle / low load / closed loop -- tune to stoich (~14.7:1)
- Full throttle / high load / open loop -- tune to safe AFR for max power (<=12.50:1 per Hondata)

For my build: if my calibration is MAP-based (more common for aggressive tunes), VE tables are where the main tuning work happens. See 6.25 for part-throttle dyno-less procedure.

6.25 Part Throttle Fuel Tuning (MAP, No Dyno) -- CAPTURED

Scope: speed/density (MAP) calibrations only. AFM calibrations skip this.

Requirements: quiet, flat, straight road, can accelerate 10->30-40 mph repeatedly.

Preparation:

- Open calibration, display low-speed cam fuel tables (F5)
- Identify cam angles (0, 10, 20, 30, 40 OR 0, 15, 30, 40, 50)
- Lock cam to FIRST angle by setting low-speed cam angle table to that value
- Save calibration as DIFFERENT filename (will restore original cam angles later)
- No need to switch o2 off -- software reads fuel trims + actual lambda

Datalogging drive:

- Start datalog + recording (F9)
- Accelerate in 1st gear, LIGHT throttle to 5000 RPM -- keep throttle steady so load column stays consistent (columns 3-4)
- Shift to 2nd, slow to ~1000 RPM
- Repeat in 2nd gear with very light throttle
- Repeat 2nd gear several more times, slightly MORE throttle each time (higher columns)
- Repeats same column OK
- **RPM range:** first + last RPM rows have incorrect lambda (tip-in / release effects). Log from 1000 RPM to VTEC+1000. Typical: 1000-6000 RPM.

Lambda Overlay display:

- Load the datalog
- Select low-speed cam fuel table (F5)
- Select 'Fuel Change' (Shift-F3) -> shows suggested fuel change

Edit fuel tables:

- Aim: suggested fuel change -> close to zero

- Zero everywhere isn't practical (temp variations, etc.) -- get close
- **Hondata Heatshield intake manifold gasket helps** by keeping IM temps stable
- Before editing: verify correct cam angle selected, uncheck 'All' so changes don't apply to other tables
- **Look for PATTERNS of changes** rather than cell-by-cell -- table must stay smooth after
- Example: select whole column 6, add 5% fuel. Then column 7, add 5%. Then 2750-8500 RPM in columns 1-5, add 2%. Etc.
- After changes: view 2D to verify load lines remain smooth and equally spaced. If not, undo and apply fewer changes to larger areas.

Iterate:

- Upload, re-test
- Repeat for same cam angle until fuel changes near zero -- typically 3-4 cycles per cam angle
- Move to next cam angle, repeat

Finish:

- Load original calibration
- Copy cam angles out
- Load tuned calibration
- Paste cam angles back

For my build: this is a genuine DIY option for part-throttle tuning if I want to contribute to my own tune between dyno sessions. Requires a MAP calibration. Good practice for understanding the tables. But I'm not doing WOT tuning without a dyno -- that's where a tuner + wideband is non-negotiable.

6.26 Tuning Starting (Cranking) Fuel -- CAPTURED

The starting process:

1. Engine cranked -> large injector pulse (fuel sticks to cold intake walls)
2. Pulse calculated from: cranking fuel table x overall fuel trim x cranking fuel trim x air temp compensation x water temp compensation
3. Fixed ignition timing while cranking (typically 5deg or 7deg BTDC)
4. As engine cranks, fuel reduced as wall deposits reach equilibrium
5. When engine speed hits 600 RPM, ECU says "engine started"
6. Last closed-loop start-term fuel trim applied to main table fuel
7. **First 10-30 seconds:** injector pulse is BLEND of cranking fuel + main fuel tables, progressively shifting to main

Troubleshooting:

Symptom	Check
Hard start	Min battery V during cranking (should be ~10.5V, not below 9.2V)
	Cranking RPM (should be >150 cold, ~200 hot)
Hard cold start	Needs more cranking fuel
Hard hot start	Too much cranking fuel OR fuel vapor lock
Fires then idles rough for seconds	Needs more cranking fuel

Adjustment process:

- If unclear rich/lean, START by adding fuel
- Increase in 10% increments, up to 30%
- Datalog each start
- Look for: lowest cranking time + quickest fast-idle time
- If adding fuel makes it worse, try subtracting in -10% decrements to -30%

?? Critical testing rule: CANNOT start cold, shut off, adjust, and restart. Running engine even briefly changes:

- Intake valve temperature (fuel atomizes differently)
- Fuel deposits on intake walls

Must leave engine several hours, preferably overnight, between tests.

Two adjustment locations:

- **Cranking fuel trim** -- if engine is hard to start at all temps except hot
- **Cranking water temp compensation** -- if only hard to start at one temperature range

Formula if adjusting water temp compensation: new trim = old trim + ((100 + old trim) x (change% / 100))

Example: add 10% more cranking fuel at 50deg: change trim from -70.0% to -67.0%.

Cranking Wall Deposits (ADVANCED -- only for experienced tuners):

- Two table groups, each with SEQ and BATCH versions:

1. **Direct ratio** -- fraction of injected fuel reaching cylinder. Higher = less injector duration needed.
2. **Wall deposit** -- fraction of fuel sucked from wall into cylinder. Higher = less injector duration needed.

- ECU fires injectors in BATCH for ~2 revolutions during start, then switches to SEQUENTIAL
- Different wall-deposit rates for batch vs sequential -> two tables per parameter

For my E85 switch (future):

- E85 needs MORE cranking fuel (~30% more) than gasoline, especially cold
- The cranking fuel trim is what I'd adjust
- Below 40degF ambient on straight E85: expect rough cranking even well-tuned
- Winter-blend E85 (lower ethanol %) helps significantly

6.27 Tuning Ignition Tables -- CAPTURED

?? Hondata's explicit warning:

> "Excessive ignition advance will damage the engine. The combustion pressure and load on the engine (especially bearing stress) increase dramatically if the engine is over-advanced. **Do not believe the fallacy that 'more is better' for ignition advance.** Too little ignition advance can also damage the engine by increasing the exhaust gas temperature, especially with turbo-charged engines. **Do not rely on the knock sensor to retard the ignition timing if the engine detonates.**"

Ignition Advance at Full Load (WOT):

- Best method: dyno
- **For naturally aspirated engines (my case):** safe to set advance NEAR max power. Aim: run the LEAST amount of timing possible.

- **Procedure:** tune for max power, then RETARD timing until losing ~1 HP. Run just-shy-of-max timing.
- Monitor knock at all times (even for NA)
- Abort dyno run if pinging audible or ECU shows knock

Ignition Advance at Part Throttle:

- Harder to determine optimally
- **Default calibrations are suitable for part throttle** -- usually don't need touching
- EGT gauge can be used for fine-tuning

Idle ignition:

- ECU uses ignition to control idle speed
- **Normal to see -10deg to +15deg during idle** (actively varying)
- Not a static value -- dynamic stabilization

Honda Knock Control:

- Only offers LONG-term adjustment of estimated fuel octane
- Does NOT use dynamic ignition timing control (unlike newer vehicles)
- **Knock sensitivity reduced in starting calibrations** so knock control doesn't interfere with tuning
- Cannot rely on knock control to save the engine -- it's a slow learner

For my 170k K20Z3: this reinforces my reliability-first stance. Run less timing than max-safe as a buffer against the slow-learning Honda knock control. Don't chase the last degree.

6.28 VTC Tuning (Variable Timing Control) -- CAPTURED

What VTC is:

- "iVTEC" = VTEC + VTC combined
- VTC: variable intake camshaft advance (0-50 crank degrees, continuous -- NOT an on/off like VTEC)
- 5 copies of each major table per cam angle = effectively 3D tables
- Cam phase is positioned by electro-hydraulic mechanism with feedback

Mechanical limits:

- Range: 0-45deg or 0-50deg depending on calibration type
- **Physical stop limits cam advance** (prevents valve-to-valve and valve-to-piston contact)
- **?? Aftermarket cams: up to manufacturer to ensure safe positioning.** Software limits don't guarantee mechanical safety.
- Delay between ECU setting and actual cam rotation: **~0.1 seconds per 10 degrees**

Tuning guidelines -- by application:

Naturally aspirated (my build):

- Cam advance to MAX just after VTEC engagement until ~6500-7000 RPM
- From 7000 RPM to redline: retard back to ~25deg (cam near 50deg at 7000)
- True for all current aftermarket cams
- Unusual cams may need different settings

Supercharged:

- Max (50deg) throughout rev range

- Only 10deg or so retard above 7000 RPM

Turbocharged:

- Less cam than stock
- More back pressure = more cam retard needed
- Small turbos + stock cats: retard fully to 0deg at 8000 RPM
- Log-style manifold: high VTEC point, low VTC angles
- Long equal-length runners: lower VTEC point, higher VTC angles (closer to NA)

Procedure:

1. Set VTEC point high (6500-7000). Only dyno low-speed cam.
2. Tune fuel + ignition at each cam angle.
3. Dyno each cam angle -> 6 dyno curves.
4. Set composite cam angle map to angles giving max power at each RPM.
5. **Bracket the composite map:** +5deg re-dyno, -5deg re-dyno. Verify optimum.
6. Fine-bracket: ±2deg for 1-1.5 HP resolution on NA (small gains).
7. Set VTEC point low (3000), repeat above for high-speed cam.

Cam Angle at VTEC:

- If cam needs to rotate a LOT at VTEC (e.g., 20deg low-cam to 45deg high-cam), you lose 500-700 RPM after VTEC as cam physically rotates
- **Solution: start advancing intake cam in low-cam table BEFORE VTEC point.** Sacrifice a few HP pre-VTEC to gain HP post-VTEC. Done right, reduces the characteristic VTEC sound change.
- See 6.30 VTEC Crossover Tuning for the Hondata-developed technique

Part-throttle cam angle: Below full NA load (column 7 and less):

- Idle and below 1000 RPM: 0deg or 5deg
- Cruising (columns 2-7, 1500-4500 RPM): 15-30deg
- Above normal cruise RPM: same value as under full load (smooths gearshifts -- cam doesn't retract to zero during shifts)

Hints:

- Cam can't rotate instantly (0.1 sec per 10deg). Don't make large cam changes over small RPM intervals.
- **Intake cam is LOCKED at 0deg for 10 seconds after hot start.** Let engine run 10+ seconds before any dyno run.

VTC Cleaning:

- **Normal behavior:** ECU briefly fully advances cam on overrun to clean VTC mechanism
- Triggers: a few times from cold start, ~5 seconds after throttle release at 2000-3000 RPM
- Shows in datalog as small spikes to 45deg or 50deg
- **Not a failure -- expected behavior.** Don't report VTC cleaning spikes as VTC error.

6.29 VTEC Point Tuning -- CAPTURED

?? Critical Hondata warnings (hard limits):

> "Do not set the VTEC point too low as the engine will lose oil pressure and possibly damage the engine. **It is not recommended to set the VTEC point below 2000 rpm.**"

> "Do not set the VTEC point too high as the high speed cam rocker arm may float on the lost motion assembly, damaging the valve spring retainers. **It is not recommended to set the VTEC point over 6500 rpm.**"

This updates my earlier floor of 4500 RPM:

- **Hondata's absolute floor: 2000 RPM** (oil pressure limit)
- **Hondata's absolute ceiling: 6500 RPM** (rocker arm float limit)

For my reliability-first approach, staying in the middle of this range (say 4500-5800) is conservative. But Hondata's envelope is wider than I had documented.

Typical VTEC points:

- **Stock engine:** 5500-5700 RPM
- **Intake + exhaust:** 4500-4800 RPM
- **Supercharged:** 3000-3200 RPM
- **Aftermarket cams:** generally HIGH VTEC point

My build plan:

- Daily map: 5400-5600 (near stock)
- Sport map (FBO pump): 4800-5200
- Sport map (FBO E85): 4800-5000

All within Hondata's 2000-6500 envelope. None too aggressive.

Determining optimal VTEC point:

- Two dyno runs: one with VTEC low (3000), one with VTEC high (6500)
- Set VTEC point at intersection of low-cam and high-cam torque curves
- **Sudden power increase after VTEC switch** -> LOWER VTEC point
- **Sudden torque dip after VTEC** -> RAISE VTEC point
- Small dip at the intersection is normal

VTEC Window (RPM + Pressure gating):

- Stops VTEC from engaging at light/medium throttle (fuel economy + driveability)
- Uses BOTH RPM and manifold pressure

Setting VTEC window:

1. Determine optimum VTEC point (procedure above)
2. Lower RPM of window = VTEC point
3. Upper RPM of window = 6000-6500 RPM
4. Pressure at lower RPM = 80 kPa (NA) or 110-120 kPa (forced induction)
5. Pressure at upper RPM = 40 kPa
6. Smooth pressure transition

Forced induction:

- Lower pressure values: 120-150 kPa at lower RPM, decreasing to 70-80 kPa at upper RPM

Small dip at VTEC (<5 lbs torque) is normal. If bigger, try bumping VTEC point up in 100 RPM increments.

6.30 VTEC Crossover Tuning (Hondata's Technique) -- CAPTURED

What it is: The optimal VTEC transition method developed by Hondata for K-series engines. Not used in factory Honda maps. Perfected for 06-09 K20 Civics (my platform).

The problem this solves:

- Optimum low-cam angle at VTEC point ? optimum high-cam angle at VTEC point
- They're often 20-35deg apart
- Cam can only rotate 10deg/0.1 sec -> ~0.3 seconds to rotate 30deg at VTEC
- During rotation: engine makes sub-optimal power = "VTEC dip"

Hondata's three-part technique:

1. **Advance cam into high-cam VTC angle BEFORE VTEC switch**
2. **Alter RPM indices to make transition as quick as possible**
3. **Keep cam angle constant across high-cam RPM rows**

Example numbers from Hondata docs:

- VTEC point: 4350 RPM
- Low cam optimum at 4350: 12deg cam angle
- High cam optimum at 4350: 37deg cam angle
- Cam needs to rotate 25deg -> takes 0.25 seconds

Low-cam VTC table optimization:

- Above VTEC point (5000-7000), resolution not needed
- Right-click RPM row -> edit RPM indices
- Create 3 fine-resolution points around VTEC:
  - 150 RPM below VTEC point
  - 50 RPM below VTEC point
  - 50 RPM above VTEC point
- For 4350 VTEC: 4200, 4300, 4400 indices
- Reduce highest RPM breakpoint from 8100 to 7000
- Set cam angle at HIGHER RPM row to match high-cam angle at same RPM (37deg in example)
- Effect: low cam stays at optimum until 4200 RPM (150 below VTEC), rotates rapidly to 37deg over a tiny RPM/time interval

High-cam VTC table optimization:

- Delete VTC rows below 4000 RPM (engine should never be on high cam below this)
- Add resolution in 6500-10000 RPM range (where cam rotates rapidly)
- Keep cam angle ADVANCED across RPM rows
- Why: stops cam from retarding to zero during gearshifts -> keeps car "on power" longer after shift

Datalog verification:

- VTC transition from 26deg low cam to 50deg high cam in ~0.3 seconds
- CAM CMD = target from tables
- CAM = measured cam angle

For my build: this is the exact technique my tuner should use for the Sport map VTEC transition. Named drop this to my tuner if they don't already know it. Massive gains in VTEC-transition smoothness and area-under-the-curve.

6.31 AFM Removal (MAP Conversion) -- CAPTURED

Context: When using MAP-based calibration, AFM is no longer used for airflow measurement BUT the ECU still needs the air temp sensor in the AFM housing.

Options:

1. Leave AFM in place (keeps temp sensor, keeps restriction)
2. Replace AFM with a standalone IAT sensor (removes restriction, relocates sensor)

Compatible IAT sensors:

- **OBDI vehicles (1992-2001 Civic/Integra)** -- 2-bolt flange, good for FI
- **RSX IAT sensor** -- barbed press-in

Connector: from harness of same vehicle, need 2-wire plug (not single-wire)

Wiring:

- Splice into AFM connector wires (red/yellow and green/black, both at one end of connector)
- Polarity not important
- Insulate, reinstall wire shield
- **Only have IAT OR AFM plugged in, never both** -- ECU won't read air temp correctly if both present
- Cover AFM plug to prevent dirt/water if removed

Sensor placement:

- NA: anywhere in intake tube
- Turbo: AFTER intercooler (near throttle or in IM)
- Supercharged: AFTER supercharger

Verification: before starting, ignition on + datalog to verify sensor reads correct temp.

No calibration changes needed -- just ensure MAP-based cal.

For my build: only relevant if I ever delete my K&N Typhoon and go to a tube-only intake with the AFM eliminated. Not in current plan. Keeping K&N = AFM stays in circuit.

6.32 Table Indexes (Re-indexing) -- CAPTURED

- Tables can be re-indexed for more resolution in specific areas (e.g., around VTEC point per 6.30) OR to expand RPM range
- **Warning:** When index is altered, all tables sharing that index must update too
- For some calibrations, can't alter index if every instance of that index isn't identified
- Detailed procedure: see "changing table indexes" topic (not yet captured)

6.33 Knock Control Tables (Advanced) -- CAPTURED

?? Warning from Hondata: only for experienced tuners.

How Honda's knock control works:

- Designed for different-quality fuels in a STOCK engine
- Factory tunes engine dyno using automated process:

1. High-octane fuel -> find MBT (mean best torque) ignition timing
2. 100 RON fuel -> find knock ignition limit (max advance before knock likely)
3. 90 RON fuel -> find low-octane knock limit

Ignition calculation:

- Normal ignition tables contain MBT values
- Additional table: knock ignition limit values
- If knock limit < MBT: ECU uses lower value (no knock, less power)
- **Key insight:** ECU may run less timing than ignition tables show at certain cells, even with no sensor-detected knock. Doesn't show as "knock retard" -- it's considered a "no knock" condition.

Knock Control Variable (K.Control):

- Adjusted by ECU from knock sensor noise + knock sensitivity
- Nominal value: estimated fuel octane as % between 100 and 90 RON
- 0% = fuel is 100+ octane
- 25% = fuel is 97.5 octane
- 50% = fuel is 95 octane
- Updated SLOWLY while driving

If K.Control > 0% AND max timing < MBT:

- Difference shows as knock retard value

Ignition formula:

```
Ignition advance = min(MBT, MBT + knock_ignition_limit - (knock_retard x knock_control))
```

Knock Tables:

Table	What it contains
Knock Ignition Limit	Max advance before knock likely. Relative to MBT ignition table. Positive = higher than MBT. Negative = lower than MBT.
Knock Air Limit	Cylinder filling limit (mg) based on knock retard. Can be increased without affecting K.Control. Makes boost more consistent.
Knock Sensitivity	Noise threshold before ECU calls it knock. Low value = more sensitive.
Knock Retard	Amount of retard (degrees) between high-octane and low-octane limits. Actual retard = table x knock control (e.g., K.Control 50% = half the retard). Only applied if knock limit < MBT.

Tuning recommendations (Hondata official):

- **Main ignition tables:** decrease timing at RPM/load points where knock ignition limit is negative. Helps factory knock control work better.
- **Knock ignition limit:** use LOW values where knock is likely. For forced induction on pump fuel, columns 9-10 should be 5deg or less, boost columns zero. ALWAYS adjust main ignition if increasing knock ignition limit values.
- **Knock sensitivity:** INCREASE sensitivity (reduce values) in columns 4+. For noisy cams/headers, reduce sensitivity (increase values).
- **Knock retard:** use SMALLER retard than stock (4-10deg). Lower columns (6 and below) will barely affect knock retard (knock ignition limit much higher than MBT there).
- **Knock control:** starts high after flash, comes down slowly as engine runs normal cycles. To decrease on

dyno: steady-RPM hold, as long as no pinging, K.Control gradually drops.

For my build: these knock tables are deep-tuner territory. I won't touch directly. What I MUST ensure: my tuner understands that **the factory knock control is slow** and won't save the engine from a marginal tune. The ignition tables themselves need to be safe, with the knock system as a secondary safety net.

6.34 Individual Cylinder Ignition -- CAPTURED

When you'd use it: unequal cylinder filling (e.g., intake manifold geometry gives more filling to cylinders 2-3 than 1-4 on a Jackson Racing supercharged setup).

Result of unequal filling: some cylinders make more cylinder pressure -> more knock tendency. Adjusting per-cylinder ignition trim evens the knock count across cylinders.

Procedure (from Hondata docs):

1. Baseline test with no cylinder trim, full-load datalog
2. Observe knock count per cylinder
3. Advance timing 3deg on all cylinders as a stress test (ONLY for illustration -- not a recommendation)
4. Observe which cylinders knock -> identify which cylinders are at risk
5. Adjust per-cylinder trim to equalize knock counts
6. Reduce all cylinders by 1deg final for reliability buffer

Alternative method: use EGT per cylinder (accepted normal method).

For my NA build: unlikely to need this unless I develop a specific per-cylinder issue. Skunk2 Ultra Street manifold + Skunk2 Alpha header are well-matched and shouldn't have major imbalance. Worth knowing exists.

6.35 Lambda Correction (AF.Corr Channel) -- CAPTURED (IMPORTANT)

The problem: The Civic Si stock oxygen sensor is semi-wideband, BUT it does NOT read accurately richer than lambda 0.88 (13:1 AF).

- Reads over 1 AFR point different with rich mixtures
- Error is CONSTANT (no exhaust gas temp or time effect)

Hondata's solution:

- Measured the difference between stock o2 and a true wideband on a dyno
- Created a correction table
- New datalog channel: **AF.Corr** (Lambda Corrected)
- Corrected channel reads very close to aftermarket wideband
- **Mixtures leaner than lambda 0.95 (14:1) are NOT corrected** (to preserve closed-loop behavior)

Disclaimer from Hondata: results may vary. They recommend dyno tuning (especially for forced induction). But corrected AF will always be LEANER than stock o2 reading -- safer to use corrected than raw.

For my build:

- Use AF.Corr in datalogs instead of raw stock o2
- AEM X-Series wideband remains the ground-truth for WOT tuning
- Stock o2 + AF.Corr is adequate for part-throttle closed-loop verification

---

6.36 Rev Hang Reduction -- CAPTURED

What rev hang is: Engine speed doesn't fall immediately when shifting (manual trans only). Some vehicles have it, some don't.

Two settings to adjust:

1. Overrun throttle:

- Parameter -> Throttle -> **Overrun throttle opening enabled** (uncheck)
- Shuts throttle plate quicker on pedal release
- **Note for 2016+ 1.5T Civic:** enabling main cruise control button reverts to stock behavior

2. Overrun Fuel Cut Delay:

- Parameter -> Fuel -> **Overrun fuel cut delay values**
- Table indexed by gear
- **Suggested values: 20-50 ms**
- Shuts injectors off quicker when throttle closes

Caveat: if vehicle isn't well-tuned, excessive closed-loop fuel trims give similar symptoms to rev hang. Fix tune first.

For my build: I've planned this in my existing advanced-features.md. Daily map = mild reduction (still clean emissions), Sport map = aggressive (sharper response). Tuner dials in during Tune #1 or #2.

6.37 Cruise Control Lambda Display (Hidden Feature) -- CAPTURED

What it is: Use cruise control Set button to show current lambda on the tachometer.

Display format: AFR x 1000 offset by 10000.

- 14.7:1 AFR = 4700 RPM shown on tach
- 13.2:1 AFR = 3200 RPM shown on tach

Conditions:

- Cruise control MAIN switch OFF
- Clutch NOT depressed

Supported vehicles:

- **All Civic Si** ? (mine qualifies)
- FK8
- 2023 Integra

For my build: free backup gauge for lambda monitoring when the LattePanda isn't booted or if I ever want an OEM-dash lambda readout without extra hardware. Worth trying after first flash just to confirm it works on my PRB. Zero cost feature.

6.38 Active Tuning -- CAPTURED (POTENTIALLY GAME-CHANGING)

What it is: ECU adjusts tables in real-time AS YOU DRIVE. **No laptop or FlashPro plugged in required** after enabling.

Tunable tables via active tuning:

- Fuel tables
- Air flow meter (AFM) tables

How it works: ECU uses target lambda tables + wideband o2 reading to alter values in a fuel-table adjustment table. Continuous, repetitive. Over time, fuel tables become tuned to engine needs.

Example from Hondata:

- Baseline fuel trim: some spread
- After 5 minutes of active tuning: reduced spread
- After 15 minutes: further reduced

For my build: this is massive. If I can enable active tuning, drive a lot, and let the ECU self-tune... I may not need 3 separate dyno sessions. BUT:

- Active tuning requires a WIDEBAND wired in (not available with stock sensor)
- Still need baseline calibrations matched to hardware
- Not a substitute for a professional dyno tune on final hardware
- But: excellent for between-hardware-changes refinement

Relationship to reliability-first: active tuning targets LAMBDA (AFR). It doesn't advance timing past dyno-set values. Safe for the engine. Good tool in my box.

6.39 Active Fuel Tuning Parameters -- CAPTURED (DETAILED CONFIG)

Enable: Active fuel tuning -> check the Enabled option.

Options (when tuning operates and what target):

Option	Purpose
Closed loop	Active in closed loop
Open loop	Active in open loop
Use fuel trims	In closed loop, uses current trims to better determine underlying fuel table values
Target Lambda	Which lambda to target. In closed loop: use ECU's calculated target (AFMCD). In open loop: can use target lambda table for finer control.

Settings (how tuning works):

Setting	Description
Delay after engine start	O2 warmup + after-start-fueling settle time
Delay after injector re-start	After overrun/TPS-out/cam-out events, o2 needs time
Change per tuning cycle	Adjustment magnitude. Default 2%. Larger = faster but rougher.
Delay in lambda reading	Compensates for lag between engine lambda and sensor reading. 150-250ms typical. Measure by noting lag after injector re-start.
Update frequency	Active fuel update delay. 25-200ms typical.
Maximum change	Max adjustment from original fuel table
Active fuel smoothing	Limits max delta between adjacent cells. Prevents jagged adjustment tables.

Maximum deltas (cam, load, RPM)	How far actual values can be from table indices before tuning applies. If cell too far from actual, no update.

Conditions (when active): minimum and maximum for various parameters. Tuning only between all min/max pairs.

Workflow:

1. Enable in calibration, upload to ECU
2. Drive for some time period
3. Optional: merge active table into main fuel table via menu operation

Notes:

- Target lambda tables ? WOT compensation tables (different purpose)
- Live tuning for active tuning table must be enabled (default on)
- Main fuel tables do NOT need live tuning
- **Active tuning resets if battery disconnected**

6.40 Active AFM Tuning -- CAPTURED

Similar to Active Fuel Tuning but for the AFM (mass flow meter) table.

Same options, similar settings: Closed loop / Open loop / Use fuel trims / Target Lambda / Delays / Change per cycle / Maximum change / Conditions.

Key difference: tunes AFM flow table -> fuel trims should reduce over time (ECU learns airflow better).

Update Active Tuning Table button: downloads AFT table from ECU into current calibration. **Must save calibration after uploading + keep it open or reopen before downloading AFT.**

Workflow:

1. Enable in calibration, upload to ECU
2. Save calibration after upload
3. Drive for some time
4. Optional: update active tuning table button -> merges changes to main tables

For my build: if I'm on an AFM calibration, this auto-calibrates my AFM to my K&N Typhoon's actual airflow. Could run this for a few weeks of daily driving after my next flash and get very accurate AFM baseline for free.

6.41 Multi-tune -- CAPTURED (NOT FOR MY PLATFORM)

Multi-tune = in-ECU switchable tune slots via cruise control buttons.

Supported vehicles:

- 2017-2021 Civic Si
- 2022+ Civic Si
- FK8 2018-2021 Civic Type R
- FL5 2023+ Civic Type R
- DE5 2024+ Integra Type S

?? My 2009 Civic Si (PRB K20Z3) is NOT supported. Multi-tune is a newer-platform feature.

How it works on supported cars:

- Cruise control MAIN off for several seconds
- Hold cruise control RESUME button
- Tachometer shows selected tune: 1000 RPM = #1, 5000 RPM = #5
- Change via cruise SET and RESUME buttons
- Exit: press CANCEL or wait several seconds

Persistence: tune number retained on key-off, NOT on battery disconnect or ECU reflash.

Tables that can vary by tune: traction control target slip, launch slip, boost by gear.

For my build: doesn't apply. My Daily/Sport switching uses the **Dual Calibration** feature (Section 6.14) -- primary/secondary on the FlashPro unit, switch via button combo on the unit.

Key distinction:

- **Multi-tune:** newer platforms, 5 slots, in-ECU, cruise control button switching, NO re-flash needed
- **Dual Calibration:** all FlashPro platforms including mine, 2 slots on FlashPro unit, button-combo switching, 90-second re-flash each switch

Dual Cal is still great for my build. Multi-tune would have been even better but isn't available.

6.42 Datalogging Overview -- CAPTURED

Two methods:

1. **To laptop via USB** (real-time)
2. **On-board datalogging** to internal memory (20 hours+)

File format: .fpdl (extension confirmed -- associated with FlashProManager by default).

If association broken: Settings dialog -> check "Associate calibration and datalog files with FlashProManager".

About Datalogs:

- FPM uses **frames** -- snapshot of all sensor values at one point in time
- All sensors datalogged per frame
- Some sensors not datalogged if engine not running
- Number of frames + total length shown below menu bar

Autosave: datalogs can auto-save when recording finishes. See datalog autosave settings (not yet captured).

6.43 Laptop Datalogging -- CAPTURED

Basic datalogging (not recording):

1. FlashPro -> vehicle OBDII
2. Laptop -> FlashPro via USB
3. Ignition on / engine start
4. Press **F10**, or Datalog menu -> Datalog, or click Datalog icon

FPM displays sensor values, error codes, table tracing in real time. **Values NOT saved** during basic datalogging

-- just monitored.

Recording (actually saving):

- Press **F9**, or Datalog menu -> Record, or click Record icon
- Saves to disk

My standard workflow: always record. F10 then F9.

6.44 On-board Datalogging -- CAPTURED

Process:

1. FlashPro -> vehicle OBDII
2. Ignition on / engine start
3. Press **Datalog button on the FlashPro unit**
4. Datalog light indicates recording

Downloading to FPM:

1. FlashPro -> computer (USB)
2. Open FlashPro window (Window menu or toolbar)
3. Select datalog -> right-click or click download button
4. **Recommended: save to computer after download**

Datalogging Rates:

Mode	Max recording time
Fast	~7.5 hours
Slow	~21 hours

For my build: the on-board feature is clutch for track days or long drives where I don't want the laptop in the car. Rotate: enable "Fast" mode before a spirited drive or dyno session. Use "Slow" for long commute monitoring.

6.45 Sensor Setup -- CAPTURED

Configure sensor display in the Sensors window:

Field	Purpose
Abbreviation	Name in Sensors window
Display Name	Name in printouts
Description	Function explainer
Type	Sensor data type
Unit	Display unit
Display Min / Max	Graph range, default in display window
Warning Min / Max	Sensor values turn green-to-red when outside range

Import / Export: sensor configurations can be saved and loaded across computers.

For my build: I'd set warning ranges for critical parameters -- e.g., knock count warning at anything > 0. ECT warning at > 215degF. IAT warning at > 150degF. Injector duty warning at > 85%. Easy visual alerts during drives.

6.46 Graphing -- CAPTURED

Open Graph window: click Graph button in toolbar (after loading a datalog or selecting from Datalog menu).

Graph window:

- Plots sensors via graph templates
- Clicking in the Graph Window updates: Sensor Window, Display Window, Error Codes Window

See: Graph Window and Graph Templates topics for deeper reference (not captured yet).

6.47 Security (FlashPro Anti-Theft) -- CAPTURED

Two security levels:

1. **Owner Information password protection**
   - Prevents owner info/registration from being altered without the password
2. **Calibration upload password**
   - Option to require password for any upload to FlashPro
   - Prevents FlashPro from being used without password
   - Also prevents FlashPro from being unlocked from current vehicle

Vehicle security (built in):

- Immobilizer CANNOT be disabled via FlashPro (prevents theft facilitation)
- FlashPro can only be locked to a vehicle that doesn't have a FlashPro already locked to it

If stolen: contact Hondata with FlashPro serial number.

For my build: I'll set a security password once the FlashPro is in the car (Phase 7 LattePanda era) so a thief can't use my FlashPro for their purposes.

6.48 Owner Information & Registration -- CAPTURED

Required for: product registration with Hondata, warranty, update notices, recall notifications.

Path: FlashPro window -> Registered Owner tab -> enter owner details.

If security password set: owner info only modifiable with password.

Critical: email address must be correct (used to retrieve lost security password).

Privacy (Hondata's stated policy):

- Only registration-relevant info transmitted
- Encrypted before transmission
- Info NOT retained by Hondata after registration

Update registration: possible if contact details change.

6.49 Security Password -- CAPTURED

Set password path: Online -> Set Security Password.

Requirements: FlashPro connected to computer (USB). No vehicle connection needed.

Entry states:

- **Entering first password:** Leave "Current Password" empty. Enter new password twice.
- **Changing password:** Enter current password, then new password twice.
- **Removing password:** Enter current password, leave new password blank. Unlocking FlashPro from vehicle also clears.

Upload security option:

- Check "Require security password when uploading" = password requested at every upload
- **Warning:** this option effectively prevents FlashPro from being used on a different vehicle (can't unlock without password)

If forgotten: resettable via registered email or phone.

6.50 Reset Password -- CAPTURED

Process (needs internet):

1. FlashPro window -> Registered Owner tab -> Reset Security Password
2. Choose email OR phone (SMS) for reset code
3. Enter 5-digit reset code
4. Leave new password blank (removes password) OR set new password

---

6.51 Purchasing a Used FlashPro -- CAPTURED

Before buying used, verify:

1. Install FlashProManager
2. Plug in FlashPro
3. Open FlashPro window, select FlashPro tab
4. Click License and Security to expand tree
5. Verify:
   - **VIN is unlocked**
   - **Security Password NOT set**

If not unlocked / password set: current owner MUST unlock / reset before you can use it.

For my build: not relevant since I bought new, but important reference if I ever buy a second unit.

6.52 Hondata Vault -- CAPTURED

What it is: cloud storage for calibrations + datalogs, accessible directly from Hondata software.

Features:

- Transfer cals between computers

- Backup if computer lost/damaged
- Public repository (add cals for other Hondata users)
- Search by keywords or tuning parameters

For my build: I should set this up for redundant backup of stock and any tuned calibrations. Better than relying solely on OneDrive.

6.53 Searching Vault -- CAPTURED

Access: Window menu.

Two search types:

- **Text Search:** any word in any calibration field (e.g., "Stock" returns cals with stock injectors, stock intake, etc.)
- **Filtered Search:** matches all criteria (e.g., Engine="B18C5" returns only B18C5 cals)

Results: list, double-click to download. Sort by clicking column heading (arrow shows current sort).

6.54 Keyboard Shortcuts (Complete List) -- CAPTURED

Table Editing:

Shortcut	Action
Shift + arrow keys	Select table cells
Ctrl + A	Select whole table
Ctrl + C	Copy selection
Ctrl + V	Paste selection
Ctrl + I	Increase selection (by user-defined step)
Ctrl + D	Decrease selection
Ctrl + J	Open Adjust window
Ctrl + P	Interpolate selection
Ctrl + M	Smooth selection
Ctrl + Z	Undo
Shift + Ctrl + Z	Redo

Graph Viewing (cursor movement):

Shortcut	Movement
Shift + Left / Right arrow	±10 seconds
Left / Right arrow	±1 second
Ctrl + Left / Right arrow	±0.1 second
Shift + Ctrl + Left / Right arrow	±0.01 second

From ini file (function keys):

- F3 = low-cam ignition, F4 = high-cam ignition
- F5 = low-cam fuel, F6 = high-cam fuel
- F7 = low-cam VTC, F8 = high-cam VTC
- F9 = Record (start datalog recording)
- F10 = Datalog (live monitoring)

6.55 Settings Window -- CAPTURED

Tabs available:

- General Settings
- Unit Settings
- Display Settings
- Lambda Overlay Settings
- Sensor Overlay Settings
- Datalog Settings
- Autotune Settings

Import/Export: buttons allow transferring settings between computers.

General Settings

- **Windows Themes:** XP+ theming on/off. Windows screen scaling enabled = clearer on high-DPI.
- **File Association:** check to associate .fpcal and .fpdl with FPM. Double-click-to-open.
- **Compress datalog files:** reduces .fpdl size.
- **Automatic check for new version:** checks on first run per day. Update from Help menu if available.
- **Include beta software versions:** I have this ENABLED in my ini. Toggle OFF for max stability.
- **Miscellaneous:** Display message after upload / Show Mod Window / Play Sound (success 2-tone, failure chime) / Send notification / Disable screen saver (GOOD -- enabled in my ini) / Calibration load use last vehicle / use FlashPro vehicle
- **USB Driver Check:** balloon hint from status bar if driver problem.

Unit Settings

- Vacuum & Pressure units (absolute vs atmospheric reference)
- Temperature units
- Lambda Unit (lambda vs AFR)

My ini shows: TempUnit=0 (Fahrenheit likely), LambdaUnit=1 (lambda).

Display Settings

- **Main Table Tracing:** shows which cells are active in real-time during datalog. 4 cells shaded by interpolation proportion if enabled. Axis trace indicator shows ECU reading values.
- **Autosize table grid:** expands cells to fit window (good on laptops).
- **Parameter Table Tracing:** same for parameter-window lookup tables.
- **Changed Value Highlight:** table window shows edited values in color (default: green=decreased, red=increased). Reset on save/load.
- **Sensor List:** "Show sensor warnings" changes display window color when sensor outside limits.

Lambda Overlay Settings

Overlay of datalogged lambda values on fuel tables.

- **Target Lambda:** used for fuel change calculation (not an ECU table -- changing doesn't affect ECU).
- **Overlay Lambda Input:** choose wideband / corrected stock o2 / uncorrected stock o2. Recommendation: uncorrected stock for closed loop, corrected or wideband for open loop.
- **Options:**
  - In closed loop apply fuel trims (more accurate fuel adjustment)
  - Calculate for each cam angle (VTC vehicles only)
  - Transition delay (time after throttle hits minimum before lambda shown -- o2 needs time to recover from overrun)
  - Maximum interpolation (distance limit between table cell and actual engine state -- filters out "too far" values)
  - Min/max throttle boundaries
  - Min/max gear
  - Min/max cam angle
  - Min/max MAP
  - Minimum samples per cell

Sensor Overlay Settings

Show ANY sensor on main tables (not just lambda). Overlay conditions filter unwanted values. Transition delay for sensor settle.

Datalog Settings

Datalog AutoSave: auto-save when recording stops or ignition off. Configurable directory, filename prefix, suffix.

Autotune Settings

Invoked: toolbar button or Ctrl+T.

- **Minimum/Maximum Lambda:** boundaries for safe tuning (prevents adjustments during injector cutoff etc.)
- **Maximum change from base map:** % cap to prevent dramatic changes (base set on save/load)
- **Make adjustments every:** time between changes (lambda settle)
- **Autotune strategies:**

1. **Closest cell:** changes one cell at a time (heaviest weighted)
2. **4 weighted cells:** changes 4 cells based on interpolation proportion
3. **Precise cell:** like #1 but only if actual MAP within 15% of load index

- **Play sounds when adjustments made:** high pitch = increase fuel, low pitch = decrease

My ini defaults:

- AutotuneMinLambda: 0.75 (~11.0 AFR floor -- aggressive)
- AutotuneMaxLambda: 1.22 (17.9 AFR ceiling)
- AutotuneMaxChange: 10%
- AutotuneFreq: 1 (second)

For my reliability-first rule, I'd TIGHTEN AutotuneMinLambda to 0.80 (12.0 AFR minimum) before enabling.

6.56 Main Window & Status Bar -- CAPTURED

Status bar indicators (bottom of app):

Indicator	Meaning
Driver	FlashPro USB driver installed (ticked)
FlashPro	FlashPro plugged into USB (ticked)
OBDII	FlashPro plugged into vehicle OBDII (ticked)
ECU	Ignition on + FlashPro can talk to ECU (ticked)
Locked	Shows FlashPro lock status to vehicle
Live	Live tuning enabled (ticked)
Datalog	Shows when datalogging from ECU
Record	Shows when recording from ECU
Modified	Current calibration has been edited

These are useful health indicators before attempting any operation.

6.57 Options Menu -- CAPTURED

- **Show Advanced Parameters:** removed in V1.9.2
- **Show Columns:** NA / Boost / All pressure columns in table window
- **Tables Follow VTEC:** live-switches table shown between low/high cam based on current VTEC state (datalog or live)
- **Tables Follow Cam Angle:** live-switches to nearest cam angle table (for VTC engines like mine)
- Settings / Sensors / Graph Templates / Keyboard Shortcuts / Language

For my K20Z3: enable both "Tables Follow VTEC" and "Tables Follow Cam Angle" so I always see the table that's currently being used.

6.58 Table Window -- CAPTURED (IMPORTANT -- primary tuning interface)

Main tables: ignition, fuel, cam angle, + supporting (cranking fuel, etc.).

Low vs High Speed Cam: tables often have separate copies for low/high cam. Window title shows which.

Major Table toolbar selection: 6 buttons:

- Low-speed ignition
- High-speed ignition
- Low-speed fuel
- High-speed fuel
- Low-speed cam angle
- High-speed cam angle

Cam Angle Selection (VTC engines like mine):

- Row of degree buttons below toolbar (for the 5 cam angle copies)
- Individual angle OR "All"
- When "All" selected:

- Percentage changes: same percentage applied to all cam angle tables
- Relative changes: same relative change to all
- Absolute changes: same value copied to all

Cell selection:

- Mouse click-drag
- Arrow keys + Space bar
- Ctrl+A = select all
- Click topmost row = whole column
- Click leftmost column = whole row
- 2D graph: left-click-drag to select range; Shift+click = row; Ctrl+click = column

Editing cells:

- Single cell: type new value
- Multiple: select + Ctrl+J Adjust window
- Ctrl+I / Ctrl+D = +/- selection by default step
- 2D graph: drag cell up/down with mouse

Quick Adjust buttons (toolbar):

- Adjustment value (absolute or percentage)
- Add button = add value to selection
- Subtract = subtract value
- Set = set all selected to value
- Fuel tables: adjustment value is percentage for add/subtract; other tables: relative change value

Special Operations:

- **Interpolate:** interpolates between highest and lowest values in selection
- **Smooth:** smooths inconsistent values

Right-click menu:

- Increase/Decrease/Adjust/Interpolate/Smooth
- Revert to last save (resets to save point)
- Import from Calibration (pull table from another .fpcal)

Live Tuning: enabled per-table via icon next to table name.

Graph View (right side):

- 2D or 3D representation
- 2D: click rectangles, drag up/down to alter values
- 3D: click selects area on grid

6.59 Changing Table Indexes -- CAPTURED

How: right-click on load/RPM value in Table Window.

Operations:

- **Edit Index:** change current index value (must be between neighbors -- indexes must ascend)
- **Insert Index:** adds new row/column; shifts other values; last index removed

- **Delete Index:** removes current; shifts others up/left; new arbitrary index added at end

Notes:

- Recommended (not critical) to set indexes before tuning
- Index values must ascend from origin; don't need to start at 0
- When index changes, ALL tables using that index must update (some may be shared)
- Fuel / ignition / knock typically share the same indexes
- Cam angle tables use their own indexes
- Load indexes shared between low and high cam tables of same type
- RPM indexes NOT shared between low and high cam
- Underlying table values auto-compensated: edit/insert uses interpolation; delete shifts cells toward deleted row/column

For VTEC crossover tuning (Section 6.30): re-indexing around the VTEC point is the critical technique. Right-click -> Edit Index -> set 4200, 4300, 4400 for a 4350 RPM VTEC point.

6.60 Proportional Tracing -- CAPTURED

Shows which table cells are being used in real-time via color gradient -- darkest on most-weighted cell (ECU interpolates between 4 cells normally). Helps see exactly what the ECU is pulling from during a datalog playback.

Axis trace indicator: small mark on table axis shows current ECU reading. Centered on axis cell when datalog value equals axis index.

6.61 Changed Value Highlight -- CAPTURED

Table cells show color to flag changes from last save/load point. Default: decreased = green, increased = red. Colors configurable in General Settings. Makes it easy to review what was edited before saving.

6.62 Sensor Overlay -- CAPTURED

Show lambda (AFR) or any other sensor value overlaid on main tables during datalog playback.

For lambda overlay, tooltip shows: RPM, load, averaged lambda, number of samples used. (Samples not integer because values applied proportionally to neighboring cells.)

6.63 Calibration Window -- CAPTURED (the parameter editor)

Edits calibration parameters (non-table settings).

Calibration Parameter Groups (all listed):

- Calibration parameters (revision, notes, gear ratios, permissions)
- Fuel parameters
- Ignition parameters
- VTEC parameters
- VTC parameters

- Closed loop parameters
- Knock control parameters
- Rev limit parameters
- Idle parameters
- Sensor parameters
- Misc parameters
- Ethanol Parameters (FLEX FUEL LIVES HERE)
- Traction Control Parameters

Parameter default/changed indicators:

- No dot = unchanged from factory default
- Red dot = edited this session
- Yellow dot = changed from default in PRIOR session

Table right-click in calibration window:

- Increase/Decrease/Adjust/Revert/Reset to default/Import from Calibration

---

6.64 Calibration Parameters (revision, notes, gear ratios, permissions) -- CAPTURED

Revision: save a version tag with calibration.

Notes: free-text notes in calibration.

Gear Ratios: for cruise control if altered from stock. Speed sensor works from 3rd-gear countershaft. Altering final drive / 3rd gear / tooth count changes speed reading. Set gear ratios + speed adjustment.

Calibration Permissions: a SECOND password separate from FlashPro security. Prompts on "Change Permissions". No recovery if lost.

- Do not show tables & parameters (hide but still upload)
- Do not show non-tuning modifications
- Do not show tuning modifications
- Do not allow upload to Vault
- Only allow upload to FlashPro serial # (restricts to specific unit)
- Password can be saved on computer where cal was created/edited

For my build: I don't need to set calibration permissions unless I'm distributing tunes. If my tuner ever sends me a protected cal, this is why.

6.65 Fuel Parameters (DETAILED) -- CAPTURED

Fuel tables (MAP-based systems): VE tables. AFM systems don't use these.

WOT lambda adjustment tables: used when ECU determines engine at WOT. Basic injection assumed stoich; WOT table adds enrichment.

? REPEATED HONDATA WARNING:

> "Important note: Do not use values above 12.50 (AF) or 0.85 (lambda), otherwise the ECU will always run closed loop."

This is the upper lean limit AND a functional limit -- above 12.50 AFR / 0.85 lambda, the ECU won't switch to open loop properly. Double reason to respect 12.50.

Injector Size: changing auto-compensates main fuel, cranking fuel, closed loop. Flow spec is at 3 bar / 43.5 psi nominal.

Fuel Pressure: compensates main fuel tables, cranking tables, closed loop for pressure differences (stock vs current, injector test pressures). Different injector brands = different test pressures. Changing these auto-adjusts tables.

Fuel Trim: compensation for main fuel (MAP), AFM, cranking. Injector-size change auto-adjusts these; normally don't touch.

Injector Opening Time (dead time): injector manufacturer supplies figures.

Overrun Fuel Cut Delay: delay between overrun conditions met and injectors off. Long = rev hang. Short = jerky at light throttle. 6MT but table may have 5 columns -- 5th gear value used for 5th AND 6th.

Fuel Table Type: AFM calibrations can optionally use MAP (VE) lookup for fuel. AFM still needs to be present but flow unused. Can also use AFM at low MAP and MAP at higher pressure -- "Minimum pressure for speed/density" crossover setting.

AFM Fuel Table: converts AFM air mass per cylinder to injector duration.

Individual cylinder fuel trim: separate fuel per injector. Only for setups with individual per-cylinder lambda probes.

Air temp compensation (IAT): three tables based on air flow rate (low/med/high). Positive = more fuel. Plus cranking IAT table.

Coolant temp compensation (ECT): two tables (below 40 kPa MAP / above). Plus cranking ECT table. Positive = more fuel.

Fuel Economy:

- Injector flow for fuel economy = reference value ECU uses
- Injector flow for dash = reference value dash uses
- Not supported on race calibrations

6.66 Fuel Pump Parameters (HPFP -- Direct Injection) -- CAPTURED

For direct injection vehicles only -- limits air charge/boost based on fuel pump duty to prevent running out of volume.

Not applicable to my K20Z3 (port injection, no HPFP). Noted for reference.

- Fuel pump target duty = where protection engages
- Fuel pump hysteresis = where protection removes
- Pump volume per duty cycle is NON-LINEAR (sine curve, not triangular) -- 90->100% duty = only ~2.5% more fuel

---

6.67 Ignition Parameters (DETAILED) -- CAPTURED

Ignition tables: base ignition timing for low/high cam. Compensations applied on top -- final timing may differ from base table.

EGR Ignition Tables: RPM x EGR%. Affects partial throttle 12-25% EGR. At WOT (0-3% EGR), first two columns barely matter. For WOT ignition changes: use Knock Ignition Limit tables.

Individual cylinder ignition trim: per-cylinder timing. Only adjust with EGT gauge for guidance.

Air temp (IAT) compensation: two tables (low load / medium-high load, crossover ~40 kPa MAP). Positive = retard. Negative = advance.

IAT retard application: two options (only ONE should be checked):

- "Ignition IAT retard applied to ignition knock limit" = only reduces timing when knock limit close to MBT (default)
- "Ignition IAT retard applied to ignition advance" = applies to commanded advance always

Coolant temp compensation (cold): retard when below operating temp. Two tables (low/high load).

Coolant temp compensation (hot): retard when overheating. Two tables.

Ignition parameters:

- Cold-start retard for catalytic converter warmup -- can disable for race cars without cat

Throttle tip-in retard:

- By RPM compensation
- By throttle compensation
- By gear compensation

Gear compensation: per-gear advance/retard. Advance = positive? No -- **positive = advance, negative = retard** in THIS table. Apply requires "Ignition IAT retard applied to ignition advance" option checked.

Cranking ignition: base ignition when cranking, indexed by ECT.

6.68 VTEC Parameters (DETAILED) -- CAPTURED

VTEC window: variable switch point based on RPM + load.

VTEC RPM window: minimum and maximum RPM for activation.

- **Twin cam engines (mine): don't go below 2500 RPM** (revised from the "2000" stated elsewhere -- 2500 is the K-series specific floor)
- Single cam: 1800 RPM floor
- R18/R20: 1000 RPM floor
- Upper: **don't exceed 6500 RPM** (rocker float / valve damage)

VTEC pressure window: minimum MAP for activation vs RPM.

VTEC conditions:

- **VTEC minimum coolant temp:** don't engage below this ECT
- **VTEC engage/disengage minimum speed:** VSS floor. Disengage < engage always.

6.69 VTC Parameters (DETAILED) -- CAPTURED

Cam angle tables: target cam angle for low and high cam. **Target should be ZERO around idle area.**

Deceleration VTC Angle (important for shift feel):

- Default: cam returns to 0deg on throttle release
- 2012 Civic Si exception: returns to 5deg above 4000 RPM
- **Problem:** cam at 0deg during shift = no power dip + slow recovery to normal 20-30deg after next gear engages
- **Solution options:**
  - **Zero VTC upon decel fuel cut** (default): cam -> 0deg on decel. Uncheck to use:
  - **VTC decel fuel cut angle** (if "Use VTC tables" not available): set intermediate angle, typically 20-30deg
  - **Use VTC tables upon decel fuel cut** (preferred if available): uses normal VTC tables during decel

Recommended settings:

- Disable Zero VTC
- Enable Use VTC tables
- Zero the idle area
- Set columns 1-2 to intermediate angle (15-35deg, typically 20-25deg)
- Result: VTC doesn't have to advance from zero when shifting

VTC Closed Loop Control: PID parameters adjustable for cam positioning. Won't fix mechanical problems.

For my build: my tuner should enable "Use VTC tables upon decel fuel cut" for smoother shifts. Keep idle area zero (don't set VTC at idle).

6.70 Closed Loop Parameters -- CAPTURED (CRITICAL FOR WIDEBAND)

Primary o2: ECU uses for closed loop control. Recommend keeping ENABLED during tuning to prevent switching-back-to-open-loop tuning differences.

Secondary o2: checks cat performance, alters closed loop target for cat optimization. Enable unless race car with no cat. (I have Berk HFC -- enable.)

WOT determination type: MAP or TPS. **MAP preferred for most vehicles, required for forced induction.** For my NA: MAP-based WOT determination.

MAP WOT determination table: manifold pressure threshold where ECU switches from closed to open loop.

TPS WOT determination: two tables (high = closed->open, low = open->closed -- hysteresis to prevent oscillation).

LAF voltage to lambda: conversion table from stock wideband o2 to lambda. **This is where the Civic Si semi-wideband calibration lives.**

Wideband lambda interface settings: cross-reference Section 6.8. Also see Section 6.72.

Closed Loop Target Lambda:

- Combination of tables by temp and load
- Actual target = richest (lowest) of: Closed Loop Target Lambda + Target Lambda Limit + Target Lambda Load (transition ~50 kPa MAP)
- **?? Hondata warning:** lean than lambda 1 (~14.6 AFR) increases EGT. Not recommended for vehicles with

catalysts.

Fuel Trim Min/Max:

- Controls STFT and LTFT percentage range ECU compensates
- AF learned value min/max = range for stored STFT used when switching from open loop to closed loop (calculated over 20 min average)

6.71 Knock Control Parameters -- CAPTURED

Knock sensor: can be disabled if not present. **Normally do NOT disable during tuning** -- if disabled, ECU uses default advance tables -> considerable ignition retard.

Light control: flashes check engine light when knock occurs. Useful while tuning.

6.72 Rev Limit Parameters (DETAILED) -- CAPTURED

Activation vs recovery RPMs: recovery should be 100-300 RPM lower than activation. Larger diff = slower, rougher limiter.

Rev limiter type: ECU uses FUEL CUT (not ignition cut) to protect cylinder walls from wash-down by unburnt fuel. Wet nitrous + fuel cut rev limit = set nitrous to shut off BEFORE rev limiter activates or engine goes lean.

Overall limiter: max engine speed under any conditions. Note: lower limiter may operate when cold or on low cam. **Does not protect against mis-shift overrev** (engine mechanically coupled to wheels).

Overrev RPM: where ECU generates fault condition. Set ~200 RPM higher than overall limiter.

Launch limiter: active when vehicle not moving (drivetrain protection). Fastest when cut RPM = recover RPM. Some vehicles have adjustable launch control (see 6.73).

Speed limiter: activates at indicated VSS. To disable: set higher than top speed.

Boost Cut: rev limit if MAP rises above threshold. Only active above 2000 RPM (so engine can start). With stock MAP (max ~1.78 bar / 11.5 psi), don't set boost cut above 1.75 bar (11 psi) to ensure voltage trigger works.

Full Throttle Shift (flat-shift):

- Detected when clutch depressed + throttle wide open
- ECU cuts fuel to drop RPM for next gear
- **RPM limit:** not critical, 2000-3000 below shift RPM suggested
- **Minimum throttle:** prevents shift limit during part-throttle / slow shifts

Pops and Bangs: FK8 and FL5/DE5 specific (not my platform).

6.73 Launch Control (Adjustable) -- CAPTURED

Concept: adjust launch RPM via cruise control WITHOUT reflashing ECU.

Initial value: from calibration on first flash.

After altering: retained until battery disconnect or ECU reflash.

Adjustment procedure:

- Clutch depressed + vehicle stationary
- Cruise control Resume = increase launch RPM (100 RPM per press)
- Cruise control Set = decrease launch RPM
- Cancel button = show current launch RPM on tachometer

Additional fuel/ignition: activates when launch limiter active + engine speed above threshold. Used to maintain turbo pressure on FI.

Parameters (launch-adjustable):

- Launch RPM disengage speed = max VSS for fuel/ignition changes
- Launch min RPM for ignition retard and fuel enrichment = where changes begin
- Launch RPM additional fuel = extra fuel (fuel units)
- Launch RPM ignition retard = retard amount (positive = retard)

?? Warning: anti-lag style settings build HEAT in exhaust. Don't use with cat. Only for seconds at a time.

For my build: launch control is NA-relevant for consistency off the line. My Sport map launch RPM = 4500. Can adjust in-car via cruise control afterward without flashing.

6.74 Hondata Mode (Sport/Eco Button) -- NOT MY PLATFORM

2017 Civic Si has Sport button -> stock power in non-sport, full power in sport. 2016+ Civic Turbo / 2018+ Accord has Eco button -> similar inverted.

Not available on my 2009 Civic Si. My Daily/Sport switching uses Dual Calibration button combo instead.

6.75 Sensor Parameters -- CAPTURED

AFM flow table: AFM voltage to air flow. Normally tuned via flow bench OR by observing fuel trims vs air mass flows.

MAP Sensor:

- Stock reads ~1.7 bar absolute (~10 psi boost)
- **Hondata 4-bar MAP sensor** reads to ~40 psi boost
- **Switching MAP sensors does NOT require table retuning**

For my NA build: no MAP upgrade needed.

Speed Adjustment: ECU receives signal from transmission 3rd-gear countershaft sensor. Calculates internal speed + dash speed. Both adjustable. Positive = increase, negative = decrease.

6.76 Idle Parameters -- CAPTURED

Idle speed: two tables indexed by ECT:

- **Idle speed (after start):** higher RPM for 20-60 sec after start
- **Idle speed (normal):** post-warmup target

Larger injectors trick: higher idle RPM improves quality. Set tables so no value < desired idle.

Idle Torque Reserve (Bosch ECUs -- newer platforms):

- Not my platform (PRB is Keihin)
- Torque reserve % at idle via ignition retard
- Too low = ECU raises RPM. Too high = over-retarded ignition. Rule: idle ignition should average ~TDC (0deg).

6.77 Throttle Parameters -- CAPTURED

?? Hondata warning: advanced tables. Incorrect settings can cause throttle non-response or sudden acceleration.

Throttle Index Table: calculates index to main throttle table (some vehicles). Row = VSS, col = throttle pedal.

Target Throttle Plate: maps pedal to plate. Row = RPM, col = pedal (or index from above). Plate opened less at low RPM for smoothness. First column values restricted to ensure close on release.

Throttle parameters:

- **Throttle dampening:** stock ECU slows throttle opening in 1st gear to prevent jerk. Can disable.

Overrun throttle: plate held slightly open on overrun. Described in Section 6.36.

Minimum throttle opening compensation: table adjusts overrun opening by ECT. All values -100% = disable overrun opening entirely.

Throttle Flow: air flow past throttle for given opening. Only alter for larger-than-stock throttle bodies. **My Phase 3 will involve bored 66mm TB -- this table may need a tuner adjustment.**

Throttle Tip-In Fuel: extra fuel on quick throttle opening. Tables for ECT compensation + throttle compensation (higher = more fuel).

Injector Overrun Cutoff:

- Restart RPM determines engine speed where injectors resume operation
- Two tables (cut / recover) for hysteresis
- Also MAP-based: tables for low/high cam with hysteresis

6.78 Boost Control Parameters -- NOT MY PLATFORM

NA engine = no boost control. Detailed docs captured for reference but not applicable to K20Z3.

Summary of boost control docs (for future if I ever go FI):

- 3-way solenoid (normally open or normally closed)
- Connection to ECU pin B2 (Civic Si) or B16 (S2000)
- **Civic Si uses pin B2** -- duty cycle controlled current source
- 10A fuse, ground, no relay, no suppression diode
- Duty cycle 15-85% typical effective range
- Frequency 10-60 Hz typical
- Activation pressure ~slightly over atmospheric
- Control methods: fixed duty / duty by RPM & gear
- Air temperature compensation (cooler = more duty)
- Quick Spool: full duty until near target boost

- Stock MAP sensor limit: ~11 psi boost

6.79 Turbocharger Parameters -- NOT MY PLATFORM

For 2016+ DI turbos. Scramble button (Res+ for FK8, Sport for 10th gen Civic Si). Boost by gear table. Not applicable to my K20Z3.

6.80 Torque Parameters -- LIKELY NOT APPLICABLE TO MY PRB

Torque limiting tables for transmission protection (CVT mostly). My 6MT doesn't need this.

Bosch ECU Torque Tables: 2016+ DI platforms. Nominal Charge / Reference Torque / Minimum Efficiency. Critical NOT to alter without paired external calculator -- throttle control disabled on mismatch. Again, not my platform.

6.81 Misc Parameters -- CAPTURED

- **Immobilizer:** CANNOT be disabled (security)
- **VSA:** enable/disable vehicle stability. **Enable for all VSA-supporting vehicles.**
- **EPS:** idle-up when steering wheel turned. Not on/off for EPS unit itself.
- **Cruise Control:** enable/disable
- **Starter relay:** push-button start vehicles only
- **? Radiator Fan:**
  - **On temp / Off temp settings** -- on should be ~3degC/5degF higher than off
  - **Can use second coolant temp sensor (radiator bottom) OR cylinder head coolant sensor** (checkbox)
  - **Stock thermostat:** opens at 85degC/185degF, fully open at 90degC/195degF. Don't set fan below thermostat fully-open temp -- fan runs constantly.
- **Air Pump [Euro-S2000]:** disable for boost control.

For my build: critical that this is where my fan tables live. My Phase 1 cooling refresh planned Koyorad radiator -- I should set fan ON ~190-195degF, OFF ~185degF (still above thermostat's 185degF fully-open).

6.82 Ethanol Parameters -- CAPTURED (? THE CRITICAL FLEX FUEL ANSWER)

? RESOLVED: THE FLEX FUEL WIRING ANSWER

Per official Hondata docs, flex fuel / ethanol control on FlashPro uses:

Input options:

- **ECT2 (second coolant temp sensor pin)** -- available on some vehicles including K-series ECUs
- **CANFlex** -- requires Hondata CANFlex module, uses CAN bus (newer vehicles only -- NOT my PRB)

For my K20Z3 PRB ECU: ECT2 is the wiring path.

Critical setup:

1. Wire the ethanol control unit's 0-5V signal output to **ECT2 pin**
2. **Disable ECT2 in misc parameters** (Section 6.81 -- uncheck it so ECU doesn't try to read it as coolant temp)

3. Configure **Ethanol percentage table** -- voltage to percentage mapping

?? Important note on ECT2: the ECU ECT2 input has an internal pull-up resistor to 5V. Some ethanol control units can't pull the voltage down enough for accurate low-ethanol reading.

Hondata's recommended per-sensor settings:

Sensor	0% voltage	100% voltage	Limitation
Zeitronix ECA	0.34V	4.97V	Will not read below 15% ethanol
Innovate ECF-1, MTX-D, ECB-1	0.00V	5.00V	Will not read below 1.5% ethanol

Standard fuel context (US):

- Pump gas: 0-10% ethanol (typically 10%)
- E85: 55-80% depending on location/season

Ethanol-adjusted tuning tables:

- **Ethanol fuel compensation:** delivers more fuel based on ethanol percentage
- **Ethanol ignition multiplier:** multiplier applied to ignition compensation table
- **Ethanol ignition compensation table:** final ignition adjustment = value x multiplier, added to commanded ignition
- **Ethanol WOT fuel compensation:** Hondata testing found DI vehicles respond BETTER to LESS fuel at WOT with high ethanol (11:1 stock goes to 11.8:1 at 85% ethanol for more power / less knock)
- **Boost by ethanol limit table:** more boost with more ethanol (but needs more fuel -- may increase or decrease depending on fuel budget)
- **Ethanol advanced mode:** disables simple 2D tables, enables 3D fuel and boost tables for precise tuning

For my build -- THIS UPDATES EVERYTHING:

Previous plan (wrong): wire flex fuel sensor to SO2 (rear O2) pin. **Correct plan (per Hondata):** wire flex fuel sensor to **ECT2 pin**, disable ECT2 in misc params.

This also means my **wideband wiring** can use SO2 (rear O2) -- wait, no, the Wideband topic (Section 6.8) said SO2 is NOT recommended. So:

Final wiring allocation for my K20Z3:

- **Wideband (AEM X-Series 30-0300): EGRL or ELD** (NOT SO2, NOT ECT2)
- **Flex fuel sensor (Continental 13577429): ECT2 pin** (with ECT2 disabled in misc)

Parts I'll need (updated from 19-Flex-Fuel-and-Fuel-System/overview.md):

- The sensor itself (Continental 13577429 -- verify output is compatible; standard GM sensor outputs frequency, not 0-5V voltage)

?? HUGE DETAIL: The Continental/GM 13577429 sensor outputs FREQUENCY, not 0-5V voltage. This is a problem:

- Hondata's ECT2 input expects 0-5V voltage signal
- The GM sensor outputs 50-150 Hz frequency signal
- **Solution: need a frequency-to-voltage converter OR use a dedicated ethanol content controller** (Zeitronix ECA, Innovate ECF-1, MTX-D, ECB-1)

The controllers Hondata recommends (Zeitronix / Innovate) are NOT just the sensor -- they INCLUDE a

controller that converts GM sensor frequency to 0-5V voltage for the ECU.

Updated BOM component for flex fuel:

- Continental 13577429 sensor (frequency output)
- Ethanol content controller (converts frequency to 0-5V): Zeitronix ECA, Innovate ECF-1, Innovate MTX-D, or Innovate ECB-1
- Choose Innovate over Zeitronix (reads down to 1.5% ethanol vs 15% -- better low-ethanol accuracy, matters for pump-gas-with-some-ethanol mode)

This is a significant update to the flex fuel build. The existing doc says Continental sensor is "$50-80" which is true for the sensor, but the TOTAL system needs an ethanol controller too (Innovate ECF-1 runs ~$150-200).

6.83 Methanol Injection -- CAPTURED (NOT PLANNED -- reference only)

Uses ECT2 pin like flex fuel. AEM 30-3300 methanol controller + Hondata wiring adapter. Voltage switches (0.65V inactive, 3.1V active) to trigger fuel/ignition changes for methanol injection. **Either flex fuel OR methanol -- not both** (both use ECT2).

Not in my plan.

6.84 Traction Control -- LIMITED ON MY PLATFORM

Hardware TC: requires Hondata Traction Control module + wheel speed sensor interface.

Software TC: uses factory VSA wheel speed data. Available on some vehicles. Ignition retard only. Works with VSA enabled or disabled.

For my 2006-2011 Civic Si: unclear if software TC is available for PRB -- would need to check in FPM with cal loaded. Hardware TC via Hondata module is option. Neither in my current plan.

6.85 Graph Window -- CAPTURED (datalog analysis)

Graph display: sensor values from live recording or past datalog.

Graph templates: pre-defined configurations of sub-graphs and sensors.

Menu options (right-click on graph):

- Zoom In / Out / Full
- Next/Previous Template
- Define Templates
- Add/Edit/Delete Comment

Keyboard navigation:

- Home / End = start/end of datalog
- Page Up/Down = viewable width left/right
- Arrow keys (±1 sec), + Shift (±10 sec), + Ctrl (±0.1 sec), + Shift+Ctrl (±0.01 sec)

Playback controls: in Graph window + Datalog menu.

Datalog -> Calibration matching: datalog can link to matching calibration IF cal was opened in FPM on same laptop AND uploaded before datalog made.

6.86 Datalog Comments -- CAPTURED

Right-click datalog at position -> Add Comment. Great for annotating specific events during a drive ("hit VTEC here", "knock at WOT 4th gear"). Comments move by editing time.

6.87 Advanced Graphs (XY, Histogram) -- CAPTURED

XY Graph: any two sensors, X vs Y.

- **Critical for AFM tuning** (AFM.v vs S.TRIM, per Section 6.23)
- Options: show mean (blue line), show SD (green line), show X for data points
- Filters: closed loop, open loop, WOT (throttle >80%), OV (overrun filter)

Histogram: bar chart of sensor frequency.

- Options: percentage vs count, cumulative
- Same filters as XY

6.88 Graph Templates -- CAPTURED

- 1-4 sub-graphs per window
- Each sub-graph: 1-4 sensors
- Import/Export buttons for sharing templates between computers

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6.89 Sensor Window & Display Window -- CAPTURED

Sensor Window: list of sensors, live-updated. Right-click for options (properties, right-justify values, group sensors).

Display Window: large font for dyno monitoring. Customizable controls:

- **Value control:** numeric sensor value
- **Text control:** fixed text
- **Image control:** bitmap only
- **Gauge control:** analog-style gauge
- **Clock control:** date/time
- **Bar control:** horizontal or vertical bar readout
- **Indicator control:** boolean or threshold warning (e.g., VTEC on/off, over-rev warning)
- **Button control:** start/stop datalog/record buttons (window must be LOCKED for clicks to respond)

Layouts saved/loaded via Display menu. Images embedded.

6.90 Error Codes Window -- CAPTURED

Displays DTCs from live datalog or past log. **Rule: fix any DTC before tuning.** Consult vehicle service manual for

diagnostic procedures.

6.91 FlashPro Window (Tabs) -- CAPTURED

Tabs:

- **FlashPro:** FlashPro + ECU info. Shows EPA/CARB restriction type.
- **Calibrations:** lists cals ON the FlashPro. Upload/download/remove. **Cal on FlashPro ? what's on ECU** -- presence doesn't indicate current ECU state.
- **Datalogging:** on-board datalogs stored on FlashPro unit
- **Registered Owner:** user details, Hondata registration
- **Jailbreak:** FK2/FK8 only (not mine)
- **History:** serial numbers of FlashPros connected to this computer. For loss/theft record. Local only, can be cleared.
- **Bluetooth:** FlashPro 2 or post-Dec-2013 FlashProOG. My unit MAY have Bluetooth depending on its manufacture date.

6.92 EPA/CARB Restrictions -- CAPTURED

Two FlashPro types:

- **Race FlashPro:** any calibration can be uploaded. Legal ONLY for racing vehicles never used on public highway within USA.
- **CARB FlashPro:** only calibrations that don't affect emissions allowed. EO number D-742-1. Legal on USA street vehicles.

For my Civic Si:

- If my FlashPro is a CARB unit (blue), I can only flash CARB-approved calibrations for USA street use
- If Race (grey) or Dealer (black), any calibration
- **Check:** FlashPro window -> FlashPro tab shows type

Key notes:

- EPA administers Clean Air Act federally
- No direct EPA approval mechanism -- they accept CARB approval
- So California-legal = federally legal
- Even no-emissions-test states must comply with Clean Air Act -- "49 state legal" concept no longer valid

6.93 OBDII Diagnostics Window -- CAPTURED

DTCs: four-digit codes starting with P (powertrain). Fix before tuning.

Readiness Codes: OBDII test completion markers.

- **Continuously monitored:** Misfire / Fuel system / Comprehensive component
- **Non-continuous (trip-monitored):** Catalyst / Heated catalyst / EVAP / Secondary air / A/C refrigerant / O2 sensor / O2 heater / EGR

Reset events that clear readiness: reprogram ECU / clear DTCs / remove ECU / disconnect battery.

Driving cycles to set readiness (rough guide):

- **Catalyst:** ~5 miles stop/go + steady cruise. May take multiple cycles if cat marginal.
- **EVAP:** cold start + stop/go warmup + 5-10 min highway steady throttle, no key-off.
- **Oxygen sensor:** warm engine + ~3.5 miles including 5-sec closed-throttle decel.
- **O2 heater:** 10-20 sec after start.
- **EGR:** warm engine + ~1 mile.

Cold start defined: ECT <120degF, IAT + ECT within 7deg of each other, min 6 hours off.

Emissions Check: simulates smog test for pre-emissions-inspection check.

Diagnostics tab: DTCs from all CAN bus modules. Older vehicles: only ECU on CAN; K-line modules not shown. Refresh searches modules (shift+click for exhaustive search ~1 min). Clear DTCs button.

For my build: after any calibration flash, I should drive the readiness cycles before any smog test. Budget ~2 tanks of gas on varied driving.

6.94 Shortcuts Window -- CAPTURED

Custom shortcuts: any action can be assigned any combination of 1-4 key strokes. Change / Remove / Reset to defaults.

6.95 Dealer FlashPro Window -- NOT MY UNIT

For Hondata authorized dealers managing customer vehicles. License counts, purchase licenses online, reflash history per-vehicle. Not applicable to end-user.

6.96 G Force Window -- CAPTURED

Shows G-sensor data from VSA-equipped vehicles where VSA data is datalogged. Range, trail, style options via right-click.

For my 2006-2011 Civic Si: VSA available, but G-force datalogging availability requires testing.

6.97 Check For Updates / Check Internet Connection -- CAPTURED

Help menu -> Check For Updates = polls Hondata server for new FPM version (needs internet). Help -> Check Internet Connection = verify Hondata communication.

6.98 Update Boot Firmware -- CAPTURED (Rare Fix)

For FlashPros with boot firmware v7 or earlier (pre-FPM 2.6.3). Some USB3 ports had connection issues. If firmware v8+, this option is hidden.

?? If boot firmware update interrupted: FlashPro BRICKED and needs return to Hondata.

Recommendation: only run if normal firmware update process is giving problems. Takes a few seconds when it

works.

For my build: FPM ini shows FlashPro=461004 (a recent firmware). Unlikely I need boot firmware update. Leave alone.

6.99 Compare Calibration -- CAPTURED (useful diff tool)

Menu: File -> Compare Calibration.

Compares current calibration to any other .fpcal. Detects changes. **Useful for:**

- Understanding what a tuner changed
- Diffing Daily vs Sport maps
- Verifying what an "official" cal has vs my working one

6.100 Sensors (Complete K20Z3 Compatibility) -- CAPTURED

For my Civic Si '06 (the row in the Sensors topic):

Supported for datalogging:

Sensor	Description
? RPM	Engine speed
? VSS	Vehicle speed sensor
? GEAR	Calculated gear from VSS x RPM
? MAP	Manifold absolute pressure
? TPS	Throttle position sensor
? TPlate	Throttle plate (DBW -- %)
? AFM Volt	Air flow meter voltage (not AFM Flow mass units)
? INJ	Injector duration (ms) -- cylinder #1
? IAT	Intake air temp
? ECT	Engine coolant temp
? VTC Actual	Actual cam phase (degrees)
? VTC Cmd	Commanded cam phase
? Lambda	Lambda from stock o2
? Target Lambda	ECU's target lambda
? Short Term Trim (STFT)	Closed-loop fuel trim
? Long Term Trim (LTFT)	Long-term fuel trim bias
? Closed Loop Status	Open/closed loop flag
? O2B Volt	Rear o2 sensor voltage
? Knock Level	Knock %, from sensor
? Knock Retard	Retard applied in degrees
? Knock Control	K.Control value (estimated octane %)

? PA	Ambient pressure
? Batt	Battery voltage
? VTS	VTEC state (boolean)
? Wide Band Lambda	FPM analog input 1 or 2 (when wired)
? Wide Band Voltage	Raw voltage from wideband
? Knock Count 1-4	Per-cylinder knock events
? Knock Limit	Ignition advance limit due to knock
? Boost Control Duty	Not useful on NA

NOT supported for datalogging on Civic Si '06:

- ? ING (no such sensor on my PRB)
- ? AFM Flow (mass units) -- only voltage available
- ? ECT2 (NOT listed as datalog channel -- BUT it's what flex fuel USES... means ethanol % datalog is ???)
- ? IAT2
- ? Lambda 2 / Target Lambda 2 / Short Term Trim 2 / etc. (not my platform)
- ? INJ 2 (no second injector reading)
- ? BP / BPCMD / Waste Gate (NA vehicle)
- ? EXVTC (exhaust VTC -- I have intake VTC only)
- ? Fuel Pressure
- ? Air Charge
- **? Ethanol Content** -- **this is concerning for flex fuel**
- ? Fuel Duty
- ? FTP (fuel tank pressure)
- ? Purge Duty
- ? TC Volts / TC Retard / IAB (traction control)

?? Flex fuel datalogging question: Sensors table doesn't show **Ethanol Content** as a datalog channel for my '06 Si. BUT Hondata Ethanol Parameters topic describes ethanol control FOR my platform using ECT2. Possibly:

- Ethanol content may show up as a NEW datalog channel when ethanol control is enabled in the calibration
- OR the ethanol value is read from ECT2 volts and datalogged as the ECT2 voltage (since regular ECT2 is disabled)
- OR the sensors matrix is outdated/incomplete

Action item: once I have flex fuel installed, verify in FPM what datalog channels appear. If ethanol content isn't directly visible, derive from ECT2 voltage (linear 0.34-4.97V -> 0-100% or whatever my specific controller outputs).

6.101 Sensor Definitions (Per-Sensor) -- CAPTURED

Summary of per-sensor topic descriptions:

Sensor	Meaning
RPM	Engine speed from crank sensor. Tachometer usually reads 0-5% high; ECU is accurate.

VSS	Calculated from transmission sensor. Driven-wheels speed -- doesn't compensate wheel slip.
Gear	Calculated by ECU from VSS x RPM.
MAP	Manifold absolute pressure after throttle plate.
BP / BPCMD	Boost pressure (turbo) / commanded boost. Not my vehicle.
AIRC	Air charge % (cylinders filled with air). >100% = under boost. Not my vehicle at my power level.
WG / WGCMD	Wastegate position (mm) / commanded. Not my vehicle.
TPedal	Throttle pedal position (%).
TPlate	Throttle plate position (%).
AFM.v	Air flow meter voltage.
AFM	AFM air mass flow (converted from voltage).
INJ	Injector duration (ms) for injector #1.
Duty	Injector duty cycle (%).
DIFP / DIFPCMD	Direct injection fuel pressure (DI only). Not my port injection.
IGN	Ignition timing (degrees BTDC).
IAT	Intake air temp (by AFM for NA, by intake manifold for turbo).
IAT2	Second IAT for turbo vehicles.
ECT	Coolant temp in cylinder head.
CAM	Intake cam position (crank degrees from base). LOCKED at 0deg for 10 sec after startup.
CAMCMD	Intake cam target. Actual lags 0.1-0.5 sec (closed-loop mechanism).
EXCAM / EXCAMCMD	Exhaust cam position/command (dual VTC vehicles). Mine is intake VTC only.
AF	Air/fuel ratio from stock lambda sensor.
S.TRIM	Short-term fuel trim (%). Normal range -10% to +10%. Outside = need part-throttle fuel retuning. Often high during shifts (overrun).
L.TRIM	Long-term fuel trim. Normal -5% to +5%.
TRIM	Combined STFT + LTFT.
Fuel Status	2 = closed loop, 1 = open (warming up), 4 = open (driving), 8 = open (error).
K.Level	Knock level (%).

Critical for my analysis:

- STFT out of ±10% -> **tuning issue** at part throttle
- LTFT out of ±5% -> **underlying consistent issue** (vacuum leak, injector drift, AFM calibration)
- Fuel Status = 8 -> ERROR, investigate
- CAM lags CAMCMD sustained > 0.5 sec -> VTC actuator failing (matches my existing guidance)

Additional sensor definitions (from later help topics):

- **K.Retard:** knock retard (degrees) actually applied to the ignition.
- **K.Control:** knock control value (%) -- estimated fuel octane as a percentage between 100 RON and 90 RON. 0% = 100+ octane, 25% = 97.5, 50% = 95.
- **Ign.Limit:** knock ignition limit (degrees). Max advance before knock likely.

- **K.Count:** knock count per cylinder. Incremented every time ECU detects a knock event. NOT affected by knock sensitivity table adjustments. **Common false-knock causes: supercharger noise, exhaust headers.** Worth noting for my Skunk2 Alpha header -- early cold-start false-knock is possible; tune sensitivity up if needed.
- **PA:** atmospheric air pressure, measured internally by ECU.
- **BAT:** battery voltage at the ECU.
- **VTS:** VTEC spool valve state -- shows when VTEC has engaged the high-speed cam (boolean).
- **Eco:** calculated fuel economy -- depends on injector size being set correctly in Fuel parameters.
- **Fuel Used:** per-trip total fuel used by the engine (ECU-calculated).
- **Frame:** datalog frame count.
- **BC Duty:** boost control duty cycle (%). Not applicable to NA.

This closes the loop on all the individual sensor definitions for datalog analysis.

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7. ECU Connectors & Pins -- CAPTURED (CRITICAL FOR WIRING)

Convention: ECU pins referenced by connector letter + pin number, e.g., C27 = pin 27 of connector C.

For My 2006-2011 Civic Si (PRB) -- The Pin Allocation Table

Pin	Stock Function	Repurpose For	Notes
A23	ELD (Electrical Load Detector)	Wideband input	ELD must be disabled in misc + factory pin removed from ECU connector
A33	ECT2 (Second coolant temp sensor)	Wideband OR Flex Fuel input	ECT2 must be disabled in misc + factory pin removed
B2	EGR output	Boost Control output	Normally unused -- open for future FI
B29	EGRL (EGR lift input)	Wideband input	Normally unused
C27	SO2 (Secondary/rear O2 sensor input)	--	NOT recommended for wideband (ECU voltage clipping). Must be disabled + pin removed if ever used

Connector Color Coding (Civic)

- **A Connector -- White**
- **B Connector -- Grey**
- **C Connector -- Green**

Physical Pin Details

- Each connector has pins numbered 1-40 organized into **5 rows**
- Top and bottom rows use **large pins**
- Middle three rows use **small pins**
- Pin numbers visible on wire side of connector once plastic cover removed

Inserting / Removing Pins

To access pins: slide the white plastic surround upward slightly to unlock pins.

Inserting pins:

- Crimp pin onto wire with a crimp tool (Molex tool with 1.6mm crimp recommended)
- Verify correct cavity before inserting (very hard to remove once in)
- Push pin in with open side of crimp UPWARD until it clicks
- Move plastic surround back to locked position

Removing pins:

- Very difficult without the correct tool
- Fine screwdriver can work as a makeshift removal tool
- Insert from ECU side, lift sprung plastic catch while pulling wire from behind
- **Don't use force on the wire** -- pin digs into plastic catch and becomes much harder to remove

Cross-Reference to Other Platforms (for context)

Platform	ELD pin	ECT2 pin	EGRL pin	SO2 pin
2006-2011 Civic Si (mine)	A23	A33	B29	C27
2012 Civic Si / ILX	A34	A44	--	--
2012-2015 R18 AT/MT	A34	A44	--	--
2014-2015 R18 CVT	A34	A46	--	--
S2000	E15	E1	B12	--
2007-2008 TSX	E15	E1	B12	--
2009-2014 TSX	A24	--	--	--

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8. DDE Server -- FPM ? Excel Integration -- CAPTURED

What it is: FlashProManager can function as a **DDE (Dynamic Data Exchange) server** -- a Windows inter-process data transfer tech. Enables Excel and other apps to pull live datalog values from FPM for further analysis.

Data I Can Pull from FPM via DDE

Server name: FlashProManager

Calibration data:

- =FlashProManager|Calibration!Name -- returns loaded calibration name

Datalog data:

- =FlashProManager|Datalog!FrameCount -- live-updates while datalogging

Sensor data:

- =FlashProManager|Sensors!Count -- number of sensors
- =FlashProManager|Sensors!Names -- array of names (Ctrl-Shift-Enter for array display in Excel)
- =FlashProManager|Sensors!Values -- array of live values
- =FlashProManager|Sensors!Units -- array of units
- =FlashProManager|Sensors!'Knock Retard' -- individual sensor value (single quotes for names with spaces)

Commands I Can Send to FPM via DDE

Calibration commands:

- Open <filename> -- load a calibration
- Notes <notes> -- change calibration notes (new line via $r)
- Lock <serial> -- lock cal to FlashPro serial number
- Save <filename> -- save current cal
- Close -- close current cal

Modification commands:

- Apply <filename> -- apply a modification to current cal (filename relative to FPM modifications directory)

Example Use Cases

- **Live dashboard in Excel** -- custom gauges, alarms, computed values from live sensor data
- **Auto-build calibration compare reports** -- script opens two cals, dumps values to Excel cells
- **Remote monitoring** -- Excel sheet pulling from FPM over DDE, shared with a tuner elsewhere

For my build: DDE is a lightweight alternative to Lua plugins for simple integrations. If I want a custom dashboard or logger that's quick to set up, this is the path. Not as powerful as plugins but easier for small jobs.

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9. Plugins and Scripting (Lua) -- CAPTURED (NEW AUTOMATION PATH)

?? Major correction to earlier docs: I previously wrote "Hondata has no public API or documented CLI." **This was wrong.** FPM supports Lua scripts packaged as Plugins -- a real programmatic automation path.

What Plugins Can Do

- Read/write any calibration parameter or table
- Analyze datalogs
- Respond to events (calibration open/close, datalog load, datalog start/stop)
- Interactive dialogs (InputQuery, ShowMessage)
- Access sensor metadata (name, type, unit, min/max)

How to Install a Plugin

- FPM -> Plugins menu -> Manage Plugins -> Install button

?? Security warning: Plugins can change ANY calibration parameter or table. **Only install from trusted sources.**

Lua Script Object Model (summarized)

Full reference in FPM help; key objects:

Application -- app, version, build, bit64, compiledate, locale, os

Device -- name, connected(), hardware(), serial(), isdatalogging(), isrecording(), datalogcount()

ECU -- ignitionon(), startdatalogging(), stopdatalogging(), startrecording(), stoprecording()

Sensor -- name(), abbreviation(), description(), min(), max(), type(), unit(), live() (real-time value)

SensorList -- count(), sensor(index) (also [] subscript)

Datalog -- filename(), framecount(), length(), timestamp(frame), framenumber(timestamp), value(Sensor, frame), sensorcount(), sensor(index)

DatalogManager -- count(), datalog(index)

Table -- name(), valuename(), dimensions(), length(), size(), type(), unit(), readonly(), GetValue(idx, idx, idx), SetValue(value, idx, idx, idx)

Calibration -- loaded(), filename(), update(), tablecount(), table(index)

ErrorCode / ErrorCodeList -- DTC query and clearing

Event Callbacks

Lua scripts can respond to:

- OnCalibrationNew(calibration)
- OnCalibrationOpen(calibration)
- OnCalibrationClose(calibration)
- OnDatalogOpen(datalog)
- OnDatalogClose(datalog)
- OnDatalogStart()
- OnDatalogStop()

Functions

- InputQuery(caption, prompt, default) -- prompt for string input, returns string or nil
- ShowMessage(message) -- show MessageBox

Implications for My Build

This is the automation path I thought didn't exist. With Lua plugins I can:

1. **Auto-analyze datalogs on close.** OnDatalogClose(datalog) fires -> script reads all frames for knock count, lean spikes, over-duty-cycle events -> writes a summary to a file or shows MessageBox.
2. **Safety guardrails on calibration save/upload.** OnCalibrationOpen/Close fires -> script inspects WOT AFR cells, timing cells, rev limits, VTEC point -> rejects out-of-envelope values BEFORE I flash.
3. **Auto-CSV export of datalogs.** OnDatalogClose -> script uses datalog:value() to iterate all frames/sensors and writes a CSV file to a known folder. Paired with cloud sync, Claude can pull logs within seconds of me ending a drive.
4. **Custom sensor alerts.** Real-time via Lua scripts monitoring Sensor.live() values and triggering UI alerts when thresholds cross.

For my build, I should write (or find a community plugin for):

- reliability-guard.lua -- enforces my safety ceilings on calibration save (WOT AFR >= 12.0 pump / >= lambda 0.82 E85, timing peaks verified, rev limit <= 8400 RPM)
- auto-csv-export.lua -- writes CSV next to .fpdl on close
- log-summary.lua -- post-drive summary of knock events, trim extremes, max IAT/ECT, injector duty peaks

This is a much better workflow than UI automation via AutoHotkey. I'll explore the Plugins community / write my own when I have more dyno data.

Other Mentions

- **Plugin Graphics** -- listed as "Work in progress. Unsupported."
- **Plugin Debugger** -- listed as "Work in progress. Unsupported."
- Writing a plugin -- referenced as another help topic (not yet pasted; worth capturing if available)

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10. Bluetooth -- CAPTURED

Which FlashPros Have Bluetooth

- **Some FlashPros manufactured after July 2013 AND all after December 2013** have built-in Bluetooth
- Compatible serial number cutoff for 2006-2011 Civic Si **US FlashPro: sold after 11/7/2013**
- Compatible cutoff for Civic Si / Type R **International: 9/27/2013**

For my unit: the INI shows my FlashPro as a recent model (FP-SI-US-461004). If sold after November 2013 it has Bluetooth. **I should check: FPM -> FlashPro window -> FlashPro tab -> expand "Hardware" -> look for "Bluetooth present" entry.**

Bluetooth Types Supported

- **V2.1 (BR/EDR)** -- Android supports unrestricted
- **BT4 (BLE -- Low Energy)** -- slower than V2.1
- **Apple devices with Bluetooth V4** -- work
- **Apple devices with V2.1** -- DON'T work (Apple requires specific chip in the hardware)

Communications Protocols

- **FlashPro binary protocol** -- datalog from FPM on laptop via Bluetooth
- **Hondata binary protocol** -- Android/iOS apps
- **ELM327 emulation** -- compatible with many existing scantool smartphone apps

Android Apps Supported

Car Gauge Lite/Pro, DashCommand, OBDAutoDoctor, OBD Car Doctor, OBD Dashboard, OBD DROIDSCAN PRO, OBDLink, ScanMaster Lite, Torque Lite/Pro -- all can read Bluetooth FlashPro.

Windows Bluetooth Stack Requirement

- FPM requires **Microsoft Bluetooth stack** (shown as "Generic Bluetooth Adapter" in Device Manager)
- Non-Microsoft stacks (chipset-manufacturer stacks) will NOT work with FPM
- If using a non-Microsoft stack, you can uninstall and let Windows re-install with the Microsoft stack (risky -- requires driver management comfort)

Bluetooth Options (FlashPro window -> Bluetooth tab)

- **Bluetooth enabled** -- enable/disable on FlashPro
- **Local Address** -- Bluetooth radio on laptop (blank = no Bluetooth OR wrong stack)
- **FlashPro Address** -- Bluetooth address of FlashPro
- **Device Name** -- FlashPro Bluetooth name
- **Tuning via Bluetooth enabled** -- datalog + change some settings over Bluetooth (uploading NOT supported)
- **Alternative pairing mode** -- FlashPro shows as a phone to pair with vehicle head units (2018+ Civic)

Bluetooth Security

- **PIN code** required on pairing (Windows can try 3 times without PIN before prompting user)
- BLE connections don't use PIN codes -> Tuning via Bluetooth DISABLED for BLE
- **Limit access by device address** -- most secure, only listed devices can connect

Connecting From Laptop

- FPM -> Online / Bluetooth menu OR Bluetooth toolbar button
- FlashPro plugged into vehicle with ignition ON
- Select Bluetooth Device window -- Refresh to find new FlashPros
- Windows remembers paired FlashPros
- Pairing code defaults to FlashPro serial number (changeable in Bluetooth tab)
- Disconnect: same menu/button

Vehicle Pairing (Hondata Mobile on Head Unit)

For 2018+ Civics running Hondata Mobile on the car's head unit:

1. Unplug FlashPro from vehicle
2. Plug into laptop, run FPM
3. FlashPro / Bluetooth -> check "Alternative pairing mode for head unit (2018+ Civic)"
4. Unplug/replug FlashPro USB
5. On vehicle radio, pair to FlashPro (code default 1234)
6. Hondata Mobile app on head unit connects to FlashPro

Not applicable to my 2009 Civic (no compatible head unit).

Bluetooth Test Mode (developers only)

Sends fixed test data:

- RPM varies, speed 100 kph, gear 4, IAT 100degF, ECT 200degF, MAP 110 kPa, Lambda 13.8 AFR, STFT +10%, LTFT -10%, Battery 12V, Ethanol 20%, etc.

For My Build

If my FlashPro has Bluetooth:

- I can datalog from a phone without a laptop -- **huge convenience for monitoring**
- Hondata Mobile app pairing for a dash display
- Android apps (Torque Pro, etc.) for custom gauges without FPM

Action item: verify Bluetooth on first FPM connection. If present, install Hondata Mobile on my phone -- free backup "gauge" that doesn't need the LattePanda.

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11. Vehicle-Specific Notes: Civic Si 2006-2011 (MY CAR) -- CAPTURED (DEFINITIVE)

Supported Models + ECU Part Numbers

Year	Region	Features	ECU Part Numbers	FlashPro Model
2006	USA	No VSA	RRB-A04 to RRB-A05, RRB-305	Civic Si Americas
2007-2011	USA	VSA	RRB-A060 to RRB-A140	Civic Si Americas
2007-2011	Mexico	--	RRB-X110 to RRB-X120	Civic Si Americas
2007-2011	Central America	--	RRB-K010 to RRB-K020	Civic Si Americas
2007-2011	South America	--	RRD-M110 to RRD-M120	Civic Si Americas
2007-2011	South America	--	RRD-M210 to RRD-M220	Civic Si Americas
2007-2011	South America	--	RRD-P110 to RRD-P120	Civic Si Americas
2007-2011	South America	--	RRD-P210	Civic Si Americas

My car: 2009, USA = **RRB-A060 to RRB-A140 ECU** family. Has VSA. **FlashPro Model: Civic Si Americas.**

Note: the CLAUDE.md file says my ECU is "PRB." The Hondata Supported Models table here uses **RRB**. These may be different part number prefixes (e.g., PRB could be a shorthand for the family, or RRB is the actual part number). **[VERIFY]** my actual ECU P/N by looking at the sticker on the ECU or in FPM's FlashPro window when connected.

Feature List (What My Calibration Supports)

General:

- ? Forced induction support
- ? Expanded tables
- ? Live tuning
- ? Active fuel tuning
- ? **Wideband lambda input**
- ? **Boost control output** (for future if I ever go turbo)
- ? **Ethanol content input (flex fuel!)**
- ? Traction control input

Fuel:

- ? AFM, MAP, OR hybrid AFM/MAP tuning
- ? Injector size & fuel pressure parameters
- ? Injector voltage compensation
- ? MAP fuel tables (VE-based)
- ? AFM fuel tables (mass flow based)
- ? Full load compensation tables (WOT)
- ? Overall fuel trim
- ? Cranking fuel trim
- ? Cranking fuel table
- ? After start fuel table
- ? Cylinder fuel trim (per-cylinder)
- ? Air temperature fuel trim
- ? Coolant temperature fuel trim

Ignition:

- ? Ignition tables (low + high cam)
- ? Tip-in retard compensation
- ? Gear ignition compensation (different timing per gear)
- ? Cranking base ignition
- ? Cylinder ignition trim
- ? Air temperature ignition trim
- ? Coolant temperature ignition trim

VTEC/VTC:

- ? VTEC RPM window (low + high)
- ? VTEC pressure window (MAP-based)
- ? VTEC conditions (min ECT, min VSS)
- ? VTEC oil pressure input enable/disable
- ? Cam angle tables (VTC target)
- ? VTC PID parameters (cam control loop tuning)

Closed Loop:

- ? Short, medium, long term trim limits
- ? Full load determination tables (MAP AND TPS methods)
- ? Wideband current to lambda conversion table
- ? Target lambda

Knock:

- ? Knock ignition limit
- ? Knock sensitivity table
- ? Knock retard table

Rev Limits:

- ? Overall rev limiter
- ? Launch rev limiter
- ? Speed limiter
- ? Boost cut (if adding FI)
- ? Full throttle shift (flat-foot)
- ? Anti-lag (for FI)

Sensors:

- ? AFM calibration
- ? MAP sensor selection (stock or Hondata 4-bar)
- ? Speed adjustment (ECU-side)
- ? Speed adjustment (dash-side)

Idle:

- ? Idle speed tables
- ? After-start idle speed table

Throttle:

- ? Target throttle plate table

- ? Overrun throttle closure (rev hang reduction)
- ? Tip-in fuel

Misc:

- ? Various sensor enable/disable (ELD, ECT2 are the critical ones for me)
- ? Various DTC enable/disable
- ? Cooling fan sensor selection (radiator bottom sensor or cylinder head sensor)
- ? Cooling fan temperature (on/off points)

Platform-Specific Notes (Hondata official)

1. **Engine oil life not functional with race calibration.** Switching from race to normal calibration -> **Maintenance Minder needs to be reset** (see Section 6.17).
2. **Trip meter not functional with race calibration.**

Implication: if I run a race calibration (non-CARB) for Sport map, the trip meter won't work and oil-life indicator gets weird. Might be worth keeping both Daily AND Sport as AFM-based / CARB-compliant if possible. Depends on what calibration the Sport map starts from.

---

12. Updating Calibrations -- CAPTURED

Why update calibrations: important that cals are uploaded via the CURRENT FPM version to properly datalog and use current parameters.

Process:

1. Last uploaded calibration stored on FlashPro -> download via Online menu -> Download
2. Re-upload via Online menu OR Ctrl+U
3. For secondary calibration: FlashPro window -> Calibrations tab -> Download button in Secondary Calibration section, then re-upload via Upload button

When this matters for me:

- After FPM version updates
- If I ever move the FlashPro to a different PC
- To verify what's actually flashed on the ECU vs what I think is flashed

---

13. Frequently Asked Questions -- CAPTURED

FlashPro Basics

"Do I need to leave the FlashPro plugged in?" No. Once reflashed you may remove the FlashPro.

"Can I leave the FlashPro plugged in?" Yes. When ignition is off, FlashPro enters power-saving mode (~10 mA draw). Will discharge battery faster than normal but should give several weeks of life on a normal-sized battery.

"Can I use the FlashPro on more than one vehicle?" FlashPro can only be locked to one vehicle at a time. Not recommended to switch between vehicles unless selling.

"What is the difference between FlashPro OG and FlashProMini?" Just physical construction. The board

inside is functionally identical. Same software, same features.

2006 Civic Calibrations -- Critical for My Build

"When should I use AFM-based vs MAP-based calibration?"

- **AFM:** stock with bolt-ons. AFM auto-compensates for different exhausts, minor mods. Works for mild cams. Doesn't work well for FI or large cams that send reversion pulses back to the AFM.
- **MAP:** required for forced induction + large cams. More flexible, more tuning needed.

For my build: I can use AFM for my full-bolt-on NA K20Z3 with Skunk2 Alpha + Skunk2 Ultra Street. If I ever went FI or large cams, switch to MAP. **AFM-based calibration is the simpler starting point.**

"What starting calibration should I use?"

- **AFM cal:** choose based on INTAKE (affects AFR the most)
- **MAP cal:** choose based on INJECTOR SIZE or specialty (supercharged, etc.)

"How do I tune fuel for AFM calibration? There are no fuel tables." AFM measures airflow; ECU injects to stoich AFR mathematically. No tables. If AFR is off at part-throttle, the AFM is reading wrong from an aftermarket intake -- edit the **AFM Flow table** to correct. At WOT, ECU applies the WOT lambda adjustment table on top of the AFM-based calculation.

"I see knock retard -- how is the knock sensor used?" Knock sensor tells ECU (over time) what octane fuel it's running. ECU can retard timing even with NO knock events because the highest-octane ignition limit table caps timing. You shouldn't normally see much knock retard for more than a short period.

"Does 'Use MAP to determine WOT' only work on MAP-based calibrations?" No. AFM calibrations ALSO use the MAP sensor -- for everything except fuel, they use the MAP sensor. The AFM is specifically for fuel calculation only.

"How accurate is the stock lambda sensor?" Fairly sensitive/responsive, but reads progressively richer as it heats up. Dyno run can show gaining 1.0-1.5 AFR points as sensor heats. **Check with your own wideband if possible.** (See Section 6.35 for AF.Corr.)

"The idle is not quite right after the reflash." Takes 3-4 cold-cycle driving cycles for ECU to "learn" the idle. Don't panic after first flash -- give it time.

---

14. Error Reporting -- CAPTURED

FPM can generate error reports when encountering problems. Some are app-level, some OS-level (Windows reports).

Privacy: Hondata asks users to send reports for bug tracking. **No personal information sent.** You can see what's being sent before transmission if concerned.

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15. Verified vs. To-Verify Summary (UPDATED)

? VERIFIED from official Hondata help + evidence:

- FPM 4.6.6.0 changelog (K20Z3 PRB unchanged in 4.6.x)

- FlashProOG physical spec (210g, 6ft cable, 0-70degC)
- 9-16V supply, <100mA active
- **ISO 15765 CAN protocol** (not K-line)
- Windows 10/11, USB, internet required
- 20-hour onboard datalog memory
- 4x faster than OBDII standard
- **Dual calibration support: primary + secondary on unit, button combo switching**
- HTML export for calibrations exists
- .fpcal format (magic bytes 41 4B)
- Calibration files are binary encrypted/compressed (~22KB)
- **Flash time: ~90 seconds** (not 20-45 as estimated)
- **PRB ECU pin allocation:** A23 ELD, A33 ECT2, B2 EGR out, B29 EGRL, C27 SO2
- **Flex fuel wiring: ECT2 (A33) for K-series Civic Si**
- **Wideband wiring: A23 (ELD), A33 (ECT2), or B29 (EGRL) -- NOT C27 (SO2)**
- **WOT AFR ceiling: 12.50:1** (Hondata official; above this ECU won't switch to open loop correctly)
- **VTEC bounds: 2500-6500 RPM** (K-series oil pressure / rocker float)
- VTC control: 0-50deg crank degrees, ~10deg per 0.1 sec rotation rate, locked at 0deg for 10 sec after startup
- **Active Tuning exists: ECU-side self-tuning for fuel + AFM tables**
- **Lua Plugins: real automation path (corrects earlier doc)**
- **DDE Server for Excel integration**
- **Bluetooth on FlashPros sold after Nov 2013** (my unit likely has it)
- Civic Si 06-11 supports: forced induction, flex fuel, live tuning, active tuning, wideband input, boost output, TC input

? TO VERIFY (action items):

1. Whether my FlashPro has Bluetooth (check FPM -> FlashPro tab -> Hardware)
2. Whether Hondata Plugins community has a reliability-guard plugin OR I write my own
3. Exact voltage-to-ethanol mapping for my chosen ethanol controller (Innovate ECF-1 vs Zeitronix ECA) once purchased
4. Whether my ECU is RRB-A060 through RRB-A140 (look at sticker on ECU or FPM connection)
5. Whether the performance tune.fpcal is AFM-based or MAP-based (affects whether it's appropriate for my current hardware state)
6. Test all major FPM features in a safe order (File -> New Calibration, Tables Follow VTEC toggle, F3-F8 shortcuts, Compare Calibration) before first flash

?? CORRECTED from earlier docs:

- ~~"Flex fuel wires to SO2 pin"~~ -> **Wires to ECT2 (A33) pin**
- ~~"FlashPro has no public API"~~ -> **Has Lua Plugins + DDE Server**
- ~~"Original FlashPro is ALL wired, no Bluetooth"~~ -> **Post-Nov-2013 units have Bluetooth**
- ~~"20-45 sec flash time"~~ -> **~90 seconds**
- ~~".fpc extension"~~ -> **.fpcal**
- ~~"VTEC floor is 4500 RPM"~~ -> **Hondata official floor 2500 RPM for K-series** (my reliability-first target 4500 is my own stricter rule)
- ~~"Wideband goes to FlashPro analog input"~~ -> **Wideband goes to ECU pin (ELD/ECT2/EGRL)**

16. Vehicle Locking -- CAPTURED (from live WebFetch 2026-04-19)

Source: https://www.hondata.com/help/flashpro/vehicle_locking.htm

Core rule: "A FlashPro may only be used on one vehicle at a time, where it is locked to the vehicle's ECU."

Transferring between vehicles:

- Unlock FlashPro from original ECU BEFORE moving to a different vehicle
- Applies to both selling the unit and swapping to another car

Vehicle replacement scenario:

- When replacing vehicle, "the FlashPro should be unlocked from the old ECU before the ECU is removed from the vehicle"

Damaged ECU edge case:

- If original ECU is damaged and can't be unlocked normally -> send BOTH old ECU + FlashPro to Hondata for unlocking

Design philosophy:

- FlashPro is NOT designed to be switched between vehicles regularly
- Multi-vehicle users: buy separate FlashPros per vehicle

For my build:

- Unlock BEFORE selling the car (if I ever sell)
- If my ECU ever gets damaged, I need to keep both ECU + FlashPro for Hondata unlock
- Don't plan to move this FlashPro to another car -- it's married to this SI

---

17. Writing a Plugin (Lua) -- CAPTURED (from live WebFetch 2026-04-19)

Source: https://www.hondata.com/help/flashpro/writing_a_plugin.htm (note: page title says "Hondata SManager" -- the guide is shared between SManager/S300 and FPM. Path substitution required: use FPM plugin directory instead of SManager's.)

Plugin Structure

Three files required:

1. **main.lua** -- primary script
2. **info.xml** -- metadata / manifest
3. **desc.rtf** -- description (currently unused by FPM but required)

Plugin Directory Path

For SManager (per Hondata docs): C:\Users\<user>\AppData\Local\Hondata\SManager\Plugins

For FPM (inferred): C:\Users\<user>\AppData\Local\Hondata\FlashPro\Plugins -- **[VERIFY]** exact path on my machine. Likely mirrors SManager's structure.

Basic Template

```
require("utilities")
```

```
function main
-- add code here
end
```

main executes when I select the plugin from the menu.

Distribution Manifest (info.xml)

```
<id>unique_identifier</id>
<version>1</version>
<name>Plugin Name</name>
<description>What it does</description>
<author></author>
<url></url>
<license>Apache 2.0 License</license>
<depends></depends>
```

id must be unique across all plugins.

Packaging

1. Zip all 3 files (main.lua, info.xml, desc.rtf) into a .zip
2. Rename extension from .zip to .fplugin
3. Share / install via FPM's Plugins menu

Plugin Ideas for My Build

Now that I have the structure, first plugins I'll write:

reliability-guard.fplugin -- fires on OnCalibrationClose or OnCalibrationNew. Reads all WOT lambda cells (low + high cam), verifies >= 0.85 lambda (12.5 AFR). Reads timing table peaks, flags if any cell > 28deg. Reads rev limit, flags if > 8400. Shows MessageBox with violations before allowing save.

auto-csv-export.fplugin -- fires on OnDatalogClose. Iterates all frames x all sensors, writes CSV next to the .fpdl. Includes calibration name and mod list as header comments for future Claude ingestion.

log-summary.fplugin -- fires on OnDatalogClose. Post-drive report: max RPM, max knock count, STFT/LTFT peaks, AFR min/max at WOT, ECT max, injector duty max. One-screen summary after every drive.

I'll build these after my first few dyno sessions so I have real data to test against.

---

18. Community Tune Landscape (from web research 2026-04-19)

Off-the-Shelf Tunes Available for My Car

HARDmotion FBO Max Tune ($199 on sale, MSRP $290)

Source: https://hardmotion.com/2006-2011-honda-civic-si-fbo-hondata-max-tune/

Calibrated for:

- 3-3.5" intake (MAP-based -- requires MAF removal)
- Race header w/ or w/o cat
- 2.5-3" race exhaust

- Optional J35/J37 throttle body

Key changes:

- **MAF removal** -- "runs without the maf and allows for a much more aggressive intake pathway"
- **VTEC engagement advanced 500-800 RPM earlier** -- smoother transitions, quicker low-end response

Claims: +30-40 HP total gain over stock with tune (14 intake + 14 header + 10 exhaust).

?? Compatibility with my current hardware:

- My K&N Typhoon intake is AFM-based (keeps the AFM sensor in place)
- HARDmotion tune is **MAP-based, removes the AFM** -- **NOT directly compatible with my current setup**
- Would require going to a tube-only intake AND doing AFM->IAT-sensor conversion (see Section 6.31)
- **Verdict: SKIP for now.** Only relevant if I later switch to a tube intake.

zoshmfg 8th Gen Si FBO Tune ($80)

Source: https://zoshmfg.com/products/8th-gen-si-full-bolt-ons-hondata-flashpro-tune

Calibrated for:

- Hybrid Racing CAI (or equivalent)
- Skunk2 Alpha Header + Exhaust (or equivalent)
- Stock K20Z3
- 1000cc or stock 310cc injectors
- Stock RBC intake manifold + stock throttle body
- 93 octane

?? RED FLAG -- One customer review notes: > "at WOT, afr shows 13.4-13.7, which is lean"

13.4-13.7 AFR at WOT EXCEEDS Hondata's 12.50:1 ceiling and violates my reliability-first rule. This tune is too lean for my engine longevity priorities. Tuner may have done this intentionally for peak power, but it's unsafe for a 170k-mile K20Z3.

Verdict: AVOID WITHOUT MAJOR MODIFICATIONS. If I wanted an off-the-shelf tune in this category, I'd need to either:

- Use it as a starting point and have a dyno tuner pull fuel/timing to safe territory
- Skip it entirely and get a custom tune from BrenTuning / HARDmotion / Church

Community base maps on 8thcivic.com forum

Community-reported experience (from the Hondata FlashPro Base Maps thread):

- **"Skunk2 Alpha + CAI + Skunk2 exhaust" base map:** conservative ignition (low values). Safer for reliability but leaves power on table.
- **Intake-only base map:** users report VTEC engagement around 4500 RPM works well. "Great pull with awesome drivability and gas mileage around 26-28 city."

FPM Built-in Base Map Library

Hondata ships 56 built-in starting calibrations per their product docs. For my platform they include:

- Stock equivalent
- Intake only (various brands -- K&N, AEM, Injen, etc.)
- Intake + header

- FBO (intake + header + exhaust + tune)
- Injector-upsize variants
- Supercharged variants (SC)
- Race (non-CARB) variants
- CARB-compliant variants where available

My starting point: since my car is K&N Typhoon + stock everything else, I want the **"Stock + K&N intake" AFM calibration** as my first flash. The other 55 base maps exist but aren't relevant until I add more hardware.

Dyno Numbers to Expect (community-validated)

State	Dyno type	WHP	WTQ	Source
Stock	Well-calibrated	170-175	~135	k20a.org consensus
Stock + CAI	Dynojet	178-188	~140	8thcivic threads
CAI + header + exhaust + base tune	Dynojet	200-210	~150	8thcivic Skunk2 threads
FBO + professional dyno tune	Dynojet	220-235	155-165	community consensus + 4Piston/HARDmotion results
FBO + E85 tune	Dynojet	230-245	165-175	projected, community estimates
Max FBO (built heads/cams)	Dynojet	240-260	170-180	beyond my scope

Reality check: the earlier reference docs said stock = 185 WHP. **Web research says 170-175 WHP is more accurate for stock on a "well-calibrated" dyno.** Difference = some shop dynos read optimistic. My expectation for FBO should be 220-230 on a honest Dynojet, 215-225 on Mustang/Dynapack.

WOT AFR + Timing Targets (community-validated)

For NA K20Z3, consensus from multiple sources:

Parameter	Conservative (my target)	Community typical	Hondata ceiling
WOT AFR (pump 93)	12.0-12.5	12.5-13.0 (0.85-0.89 lambda)	12.50 (won't open-loop above)
WOT ignition (peak)	24-26deg	24-26deg proven safe; 28deg borderline	"Don't rely on knock sensor"
VTEC engagement (stock)	5400-5600	5700	6500 max
VTEC engagement (FBO)	4800-5200	4475-5100 common	2500 min

Key insight from community (k20a.org / hpacademy): > "One K-series engine on the dyno showed timing at WOT was 24-26 degrees, and while a pull at 28 degrees made a couple more HP, it wasn't worth the risk."

This reinforces my reliability-first rule. 24-26deg is the street-safe ceiling for pump 93. Leave 28deg for dyno-verified special calibrations only.

Tune Sources to Consider for My Build (ranked by fit)

1. **BrenTuning remote** -- community-trusted, $300-450 per revision, works from my datalogs. **Best fit for my reliability priorities.**
2. **Custom dyno session** at Church / 4Piston / HARDmotion for Tune #3 (flex fuel) -- $500-800
3. **FPM built-in base map** for initial flash (free) -- "Stock + K&N intake" AFM variant
4. **Community base map from 8thcivic** -- free but conservative; good fallback
5. ~~HARDmotion FBO Max Tune~~ -- incompatible with my AFM-based setup

6. ~~zoshmfg FBO tune~~ -- reports lean WOT AFR, violates my safety ceilings

19. Hondata Vault -- Web Research Note

The Hondata Vault help topic URL I tried (vault_use.htm) returned generic site nav, not the actual help content. The Vault exists and is covered in Sections 6.52-6.53 of this reference from user pastes, but detailed usage docs (registration, login, file operations, calibration details) remain uncaptured.

What I know:

- Accessible from Window menu -> Vault Search
- Two search types: Text Search + Filtered Search
- Results sortable by clicking column headings
- Used for uploading/downloading calibrations + datalogs to cloud

Action item: browse hondata.com/help/flashpro/ frame-by-frame in a regular browser to find the actual Vault usage URL, then capture. Not critical for my near-term work.

7. Topics Still to Capture

Topics I still need from the user's paste workflow (priority order):

1. **ECU Connectors** -- pin locations referenced by the Wideband doc (EGRL, ECT2, ELD). Resolves wiring path for Phase 6 flex fuel + Phase 17 wideband.
2. **Tuning Guide** -- the core how-to-tune doc
3. **Fuel Tables** -- every fuel parameter explained
4. **Ignition Tables** -- every ignition parameter
5. **VTEC** -- engagement settings detail
6. **Cam Angle (VTC)** -- phase target tables
7. **Closed Loop Parameters** -- contains the wideband settings per 6.8
8. **Idle Control**
9. **Rev Limits** -- per-gear details
10. **Launch Control / Flat-Foot Shift / 2-Step**
11. **Torque Management**
12. **Fan Control**
13. **A/C Control**
14. **EVAP / Canister Purge**
15. **Analog Inputs / Gauges** (generic, not wideband-specific)
16. **Sensors** -- complete parameter list with units
17. **Datalogger** -- full datalog reference
18. **Graphs / Histograms**
19. **Autotune** -- big time-saver, need to understand
20. **DTC / CEL Options**
21. **Race Settings / Race Calibration**
22. **Updates** -- firmware update process details
23. **Security / VIN Lock / Registration**

24. **Traction Control** (on my platform?)
25. **Keyboard Shortcuts** (partial list already -- Section 10)
26. **Troubleshooting**
27. **On-Board Datalogging** -- 20-hr unit-side logs
28. **Flex Fuel** -- explicit topic if one exists

8. FPM Config File (FlashPro.ini) -- What It Tells Me

Source: C:\Users\extra\AppData\Local\Hondata\FlashPro\FlashPro.ini

Selected notable settings (full file preserved for reference):

User preferences already configured

Setting	Value	Meaning
FileDir	C:\Users\extra\OneDrive\Documents\FlashPro Calibrations	Default folder for .fpcal files
DatalogDir	Same as above	Default folder for datalogs
PressureUnit	4	Pressure unit (4 = likely kPa)
TempUnit	0	Temperature unit (0 = likely Fahrenheit)
LambdaUnit	1	Fuel unit display (1 = likely lambda, not AFR)
LanguageCode	en	English
AutomaticUpdates	1	FPM auto-checks for updates
IncludeBetaUpdates	1	Beta updates enabled -- worth considering toggling off for reliability
CompressDatalogs	1	Datalogs saved compressed
DisableScreenSaverWhenDatalogging	1	Good practice

Autotune defaults (from ini)

Setting	Value
AutotuneMinLambda	0.75
AutotuneMaxLambda	1.22
AutotuneMaxChange	10
AutotuneFreq	1

These are the default guardrails if I use FPM's Autotune feature. Lambda floor of 0.75 = ~11.0 AFR on gasoline -- aggressive but within safe WOT range. Max change of 10 is a percentage change limit per iteration.

Target Lambda Display Overlays (user-set)

Setting	Value	Meaning
TargetLambdaLow	1.00	Cruise/low-load target = stoich
TargetLambdaMed	0.918	Mid-load target = ~13.5 AFR (mild enrichment)

TargetLambdaHigh	0.850	High-load target = ~12.5 AFR (WOT enrichment)

These appear to be the user-configured overlay targets for the lambda display feature. **Note: 0.850 lambda at WOT = 12.5 AFR, which matches my reliability-first recommendation for pump gas (12.0-12.5 range).**

Recent files in ini

1. performance tune 1.html (HTML export of performance tune -- file no longer exists on disk but was exported)
2. performance tune.fpcal
3. Factory Flash - Stock Tune.fpcal

The HTML export is the critical workflow -- FPM can export a tune as HTML, which IS readable. File was apparently deleted/moved; can be re-exported at any time.

Window sizes (from ini)

Main window 1200x900 at position (550, 135) -- user's current FPM layout. Multi-display form at 1350x252.

---

9. Calibration Files (.fpcal) -- What I Know From Evidence

Observed files

File	Size	Created	Notes
Factory Flash - Stock Tune.fpcal	22,431 bytes	2026-04-19 20:58	Stock OEM calibration backup
performance tune.fpcal	22,842 bytes	2026-04-18 19:53	Performance tune (origin unknown -- to confirm with user)

Format observations

- **Extension:** .fpcal (NOT .fpc as older community docs sometimes say -- I had this wrong in earlier reference docs, correcting here)
- **Magic number:** First 2 bytes are 41 4B = ASCII "AK"
- **Size:** Both observed files are ~22KB. This is small -- file is likely encrypted+compressed because raw parameter tables for a K-series ECU would be many tens of KB uncompressed.
- **Binary encrypted/compressed:** No readable strings in the file. Cannot be parsed outside FPM.
- **Difference between stock and performance:** ~411 bytes. Suggests meaningful calibration changes (populated non-default cells, different tables) rather than just metadata changes.

What can be exported (machine-readable)

- **HTML report** -- File -> Save As HTML (or similar menu -- need to confirm exact path). Dumps all table values as readable HTML. performance tune 1.html in the recent-files list confirms this feature exists.
- **CSV datalog export** -- File -> Export Datalog -> CSV. For log files, not calibrations.
- **Calibration comparison** -- FPM has compare feature that shows cell diffs between two .fpcal files ([VERIFY exact menu])

---

10. Keyboard Shortcuts (from FlashPro.ini)

The ini file exposes every keyboard shortcut currently mapped. Key ones:

Action	Key Code	Note
actIgnitionLowSpeed	114	F3 -- low-cam ignition table
actIgnitionHighSpeed	115	F4 -- high-cam ignition table
actFuelLowSpeed	116	F5 -- low-cam fuel table
actFuelHighSpeed	117	F6 -- high-cam fuel table
actCamLowSpeed	118	F7 -- low-cam VTC table
actCamHighSpeed	119	F8 -- high-cam VTC table
actViewGraph2D	120	F9 -- 2D graph view (also actRecord -- datalog start)
actViewGraph3D	121	F10 -- 3D graph view (also actDatalog)
actEditUndo	16474	Ctrl+Z
actEditRedo	24666	Ctrl+Shift+Z
actEditCopy	16451	Ctrl+C
actEditPaste	16470	Ctrl+V
actEditSelectAll	16449	Ctrl+A
actEditIncreaseSelection	16457	Ctrl+I -- increase values in selection
actEditDecreaseSelection	16452	Ctrl+D -- decrease values in selection
actInterpolate	16464	Ctrl+P -- interpolate between cells
actSmoothSelection	16461	Ctrl+M -- smooth selected cells
actAdjust	16458	Ctrl+J -- adjust values
actFileNew	16462	Ctrl+N
actFileSave	16467	Ctrl+S
actFillFuelTables	16454	Ctrl+F -- fill fuel tables
actAutotune	16468	Ctrl+T -- Autotune
actOnlineUpload	16469	Ctrl+U -- upload to ECU (FLASH)
actFileExit	32883	Alt+F4

Implication for my workflow

- **F3 / F4 / F5 / F6 / F7 / F8** = jump directly to ignition / fuel / VTC tables, low-cam vs high-cam. Critical shortcut for any table editing session.
- **F9** = start datalog recording. Single tap in car.
- **Ctrl+I / Ctrl+D** = bump selected cells up/down by a user-defined step. Faster than manual entry.
- **Ctrl+P** = interpolate between selected cells. Smooths out any tables I manually edit.
- **Ctrl+U** = flash to ECU. Know this one cold.
- **Ctrl+T** = Autotune. Worth exploring what this actually does before I use it.

11. Cross-References to Existing Repo Docs

The six reference files from the 2026-04-18 and 2026-04-19 work remain valid as context and overlap with this master file. This file is the **authoritative** reference going forward; the others are supplementary.

Existing file	Scope	Relationship to this master
hardware-software-deep-reference.md	Hardware + software deep dive	Being superseded by this file as topics are verified
calibrator-tables-reference.md	Table-by-table reference	Being validated against official help as user pastes
datalog-analysis-guide.md	Datalog reading guide	Still valid -- pattern-focused
advanced-features.md	Launch / flat-foot / per-gear	Needs cross-check vs. dual-cal discovery
troubleshooting-and-faq.md	Issues + DTCs	Still valid -- operational
tuner-directory.md	Shop recommendations	Still valid -- unchanged by software research
tuning-workflow-and-maps.md	Daily/Sport + reliability rules	Anchor document -- unchanged
overview.md	Summary overview	Stub -- can stay or be replaced

Rule: if this master reference conflicts with an older doc, this master wins (because it's verified against Hondata's own help). Any [VERIFY] item in this file that later gets confirmed should be updated to remove the flag AND the corresponding older-doc content should be updated to match.

---

12. Verified vs. To-Verify Summary

VERIFIED from Hondata official help + config evidence:

- FPM 4.6.6.0 release date and changelog
- FlashProOG physical spec (210g, 6ft cable, 0-70degC operational)
- Supply voltage 9-16V, <100mA active
- ISO 15765 CAN protocol (not K-line)
- Windows 10/11 support, USB any version, internet required
- 20 hours onboard datalog memory (unit-side, independent of PC)
- 4x faster datalog than standard OBDII
- Cam angle aware table editing
- Dual calibration support exists as a feature (meaning TBD)
- HTML export for calibrations exists
- .fpcal format (not .fpc) with 41 4B magic number
- Calibration files are binary encrypted/compressed (~22KB)
- Keyboard shortcut mappings (from ini)

TO VERIFY (priority order):

1. ~~**What "Dual calibration support" actually enables**~~ -- **RESOLVED:** dual-cal is real, on-unit, primary + secondary slots, switch via button combo (Program alone = primary; hold Datalog + Program = secondary). Huge finding.
2. **Flex fuel sensor wiring pin on PRB ECU** -- likely EGRL or ECT2 (not rear O2 because wideband wants ECT2 and SO2 is unusable). Need ECU Connectors topic.
3. **Exact menu path for HTML calibration export** (shown in recent files list so it exists; just need the menu path)

4. **Autotune safety** -- what Ctrl+T does, whether it respects reliability ceilings
5. **Full parameter list for PRB datalog**

CORRECTED from earlier docs (these older claims were WRONG):

- ~~"Original FlashPro cannot switch calibrations on-unit"~~ -- wrong, dual-cal slots exist
- ~~"Flash time ~20-45 seconds"~~ -- wrong, ~90 seconds
- ~~"Wideband wires to FlashPro analog input"~~ -- wrong, wires to ECU input pin (ECT2, ELD, EGRL)
- ~~".fpc file extension"~~ -- wrong, it's .fpcal
- ~~"Original FlashPro uses K-line"~~ -- wrong, uses ISO 15765 CAN (newer, faster)

Last updated: 2026-04-19 Maintainer: LxveAce, with Claude assistance Status: LIVING DOCUMENT -- grows as Hondata help topics are captured

Tunes -- Live Tracking

> Every calibration I have, with screenshots, parameter extracts, safety assessments, and revision notes. This file is constantly updated as I create new tunes, consult on existing ones, or change parameters after dyno sessions.

Living document -- entries get added, modified, and revised. History is preserved in git (use git log docs/reports/tunes.md or directly browse commits if I ever want to see what changed when).

How Entries Are Structured

Each tune gets its own section with:

1. **Tune header** -- name, source (tuner / self-made / community base map), hardware it's calibrated for, date, file path
2. **Safety pre-check** -- is it matched to my actual hardware? Is the stock backup saved?
3. **Tab-by-tab capture** -- screenshots or text extracts of every tab/table I've reviewed
4. **Parameter table** -- key values I care about (WOT AFR, timing peaks, VTEC, VTC, rev limit)
5. **Assessment** -- does this tune pass the reliability-first ceilings from tuning-workflow-and-maps.md?
6. **Revision log** -- what changed and when
7. **Next actions** -- what I'd do with this tune before driving on it

---

Safety Reference (Quick Ceilings)

From tuning-workflow-and-maps.md -- every tune I log here gets compared to these:

Parameter	Hard Ceiling	Notes
Knock count at WOT	0 (always)	Any is too much on 170k motor
Pump gas WOT AFR	>= 12.0	Never leaner
E85 WOT lambda	>= 0.82	Never leaner
Injector duty cycle	<= 85% sustained	Headroom is safety
Rev limiter	<= 8400 RPM (stock internals)	+200 over stock max
VTEC engagement	>= 4500 RPM	Below this, cam doesn't match airflow
ECT sustained	<= 210degF under WOT	Above = cooling system issue

---

Tune 1 -- Factory Flash - Stock Tune

Header

Field	Value
Filename	Factory Flash - Stock Tune.fpcal
Location	C:\Users\extra\OneDrive\Documents\FlashPro Calibrations\Factory Flash - Stock Tune.fpcal

Source	Read from ECU (factory OEM calibration)
Hardware calibrated for	Completely stock 2009 Civic SI (FG2, K20Z3) -- ZERO aftermarket
Date saved	2026-04-19 20:58:51
File size	22,431 bytes
Flashed to ECU?	This IS what's on the ECU (never been modified yet)
Purpose	Factory reset / insurance. If anything goes wrong with a performance tune, I flash this back.

Safety pre-check

- ? **Matched to hardware** -- this IS the hardware state the ECU shipped with
- ? **Saved as backup** -- that's its entire purpose
- ? **Stock ceilings** -- Honda's engineers set this. It's the safest possible calibration by definition.

Parameter extract

(To be populated if user exports HTML report of stock tune for reference. Optional -- this tune should never need changing.)

Assessment

PASS on all reliability ceilings. This is the ground-truth reference. Any deviation in other tunes that moves AFR leaner than this, timing more advanced than this, or VTEC lower than this should be justified by either (a) hardware that changes the engine's breathing, or (b) a specific dyno-verified finding.

Revision log

- 2026-04-19: Initial read from ECU. Saved to FlashPro Calibrations folder.

Next actions

- Do NOT modify
- Do NOT flash experimentally
- If anything feels wrong on any other tune, this is the escape hatch

---

Tune 2 -- Performance Tune 1 (Sport) -- FLASHED & DRIVEN, KNOCK DETECTED

-> Full dedicated profile: tune-profiles/performance-tune-1.md

Header (short form -- see profile for full detail)

Field	Value
Filename	performance tune.fpcal
Location	C:\Users\extra\OneDrive\Documents\FlashPro Calibrations\performance tune.fpcal
Date saved	2026-04-18 19:53:05
File size	22,842 bytes
Flashed to ECU?	YES -- flashed and driven 2026-04-20

Purpose	Sport map (performance mode)

Safety pre-check (summary)

- ? **Matched to hardware?** -- NO. Tune is FBO-targeted; car has only intake (stock headers + exhaust).
- ? **Stock backup exists** -- Factory Flash - Stock Tune.fpcal 2026-04-19 20:58.
- ? **Knock count at WOT = 0** -- FAILED on 2026-04-20 drive.

Assessment (summary)

FAIL on 3 of 11 reliability ceilings: WOT AFR (cells 12.54-13.34 = exceed Hondata 12.50 open-loop floor), WOT timing (30-33deg on pump gas = too aggressive), VTEC engage (4200 < my 4500 rule). **See full profile for specific cell-level fix directions.**

Revision log (summary)

- 2026-04-18 19:53: File saved
- 2026-04-20: Flashed as Sport map and driven -- knock detected
- 2026-04-20: 48 FPM tab screenshots transcribed into full profile doc

Next actions (summary)

1. **Datalog the next drive** per protocol in profile -> Datalog Capture Protocol
2. Paste CSV + pull notes to Claude for cell-level correction list
3. Re-flash with corrections, re-datalog, verify knock = 0

Template for Future Tunes

When I create, receive, or analyze a new tune, add a section like this:

```
## Tune N -- [Name]

### Header
| Field | Value |
|-------|-------|
| Filename | xxx.fpcal |
| Location | [path] |
| Source | [self / community / tuner name + date] |
| Hardware calibrated for | [exact mod list] |
| Date saved | YYYY-MM-DD HH:MM |
| File size | xxx bytes |
| Flashed to ECU? | Yes/No |
| Purpose | [daily / sport / e85 / test / etc] |

### Safety pre-check
- [ ] Matched to current hardware
- [ ] Stock backup verified
- [ ] Reliability ceilings confirmed (knock 0, AFR >= 12.0 pump / lambda >= 0.82 E85, inj duty <= 85%, rev <= 8400, V...

### Parameter extract
[Table-by-table values from HTML export or screenshots]

### Assessment
```

```
[PASS / FAIL / PENDING, with reasoning]

### Revision log
- YYYY-MM-DD: [what changed]

### Next actions
[Specific things to do with this tune]
```

Comparison Matrix (As Tunes Accumulate)

Once I have 3+ tunes in here, maintain a side-by-side view:

Parameter	Stock	Performance Tune 1 (Sport)	[Future Daily Map]	[Future E85 Map]
WOT AFR low-cam (leanest cell)	~12.5	12.54-13.34 ?	TBD (<= 12.0)	TBD (? >= 0.82)
WOT AFR high-cam (leanest cell)	~12.5	12.54-13.34 ?	TBD (<= 12.0)	TBD
Peak timing high-cam	~22deg	30-33deg ?	TBD (<= 24deg)	TBD (<= 30deg)
VTEC engagement	5800	4200 ?	TBD (4800+)	TBD
Rev limit	8200	8400 ?	7800	8400
Launch control	off	5000 RPM	off	off
Boost cut	N/A	disabled ?	disabled	disabled
Full throttle shift	off	off ?	off	off
Flex fuel	off	off (table inert)	off	on

(Fills in as tunes get analyzed.)

Known Off-the-Shelf & Community Tunes -- Reference Library

These aren't on my FlashPro yet -- they're tunes I've researched that could be starting points. Full writeup + compatibility analysis is in fpm-master-reference.md Section 18.

Ranked by fit for my current build

#	Tune source	Cost	Hardware target	Reliability flag	Fit for me?
1	FPM built-in: "Stock + K&N intake" AFM	Free	K&N / equiv CAI + stock everything else	Hondata-shipped = conservative safe	? YES -- first flash starting point
2	BrenTuning remote (custom)	$300-450	Tell them my hardware	Custom to mine = safe if I communicate reliability-first	? YES -- Tune #1 and #2 sessions
3	Church / 4Piston / HARDmotion in-shop	$500-800	Custom to mine	Live dyno verification	? YES -- Tune #3 (flex fuel)
4	8thcivic community base map (Skunk2 + CAI + exhaust)	Free	Matches Phase 2 state	Conservative ignition (community-reported low values)	? MAYBE -- fallback if remote tuner unavailable
5	HARDmotion FBO Max Tune	$199-290	3-3.5" intake (MAP-based, NO AFM) + race header + 2.5-3" exhaust + optional bored TB	--	? NO -- MAP-based, my K&N keeps AFM

6	zoshmfg 8th Gen Si FBO tune	$80	Hybrid Racing CAI + Skunk2 Alpha + Skunk2 exhaust	?? 13.4-13.7 WOT AFR per review = lean, violates 12.50 ceiling	? NO -- unsafe per my reliability rule

Community consensus parameter targets for NA K20Z3 FBO

From web research (k20a.org, 8thcivic.com, Hondata forum):

Parameter	Safe value	Notes
WOT AFR (pump 93)	12.0-12.5 (lambda 0.82-0.85)	Hondata hard ceiling 12.50. My reliability rule 12.0-12.2.
WOT ignition peak	24-26deg	28deg gains 2 HP but "not worth the risk" per dyno reports
VTEC engagement (stock)	5400-5700	Hondata floor 2500, ceiling 6500
VTEC engagement (FBO)	4800-5200	Community commonly lands 4475-5100
Rev limit	8200-8400 max	My hard ceiling at 170k stock internals

Expected dyno numbers (Dynojet, honest shop)

Hardware state	WHP	WTQ
Stock	170-175	~135
Stock + K&N	178-188	~140
+ header + HFC + base tune	200-210	~150
Full bolt-on + pro tune (Tune #2)	220-235	155-165
FBO + E85 (Tune #3)	230-245	165-175

Reality check on stock baseline: web research says stock K20Z3 is 170-175 WHP on honest Dynojet, not the 185 I had in earlier docs. Some dynos read optimistic. My actual baseline pulls will show the real answer.

Workflow Reminders

When LxveAce sends screenshots of a tune's tabs:

1. Identify which tune (by filename) is being shown
2. Find or create that tune's section
3. For each screenshot: transcribe the parameter values into the parameter extract table
4. Check each value against the safety ceilings
5. Update the assessment
6. Log what was learned in the revision log
7. Suggest concrete next actions

When LxveAce creates a new tune:

1. Create a new section with the template above
2. Ask for the hardware state + flash status + source immediately
3. Don't assess safety until hardware match is confirmed

When LxveAce modifies an existing tune after dyno:

1. Add a revision log entry dated to the change
2. Update the parameter extract with the new values
3. Re-assess against ceilings (any ceilings broken? tightened?)
4. Compare to the prior version -- what changed, what didn't

Last updated: 2026-04-19 Maintained by: LxveAce + Claude, continuously

Performance Tune 1 -- Full Profile

> **Living document.** This is the dedicated profile for performance tune.fpcal -- the Sport calibration I flashed and drove on 2026-04-20. Every parameter transcribed from the 48 tab screenshots taken in FPM on 2026-04-20. Updated every time the tune is modified.

Status as of 2026-04-20: FLASHED, DRIVEN, **KNOCK DETECTED** -- see Knock Diagnosis below. Do not drive hard on this tune again until corrections are made.

Header

Field	Value
Filename	performance tune.fpcal
Location	C:\Users\extra\OneDrive\Documents\FlashPro Calibrations\performance tune.fpcal
HTML export	performance tune 1.html (same folder)
Source	Received from LxveAce / personal copy (pre-existing file prior to repo build)
Hardware targeted (inferred)	FBO -- header + exhaust + intake + tune. Car does NOT currently match this.
Car's actual hardware (2026-04-20)	K&N Typhoon intake + Exedy Stage 1 clutch + lightweight flywheel + Acuity short shifter + DWS06 tires + front slotted rotors + pads installed today. Stock headers, stock exhaust, stock injectors, stock fuel pump, 170k miles.
Date file saved	2026-04-18 19:53:05
File size	22,842 bytes (411 larger than stock's 22,431)
Flashed to ECU?	YES -- flashed and driven on 2026-04-20
Purpose	Sport calibration -- performance mode

Safety Pre-Check

Check	Status	Notes
Matched to current hardware	? NO	Tune is FBO-targeted. Car has only intake. Header + exhaust absent = less airflow than tune expects = richer than tune expects at low RPM, but lean spots at high RPM because fuel map was built for freer-flowing exhaust
Stock backup saved	?	Factory Flash - Stock Tune.fpcal dated 2026-04-19
Knock count at WOT = 0	? FAILED	Knock sensors triggered on 2026-04-20 drive
Pump gas WOT AFR >= 12.0	?? AT-RISK	WOT lambda tables show cells at 12.54-13.34 AFR. Above Hondata 12.50 ceiling = ECU may not enter WOT mode fully. See Fuel Analysis
Injector duty <= 85%	? UNKNOWN	No datalog captured yet. Must measure.

Rev limiter <= 8400	?	Main 8400 / Recovery 8200
VTEC engagement >= 4500	?	4200-5500 RPM engage window (engage edge 4200 is below 4500, but upper bound matches FBO convention)
ECT sustained <= 210degF	? UNKNOWN	No datalog. Measure.

Full Parameter Extract

Header / identity

Cal window tree structure confirmed:

- **Fuel** (WOT lambda low, WOT lambda high, Cranking fuel, Cylinder trim, IAT comp, ECT comp, Active fuel tuning low, Active fuel tuning high, Active Tuning)
- **Ignition** (Ignition low, Ignition high, Cylinder trim, IAT comp, ECT comp)
- **VTEC** (engagement table)
- **VTC** (Cam angle low, Cam angle high, Cam angle index low, Cam angle index high)
- **Closed Loop** (Lambda)
- **Knock Control** (Knock ignition limit low, Knock ignition limit high, Knock sensitivity low, Knock sensitivity high, Knock retard low, Knock retard high)
- **Rev Limits**
- **Sensors** (AFM)
- **Boost Control** (disabled -- NA)
- **Idle**
- **Throttle** (TPlate Normal)
- **Misc**
- **Traction Control**
- **Ethanol** (Ethanol ignition compensation)

Rev Limits (screenshot 165507)

Parameter	Value	Safety
Rev limiter (main)	8400 rpm	? At my hard ceiling
Rev limiter recover	8200 rpm	? 200 RPM hysteresis
Launch RPM (adjustable)	5000 rpm	--
Launch disengage speed	3 mph	--
Speed limiter	180 mph	? Effectively removed
Boost cut enabled	No	? Correct for NA
Boost cut pressure	168 kPa (stored but disabled)	--
Full throttle shift	Disabled	? Correct -- stock internals
Anti-lag min RPM	15000	? Disabled (above rev limit)

Launch additional fuel	0	--
Launch ign retard	0deg	--

VTEC Window

Parameter	Value
Engage RPM	4200
Disengage RPM	5500 (actually "min hold RPM" -- below this at part throttle, VTEC drops out)
Engage min ECT	140degF
Engage min speed	6 mph
Disengage min speed	3 mph

?? Note: 4200 engage is below my reliability rule of >= 4500. This is aggressive-low for FBO hardware. On my current stock-header car it will engage where the cam is making LESS airflow than the tune expects.

VTC -- Cam Angle Tables

Table	Peak value	At RPM
Cam angle low	~40deg	4300-4400
Cam angle high	~48-49deg	5500-5750
Cam angle index low	--	stock-like phasing
Cam angle index high	--	stock-like phasing

Comments: Aggressive cam advance (high 40s crank degrees) at peak VTEC RPM -- consistent with a tuner trying to extract top-end power on a full FBO car.

Fuel -- WOT Lambda Adjustment Tables

This is the most concerning table set in the entire tune.

Both WOT lambda adjustment LOW and HIGH cam tables contain cells with values **12.54 to 13.34 AFR**.

Why this is a problem (the Hondata 12.50 AFR ceiling):

- Hondata's WOT fuel mode entry requires the commanded AFR to be richer than 12.50:1.
- Cells leaner than 12.50 cause the ECU to stay partially in closed-loop (targeting ~14.7 stoich) instead of entering open-loop WOT enrichment.
- **Result:** at those WOT cells, the car runs **leaner than 14.7** for brief moments, which is extremely knock-prone at high load.

Specific worst cells observed (approximate from screenshots -- to be re-transcribed from HTML export):

- Low-cam high-RPM high-load cells: 12.80-13.10 range
- High-cam high-RPM high-load cells: 12.54-12.95 range

Target after correction: every WOT cell <= **12.20 AFR** (lambda 0.83) per my reliability rule. Factor of safety on a 170k motor demands richer than Hondata's floor, not at it.

Fuel -- Support tables

Table	Observation
Cranking fuel	Standard multi-cell ECT-indexed -- not examined for deviations
Cylinder trim (fuel)	Baseline zeros -- no per-cylinder trim adjustment
IAT compensation (fuel)	Standard -- richer when intake air is hot
ECT compensation (fuel)	Standard -- richer when cold
Active fuel tuning low/high	Enabled -- these are the autotune delta tables; they accumulate corrections from wideband feedback
Active Tuning	Enabled

Ignition -- Main Tables

Table	Peak WOT values
Ignition low (0deg, 10deg cam)	Peaks in low 20sdeg -- conservative at low cam
Ignition high (at 30deg, 40deg cam)	30-33deg at 5000-6000 RPM, high load cells

?? 30-33deg WOT timing:

- Community consensus for NA K20Z3 pump 93: **24-26deg WOT** is safe; 28deg is borderline; 30deg+ needs E85 or flawless fuel.
- On pump 91 (Indiana winter gas) this is **overtly dangerous**.
- Combined with the leaner-than-12.50 AFR spots = guaranteed knock under load.

Ignition -- Support tables

- Cylinder trim: stock distribution
- IAT compensation: standard pull when IAT rises
- ECT compensation: standard

Knock Control

Table	Observation
Knock ignition limit low	Up to 45deg at -26deg cam angle (ceiling; actual demand never approaches this)
Knock ignition limit high	Up to 45-47deg at -26deg cam angle, drops sharply at high-cam/high-load cells (as expected)
Knock sensitivity low	Aggressive thresholds -- 22-90% threshold curve across RPM
Knock sensitivity high	Aggressive thresholds -- 22-95%
Knock retard low	Up to 25deg pull at -26deg cam angle
Knock retard high	Up to 20deg pull at -26deg cam angle

Assessment: Knock detection and response are aggressive and correctly configured. When knock happens, the ECU WILL pull timing hard. The problem isn't knock detection -- it's that the base timing + AFR combination **causes knock in the first place**. Pulling timing via knock retard is a safety net, not a substitute for a safe base map.

Idle

ECT	Normal idle (rpm)	After-start idle (rpm)
-13degF	1500	1800

5degF	1500	1800
32degF	1500	1800
86degF	1500	1800
113degF	1500	1500
149degF	1050	1050
176degF	750	1000
194degF	750	1000

Note: Cold idle at 1800 RPM after-start is high but within spec. Warm idle at 750 RPM is normal for stock K20Z3.

Throttle

Parameter	Value
1st gear throttle dampening	? Enabled (smooths clutch engagement)
Overrun throttle opening (rev hang reduction)	? Enabled (quicker RPM drop on shifts)
Tip-in fuel ECT comp	-22degF: +50%, 32degF: +50%, 176degF: +50%
Overrun injector cutoff	Configured with MAP + RPM thresholds

Traction Control

Table present -- not examined in detail. Likely stock-like / disabled.

Ethanol

Ethanol ignition compensation table present but flex fuel hardware NOT installed on car, so this table is effectively inert until a flex fuel sensor is wired.

Knock Diagnosis -- Why the Knock Sensors Triggered

Symptom

On 2026-04-20 drive after flashing this tune, knock count > 0 in ECU reading. Exact count/RPM/cell needs a datalog to pinpoint.

Preliminary root cause (before datalog review)

Two compounding factors:

1. **WOT AFR target is leaner than Hondata's 12.50 open-loop entry threshold** -- ECU stays partially in closed-loop during WOT, so brief moments of AFR lean spike above 12.50 (even toward stoich) occur. On a 93-octane tune in Indiana's 91-octane summer gas, this alone is enough to knock.
2. **WOT ignition timing 30-33deg at high-load / 5000-6000 RPM is aggressive for pump gas, and UNMATCHED for my current hardware.** The tune was built for an engine breathing through a Skunk2 Alpha header + HFC + cat-back. Mine still has stock headers + full stock exhaust = more back-pressure = more residual exhaust gas = **hotter combustion = more knock** at the same timing/AFR.

Contributing factor -- hardware mismatch

The car is running the Sport calibration as if it had FBO hardware, but it only has:

- ? K&N Typhoon intake
- ? Stock header (restrictive)
- ? Stock cat-back (restrictive)
- ? Stock injectors (fine for NA -- not an issue here)

The tune assumes ~220-235 WHP airflow. The car is probably making ~180-190 WHP worth of airflow. The timing is calibrated for the freer-flowing state, and when the engine is making less airflow but the tune is adding more timing than needed, knock happens.

Fix direction (to be confirmed after datalog review)

Priority	Action	How
1	Richen every WOT cell leaner than 12.20 AFR to 12.0-12.2	WOT lambda adjustment low + high tables -- find every cell >= 12.30 and pull it to 12.10
2	Pull 3-5deg of timing from high-load cells in Ignition high	30deg -> 25-26deg. Specifically the cells at 5000-6500 RPM, high load, for all cam-angle slices
3	Raise VTEC engagement to >= 4500	4200 -> 4500 to give cam airflow room to match tune expectation
4	Re-flash and re-drive with datalog running	Verify knock count = 0 across multiple pulls
5	Long-term: build a Daily calibration	This tune is the Sport map. I need a sister Daily map that's conservative (<= 24deg WOT timing, >= 12.0 AFR, VTEC 4800, rev 7800). Two calibrations on the FlashPro -- primary Daily, secondary Sport.

Datalog Capture Protocol

For when I take the car out to datalog (no internet needed in the moment -- this section travels offline with the repo):

Before starting

1. **Engine fully warmed** -- ECT >= 180degF. No pulls before thermostat fully open.
2. **FlashPro connected via USB cable** (keep cable loose, don't strain port)
3. **FPM open with Datalog -> Record armed** (red record icon)
4. **Confirm sensors streaming in FPM** -- RPM, MAP, IAT, ECT, AFR, knock count all moving
5. **Fuel in tank:** ¾+ ideally (keeps pump cool, consistent pressure)
6. **Safe, legal, empty road** -- 3rd gear WOT pulls to 7000 RPM require 40-70 mph window

What to log -- specific pulls I need

Pull Set A -- 3rd gear WOT sweeps (primary data)

- **Gear:** 3rd
- **Start RPM:** 3000 (clutch fully out, rolling)
- **End RPM:** 7000-7500 (cut 500 RPM before rev limiter -- don't hit the fuel cut on data runs)

- **Throttle:** 100% hammered from start to end, no lift
- **Count:** 3 pulls minimum (so I can see if knock is repeatable or random)
- **Cool between pulls:** 30-60 seconds cruise at low load after each pull so IAT settles back near ambient and ECT doesn't climb

Pull Set B -- 4th gear WOT sweeps (load validation)

- **Gear:** 4th
- **Start RPM:** 3000
- **End RPM:** 6500-7000
- **Throttle:** 100%
- **Count:** 2 pulls
- **Why:** 4th loads the engine harder than 3rd at the same RPM -- MAP readings are higher, so any fuel/timing issue is more visible

Pull Set C -- Part throttle cruise (drivability + knock at light load)

- **Gear:** 5th or 6th
- **RPM:** 2000-3500
- **Throttle:** 25-50%, steady cruise for 30-60 seconds
- **Why:** most real-world driving is here. If there's knock in cruise, it will show up in this log.

Pull Set D -- Idle + tip-in

- **Stopped at idle** for 60 seconds after warm-up (proves idle stability)
- **Light tip-ins** from idle to 3000 RPM in neutral, a few times (tests throttle tip-in fuel and trims)

What to note in a separate text file (I'll create pull-notes-YYYY-MM-DD.txt alongside each log)

For each pull, I record:

- **Ambient temp** (outside thermometer / phone weather)
- **Fuel octane** (91 / 93)
- **Fuel brand + station** (different stations have different ethanol content and additive quality)
- **Odometer reading**
- **Wind direction** (for any 1/4-mile timing comparisons later)
- **Road surface** (flat, uphill, downhill -- loads differently)
- **Any audible ping** I heard during the pull (even if ECU didn't catch it)

After the drive

1. **Do NOT shut off FPM before saving the log** -- Datalog -> Save As, name it performance-tune-1_YYYY-MM-DD_pull-N.fpdl
2. **Also export to CSV** -- File -> Save As -> CSV (or use the Export function depending on FPM version). CSV is the format I send to Claude for analysis.
3. **Save logs to:** C:\Users\extra\OneDrive\Documents\FlashPro Calibrations\Datalogs\ (create folder if missing)
4. **Back at the desk:** paste CSV contents + pull notes to Claude for analysis.

Sensors that MUST be in the log

FPM's default datalog preset captures most of these, but verify in Tools -> Datalog Parameters before the drive:

Sensor	Why
RPM	x-axis of every analysis
MAP (kPa or bar)	Load axis -- shows when WOT enters open-loop
TPS (%)	Confirms actual throttle position -- proves I was WOT
VSS (mph)	Speed -- proves gear position
Gear (calculated)	Confirms gear during pull
IAT (degF)	Intake air temp -- knock driver
ECT (degF)	Coolant temp -- knock driver if too high
AFR (commanded)	Target the ECU is asking for
AFR (measured)	Actual lambda (from factory O2 or wideband if installed)
Short-term fuel trim	How much the ECU is correcting fuel
Long-term fuel trim	Baseline offset
Cam angle (VTC actual)	Proves cam is reaching its target
Cam angle (VTC target)	What the tune asked for
VTEC solenoid state	On/Off -- proves cam changeover
Ignition timing	Actual degrees delivered to plugs
Knock count	THE CRITICAL ONE -- any > 0 at WOT is a flag
Knock retard	Degrees the ECU pulled in response to knock
Injector duty cycle	Must stay <= 85% -- proves injector sizing is enough
Fuel pressure (if sensor available)	-- stock car may not have this
Battery voltage	Low voltage = injectors can't open as fast = lean spikes
Barometric pressure	Altitude compensation check

Offline analysis fallback

Since I won't have internet during the drive and for some time after: **the entire diagnostic playbook is written in 10-Hondata-FlashPro/datalog-analysis-guide.md** in this repo. It walks through what to look for in a log, in what order. I can self-triage the data before sending to Claude.

When I do get back online, I paste:

1. The CSV log (as text)
2. The pull-notes file
3. The weather + octane + any audible notes

And Claude produces specific cell-level tune corrections.

Comparison -- This Tune vs Safety Ceilings

Parameter	This Tune (Performance Tune 1)	My Reliability Ceiling	Pass/Fail
WOT AFR minimum	~12.54 (leanest cell)	>= 12.0	? FAIL

WOT timing maximum	30-33deg	<= 26deg (pump)	? FAIL
VTEC engage	4200 RPM	>= 4500	? FAIL (barely)
Rev limiter	8400	<= 8400	? PASS
Boost cut	Disabled	N/A (NA car)	? PASS
Full throttle shift	Disabled	Disabled	? PASS
Speed limiter	180 mph	removed	? PASS
Launch RPM	5000	sensible	? PASS
Idle (hot)	750 rpm	700-800	? PASS
Anti-lag	Disabled	Disabled	? PASS
Knock retard response	Aggressive (up to 25deg pull)	aggressive	? PASS

Overall: 3 FAIL / 8 PASS. The failures are exactly where my reliability rule demands margins -- AFR and timing on a 170k motor.

Datalog Analysis -- 2026-04-20 Pull Set 1

Tune applied during these pulls: performance tune.fpcal (this profile's subject) -- preserved snapshot in tune-profiles/archive/2026-04-20_performance-tune.fpcal.

Stock baseline reference: Factory Flash - Stock Tune.fpcal preserved at tune-profiles/archive/2026-04-20_Factory-Flash-Stock-Tune.fpcal.

Logs captured (all in C:\Users\extra\OneDrive\Documents\FlashPro Calibrations\):

- stock tune idle.csv -- 46 seconds of stock-tune idle, fully warm
- 3000 rpm tap.csv -- tip-in from idle up to ~4400 RPM, neutral, 24s
- 3rd Gear Pull 1.csv -- cruise only (no actual pull)
- 3rd Gear Pull 2.csv -- 2nd-to-6th gear pulls, 33s, peak 7568 RPM
- 3rd Gear Pull 3.csv -- 4th-to-6th gear pulls, 17s, peak 7621 RPM
- 4th Gear Pull 1.csv -- 4th-5th-6th sweep, 19s, peak 7367 RPM
- 4th Gear Pull 2.csv -- full sweep, 32s, peak 7389 RPM
- 6th Gear Cruise.csv -- 63s steady cruise, 2770-2997 RPM

Headline findings

Metric	Observed	Ceiling	Status
Knock count delta across ALL pulls	+5 events total (all Cylinder 1)	0	? FAIL
Max knock retard observed	10deg (4th Pull 2)	0	?? ECU SAVED THE MOTOR
Peak injector duty	74%	<= 85%	? PASS
Peak IGN actual at WOT	34deg @ 5300-5400 RPM high cam	<= 26deg (pump)	? FAIL
WOT AFR actual at high cam	12.0-13.4 range	>= 12.0	?? MARGINAL
WOT AFR commanded at high cam	12.54-12.80	>= 12.0	?? AT HONDATA FLOOR
ECT max	194degF	<= 210degF	? PASS

IAT max during WOT	91degF	--	? good ambient
Battery voltage range	13.7-14.5V	--	? charging healthy
STFT range at idle (stock tune)	+14 to +17	±10 typical	?? lean bias
STFT range at WOT	-15 to +10	--	? normal open-loop swings
LTFT	-3 to +1	±5	? tight

Knock event breakdown (the most important finding)

Knock count progression across the session:

Log	K.Count start	K.Count end	Delta	Cylinders
3rd Pull 1 (cruise)	1	1	0	--
3rd Pull 2	1	1	0	--
3rd Pull 3	1	1	0	--
4th Pull 1	1	2	+1	Cyl 1
4th Pull 2	3	7	+4	Cyl 1 (all)
6th Cruise	7	7	0	--

Pattern: 4th gear WOT consistently triggered knock on Cylinder 1 specifically. 3rd gear pulls did not, despite hitting the same RPM and almost the same MAP. The difference is engine load -- 4th gear pulls the engine harder at the same RPM, and that's where knock emerges.

Max K.Retard = 10deg in 4th Pull 2 = the ECU pulled up to 10deg of timing to protect the engine. This is the knock control system doing its job, but **the base map shouldn't be demanding this much correction**.

WOT AFR behavior -- actually tracking commanded pretty closely

Pulling WOT samples (TPedal >= 80%, VTEC on, 4500+ RPM) from all pulls:

RPM zone	AFR commanded	AFR actual	Comment
4500-5500	12.63-12.80	12.47-12.88	Within ±0.25 of target ?
5500-6000	12.54	11.76-12.47	Richer than commanded (safe)
6000-6500	12.54	12.35-13.23	Transient lean spikes to 13.2 -- flag
6500-7000	12.54	13.12-13.25	Consistently 0.6 lean of target
7000-7500	12.54	13.21-13.47	Leaner than commanded -- probably fuel pump/injector headroom limit

Observation: above 6500 RPM the actual AFR drifts 0.6-0.8 lean of commanded. This is the fuel system running out of headroom -- stock injectors + stock fuel pump can deliver the commanded, but only at lower RPM. At top-end the injectors are hitting 70-74% duty and the pressure/flow can't keep up. **This is hardware-limited, not tune-limited.** It's also why the Phase 6 DW300C pump and EV14 550cc injector upgrade exists in the mod plan -- this is the evidence.

Timing behavior -- aggressive but within knock limit (mostly)

Peak ignition timing during the pull set:

RPM	CAM	Peak IGN observed
3000	0deg (low cam)	26-30deg
4000	0deg (low cam)	30-32deg
4500	43deg (high cam)	32deg
5000	45-46deg (high cam)	32-33deg
5400	43-46deg (high cam)	34deg <- peak
5700	44deg (high cam)	33deg
6000	42deg (high cam)	32deg
6500	40deg (high cam)	31deg
7000	36-38deg (high cam)	30deg
7500	30-33deg (high cam)	29-30deg

34deg at 5400 RPM under full load on 91-octane pump is the spike that triggered knock in 4th gear. Tune commands it, no knock retard intervenes until Cyl 1 complains.

Idle baseline (stock tune -- separate log)

From stock tune idle.csv, fully warm:

- RPM: 730-772 (stable)
- ECT: 201-203degF (thermostat fully open -- within spec, slightly hot side of normal)
- IAT: 126-127degF (heat-soaked, expected at closed-throttle idle)
- **STFT: +14 to +17 <- notable**
- LTFT: 0-1
- K.Count: 0

STFT +14-17 at idle on the STOCK tune is a real finding. The ECU is adding 14-17% fuel to reach stoich at idle, which means the AFM is under-reporting airflow. Most likely cause on my car: the K&N Typhoon CAI flows more air than the OEM paper filter, but the stock tune's AFM calibration was built around the OEM filter's restriction curve. Not dangerous, but means any tune built on top of stock AFM scaling is starting from a 14-17% lean bias in that cell.

Implication: Performance Tune 1 probably inherited this same AFM calibration. If the tuner didn't rescale the AFM for the K&N, there's a built-in lean bias that explains some of the high-RPM AFR drift above.

Transient behavior -- 3000 RPM tap

From 3000 rpm tap.csv (neutral, tip-ins):

- Peak TPedal: 54% (mild throttle blip, not WOT)
- Peak RPM: 4406
- IGN spiked to 44deg briefly at ~2500 RPM low-cam off-throttle (normal overrun timing)
- K.Level peaked 62, K.Retard peaked 8deg -- **mild knock event during a throttle blip**
- K.Count: 0 change (retard was pre-emptive, not reactive)
- No sustained load so no real stress

Overall assessment

The tune is not blowing up the engine. The knock control is working and kept K.Retard to 10deg max on the worst event. Fuel system is at the edge of its capacity above 6500 RPM but not over it. Temps stayed safe. Injector duty is in the green. No catastrophic failure mode.

But the tune is violating the reliability rule. Any knock count > 0 on a 170k motor is a rule break. The rule isn't "detune until the ECU can compensate" -- the rule is "build a map that doesn't need the compensation."

Specific cell-level correction list

Priority ordered. These are the changes to make in FPM before the next flash:

1. Pull timing in the knock zone (highest priority)

- **Table:** Ignition high
- **Cam angle slice:** 40deg (and 30deg)
- **Cells:** RPM 5000-5750, load (MAP kPa) -14.8 through +0.2
- **Current value:** 30-34deg
- **New value:** 27-30deg (pull **3deg across the block**)
- Why: this is exactly where Cyl 1 knocked in 4th Pull 2

2. Add fuel to Cylinder 1 specifically

- **Table:** Cylinder trim (fuel)
- **Cylinder 1:** add **+2% to +3%**
- Why: Cyl 1 is the only cylinder showing knock events. A small enrichment bias protects it without affecting the others or drivability

3. Raise commanded AFR richness in the top-end (above 6500 RPM)

- **Table:** WOT lambda adjustment high
- **Cells:** RPM 6500-8000, WOT load
- **Current commanded:** 12.54
- **New commanded:** 12.20 (lambda 0.83)
- Why: Two reasons -- (a) aligns with my reliability rule of >= 12.0 pump floor with safety margin; (b) commanding richer helps offset the observed 0.6-0.8 lean drift from fuel system hardware limit. Actual AFR should land near 12.7-12.9 after the change.

4. Raise VTEC engage to 4500 RPM

- **Table:** VTEC window
- **Engage RPM:** 4200 -> **4500**
- Why: reliability rule. At 4200 the cam is making airflow the tune expects from higher up.

5. Recalibrate AFM for K&N Typhoon intake (stretch goal, not immediate)

- **Table:** AFM (sensors branch)
- **Issue:** stock AFM curve under-reads by ~14-17% at idle load based on STFT
- **Fix:** add +14% offset across the low-flow region, smaller correction up through mid-RPM
- This is a tuner's job on a dyno -- not a first-flash fix

Re-flash + re-test protocol

After applying corrections 1-4 above:

1. Flash the modified calibration as PRIMARY
2. Keep this current performance tune.fpcal as SECONDARY (escape hatch -- switch back with button combo if the new cal misbehaves)
3. Repeat exactly the same pull set (3rd Pulls 1-3, 4th Pulls 1-2, 6th Cruise) on the same road, same fuel, same weather if possible
4. Verify:
   - Knock count delta = 0 across the full session
   - K.Retard max <= 2deg (small corrections OK; 10deg is not)
   - Peak IGN <= 30deg anywhere
   - AFR floor >= 12.0 at all cells
5. If clean: this becomes the new baseline. Log it.

Revision Log

Date	Change	Reason
2026-04-20	Tune flashed as Sport map and driven	First real-world test
2026-04-20	Knock count detected	Root cause analysis initiated
2026-04-20	Full parameter transcription captured from 48 FPM screenshots	Build baseline record before any changes
2026-04-20	Profile document created	Dedicated home for this specific tune
2026-04-20	8 datalogs captured (idle, 3k tap, 3 x 3rd pulls, 2 x 4th pulls, 6th cruise)	First real-world data set
2026-04-20	Datalog analysis section added -- 5 cell-level corrections proposed	Knock isolated to Cyl 1 @ 5000-5700 RPM high cam, 34deg timing
2026-04-20	Archived snapshot of .fpcal tune file in tune-profiles/archive/	Version-preserve the exact tune that produced these logs

Next Actions

Immediate -- cell-level fix list ready

-> See fix-list-2026-04-20.md for the concrete per-cell corrections with annotated screenshots of every tab to edit.

Summary of corrections:

1. Pull 5-7deg timing at Ignition high 30deg and 40deg cam tabs, cells 5000-6000 RPM x MAP cols 5-7 (critical -- root cause of Cyl 1 knock)
2. Richen WOT lambda from 12.54 to 12.00-12.20 at 5500+ RPM in WOT lambda adjustment high
3. Bias Cyl 1 fuel trim +3% as a belt-and-suspenders until Phase 15 ignition refresh is done
4. Raise VTEC engagement 4200 -> 4500 RPM
5. Defer AFM recalibration for K&N Typhoon to the dyno session

Also worth investigating (not a tune edit): Cyl 1-only knock pattern on a 170k motor could be hardware-biased -- worn coil #1, worn plug #1, injector flow drift, or carbon asymmetry. The Phase 15 Ignition Refresh (OEM coils

30520-RWC-A01 + NGK plugs) addresses the two most likely causes directly. Add a compression + leak-down test to Phase 0 maintenance as a $50 sanity check before the next dyno pull.

Short-term (within next week)

1. Apply the fix list above, save as performance-tune-2.fpcal
2. Re-flash, repeat 3rd-gear pull A, verify K.Count.1 delta = 0
3. Archive the new .fpcal and start performance-tune-2.md profile doc

Medium-term (before Phase 2 mods)

1. Build a conservative **Daily** calibration as primary (25deg WOT max, 12.0 AFR floor, VTEC 4800, 7800 rev)
2. Corrected version of this tune becomes the **Sport** secondary calibration
3. Both calibrations load to FlashPro unit -> button-combo switching in-car

Long-term (after FBO hardware install)

1. Re-dyno on Skunk2 Alpha + HFC + cat-back to properly calibrate airflow curves
2. Revisit every WOT cell -- the airflow model will have changed
3. Consider E85/flex fuel calibration as third cal

Document created 2026-04-20 -- LxveAce + Claude Updated every time this tune's parameters change, every time a datalog reveals something new, every time a flash is performed

Performance Tune 1 -- Fix List (2026-04-20)

Tune file: archive/2026-04-20_performance-tune.fpcal **Datalogs:** datalogs/2026-04-20_pull-set-1/*.csv **Symptom :** Cylinder 1 knock during 4th-gear WOT pulls at 5000-5700 RPM, peak K.Retard 10deg. No other cylinder retarded more than 2deg.

This document is a concrete, cell-by-cell edit list for the next revision of the tune. Every change below has an image reference so you can open the exact tab in FlashProManager, click the exact cell, and type the new value.

Navigation reference

Full calibration tree (reference shot):

!Calibration tree navigation

Left pane in FlashProManager is grouped as: **Fuel · Ignition · VTEC · VTC · Closed Loop · Knock Control · Rev Limits · Sensors · Boost Control · Idle · Throttle · Misc · Traction Control · Ethanol**

Every correction below names the tree node to click.

Axis refresher (applies to every ignition + WOT-lambda table)

- **Rows = RPM** (500-9000 in 250-500 RPM steps).
- **Columns = MAP load** (inHg gauge). Column 1 = -26.4 (light cruise), Column 10 = +0.2 (WOT). WOT hits columns 6-10.
- **Cam-angle tabs** (All · 0deg · 10deg · 20deg · 30deg · 40deg) switch between 3D slices. The final engine output is interpolated between the two nearest cam slices based on actual VTC position.

The knock happened when VTC cam was high (30-40deg slice) at 5000-5700 RPM and load near column 5-6. That's where we edit.

Correction 1 -- Pull timing in the knock cells (critical, do first)

Root cause of the Cyl 1 knock. Datalog shows peak IGN = 34.4deg at 5500 RPM, high cam, load column 5-6. K20Z3 community-safe ceiling at this RPM/load/cam is 26-28deg. We are 6-8deg over, and it knocked exactly as predicted.

Tab 1: Ignition -> Ignition high -> 30 degrees cam angle tab

!Ignition high at 30deg cam angle

Cells to edit (row = RPM, col = MAP index):

RPM	Col 5 (-14.8)	Col 6 (-11.3)	Col 7 (-8.0)
5000	42.30 -> 36.00	38.00 -> 33.00	34.30 -> 30.00
5250	42.90 -> 36.00	37.90 -> 33.00	35.00 -> 30.00

5500	42.80 -> 36.00	38.00 -> 33.00	35.70 -> 30.00
5750	42.40 -> 35.50	37.70 -> 32.50	35.10 -> 29.50
6000	42.20 -> 35.00	37.00 -> 32.00	34.00 -> 29.00

Net effect: -6deg in the hottest cell, tapering to -5deg by 6000 RPM. Leaves timing well inside the community-safe envelope at high cam.

Tab 2: Ignition -> Ignition high -> 40 degrees cam angle tab

!Ignition high at 40deg cam angle

Cells to edit:

RPM	Col 5 (-14.8)	Col 6 (-11.3)	Col 7 (-8.0)
5000	45.00 -> 37.00	38.50 -> 33.00	33.00 -> 30.00
5250	42.00 -> 37.00	37.50 -> 33.00	34.50 -> 30.00
5500	44.50 -> 37.00	37.00 -> 33.00	34.50 -> 30.00
5750	43.70 -> 37.00	38.80 -> 32.50	35.90 -> 30.00
6000	43.50 -> 36.00	37.50 -> 32.00	33.50 -> 29.00

The 40deg cam slice had the most egregious values -- 44.50deg at 5500 RPM high load is pure fantasy for a 170k K20Z3 on 91 octane. -7deg here.

Verify the lower cam slices are already sane

Check these tabs quickly -- they should already be well under 30deg in the knock zone, but confirm visually:

- **Ignition high -> 0deg cam angle tab** (pre-VTEC, low-cam fallback):

!Ignition high at 0deg cam

- **10deg cam angle tab:**

!Ignition high at 10deg cam

- **20deg cam angle tab:**

!Ignition high at 20deg cam

If any cell in the 5000-6000 RPM x col 5-7 block exceeds 30deg in the 20deg tab, pull it to 28deg. The 0deg and 10deg tabs are below VTEC engagement in practice, so they're less critical.

Correction 2 -- Richen WOT target from 12.54 to 12.20 at high RPM

Datalog shows actual AFR drifts 0.6-0.8 lean of commanded above 6500 RPM, indicating the stock fuel system is at capacity. Running a leaner target on top of that is what tipped Cyl 1 past the knock line. Richen the target until the Phase 12 DW300C pump and EV14 550cc injectors are in.

Tab: Fuel -> WOT lambda adjustment high

!WOT lambda adjustment high

Cells to edit -- all cells RPM >= 5500, cols 5-10:

RPM	Col 5-10 current	Col 5-10 new
5500	12.54	12.20
5750	12.54	12.20
6000	12.54	12.10
6250	12.54	12.10
6500	12.54	12.00
6750	12.54	12.00
7000	12.54	12.00
7400+	12.54	12.00

Leave the 4000-5250 RPM block alone (already 12.80 -- fine).

Also check: Fuel -> WOT lambda adjustment low

!WOT lambda adjustment low

This is the low-cam slice (pre-VTEC). Should already be richer -- confirm no WOT cells above 12.50. If any are leaner, pull them to 12.20.

Correction 3 -- Bias Cyl 1 fuel +3% until hardware is verified

This is a band-aid, not a fix. It hides the knock symptom while we figure out whether Cyl 1 is short on fuel because of an injector, a coil, a plug, or just because the spark arrived late. It costs us nothing at this mileage to be conservative with #1 until the Phase 15 ignition refresh is done.

Tab: Fuel -> Cylinder trim -> Individual cylinder fuel trim

!Individual cylinder fuel trim

Change:

Cylinder	Current	New
1	0.0	+3.0
2	0.0	0.0
3	0.0	0.0
4	0.0	0.0

Revisit this cell **after** Phase 15 Ignition Refresh. If Cyl 1 stops knocking with timing reduction and fresh plugs/coils, remove the trim.

Correction 4 -- Raise VTEC engagement from 4200 -> 4500 RPM

Personal rule: VTEC stays at or above 4500 RPM until we've proven the low-cam tune is clean. Current

engagement sits at 4200 which is lower than my build doc allows. Small fix, big peace of mind.

Tab: VTEC -> VTEC window (the node directly under "VTEC" in the tree -- sibling of VTC, not a child).

Change VTEC engagement threshold from 4200 RPM to 4500 RPM. Disengagement hysteresis stays at whatever Hondata's default gap is (typically 500 RPM lower, so 4000 RPM disengage -- leave alone).

No screenshot of this tab exists in the 2026-04-20 capture set -- it wasn't one of the 48 tabs photographed. Navigate to it in FPM and match values. If the tab structure differs from what this doc implies, document it in a new screenshot next session.

Correction 5 -- AFM recalibration for K&N Typhoon intake (tuner-territory)

The stock AFM calibration expects stock intake airflow characteristics. The K&N Typhoon CAI moves more air per unit voltage, which shows up as LTFT +14% and STFT +14-17% at idle on the stock tune. The ECU is compensating via fuel trims, which masks the real MAP/MAF curve and degrades every downstream calculation (timing, VTC target, knock thresholds).

This is not a DIY cell edit. The right approach is:

1. Log a steady-state cruise pass on the current tune at multiple load/RPM points.
2. Use the STFT+LTFT delta to generate an AFM scaling correction curve.
3. Apply the correction in **Sensors -> AFM** (scalar or table, depending on FPM version).
4. Re-log and verify trims return to ±5%.

Defer this until the actual dyno session. A tuner with a sniffer will knock this out in 20 minutes. Attempting it blind from road data risks overcorrecting.

In the meantime: the +14% LTFT is annoying cosmetically but not dangerous -- the ECU is successfully compensating. The knock was not caused by this.

Related but not editing the tune file: Closed loop sanity

!Closed Loop -> Lambda

Noted for reference: Wideband voltage input is set to **None**. Once the AEM X-Series UEGO (Phase 17) is installed, set this to the correct analog input and populate the voltage->lambda table per the AEM spec sheet. Not relevant to the current knock fix -- just a bookmark for Phase 17 work.

Execute order

1. Open archive/2026-04-20_performance-tune.fpcal in FlashProManager.
2. File -> **Save As** -> performance-tune-2.fpcal (never edit the archive copy).
3. Apply **Correction 1** (the two ignition tables). This alone should stop the knock.
4. Apply **Correction 2** (WOT lambda high-cam richen).
5. Apply **Correction 3** (Cyl 1 +3% fuel trim).
6. Apply **Correction 4** (VTEC 4500 RPM engage).

7. Skip **Correction 5** until dyno day.

8. Flash, cold-start, verify idle trims haven't changed (baseline sanity).

9. Repeat pull set A (3rd gear 2500->7800) -- a single pull is enough to confirm K.Retard stays at 0 on Cyl 1. If it does, we're done with this revision.

10. Archive the new .fpcal in tune-profiles/archive/YYYY-MM-DD_performance-tune-2.fpcal, add a new row to tunes.md, and start revision 2's profile doc.

What I'm deliberately NOT changing

- **Rev limits** -- already at 8200/8400 which is fine for this build.
- **Idle speed** -- idle quality was acceptable in the datalog.
- **Cranking fuel** -- no cold-start issues reported.
- **Air-temperature fuel compensation** -- IAT stayed 95-110degF during logging, well inside the sane region of that table.
- **Knock sensitivity** -- the ECU's knock control *worked correctly*, catching Cyl 1 at 10deg retard. Don't blunt it. The only reason the engine survived this pull is that knock control was doing its job. Keep it exactly as is.

Last updated: 2026-04-20

Wideband AFR Gauge -- AEM X-Series

Status: Planned -- Install BEFORE first dyno tune, MANDATORY before E85

Why I Need This

FlashPro datalogs the car's existing narrowband O2 sensor, which is plenty accurate *in the closed-loop cruising range* (~14.0-15.0 AFR). It does NOT give useful data outside that range -- which means during WOT, hard transients, flex-fuel blending, or E85 operation, the ECU is essentially guessing and the narrowband reads pegged.

For any real tuning -- especially flex fuel -- I need a **wideband O2 sensor** that reads 10:1 to 20:1 AFR accurately. This is non-negotiable for E85 work. The tuner needs it, I need it to verify their work, and I want to see real-time numbers on the LattePanda touchscreen eventually.

Part I'm Going With

AEM X-Series Wideband UEGO Gauge (30-0300)

Field	Value
Part Number	AEM 30-0300
Sensor	Bosch LSU 4.9
Response Time	<1s full range (fastest in class)
Display	52mm (2-1/16") analog + digital
Output	0-5V analog (to FlashPro) + OBD-II serial
Accuracy	0.1 AFR
Price	~$230

Why the AEM X-Series

I compared this to:

Gauge	Price	Sensor	Notes
AEM X-Series 30-0300	~$230	Bosch LSU 4.9	My pick. Fastest response. Clean mount. Well-integrated with Hondata.
Innovate LC-2	~$200	Bosch LSU 4.9	Older gauge design. Slower digital display. Cheaper bracket quality.
Innovate MTX-L Plus	~$220	Bosch LSU 4.9	Good. Slightly uglier display.
AEM X-Series OBD-II	~$290	N/A (reads OBD-II narrowband)	Do NOT buy this one for tuning. It only displays the stock narrowband.
PLX Devices SM-AFR	~$180	Bosch LSU 4.9	Budget. Slower response. Fine for monitoring only.

The AEM X-Series is overwhelmingly the community's pick on 8thcivic and in the Hondata forum for FlashPro users. Not going to overthink it.

Installation -- Three Parts

1. Bung Welding

I need a 18mm O2 bung welded into the exhaust **upstream** of the catalytic converter. Two common options:

- **In the aftermarket header collector** -- best position, fastest response, most accurate. Skunk2 Alpha has a provision for this. This is what I'm doing.
- **In the mid-pipe before the cat** -- acceptable if I can't get to the collector.

Post-cat mounting is fine for long-term sensor life (less thermal stress) but less responsive to transient changes. I'll accept the hotter placement for tuning accuracy and just budget for a new sensor every 2-3 years of heavy driving.

2. Wiring to FlashPro

The AEM X-Series outputs a 0-5V analog signal that I'll wire to one of the FlashPro's analog inputs. This lets the wideband reading appear directly in the FlashPro datalog alongside RPM, timing, fueling, etc. -- which is what the tuner will actually use.

FlashPro analog input wiring: Hondata publishes a diagram in their FlashProManager help docs. The 8th gen SI FlashPro has two analog inputs available. Wideband goes on one; future flex fuel sensor goes on the other (or via the rear O2 signal wire -- see 19-Flex-Fuel-and-Fuel-System/).

3. Display Mounting

Where the gauge physically goes:

- **A-pillar pod** -- cleanest. Autometer A-pillar pod for 06-11 Civic, fits one gauge. ~$40.
- **Vent pod** -- uses the passenger-side dash vent. Doesn't block airflow to me. ~$30.
- **LattePanda display (later)** -- once the LattePanda is in, I can log the AEM reading to the touchscreen via the same 0-5V input through an Arduino ADC. The physical gauge becomes redundant. I'll keep it for analog redundancy.

Parts List

#	Item	Part/Source	Price
1	AEM X-Series Wideband	AEM 30-0300	~$230
2	18mm O2 bung (stainless)	Universal weld-in	~$10
3	Bung welding	Muffler shop, 15 min job	~$20-40
4	A-pillar gauge pod	Autometer 20400 (black)	~$40
5	18 AWG wire (power, ground, signal)	~15 ft each	~$10
6	Inline fuse holder + 5A fuse		~$5
	Total		~$315-335

Buy Links

Item	Source	Link
AEM 30-0300	Summit Racing	summitracing.com -- search AEM 30-0300
	Amazon	Search "AEM 30-0300"
	Hondata forum / Direct	aem.com

When To Install

Must be installed before:

- First dyno tune (tuner will require it -- most won't tune without one)
- Any E85 map work
- Any attempt to run richer/leaner AFR targets manually

Can be installed alongside:

- Headers (bung welded into header collector)
- Any exhaust work where the system is off the car

Replacement interval: The Bosch LSU 4.9 sensor is rated ~50,000 miles of exposed life. On E85 it can go sooner (ethanol is rough on the sensor). Budget $90 for a replacement sensor every 2-3 years of active use. Gauge itself is a buy-once item.

What This Enables

Once wired into FlashPro's analog input, the tuner can see real-time AFR alongside every other parameter in the datalog. This is what lets me:

- Verify the 93-octane tune is hitting target AFR in both valve-open and valve-closed cutout modes
- Validate the E85 calibration interpolates correctly as the flex fuel sensor reports ethanol content changes
- Catch lean-out conditions under WOT before they cause damage
- A/B compare before/after changes with real data instead of butt dyno

It's the single most important tuning tool on the car after FlashPro itself.

See Also

- 10-Hondata-FlashPro/ -- FlashPro setup that this feeds into
- 19-Flex-Fuel-and-Fuel-System/ -- ethanol sensor shares FlashPro analog inputs
- docs/baseline-logs.md -- logging plan

Last updated: 2026-04-18

Wideband AFR Install Guide -- AEM X-Series 30-0300

Status: Pending -- install tied to header install (Phase 2B / 1G)

The bung welding step physically happens DURING my Skunk2 Alpha header install (see 13-Headers/install-guide.md Phase H). The rest of this job -- sensor, gauge, wiring, FlashPro config -- can be done in the days after the header swap, before I hand the car over for a dyno tune.

Reality Check Before I Start

Mid-complexity install with two skill demands: welding (or hiring it out) and careful low-voltage wiring. The wiring is where most DIY installs fail and then spend a week chasing phantom AFR noise in datalogs.

- **Bung welding:** 30 min at a muffler shop, $20-40 cash. I'm NOT welding stainless exhaust tube myself unless I have a proper MIG/TIG setup with stainless filler. Leaky bung weld = lean-reading sensor = engine damage. Outsource this.
- **Wiring:** 2-3 hrs careful work. Crimp, solder critical joints, heat-shrink, twisted-pair signal, star-ground topology.
- **Total hands-on time:** ~3 hrs if welding is outsourced and header install is already happening.
- **Reliability note:** NEVER tune without a working wideband. Every fuel table decision depends on this sensor. A bad wideband reading is worse than no wideband reading -- it produces confidently wrong tuning changes.

Prerequisites / Cross-references

Must be done FIRST or AT THE SAME TIME

Requirement	Why
Header install in progress (see 13-Headers/install-guide.md Phase H)	The bung welds into the Skunk2 Alpha collector while the exhaust section is off the car. Retrofitting later is a pain.
FlashPro paired + first flash complete (see 10-Hondata-FlashPro/overview.md)	I need FPM working before I can enable and scale Analog Input 1.
FlashPro analog pinout confirmed (see 10-Hondata-FlashPro/hardware-software-deep-reference.md Section 2)	Wrong pin = no signal or ECU damage. Verify from Hondata's published diagram before wiring.

Must be done BEFORE any dyno tune

Rule	Why
Wideband MUST be installed, calibrated, and verified before dyno time (see 10-Hondata-FlashPro/tuning-workflow-and-maps.md)	The tuner will not (and should not) touch fuel tables without real AFR data. Showing up to a dyno session without a working wideband wastes the appointment.
Commanded vs actual AFR verified in datalog	If the wideband reads 0.5+ AFR off commanded at steady-state cruise on a stock-ish tune, something is wrong (exhaust leak, bung placement, sensor aging) and the tuner needs to know BEFORE pulling on throttle.

Shared analog input budget

My FlashPro (original, not FlashPro 2) has **2 analog inputs**. The allocation is locked:

- **Analog Input 1 -> Wideband AFR (this mod)**
- **Analog Input 2 -> Flex fuel ethanol sensor** (see 19-Flex-Fuel-and-Fuel-System/)

There is no third input. If I ever want oil pressure or boost logging later, I'm either upgrading to FlashPro 2 or reading those values through a separate datalogger.

Tools Required

For the bung weld (if DIY -- otherwise muffler shop does this):

- MIG or TIG welder capable of clean stainless welds
- 308L or 316L stainless filler rod/wire
- Step drill or hole saw sized for the 18mm bung
- Grinder with stainless-safe flap disc
- Welding helmet, gloves, leather

For sensor install:

- O2 sensor socket (22mm, with wire slot) -- already own from header job
- Torque wrench, 30-40 ft-lb range
- Copper anti-seize (NOT aluminum, NOT nickel -- copper handles exhaust temps better)

For wiring:

- 18 AWG automotive primary wire (red, black, and one signal color -- I'm using green)
- Quality crimp terminals (butt connectors, ring terminals for ground)
- Solder + flux (for critical joints only -- crimps for everything else)
- Adhesive-lined heat-shrink tubing, multiple diameters
- Heat gun (NOT a lighter -- uneven heat melts the adhesive unevenly)
- Wire strippers
- Multimeter (continuity + DC voltage modes mandatory)
- Add-a-fuse tap, mini blade style (FG2 interior fuse box uses mini blades per CLAUDE.md)
- Zip ties (high-temp rated for anything within 6" of exhaust)

Parts List

From overview.md, with quantities and placement notes:

#	Item	Part Number	Qty	Notes
1	AEM X-Series Wideband UEGO Gauge Kit	AEM 30-0300	1	Includes gauge, Bosch LSU 4.9 sensor, weld bung, bung plug, wiring harness
2	Extra 18mm O2 bung (backup)	Universal stainless weld-in	1	Kit usually ships with one -- this is my backup if the first one gets tacked in the wrong spot
3	A-pillar gauge pod	Autometer 20400 (black, 06-11 Civic)	1	Single 2-1/16" gauge

4	Mounting hardware for pod	Comes with pod	--	
5	Add-a-fuse tap (mini blade)	Littelfuse FHM200BP or similar	1	For ACC(A) fuse #8 7.5A switched 12V
6	Inline fuse holder + 5A fuse	Generic	1	Protects the gauge power wire past the tap
7	18 AWG primary wire	15 ft red, 15 ft black, 15 ft green	--	Signal run needs to be long enough to route properly
8	Copper anti-seize	Permatex 80078 or similar	1 tube	Exhaust threads only
9	Adhesive-lined heat-shrink assortment	Generic	1 pack	3/16" through 1/2" diameters

Kit contents in the AEM 30-0300 box: X-Series 52mm gauge, Bosch LSU 4.9 sensor (pre-calibrated), 18mm weld-in bung + plug, wiring harness with AO1/AO2 outputs, sensor extension harness, mounting bezel.

Step-by-Step

Phase A: Bung Welding (during header install -- see 13-Headers/install-guide.md Phase H)

This happens while the Skunk2 Alpha + Berk HFC assembly is OFF the car during my header session.

1. **Mark the bung location.** On the Berk HFC section, 12-18" downstream of the Skunk2 Alpha collector outlet. The sensor must be:
   - **Post-collector** -- merged flow from all primaries, laminar, cylinder-averaged AFR
   - **Post-HFC** -- high-flow cat, so sensor sees exhaust hot enough for the LSU 4.9
   - **NOT inside any catalytic converter substrate** -- cat chemistry skews O2 readings
   - **In a straight section** -- bends cause swirl and noisy readings
2. **Angle the bung UP.** At least 10 degrees above horizontal (ideally 15-45 degrees) so condensate drains AWAY from the sensor tip on cold starts. The LSU 4.9 will crack if water hits it while heated. Top or upper side of the pipe, NEVER bottom.
3. **Drill the hole.** Step drill sized to snug-fit the bung shank. Deburr.
4. **Weld the bung on.** Muffler shop, $20-40, 30 minutes. Hand them the bung, point at the marked spot, tell them "post-cat, angled up, full penetration weld, no leaks."
5. **Install the bung plug.** The AEM kit ships with a threaded plug. **LEAVE THE PLUG IN during initial header heat cycles and the first 50-100 miles of header break-in.** The LSU 4.9 is fragile to thermal shock and curing ceramic/oil fumes on a fresh header fire-up. Plug first, sensor after bed-in.

Phase B: Sensor Install (after header break-in, before first tune)

6. **Use the NEW LSU 4.9 from the AEM kit.** Do NOT reuse an old narrowband or used wideband. The kit sensor is pre-calibrated to the gauge electronics.
7. **Apply copper anti-seize to the sensor threads -- SPARINGLY.** Thin smear on threads only. Do NOT get anti-seize on the sensor element (the slotted tip) -- it will contaminate the sensor permanently. Copper only; aluminum anti-seize degrades at exhaust temps.
8. **Hand-thread the sensor into the bung** until snug. If it doesn't start smoothly, back out -- cross-threading wrecks the bung.

9. **Torque to 33 ft-lb** with the O2 socket. See docs/torque-specs.md.
10. **Route the sensor harness** away from hot exhaust. Minimum 2" clearance. Zip-tie to rigid exhaust-adjacent points (subframe, brackets) -- NEVER to the exhaust itself or to anything moving independently of the engine.
11. **Route the harness into the engine bay** following the stock O2 harness path to the firewall grommet.

Phase C: Gauge Mounting

12. **Pick a mount location.** A-pillar pod (Autometer 20400), dash-top, or vent pod. A-pillar is cleanest.
13. **Temporary mount first.** Before committing to drilling or permanent wiring, sit in the driver's seat with the gauge held in the prospective location and verify: readable in under 0.5s glance, does NOT block view of road/gauges/mirrors, cable run to firewall is feasible.
14. **Run wiring through the existing firewall grommet** on the driver's side near the brake booster. Do NOT drill a new hole -- new firewall holes invite water and noise.
15. **Install the gauge in the pod** per the pod instructions.

Phase D: Wiring

Every wire decision on this gauge matters because the 0-5V analog signal going to FlashPro is low-level and susceptible to EMI from the ignition system.

16. **Power -- switched 12V.** Tap the ACC(A) fuse #8 (7.5A) in the interior fuse box per CLAUDE.md. Mini blade add-a-fuse: load side keeps the original 7.5A fuse, hot side gets a NEW 5A fuse protecting my circuit. **DO NOT tap constant 12V (BACK UP fuse #20)** -- the LSU 4.9 heater pulls ~2A and will drain the battery overnight.
17. **Ground -- chassis, not battery.** Clean, paint-free bare metal stud near the gauge mount (FG2 has one behind the driver's dash near the kick panel). Sand paint, star washer, ring terminal. Routing ground to battery negative directly introduces alternator/starter noise.
18. **Signal wire -- 0-5V analog output to FlashPro Analog Input 1.** AEM gauge harness "AO1" -> FlashPro AI1 signal pin (exact pinout per 10-Hondata-FlashPro/hardware-software-deep-reference.md Section 2). Use 18 AWG, keep short, route AWAY from ignition coils, spark plug wires, and alternator harness.
19. **Signal ground -- star ground, NOT daisy chain.** This is where most DIY installs fail. The AEM harness has a dedicated analog/signal ground wire separate from chassis ground. Route it back to the SAME reference point as the FlashPro's analog input ground. All analog grounds meet at ONE point. Daisy-chaining introduces ground-loop drift and noise in the datalog.
20. **Shield wire (if present).** If the AEM harness has a foil shield: ground it at the GAUGE END ONLY, leave the FlashPro end floating. Grounding both ends creates a ground loop.
21. **Twisted pair the signal + signal-ground wires.** ~1 turn per inch for the entire run. Cancels inductively-coupled EMI from ignition coils.
22. **All crimps and critical joints get heat-shrink** -- adhesive-lined tubing. For the signal wire joints, I'm soldering AND heat-shrinking, not just crimping. A corroded crimp in the signal path introduces resistance that skews voltage.
23. **Fuse the power wire** with an inline 5A fuse after the add-a-fuse tap -- belt-and-suspenders.
24. **Multimeter check before powering on:** continuity YES from gauge power to fuse tap load side, gauge ground to chassis stud, signal output to FlashPro AI1. Continuity NO from gauge power to ground (no short) and signal to ground (no short).

Phase E: FlashPro Configuration

25. **Plug laptop into FlashPro, connect FlashPro to OBD2 port, key to ON (engine off is fine for config).**
26. **Open FlashProManager.** Load my current calibration (or read from ECU first if I haven't -- see 10-Hondata-FlashPro/hardware-software-deep-reference.md Section 2 for the Read from ECU workflow).
27. **Navigate to Parameters -> Analog Inputs -> Analog Input 1** (menu path may vary by FPM version).
28. **Enable AI1.** Check the box or flip the toggle.
29. **Set the scale.** The AEM X-Series 30-0300 default gasoline calibration is:
    - **0V = 10.0 AFR**
    - **5V = 20.0 AFR**
    - Linear in between
    - Verify this against the AEM 30-0300 manual -- some revisions use a slightly different curve, and I want the FPM setting to EXACTLY match what the gauge is outputting.
30. **Label the input "Wideband AFR"** so it shows up clearly in the datalog column header.
31. **Save the calibration** with a descriptive filename (e.g. ADDED_WIDEBAND_AI1_20260420.fpc).
32. **Flash the updated calibration to the ECU.** ~20-45 seconds.
33. **Verify in Parameter Monitor.** Open FPM's live parameter view with the key on (engine can be off -- gauge still outputs a voltage). The Analog Input 1 parameter should show a plausible AFR value. With engine off and sensor in free air, it should read near 20.9 (atmospheric oxygen equivalent).

Phase F: Calibration Verification

34. **Free-air calibration check.** LSU 4.9 ships pre-calibrated but I verify. Remove sensor from bung (plug back in), expose to ambient air 60 sec. Gauge should read ~**20.9 AFR** (atmospheric oxygen). FPM Parameter Monitor AI1 should match. If off by >0.5 AFR, run AEM's free-air calibration per the 30-0300 manual.
35. **Reinstall sensor.** Re-torque 33 ft-lb. Set the bung plug aside -- I'll need it if I ever pull the sensor.
36. **Start the engine.** Let idle until sensor heats up (AEM heater countdown, ~30-60 sec).
37. **Verify closed-loop cruise AFR.** Steady-state 2000-3000 RPM, flat road. Gauge tracks ~**14.7 AFR** stoichiometric. Datalog AI1 matches within 0.2 AFR.
38. **Verify WOT AFR.** Brief WOT 3000->redline in 3rd on a safe open road. Gauge drops into **12.0-13.0 AFR** (enrichment). Datalog AI1 matches gauge within 0.2 AFR.
39. **Gauge vs datalog AI1 mismatch** means scaling error in FPM or voltage drop from a bad ground. Chase it before calling wiring done.

Phase G: Sanity Check Before Tune

Before I hand the car over for a dyno session, I want to see a clean datalog that confirms everything is working.

40. **Record a 10-minute mixed-drive datalog.** Idle, cruise, part-throttle, one or two brief WOT pulls.
41. **Export to CSV** (see 10-Hondata-FlashPro/hardware-software-deep-reference.md Section 2).
42. **Compare "Commanded AFR" vs "Analog Input 1 (Wideband AFR)" in the log:**
    - Closed-loop cruise (light throttle, warm): commanded 14.6-14.8, wideband tracks within 0.3 AFR. 100-200ms lag is normal (sensor is downstream of injection).
    - Warmup (cold): commanded richer (13-14), wideband tracks.
    - WOT: commanded drops to low 12s on stock/base tune, wideband follows.
43. **Red flags that mean STOP and DEBUG before tuning:**
    - Wideband leaner than commanded at cruise -> exhaust leak upstream, or bad bung location
    - Wideband richer than commanded at cruise -> sensor aging/contaminated or free-air cal drifted

- Noisy/spiky readings (>0.5 AFR swings at cruise) -> EMI, bad ground, or non-twisted-pair run
- Reads pegged at 10.0 or 20.0 -> wiring issue or dead sensor

First Drive Checklist

Before I consider this mod "done":

- [ ] Gauge reads sensible AFR at idle (14-15 warm, possibly 12-13 cold)
- [ ] Gauge reads ~14.7 at steady cruise
- [ ] Gauge reads 12-13 at WOT
- [ ] FPM datalog AI1 matches gauge within 0.2 AFR at multiple operating points
- [ ] No EMI noise spikes visible in datalog during ignition events
- [ ] No CELs thrown by the car (the wideband is a passive add -- shouldn't trigger any stock sensor DTCs, but I'm confirming)
- [ ] Sensor harness is not rubbing or melting on any exhaust surface
- [ ] Gauge does not obstruct driver sightline
- [ ] Add-a-fuse tap is seated firmly, fuse box cover closes

Common Mistakes

1. **Bung downstream of HFC but upstream of an exhaust leak.** A pinhole leak between header and sensor admits fresh air = false "lean" reading. Tuner adds fuel to compensate; fixing the leak later leaves the engine dangerously rich. Seal every flange and smoke-test before trusting the wideband.
2. **Signal ground tied to battery negative.** Alternator noise on the 0-5V signal. Star-ground to chassis with FlashPro analog ground as the star point.
3. **Running the header during break-in without the bung plug installed.** Bare sensor exposed to curing coating/residual oils shortens sensor life. Plug in for 50-100 miles of heat cycles.
4. **Mounting the gauge in the driver sightline.** Fails the "glance in under 0.5s" test. Temporary mount first, always.
5. **Mounting the bung on the bottom of the pipe.** Condensate pools and cracks sensor ceramic. Top/upper side only, angled up 10+ degrees.
6. **Using aluminum anti-seize.** Breaks down at exhaust temps and can contaminate the sensor. Copper only.
7. **Skipping the free-air check.** 60 seconds of verification before install saves a wasted tuning session.
8. **Twisted-pairing power and ground instead of signal and signal-ground.** Common reversal -- the twisted pair protection belongs on the low-voltage signal, not the robust power wire.

Integration with Dyno Session

The tuner will want this gauge INSTALLED, VERIFIED against a datalog, and REPORTING TO FLASHPRO AI1 before the car goes on the dyno. Showing up with a wideband that "I just put in this morning" wastes the first 30 minutes of paid dyno time on my verification work.

When booking the dyno session:

- Confirm the tuner accepts AEM X-Series 30-0300 as the logging wideband (almost all do)

- Confirm they read via FlashPro datalog AI1 (if they insist on their own dyno-room wideband, I still want mine logging so I have the data in my FPM format for later)
- Offer to send a sample datalog ahead of time -- demonstrates clean wiring

See Also / Cross-references

- overview.md -- parts list, rationale, buying links
- ../13-Headers/install-guide.md Phase H -- the bung welding step happens HERE, during the header install. Single most important cross-reference on this page.
- ../10-Hondata-FlashPro/hardware-software-deep-reference.md Section 2 -- FlashPro analog input pinout, AI1/AI2 allocation, FPM config workflow
- ../10-Hondata-FlashPro/tuning-workflow-and-maps.md -- how the wideband feeds the daily/sport tune workflow, and why no tune happens without it
- ../19-Flex-Fuel-and-Fuel-System/ -- takes Analog Input 2, completing the FlashPro analog budget
- ../docs/torque-specs.md -- 33 ft-lb for the wideband sensor, 25 ft-lb for any exhaust flanges disturbed during install
- ../docs/baseline-logs.md -- the datalog protocol that depends on this sensor being live

Last updated: 2026-04-20

Valved Exhaust -- My DIY 3" QTP Cutout Bypass

Status: Planned (Not Yet Purchased)

The Idea

I want to keep my OEM resonator and muffler for quiet daily driving, but have a valve-controlled bypass that dumps exhaust before the resonator/muffler -- basically a straight-pipe experience at the flip of a switch. A relay controls the valve, tied to my Hondata FlashPro calibration selection (daily = closed/quiet, sport = open/loud).

No bolt-on valved exhaust exists for the 8th gen Civic SI. I looked everywhere. This has to be a DIY build using universal components, which honestly is fine -- gives me more control over the design.

My Design

Exhaust Path (Updated: 3" Dump with Step-Down)

```
4-1 Header -> High-Flow Cat -> 3" Pipe -+-> [QTP QTEC30 CUTOUT] -> 3" Dump Pipe -> Exit Tip
                                        |
                                        \-> [3" to 2.5" Reducer Cone]
                                                    |
                                            2.5" Pipe -> Resonator -> Rear Muffler -> Stock Tailpipe
```

How It Works

- **Valve CLOSED (Daily):** Cutout butterfly is shut. 100% of exhaust flows through the reducer cone, down to 2.5", through the resonator and muffler. Quiet, stock-like.
- **Valve OPEN (Sport):** Cutout butterfly opens. Exhaust takes the path of least resistance -- dumps out the 3" straight pipe. Bypasses resonator and muffler completely. Very loud.
- **Catalytic converter stays in the flow path at all times** (emissions legal)

Pipe Diameter Strategy

Section	Diameter	Reason
Header collector to cutout	3"	Maximum flow for sport mode. Matches 4-1 header collector output.
Sport dump pipe (cutout to exit)	3"	Full flow, no restriction when valve is open
Daily path (cutout to resonator)	3" -> 2.5" reducer	Steps down to match resonator/muffler inlet size
Resonator -> muffler -> tailpipe	2.5" / stock	Retains stock components, no modification needed

Why I'm going 3" on the dump side:

- My 4-1 header collector outputs at 2.5-3". Carrying 3" to the dump maximizes flow for the sport path.
- On an NA K20Z3 at 220-240 WHP, 3" won't hurt low-end torque (common myth). Exhaust gas velocity matters less than minimizing restriction at this power level.
- 3" sounds deeper and more refined than 2.5" when the valve is open
- Gives me headroom if I ever go forced induction

Why I'm stepping down to 2.5" for the muffled path:

- The stock resonator and muffler inlets are ~2-2.5". You can't feed 3" into them.

- A gradual reducer cone (3" to 2.5", ~6" long) creates a smooth transition with minimal turbulence
- The muffled path doesn't need maximum flow -- it's the quiet mode

Note: The FG2 SI has ONE resonator (small canister, ~under rear seats) and ONE rear muffler (large canister, under trunk) -- NOT two mufflers.

The Valve I'm Going With

QTP QTEC30 -- 3" Electric Exhaust Cutout (~$240-260)

- Stainless steel butterfly valve, 12V DC gear motor
- **3" bore** (upgraded from the originally planned 2.5" QTEC25)
- Motor mounts on a bracket away from the hottest pipe area
- Opens/closes in 2-3 seconds
- Proven product, thousands of installs -- this is the move

QTP Cutout Positioning

I'll place the cutout **after the high-flow cat, before the resonator**, with at least 14" between the header collector and the cutout (QTP's thermal recommendation for motor protection).

Control Methods

1. **Phase 1:** Simple DPDT toggle switch on the dash ($5)
2. **Phase 2:** LattePanda Arduino GPIO -> relay -> QTP motor (tied to tune selection)
3. **Manual override toggle retained** for Phase 2 (works even with PC off)

Parts List with Buy Links

#	Item	Product	Price	Link
1	Electric Cutout Valve	QTP QTEC30 (3")	~$217-269	Summit Racing -- $217
				Amazon
				QTP Direct
				Real Street Performance -- $232
2	3" Stainless Pipe	Mandrel-bent sections for dump pipe	~$40	Muffler shop / Amazon
3	3" to 2.5" Reducer	Stainless reducer cone (~6" long)	~$20	Muffler shop / Amazon
4	2.5" Adapter Pipe	Connect reducer to stock resonator inlet	~$15	Muffler shop
5	3" Exhaust Tip	Stainless rolled tip for dump exit	~$30	Amazon / Summit Racing
6	Gaskets & Clamps	3" and 2.5" band clamps, gaskets	~$25	Amazon / Summit Racing
7	Exhaust Hangers	Rubber hangers + brackets	~$15	Amazon / AutoZone
8	Wiring & Control	Toggle switch, high-temp wire, connectors	~$50	Amazon
9	Heat Protection	DEI fire sleeve, silicone tape	~$40	Amazon / Summit Racing
	Total		~$452-504	
	Muffler shop labor (if not welding)		$150-250	

See Also

- brainstorm.md -- Full research, alternatives I looked at, valve options, heat/waterproofing
- purchasing.md -- Original detailed parts list (to be updated for 3" spec)

Last updated: 2026-04-16

Valved Exhaust Install Guide -- QTP QTEC30 Custom 3" Build

Status: Planning

Not purchased yet. This guide is the plan I'll work from once parts are in hand and I've booked the muffler shop appointment. I'll update timings and gotchas after the actual install.

Reality Check Before I Start

Custom build. No bolt-on valved exhaust exists for the 8th gen Civic SI (2006-2011). I researched everything in brainstorm.md -- Varex, Fi, Armytrix, IPE, Corsa, Borla -- nothing fits the FG2. The DIY QTP cutout path isn't a shortcut, it's the only path.

Time budget: Muffler shop fab 4-6 hours (drop off morning, pick up end of day) + DIY electrical 3-4 hours at home next day + 30-minute first drive. Budget a full weekend.

Cost: ~$485 parts (see purchasing.md) + $150-250 shop labor.

Why this matters: This mod unlocks the Daily/Sport duality the entire tuning philosophy is built around (see 10-Hondata-FlashPro/tuning-workflow-and-maps.md). Every tune after this ships as two calibrations. Getting the cutout right is the foundation for Tune #1.

Prerequisites -- Do These First

MUST BE DONE BEFORE HEADERS

Hard ordering rule. Install the cutout BEFORE the Skunk2 Alpha header (-> 13-Headers/). The cutout position depends on downstream geometry -- if headers go in first, the Skunk2 Alpha + Berk HFC can shift the cat flange enough to put the cutout in the wrong spot. Install on the stock manifold + cat now (stable known geometry); headers tie into the existing cutout flange later. The 13-Headers guide enforces this. Do not reverse.

FlashPro First Flash Complete

FlashPro paired, community base map loaded, stock cal saved. I need a known-good calibration running before the cutout install so closed-loop O2 can correct the small AFR shift between valve states on a single map until Tune #1. See 10-Hondata-FlashPro/.

Interior Fuse Box Access

Switch needs a **switched 12V source** so the cutout can't accidentally open with the car sitting in the driveway. Per CLAUDE.md: driver-side lower left dash, behind removable panel. Target fuse **#8 ACC(A) -- 7.5A** (live in ACC and RUN only). Mini blade add-a-fuse; hot side gets a new 5A fuse for the trigger circuit, load side takes the original 7.5A. The motor does NOT run off this fuse -- that's a dedicated 10 AWG run from the battery. ACC(A) only carries the sub-1A relay-coil trigger.

Other Cross-References

Prerequisite	Why
10-Hondata-FlashPro/ first flash done	Base map available for leak-check drive
11-Wideband-AFR/ installed or at least ordered	AFR datalog pre- and post-install confirms nothing got worse
09-Cooling-System/ refresh ideally done	Don't stack unknowns on an aging cooling system
Stock cat health verified (no P042X codes)	Post-cat bypass assumes a healthy cat stays in the flow path

Do NOT Do This Before the Cutout

- **Headers.** Already covered above. Reordering breaks the geometry.
- **Intake manifold or bored TB.** Those are Tune #2 hardware -- way downstream of this step.

Parts List

Full details with buy links in purchasing.md. Quick summary of what needs to be on the shelf before muffler-shop day:

Core exhaust hardware (the shop needs these in-hand)

#	Part	Spec
1	QTP QTEC30 -- 3" electric cutout valve	Stainless butterfly, 12V DC gear motor, ~5-8A draw, polarity-dependent direction
2	3" stainless mandrel-bent straight, ~12"	Dump pipe main section
3	3" stainless mandrel 45-degree bend	Dump pipe exit angle toward rocker panel
4	3" -> 2.5" stainless reducer cone, ~6" long	Smooth transition into muffled path
5	2.5" adapter pipe	Connects reducer cone to stock resonator inlet
6	3" stainless rolled exhaust tip	Vibrant or Magnaflow universal, rolled for water runoff
7	3" and 2.5" exhaust gaskets (3 total)	Standard graphite-faced
8	3" and 2.5" band clamps (4)	For any non-welded joints the shop chooses
9	Rubber exhaust hangers + mount brackets (2)	For the dump pipe and the new reducer-cone section

Electrical (DIY side, I handle these at home)

#	Part	Spec
10	Normally-open 12V automotive relay, 30A-rated	Bosch-style 5-pin or equivalent. Critical: NO, not NC -- details in Phase D.
11	30A in-line fuse holder + 15A fuse	On the power feed from battery to relay
12	10 AWG silicone-insulated wire	Power + ground runs to relay. ~10 ft total.
13	14 AWG silicone-insulated wire, 20 ft	Motor leads to cutout + trigger wire through firewall
14	SPST illuminated rocker switch	Dash or center console mount. 12V-lit so I can see state from the driver's seat.
15	DEI fire sleeve, 3 ft	Any wire passing within 12" of exhaust
16	Deutsch DT 2-pin connector sets (2)	Weatherproof disconnect near cutout motor
17	Dielectric grease	On all underbody connections

18	3/8" split wire loom, 10 ft	Routing protection along frame rail
19	Crimp terminals, assorted ring + spade	For relay + battery terminals
20	Heat-shrink tubing assortment	Over every crimp, no electrical tape under the car
21	Fuse tap for ACC(A) mini blade	Interior fuse box connection for trigger
22	Zip ties + wiring clips	Proper routing, no dangling wire

Why 10 AWG on the power run, not 14: The QTEC30 pulls 5-8A peak while actuating against exhaust pressure. With ~8 ft round-trip wire length and 15A fuse protection, 10 AWG keeps voltage drop under 2% under load. 14 AWG works on paper but sags with heat and cycling. Undersized wire is non-negotiable failure territory.

Tools Required

At the muffler shop (they have these)

- TIG or MIG welder with stainless capability
- 3" and 2.5" pipe notcher / bead roller
- Pipe bender (3" mandrel)
- Pressure tester with smoke or soap solution
- Lift + fabrication table

At home (DIY electrical)

- Multimeter (continuity + 12V DC)
- Crimper -- ratcheting, for insulated terminals
- Heat gun (for shrink tubing)
- Wire strippers 10-22 AWG
- 10mm + 12mm sockets (for battery terminal, relay bracket)
- Drill + step bit (for switch mounting hole in dash blank -- NO firewall drilling, see below)
- Deutsch DT connector crimper (proper tool, not generic crimps)
- Torque wrench, 10-40 ft-lb range (for band clamps + battery terminal)
- Fuse tap install tool (usually comes with the add-a-fuse kit)
- Trim removal tools (for dash panel + under-dash access)
- Inspection mirror + headlamp (underbody wire routing)
- Zip-tie tensioner

Do not substitute. Deutsch connectors crimped with generic terminals fail open at the worst time. A proper DT crimper is $40 and owns the job for life.

Muffler Shop Planning

The shop does the welding I can't. I'm still the engineer -- showing up with a drawing, the parts, and a written spec sheet.

Finding the right shop

Looking for: stainless welding capability (mild-steel weld on stainless = galvanic corrosion in 2 years), custom fab reputation (track/show build photos, not just "swap on a Magnaflow"), willingness to work off my drawing (any "let us do it our way" pushback = wrong shop), and pressure testing included.

Design brief I bring to the shop

One page, printed:

1. **Final layout drawing** -- header -> cat -> 3" pipe -> cutout -> Y-branch -> dump pipe + reducer to stock path. Same diagram from overview.md.
2. **Cutout position** -- after the high-flow cat, before the resonator, **minimum 14" from header collector flange to cutout body** (QTP thermal spec to protect the motor).
3. **Dump pipe** -- 3" straight, 45-degree bend exiting forward of the rear axle, rolled tip. **Not toward any tire.**
4. **Reducer cone** -- 3"->2.5", ~6" long, smooth transition into OEM resonator inlet.
5. **Hangers** -- add 2 rubber hangers: cutout body + dump pipe near exit.
6. **Ground clearance** -- dump exit no lower than the lowest stock exhaust point (measured at resonator).
7. **Wideband bung -- NOT YET.** Goes in the Skunk2 Alpha collector during header install (13-Headers/). Premature bung = wrong location.

Quote + agreements

- $150-250 expected. Under $100 = they missed something, walk. Over $350 = second quote.
- Confirmed design in writing, same-day turnaround if possible, 30-day weld warranty, pressure test included, electrical is mine (they do not wire the motor).

Step-by-Step

Phase A: Pre-Install Planning (at home, day before shop)

1. **Measure the stock mid-pipe** from cat-back flange to resonator inlet. Record length, bend clock positions. Photograph from 4 angles -- reference for the shop.
2. **Plan cutout orientation.** QTEC30 motor mounts **up and slightly outboard** -- not dangling where debris can hit it, not facing the hottest pipe. Sketch it on the design brief.
3. **Plan the wire routing.** Battery -> relay (engine bay) -> through existing driver-side firewall grommet (NO new hole) -> along center tunnel under carpet -> switch. Separate path: relay -> frame-rail grommet or subframe -> cutout motor connector.
4. **Pre-make the motor harness at the bench.** Deutsch DT terminated, fire sleeve already applied. Install day becomes plug-and-route, not underbody crimping.
5. **Fuse tap check.** Confirm ACC(A) = fuse #8. Pull and meter: key OFF = dead, key ACC = hot. If the chart doesn't match, stop and verify.

Phase B: Remove Stock Mid-Pipe (morning of shop day)

6. Let the car sit overnight -- cold exhaust bolts back out, hot ones shear.
7. Lift front onto 4-ton jack stands on subframe pinch welds. Never work under a jack alone.
8. **Disconnect battery (-) terminal.**
9. Penetrant on cat-to-B-pipe and resonator flange bolts. Let sit 15 minutes.

10. **Cat-to-B-pipe flange:** 2 spring-loaded bolts, 25 ft-lb per docs/torque-specs.md. Compress springs and unbolt -- rag over the springs so they don't launch.
11. **Resonator flange** at the rear -- confirm with the shop whether they want the full section including the resonator or just the mid-pipe.
12. Support on the hangers, slide back to clear the cat flange, remove.
13. Inspect flange faces for pitting. Order fresh donut gaskets if bad.
14. Hand mid-pipe + parts kit + design brief to the shop. Do not hover.

Phase C: Muffler Shop Fabrication (shop work, 4-6 hours)

Documented so I know what to verify at pickup.

15. Cut a 3" stainless section from the cat flange to calculated cutout position. Minimum 14" cat-to-cutout for heat isolation.
16. Fabricate Y-junction (or tee) downstream of the cutout -- one leg is the 3" dump pipe, the other is the reducer cone feeding the OEM path. Geometry: path of least resistance with valve open = dump pipe.
17. QTEC30 welded inline, motor oriented up-and-outboard per brief. V-band clamps at flanges recommended (serviceability -- I can pull the cutout later without cutting pipe).
18. Dump pipe: 3" straight + 45-degree bend + rolled tip. Exit clears rocker panel, points down-and-rearward, away from tires.
19. Reducer cone 3"->2.5" welded to the muffled side, feeding stock resonator inlet with a fresh 2.5" gasket.
20. Two new rubber hangers: cutout body + dump pipe near exit.
21. **Pressure test.** Cap cat flange, pressurize, soap all welds. Bubbles = re-weld before it goes on the car.
22. Reinstall. Cat-to-B-pipe spring bolts **16 ft-lb**, resonator/muffler flange bolts **25 ft-lb**, cutout flange bolts **25 ft-lb**. Fresh donut gaskets.
23. **Pickup inspection** -- flashlight under the car. Cutout motor up/outboard. Minimum 3" ground clearance at dump exit. No weld splatter on wires/lines/tank. All hangers engaged. V-bands torqued. Motor pigtail protected and reachable.
24. **Drive home sounds stock.** Valve is unpowered -- and because I'm using a normally-open relay wiring, no power = valve stays CLOSED. Exact fail-safe behavior I want.

Phase D: Electrical Wiring -- The Fail-Safe Is Everything (3-4 hours, garage)

The single rule in this phase:

> **Relay must be NORMALLY-OPEN (NO). Default state -- power off, switch off -- must be VALVE CLOSED.**

If it's wired NC, any failure (blown fuse, dead battery, chafed wire, dead switch) opens the valve. Loud exhaust at 2 AM because of a corroded crimp is not a failure mode I'm accepting. Fail-safe direction is QUIET. With NO wiring driving the motor OPEN when energized, loss of power = valve returns to closed. The QTEC30's gear motor is non-back-drivable, so the wiring also gives a CLOSE-direction pulse at switch-off -- but default with no power is still closed.

D1. Battery + main power run

25. Battery already disconnected (step 8).
26. **Mount the relay** on a metal bracket within ~12" of the battery. Dry, shielded from spray, not above the exhaust manifold side.
27. **10 AWG from battery (+) through a 15A in-line fuse within 6" of the battery terminal** to relay common.

Ring terminal, crimped + heat-shrunk. Torque battery nut.

28. **10 AWG ground from relay ground pin** to a scraped-paint chassis point with star washer. Not a painted bolt, not a body seam.

29. Route power + ground along factory wire channels. Zip-tie every 6-8". Clear of exhaust and moving suspension.

D2. Motor leads -- POLARITY MATTERS

30. QTEC30 motor is polarity-dependent. **Bench-test off the car** with a 12V source. Note which lead is positive when the valve OPENS. Label both leads.

31. Route 14 AWG motor leads (fire sleeve within 12" of exhaust) from relay output, through frame-rail grommet, along subframe, to the motor pigtail.

32. **Deutsch DT 2-pin connector** at the motor, dielectric grease in the cavity. Positive lead goes to the motor terminal that drives OPEN.

33. Wrap motor housing with self-fusing silicone tape for a secondary water barrier.

D3. Firewall pass-through -- NO DRILLING

34. **Use the existing driver-side firewall grommet** where the OEM harness enters. Lift the lip, push an awl through, feed the single 14 AWG trigger wire (signal only -- power stays in the engine bay), reseal with silicone.

35. **Drilling a new firewall hole is forbidden.** Any unsealed penetration starts rust. The factory grommet has capacity for one thin signal wire.

D4. Switch trigger circuit

36. Route trigger wire along the driver-side center tunnel under the carpet to the switch.

37. Switch gets switched 12V from the **ACC(A) add-a-fuse tap** (fuse #8 -- 7.5A original stays in the load side; 5A new fuse on the tap side for the trigger circuit).

38. Switch output -> relay coil (+) via the 14 AWG trigger wire through the firewall. Coil (-) to chassis ground.

39. Switch ON = coil energized = relay throws = motor drives OPEN.

40. Switch OFF or key removed = coil dead = motor drive disconnected = valve returns to / stays CLOSED.

D5. Protect every connection

41. Heat-shrink on every crimp, no exceptions.

42. Dielectric grease on every underbody connection.

43. Fire sleeve on every wire within 12" of exhaust.

44. Split loom on every other wire for debris protection.

45. Zip-tie every 6-8".

Phase E: Switch Mounting

46. **Location:** center console blank panel or unused dash switch cutout (the FG2 has a blank switch panel near the steering column on some trims).

47. Drill with a step bit sized exactly for the SPST rocker body. Deburr.

48. **Weatherproof 12V rocker, illuminated.** LED pin wired so it lights only when the switch is ON -- dashboard confirmation of commanded valve state.

49. **Label it.** Label-maker, "EXHAUST SPORT" or "VALVE OPEN." Legible. If someone else ever drives the car, they should know what that switch does.

Phase F: Initial Test (key ON, engine OFF)

50. Reconnect battery, torque terminal to spec.
51. Key to ACC, switch OFF. Nothing should click. LED dark. Valve silently closed.
52. Switch ON. Relay clicks within 50ms. Motor runs 2-3 seconds then stops at the QTEC30 internal limit. LED lights.
53. Switch OFF. Relay clicks off. Motor runs briefly closing. LED dark.
54. **Key-OFF fail-safe check.** Leave switch in ON, cycle key OFF. ACC(A) feed drops, relay drops, valve closes. If valve stays open with the key pulled, the fail-safe is wired wrong -- stop and fix before driving.
55. **Visual valve confirmation.** Someone flips the switch while I watch under the car with a flashlight. Open = butterfly parallel to flow (dump pipe open). Closed = perpendicular (dump sealed, all flow to OEM path).

Phase G: First Drive Test

56. Base calibration already flashed and confirmed.
57. Switch OFF, start engine. Idle sounds stock (OEM resonator + muffler fully in the path). AFR idle 14.5-14.7 closed-loop. **Walk around listening for leaks** -- especially new cutout flanges, reducer cone welds, cat-to-B-pipe joint. Ticking or hissing = leak.
58. Drive gently -- parking lot, then side street, switch still OFF. No drone, no new resonance, no cabin smell.
59. At 25 mph on a quiet street, switch ON. Valve opens, note transforms immediately to deep straight-piped 3". Quick 2nd-gear WOT blip, 5-7 seconds, not extreme. AFR at WOT ~12.0-12.5 on base cal, maybe 0.2-0.3 leaner than closed-valve WOT from reduced backpressure -- closed-loop trims handle it on a single map.
60. Switch OFF at speed. Valve closes in 2-3 seconds, note returns to stock-quiet.
61. Home, cool down, lift, flashlight underneath. Soot around the cutout = leak (small amount normal, shouldn't grow). Hangers seated. Wire routing still off exhaust.
62. **AFR datalog for both valve states** on FlashPro. Save. Baseline for Tune #1.

Tuning Integration

Single base map handles both valve positions via closed-loop O2 correction. Closed = more backpressure = slightly richer baseline; open = less backpressure = 0.2-0.5 leaner at WOT. Closed-loop trims compensate at part-throttle; at WOT, open-loop fueling is identical in both states, so the small AFR delta is acceptable on a single map for the first few weeks. No dyno just for the cutout.

Dual-map tuning comes at Tune #1 (see 10-Hondata-FlashPro/tuning-workflow-and-maps.md), after both the cutout AND the headers + Berk HFC are installed (the header AFR shift is much bigger than the cutout shift, and I'm not paying for dyno time twice). Tune #1 outputs two calibrations:

- **Daily:** valve closed, conservative timing, AFR 12.0-12.2, stock VTEC, stock rev limit.
- **Sport:** valve open, max-safe timing (0 knock verified + wideband confirmed), AFR 11.8-12.0, VTEC lowered to 4800-5200, rev limit +100-200.

Neither ships without hitting every reliability ceiling in the tuning workflow doc. "Reliability over power, always" applies to Sport too -- it's "stop leaving free power on the table," not "remove safety margins."

First Drive Checklist

- [ ] Engine starts normally, no new codes

- [ ] Closed mode: idle sounds stock, no resonance, no cabin exhaust smell
- [ ] Open mode: dramatic 3" straight-piped note, no weird rattle or drone
- [ ] Switch transitions smoothly in 2-3 seconds, both directions
- [ ] Key OFF with switch ON -> valve physically closes (fail-safe verified)
- [ ] Switch LED tracks state correctly (lit = open, dark = closed)
- [ ] No exhaust leaks at any new flange or weld (soapy-water test the flanges)
- [ ] AFR datalog captured for closed state (quiet commute, 20 min)
- [ ] AFR datalog captured for open state (spirited back-road drive, 20 min)
- [ ] Daily driving comfort acceptable (closed mode = can have a phone call at 70 mph)
- [ ] Sport driving satisfaction (open mode = grinning at every throttle stab)
- [ ] No new check engine codes after 100 miles of mixed driving

Common Mistakes to Avoid

Mistake	Consequence	My safeguard
Normally-closed (NC) relay instead of NO	Valve fails OPEN when power drops. Wakes the neighborhood when battery dies overnight.	Specifying NO relay in the parts list, verifying coil state with a multimeter before wiring.
Swapped motor polarity	Switch says OPEN, valve drives CLOSED (and hits its stop, burning out the motor).	Bench-test motor direction before install. Label leads. Verify visually in Phase F.
Undersized power wire (14 AWG on the main run)	Voltage drop under load, motor stalls partway open, gear motor burns out at 60% travel.	10 AWG minimum, non-negotiable.
Drilling a new firewall hole	Rust ingress within 2 years. Insurance-annoyance. Pride hit.	Use OEM grommet. One signal wire fits.
Cutout mounted too low	First speed bump tears the dump pipe off.	Ground clearance spec in design brief. Shop measures before final weld.
Skipping the shop pressure test	Invisible weld leak detected 500 miles later when the cat code trips.	Refuse pickup until pressure test completed in front of me.
Dump exit pointed at the rear tire	Blows debris + hot gas into tire sidewall. Tire damage + ugly soot.	45-degree bend points forward-and-down, outside of tire arc.
Not saving AFR datalogs from both valve states	First dyno session starts blind -- tuner has to re-characterize what I already lived through.	Two 20-min drives, save both logs, bring them to Tune #1.
Running wire along exhaust without fire sleeve	Insulation melts in a season. Short-to-ground. Blown fuse minimum, fire maximum.	DEI fire sleeve on anything within 12" of exhaust. Period.
Butt splices under the car	Corrode open in one winter. Switch stops working.	Deutsch DT connectors at every underbody disconnect. Heat-shrink over every crimp.

Future Integration: LattePanda + Arduino

Phase 2 replaces the dashboard toggle with software-driven control via the LattePanda 3 Delta + Arduino co-processor:

- **Tune-linked valve state.** FlashProManager on the LattePanda reports the active cal name; a small Python script reads it and commands Arduino GPIO (Daily = LOW/closed, Sport = HIGH/open).

- **Arduino GPIO drives the same relay coil I'm using now.** Phase 1 wiring doesn't change -- I'm just replacing the manual switch as the trigger source with an Arduino pin through an optoisolator.
- **Manual override retained.** Physical rocker is wired in PARALLEL with the Arduino output, not in series. LattePanda off, crashed, booting -- the switch still works. Non-negotiable: I will not make a physical feature depend on a computer booting.
- **Still NO relay, still fail-safe quiet.** LattePanda dies + switch OFF = valve closed. Either source commands open = valve opens (harmless redundancy). Both closed = closed.

Phase 2 belongs to 20-Permanent-LattePanda-Install. Phase 1 ships fully functional on the manual switch alone.

See Also

- overview.md -- Parts rationale, pipe diameter strategy, design decisions
- brainstorm.md -- Research, alternatives I ruled out, heat/wiring considerations
- purchasing.md -- Full parts list with vendor links and prices
- ../13-Headers/install-guide.md -- **Install this AFTER the cutout. The install order is enforced.**
- ../10-Hondata-FlashPro/tuning-workflow-and-maps.md -- The dual-map framework this unlocks + reliability ceilings that bound Sport-map tuning
- ../11-Wideband-AFR/ -- AFR logging setup. Needed for AFR delta characterization between valve states.
- ../docs/torque-specs.md -- All exhaust flange and battery terminal torque values
- ../CLAUDE.md -- Owner/vehicle reference + fuse box map (ACC(A) fuse #8)

Last updated: 2026-04-20

Headers -- I'm Going with the Skunk2 Alpha 4-2-1

Status: Research Complete, Not Yet Purchased

My Pick

Skunk2 Alpha 4-2-1 Header with Stepped Primaries -- after a ton of research, this is honestly the best-performing header you can actually buy for the FG2.

Field	Value
Part Number	412-05-1930 (FG2 Coupe)
Design	4-2-1 with stepped primaries (1.625" -> 1.75")
Material	304 stainless steel, TIG-welded
Collector Outlet	2.5"
Cat Included	No -- I'll pair it with a Berk Technology HFC
Header Price	~$700-800
Total (header + HFC)	~$1,000-1,200
Availability	In production, readily available from authorized retailers

Buy Links

#	Item	Product	Price	Link
1	Header	Skunk2 Alpha 412-05-1930	~$700-800	Hybrid Racing
				HARDmotion
				JHPUSA
				Skunk2 Direct
				King Motorsports
2	High-Flow Cat	Berk Technology HFC (BT1101-HFC-MET)	~$300-400	Berk Technology (direct)
				Vivid Racing
				Ballade Sports
				Speed Freaks USA
	Total		~$1,000-1,200	

Why I Chose the Skunk2 Alpha

Stepped Primaries -- The Key Differentiator

The Alpha's **1.625" to 1.75" stepped primary tubes** (dimensions are community-reported from dyno and fitment data -- Skunk2 doesn't officially publish primary tube dimensions) are what separate it from every other header on the market:

- Starts narrow -> maintains exhaust gas velocity at low-mid RPM (keeps scavenging strong)
- Opens up -> flows more volume at high RPM when the engine pushes more exhaust

- PLM and DC Sports use constant 1.625" primaries -- this design difference accounts for the 5-8 WHP gap

Every Header for the FG2, Ranked

Header	Design	Peak WHP Gain	Price	Available?	My Verdict
Skunk2 Alpha	4-2-1 stepped	10-18	$700-800	Yes	This is the one
PLM Tri-Y V2	4-2-1 constant	8-13	$350-500	Inconsistent	Budget alternative
DC Sports	4-2-1 constant	8-12	$400-550	Yes	Decent mid-tier
Hytech 4-1	4-1 merge collector	12-20	N/A	No (defunct ~2014)	Unobtainable
Megan Racing	4-2-1	6-10	$250-350	Intermittent	Not recommended
OBX	4-2-1	5-8	$150-250	Yes	Avoid

No JDM brands (Spoon, Toda, Mugen, J's Racing) ever made a header for the 8th gen SI. **No reputable 4-1 header** exists for the FG2 in 2025-2026.

Why I'm NOT Going 4-1 (For This Build)

I did a deep dive on 4-1 vs 4-2-1 (see 4-1-header-research.md). Here's what I concluded:

A properly designed 4-1 with merge collector would make **2-5 more peak WHP above 7000 RPM** but **lose 3-8 lb-ft of torque from 3000-5500 RPM.**

For my build specifically:

- My **Skunk2 Ultra Street manifold** already provides excellent top-end airflow. A 4-1's top-end advantage is diminished when paired with a manifold that already breathes well up top.
- My **3" valved exhaust** eliminates downstream backpressure. The 4-2-1's slight flow disadvantage at peak RPM is negated.
- I spend far more time in the **3000-6000 RPM** range than at 8000+ RPM. Area under the curve > peak numbers.
- Professional K-series tuners overwhelmingly recommend the Alpha for full bolt-on street builds.
- The 4-1 only becomes the right choice with race cams, ported heads, and a build that lives above 6500 RPM. That's not my car.

Synergy with My Build

The Skunk2 Alpha + Skunk2 Ultra Street intake manifold are complementary:

- Ultra Street's medium-length runners build mid-range torque
- Alpha's stepped primaries maintain exhaust velocity across the same RPM range
- Together: **broad, fat torque curve from 3000-8000 RPM** with strong pull to redline
- Most commonly tuned header on the 8th gen SI -- my Hondata tuner will have extensive experience with this exact combo

Full Bolt-On Power Expectations

Configuration	WHP (Dynojet)	WTQ (Dynojet)
Stock baseline	175-185	130-138

Current (intake only, no tune)	178-188	132-140
Full bolt-on + Hondata tune	220-235	160-170

To push beyond 240 WHP I'd need aftermarket cams (Skunk2 Stage 2), race fuel, or both. Not in the plans right now.

Catalytic Converter Pairing

Cat	Type	Price	Notes
Berk Technology HFC	200-cell high-flow, bolt-on	~$300-400	Most popular pairing, retains emissions legality
PLM High-Flow Cat	200-cell, bolt-on	~$150-250	Budget option
Test pipe (catless)	Straight pipe	~$50-100	Off-road only. Not street legal. Requires Hondata CEL fix.

I'm going with the Berk. Most popular pairing with the Alpha for a reason -- it just works.

Stock Exhaust Clarification

The FG2 SI has **one resonator and one muffler** -- NOT two mufflers. The resonator is a small canister welded into the mid-pipe (~under rear seats). The rear muffler is the large canister under the trunk.

My Install Order

Headers go in AFTER the valved exhaust cutout is working and tuned.

1. Install QTP cutout, get both tunes working (valve open and closed)
2. Install Skunk2 Alpha header + Berk HFC
3. Retune both Hondata calibrations on dyno (strongly recommended)
4. Verify AFR in both valve positions with new headers
5. Re-torque all connections after 100 miles of heat cycling

Installation Notes (170k Miles)

At 170k, I'm expecting seized studs. Here's my plan:

- **PB Blaster the manifold studs 3-5 days before install** -- heat cycle + spray daily
- Use double-nut method for stud removal, NEVER vise grips
- Have new M10x1.25 stainless studs and copper-plated flange nuts ready
- Budget 6-10 hours DIY or 3-5 hours professional
- **I'm leaning toward professional install** due to seized stud risk ($300-500 labor). If a stud snaps and I don't have the tools to deal with it, it becomes a much bigger problem.
- May need to lower subframe ~1" for clearance during manifold removal

See Also

- brainstorm.md -- Full exhaust system research, stock layout, pipe diameters, all headers compared, sound character
- 4-1-header-research.md -- Deep dive on 4-1 merge collector design, scavenging theory, Hytech analysis, 4-1

vs 4-2-1 trade-offs

- Future: purchasing.md -- Complete parts list (at purchase time)
- Future: install-guide.md -- Step-by-step instructions

Last updated: 2026-04-16

Header Install Guide -- Skunk2 Alpha + Berk HFC

Status: Planning -- install deferred until exhaust cutout working and tuned

Reality Check Before I Start

This is the single scariest job on my entire build. It's not the technical difficulty -- the actual swap is intermediate-level wrenching. It's the **170k miles of seized exhaust manifold studs**. If I snap one and can't extract it, I go from "weekend job" to "head off the engine and machine shop visit." I'm doing 3-5 days of prep work specifically to avoid that outcome.

I'm strongly considering paying a shop the $300-500 for this one. The ROI on shop labor is: if they snap a stud, THEY own drilling and extracting it. If I snap a stud DIY, I own it. A Honda specialist shop has seen this a thousand times and has the tools.

Decision point before I start: I get my final quote from a Honda specialist shop (one that's done Skunk2 Alpha installs on 06-11 SIs). If under $500 and they have good reviews for header work specifically, I hire it out. If not, DIY and I accept the risk.

Prerequisites -- Don't Skip

Cross-references -- things that need to be done FIRST or AT THE SAME TIME

Do First	Why
Phase 0 maintenance complete	Clean engine baseline means if something goes wrong after install, it's header-related not some other lurking issue
Valved exhaust (QTP cutout) installed and working	Cutout position changes based on header/HFC flow; installing headers first locks me into a downstream geometry I may regret
Wideband AFR bung welded into header collector	Do this at the muffler shop DURING the header installation. Retrofitting a bung into an installed header is a pain.
Cooling system refresh	Headers add heat. I want fresh cooling headroom before this goes in.
PB Blaster on manifold studs -- 3-5 days ahead	Seized-stud prevention, starting well before install day

Do AT THE SAME TIME (car is already apart)

Also Do	Why
Replace upstream O2 sensor	New sensor is ~$100 OEM Denso. Old sensor threads will unscrew with the manifold. Anti-seize new threads.
Replace cat-to-B-pipe donut gasket	It's going to get disturbed anyway.
Apply ceramic coating or thermal wrap to header	Skunk2 Alpha runs HOT in the FG2 bay; protects alternator harness over time (community-reported issue). Jet-Hot or Swain coatings are ~$200-350. DEI titanium wrap is ~$50 as an alternative.

Timeline

Day ?5 through Day ?1: PB Blaster Prep

Day	Action
?5	Apply PB Blaster or Seafoam Deep Creep to all 8 manifold studs from engine-bay side. Let soak overnight.
?4	Start engine, warm to full temp. Shut off. Apply more penetrant while hot (thermal cycling breaks corrosion bonds). Let cool overnight.
?3	Daily cycle: penetrant -> heat -> penetrant -> cool
?2	Same
?1	One more application. Remove heat shield bolts (they seize too) -- spray those too.
Day 0	One final spray 30 min before starting.

Have extra PB Blaster on hand. Not expensive, not worth being caught short.

Day 0: Install

Total Time Budget

- **DIY (first-timer on high-mileage car):** 8-10 hours -- plan a full Saturday, nothing else that day
- **DIY (experienced, no seized studs):** 4-6 hours
- **Shop (Honda specialist):** 3-5 hours

Tools Required

Critical:

- Torque wrench (1/2" drive, 30-150 ft-lb range)
- 14mm, 17mm, 19mm sockets + matching wrenches
- **O2 sensor socket** (22mm, with wire slot) -- $10 at AutoZone, worth owning
- **Breaker bar** (18"+) -- for the manifold-to-head nuts
- Stud extractor set -- Irwin Hanson Bolt-Grip set -- for if (when) a stud breaks
- **Double-nut setup** (2x M10x1.25 nuts) -- standard stud removal technique
- Penetrating oil (PB Blaster + Seafoam Deep Creep, both)
- Jack + 4-ton jack stands -- subframe may need to drop
- Mechanic's creeper + a good droplight or headlamp

Also useful:

- Inspection mirror + flashlight
- Parts magnet
- Drip pan (coolant may come out if thermostat housing is disturbed)
- Sharpie for marking bolt positions before removal

Parts List (Everything I Should Have Before Starting)

#	Item	Source	Status
1	Skunk2 Alpha header 412-05-1930	Hybrid Racing / HARDmotion / JHPUSA	--

2	Berk Technology HFC BT1101-HFC-MET	Berk direct / Ballade Sports	--
3	Exhaust manifold gasket, OEM 18115-RRB-A01	Honda dealer only -- aftermarket blows	--
4	ARP exhaust manifold stud kit 208-4301	Summit Racing	--
5	Copper-plated flange nuts M10x1.25 (8x)	ARP 200-8321 or equivalent	--
6	Donut gasket, cat-to-B-pipe	O'Reilly, AutoZone	--
7	Fresh upstream O2 sensor (Denso 234-9049)	RockAuto	--
8	O2 sensor anti-seize (Permatex, NOT copper)	AutoZone	--
9	Copper anti-seize (exhaust threads)	AutoZone	--
10	Skunk2 O2 extender wire	K Series Parts	--
11	Heat protection for header (wrap OR ceramic coating)	DEI / Jet-Hot	--

Step-by-Step

Phase A: Lift and Access (30-45 min)

1. Let engine fully cool (overnight if possible).
2. Disconnect battery (-) terminal. Exhaust work near alternator wiring = bad surprises without disconnection.
3. Lift front of car, set on 4-ton jack stands on subframe pinch welds. NEVER work under a jack alone.
4. Remove driver-side front wheel and wheel well liner for access.
5. Remove engine-bay strut brace (if mine ends up fitting post-coilovers, not relevant until then -- see 18-Strut-Bar/).
6. Remove the heat shield over the stock manifold (4-6 bolts, they're seized, more PB Blaster).

Phase B: Disconnect Upstream O2 (10 min)

7. Unclip the upstream O2 sensor connector from the harness (up by the ECU side of the bay).
8. Using the O2 socket, break the sensor loose -- if it spins, great. If not, more penetrant and come back in an hour.
9. Unscrew fully, set aside. I'll install the NEW O2 in the new header.

Phase C: Disconnect Cat from Stock Manifold (15-20 min)

10. Under the car, find the 2 spring-loaded bolts connecting stock manifold to the cat inlet flange.
11. Penetrant, then compress and unbolt. The springs go flying if I'm not careful -- hold a rag over them.
12. Set the cat hanging on its own brackets/rear bolts (don't disconnect the cat-to-B-pipe flange yet unless needed for clearance).

Phase D: Remove Stock Manifold -- The Scary Part (30-90 min, depending on studs)

13. **Inspect each of the 8 manifold-to-head nuts.** Penetrant one more time. Let sit 10 minutes.
14. Start with the EASIEST-looking nut (typically a middle one, more protected from heat cycling).
15. Work SLOWLY. Back and forth -- loosen 1/8 turn, tighten 1/16, loosen 1/4, tighten 1/8. This walks the corrosion out without shearing the stud. PATIENCE.
16. If any nut feels like it's stripping or the stud is turning out with the nut -- STOP. Apply more penetrant. Try heating the nut with a small butane torch (careful of nearby plastic/wiring).

17. Repeat for all 8. If a stud breaks: STOP. Move to extraction procedure (below).
18. Once all 8 nuts are off, wiggle manifold free. It may need to be flipped to come out past the cat converter area.

Phase E: Install New Studs + Prep Head Surface (20-30 min)

19. **If using ARP studs:** remove any remaining stock stud stubs with the double-nut technique. Then install ARP studs per ARP's thread-prep instructions.
20. Scrape old gasket material off the head mating surface. A plastic razor is safer than metal. Do not gouge the aluminum head.
21. Clean surface with brake cleaner + lint-free cloth.
22. Check head mating surface for flatness with a straight edge -- if warped, this is a much bigger job. Expected outcome: flat.

Phase F: Install Skunk2 Alpha (30-45 min)

23. Place new OEM 18115-RRB-A01 gasket over the studs, making sure orientation is correct (the intake/exhaust port shapes are asymmetric).
24. Lift Skunk2 Alpha into position -- heavy and awkward. A second pair of hands helps; otherwise, a floor jack with a 2x4 block supports from below.
25. Thread all 8 copper-plated nuts hand-tight first. All 8 before torquing.
26. Torque in 3 passes, star pattern:
    - Pass 1: 10 ft-lb
    - Pass 2: 18 ft-lb
    - Pass 3: **23 ft-lb (final spec)**
27. Install Skunk2 O2 extender wire, then the fresh Denso O2 sensor with anti-seize on threads only (NOT on the sensor element).
28. Torque O2 sensor to 33 ft-lb.

Phase G: Connect Header to Berk HFC (20-30 min)

29. Slide HFC onto header collector. Verify O-ring/gasket between them is correctly positioned (Berk ships with spec gasket).
30. Install flange bolts. Torque to 25 ft-lb.
31. Connect HFC outlet to existing exhaust (or to QTP cutout / dump setup if exhaust is already done).
32. Fresh donut gasket at every flange junction. NEW crush washers on any banjo fittings.

Phase H: Wideband Bung Install (IF doing this during session)

33. While the system is on the car but before sealing up, mark the wideband bung location in the collector (post-HFC is the usual spot -- ask tuner's preference).
34. Remove collector-to-HFC section, take to muffler shop for bung welding. 30 minutes at a local shop, $20-40.
35. Reinstall with fresh gasket.

Phase I: Reassembly (30-45 min)

36. Reinstall heat shield (or install thermal wrap / ceramic coating if planned).
37. Reinstall wheel well liner, wheel.

38. Reconnect battery (-).
39. Torque lug nuts to 80 ft-lb.

First Start + Break-In

40. Key to ON (not start) for 5-10 seconds. Check for fuel or coolant leaks visible in engine bay.
41. Start engine. Let idle 2 minutes. **Walk around car sniffing for exhaust leaks.** A slight smell of burning off coating is NORMAL for 10-20 minutes on a new header.
42. Listen for any ticking / hissing from manifold area = leak = torque down again or re-gasket.
43. Check all flange junctions for leaks again.
44. **First 50 miles: drive gently. NO WOT pulls.** Let everything heat cycle and seat.
45. After 50 miles: re-torque all exhaust bolts/nuts -- they'll have relaxed slightly with heat cycles.
46. After 100 miles: one more re-torque pass. Should be stable by then.

Post-Install Tuning (Cross-reference 10-Hondata-FlashPro/tuning-workflow-and-maps.md)

DO NOT WOT the car on the stock calibration or old tune after a header install. The airflow characteristics have changed significantly; running aggressive timing on stale maps is how engines get hurt.

Safe procedure:

1. Drive gently home from install.
2. Flash a conservative base map (community Skunk2 Alpha + intake map, or previous-session calibration with timing pulled 2 deg as safety buffer).
3. Datalog 3 sessions of mixed driving (~1-2 days of commuting). Confirm zero knock, stable AFR, reasonable fuel trims.
4. Book dyno session (Tune #1 -- see tuning-workflow-and-maps.md).
5. Tuner dials in daily + sport maps on dyno with wideband + knock monitoring.

If a Stud Breaks -- Recovery Path

Case 1: Stud snapped flush with the head or above

- **Extractor kit:** Irwin Hanson Bolt-Grip (spiral-flute extractor). Drill slightly, tap the extractor in with a hammer, back out with a wrench. Success rate ~60-70% if corrosion isn't extreme.

Case 2: Stud snapped BELOW the head surface

- Much harder. Need to drill the remaining stud, extract with easy-out. If easy-out breaks off inside (hardened steel), I'm at the machine shop.
- Machine shop cost: $200-400 to drill and re-tap if clean, $500-1000+ if they have to pull the head.

Case 3: Threads are damaged in the head

- **Heli-Coil** or **Time-Sert** repair kit. Re-threading the hole with a larger insert. Restores to full strength if done right.

- DIY-able with a careful hand and $60 kit, but professional shop is safer.

This is the scenario the $300-500 shop quote is insuring against. If I DIY and hit Case 2 or 3, I'm paying the shop anyway AND I've lost my weekend.

See Also

- overview.md -- the parts I picked and why
- brainstorm.md -- complete research, all header options compared
- 4-1-header-research.md -- why I didn't go 4-1
- ../12-Valved-Exhaust/ -- must be installed first
- ../11-Wideband-AFR/ -- bung goes in during this install
- ../09-Cooling-System/ -- fresh cooling before more heat
- ../10-Hondata-FlashPro/tuning-workflow-and-maps.md -- the tune this enables
- ../docs/torque-specs.md -- all torque values used here

Last updated: 2026-04-18

Intake Manifold & Throttle Body -- Skunk2 Ultra Street + Bored Stock TB

Status: Planned (Not Yet Purchased)

What I'm Going With

Skunk2 Ultra Street Intake Manifold (307-05-0600) paired with a **bored stock K20Z3 DBW throttle body (66mm).**

Why I Chose the Ultra Street

The stock RRC manifold (on my 06-11 SI) has long runners designed for low-end torque and fuel economy. Once I add headers, exhaust, and a tune, the RRC becomes the airflow bottleneck. The Ultra Street shortens the runners, enlarges the plenum, and moves the powerband upward -- exactly what I need for a full bolt-on NA K20Z3 targeting 220-240 WHP.

Spec	Stock RRC	Skunk2 Ultra Street
Runner Length	~12.5"	~8.5-9"
Plenum Volume	~2.5L	~1.8L (expandable with spacers)
Runner Diameter	~45mm	~55mm
Throttle Body Bore	62mm (stock DBW)	71mm throttle body opening
Power Focus	Low-mid (2500-5500 RPM)	Broad pull from ~4000-8000+ RPM
Idle Quality	OEM smooth	Slightly rougher (tunable with Hondata)

Skunk2 Manifold Lineup -- Why Ultra Street Is the Sweet Spot

	Pro Series	Ultra Street	Ultra Race
Runner Length	~10-11"	~8.5-9"	~5-6"
Plenum Volume	~3.0L	~1.8L	~3.5L+
Power Focus	Mid-range	Street/sport balance	Top-end only (7000-9000 RPM)
Idle Quality	Good	Acceptable	Poor
Best For	Daily + light mods	Full bolt-on street build	Track / built motor

The Ultra Street is the right one for my build. The Pro Series leaves performance on the table -- not aggressive enough for what I'm doing. The Ultra Race kills low-end torque and idle quality for a street car. I'm not building a track car, so the Ultra Street hits the sweet spot.

Part Information

Field	Value
Intake Manifold	Skunk2 Ultra Street

Part Number	307-05-0600 (black) / 307-05-0605 (silver)
Port Pattern	PRB/PRC (direct fit to K20Z3 head)
Material	Cast aluminum, black powder coat
Price	~$550-620

Throttle Body -- CRITICAL: DBW Compatibility

The Problem

My K20Z3 uses **drive-by-wire (DBW)** -- an electronically controlled throttle body with an internal motor and position sensors. **Most aftermarket K-series throttle bodies are cable-throttle only and WILL NOT work.** I learned this early and I'm glad I did, because it would be an expensive mistake.

Throttle Body	DBW Compatible?	Notes
Stock K20Z3 62mm	Yes (OEM)	Baseline -- restricts Ultra Street's potential
Bored stock 64-66mm	Yes	Best option -- retains full DBW, proven
Skunk2 Pro Series 68mm	NO (cable only)	Would require full DBW-to-cable conversion
Blox 68mm DBW	Possibly	Must verify exact part is DBW-specific
Ktuned 72mm	NO (cable only)	Cable only
Hybrid Racing	NO (cable only)	Adapter-based, cable conversion needed

My Choice: Bored Stock Throttle Body (66mm)

Several Honda specialist shops (Drag Cartel, HeelToe Auto, and others) will bore the stock K20Z3 DBW throttle body from 62mm to **64-66mm**.

- **66mm is the maximum safe bore** -- the housing walls get too thin beyond that
- Retains full DBW function -- no wiring changes, no adapter hassles
- Proven reliability, used by hundreds of K20Z3 builds
- **Cost:** ~$100-175 (boring service, I ship my TB)
- **The stock 62mm TB isn't a major restriction** at the 220-240 WHP level, but a bored TB eliminates it as any bottleneck and lets the Ultra Street breathe freely

Honestly this is the move. No headaches, no wiring nightmares, just a slightly bigger hole in a part I already have.

Does a Larger TB Actually Help?

Power Level	Stock 62mm	Bored 66mm	68mm+
Stock-220 WHP	Fine, not a restriction	Marginal gain (1-3 WHP)	Unnecessary
220-240 WHP	Mild restriction up top	Measurable gain (3-7 WHP)	Diminishing returns
240+ WHP	Noticeable restriction	Good	Needed

For my build (targeting 220-240 WHP): the bored 66mm is the sweet spot. Going larger than 68mm on an NA K20Z3 loses throttle response at low RPM for minimal gain. Not worth it.

Power Gains (With Supporting Mods + Tune)

- **Ultra Street alone (no tune):** 3-7 WHP, may lose low-end torque. I'm not running it without a tune.
- **Ultra Street + 66mm TB + full bolt-ons + Hondata tune: 15-22 WHP over stock manifold** on a similarly-modded car
- **Peak power shifts** from ~6800-7200 RPM (stock) to ~7400-7800 RPM with the Ultra Street

Installation Notes

- **Difficulty:** Moderate. 3-5 hours first time.
- **Vacuum lines:** I need to take photos of ALL vacuum line routing before removal. Label everything. The Ultra Street has provisions but routing differs from stock RRC.
- **Coolant bypass:** The stock RRC has a coolant passage (throttle body heater). Block off the ports on the engine side and run a short bypass hose between them. Takes 5 minutes.
- **MAP sensor:** Relocates to the Ultra Street plenum. Stock sensor works, just moves.
- **IACV:** Ultra Street retains IACV provision -- important for idle on a street car.
- **Fuel rail:** I need to confirm the Ultra Street is compatible with the stock K20Z3 fuel rail, or check if Skunk2 includes one.
- **Fuel injectors:** Stock 310cc injectors are adequate to ~230 WHP. No upgrade needed.
- **No permanent modifications** to my car are required. Fully reversible.

Tuning -- MANDATORY

The intake manifold swap and Hondata tune are inseparable. I will NOT drive aggressively on the stock calibration.

After installing:

- Load a conservative base map (Hondata community or tuner's starting point)
- Drive gently to the dyno shop
- Key calibration changes:
  - Fuel tables: Full recalibration (4000-8000 RPM range especially)
  - VTEC point: Advance to ~5200-5400 RPM (from stock ~5800)
  - Ignition timing: Can advance in mid/upper RPM with better airflow
  - Idle target: May need slight increase (750 -> 800 RPM) for larger plenum
  - Tip-in enrichment: Adjust to prevent lean stumble from larger plenum volume

Budget Alternative: RBC Manifold Swap

I looked at the OEM RBC manifold (from 02-06 RSX Type S) as a budget option. It's the most popular budget intake manifold for the K20Z3 -- 70-80% of the Ultra Street's gains at 30-40% of the cost. I seriously considered it.

Factor	RBC (Used)	Skunk2 Ultra Street
Price	$100-250	$550-620
Power Gain (w/ tune)	10-18 WHP	15-22 WHP
DBW Compatibility	Needs $60-100 adapter plate	Designed for aftermarket TB
Community Support	Massive (thousands of builds)	Large

Ultimately I decided the Ultra Street is worth the extra money for the cleaner install and slightly better gains. But the RBC is a genuinely great option for anyone on a tighter budget.

Where I'm Buying

#	Item	Product	Price	Link
1	Intake Manifold	Skunk2 Ultra Street 307-05-0600	~$565-620	Hybrid Racing
				TF-Works -- $565
				JHPUSA
				Headgames Motorworks
2	TB Boring Service	Stock K20Z3 DBW TB bored to 66mm	~$100-175	Contact Drag Cartel or HeelToe Auto -- send my stock TB
3	TB adapter plate	If needed for 66mm on 68mm inlet	~$30-50	Sourced with manifold purchase
4	Dyno tune	Mandatory after install	~$400-600	Local K-series Hondata tuner
	Total		~$1,095-1,445	

TF-Works at $565 looks like the best price I've found for the manifold.

Note on part number: The Ultra **Street** is 307-05-0600 (black) or 307-05-0605 (silver). The Ultra **Race** is 307-05-8000. These are different manifolds -- the Ultra Street has longer runners and is the correct one for my build.

Last updated: 2026-04-16

Intake Manifold Install Guide -- Skunk2 Ultra Street + Bored 66mm TB

Status: Planning -- Phase 3

This is the "full bolt-on" finisher. By the time I'm here the car is already tuned on exhaust + headers; the IM + bored TB unlock the top end the RRC manifold was capping. The install itself is intermediate-level wrenching, but there are two catches that will sink a weekend: the **PRB-to-RBC head mismatch** (the Skunk2 Ultra Street is PRB-pattern and my K20Z3 is an RBC head -- adapters are mandatory), and the **2-week lead time on the bored throttle body**.

Reality Check

- **DIY time:** 3-5 hours first time, intermediate+ skill level. Not scary like the headers were -- no seized studs -- but many small lines (vacuum, coolant, fuel, harness) and getting them all back right is what separates a clean install from a 2am vacuum-leak hunt.
- **Coolant drain required:** Partial drain to free TB heater lines and the water-bypass adapter circuit. ~1/2 gallon captured. I'm doing it AFTER the full cooling refresh (09-Cooling-System), so I'm just partially draining fresh Honda Type 2 and putting it back.
- **Throttle body out 1-2 weeks ahead.** Drag Cartel or HeelToe Auto to bore 62mm -> 66mm. Ship 2-3 weeks ahead and confirm tracking before I order the manifold.
- **Dyno retune is mandatory.** Tune #2 -- "full bolt-on." I do NOT drive WOT on the headers-only calibration. Engine-damage territory.
- **Reliability-first rule:** adapters on the RBC head are non-negotiable. Skipping the water-bypass adapter or the DBW TB adapter is not a shortcut -- it's a guarantee of a leak or misalignment.

Prerequisites / Cross-references

Must Happen Before

Do First	Why
13-Headers/install-guide.md complete AND Tune #1 logged	Phase 2 baseline. Changing too many variables between tunes breaks the A/B.
09-Cooling-System/install-guide.md complete	Fresh coolant, bled. I'm partially draining a freshly-serviced system, not fighting failing hoses.
11-Wideband-AFR/ AEM X-Series logging	Mandatory before Tune #2.
TB shipped + bored + returned before install weekend	1-2 week lead. No install scheduled until tracking shows TB back.
Skunk2 water-bypass adapter in parts bin	CRITICAL -- PRB manifold's coolant ports don't line up with RBC head. Without this, the IM does not seal.
Skunk2 DBW TB adapter 309-05-0120 in parts bin	CRITICAL -- my bored DBW TB is RBC pattern; Skunk2's TB flange is PRB. Without this, the TB flange misaligns.

Fresh OEM IM gasket	No reusing 170k-mile gasket. ~$12.
FlashPro ready for Tune #2	Tune #1 .fpc archived, conservative base map loaded in case dyno gets delayed.

Can Happen At The Same Time

Also Do	Why
Fuel rail + injector swap (if bundling 19-Flex-Fuel)	Rail is off, injectors are out -- shared labor hours. Skip this window and I redo the same disassembly later.
Fresh PCV valve	Plastic, cheap, 170k old, easy with IM off
Clean throttle body internals	Carbon sludge behind the blade -- brake cleaner + rag
Fresh MAP sensor O-ring	$3, only accessible with IM removed

Tools Required

Critical:

- 10mm, 12mm, 14mm sockets on 3/8" ratchet
- Long extensions (3", 6", wobble) -- IM bolts sit deep
- Torque wrench, 3/8" drive, 10-80 ft-lb range
- Torque wrench, 1/4" drive, 20-200 in-lb range (rail, TB, PCV)
- Coolant drain pan, clean, labeled
- Hose clamp pliers
- **Fuel line quick-disconnect tool** (Lisle 39400 5/16" QC, ~$8)
- Vacuum line labels (masking tape + Sharpie) -- and phone photos of every joint
- Brake cleaner -- large can, for head mating surface
- Plastic razor -- NEVER metal on aluminum head
- Fire extinguisher in arm's reach (fuel rail work)
- Headlamp + inspection mirror

Nice to have:

- Parts organizer tray -- 20+ fasteners come off
- Nitrile gloves
- Intake port covers (blue tape) -- keep debris out of open runners

Parts List

#	Item	Part Number	Notes
1	Skunk2 Ultra Street Intake Manifold (black)	307-05-0600	~$565-620; TF-Works, Hybrid Racing, JHPUSA
2	Skunk2 water-bypass adapter (PRB->RBC coolant)	Skunk2 direct / Hybrid Racing	MANDATORY
3	Skunk2 DBW TB adapter 309-05-0120	TF-Works, Hybrid Racing	MANDATORY

4	Bored K20Z3 DBW throttle body (62->66mm)	Drag Cartel or HeelToe Auto	~$150, 1-2 wk lead
5	OEM Honda IM gasket	17105-RBC-004	Fresh, ~$12
6	Adapter gaskets	Ship with Skunk2 adapters	In kit
7	Honda Type 2 coolant top-off	OL999-9011	~1/2 gallon
8	(If bundling) Acuity 1913-PPL + Bosch EV14 550cc (0280158117) + OBD2 pigtails	See 19-Flex-Fuel	Optional

Part number discipline: 307-05-0600 is the black Ultra Street **Street** (long-runner). 307-05-8000 is the Ultra **Rac e** -- DIFFERENT manifold, wrong for street. **309-05-0120** is the DBW adapter specifically -- there's a cable-throttle variant too, wrong part.

Step-by-Step

Phase A: Send TB Out (2 weeks ahead)

Do this BEFORE ordering the manifold. If the bore takes 3 weeks instead of 1, I want to know before Saturday's locked.

1. Cold engine. Battery (-) off.
2. Loosen K&N intake tube clamp at TB, pull tube back.
3. Unplug DBW harness connector from TB.
4. Disconnect the 2 small coolant lines on TB heater ports. Small pan ready -- ~100ml coolant drips. Plug hose ends with a golf tee to stop siphoning.
5. Remove 4 bolts securing TB to stock IM (10mm). Lift TB off.
6. Bag TB-to-IM gasket, bolts, brackets. Reinstall intake tube loosely on the stock IM so nothing falls in.
7. **Box carefully** -- DBW internals are the expensive part. Ship to Drag Cartel (dragcartel.com) or HeelToe (heeltoeauto.com) with: "Bore to 66mm, retain DBW, K20Z3 stock body, return ASAP."
8. Car is undriveable during this window. Park it. Commute differently.
9. **Do NOT proceed until bored TB is back, test-fit, bore measured with calipers.**

Phase B: Install-Day Prep

10. Cold engine, overnight. Spill mats down -- Honda Type 2 stains concrete.
11. Battery (-) off.
12. **Partial coolant drain.** Open rad cap slowly. Open radiator petcock until ~1/2 gallon is in the pan -- drops level below IM coolant ports. Close petcock. Save the drained coolant in a sealed container; it's fresh from 09-Cooling-System and goes back in.
13. Remove engine cover (4x 10mm caps + bolts).
14. Loosen K&N intake tube at TB, slide off. Cap both ends with a plastic bag + rubber band.
15. **Take photos.** Phone on macro: overhead of the whole manifold, sides of every vacuum line, MAP, IACV, coolant lines, fuel rail, injector harness, PCV. I will thank myself at reassembly.

Phase C: Disconnect Stock IM

16. **Fuel pressure relief.** Pull PGM-FI Main Relay 2 (fuel pump relay) from under-hood aux box. Start, idle until stall (15-60 sec). Crank 2-3 sec after stall to bleed rail.

17. Reinstall relay.

18. **Fuel rail disconnect.** Rag wrapped around the Honda QC at the rail feed. Lisle 5/16" QC tool -- slide over line, press until retainer releases, gently pull line off. Residual fuel drips. Do NOT pry with a screwdriver.

19. Cap the fuel line end with a clean rubber cap. Route clear.

20. Unclip 4 injector harness connectors.

21. **Label every vacuum line** with masking tape + Sharpie before disconnecting: brake booster, PCV, EVAP purge, MAP. Photo first, then label, then disconnect. Don't yank plastic nipples on a 170k IM.

22. MAP sensor: unplug, remove 1-2 screws, set aside. Reuses with Skunk2.

23. IACV: unplug, set aside. Ultra Street has an IACV provision for street idle quality.

24. **Coolant lines off the IM.** Stock IM circulates coolant through its bypass + TB heater path. Loosen hose clamps with the pliers, work each hose off. Cap each hose with a golf tee.

25. If NOT bundling 19-Flex-Fuel: remove 2x M8 fuel rail bolts (108 in-lb / 9 ft-lb), lift rail with injectors seated, set aside on a clean rag under another rag. No dirt into injector bores.

26. If bundling with 19-Flex-Fuel: leave stock rail in place, bench-swap in Phase H.

27. **IM bolts -- REVERSE torque sequence.** Starting from outer bolts working inward (opposite of install order), break each bolt 1/4 turn. Then loosen fully. Outer-to-inner prevents warping the flange or the head surface.

28. Lift stock IM straight off. It's heavier than it looks.

29. **Immediately cover open runners** with blue painter's tape or rag stuffed in each port. Nothing falls in -- not a bolt, not a washer.

Phase D: Head Prep

30. Old gasket material: **plastic razor only**, no metal. K20Z3 head is aluminum -- gouges become leak paths forever.

31. Brake cleaner + lint-free cloth. Let flash off.

32. Straight edge check on the mating surface. Expected: flat. Warped = head-off job, above my scope.

33. Keep runner ports covered until the new IM is seconds from going on.

Phase E: Install Adapters (CRITICAL)

This is the step that matters most. The Ultra Street is a PRB-pattern manifold -- it was designed for the K20A2/K20A3 head (02-06 RSX, EP3 SI), not the RBC head on my 06-11 K20Z3. The two heads differ in coolant port geometry and TB bolt pattern. These adapters are mandatory and sold separately.

34. **Water-bypass adapter first.** Unbox the Skunk2 water-bypass adapter: plate, gasket (or O-rings), hardware.

35. Dry-fit to the RBC head coolant port. Confirm orientation -- adapter re-routes the RBC coolant circuit to an external bypass line since the PRB manifold has no matching internal coolant passage.

36. Supplied gasket between adapter and head. No RTV unless Skunk2's instructions specify -- default dry.

37. Hand-thread all adapter bolts, torque to **Skunk2's published spec** (typically 8-10 ft-lb -- verify on the sheet).

38. Route the bypass hose per Skunk2's diagram. Usually loops to thermostat housing or block bypass port. Fresh hose clamps.

39. **DBW TB adapter 309-05-0120.** Mounts between the Ultra Street's PRB TB flange and my bored DBW TB's

RBC bolt pattern. Dry-fit to the Ultra Street TB flange with supplied gasket.

40. 4 supplied bolts hand-tight, then torque per Skunk2 spec (~15-17 ft-lb). Verify on the sheet.

41. The bored TB mounts to this adapter's outer face in Phase G. The adapter is installed on the IM at this stage for easier access.

Phase F: Install Skunk2 Ultra Street IM

42. **Fresh OEM IM gasket** (17105-RBC-004) over the head studs/bolt holes. Orientation matters -- runner shapes are asymmetric. Sit flush all the way around.

43. **Remove runner-port covers** from Phase C -- last thing before IM goes on. Zero debris in head runners.

44. Lift Skunk2 Ultra Street into position. With both adapters already mounted, it's bulky but manageable single-handed.

45. Hand-thread ALL IM bolts before tightening any. If one doesn't start by hand, back out and check for cross-threading -- don't muscle it.

46. **3-pass star-pattern torque, center-outward:**

- Pass 1: 6 ft-lb (seating)
- Pass 2: 10 ft-lb
- Pass 3: **12-16 ft-lb final** (Skunk2 aftermarket IM spec; OEM stock is 17 ft-lb but Ultra Street's aluminum flange doesn't need that and over-torque cracks mounting ears). **Honor Skunk2's sheet over mine if they differ.**

47. Reconnect coolant lines per photos/labels. NEW Hybrid Racing T-bolt clamps or fresh spring clamps -- no reusing stretched 170k clamps.

48. Reconnect vacuum lines per labels. PCV, brake booster, EVAP, MAP nipple, emissions. **Every line accounted for -- an unplugged port is a guaranteed lean-miss idle nightmare.**

49. MAP sensor with fresh O-ring, hand tight + ~7 ft-lb. Harness connected.

50. IACV on Ultra Street's provision. Harness connected.

Phase G: Install Bored 66mm TB

51. Unwrap the returned bored TB. Visual: smooth bore, no burrs, DBW blade operates freely.

52. Fresh TB-to-adapter gasket (supplied with 309-05-0120).

53. Bolt TB to the outer face of the adapter. 4 bolts hand-tight, then torque **8-10 ft-lb** (Skunk2 adapter flange spec).

54. Reconnect TB heater coolant lines. Fresh clamps.

55. DBW harness connector -- retention tab clicks.

56. K&N intake tube back onto TB inlet. Clamp snug + 1/4 turn, no cranking.

Phase H: Install Fuel Rail + Injectors

Path 1 -- Staying stock (IM swap only):

57a. Fresh lower-injector O-rings on each of 4 stock injectors. Lube with engine oil (no silicone grease). 58a. Fresh upper O-rings if any look pinched. 59a. Line up 4 injectors with Ultra Street bores. Press rail straight down, even pressure. Verify each injector bottoms and no O-ring rolled. 60a. 2x M8 rail bolts. **Torque 108 in-lb / 9 ft-lb** (crushing aluminum, gentle). 61a. Honda QC back onto rail feed until retainer clicks. Tug-test -- cannot pull by hand. 62a. Reconnect 4 injector harness connectors.

Path 2 -- Bundling with 19-Flex-Fuel (swap to Acuity 1913-PPL + EV14 550cc + OBD2 pigtails):

57b. Follow Stage 3 (Pre-Assemble Acuity Rail + EV14) and Stage 4 (Install Rail onto IM) in the flex-fuel guide.

58b. Torque Acuity rail M8 screws to **10 ft-lb** (Acuity spec). 59b. Full pump/sensor/gauge hardware only if same session -- otherwise stock Honda QC connection to the Acuity rail inlet, save pump/sensor for later.

Phase I: Refill + Bleed Cooling

60. Walk the bay: every hose connected, every clamp torqued, every vacuum line seated, no tools left on the engine.
61. Close open coolant drains.
62. Pour the saved coolant from Phase B back into the radiator. Top off with fresh Honda Type 2 to MAX COLD at the neck.
63. **Full bleed procedure -- follow 09-Cooling-System/install-guide.md Phase G.** Lisle funnel on rad neck, open bleed screw on upper hose junction, heater FULL HOT, idle + 2000 RPM cycles, watch funnel until bubbles stop and upper hose is fully hot. #1 DIY-caused K20 failure -- don't short it.
64. Top off reservoir to MAX COLD.

Phase J: First Start + Tune #2 Prep

65. Battery (-) reconnected, terminal torqued.
66. **Fuel prime.** Key ON/OFF/ON/OFF/ON -- listen for the pump to prime 2-3 sec each cycle. No start yet.
67. **5-minute leak inspection.** Flashlight walk. Every fuel fitting dry? Every vacuum line seated? No coolant wet? Smell: no fuel, no coolant. Any leak = address now.
68. Start. Cranking ~2-3 sec longer than normal (fuel rail refilling, IACV learning). First idle may be rougher -- larger plenum, unlearned trims. Normal.
69. Idle 5 min minimum. Watch:
    - Temp gauge reaching normal in ~5 min
    - No CEL
    - No vacuum hiss
    - No fuel or coolant smell
    - Heater blows HOT (heater loop bled)
70. Shut off. Cool 15 min. Recheck coolant at rad neck and reservoir. Top off.
71. **Short partial-throttle drive ONLY.** Around the block, <3500 RPM, no aggressive throttle, no VTEC. 5-10 minutes. Not a shakedown -- a "does it run" drive.
72. **Datalog baseline.** FlashPro connected, record .fpdl of idle + partial-throttle. STFT, LTFT, MAP, IAT, knock counts, ECT. This is the pre-Tune #2 baseline for my tuner.
73. **Book dyno for Tune #2.** Full bolt-on calibration: Ultra Street + 66mm TB + headers + exhaust + intake. Expect **~10-20 WHP pickup** from IM + TB alone on top of headers-tuned baseline (target 220-240 WHP on pump gas). Bring baseline datalog + archived Tune #1 .fpc.
74. **DO NOT WOT until Tune #2 is dialed in.** Gentle partial-throttle only between install and dyno. No freeway pulls, no VTEC. Treat it like break-in.

Post-Install Tuning

Tune #2 -- "Full Bolt-On" replaces Tune #1 (headers-only) as my baseline. See 10-Hondata-FlashPro/tuning-workflow-and-maps.md for the Daily + Sport two-map protocol.

- **Fuel tables:** Full recal across 4000-8500 RPM. Larger plenum shifts MAP at low RPM (tip-in enrichment) and

runs lean at top without more pulse. Tuner dials.

- **Ignition timing:** +1 to +2deg over Tune #1 at peak torque RPM, knock-monitored on dyno.
- **VTEC:** Advance to ~5200-5400 RPM. Ultra Street's powerband wants VTEC earlier.
- **Idle target:** 750 -> ~800 RPM. Larger plenum idles rougher; the bump smooths without hurting driveability.
- **Tip-in enrichment:** Richer transient to prevent lean stumble on throttle open (larger plenum = more air to fill before MAP responds).
- **Expected:** +10-20 WHP over Tune #1. Peak shifts from ~6800-7200 to ~7400-7800 RPM. Low-end torque softens slightly below 4000 -- expected and acceptable.
- **Ceiling:** Still 220-240 WHP on 93 octane with margins. No chasing the dyno top number. Reliability over power, always.

Expected from IM + TB alone (over headers-tuned): +10-20 WHP. Combined total-from-stock with Phase 2: ~40-60 WHP at the wheels.

First Drive Checklist

- No vacuum hiss at idle (stethoscope)
- Idle stable 750-800 RPM, no hunting
- No misfires at idle or low RPM
- No coolant or fuel leaks (visual + smell)
- No CEL
- Temp normal in ~5 min, stays there
- Heater HOT at full hot knob
- STFT within ±5%, LTFT within ±8%
- No knock counts at partial throttle
- Clean throttle response, no tip-in stumble
- **WOT only after Tune #2 is flashed and logged**

Common Mistakes (Don't)

1. **Skipping the water-bypass adapter.** PRB manifold has no coolant circuit matching the RBC head. Bare mount = IM won't seal OR coolant circuit is broken (no heater, overheat risk). Mandatory, not optional.
2. **Skipping the DBW TB adapter 309-05-0120.** Bored K20Z3 DBW TB is RBC pattern; Ultra Street TB flange is PRB. Without adapter, TB flange misaligns and won't seal. Vacuum leak + un-tuneable idle on day one.
3. **Reusing the stock IM gasket.** 170k miles + 1 removal cycle = torn. $12 for fresh OEM. No excuses.
4. **Unlabeled vacuum lines.** "I'll remember" is how a clean install turns into a 3-hour vacuum-leak chase. Label every line, photo every joint.
5. **Over-torquing IM bolts.** Ultra Street is cast aluminum but bolt bosses aren't infinite. Past 16 ft-lb risks cracking a mounting ear. Honor Skunk2's spec.
6. **WOT before Tune #2.** Aggressive throttle on Tune #1 with a different IM is how engines get hurt. Larger plenum + bigger TB = different MAP behavior = old tune is wrong at WOT. Partial throttle only until retune.
7. **Prying the stock fuel QC instead of using the Lisle tool.** Prying cracks the plastic retainer -- new fuel line is a dealer $40 part.

8. **No key-cycle fuel prime before first start.** Cold-crank with empty rail = stressed starter + lean first seconds. Three ON/OFF cycles, THEN crank.
9. **Leaving IACV unplugged.** Fighting idle at every cold start. It's a 30-second connection.
10. **Skipping the coolant bleed.** Same K20 head-warping trap 09-Cooling-System warns about. Follow Phase G of that guide fully.

Skunk2 Ultra Street vs RBC Compatibility

The Skunk2 Ultra Street 307-05-0600 was designed for the **PRB casting family** -- K20A2/K20A3 on 02-06 RSX Type-S and EP3 Civic SI. My 2009 Civic SI K20Z3 uses the **RBC head casting** (same family as all 06-11 FA5/FG2 SI).

The two casting families differ in two ways that block a direct Ultra Street install on an RBC head:

1. **Coolant port geometry.** The PRB manifold includes an internal coolant passage that mates to PRB head coolant ports. The RBC head has coolant ports in different locations. Bolting the Ultra Street directly leaves the RBC head's coolant ports open to atmosphere (leak) AND provides no path for the bypass circuit (TB heater + head bypass flow). The **Skunk2 water-bypass adapter** caps the RBC head coolant ports at the IM mating surface and routes an external bypass hose around the Ultra Street.
2. **Throttle body bolt pattern.** The PRB TB flange on the Ultra Street uses the 02-06 RSX pattern. My K20Z3 DBW TB (bored to 66mm but same bolt pattern as stock) uses the RBC/06-11 SI pattern. These do not interchange. The **Skunk2 DBW TB adapter 309-05-0120** bolts to the Ultra Street's PRB TB flange on one side and accepts the RBC DBW pattern on the other.

Both adapters sold separately by Skunk2, both mandatory for an RBC-head install, both frequently missing from first-time buyers' orders because the main listing doesn't always flag the RBC requirement. TF-Works and Hybrid Racing have them in stock -- buy with the manifold in one order.

(Cross-reference: this is also called out in 19-Flex-Fuel-and-Fuel-System/acuity-fuel-rail-and-flex-fuel-build.md §1 -- the Acuity rail bolts directly to the Ultra Street, but only once these adapters are in place.)

See Also

- overview.md -- parts decision rationale, Ultra Street vs Pro vs Ultra Race comparison, DBW decision (source of truth)
- ../19-Flex-Fuel-and-Fuel-System/acuity-fuel-rail-and-flex-fuel-build.md -- bundle window for Acuity 1913-PPL + Bosch EV14 550cc during this install; IM-compatibility warnings also there
- ../10-Hondata-FlashPro/tuning-workflow-and-maps.md -- Tune #2 full bolt-on Daily + Sport protocol
- ../13-Headers/install-guide.md -- Phase 2 prerequisite; Tune #1 must exist before this install
- ../09-Cooling-System/install-guide.md -- Phase G bleed procedure
- ../docs/torque-specs.md -- IM 17 ft-lb OEM / 12-16 ft-lb Skunk2, TB 17 ft-lb OEM / 8-10 ft-lb adapter, fuel rail 9 ft-lb

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Last updated: 2026-04-20

Pulleys & Harmonic Balancer -- ATI Super Damper + NST Accessories

Status: Research Complete, Not Yet Purchased

The Plan

I'm going with the **ATI Super Damper (918477)** for the crank -- honestly the gold standard harmonic damper for K-series, no question. Pairing it with **NST (NonStop Tuning) alternator + idler pulleys** from their 3-piece kit. I get lightweight accessory underdrive with an overdrive alternator, which matters because I'm planning a LattePanda in-car PC that'll be pulling power constantly.

This combo gives me proper harmonic damping (which I really want at 170k miles), weight reduction off the rotating assembly, and maintained alternator output. No compromises.

Shopping List with Buy Links

#	Item	Product	Price	Link
1	Crank Pulley / Harmonic Damper	ATI Super Damper 918477	~$352-464	Drag Cartel -- $352
				Hybrid Racing -- $360
				K Series Parts -- $464
				JEGS
2	Alternator + Idler Pulleys	NST 3-Piece Kit NST22015K (use alt + idler, skip NST crank)	$268	NonStop Tuning -- $268
				Vivid Racing
				HeelToe Auto (multiple colors)
3	Serpentine Belt	Stock OEM or Gates K070663	~$20-30	Any auto parts store
	Total		~$640-760	

Note on the NST kit: I'm buying the full 3-piece kit ($268) but only using the alternator and idler pulleys. The NST crank pulley becomes a spare since the ATI replaces it. Unfortunately, NST doesn't sell the alternator and idler individually -- the kit is the only way to get them. Even so, the total is worth it for what I consider the best possible setup.

Cheapest ATI source I've found: Drag Cartel at $352.47 (sale from $379).

Why I Chose This Setup

ATI Super Damper 918477 -- Crank Pulley

Spec	Detail
Part Number	918477

Type	Elastomer-ring harmonic damper
Materials	Aluminum outer shell, steel hub, tuned elastomer ring
Total Weight	~3.6 lbs (stock is ~5+ lbs -- saves ~1.5 lbs)
Diameter	5.67" (5-11/16")
Belt Drive	6-rib serpentine (retains stock belt setup)
Timing Marks	360-degree laser-etched
Underdrive	Slight (smaller diameter than stock)
Safety Rating	SFI 18.1 certified (NHRA/IHRA approved)
Power Rating	Up to 600 HP
Harmonic Damping	YES -- elastomer ring between hub and inertia weight
Compatibility	All K-series (K20A, K20A2, K20Z3, K24)

Why ATI is the best option:

- ATI has made harmonic dampers for NASCAR, NHRA, Formula Drift, and every form of professional motorsport for decades -- these guys know what they're doing
- The 918477 is specifically tuned for K-series harmonic frequencies
- **Lighter than stock** while providing **better damping than stock** -- no compromises
- My stock damper's rubber isolator has had 170k miles of heat cycles. It's almost certainly providing reduced damping at this point. The ATI elastomer is engineered for far longer service life and higher heat resistance.
- 360-degree laser-etched timing marks are invaluable for tuning with a timing light
- SFI 18.1 certified for competition use

Why I ruled out the Fluidampr 570601: The Fluidampr is technically superior damping (viscous fluid, self-tuning across all RPMs) but weighs **6.4 lbs** -- heavier than stock. It adds rotational inertia, partially negating the snappier throttle response I'm after. It also costs $480-500. For my bolt-on NA K20Z3, the ATI provides more than enough protection while being lighter and cheaper. The Fluidampr would be the move if I were running race cams and living above 8000 RPM, but that's not my build.

NST Accessory Pulleys -- Alternator + Idler

Spec	Detail
Kit Part Number	NST22015K
Material	6061-T6 aircraft aluminum, CNC machined, hard anodized
Alternator Pulley Weight	2.8 oz / 80g (stock is significantly heavier)
Idler Pulley Weight	9.5 oz / 275g
Alternator	OVERDRIVE -- spins faster than stock
Idler	Lightweight replacement, uses stock OEM belt
Underdrive	~15% on accessories except alternator
Colors Available	Black, Red, Green, Purple, Raw

Why I went with NST over Buddy Club or others:

- **Overdrive alternator pulley** -- this is what sold me. Every other kit (Buddy Club, Unorthodox Racing)

underdrives the alternator, reducing charging output. NST overdrives it, spinning the alternator faster than stock. I need this because my planned LattePanda in-car PC adds 15-30W of continuous electrical draw. An underdriven alternator at idle could cause voltage drops, and I don't want to deal with that.

- CNC machined from 6061-T6 with tight tolerances
- Dyno-tested: +3.4 WHP / +4.0 WTQ peak at Skunk2's facility (Import Tuner Magazine verified)
- Designed to work with the stock OEM belt (no hunting for a different belt)

About Belt Retention (Raised Lips)

I went down a rabbit hole on this, so here's what I found. Serpentine pulleys use **multi-rib grooves** (the V-shaped channels) as the belt retention system. The ribs on the belt interlock with the grooves on the pulley -- that's what prevents slip and lateral walk. Raised flanges on the outer edges are only found on **smooth/flat idler pulleys** (which have no ribs), not on ribbed pulleys.

The stock automatic tensioner's idler pulley already has flanged edges because it's a smooth-face pulley. The ribbed pulleys (crank, alternator) don't need flanges because the groove profile does the job.

Belt walk is caused by:

- Misalignment between pulleys (not a concern with bolt-on replacements designed for the K20Z3)
- Worn/stretched belt (just replace the belt when installing new pulleys)
- Worn automatic tensioner (inspect at install -- replace if it doesn't spring back firmly)

Bottom line: With properly machined pulleys (ATI + NST), a fresh serpentine belt, and a functioning auto-tensioner, belt walk isn't a concern. The grooves ARE the retention system.

What Underdrive Pulleys Actually Do

The K20Z3 uses a single serpentine belt to drive all accessories. These accessories consume engine power (parasitic loss):

Accessory	Parasitic Loss at WOT
Alternator	1-3 HP
A/C Compressor	3-8 HP (when engaged)
Power Steering Pump	1-3 HP
Water Pump	1-2 HP
Total	~6-16 HP

Underdrive pulleys spin accessories ~15% slower, recovering 2-5 HP. The lightweight aluminum construction also reduces rotational inertia, improving throttle response. Community consensus: **~80-85% of the felt gain comes from weight reduction** (faster revving), ~15-20% from actual underdrive.

Realistic Power Gains

Configuration	Est. WHP Gain	Harmonic Damping	Cost
ATI crank + NST accessories (my build)	+3-6 WHP	YES (ATI)	~$640-760
Buddy Club P1 complete kit	+5-8 WHP	NO (damper deleted)	~$280
NST complete kit alone	+3-5 WHP	NO (damper deleted)	~$268
ATI crank only	+1-2 WHP	YES (ATI)	~$352-464

My honest take: Pulleys are one of the smaller gains in the bolt-on catalog. The 3-6 WHP is real but modest compared to headers (+10-18 WHP). The main benefit is **feel** -- improved throttle response and faster revving. My car will feel more eager and responsive, which honestly enhances the driving experience beyond what the dyno numbers suggest.

Why I'm NOT Deleting the Harmonic Damper

A lot of aftermarket crank pulleys (Buddy Club, NST, Unorthodox Racing) are solid billet aluminum with **no rubber isolator and no damping.** They delete the harmonic damper entirely.

At 170k miles, I'm not willing to take that risk:

- My crankshaft has already accumulated 170k miles of fatigue cycles
- I'm planning mods (headers, aggressive timing, higher RPM operation) that increase torsional vibrations
- Uncontrolled vibrations accelerate main bearing wear, timing chain stretch, and crankshaft fatigue
- The ATI provides better-than-stock damping while still being lighter -- there's literally no reason to delete the damper when this option exists

The K-series community debates this endlessly. No confirmed crankshaft failures from billet pulleys exist. But the ATI costs ~$100-200 more than a billet pulley and eliminates the risk entirely. At 170k miles, that's cheap insurance.

K20Z3 Pulley System Reference

Pulley	Function	Stock Diameter	Stock Weight
Crank (Harmonic Balancer)	Drives belt, dampens vibrations	5.75" (146mm)	~5+ lbs
Alternator	Charges battery/electrical	~2.5"	Heavy (steel)
A/C Compressor	Drives A/C (has clutch)	~4.5"	Not replaced
Power Steering	Drives PS pump	~5.0"	Not replaced
Idler	Routes/tensions belt	~3.0"	Moderate
Auto Tensioner	Maintains belt tension	Spring-loaded	Not replaced

What I'm replacing: Crank, alternator, and idler. The A/C, power steering, and auto tensioner stay stock.

Compatibility Notes

Alternator Charging with LattePanda

The NST overdrive alternator pulley spins the alternator **faster** than stock, maintaining full charging capacity even with the LattePanda drawing 15-30W continuously. I'll monitor battery voltage after install -- should remain above 13.8V at idle with all accessories on.

A/C Function

The slight underdrive means the A/C compressor spins ~15% slower. At idle in extreme heat (100F+), cooling may be marginally reduced. Above ~1500 RPM, no perceptible difference. I can live with that.

Power Steering

Slightly heavier steering at parking-lot speeds. At any speed above ~5 MPH, unnoticeable. I actually prefer slightly heavier steering feel anyway.

Drive-by-Wire

Zero effect -- throttle is electronically actuated, not connected to the pulley system.

Installation

Difficulty: Easy-Moderate (2-3 hours DIY)

I'm planning to do this myself. Straightforward job.

Tools Needed

- Impact wrench or long breaker bar + cheater pipe (crank bolt is ~180+ ft-lbs)
- Harmonic balancer puller (rent free from AutoZone)
- 19mm socket (crank bolt)
- Torque wrench
- Standard socket set for alternator and idler bolts

Steps

1. Remove serpentine belt (release auto tensioner with wrench)
2. Remove crank bolt (19mm, impact wrench recommended)
3. Use harmonic balancer puller to remove stock crank pulley -- **DO NOT pry** (damages crank seal)
4. Install ATI Super Damper, torque crank bolt to spec
5. Swap alternator pulley to NST unit
6. Swap idler pulley to NST unit
7. Install new serpentine belt (stock OEM or Gates K070663), verify routing
8. Start engine, verify belt tracking, listen for noise
9. Re-check belt tension after 100 miles

Last updated: 2026-04-16

Pulleys + Harmonic Balancer Install Guide -- ATI Super Damper + NST Alt/Idler

Status: Planning -- Phase 4

Reality Check Before I Start

This job looks simple -- swap three pulleys, run a new belt, drive. In reality, the crank bolt is the scariest fastener on this build that isn't an exhaust stud. Factory spec is **131 ft-lb plus a 90-degree yield angle**, and on a 170k-mile engine with original thread-locker the breakaway torque typically lands between **200 and 300 ft-lb**. A 120 ft-lb cordless impact will laugh at this bolt and round the socket before it cracks the bolt.

Budget **3-5 hours, intermediate-level** wrenching. I need either a proper crank pulley holder with a long breaker bar, or a **600+ ft-lb air or high-torque cordless impact**. No heat on this bolt, and no sub-500 ft-lb impacts -- both paths end in damaged crank threads or a stretched bolt left behind.

The **yield-bolt sequence is non-negotiable**. The replacement 90017-RWC-A01 is torque-to-yield, single-use. Reusing the old bolt is how dampers walk off the crank nose and destroy engines. I follow the Honda FSM sequence exactly: hand tight -> 37 ft-lb -> 90 degrees of angle. Done. I do not re-verify with a torque wrench.

The other non-negotiable: **the NST crank pulley from the NST22015K-Purple kit does NOT go on this car.** The ATI Super Damper replaces the crank pulley entirely. I'm using ONLY the NST alternator and idler from that kit -- the NST crank piece goes in the spare bin.

Prerequisites / Cross-references

Must be done first

Prerequisite	Why
Phase 3 full bolt-on tune complete	Pulleys are mechanical only, no retune needed, but a tuned baseline means any post-install shift is attributable to the pulley swap
New crank bolt 90017-RWC-A01 in hand	Torque-to-yield, single-use. Do not start if this bolt is not on the bench
New crank seal 91214-RZY-A01 in hand	Timing cover is exposed while the damper is off. $15 now prevents a second teardown when the 170k-mile seal starts weeping
Engine fully cold	Overnight ideal. I don't want coolant splashing if a hose is bumped

Must NOT be done / known conflicts

- **NST crank pulley MUST NOT be installed.** ATI Super Damper replaces it. Installing both deletes damper function or creates mechanical interference. Coordinate with 21-Purple-Cosmetics/overview.md -- the purple NST kit is purchased as a 3-piece, but only alt + idler are installed.
- **Do not reuse the old crank bolt.** Yield bolts are stretched past their elastic limit by design. They cannot provide correct clamp load a second time regardless of how clean they look.

Opportunistic bundling

If I'm pulling the timing cover for 05-Timing-Chain-Maintenance/ or the VTC actuator under 06-Ignition-Refresh/, this pulley job is already half-done and should be combined. Crank bolt comes out either way -- doing it twice wastes yield bolts and labor.

Tools Required

Critical -- do not start without these

Tool	Purpose
Crank pulley holder tool	Holds crank stationary for breaker-bar method. AutoZone loan-a-tool or Evergreen HCPH1 (~$35)
OR 600+ ft-lb impact wrench	Alternative to holder + breaker bar. Genuine 600+ ft-lb rated -- Milwaukee 2767, Ingersoll Rand 2235. A Harbor Freight "600 ft-lb peak" unit is not the same thing
19mm impact-rated socket	Crank bolt head, 1/2" drive. Chrome sockets shatter under these loads
18"+ breaker bar	24"+ preferred for the breaker-bar method
150 ft-lb torque wrench, 1/2" drive	Final torque pass at 37 ft-lb
Angle gauge	90-degree rotation after 37 ft-lb. ~$20 at Harbor Freight. You cannot eyeball this
Harmonic balancer puller (3-jaw)	If stock damper is seized. AutoZone loan-a-tool. Pulls from hub center, NOT outer inertia ring
ATI install tool	Threaded rod + thrust bearing washer to draw the damper onto the crank nose. No hammering, ever
Serpentine tensioner tool	Lisle 59250 or 3/8" breaker bar in the tensioner's square drive
Lisle alternator pulley tool (59300)	Inner hex holds alt shaft, outer socket turns pulley nut
Seal driver	Proper seal driver for the 91214-RZY-A01

Standard tools

Metric sockets 10/12/14/17/19mm, 3/8" and 1/2" ratchets + extensions, metric wrenches, shop light, rags, brake cleaner, floor jack + stands + chocks, drip pan, paint marker.

Parts List

#	Item	Part Number	Notes
1	ATI Super Damper	ATI 918477	K-series harmonic damper, SFI 18.1, 6-rib serpentine. ~$352 at Drag Cartel
2	NST Alternator Pulley	from NST22015K-Purple	Overdrive alt, 6061-T6, purple anodize
3	NST Idler Pulley	from NST22015K-Purple	Lightweight idler, 6061-T6, purple anodize
4	NST Crank Pulley (from kit)	DO NOT INSTALL	ATI replaces this. Spare bin

5	Serpentine Belt	Honda OEM or Gates K070663	Fresh belt every pulley swap
6	Crank Bolt (ONE-TIME-USE YIELD)	90017-RWC-A01	NEW, single-use, non-negotiable
7	Front Crank Seal	91214-RZY-A01	Replace while apart, ~$15

Total parts cost: ~$640-760.

Step-by-Step Procedure

Phase A: Pre-Install Prep (20-30 min)

1. Engine **fully cold**. Overnight is ideal.
2. Level ground, rear wheels chocked.
3. **Disconnect battery negative.** Alternator pulley comes off; I don't want anything live nearby.
4. Raise passenger side. Jack stands on the pinch weld.
5. Remove passenger front wheel.
6. Remove inner fender liner (plastic pop-clips + 10mm bolts). Exposes the accessory drive from the wheel well.
7. Remove lower engine splash shield if present.
8. **PHOTOGRAPH the serpentine belt routing.** From wheel well and from above, multiple angles. I'll need this at Phase I. Also confirm the routing diagram on the underside of the hood is legible.

Phase B: Serpentine Belt Removal (10-15 min)

9. Locate the auto-tensioner (spring-loaded pulley).
10. With the tensioner tool or a 3/8" breaker bar in the tensioner's square drive, rotate **clockwise** against spring pressure to slacken the belt.
11. Hold tensioner relaxed, slip the belt off the most accessible pulley (alternator or idler).
12. Slowly release the tensioner. It should spring back firmly. **If the tensioner is weak, bouncy, or doesn't return firmly, replace it now** (Honda 31170-RRB-A02) -- I don't want to find this after everything is buttoned up.
13. Work the belt off the remaining pulleys by hand.
14. Inspect the old belt. Glazing, cracks across the ribs, missing chunks, or oil contamination mean something caused it -- investigate alignment before installing new pulleys.

Phase C: Break the Crank Bolt (THE HARD PART) (15-60 min)

This is where most DIY pulley swaps stall. Factory 131 ft-lb + 90-degree yield, heat-cycled 170k miles. Breakaway **200-300 ft-lb** is normal.

Three options in order of preference:

15a. **Option 1 -- Crank pulley holder + long breaker bar (preferred).**

- Install the holder onto the stock damper per its instructions.
- Brace the holder against the subframe or a frame rail -- NOT against the oil pan.
- 19mm impact socket on the bolt, 24"+ breaker bar. Steady constant pressure counter-clockwise. Do not

jerk -- shock loading breaks things.

- Bolt breaks with a loud pop. Normal.

15b. **Option 2 -- 600+ ft-lb impact wrench.**

- 19mm impact socket, short bursts. Crank's rotational mass resists the impact; no holder needed with a real impact.
- If the impact hums 10 seconds with no movement, stop. It isn't rated high enough. Switch to Option 1.

15c. **Option 3 -- Starter-bump (NOT RECOMMENDED unless experienced).**

- Wedge the breaker bar against the subframe, bump the starter briefly to use engine torque to break the bolt.
- Can work; also can destroy things quickly if the bar slips. I'm not doing this unless Options 1 and 2 are unavailable.

16. **DO NOT heat the crank bolt.** Heat is for exhaust studs. Localized heat on the crank nose risks annealing steel and cooking the crank seal right behind the damper.
17. **DO NOT use a sub-500 ft-lb impact.** Either fails to break the bolt or partially loosens it into an unknown state.
18. Once loose, thread the bolt out by hand. **Throw this bolt away immediately.** No matter how good it looks.

Phase D: Remove the Stock Damper (10-30 min)

19. With the bolt out, the damper should slide off the crank nose by hand. Wiggle back and forth pulling straight out. Crank has a Woodruff key -- do not rotate while pulling.
20. If seized on (likely at 170k), use the 3-jaw puller:
    - Jaws grip the **hub** (inner metal), NOT the outer elastomer ring or inertia weight. Pulling on the outer ring separates the damper.
    - Center bolt pushes against the crank nose. Use a protective cap or soft shim between puller bolt and crank threads -- the crank threads must stay perfect for the new bolt.
    - Tighten slowly. Damper pops free with moderate force.
21. Find and remove the **Woodruff key** in the crank keyway. Magnet or needle-nose pliers. Put it in a dedicated container. **Do not lose it.** The new damper needs this key to index. If stuck, tap it out gently -- do NOT pry with anything that gouges the keyway.
22. Inspect crank nose for damage, galling, corrosion. Brass brush + brake cleaner. Must be smooth, dry, clean before the new damper goes on.

Phase E: Replace the Crank Seal (15-20 min)

With the damper off, the timing cover seal is fully accessible. Replace it. The 170k-mile seal is hardened and almost certainly weeping microscopically.

23. Small pick or seal puller to remove the old seal.
24. **Do not gouge the crank nose or the seal bore.** Any scratch is a future leak path. Angle the pick toward the seal, not the crank or the aluminum cover.
25. Clean the seal bore thoroughly with brake cleaner and a lint-free rag.
26. Light film of fresh engine oil on the lip of the new seal.
27. Drive the new seal squarely with a proper seal driver. Seat it **flush with the face of the timing cover** -- not recessed, not proud.

28. **No hammer directly on the seal.** Proper driver only. Socket-as-driver works only if it's exactly the seal's OD.

Phase F: Install the ATI Super Damper (15-20 min)

29. Confirm the Woodruff key is seated flat in the crank keyway, fully engaged.
30. Align the keyway in the ATI hub with the Woodruff key.
31. Slide the damper onto the crank by hand. It binds before fully seating -- normal. The crank-nose interference is snug by design.
32. **Use the ATI install tool (threaded rod + bearing washer).**
    - Thread the rod into the crank's center thread.
    - Bearing washer and nut over the rod, against the damper face.
    - Tighten the nut to draw the damper on smoothly and squarely.
    - Seated when it bottoms against the timing cover boss / crank spacer. Hard stop -- do not force past.
33. **NEVER HAMMER THE DAMPER.** Hammer blows crack the ATI hub, knock the elastomer out of round, and bend the crank nose. This is how a $350 damper becomes scrap.
34. Remove the install tool. Damper should be flush against the timing cover and rotate freely with the crank.

Phase G: New Crank Bolt -- YIELD TORQUE SEQUENCE (10 min)

Non-negotiable. Read twice.

35. Verify I'm using the **NEW 90017-RWC-A01.** Not the old one. The new OEM yield bolt.
36. Inspect threads on the new bolt and in the crank. **Clean and DRY.** No oil, no anti-seize, no thread locker. Factory spec assumes dry friction coefficients. Lubricant dramatically over-clamps and can snap the bolt.
37. Thread by hand until it contacts the damper face.
38. **Pass 1:** Torque to **37 ft-lb** with a calibrated torque wrench.
39. **Pass 2:** Rotate an additional **90 degrees** using the angle gauge. Paint-marker the bolt head at the 0-degree position if it helps me track.
40. **DO NOT re-check with a torque wrench after the angle pass.** The bolt is past its yield point -- torque readings are meaningless. It has correct clamp load by design. Touching it with a torque wrench is how DIYers talk themselves into "just a little more" and snap the bolt.
41. Wipe the damper face clean. Crank bolt is done.

Phase H: Install NST Alt + Idler Pulleys (20-30 min)

42. **Alternator pulley.** Lisle 59300 tool -- inner hex holds alt shaft, outer socket turns pulley nut.
    - Remove stock nut. Slide stock pulley off.
    - Slide NST overdrive alt pulley on.
    - Torque nut per NST: typically **65-80 ft-lb**. Blue (242) thread locker on threads.
43. **Idler pulley.** Remove stock idler bolt (14mm or 17mm center). Slide stock idler off. Install NST idler. Torque center bolt per NST: typically **30-40 ft-lb**. Blue thread locker.
44. **DO NOT INSTALL THE NST CRANK PULLEY.** ATI is on the crank. The NST crank piece goes in the spare bin. If I hit a brain-fart and installed it -- remove it now.
45. Rotate both new pulleys by hand. Spin freely, no binding, no lateral play, no bearing noise.

Phase I: Serpentine Belt Install (15 min)

46. Reference the routing photo from Phase A.

47. Start by wrapping the new belt around the crank (ATI) pulley -- secure starting point at the bottom.
48. Route up through the accessory pulleys per the routing diagram (typically crank -> A/C -> idler -> power steering -> water pump -> alternator -> tensioner, but confirm against hood sticker).
49. Last pulley is typically the tensioner or the alternator. Rotate tensioner to slack, slip the belt over the final pulley, slowly release.
50. **Verify:** belt centered on every pulley, no twists, ribs engaged with the grooves on ribbed pulleys, flat back on smooth idlers.
51. Rotate the crank by hand (19mm, gentle clockwise) two full revolutions. Belt tracks centered through full rotation -- no rubbing, no slapping.

Phase J: First Start + Verification (15-20 min)

52. Reconnect battery negative.
53. Reinstall fender liner and splash shield (finger tight for now).
54. **Key ON, do not start, for 5 seconds.** Listen for abnormal electrical sounds. Dash lights come up normally.
55. **Start the engine.** Fires within 2 seconds as usual. If not, kill it and check the alternator connection -- that's the only electrical thing I touched.
56. **Idle 30 seconds.** Listen for:
    - Belt squeal (misalignment, contamination, undertension)
    - Chirp (worn tensioner or bad bearing)
    - Grind (bad pulley bearing)
    - Knock or rattle from the damper area (damper unseated or bolt under-torqued -- shut off immediately)
57. Dash charging voltage indicator normal. If battery light comes on, the NST alternator pulley install is suspect.
58. **Rev briefly to 2,000-3,000 RPM** a couple times. Watch belt from the wheel well. Stays centered on every pulley.
59. Shut down. 5 minutes to settle.
60. **Visual re-inspection:** everything tight, no oil at the new crank seal, no belt walk, damper still seated, no tools left in the engine bay.
61. **Short shakedown: 5 miles at low-to-moderate RPM.** Home, re-inspect -- belt still tracking, tensioner in normal position.
62. Reinstall wheel, torque lug nuts to **80 ft-lb**, star pattern.

Post-Install Break-In

Interval	Action
First 100 miles	Drive gently. Avoid sustained WOT. Let the new belt bed in
100-mile inspection	Visual: belt tracking, tensioner position, damper seating, no oil at seal or bolt
500-mile inspection	One more visual. Should be clean

No retune required. Mechanical-only mod. Slight underdrive from NST alt + idler and smaller ATI diameter won't measurably shift AFR or timing. I should still **datalog a baseline after install** per docs/baseline-logs.md as a record of the as-installed state.

First Drive Checklist

- [ ] No belt squeal, chirp, or grind at idle or under load
- [ ] No damper noise (thump, knock, rattle) at any RPM
- [ ] Alternator charging -- dash or OBD2 reads 13.8-14.4V at idle with accessories on
- [ ] A/C functions normally
- [ ] Power steering functions normally
- [ ] Water pump circulating (heater hot after 5 min)
- [ ] No oil leaks at crank seal, timing cover, or damper face
- [ ] Belt centered on every pulley through a short drive
- [ ] No new OBD2 codes after shakedown

Common Mistakes (Do Not Do These)

Mistake	Consequence	Prevention
Reusing the yield crank bolt	Bolt loosens in service -> damper walks off crank -> destroyed engine. Documented repeatedly on 8thcivic and k20a.org	Buy 90017-RWC-A01 new. Trash the old one immediately
Installing the NST crank pulley despite having the ATI	Mechanical interference or deleted damper function on a 170k crank	NST crank goes in the spare bin. Only alt + idler from the kit
Hammering the ATI damper onto the crank	Cracked hub, broken elastomer bond, bent crank nose	ATI install tool (threaded rod + bearing washer) only
Skipping the angle torque (doing just 37 ft-lb)	Under-clamped bolt -> walks loose in service	Full sequence: 37 ft-lb then 90-degree rotation with angle gauge
Oil or anti-seize on the crank bolt threads	Over-clamp at the same indicated torque -> snapped bolt or damaged crank threads	Clean dry threads per FSM
Damaging the crank seal during removal	New seal leaks from day one	Pick away from the crank nose, never pry against it
Skipping the crank seal replacement	Weeps within 1-2k miles, second teardown to fix	Replace every time the damper is off. $15
Using a 300 ft-lb cordless impact on the crank bolt	Impact hums, bolt doesn't move, socket rounds the head	600+ ft-lb rated only, or holder + 24" breaker bar
Heating the crank bolt	Annealed crank steel, cooked crank seal	Never. This is not an exhaust stud
Losing the Woodruff key	Damper spins on crank nose under load = catastrophic	Dedicated container the moment it comes out

Why ATI Over Aluminum Underdrive Crank Pulleys

A lot of K-series builders run solid billet aluminum crank pulleys -- Buddy Club P1, NST's own crank piece, Unorthodox Racing. Cheaper, lighter, marginally more underdrive effect. **They also delete the harmonic damper entirely.**

Harmonic dampers aren't decorative. The K20Z3 crankshaft has natural torsional frequencies hit within the operating range, and the stock damper's elastomer ring (or the ATI's) absorbs those vibrations before they

destroy main bearings and stretch timing chains. At **170k miles on my crank**, with headers, aggressive timing, and higher sustained RPM planned, deleting damper function to save a few dollars and maybe gain a whp is exactly the trade-off my reliability rule forbids.

No one has published a confirmed "billet pulley killed my K20" case. The failure mode is slow and silent: accumulated bearing wear over 20-50k miles, chain stretch, eventual rod knock. By the time damage is visible, the cause is unattributable. The ATI is the insurance policy.

Why NST Purple Alt + Idler (It's an Aesthetic Call)

Realistic power gain from underdriving the alternator and idler is **under 2 WHP**, and on NST's overdrive alt it may be net-zero or slightly negative at higher RPM. Weight reduction on the rotating assembly is real but tiny.

I'm buying these for the purple anodize, full stop. They fill the alt + idler slots of my 21-Purple-Cosmetics/ theme -- reputable brand, correct fitment, CNC 6061-T6. The "underdrive power gain" is a rounding error; the cosmetic payoff is the real reason. I own that honestly.

The one genuine functional benefit of the NST alt is that it **overdrives** the alternator rather than underdriving it -- matters because my planned LattePanda in-car PC draws 15-30W continuously, and I want the alternator spinning faster than stock at idle, not slower.

See Also / Cross-references

- overview.md -- parts selection rationale, ATI vs Fluidampr, NST vs Buddy Club
- ../21-Purple-Cosmetics/overview.md -- NST22015K-Purple kit notes, crank-pulley conflict
- ../05-Timing-Chain-Maintenance/overview.md -- same yield-bolt procedure; bundle if chain is getting done
- ../06-Ignition-Refresh/overview.md -- VTC actuator shares the crank-bolt teardown; bundle opportunistically
- ../10-Hondata-FlashPro/tuning-workflow-and-maps.md -- no retune required, but log a post-install baseline
- ../docs/torque-specs.md -- build-wide torque reference including crank-bolt yield sequence

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Last updated: 2026-04-20

Cylinder Head -- Aftermarket Drop-In Head + Valvetrain

Status: Research Complete, Not Yet Purchased

The Decision

I decided to go with an **aftermarket drop-in head**, not a stock-head refresh. The "go all out" call on the head side of this build -- while keeping reliability-first intact and not buying absolute top-tier race parts.

What The Research Actually Says (Important Counter-Point)

Before the recommendation: **at my target of 220-240 WHP pump / 230-250 WHP E85, a refreshed stock RBC head would hit the number**. The stock K20Z3 head flows ~290 CFM at stock valve size, which supports ~260 WHP NA all day when paired with headers, IM, and tune. The FBO package doesn't run out of head at 250 WHP.

The drop-in head is a ceiling-expansion decision. It buys ~20-30 WHP of future headroom, makes the next upgrade step (bigger cams, higher CR pistons, aggressive E85 tune) bolt-on rather than teardown-again, and gives me the option to move beyond 270 WHP NA later without pulling the engine a second time.

Given I'm going all-in on this build, the drop-in path is the right move. Documented here because I want future-me (or anyone reading this) to understand the trade.

Critical Casting Compatibility Check (don't skip this)

The K20Z3 is an RBC casting, NOT PRB. This matters for head selection:

- **RBC** (K20Z3 Civic Si 06-11 + K24A2 US TSX 04-08): the casting I have
- **PRB** (K20A2/K20Z1 Acura RSX Type-S 02-06, JDM DC5 Integra R): different coolant routing, different IM bolt pattern

Several popular drop-in heads are **PRB-casting-based** in their base product:

- 4Piston Pro 156v2 **bare CNC** head -- PRB base
- Iconic PRB K20 CNC Head -- PRB-only
- Some Drag Cartel listings

What works for RBC (K20Z3):

- ? **4 Piston Pro 156v2 All Motor Package** (assembled) -- RBC-compatible per 4P's fitment list
- ? 4 Piston Pro 163 -- same fitment envelope
- ? KMOD Stage 2 K20 Head -- RBC or PRB core accepted
- ? P2R RBC K-Series CNC Head -- specifically RBC
- ? Portflow custom work -- RBC core accepted via core exchange

Before buying anything, I will email the vendor with "This is going on a K20Z3 RBC-casting block -- is this part correct for that casting?" and get the confirmation in writing. Mis-orders on this exact casting question are the #1 forum complaint.

Options Surveyed

4 Piston Racing -- Industry Benchmark

Danville, IN. Luke at 4Piston is considered the top K-series head builder in the US.

Product	Price	Flow	Target Power	Included
Pro 156v2 All Motor Package (RBC-compatible assembled)	$2,740	~360 CFM stock-size valve	220-270 NA WHP	CNC port, NEWEN 5-angle valve job, +1mm-capable oversized steel intake seats, manganese bronze guides, Ferrea valves + Supertech springs + Ti retainers + locks, fully assembled
Pro 156v2 bare CNC	$1,350	360 CFM	~270 NA WHP	CNC work only -- you supply valvetrain (?? PRB-casting base -- verify for RBC)
Pro 163 CNC	$1,850	400 CFM	260-320+ NA	Larger 50deg 37mm intake valve (7g lighter), NEWEN valve job
4P Reaper CNC	$3,100	440+ CFM	Race only	All-motor record holder; overkill for street

Lead time: 4Piston is historically backordered. Email Luke@team4piston.com for current lead time before ordering.

Cams: Not included. Assumes you bring Skunk2 Tuner Stage 2 / Drag Cartel OG / Blackworks / 4P cams.

KMOD Performance -- Best Value

Kevin at KMOD has built the reliable-budget-tier reputation in the K-series drag-racing world.

Product	Price	Flow	Target Power
Stage 1 K20	~$750-900	310-320 CFM	Up to 300 NA
Stage 2 K20	$1,000-1,250	348 CFM @ 0.650" lift	Up to 360 NA
Stage 3 K20	higher tier	CNC race-port	Race

Valvetrain: Not included. Supertech 82lb dual + Ti retainer kit is a separate ~$500 add.

Lead time: 2-4 weeks -- fastest of the big three.

Stage 2 + Supertech kit + labor: ~$1,750 all-in -- the closest parallel to 4P Pro 156v2 All Motor at 60% of the cost.

Skunk2 (Pro-Series Head -- effectively discontinued)

Skunk2's current official "cylinder head" page is literally a referral: **"Skunk2 partners with Portflow for cylinder head machining. Contact Portflow directly."** The old Pro1 / Pro2 assembled heads are not currently shelf-stocked.

Skunk2 does still sell head-related components: Pro head studs (661-05-0001), Ultra-Series valves, spring/retainer kits, adjustable cam gears (304-05-0300), and the full Tuner/Ultra cam line.

Takeaway: If I wanted "Skunk2-branded drop-in," the path is send a core to Portflow for a Skunk2-spec build -- see next.

Portflow Design (Tom, Harbor City CA)

Skunk2's porting partner. 25+ years of K-head work. Flow numbers are among the best in the industry.

Tier	Price	Notes
Basic port + 3-angle valve job	~$1,000	Head only
Stage 3 competition port	~$2,200	Full CNC-equivalent hand porting, oversized intake valves, race-grade valve job

Lead time: 2-6 months historically. Slowest option. No online ordering -- phone only (310-257-0351).

Valvetrain: Separately sourced (Portflow will install Supertech or Ferrea kits you supply).

Drag Cartel

CNC-focused, drag-racing heritage.

Product	Price	Target	Included
Street/Strip CNC Head	$2,249	260-300 NA	Head only
Street/Strip + OG Cams Combo	$4,999	280-320 NA	Head + Supertech dual springs + Ti retainers + Drag Cartel OG cams
Pro Series / Bespoke K	custom	Race	Custom

Over-spec for my 240 WHP target at ~$5,000. Saved here for reference.

Others Surveyed

- **Iconic Auto-Design PRB K20** -- PRB-only, not for K20Z3 without custom work
- **Humble Performance Street Spec** -- ~$1,400, well-reviewed in 9thcivic scene
- **P2R / Power Rev Racing RBC K-Series CNC Ported Head** -- RBC-specific, ~$1,600-1,900
- **King Motorsports** -- premium white-glove service, ~$1,500-3,000
- **K-Tuned** -- doesn't make a head product (plumbing only)
- **Bisimoto** -- custom full builds only, no off-the-shelf head

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My Pick: 4 Piston Pro 156v2 All Motor Package

$2,740 assembled, RBC-compatible, valvetrain included.

Why:

1. **Assembled with matched components** -- Ferrea valves + Supertech 82lb dual springs + Ti retainers + locks, all paired by the shop that built the head. No "mix and match" risk.
2. **CNC-ported with NEWEN 5-angle valve job** -- repeatable, documented flow numbers (~360 CFM)
3. **Manganese bronze guides + oversized steel intake seats** -- durable for E85 moisture, good for 200k+ more miles
4. **Industry-leading shop reputation** -- 4Piston has held All-Motor records for a decade
5. **RBC-compatible** (confirmed before order -- see section above)
6. **Lead time bite-sized** -- assembled packages ship faster than custom CNC work

Alternatives I considered and rejected:

- **KMOD Stage 2 + DIY valvetrain** ($1,750): saves $1,000 but requires me to source Supertech separately, do

my own install of springs/retainers, and trust the match. Also Stage 2 is the right head, but I'm paying 4Piston for the assembled certainty.

- **Drag Cartel Street/Strip + OG combo** ($4,999): over-spec. The 5-15 WHP extra over 4P 156v2 isn't worth the premium for 250 WHP.
- **Portflow** ($1,500-2,200): I'd save ~$500 but accept 2-6 month lead time. Not acceptable given my mod timeline.

Head-Related Hardware (additional budget)

With any drop-in head, these are the "you'll also need" parts:

Head Gasket

Option	P/N	Thickness	Price	When
OEM Honda	12251-RBC-004	.030"	$60	Stock-bore, Path-A (refresh); baseline
Cometic MLS (my pick)	C4561-030	.030"	$85	Stock-bore, RECOMMENDED with ARP studs
Cometic MLS thicker	C4561-036	.036"	$90	If deck was surfaced, to restore deck height

Going **Cometic C4561-030** because I'm using ARP head studs -- MLS tolerates higher clamp load better than OEM single-layer, and $25 is cheap insurance.

Head Studs -- ARP 208-4701

- **Part number: ARP 208-4701** (K-series Pro Series, 8740 chrome-moly, 190,000 PSI)
- Price: ~$185-210
- **Don't order 208-4303** -- that's the B-series GSR kit, wrong engine
- Torque procedure: 40 / 60 / 80 ft-lb in 3 passes, spiral from center outward, with ARP Ultra-Torque lube (included)
- **Skip the 625+ upgrade** -- standard 208-4701 handles 500+ WHP on E85. 625+ ($400-500) is for 500+ WHP turbo builds, not this.

Complete Top-End Gasket Set

Part	P/N	Price	Notes
Valve cover gasket kit (incl. plug tube seals + grommets)	12030-PNC-000	$40	Mandatory -- the kit, not bare gasket
Intake manifold gasket (head-to-IM)	17115-RAA-A01	$15	Fresh every time IM comes off
Exhaust manifold / header gasket	18115-RRB-A01	$20	OEM only -- aftermarket blows
Front cam seals	91213-PLA-003	$15 ea x 2 = $30	Both cams
Cam cap half-moon plug (rear)	90041-PNA-000	$8	Often leaks at 170k
Cam chain case oil seal	91319-P0Y-003	$12	Replace on any timing chain disturbance

Timing Chain Bundling (labor efficiency)

With the head off, the timing chain area is exposed. **This is the time to do the full chain service** (see ../05-Timing-Chain-Maintenance/overview.md for the full procedure).

Part	P/N	Price
Timing chain (14401-PNA-004)		$75
Chain tensioner (14510-PRB-A01)		$65
Chain arm + fixed guide		$80
VTC actuator (14310-RBC-003)		$280 (covered under Phase 6 Ignition Refresh budget)

Cams + Cam Gears (optional but high-ROI)

The cams pair with the head. Research community consensus for 220-250 WHP FBO NA:

Option	Price	Notes
Skunk2 Tuner Stage 2 (305-05-0220)	~$500-600	Community default. Published 22-25 HP gain.
Drag Cartel OG 3.2 Street	~$650	Slightly more aggressive, similar power
Blackworks	~$550	Alternative cam brand
Skunk2 Pro adjustable cam gears (304-05-0300)	~$270	Allows dyno-dialed cam timing. Strongly recommended with any aftermarket cam.

My pick: **Skunk2 Tuner Stage 2 + Skunk2 Pro cam gears**. Total ~$770-870.

Install Considerations

Engine-Pull vs Head-Off In-Car

Head-off in-car is doable on the FG2. Transverse mount is tight but clearance exists. Remove IM + fuel rail + headers, remove timing chain cover via passenger fender well, remove valve cover + cams + head bolts, lift head straight up with two-person hoist. Hood and wiper cowl stay on.

Time estimates:

- First-time DIY in-car: 3-5 days weekend work
- Experienced indie shop: 12-16 hours = **$1,200-1,800** at $100-150/hr
- Engine pull adds 8-12 hours (~$800-1,200) -- **not warranted** since bottom end stays stock

Tools beyond what I already have:

- Engine hoist rental (for the head lift, ~45 lb)
- Honda timing chain holder / cam alignment tool (Lisle 37950, $40)
- ARP torque wrench 80 ft-lb capable (already have)
- VTC actuator wrench ($50 or shop-made)

Tune Implications -- Mandatory Dyno Retune

A drop-in head changes:

- Volumetric efficiency curve (more flow at higher lift / longer duration)
- VTC engagement window (cams dictate VTEC worth point)
- Ignition timing requirements (faster burn = less advance in peak region, sometimes more in transition)
- Fuel requirements (more air = more fuel, injector duty math changes)

Budget $500-800 for a proper dyno retune at a Hondata-capable shop. See

10-Hondata-FlashPro/tuning-workflow-and-maps.md.

Do NOT drive on the street before retune. Fuel maps for new cams/head on old tune = lean WOT = cracked piston. Gentle 10-mile idle/break-in loop to dyno, then retune.

Cross-Mod Bundling Opportunity (the "while you're in there" list)

With the head off, these bolt-on mods consolidate labor:

- 12-Valved-Exhaust/ + 13-Headers/ -- Skunk2 Alpha header install uses the same teardown
- 14-Intake-Manifold/ -- Skunk2 Ultra Street IM + bored 66mm TB installs at head reinstall
- 15-Pulleys-and-Harmonic-Balancer/ -- ATI damper uses same crank-pulley-off access
- 06-Ignition-Refresh/ -- VTC actuator 14310-RBC-003, plugs, coils
- 19-Flex-Fuel-and-Fuel-System/ -- EV14 injectors swap at fuel rail reinstall
- 09-Cooling-System/ -- Koyorad + hoses + thermostat + WP, coolant already drained
- 17-Wideband-AFR/ -- AEM X-Series bung welds into Skunk2 Alpha collector during header install

One teardown, eight mods. This is why docs/mod-order-and-maintenance.md phase-sequences everything.

Cost Summary

"Just the head work" (drop-in head + required hardware + install + tune)

Line item	Cost
4P Pro 156v2 All Motor Package	$2,740
Cometic C4561-030 head gasket	$85
ARP 208-4701 head studs	$200
Valve cover gasket kit 12030-PNC-000	$40
IM gasket 17115-RAA-A01	$15
Exhaust manifold gasket 18115-RRB-A01	$20
Front cam seals 91213-PLA-003 x 2	$30
Cam cap half-moon + oil seal	$20
Skunk2 Tuner Stage 2 cams	$550
Skunk2 Pro adjustable cam gears	$270
Misc hardware / bolts / Moly 60 / Hondabond	$100
Break-in oil (Driven BR30 or Brad Penn, 5 qt)	$60
Parts subtotal	~$4,130
DIY install (in-car):	$0
Shop install (in-car):	+$1,200-1,800
Dyno retune	+$500-800
DIY total	~$4,630-4,930
Shop total	~$5,830-6,730

Comparison: Path A (refresh stock head) -- rejected but documented

Same cams + cam gears + gaskets, but keeping stock RBC head with:

- Supertech 82lb dual springs + Ti retainers + seats kit: $598
- OEM Honda valve stem seals: $40
- OEM valves if worn: $240-480
- Machine shop (5-angle valve job, resurface, pressure test): $400-600

Path A parts total: ~$1,900-2,500 + dyno tune. Saves ~$1,700-2,000 over drop-in. Hits 220-250 WHP target. Doesn't expand future ceiling.

Why I rejected Path A despite lower cost: The money saved isn't enough to buy me future-proof. The drop-in opens the door to 270+ WHP later with just a cam swap -- no second head-off teardown needed. Committed to "go all out" on this build; the drop-in earns its premium on ceiling, not on current target.

Risks I'm Accepting

1. **Bottom end has 170k miles.** A fresh head won't stress the bottom end beyond stock levels (still 250 WHP, not 400). But tell-tale bottom-end noise (rod knock, main bearing whine, piston slap) will be less masked by a quieter top end. Accept and monitor.
2. **Port-matching downstream is required.** Head -> IM port floor, header -> head flange. A 0.5mm step is common and costs 3-5 WHP. 20 minutes with a porting burr at install fixes it -- see 13-Headers/install-guide.md.
3. **Chamber CC may shift.** If the ported head unshrouds aggressively, CC climbs from stock ~35cc to ~37-38cc, dropping static CR from 11.0:1 to ~10.5-10.7:1. Costs 2-4 WHP NA. 4Piston will document as-machined chamber CC on request.
4. **Lead time.** Email Luke at 4Piston first. If lead time pushes out the install date unacceptably, KMOD Stage 2 is the faster fallback.
5. **E85 seat compatibility.** Confirm with vendor that seats are hardened for E85 moisture. Modern Supertech / Ferrea / OEM Honda valves all are. Just verify before clicking buy.
6. **"Head done but not tuned" is the danger window.** Do not drive more than gentle 10-mile break-in between install and dyno. Closed-loop learning on wrong cam/head data corrupts trims.

Cross-References

- 05-Timing-Chain-Maintenance/overview.md -- chain service is free labor during head swap
- 06-Ignition-Refresh/overview.md -- VTC actuator install bundles
- 08-Engine-Mounts/overview.md -- passenger mount removed for head-off access
- 09-Cooling-System/overview.md -- Koyorad + WP bundle during head job
- 13-Headers/overview.md -- Skunk2 Alpha install with head job
- 14-Intake-Manifold/overview.md -- Ultra Street IM + 66mm TB install with head reinstall
- 10-Hondata-FlashPro/tuning-workflow-and-maps.md -- post-install dyno session
- 20-Built-Motor-Bottom-End/overview.md -- the next phase IF I go past 270 WHP
- docs/torque-specs.md -- ARP stud torque sequence, cam cap torque
- docs/maintenance-parts-catalog.md -- add head P/Ns to catalog

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Sources

- 4 Piston Racing -- K-Series
- 4 Piston Pro 156v2 All Motor Package
- 4 Piston Pro 163
- Skunk2 -- Portflow Cylinder Heads referral
- Skunk2 Tuner Stage 2 K-series
- Skunk2 Pro Series Adjustable Cam Gears 304-05-0300
- KMOD Stage 2 K20 Cylinder Head
- Drag Cartel Street/Strip CNC Head
- Drag Cartel Top-End Combo
- P2R RBC K-Series CNC Ported Head
- ARP 208-4701 K-Series Head Studs
- Cometic C4561-030 K20Z3 .030" MLS
- OEM K20Z3 Head Gasket 12251-RBC-004
- Supertech 82lb Dual + Ti Retainers for K-Series (for reference)
- k20a.org -- K20Z3 vs K20A2 Head (RBC vs PRB)
- 8thcivic -- Official All-Motor Dyno Chart
- 9thcivic -- K20Z3 Head Swap 100% Stock Bottom End
- FunctionTheory -- K-Swap Guide Part One (RBC/RRC/PRB)

Last updated: 2026-04-20

Cylinder Head Install Guide -- 4Piston Pro 156v2 All Motor + ARP 208-4701 + Skunk2 Stage 2

Status: Planning -- Phase 4.5 (bundle with headers, IM, VTC, pulleys, cooling)

This is the highest-stakes install on the entire build. A timing chain or header job going sideways costs me a weekend. A head job going sideways costs me the engine. I'm reading this guide six times before I touch a wrench, and re-reading Phase G (head stud torque) the morning of.

Reality Check Before I Start

Head-off-in-car, not engine-pull. Bottom end stays put, trans stays bolted, subframe stays in place. The rest is not weekend-DIY territory without prior head-gasket experience.

Skill Tier	Time Estimate	Notes
First-time DIY, no head-gasket experience	12-20 hours over 3-5 days	Not a single-day job. Mistakes compound.
DIY with prior head-gasket experience	8-12 hours over a long weekend	Still slow -- cam degreeing adds time
Honda specialist shop	10-14 hours @ $100-150/hr = $1,200-2,100	Includes block prep, stud install, break-in

The shop option is legitimate here. Unlike the header job, the worst-case on this install is a bent valve or warped head -- pulling the engine and paying machine-shop rates. If I don't already know what a straight-edge flatness check on an aluminum deck feels like, **pay the shop.**

What tips me DIY: I've planned this as a one-shot teardown with headers, IM, VTC, pulleys, and cooling bundled in. One careful week beats three weekend jobs stacked. What tips me shop: this is the one mod where "shop owns it if it breaks" has real teeth.

Decision checkpoint: quote from a K-series specialist (not a generic indie). Under $2,000 labor for the bundle + they've done 4P Pro 156v2 before = hire out. Otherwise DIY with the ARP torque sheet taped to the fender.

Prerequisites / Cross-references -- BUNDLE TEARDOWN

Not a standalone mod. The labor to get the head off is the same labor to do six other things. Pulling the head twice is the most expensive mistake I could make on this build.

Must be planned into the same teardown

Bundled Mod	Cross-reference	Why Bundle
Timing chain service	../05-Timing-Chain-Maintenance/	Chain cover is off anyway. Measure tensioner rod, replace chain/arm/guides/tensioner if >=10 mm extension. At 170k, assume full replacement.
VTC actuator 14310-RBC-003	../06-Ignition-Refresh/	Lives directly under the timing cover. Replace every time. -003 part only -- no -002.

Headers (Skunk2 Alpha + Berk HFC)	../13-Headers/install-guide.md	Exhaust manifold studs get pulled during head removal. Header install happens before head reinstall -- same teardown.
Intake manifold (Skunk2 Ultra Street + 66mm TB)	../14-Intake-Manifold/overview.md	IM bolts to the head. Install at head reinstall, not separately.
Cooling system refresh (Koyorad + hoses + WP + thermostat)	../09-Cooling-System/overview.md	Coolant is already drained. Water pump accessible. Do not skip.
Pulleys and harmonic balancer (ATI Super Damper)	../15-Pulleys-and-Harmonic-Balancer/overview.md	Crank pulley is off for timing cover access. NEW crank bolt yield procedure happens once, not twice.
Ignition refresh (plugs + OEM coils)	../06-Ignition-Refresh/overview.md	Coils and plugs come out to clear cams and valve cover. Replace with fresh parts before reinstall.

Confirmed before ordering anything

- [ ] **RBC fitment on 4P Pro 156v2 All Motor Package** -- email reply in writing from Luke at 4Piston confirming this specific assembled package fits K20Z3 RBC casting
- [ ] Cometic C4561-030 ordered (not C4561-036 -- the thicker gasket is for decks that were surfaced, which mine wasn't)
- [ ] ARP 208-4701 ordered (not 208-4303 -- that's B-series GSR, wrong engine)
- [ ] Skunk2 Tuner Stage 2 cams 305-05-0220 + Skunk2 Pro adjustable cam gears 304-05-0300 ordered and in hand
- [ ] Fresh OEM gaskets: IM 17115-RAA-A01, exhaust 18115-RRB-A01, valve cover kit 12030-PNC-000, front cam seals 91213-PLA-003 x2, half-moon plug, oil seal
- [ ] Cooling system parts staged (Koyorad HH060063 + OEM hoses + thermostat + water pump + coolant)
- [ ] Header bundle staged (13-Headers/install-guide.md parts list complete)
- [ ] IM bundle staged (Skunk2 Ultra Street + bored 66mm TB + RBC-to-Ultra-Street adapters as needed)
- [ ] Dyno session pre-booked for ~14 days after install day
- [ ] Break-in oil on the shelf (Driven BR30 or Brad Penn, 5 qt) + fresh synthetic for post-break-in

If any of these are NO, I'm not starting the teardown.

Tools Required

Not the header tool list. Several are single-use specialty tools. Budget accordingly -- the shop option starts looking rational right about here.

Absolutely Required

Tool	Purpose	Cost / Source
Lisle 37950 cam lock	Locks both cams at TDC cyl 1. Non-negotiable.	~$40
ARP stud installer tool	5-10 in-lb install torque on ARP studs	~$25
ARP Ultra-Torque lube	Ships with 208-4701 kit. Do not substitute.	Included
Torque wrench, 30-150 ft-lb, 1/2"	ARP final-pass head stud torque (75-80 ft-lb typical)	Calibrated <1 yr
Torque wrench, 10-80 ft-lb, 3/8"	Cam caps, IM bolts, header nuts, crank pulley first-pass	Own
Torque wrench, 20-200 in-lb, 1/4"	Valve cover (87 in-lb), timing cover, coil hold-down	Own

Torque angle gauge	Crank bolt +90deg, possibly ARP final angle	~$20
Crank pulley holder tool	Crank bolt breakaway at 181 ft-lb + 170k heat cycles	~$35
Breaker bar, 24"+	Crank + header nut breakaway	Own
Valve spring compressor	Only if troubleshooting a dropped keeper	~$60 rental
Straight edge (18"+, precision)	Deck flatness check, 0.002"/6" max warp spec	~$40 Starrett 385
Feeler gauge set (to 0.0015")	With straight edge	Own
Micrometer (0-1") + depth gauge	Verify ARP stud protrusion is consistent	Own or rent
Engine crane OR 2 people + head cradle	4P assembled head ~45 lb -- one-person lift risks dropping onto deck	Crane rental ~$35/day
Plastic razor blades (NOT metal)	Gasket residue removal without gouging aluminum	~$10 pack
Compressed air + safety glasses	Blow stud holes clean -- oil in hole = false torque reading	Shop compressor
Lint-free rags + shop vac thin nozzle	Final deck wipe + bore debris removal	Own

Also Needed (Bundled Mods Draw On These)

Tool	Cross-reference
Crank pulley harmonic balancer puller	../15-Pulleys-and-Harmonic-Balancer/
O2 sensor socket (22mm slotted)	../13-Headers/install-guide.md
Coolant funnel kit + bleed procedure	../09-Cooling-System/overview.md
Fuel line disconnect tool + rag for fuel spill	IM removal

All Fresh Gaskets (none reused, ever)

See Parts List below.

Parts List

The Head Package

Part	P/N	Qty	Price	Notes
4Piston Pro 156v2 All Motor Package (RBC-compatible, assembled)	4P-PRO-156V2	2-AMP	$2,740	Ferrea valves + Supertech 82lb dual springs + Ti retainers + locks, NEWEN 5-angle valve job, ~360 CFM. Confirm RBC fitment before clicking buy.
Cometic MLS head gasket .030"	C4561-030	1	$85	MLS layer -- NO sealer.
ARP head stud kit (K-series K20/K24)	ARP 208-4701	1	~$200	8740 chromoly, 190k PSI. Ships with ARP Ultra-Torque lube. DO NOT ORDER 208-4303 -- that is B-series GSR, wrong engine entirely.
Skunk2 Tuner Stage 2 cams (intake + exhaust)	305-05-0220	1 pair	~$550	Community default for FBO NA K20Z3. 22-25 HP published.
Skunk2 Pro adjustable cam gears	304-05-0300	1 pair	~$270	Required with aftermarket cams for dyno-dialed cam timing.

Gaskets + Seals (All New)

Part	P/N	Qty	Price
Valve cover gasket kit (incl. plug tube seals + grommets)	12030-PNC-000	1	$40
Intake manifold gasket (head-to-IM)	17115-RAA-A01	1	$15
Exhaust manifold / header gasket	18115-RRB-A01	1	$20
Front cam seals	91213-PLA-003	2	$30
Cam cap half-moon plug (rear)	90041-PNA-000	1	$8
Cam chain case oil seal	91319-P0Y-003	1	$12
Front crank main seal	91214-RZY-A01	1	$18

Consumables

Item	P/N	Notes
Break-in oil	Driven BR30 or Brad Penn 10W-30, 5 qt	First fire only, drained after cam break-in
Oil filter	15400-PLM-A02	New
Moly 60 assembly grease	Honda 08798-9010	Cam journals, VTC vanes, crank nose
ARP Ultra-Torque lube	Included with 208-4701	Stud threads + stud top + washer underside
Hondabond HT	Honda 08718-0004	Timing cover corners, cam cap corners
Honda Type 2 coolant OL999-9011	--	System refill
Brake cleaner (non-chlorinated) + lint-free rags	--	Deck cleanup

Bundled (Covered in Other Mod Docs' Parts Lists)

- Timing chain kit -- see ../05-Timing-Chain-Maintenance/overview.md
- VTC actuator 14310-RBC-003 -- see ../06-Ignition-Refresh/overview.md
- Header bundle (Skunk2 Alpha 412-05-1930 + Berk HFC + ARP 208-4301 + header gasket + copper nuts) -- see ../13-Headers/install-guide.md
- IM bundle (Skunk2 Ultra Street 307-05-0600 + bored 66mm TB + adapter as needed) -- see ../14-Intake-Manifold/overview.md
- Cooling bundle (Koyorad + hoses + thermostat + WP) -- see ../09-Cooling-System/overview.md
- Pulleys + NEW crank pulley bolt 90017-RWC-A01 -- see ../15-Pulleys-and-Harmonic-Balancer/overview.md
- Fresh plugs (NGK ILZKR7B-11S x4) + fresh coils (30520-RWC-A01 x4) -- see ../06-Ignition-Refresh/overview.md

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Step-by-Step -- Major Phases

12 phases (A through L), each with a checklist. I do not skip ahead. I do not start Phase D before Phase C is verified.

Phase A: Pre-Teardown (Day ?7 through Day 0)

Everything here is preventive.

1. **All parts in hand.** Unbox, count, verify part numbers. I do not discover I got 208-4303 instead of 208-4701 on teardown day.
2. **Unbox and inspect 4P head.** Verify assembled 156v2 All Motor Package, RBC-compatible. Inspect combustion chambers, valves, seats. Photograph the 4P documentation card (chamber CC, flow numbers, valve job spec) and file it in the repo.
3. **Torque wrench calibration.** Any wrench older than 12 months since calibration = send out or replace. Wrong head stud torque kills engines.
4. **PB Blaster on exhaust manifold studs** starting Day ?7, per ../13-Headers/install-guide.md schedule. Thermal cycling + penetrant daily.
5. **PB Blaster on IM bolts** Day ?3 through Day 0.
6. **Clear the garage.** Engine crane space, parts staging shelves, labeled zip-lock bags + Sharpie.
7. **Review ARP 208-4701 instruction sheet** end to end. Current spec is typically 30/50/75 ft-lb three passes with Ultra-Torque, star pattern from center. **Use the sheet that shipped with MY kit** -- ARP spec changes between revisions.
8. **Review Skunk2 Tuner Stage 2 + Pro cam gear notes.** Know whether I'm degreeing static or letting the tuner dyno-degree.
9. **Confirm dyno appointment** 10-14 days out.

Phase B: Disassembly (4-6 hours)

1. Battery negative off. Car sits overnight -- capacitor drain, fuel pressure bleed.
2. Drain coolant (../09-Cooling-System/overview.md bleed procedure in reverse). Drain oil.
3. Engine cover + K&N intake tube off (label/photograph every vacuum line).
4. Ignition coils (9 ft-lb) + plugs out -- these get replaced with Ignition Refresh parts anyway.
5. Valve cover off: 9 M6 bolts in spiral-inward pattern. Lift, inspect cams.
6. Fuel rail pressure bleed (Schrader valve, rag over it, depress with small screwdriver). Unbolt rail (9 ft-lb), disconnect fuel line, lift rail with injectors attached onto a clean rag. Do NOT let injector tips touch the ground.
7. IM off: see ../14-Intake-Manifold/overview.md Phase C. Disconnect TPS/IAT/MAP/IACV/brake booster/PCV. Unbolt IM star pattern 17 ft-lb release.
8. Exhaust manifold nuts off: see ../13-Headers/install-guide.md Phase D. Slow, back-and-forth, more PB Blaster. Stop if a stud feels like it's turning with the nut.
9. Crank pulley off: see ../15-Pulleys-and-Harmonic-Balancer/overview.md Phase E. **Discard the old bolt -- 90017-RWC-A01 is one-time-use.**
10. Timing chain cover off: see ../05-Timing-Chain-Maintenance/overview.md Phase F. 13 bolts of varying lengths -- document bag layout.
11. **Rotate crank clockwise to TDC cyl 1.** Crank pulley mark to pointer, both cam gear triangle marks point straight up parallel to deck.
12. **INSTALL LISLE 37950 CAM LOCK TOOL.** Non-negotiable. Cams will spring-rotate under valve spring pressure without it.
13. Measure tensioner rod protrusion. Record. At 170k, assume full chain replacement.
14. Remove tensioner, chain arm, fixed guide, chain off cam gears, crank gear off.
15. **Cam caps off in REVERSE torque sequence** (FSM sequence, three passes outside-in). Wrong order = bent cam from uneven spring load.
16. Lift cams out. Set on clean rag with paint-mark orientation notes.

17. **Head bolts out in REVERSE torque sequence** -- three passes, outside-in. Don't break torque on one bolt while others hold full load. Patience.
18. **Head lift off -- two people or engine crane with sling.** ~45 lb plus valvetrain. Lift STRAIGHT UP. Tilting drags dowels across the gasket surface.
19. Set head combustion-chamber-up on a head stand or clean wood blocks. Never on its deck surface.

Phase C: Block Prep (1-2 hours)

Every second here pays off at first-start.

20. **Plastic razor to remove old gasket material.** NEVER metal. Steel gouges aluminum = permanent leak path.
21. Rags stuffed in cylinder bores and oil passages. Do not skip.
22. Wipe deck with brake cleaner + lint-free rag until rag comes away clean.
23. **Straight edge across the deck in 6 directions:** longitudinal center, both diagonals, and at each cyl boundary. Feeler gauge under. **Max 0.002" warp over any 6" span.** More than that = machine shop.
24. **Compressed air into each stud hole** -- blow out oil/coolant/debris. Oil in a blind stud hole hydro-locks the stud and fakes out torque readings. #1 ARP install mistake.
25. Shop vac thin nozzle into each bore. Pull the rags from step 21.
26. Final deck wipe with brake cleaner. Let flash off.
27. Inspect block dowels -- both present, not bent, not mushroomed. Damaged = machinist.

Phase D: Install ARP 208-4701 Head Studs

28. **Re-read the ARP instruction sheet that shipped with the kit.** The sheet is truth, not this doc.
29. Clean each stud with brake cleaner + dry rag. No packing grease, no shipping oil.
30. Clean each block stud hole: bottle brush, brake cleaner, compressed air, twice. Cotton swab at the bottom should come up dry.
31. **Drop of Ultra-Torque on the bottom 3-4 threads of each stud.** Not a coating. Lets the stud seat without galling but doesn't fake out top-end torque.
32. Thread each stud in by hand until it bottoms. Back off a hair.
33. ARP stud installer tool, snug to **5-10 in-lb**. Bottomed, not stretched.
34. Micrometer or depth-gauge check all 10 studs -- protrusion within 0.010" of each other. Outlier = not seated, pull and re-inspect.
35. Verify top threads on each stud are clean and undamaged.

Phase E: Install Cometic C4561-030 Head Gasket

36. Final deck wipe with brake cleaner. Zero residue, zero fibers.
37. **"TOP" / part-number side of the Cometic UP.** MLS has a correct orientation -- reversed = compression leak.
38. Lower gasket over studs. Align block dowels through gasket dowel holes. Sits flat, zero edge gap.
39. **NO sealer, NO Hondabond, NO copper spray.** Cometic MLS is dry-install. Sealer on MLS creates a leak.
40. Verify coolant passages, oil passages, and bolt holes align to block features. Gasket installed backwards blocks coolant to one bank -- yes, this happens, the asymmetry is subtle.

Phase F: Install 4P Pro 156v2 Head

41. Final wipe of the 4P head deck, combustion chambers clean, valves seated.

42. **Two people OR engine crane with sling.** No solo-lift of 45 lb onto studs at chest height.
43. Lower head straight down. Feel for block dowels catching head dowel sockets.
44. Verify SQUARELY seated -- drops the last 1/4" under its own weight, studs protrude evenly through bolt holes.
45. **Do NOT rock or wiggle once down.** MLS is sensitive. If it doesn't drop flat first try, lift straight up and re-align.

Phase G: Head Stud Torque Sequence (CRITICAL)

This phase is where engines get killed. Read every word.

46. **Ultra-Torque on each stud:** one drop on top threads, one drop on the washer surface the nut rides on. Two drops per stud, not a glob.
47. ARP washers -- beveled side up (chamfer facing the nut).
48. ARP nuts finger-tight on all 10 studs.
49. **Follow the ARP sheet that shipped with MY kit.** Typical current spec:
    - **Pass 1:** 30 ft-lb, star from center outward
    - **Pass 2:** 50 ft-lb, same sequence
    - **Pass 3 (final):** 75 ft-lb (verify -- some revisions spec 80 ft-lb)
    - If the sheet says angle-torque instead, use the angle gauge exactly.
50. **Star pattern from center outward.** Cyl 2/3 bolts first, working out to cyl 1 and cyl 4. Even gasket loading, no warp.
51. Between passes, no binding. Smooth progression.
52. **After final pass:** let sit 10-15 minutes, retorque the final pass once more. Gasket takes a minute to fully compress.
53. Paint-pen dot each nut aligned to a dot on the stud top -- visual "has this moved?" reference for break-in re-check.

Phase H: Install Skunk2 Tuner Stage 2 Cams

54. **Moly 60 on each cam journal.** All five surfaces per cam. Generous -- cams run dry 2-3 sec on first fire.
55. Moly 60 on each cam lobe nose. Stage 2 cams wipe out in 20 seconds without it.
56. **Cam orientation per FSM TDC cyl 1 diagram:** intake triangle mark up-and-slightly-exhaust-side, exhaust triangle mark up-and-slightly-intake-side -- mirrored, both generally UP, parallel to deck. Both cams parked between lobe pushes.
57. Lay cams in. Skunk2 Stage 2 labeled I (intake) / E (exhaust) -- do not mix.
58. Install Skunk2 Pro adjustable cam gears loose -- final torque at chain install.
59. Cam caps in OEM sequence, three passes, **inside out** (cap 3 first, then 2+4, then 1+5), **final 9 ft-lb (108 in-lb)**. Over-torque crushes cam into journal; under-torque wrong clearance.
60. Rotate each cam slightly by hand -- spins free, no binding. Tight feel = STOP.
61. **Lisle 37950 back on** now that caps are torqued.
62. Cam degreeing decision: Skunk2 Pro gears are made for dyno-degreeing. Most FBO builds let the tuner degree on the dyno -- don't pre-degree unless the Stage 2 instruction says to.

Phase I: Chain, VTC Actuator, Tensioner

63. **VTC actuator 14310-RBC-003:** see ../06-Ignition-Refresh/overview.md and

../05-Timing-Chain-Maintenance/overview.md Phase I. Moly 60 on vanes, slide onto intake cam nose, hold cam on the flats (not a lobe), center bolt **83 ft-lb dry, no impact**.

64. **Route new chain** per ../05-Timing-Chain-Maintenance/overview.md Phase J: single colored link to crank mark, two colored links (9 apart) to cam gear marks. Verify cam gear triangles still point up AND colored links align.
65. Fixed chain guide in, then pivoting chain arm.
66. **Tensioner piston reset:** compress with retainer pin through lock hole. Verify pin holds.
67. Tensioner installed with bolts at **8.7 ft-lb**. **Pin STAYS IN until everything is bolted down.**
68. **Pull retainer pin.** Piston extends, chain tensions.
69. **Rotate crank 2 full turns clockwise by hand** (old bolt finger-tight for leverage).
70. **Verify cam marks return to TDC cyl 1.** Not aligned = chain off a tooth -- STOP, re-route.
71. **Remove Lisle 37950** once timing is verified correct.
72. Final torque Skunk2 Pro cam gear center bolts per Skunk2 spec (typically 37 ft-lb with blue Loctite). Discs stay at static -- tuner dyno-adjusts.

Phase J: Reassembly

73. **Timing cover:** press new crank seal in, Hondabond HT bead per ../05-Timing-Chain-Maintenance/overview.md Phase L, install within 5 min, bolts crisscross from center out **8.7 ft-lb**.
74. **Crank pulley:** NEW bolt 90017-RWC-A01, Moly 60 on crank nose, **131 ft-lb -> loosen fully -> 37 ft-lb -> +90deg** per ../15-Pulleys-and-Harmonic-Balancer/overview.md Phase G. Do not reuse old bolt.
75. **Valve cover:** fresh 12030-PNC-000 kit + plug tube seals + grommets. Hondabond HT dabs at the 4 cam cap corners. **7.2 ft-lb (87 in-lb)** spiral out from center.
76. **Plugs:** fresh NGK ILZKR7B-11S, anti-seize sparingly (or none), **13 ft-lb**. Never muscle iridium.
77. **Coils:** fresh 30520-RWC-A01 x4, dielectric grease on boots, hold-down **9 ft-lb**.
78. **Headers:** per ../13-Headers/install-guide.md Phase E-F. New 18115-RRB-A01, 3-pass star **23 ft-lb**.
79. **IM with RBC-compatible adapters:** per ../14-Intake-Manifold/overview.md. New 17115-RAA-A01, 17 ft-lb star from center. Reconnect vacuum lines per teardown photos. Coolant bypass block-off plate on.
80. **Fuel rail + injectors:** rail bolts 9 ft-lb, fuel line quick-connect, Schrader cap back on.
81. Intake tube + K&N filter on, vacuum lines reconnected per photos.
82. **Coolant refill + bleed** per ../09-Cooling-System/overview.md. Honda Type 2 OL999-9011. Bleed properly -- air in the system hurts first-start.
83. New oil filter 15400-PLM-A02, **break-in oil** (Driven BR30 or Brad Penn 10W-30) 4.4 qt.
84. Battery reconnect. Key ON -- verify dash, no CELs, fuel pump primes.

Phase K: First-Start Protocol (Cam Break-In)

This phase is the difference between 200k-mile cams and 2,000-mile cams.

85. **Oil pump prime:** pull the main relay (driver footwell), crank 10 sec, rest 30 sec, crank 10 more sec. Spins the oil pump without firing -- pressurizes the top end before cams try to run.
86. **Reinstall main relay.** Verify fuel pressure primes on key-ON.
87. **Start the engine.** Should fire within 3 seconds. Long cranking = diagnose, don't keep cranking.
88. **IMMEDIATELY elevate to 2000-2500 RPM and HOLD for 20 minutes.** Cam break-in. Cams need splash AND RPM-driven lobe loading to seat. **Do NOT idle during break-in.**

89. **Watch coolant temp continuously.** ECT above 220degF = shut down, diagnose bleed/thermostat/air pocket.

90. **Listen for anomalies:** cam-lobe tap (wrong lash), valvetrain clatter (dropped keeper), exhaust leak hiss, coolant smell, oil smoke (pinched gasket). Any = shut down, fix.

91. **After 20 min:** idle 2-3 min, then shut down.

92. **Drain break-in oil + filter.** Break-in deposits metal fines -- do not circulate. Refill with fresh break-in oil or 5W-30 synthetic per tuner's call.

93. Let engine cool fully before any further driving.

Phase L: First Drive + Mandatory Dyno Retune

94. **First 100 mi: partial throttle only.** No WOT, no VTEC crossovers if I can help it. Heat cycle, stud settle, gasket seat.

95. **Flash a conservative base map** before first drive -- community FBO + 4P + Stage 2 map with timing pulled 2deg safety, or previous cal with safety pull. **Do NOT drive on the old pre-head cal.** Wrong cam/head on old fuel map = lean WOT = cracked piston.

96. **Re-torque valve cover at 100 mi** per ../05-Timing-Chain-Maintenance/overview.md Phase P. 7.2 ft-lb, 10 minutes.

97. **Optional but smart at 100 mi:** compression + leak-down test. All 4 cyl within 10% of each other, leak-down under 10% each. Fail = do not go to dyno yet.

98. **Confirm dyno session** (booked in Phase A). Mandatory $500-800 at a Hondata-capable shop.

99. **At the dyno:** tuner re-baselines VE, rebuilds fuel 4000-8000 RPM, max-safe timing with zero knock, VTC crossover for new cam/head, degrees Skunk2 Pro cam gears for the flow curve, produces BOTH daily + sport maps per ../10-Hondata-FlashPro/tuning-workflow-and-maps.md.

100. **Do not drive hard between install and dyno.** That 10-mile window is where engines get hurt.

Why 4P Pro 156v2 All Motor Package (RBC-Compatible)

$2,740 assembled is expensive. Here's why I'm not saving money here.

Ferrea valves + Supertech 82lb dual springs + Ti retainers + locks matched and installed by the shop that built the head. No mix-and-match risk. I don't trust myself to spec a valvetrain and set spring height without a machine shop; 4P delivers shop-grade precision.

NEWEN 5-angle valve job -- CNC-repeatable seat geometry. Smoother throat-to-chamber transition than a 3-angle. ~360 CFM at stock valve size vs ~290 CFM for a stock refreshed head -- 25% more flow, which is exactly what Skunk2 Ultra Street + Stage 2 cams will demand on the intake side.

Manganese bronze guides + oversized steel intake seats -- durable for E85 moisture over the next 200k miles. Important because I'm going E85 in Phase 7; stock aluminum guides get chewed up on ethanol-rich fuel over time.

RBC-compatible assembled package. Bare 4P Pro 156v2 CNC is PRB-base and would need custom RBC work. The All Motor Package assembled version is RBC-compatible per 4P's fitment list. **Verified in writing from Luke before I click buy.**

Flow match to the build: 360 CFM is what Skunk2 Ultra Street IM + bored 66mm TB + Skunk2 Alpha + Stage 2

cams want. Stock 290 CFM would bottleneck the rest of the FBO. 4P Pro 163 at 400 CFM is ceiling I don't need for 250 WHP.

Pre-Order Verification Email Template

Before clicking buy on the $2,740 package, I send this to **Luke@team4piston.com**:

> Subject: RBC Fitment Confirmation -- Pro 156v2 All Motor Package for 2009 Civic Si K20Z3 > > Hi Luke, > > I'm planning to order the 4P Pro 156v2 All Motor Package (assembled -- Ferrea valves + Supertech springs + Ti retainers) for my 2009 Civic Si (FG2, K20Z3, RBC casting). > > Confirming before ordering: is the Pro 156v2 **All Motor Package** (assembled, not bare CNC) direct-fit on K20Z3 RBC? I know the bare CNC Pro 156v2 is PRB-base; I want the All Motor Package RBC compatibility in writing before buying. > > Build context: 220-250 WHP NA, pump then E85. Skunk2 Ultra Street IM, bored 66mm DBW TB, Skunk2 Alpha + Berk HFC, Skunk2 Tuner Stage 2 cams + Pro adjustable cam gears, ARP 208-4701 studs, Cometic C4561-030. 100% stock bottom end. > > Questions: (1) Drop-in fit on K20Z3 RBC, no casting work? (2) Current lead time? (3) As-machined chamber CC documentation? (4) Known issues pairing with Skunk2 Tuner Stage 2? > > Thanks, > LxveAce

Written reply confirming RBC fitment = archive the email, file in repo, order. No verbal "yeah it'll fit."

First Drive Checklist

Gentle 10-mile loop after break-in, then verify every item. Any NO = no dyno yet.

- [ ] Smooth idle (slight Stage 2 cam lope acceptable, not rough)
- [ ] No cam-lobe tapping or valvetrain clatter under the valve cover
- [ ] No exhaust leak hiss at header-to-head flange
- [ ] Oil pressure reads normal range at idle + 2500 RPM
- [ ] No coolant weep at head-to-block deck
- [ ] No oil weep at valve cover, timing cover, or front crank seal
- [ ] Coolant temp stabilizes at thermostat spec (~185-195degF) -- no runaway heat
- [ ] No CEL, no pending OBD2 codes
- [ ] Datalog at idle + 2500 RPM shows stable fuel trims (±5%), commanded vs actual cam phase within 2deg
- [ ] No compression loss (even firing, no startup smoke)
- [ ] Spark plugs clean and even when pulled (optional at 100 mi)

Common Mistakes

Every one of these is a documented forum failure mode.

Mistake	Consequence	Prevention
Ordering bare 4P Pro 156v2 CNC for an RBC engine	PRB-base casting won't mate to K20Z3 RBC deck + coolant routing	ONLY the All Motor Package is RBC-compatible. Confirm in writing via Luke.
Ordering ARP 208-4303 instead of 208-4701	208-4303 is B-series GSR -- wrong engine entirely	Verify at click time. 208-4701. Not 208-4303.

Reusing old OEM TTY head bolts instead of ARP studs	Stretches past yield, clamp drops, gasket blows	ARP 208-4701 every time. $200 is cheap insurance.
Dry-assembling ARP studs without Ultra-Torque	False torque reading, galled block threads	Ultra-Torque ships with kit. One drop bottom, one drop top, one drop washer. Not optional.
Oil in stud holes during install	Hydro-lock, false torque, clamp far under spec	Compressed air + brush + brake cleaner + cotton swab. Dry holes.
Skipping cam break-in RPM hold	Cam lobes wiped in 20 seconds of idle	20 min at 2000-2500 RPM first start. Non-negotiable.
Head studs not following ARP star pattern	Uneven clamp, warped head	3-pass star from center outward per the sheet shipped with MY kit.
Skipping angle gauge when ARP calls for it	+90deg is ~12-15 ft-lb actual, not visual	$20 angle gauge, use it.
Rotating crank with chain off, no cam lock	Valve-to-piston contact, bent valves	Lisle 37950 in BEFORE chain out. Stays in until verified correct after 2 rotations.
Pre-degreeing cams wrong before dyno	Tuner wastes 2 dyno hours re-degreeing	Let the dyno tuner degree Skunk2 Pro gears -- that's what they're for.
Sealer on the Cometic MLS gasket	Creates leak path between layers	Dry install only. No Hondabond, no copper spray.
Reusing crank pulley bolt 90017-RWC-A01	TTY stretch, loosens, crank damage	New bolt every time. ~$12.
Reusing OEM exhaust manifold gasket	Crush pattern wrong for new geometry, leaks	Fresh 18115-RRB-A01. OEM only.
Driving to dyno on pre-head tune	Lean WOT = cracked piston	Conservative base flash before first drive. 10-mile gentle loop.

Confirm-Before-Buy Checklist

Read this list the morning of the day I click buy on the $2,740 package:

- [ ] **RBC fitment confirmed via email from Luke@team4piston.com** for the 4P Pro 156v2 All Motor Package (assembled)
- [ ] **All parts in hand** or firm shipping ETAs that fit the install week:
  - [ ] 4P head package
  - [ ] Cometic C4561-030 gasket
  - [ ] ARP 208-4701 stud kit (VERIFIED P/N, not 208-4303)
  - [ ] Skunk2 Tuner Stage 2 cams 305-05-0220
  - [ ] Skunk2 Pro adjustable cam gears 304-05-0300
  - [ ] All fresh OEM gaskets (valve cover kit, IM, exhaust, cam seals, half-moon)
  - [ ] Timing chain bundle (see 05-Timing-Chain-Maintenance/)
  - [ ] VTC actuator 14310-RBC-003 (-003, not -002)
  - [ ] Header bundle (see 13-Headers/)
  - [ ] IM bundle (see 14-Intake-Manifold/)
  - [ ] Cooling bundle (see 09-Cooling-System/)
  - [ ] NEW crank bolt 90017-RWC-A01
  - [ ] Break-in oil + filter

- [ ] **Lisle 37950 cam lock tool** on my bench, not in a cart somewhere
- [ ] **ARP stud installer tool** on my bench
- [ ] **Torque wrenches calibrated** within last 12 months (or replacement bought)
- [ ] **Straight edge + feeler gauges** for deck flatness check
- [ ] **Machine shop backup identified** in case deck warp check fails or a bolt hole strips -- specific shop name, phone number, ballpark quote for resurface ($400-600) and for thread repair ($200-400)
- [ ] **Dyno session pre-booked** for ~14 days after install day with a Hondata-capable K-series tuner
- [ ] **Shop option quote** on the table: Honda specialist shop quote for bundled install. If under $2,000 and the shop has done 4P head installs before, seriously reconsider DIY at this complexity level.
- [ ] **Clear weekend + vacation days** scheduled. This is not the job I crash-squeeze into a Saturday.

See Also / Cross-references

- overview.md -- the research behind picking 4P Pro 156v2 All Motor, cost breakdown, alternatives rejected
- ../05-Timing-Chain-Maintenance/overview.md -- full chain service procedure (measure tensioner rod, replace chain/arm/guides)
- ../05-Timing-Chain-Maintenance/install-guide.md -- Phase F through Phase L of timing cover + chain work references here
- ../06-Ignition-Refresh/overview.md -- VTC actuator 14310-RBC-003 (revised part), fresh plugs, fresh OEM coils
- ../09-Cooling-System/overview.md -- coolant drain + refill, Koyorad + hoses + thermostat + water pump bundle
- ../13-Headers/install-guide.md -- Skunk2 Alpha header install (exhaust stud prep + install happens during same teardown)
- ../14-Intake-Manifold/overview.md -- Skunk2 Ultra Street IM + bored 66mm TB install at head reinstall
- ../15-Pulleys-and-Harmonic-Balancer/overview.md -- ATI Super Damper + NEW crank pulley bolt 90017-RWC-A01 + yield torque procedure
- ../10-Hondata-FlashPro/tuning-workflow-and-maps.md -- mandatory dyno retune, daily/sport two-map requirement, reliability-first ceilings
- ../docs/torque-specs.md -- every fastener spec cross-referenced here
- ../docs/maintenance-parts-catalog.md -- head P/Ns added to build-wide catalog

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Last updated: 2026-04-20

Suspension -- Coilovers + Full Refresh

Status: Research Complete, Not Yet Purchased

My Build Philosophy

At 170,000 miles, every rubber bushing, shock, and joint in my suspension is degraded. Rather than just replacing worn parts with OEM equivalents, I'm treating this as an opportunity to do a complete suspension overhaul -- coilovers for adjustable ride height and damping, fresh polyurethane bushings throughout, upgraded sway bars, and proper alignment correction for a lowered car.

This fits my build's daily/sport duality perfectly: coilovers with adjustable damping can be softened for daily commuting and stiffened for spirited driving, just like my valved exhaust switches between quiet and loud.

FG2 Suspension Layout -- Every Component

Front Suspension (MacPherson Strut)

Component	Qty	What It Does	170k Status
Strut assemblies (shock + spring combined)	2	Controls body motion, dive, roll -- the single most important suspension component	Degraded. OEM shocks last 60-100k miles. At 170k, mine are providing maybe 50-60% of original damping.
Strut top mounts (bearing + rubber isolator)	2	Allows strut to rotate during steering, isolates road harshness from chassis	Rubber hardened, bearing likely worn. I've been hearing creaking when turning.
Bump stops + dust boots	2	Prevents harsh bottoming out, protects shock shaft from debris	Likely cracked or detached from strut body
Lower control arms (LCA)	2	Primary structural link -- connects wheel to subframe	Arms themselves are fine (steel). Bushings are the issue.
LCA compliance bushings	4 (2/arm)	Allows controlled arm movement while isolating vibration. Front bushing + rear bushing per arm.	First-to-fail item. Almost certainly cracked, torn, or mushy. Causes clunking and steering vagueness.
Lower ball joints	2	Pivot point connecting LCA to steering knuckle. Allows vertical + rotational movement.	Need to check for play. 170k is past typical ball joint life (80-120k).
Tie rod ends (outer)	2	Connects tie rod to steering knuckle. Transmits steering input to wheel.	Likely worn -- I need to check for play by grabbing tire at 3/9 o'clock and rocking.
Tie rod ends (inner)	2	Connects steering rack to tie rod. Less wear-prone than outers.	Inspect. Replace if play detected.
Sway bar (anti-roll bar)	1	Connects left and right suspension to resist body roll in corners	SI-specific (thicker than base Civic). Bar itself is fine -- bushings wear out.
Sway bar bushings	2	Frame-mounted rubber bushings holding sway bar to chassis	Hardened and cracked at 170k. Causes clunking.
Sway bar end links	2	Connects sway bar ends to each strut/LCA. Ball joints at each end.	Bushings likely worn. Clunking over bumps.

Wheel bearings/hubs	2	Sealed bearing units allowing wheel rotation. 5-lug pattern.	Sealed, non-serviceable. Replace only if noise/play detected.
Steering knuckles	2	Structural -- holds wheel bearing, connects to ball joints and tie rods	Fine unless physically damaged

Rear Suspension (Multi-Link)

Component	Qty	What It Does	170k Status
Shock absorbers (separate from springs)	2	Controls rear body motion and rebound	Degraded. May be visibly leaking fluid.
Coil springs	2	Supports vehicle weight, determines rear ride height and rate	SI-specific (stiffer than base Civic). Springs rarely fail but I'll check for sag.
Shock top mounts	2	Upper attachment point for shock. Rubber isolation.	Rubber compressed/cracked at 170k
Bump stops	2	Prevents excessive rebound travel	Likely deteriorated
Lower control arms	2	Primary rear suspension link. Controls wheel position.	Arms fine -- bushings wear. Honda TSB exists for revised geometry (stamped "C").
Lower control arm bushings	4	Compliance bushings at forward and rear attachment points	Worn -- causes rear instability and creaking
Upper control arms	2	Secondary link. Controls camber geometry. Stock has NO camber adjustment.	Arms fine. No factory rear camber adjustment -- aftermarket adjustable arms are critical if I'm lowering the car.
Upper control arm bushings	4	Attachment point bushings	Worn -- causes camber drift and uneven tire wear
Trailing arms	2	Longitudinal members controlling fore/aft wheel position	SI has reinforced trailing arms. Structure is fine.
Trailing arm bushings	4	Front and rear bushings per trailing arm	Known high-mileage failure point. Hardened, cracked. Causes creaking and rear instability.
Sway bar	1	Rear anti-roll bar. Stock SI rear bar is small/weak.	Bar is fine -- bushings wear. Upgrading the rear bar is one of the highest-value handling mods I can do.
Sway bar bushings	2	Frame-mounted bushings	Deteriorated at 170k
Sway bar end links	2	Connects sway bar to rear suspension arms	Bushings worn
Wheel bearings/hubs	2	Sealed bearing units	Replace only if noise/play
Rear knuckles	2	Structural -- holds hub, connects all rear links	Fine unless damaged

Component-by-Component: Options, Research, and Buy Links

1. Coilovers -- Going with coilovers, no question

Coilovers replace the front strut assemblies (shock + spring) and rear shocks + springs with a single adjustable unit per corner. Ride height adjustment and (on most) damping adjustment.

What I was looking for:

- Monotube design (better heat dissipation than twin-tube)

- 30+ clicks of damping adjustment (more = finer tuning)
- Pillow ball front camber plates (allows camber adjustment without separate bolts)
- Rebuildable/revalveable (extends lifetime, allows future tuning)
- Appropriate spring rates for street/sport use (6-8 kg/mm front, 4-6 kg/mm rear)

Coilover	Part Number	Spring Rates	Damping	Drop Range	Price	Notes
BC Racing BR	A-18-BR	6K/6K (custom free)	30-click	1-3"	~$1,195	Community #1 pick. Pillow ball camber plates included. Rebuildable. Massive aftermarket support. Best value for the money.
Fortune Auto 500	FA500-FA	Custom to order	24-click digressive	1-3"	~$1,978	Premium. 5-year warranty. Rebuildable/revalveable. "Buy once cry once." Best ride quality in this class. 4-8 week build time.
Tein Flex Z	VSB14-CUAA1	Proprietary	16-click	1-3"	~$1,045	Solid mid-range. Includes upper mounts with camber. Good daily comfort. Less adjustability than BC.
KW V1	10250052	Fixed	None (pre-set)	1-2.6"	~$1,694	German quality. Excellent ride. No damping adjustment at this price = hard sell vs BC or Fortune Auto.
Skunk2 Pro-ST	541-05-8750	8K/6K	12-click	Adjustable	~$1,147	Decent option. Less community support than BC Racing.
K-Tuned K1 Street	--	8K/6K	36-click	Adjustable	~$1,059	Good specs on paper. Less real-world feedback than BC Racing.
Godspeed Mono-SS	--	--	16-click	Adjustable	~$675	Budget. Ride quality complaints come up a lot.
Truhart StreetPlus	--	--	36-click	Adjustable	~$619-699	Budget. Mixed reviews.

Buy Links -- BC Racing BR (A-18-BR):

Retailer	Price	Link
K Series Parts	~$1,195	K Series Parts
BC Racing Direct	~$1,195	BC Racing
Redline360	~$1,195	Redline360

Buy Links -- Fortune Auto 500 (FA500-FA):

Retailer	Price	Link
Fortune Auto Direct	~$1,978	Fortune Auto

Buy Links -- Tein Flex Z (VSB14-CUAA1):

Retailer	Price	Link
Springrates.com	~$1,045	Springrates

My pick: BC Racing BR (A-18-BR) at ~$1,195. Best value, most community support, 30-click damping, pillow ball camber plates included, rebuildable, custom spring rates at no extra charge. The 8th gen Civic SI community overwhelmingly recommends these, and after going through all the options, I agree.

If my budget allowed and I wanted the absolute best ride quality: Fortune Auto 500 at ~$1,978 with 5-year warranty and custom valving. But that's nearly double the price for diminishing returns on a street car.

2. Rear Camber Arms -- Non-Negotiable with Coilovers

The FG2 has **zero factory rear camber adjustment.** When I lower the car, the rear wheels tilt inward (negative camber) beyond acceptable limits. Without adjustable rear camber arms, my rear tires will eat through the inner edges in months. Not optional.

Product	Part Number	Adjustment	Construction	Bushing Type	Price	Notes
Skunk2 Pro Series	516-05-0625 (Gold)	6+ degrees total	Forged steel + 7075 aluminum	Polyurethane + spherical	~$333-350	Gold standard. Telescoping design -- adjust camber without removing the arm. Best quality.
Skunk2 Pro Series	516-05-0620 (Hard Anodized)	6+ degrees total	Same as above	Same	~$333-350	Same thing, different finish
Godspeed Gen3	AK-043-G3	Adjustable	CNC 6061 aluminum	Spherical bearings	~$238	Best budget option. 40% stiffer, 30% lighter than stock. Lifetime warranty.
Godspeed Gen2	AK-043	Adjustable	CNC 6061 aluminum	Polyurethane	~$179	Budget. Gen3 with spherical bearings preferred over Gen2.
Hardrace	--	Adjustable	Steel/aluminum	Hardened rubber or pillow ball	~$150-200	Solid option. Check buyhardrace.com for current 8th gen catalog.
SPC	--	Adjustable	--	--	~$150-200	Check SPC catalog for current part number

Buy Links -- Skunk2 Pro Series (516-05-0625):

Retailer	Link
Skunk2 Direct	Skunk2
K Series Parts	K Series Parts
Hybrid Racing	Hybrid Racing

Buy Links -- Godspeed Gen3 (AK-043-G3):

Retailer	Link
K Series Parts	K Series Parts

My pick: Skunk2 Pro Series (516-05-0625) at ~$333. The telescoping adjustment design (no removal needed for camber changes) is worth the premium over Godspeed. This is a "set it and maybe adjust later" part -- I want quality here.

Budget alternative: Godspeed Gen3 (AK-043-G3) at ~$238 with spherical bearings. Still a solid option if I need to save.

3. Front Camber Adjustment

The BC Racing BR coilovers include pillow ball camber plates for the front -- those give me ~2 degrees of front camber adjustment. But additional fine-tuning with camber bolts is useful and cheap.

Product	Part Number	Adjustment	Price	Notes
SPC EZ-Cam 14mm	81260	+/- 1.75 per bolt	~$25-36/pair	Industry standard. 1 bolt/side = +/- 1.75. 2 bolts/side = +/- 3.
Ingalls 81260	81260	Same	~$25-36	Same product -- SPC and Ingalls are the same manufacturer.

Buy Links:

Retailer	Link
HARDmotion	HARDmotion -- $25
Amazon	Search "SPC 81260"

My pick: SPC 81260 at ~$30. Cheap insurance for fine-tuning front camber. Nice-to-have since the BC Racing camber plates handle most adjustment, but for $30 there's no reason not to.

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4. Full Suspension Bushing Kit -- Must-Have at 170k Miles

At 170,000 miles, every rubber bushing in my suspension is shot -- hardened, cracked, or torn. Replacing them all with polyurethane is the second biggest "feel" upgrade after coilovers. **I'm doing this while the suspension is apart for the coilover install -- no way I'm taking it all apart twice.**

Product	Part Number	Material	What's Included	Price	Notes
Energy Suspension Master Kit	16.18114G (Black)	Hyper-Flex polyurethane	Front LCA bushings, rear LCA bushings, knuckle bushings, front sway bar bushings, rear sway bar bushings	~$278	Most complete kit. One box, everything I need.
Energy Suspension Master Kit	16.18114R (Red)	Same	Same	~$278	Same kit, red color
Prothane Complete Kit	82020 (Red)	Polyurethane	Front/rear control arm bushings, sway bar bushings	~$248	Good kit. Does NOT include front bushing on rear trailing arm -- supplement with Hardrace 6926. Includes Super Grease. Lifetime warranty.
Prothane Complete Kit	8-2020-BL (Black)	Same	Same	~$248	Same kit, black color

Buy Links -- Energy Suspension (16.18114G):

Retailer	Link
HARDmotion	HARDmotion -- $278
Amazon	Search "Energy Suspension 16.18114"

Buy Links -- Prothane (82020):

Retailer	Link
Prothane Direct	Prothane
Amazon	Search "Prothane 82020"

My pick: Energy Suspension 16.18114G at ~$278. Most complete coverage in a single kit. Going with black because red bushings peeking out aren't my thing.

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5. Rear Trailing Arm Bushings -- Must-Have at 170k Miles

These are a **known high-mileage failure point** on the 8th gen Civic. The Energy Suspension master kit covers most bushings, but the front bushing on the rear trailing arm may need to be purchased separately depending on kit coverage.

Product	Part Number	Material	Price	Notes
Hardrace Hardened Rubber	6926	Hardened rubber	~$143-184	My preference for street. No greasing required. Less NVH than spherical. 2-piece set covers both sides.
Energy Suspension	Included in 16.18114	Polyurethane	Included	Need to check kit contents for trailing arm coverage
Prothane 8319	8319	Polyurethane	~$50-70	Rear control arm bushing kit. Supplements Prothane 82020 which misses the front trailing arm bushing.

Buy Links -- Hardrace (6926):

Retailer	Link
Hardrace USA	Hardrace -- $143-184

My pick: Hardrace 6926 at ~$150. Hardened rubber is the sweet spot for a street car -- poly bushings in trailing arms can transmit harsh NVH that gets old quick on a daily driver. If I'm buying the Energy Suspension master kit, I'll check if it covers the trailing arm bushings first to avoid double-buying.

6. Sway Bars -- Highly Recommended

Upgraded sway bars reduce body roll during cornering. The stock SI rear bar is undersized -- upgrading it is one of the highest-value handling mods on the 8th gen platform.

Community consensus: Start with a rear bar upgrade. The stock front bar is reasonably sized. The stock rear bar is honestly weak.

Rear Sway Bars

Product	Part Number	Size	Adjustable?	End Links?	Price	Notes
Progress Technology	62.1009	22mm	No	Not included	~$220	Most popular rear bar. Cold-formed CNC alloy steel. Made in USA. Dramatically reduces rear body roll.
Progress Technology	62.1025	24mm	3-way	Included	~$349	My pick. 3 stiffness settings. Comes with end links.
Eibach Anti-Roll	4031.312	--	2-way	Not included	~$269	2-way adjustable. Good quality.

Front Sway Bars

Product	Part Number	Size	Price	Notes
Progress Technology	61.1006	27mm	~$288	Upgrade from stock SI bar. Cold-formed CNC alloy steel.
Eibach	--	--	~$217	Front bar only

Front + Rear Combo

Product	Part Number	Price	Notes
Eibach Anti-Roll Kit	4031.320	~$400-486	Front + rear combo. 2-way adjustable rear.

Buy Links -- Progress 22mm Rear (62.1009):

Retailer	Link
Progress Technology Direct	Progress -- $220

Buy Links -- Progress 24mm Adjustable Rear (62.1025):

Retailer	Link
Progress Technology Direct	Progress -- $349

Buy Links -- Progress 27mm Front (61.1006):

Retailer	Link
Progress Technology Direct	Progress -- $288

Buy Links -- Eibach Front + Rear Kit (4031.320):

Retailer	Link
Redline360	Redline360

My pick: Progress 24mm adjustable rear bar (62.1025) at ~$349. The 3-way adjustability matches my daily/sport philosophy perfectly -- softer setting for commuting, stiffer for spirited driving. And it includes end links, which saves me a separate purchase. I'll add the Progress 27mm front (61.1006, ~$288) if budget allows for the complete front/rear upgrade.

7. Sway Bar End Links

On a lowered car, stock-length end links pre-load the sway bar at rest (the geometry changes). Adjustable-length end links let me dial out that pre-load for proper sway bar function.

Product	Part Number	Adjustable?	Price	Notes
Whiteline Adjustable	KLC139	Yes (+/- 15mm)	~$154/pair	Ball joint ends. Properly sets sway bar preload on lowered cars.
Progress End Links	67.10.062	--	~$128	3-piece kit. Included with Progress 62.1025 adjustable rear bar.
Moog (OEM replacement)	K80768 (R) / K80769 (L)	No	~$25-40 each	Good quality OEM replacement. Fine if dropping under 1.5".

Buy Links -- Whiteline (KLC139):

Retailer	Link
ProCivic	ProCivic

My plan: If I buy the Progress 24mm adjustable rear bar (62.1025), it includes end links. For the front, Moog K80768/K80769 at ~$35/each are fine for drops under 1.5". Whiteline KLC139 if I go over 1.5".

8. Roll Center Adjusters -- Must-Have if Dropping 1.5"+

When you lower a MacPherson strut car, the lower ball joint angle changes, moving the roll center below the ground plane. This actually makes the car handle **worse** -- more body roll, more weight transfer, and bump steer. Roll center adjusters (extended lower ball joints) restore proper geometry.

Product	Part Number	Extension	Price	Notes
K-Tuned Roll Center Adjusters	KTD-RCA-611	10mm	~$190	Stiffer durometer bushings. Restores geometry on lowered cars. Reduces bump steer.
K-Tuned Lower Ball Joints	KTD-LBJ-611	Standard	~$139	Standard replacement. No extension.

Buy Links -- K-Tuned RCA (KTD-RCA-611):

Retailer	Link
K Series Parts	K Series Parts -- $190

My plan: K-Tuned KTD-RCA-611 at ~$190. Required if I'm dropping 1.5" or more. If I land at 1-1.5", still strongly recommended. Under 1", not needed.

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9. Extended Tie Rod Ends -- Nice-to-Have for Lowered Cars

Same concept as roll center adjusters, but for the steering rack. Reduces tie rod angle on lowered cars to reduce bump steer.

Product	Part Number	Price	Notes
K-Tuned Extended Tie Rods	KTD-TRO-611	~$200/pair	Mount upside-down through knuckle. Includes stiffer durometer bushings. Pairs with roll center adjusters for complete geometry correction.

Buy Links -- K-Tuned (KTD-TRO-611):

Retailer	Link
K Series Parts	K Series Parts -- $200

My take: Nice-to-have. Pairs with roll center adjusters for complete geometry correction on cars dropped 1.5"+. Not necessary under 1.5" drop. I'll decide based on where I set my ride height.

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10. Tie Rod Ends (OEM Replacement) -- Must-Have at 170k Miles

I'm replacing the outer tie rod ends regardless of lowering. At 170k miles these are wear items, period. I'll inspect the inners too.

Product	Part Number	Position	Price	Notes
Moog Outer (Right)	ES800373	Front outer, passenger	~$25-40	OEM-equivalent quality
Moog Outer (Left)	ES800374	Front outer, driver	~$25-40	OEM-equivalent quality
Moog Inner (Passenger)	EV800246	Front inner, passenger	~$25-40	Replace if play detected
Beck Arnley Inner	101-7783	Front inner	~$25-40	Alternative to Moog

Buy Links: Available at AutoZone, RockAuto, Amazon -- search by part number.

My plan: Replace all 4 tie rod ends (2 outer, 2 inner) at ~$100-160 total. I'm already tearing the front suspension apart for coilovers. New tie rods are cheap insurance for steering precision and are required for a proper alignment anyway. No reason to reuse 170k-mile tie rods.

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11. Lower Ball Joints -- Must-Have at 170k Miles

I'll check for play by jacking the car up and checking for lateral movement at the wheel. At 170k, these are likely worn.

Product	Part Number	Price	Notes
Moog	K500149	~$30-60/each	OEM-equivalent quality. Two needed (left + right).
K-Tuned	KTD-LBJ-611	~$139/pair	Standard replacement. Higher quality than Moog.
K-Tuned Roll Center	KTD-RCA-611	~$190/pair	Extended. Replaces ball joints AND corrects roll center. Best option if lowering 1.5"+.

My plan: If I'm dropping 1.5"+, I'll get the K-Tuned RCA (KTD-RCA-611) -- it replaces ball joints and corrects roll center in one part. Two birds, one stone. If dropping less, Moog K500149 at ~$50/each is fine.

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12. Rear Shock Top Mounts -- Replace During Install

Most coilovers reuse the OEM rear top mount. At 170k, the rubber is compressed. I'm replacing with fresh OEM Honda mounts while I'm in there.

Product	Part Number	Price	Notes
Honda OEM Rear Mount	52675-SNA-A03	~$20-35/each	Replace both sides. Cheap insurance.

My plan: Replace both at ~$50 total. Doing this during coilover install. Not worth skipping for $50.

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13. Upgraded Sway Bar Bushings

If keeping stock sway bars (or any bar without included bushings), upgrade the bushings to polyurethane.

Product	Part Number	Price	Notes
Energy Suspension Front (27mm)	16.5148	~$20-30	Size must match sway bar diameter. 27mm fits stock SI front bar and Progress 27mm.
Included in master kit	16.18114	Included	The Energy Suspension master kit includes sway bar bushings.
Included with Progress bars	--	Included	Progress sway bars come with poly bushings.

My plan: If I buy the Energy Suspension master bushing kit (16.18114) AND Progress sway bars (which include bushings), I'm fully covered. No separate purchase needed.

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14. Professional Alignment -- Mandatory

After any suspension work, a 4-corner alignment is non-negotiable. I'm not skipping this.

Service	Price	Notes
Standard 4-corner alignment	~$80-120	Basic. Acceptable for OEM-spec cars.
Performance alignment	~$120-200	What I want. Technician adjusts to performance specs (more negative camber, adjusted toe).
Corner balancing + alignment	~$200-400	Track prep. Weighs each corner and adjusts spring preload for even weight distribution.

My target alignment specs for street/sport use:

Parameter	Front	Rear
Camber	-1.0 to -1.5	-1.0 to -1.5
Toe	0 to +0.05 (very slight toe-in)	0 to +0.05 (very slight toe-in)
Caster	Max available (stock geometry)	N/A

Important: I need to bring the car to the alignment shop with a half tank of gas, no extra weight in the trunk, and tell the tech my target specs. A lot of shops will default to factory specs which are too conservative for a performance setup.

Optimal Setup -- My Complete Recommended Build

Must-Have (Buy with coilovers)

#	Component	Product	Part Number	Price
1	Coilovers	BC Racing BR	A-18-BR	~$1,195
2	Rear camber arms	Skunk2 Pro Series	516-05-0625	~$333
3	Front camber bolts	SPC EZ-Cam 14mm	81260	~$30
4	Full bushing kit	Energy Suspension Master	16.18114G	~$278
5	Rear trailing arm bushings	Hardrace hardened rubber	6926	~$150
6	Outer tie rod ends	Moog ES800373 + ES800374	--	~$70
7	Inner tie rod ends	Moog EV800246 x2	--	~$70
8	Rear shock top mounts	Honda OEM	52675-SNA-A03 x2	~$50
9	Professional alignment	4-corner performance	--	~$150
	Must-Have Total			~$2,326

Highly Recommended (Same install session)

#	Component	Product	Part Number	Price
10	Rear sway bar	Progress 24mm adjustable	62.1025	~$349
11	Front sway bar	Progress 27mm	61.1006	~$288
12	Roll center adjusters	K-Tuned	KTD-RCA-611	~$190
	Highly Recommended Total			~$827

Nice-to-Have (Phase 2 or if budget allows)

#	Component	Product	Part Number	Price
13	Adjustable front end links	Whiteline	KLC139	~$154
14	Extended tie rods	K-Tuned	KTD-TRO-611	~$200
	Nice-to-Have Total			~$354

Grand Total (Everything)

Tier	Cost
Must-Have	~$2,326
Highly Recommended	~$827
Nice-to-Have	~$354
Grand Total	~$3,507

Installation Notes

Difficulty: Advanced -- 12-20 hours DIY, or $800-1,500 professional labor

This is a full suspension teardown and rebuild. I'm planning to do it myself, so here's what I'm looking at:

- **Jack stands are mandatory.** I'll be under the car for extended periods.
- **Spring compressor is NOT needed** -- coilovers come pre-assembled. I'm removing complete strut assemblies, not compressing springs.
- **Bushing removal requires a press** for most bushings. I'll need a shop press or bushing removal tool set.
- **Rear trailing arm bushing removal** is notoriously difficult. A lot of DIYers have a shop press out the old bushings and press in new ones. I might go that route.
- **I'm doing ALL bushing work, ball joints, tie rods, and coilovers in one session.** The car is already apart -- doing this in phases would be insane.
- **Label everything.** I'll photograph every connection before disassembly.
- **Anti-seize on all bolts** during reassembly.
- **Torque all fasteners to spec** -- no guessing.
- **Alignment immediately after.** I won't drive more than necessary before getting aligned.

My plan of attack:

1. Buy all parts before starting
2. Remove front and rear suspension one side at a time
3. Press out old bushings, press in new poly bushings
4. Install coilovers, new ball joints, tie rods, sway bars
5. Install rear camber arms
6. Set approximate ride height (adjust later)
7. Drive carefully to alignment shop
8. Get performance alignment

9. Fine-tune ride height and damping over the next week
10. Re-torque all fasteners after 100 miles

See Also

- 18-Strut-Bar/overview.md -- Front strut tower brace (separate from coilovers)
- docs/mod-order-and-maintenance.md -- Full build order and maintenance plan

Last updated: 2026-04-16

Suspension Install Guide -- BC Racing BR + Bushings + Sway Bars + Alignment

Status: Planning -- Phase 5

This is the biggest, most invasive mechanical job on my entire build. Coilovers at all four corners, polyurethane bushing refresh, front and rear sway bars, roll center correction, tie rods, ball joints, and a mandatory professional alignment at the end. Every wear item I can touch gets touched.

Reality Check

2-3 full day DIY job if I do everything in one session, which is the only sane way to approach it -- the car is already apart. Shop labor runs $800-1,500. I'm doing it myself, but I'm not fooling myself on scope.

Time breakdown, realistic:

Sub-project	DIY time	Notes
Front coilover install (both sides)	4-6 hrs	Straightforward strut swap, but ball joint separation is the wildcard
Rear coilover + Skunk2 camber arms	4-6 hrs	Trunk liner access for top hats adds time
Energy Suspension master bushing kit	6-10 hrs	Requires hydraulic press for LCA bushings -- shop trip or own tool
Hardrace trailing arm bushings	3-5 hrs	Requires press. $150-300 to outsource, often worth it
Progress sway bars F+R	2-3 hrs	Bolt-in, easy win
K-Tuned roll center adjusters	1-2 hrs	Ball joint work
Tie rods + ball joints	2-3 hrs	Ball joint separator/press needed
Torque-down + load settle	1 hr	Critical -- see below
Alignment (at shop)	1-2 hrs	Not DIY, don't try

Realistic total: 20-35 hours across 2-3 full days, PLUS one press-shop trip PLUS one alignment rack appointment. Two weekends or a long holiday weekend is the right budget.

Two things I cannot DIY:

1. Pressing OEM rubber bushings out of steel control arms. Takes 10+ tons and proper fixtures. Either take arms to a machine shop with a 20-ton hydraulic press, or buy pre-bushed complete arms and skip pressing.
2. 4-corner performance alignment. A Hunter HawkEye rack with a trained tech is $150 well spent. Tire wear on a mis-aligned lowered car starts in days.

Book the alignment appointment BEFORE teardown. Find a shop that does performance alignments -- not "we align to factory spec" but "we'll hit your target numbers."

Prerequisites / Cross-References

No tune dependency. Suspension doesn't touch the ECU. Install any time after Phase 1-4.

Prereq	Cross-reference	Why
Brake refresh complete	04-Brakes/	Wheels are off anyway. Do them first or same session.
Alignment shop booked	External -- call first	Don't start teardown without a rack reservation.
Bushing press plan decided	Shop list or DIY	Either a 20-ton press shop or pre-bushed arms. No improvising with a bottle jack.
Dry indoor workspace	My garage	2-3 days of parts on the floor.
Strut bar install stays deferred	18-Strut-Bar/install-guide.md	Do NOT install the Megan bar this session. Pillow-ball camber plates change strut-top geometry; final fitment is only knowable post-alignment.

Short version on the strut bar: BC Racing BR camber plates change the strut-top stud pattern and top-deck height. Forcing the Megan bar onto stock-geometry assumptions is how tower welds crack. Coilovers this session; bar install is a separate session after alignment.

Tools Required

Coilover install

- 1/2" drive torque wrench, 30-250 ft-lb
- 3/8" drive torque wrench, 10-80 ft-lb
- 14mm, 17mm, 19mm, 21mm sockets + matching wrenches
- Breaker bar, 18"+ -- for stubborn lower control arm bolts
- Ball joint separator (pickle fork or press-type)
- Tie rod puller
- 4-ton jack stands (4 of them) -- car fully elevated with all 4 wheels off
- Floor jack, 2-ton minimum
- Spring compressor -- NOT needed; coilovers ship pre-assembled. Explicitly don't buy one.

Bushing work

- **Hydraulic press (20-ton)** -- I don't own one. Shop pressing runs $50-100 per bushing. For LCA and trailing arm bushings, that's the plan.
- Bushing driver set (optional) -- sway bar bushings only.

Sway bar install

- 14mm + 17mm sockets, grease for the poly bushings.

Alignment

- A drive to the shop. Not DIY.

Parts List

Everything from overview.md, consolidated. Before I start, I verify every box on this list is physically in my garage

-- no "I'll order that when I get to it" mid-install.

#	Part	Part Number	Qty	Price	Source
1	BC Racing BR coilovers (front + rear set, pillow-ball camber plates included)	A-18-BR	1 kit	~$1,195	K Series Parts / BC Racing direct / Redline360
2	Skunk2 Pro Series rear camber arms	516-05-0625	1 pair	~$333	Skunk2 direct / K Series Parts / Hybrid Racing
3	Energy Suspension master bushing kit (black)	16.18114G	1 kit	~$278	HARDmotion / Amazon
4	Hardrace rear trailing arm bushings	6926	1 set (both sides)	~$150	Hardrace USA
5	Progress front sway bar (27mm)	61.1006	1	~$288	Progress Technology direct
6	Progress rear sway bar (24mm, 3-way adjustable, end links included)	62.1025	1	~$349	Progress Technology direct
7	K-Tuned roll center adjusters (10mm extended ball joints)	KTD-RCA-611	1 pair	~$190	K Series Parts
8	Moog outer tie rod end, passenger	ES800373	1	~$35	RockAuto / Amazon
9	Moog outer tie rod end, driver	ES800374	1	~$35	RockAuto / Amazon
10	Moog inner tie rod end	EV800246	2	~$70	RockAuto / Amazon
11	Honda OEM rear shock top mount	52675-SNA-A032		~$50	Honda dealer
12	SPC EZ-Cam front camber bolts	81260	1 pair	~$30	HARDmotion / Amazon
13	Professional 4-corner performance alignment	--	1	~$150	Local performance shop

Consumables:

- New cotter pins for castle nuts (tie rods, ball joints) -- never reuse
- Anti-seize (copper) for strut top nuts, ball joint threads, any fasteners I'll touch again
- Blue Loctite 242 for sway bar bracket bolts, camber arm lock nuts
- Grease for sway bar bushings (most poly bushings ship with a little tube -- have extra)
- Fresh DOT 4 brake fluid if I disturb any brake lines beyond routing (I shouldn't, but have it on hand)

Step-by-Step -- Split by Sub-Project

Order: front coilovers -> rear coilovers + camber arms -> bushings (press shop between those two if needed) -> sway bars -> roll center adjusters -> tie rods/ball joints -> torque down loaded -> drive to alignment.

Sub-Project A: Front Coilover Install (4-6 hrs)

1. Park on level ground. Loosen front lug nuts 1/2 turn.
2. Jack up front, set on 4-ton jack stands under the frame rails (NOT the LCAs -- those are coming off).
3. Pull both front wheels.
4. Start passenger side (more room):
   - **Remove sway bar end link** from strut body. 14mm on the stud flats while cracking the 17mm nut.
   - **Remove ABS sensor cable clip** from strut. Do NOT cut the wire.
   - **Remove brake line bracket** from strut. Do NOT disconnect the hose itself.
   - **Separate tie rod end from knuckle.** Cotter pin out, castle nut backed off until flush with stud top. Tie rod puller to separate.

- **Separate lower ball joint from knuckle.** Pinch bolt out (40 ft-lb spec but 170k mi seized, breaker bar). **Pinch bolt is a stretch bolt -- new one required on reinstall.** Dead-blow the knuckle to separate from the stud.
- **Support knuckle** with bungee or wire to the strut spring perch so it doesn't rip the brake hose.

5. **Top of strut** -- 3 nuts per tower under the hood. Loosen all 3, hold strut from below, remove the last one.
6. Drop the strut assembly out the top of the wheel well. Heavy -- two hands.
7. **Prep BC Racing BR front coilover.** Verify BC's pre-set ride height against their spec sheet. Confirm pillow-ball camber plate orientation (front-facing marking on BC plates). Torque top hat assembly per BC's printed sheet -- typically 40-50 ft-lb.
8. **Install coilover bottom-up.** Feed up through wheel well. Thread all 3 top nuts hand-tight before torquing any. Torque top nuts to **29 ft-lb** per docs/torque-specs.md OR BC's alternate spec if they deviate.
9. **Reconnect bottom** in reverse:
   - Ball joint stud into knuckle. **New pinch bolt, 40 ft-lb**.
   - Tie rod into knuckle, new castle nut, **40 ft-lb**, new cotter pin (turn up to next slot, NEVER back off).
   - Brake line bracket + ABS sensor clip reattached.
   - Sway bar end link -- leave loose for now.
10. **CRITICAL: Do NOT torque strut-to-knuckle bolts or any lower suspension pivot bolts yet.** Stay loose until car is loaded. Torquing unloaded = bushing binding = premature wear.
11. Repeat on driver side.

Sub-Project B: Rear Coilover Install + Skunk2 Pro Series Rear Camber Arms (4-6 hrs)

Rear is easier (no ball joint, no tie rod) but harder in one way -- top hats live under the trunk liner.

12. Move jack stands to rear. All 4 wheels off simultaneously -- fronts back on jack stands at frame rails, rears on stands at the rear subframe pinch welds.
13. Pull rear wheels.
14. **Access rear strut top hat** -- pull trunk liner and trim covering each shock tower (push-in clips + 10mm bolts).
15. One side at a time, shock and camber arm together.
16. **Remove rear shock lower bolt** (40 ft-lb, breaker bar).
17. **Remove rear shock upper nut** from inside the trunk. Shock drops out.
18. **Remove OEM rear upper control arm** -- one bolt at inner pickup, one at knuckle.
19. **Install Skunk2 Pro Series 516-05-0625:** Pre-set length to MATCH stock eye-to-eye -- alignment shop fine-tunes. Mounting bolts to **47 ft-lb**. Lock nuts on telescoping section: **35 ft-lb** per Skunk2 spec.
20. **Install new OEM rear shock top mount** (Honda 52675-SNA-A03) onto the BC rear coilover. 170k-mile mounts are compressed -- no reason to reuse.
21. **Install BC Racing BR rear coilover:** Pre-set height per BC's spec. Feed up into rear tower, upper nut from trunk side, torque to **22 ft-lb** (or BC's spec). Lower mount onto rear control arm -- lower bolt stays hand-tight until car is loaded.
22. Repeat other side.

Sub-Project C: Energy Suspension Master Bushing Kit 16.18114G (6-10 hrs total -- or "send to shop for one day")

The 16.18114G kit covers front LCA bushings, sway bar bushings, and front subframe bushings. Each one is a different level of effort.

Front Lower Control Arm Bushings -- PRESS REQUIRED:

23. Front end already apart from Sub-Project A. Unbolt front LCAs entirely: front pivot bolt (54 ft-lb spec), rear pivot bolt to subframe (76 ft-lb), ball joint already disconnected.
24. Take LCAs to a press shop. $50-100/bushing outsourced, $200-400 both arms complete. **Alternative: pre-bushed complete LCAs** (~$300-500 pair) saves the shop trip.
25. Reinstall LCAs -- both pivot bolts hand-tight only. Leave loose until loaded.

Sway Bar Bushings (front and rear):

26. Unbolt the 2 D-cup brackets per bar (14mm). Swap old rubber for new Energy Suspension poly D-cups. Grease the inside face.
27. Repeat on rear bar.
28. Torque bracket bolts to **29 ft-lb**, blue Loctite on threads.

Front Subframe Bushings:

29. Requires dropping the front subframe. **Skip this session** unless the subframe is already dropped for a header install (13-Headers/install-guide.md). Not obligated to use every piece of the kit in one sitting.

Sub-Project D: Hardrace Trailing Arm Bushings 6926 (3-5 hrs or outsource)

The most notorious 8th gen Civic suspension item. OEM rubber hardens and creaks. Hardrace 6926 replaces with hardened rubber -- better NVH than poly for a street car.

30. **Press required.** Take trailing arms to a press shop -- $150-300 total outsourced, **often worth it.** Or buy pre-bushed complete Hardrace trailing arms and skip pressing.
31. If removing arms myself: unbolt both ends (both **54 ft-lb** spec, breaker bar). Arm comes out.
32. At the press shop: verify orientation (void/directional indicator points fore-aft correctly).
33. Reinstall -- both bolts hand-tight only. Leave loose until loaded.

Sub-Project E: Progress Sway Bars (front 27mm + rear 24mm adjustable) (2-3 hrs)

Done after coilovers and bushings so that the coilovers' end link geometry is known.

Front (Progress 61.1006, 27mm):

34. Stock bar is still on at this point. Unbolt and remove.
35. **Use Progress-supplied bushings** -- the Energy Suspension D-cups I just installed are sized for the stock bar and won't fit the thicker 27mm Progress bar. Grease liberally.
36. Slide Progress bar into position, bolt up with new bushings.
37. New end links -- **do NOT reuse 170k-mile OEM end links** (worn ball joints, clunking).
38. Bracket bolts **29 ft-lb** (blue Loctite), end link nuts **29 ft-lb** but leave snug until loaded.

Rear (Progress 62.1025, 24mm, 3-way adjustable):

39. Same procedure -- stock bar out, Progress 24mm in with its supplied bushings.
40. **End link attachment:** Progress 62.1025 has 3 holes per end (soft/medium/stiff). **Start MIDDLE.** Adjust stiffer after shakedown if understeery, softer if snappy. Stiffer rear = more oversteer.
41. End links supplied in the 62.1025 kit -- use them. **29 ft-lb**, leave loose until loaded.

Sub-Project F: K-Tuned Roll Center Adjusters (1-2 hrs)

42. KTD-RCA-611 is a 10mm extended lower ball joint. Replaces stock ball joint with a taller one, moving the

stud up 10mm. On a lowered car this restores the LBJ angle toward OEM and moves the roll center back up off the ground -- fixes the handling penalty lowering otherwise introduces.

43. LCAs are off the car for bushing press work. NOW is the time to swap ball joints -- same press shop handles both jobs in one visit.

44. Old ball joint pressed out, new K-Tuned pressed in. Use supplied retaining hardware (c-clip or threaded collar per K-Tuned spec).

45. **Install on the CORRECT SIDE.** Side-specific due to taper direction. Verify K-Tuned markings before pressing.

46. Reinstall LCA with new ball joint. Knuckle connects to the new extended stud. **New pinch bolt, 40 ft-lb.**

Sub-Project G: Tie Rods + Ball Joints (2-3 hrs)

Overlaps with Sub-Projects A and F but conceptually its own job.

47. **Ball joints** -- already handled in Sub-Project F (K-Tuned RCA replaces ball joint AND corrects roll center).

48. **Outer tie rod ends (Moog ES800373 passenger, ES800374 driver):**

- Before removing old outers, **count the threads** or measure length from jam nut to end. Gives a ballpark alignment starting point so the car drives to the rack without tramlining.
- Loosen jam nut, thread off old, thread on new Moog to the SAME length.
- Outer into knuckle, new castle nut to **40 ft-lb**, new cotter pin.

49. **Inner tie rod ends (Moog EV800246 x2):** Only replace if play detected. Requires an inner tie rod tool (big crow's-foot on 1/2" drive). Torque per Moog's spec (40-55 ft-lb).

Sub-Project H: Torque Down + Load Car (1 hr)

This step makes or breaks suspension lifespan. Bushings must be torqued at their NEUTRAL (loaded) position. Torquing on jack stands twists the bushing at its installed angle; every compression cycle after that adds twist load. Bushings wear out in months instead of years.

50. Audit every fastener:

- Fully torqued (doesn't pivot through travel): strut tops, sway bar brackets, camber arms, RCA pinch bolt, tie rod castle nuts.
- Hand-tight / snug (pivots with suspension): all LCA pivots, trailing arm pivots, sway bar end links.

51. Lower car to ground. All 4 wheels down.

52. **Roll back and forth ~10 feet** to settle suspension into static loaded position. Bushings rotate to neutral.

53. NOW torque the hand-tight fasteners:

- LCA front pivot: **54 ft-lb**
- LCA rear pivot (to subframe): **76 ft-lb**
- Trailing arm bolts (both ends): **54 ft-lb**
- Rear lower shock bolt: **40 ft-lb**
- Sway bar end link nuts: **29 ft-lb** front AND rear

54. Double-check strut top nuts -- **29 ft-lb** front, **22 ft-lb** rear.

Sub-Project I: Mandatory Alignment

55. Drive gently to the alignment shop. Car may pull, track off-center, feel crooked -- normal with fresh suspension. Baby it.

56. Request a **4-corner performance alignment** with specs matched to my driving style:

Parameter	Daily setup	Sport setup
Front camber	-1.0 deg	-2.0 deg
Rear camber	-1.5 deg	-2.0 deg
Front toe	slight toe-in (+0.05)	0 (zero toe)
Rear toe	slight toe-in (+0.05)	0 (zero toe)
Front caster	max available	max available

57. Ask the tech for the **printed alignment sheet** with before/after numbers. Keep with build records as baseline.
58. Pay the ~$150. Even enthusiasts don't DIY a 4-corner performance alignment.

Ride Height Setup (pre-alignment)

Verify ride height is sane and even corner-to-corner before driving to the shop.

59. After torque-down and roll-settle, measure fender-to-hub-center at all 4 corners:
    - Front: 25.5-27" (daily-usable drop, not slammed)
    - Rear: 26-27.5" (~0.5" higher than front -- prevents rake-forward look)
60. If one corner is off by more than 0.5", adjust that corner's BC lower perch with the supplied spanner wrenches. One full turn ~ 3mm height change.
61. **Adjust all four corners together** to avoid cross-loading the chassis.
62. Document final heights. Reference for future alignment checks, sag detection, and the strut bar clearance check in 18-Strut-Bar/install-guide.md.

First Drive Checklist

After alignment, short easy drive before highway. Listening for:

- **No clunks** over bumps -- unseated bushing or loose fastener
- **No rubbing** at full steering lock -- parking lot, lock-to-lock, listen for tire-to-liner contact
- **Steering centered** -- wheel straight when driving straight
- **Ride firmer but not harsh** -- BC middle damping should be noticeably stiffer, not painful
- **No alignment pull** -- car tracks straight hands-off briefly on empty road
- **Brake pedal feel unchanged**
- **Sway bar function** -- parking lot circles should show obviously reduced body roll

If any fails, stop and diagnose. Don't highway-drive a suspension that isn't behaving right.

Post-Install Settling

Poly and rubber bushings settle in over the first 100-500 miles. Spring perches seat slightly.

- **100 miles:** re-check torque on all lower fasteners (LCA pivots, trailing arms, end links, brackets).
- **500 miles:** re-measure ride height. If any corner settled >0.25", adjust perches. If pulling or uneven tire wear

appears, **go back for re-alignment** -- most shops offer a free 30-day recheck.

- **1000 miles:** stable.

Common Mistakes

- **Torquing bushings unloaded = bushings bind = short life.** The single most important rule. LCA pivot bolt torqued on jack stands = bushing twisted every compression cycle, tears internally within a year. **Always torque loaded, after rolling to settle.**
- **Reusing 170k-mile sway bar end links.** Worn ball joints = clicking from day one. New sway bar deserves new end links.
- **Skipping alignment.** Lowered car with wrong rear camber eats the inner shoulder off a rear tire in 2 weeks. No exceptions.
- **Skunk2 camber arm pre-set wrong.** Set length wildly off stock and the shop can't get spec. Match stock eye-to-eye, let the tech fine-tune.
- **K-Tuned RCA on wrong side.** Side-specific due to taper direction. Verify markings before pressing.
- **Subframe bushings attempted without dropping the subframe.** Not accessible otherwise. Skip or combine with a header session.
- **Pillow-ball camber plate without strut bar clearance check** -- forcing the Megan bar on afterward cracks tower welds. See 18-Strut-Bar/install-guide.md.
- **Wrong-diameter sway bar bushings.** Energy Suspension D-cups are sized for the stock bar; Progress 27mm needs Progress-supplied bushings. Wrong diameter = rattling bar.

Strut Bar Decision

The Megan Racing Race Spec bar (MR-SB-HC06FU-1P) is in the box on the shelf. **NOT going on during this session.** See 18-Strut-Bar/install-guide.md for full reasoning. Summary:

- BC Racing BR pillow-ball camber plates replace the stock upper strut mount entirely -- top-deck height and stud pattern may differ from stock
- Bar fitment is only knowable after coilovers are installed AND alignment is done (final camber moves the pillow-ball plate slightly)
- Attempt fitment only after this session is closed out -- separate install day
- Accept that the bar may not fit with the BR plates; be prepared to shop alternatives (Cusco Type OS Ti, Ultra Racing URTW2-003, or custom)

No measuring or pre-fitting the strut bar during this suspension session.

See Also

- overview.md -- parts research, rationale, budget tier breakdown
- 18-Strut-Bar/install-guide.md -- **strut bar install happens AFTER alignment, with post-coilover clearance check first**
- 04-Brakes/ -- brake refresh happens at the same time, wheels are off anyway
- 13-Headers/install-guide.md -- if headers are being installed in the same window, the subframe drop there

opens access for the Energy Suspension front subframe bushings

- docs/torque-specs.md -- every torque value referenced here, OEM-spec, cross-checked

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Last updated: 2026-04-20

Front Strut Bar -- Megan Racing Race Spec

Purchased, INSTALL DEFERRED (wait until after coilovers)

Part Information

Field	Value
Part	Megan Racing Strut Bar Honda Civic & Si (06-11) Front - Race Spec - Polished Upper
Part Number	MR-SB-HC06FU-1P
Type	Upper front strut tower brace
Material	Polished aluminum
Fitment	2006-2011 Honda Civic & Civic SI
Actual Cost	Included in $960.43 clutch/flywheel/strut bar bundle

Why I Want This

A front strut tower brace connects the two front strut tower tops to stiffen up the chassis. Less body flex during cornering means better front-end response and sharper turn-in. On the 8th gen Civic with its MacPherson strut front suspension, a strut bar is one of the most effective and easiest chassis stiffening mods you can do.

Why I'm Not Diagnosing the Clearance Issue Right Now

Cross-reference to 17-Suspension/. I'm going BC Racing BR coilovers as part of my full suspension overhaul. The BR includes **pillow-ball camber plates** that replace the stock upper strut mount entirely -- the strut top geometry on my car is going to change. Specifically:

- Stock upper strut mount = thick rubber isolator with a steel top cap, ~1" tall
- BC Racing pillow-ball camber plate = lower profile spherical bearing housing, different bolt pattern, different top-deck height

Any diagnosis I do on the Megan Racing bar TODAY against the STOCK strut tops is wasted effort, because the thing that fits (or doesn't fit) will be physically different once the BR is installed. Whatever is currently interfering may become a non-issue, or a different interference may appear where there wasn't one. There's no way to know until the coilovers are in.

So: the strut bar sits in the box until coilovers go on. Revisit fitment then.

The Two Likely Outcomes After Coilovers

Case 1: It Fits

Great. 20-minute install (2 bolts per side + center adjustment), done.

Case 2: It Still Doesn't Fit

Once the coilovers are in and the strut tops are at their new geometry, I'll know exactly what's interfering because I can measure it against a known reference. Likely suspects at that point (after ruling out the stock mount as a

variable):

- **Brake master cylinder / booster** -- some aftermarket Civic strut bars clear on the sedan (FA5) but not the coupe (FG2). The BMC/booster sits in the path of some designs.
- **Bar was for the sedan** -- Megan Racing occasionally cross-lists parts. I'll verify the MR-SB-HC06FU-1P part number is specifically the coupe unit.
- **I need a different bar entirely** -- possibilities: Cusco Type OS, Ultra Racing front strut, or a custom bent one. The BC Racing camber plates have a specific bolt pattern that any replacement brace needs to match.

What I'm NOT Going to Do

- ? Grind, bend, or cut the Megan bar to force fit against the stock mount. The coilovers will change the situation, and any modifications now will likely be wrong for the final geometry.
- ? Return it for refund. It's bundled with my clutch purchase and I want to give it a fair chance once the strut tops change.
- ? Buy a replacement strut bar today. No data yet on what actually fits my final suspension config.

Notes

- No tuning impact -- purely mechanical/chassis
- This is the cleanest example in my whole build of "don't solve problem X if you're about to do mod Y that changes the problem entirely"
- Status updates to this doc when coilovers go in -- I'll know in one afternoon whether this bar works or I'm shopping for a different one

See Also

- 17-Suspension/overview.md -- the coilover build that changes the strut tops
- docs/mod-order-and-maintenance.md -- strut bar is listed as deferred-until-coilovers there too

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Last updated: 2026-04-18

Strut Bar Install Guide -- Megan Racing Race Spec (Deferred Until Post-Coilover)

Status: DEFERRED -- not blocked. Do NOT install until coilovers are in.

The Megan Racing bar (MR-SB-HC06FU-1P) is in the box on my shelf. I'm not installing it and I'm not troubleshooting fitment yet either. This is the cleanest "right part, wrong time" situation in my entire build. The part was bundled into the $960.43 clutch/flywheel/strut bar purchase from shop.redline360.com, so it's paid for -- but it's NOT going on before the coilovers.

Reality Check

The install itself is 30-45 minutes. Lift hood, shop towel across strut towers, drop bar on, torque 4 bolts, verify clearance, close hood. If the part fits the final suspension geometry, this is one of the easiest mods on my list.

The work is not the install -- it's the decision of WHEN. I almost test-fit the bar against my stock strut mounts a few weeks back and only stopped because I remembered the coilovers were coming. Any data I collect on clearance TODAY is wrong data.

Why Deferred

Cross-reference 17-Suspension/overview.md -- my coilover pick is the BC Racing BR (A-18-BR), which includes **pillow-ball camber plates** that replace the stock upper strut mount entirely. Stock mount = thick rubber isolator with steel top cap, ~1" tall, 3-nut pattern. BR camber plate = lower-profile spherical bearing housing, different top-deck height, adjustable bolt spacing via the camber slot.

A strut bar interacts with exactly that surface: the mounting studs, the top-cap height (which sets how tall the bar's end brackets sit), and clearance to adjacent engine-bay components. Every one of those variables changes when the BR goes in. Three outcomes are on the table, all unknowable until the coilovers are installed and aligned:

- Bar fits perfectly post-coilover.
- Bar needs a spacer under one or both end brackets for the new top-deck height.
- Bar doesn't fit at all -- shop for a different bar that's known to clear pillow-ball camber plates.

Reliability-first rule: installing chassis bracing that doesn't account for final suspension geometry leads to a rattling bar, bolts loosening over time from a wrong load path, or worst case, pre-loading the towers in a direction they weren't designed to carry. The 8th gen strut tower sheet metal isn't heavy gauge. I'm not finding out the hard way that a forced-fit bar cracked a weld.

Prerequisites / Cross-References

Before I touch this install, ALL of these must be true:

Prerequisite	Cross-reference	Why
BC Racing BR coilovers installed on all 4 corners	17-Suspension/	Strut tops are the physical thing the bar bolts to. Needs to be the FINAL geometry.
4-corner performance alignment complete	17-Suspension/ -- alignment step	Final camber setting changes pillow-ball orientation slightly. Do fitment AFTER alignment.
Pillow-ball camber plate clearance measured	This doc, pre-install step	Record distance from top of camber plate to firewall/cowl panel BEFORE opening the MR bar's box.
Engine bay cool	Obvious	I'll be reaching across the bay.

If any of those are false, the bar stays in the box. No exceptions.

Tools Required

Modest -- no torch, no breaker bar, no jack stands.

- 14mm socket + wrench (end bracket fasteners -- verify against MR's hardware)
- 17mm socket + wrench (OEM-pattern strut-top nuts -- BR camber plates may differ; verify)
- 3/8" drive torque wrench, 10-50 ft-lb range
- Feeler gauge or small steel ruler (clearance measurement)
- Clean shop towel (paint protection)
- Flashlight and phone camera (document interference points)

Parts List

From overview.md -- everything I need is in the box as shipped:

#	Item	Part Number	Quantity	Source
1	Megan Racing Race Spec Front Upper Strut Bar, polished aluminum	MR-SB-HC06FU-1P	1	Included in clutch/flywheel bundle from shop.redline360.com
2	End brackets (integrated with bar assembly)	--	2	In box
3	Strut-top mounting hardware	--	Per Megan's instructions	REUSE OEM bolts per Megan's instructions -- verify this is still correct once BC Racing camber plates are in, since the camber plates may use different-spec studs.

Hardware check to do before install day: confirm that the BC Racing BR camber plate stud spec is compatible with the bolt holes on the MR end brackets. If not, I may need longer studs or different nuts -- but that's a post-coilover conversation, not a pre-coilover one.

Pre-Install Fitment Check (DO FIRST, BEFORE REMOVING ANY BOLTS)

The single most important step on the page. I am NOT removing any existing fasteners until the bar test-fits

without being forced.

1. Hood up. Shop towel across both strut towers.
2. Lift the bar out of the box. Note orientation -- MR bars have an asymmetric bend to clear the brake booster on LHD cars. Identify driver side.
3. Rest the bar across both towers WITHOUT threading any fasteners. Let it sit on its own.
4. Check clearance visually and with a ruler to every adjacent component:
   - **Brake master cylinder / booster** (driver side -- biggest clearance offender on 8th gen coupes)
   - **Fuel rail cover / intake manifold top**
   - **Cowl panel** (plastic trim below windshield)
   - **Intake tube** -- my K&N Typhoon runs hot-side; confirm no contact
   - **ABS module** (passenger side corner)
   - Any wiring harnesses or vacuum lines across the bay
5. Close hood slowly. Stop at any resistance. Check for hood-pad contact.
6. Photograph any interference points.

Decision point:

- No interference, bar sits flat: proceed.
- Minor interference (hood pad touching, one wire rubbing): proceed, plan to trim pad foam or reroute wire.
- Severe interference (bar won't sit flat, booster in the way, bottoms out on cowl): STOP. See alternatives at bottom.

Step-by-Step (post-coilover, assuming fitment check passed)

Phase A: Prep (5-10 min)

1. Engine cold, hood up and propped.
2. Remove intake tube or airbox if needed for passenger-side strut-top access. My K&N Typhoon's MAF section may need to come out for socket clearance -- see 01-Cold-Air-Intake/ for the removal procedure. Reinstall after.
3. Shop towel across both towers -- aluminum bar will scratch paint instantly on contact.
4. Dry-fit one more time with the towel in place.

Phase B: Strut-Top Fastener Prep (5-10 min)

Factory 8th gen SI uses 3 nuts per strut tower. The BR camber plate may use the same pattern or a 2-bolt camber-slot design -- I'll know once the coilovers are in.

5. **DO NOT remove all strut-top nuts on both sides simultaneously.** The suspension loads against those fasteners; removing them all lets the strut push down into the knuckle. One side at a time, one nut at a time.
6. Loosen -- do NOT fully remove -- the nut(s) that the MR end bracket will bolt to. Most MR designs slip the end bracket over an existing strut-top stud and reuse the OEM nut.
7. If the MR design uses its own pass-through hardware instead of the strut-top studs, leave the strut-top nuts alone. Megan's instruction sheet is the authority -- verify before touching anything.

Phase C: Bar Placement (5 min)

8. Lower the bar onto both towers in the orientation confirmed during fitment check.

9. Align pass-through holes with the prepared fasteners.
10. Thread fasteners on BOTH sides loosely, hand-tight, equal thread engagement.
11. Verify the bar sits flat. Any gap between end bracket and tower means the geometry is fighting me -- tightening fasteners would pull the chassis into the bar's shape. That's the exact outcome I'm trying to avoid. STOP if I see a gap.

Phase D: Torque Sequence (5-10 min)

12. Center the bar -- equal distance to any engine-bay reference point from each end. A shifted bar loads the towers asymmetrically.
13. **Target torque: 30-35 ft-lb.** Verify against Megan's printed sheet -- their spec wins if it differs. Factory strut-top nut spec is 29 ft-lb (docs/torque-specs.md), so this range is consistent.
14. Three passes alternating sides between bolts:
    - Pass 1: ~15 ft-lb
    - Pass 2: ~25 ft-lb
    - Pass 3: final 30-35 ft-lb
15. Grab the bar at center, wiggle up-down. Should feel completely rigid. Any flex = loose fastener; find it.

Phase E: Clearance Verification (5 min)

16. Hood closes and latches fully, no resistance.
17. Open hood, confirm no contact between bar and brake lines, harnesses, vacuum lines, coolant hoses, or hood pad.
18. Key on (engine off). Steering lock-to-lock both ways -- no wheel-well liner interference, no rubbing. If the bar somehow affects steering, STOP.
19. Reinstall any intake components that came off for access.

First Drive Checklist

Short low-speed drive around the block. Listening and feeling for:

- No squeaks or rattles from the strut-tower area over bumps
- No knocking when loading the front suspension (driveway entry, speed bumps)
- Steering feels the same or slightly crisper on turn-in -- chassis bracing does not change steering effort. If it feels heavier, something is pre-loaded wrong.
- Brake pedal feel unchanged. If the bar is contacting the booster, pedal feel changes -- that means I missed a clearance check.

If all four pass: drive normally and re-check fastener torque after 100 miles.

Common Mistakes

- **Forcing the bar into place when it doesn't naturally fit.** Prying, bending, or cranking fasteners to bridge a gap pre-loads the towers against each other. Cracked welds at the tower-to-apron junction are a known outcome.
- **Removing all strut-top nuts on both sides at once.** Unloads the suspension in an uncontrolled way.
- **Skipping the fitment check before coilovers are installed.** I almost did this. The data would have been

garbage.

- **Torquing in wrong sequence.** Single-pass of one side then the other pulls the bar sideways as each side seats. Always alternate.
- **Not re-checking torque after 100 miles.** Chassis flex seats everything; fasteners relax slightly.

If Fitment Fails Post-Coilover

No grinding or bending the MR bar to force it. If it truly doesn't fit with the BR camber plates, the Megan goes on the shelf as a spare or gets resold and I shop for a known-good alternative:

- **Cusco Type OS Ti 2-point front strut bar** -- designed for JDM cars running aggressive camber, known to clear most pillow-ball plates. Higher cost, proven fitment.
- **Ultra Racing Hard URTW2-003** -- community reports of clearance on lowered FG2 coupes. Mid-priced.
- **Custom triangulated bar from a specialist fab shop** -- nuclear option. Match the camber-plate bolt pattern exactly. $300-500, overkill for street, guaranteed to fit.

Decision happens after the coilovers are in, after alignment, and after one more try with the MR bar against the final geometry.

See Also

- 17-Suspension/ -- **MUST be installed first.** Full coilover overhaul including the BC Racing BR pillow-ball camber plates that define the geometry this bar bolts to.
- 01-Cold-Air-Intake/ -- the K&N Typhoon intake may need temporary removal for tool access to the passenger-side strut top during install.
- docs/torque-specs.md -- strut-top nut torque reference (29 ft-lb factory), hardware torque cross-check.
- overview.md -- why the MR bar was chosen, the two possible fitment outcomes, and the decision not to modify the bar for stock geometry.

Last updated: 2026-04-20

Flex Fuel and Fuel System Research -- 2009 Honda Civic SI (FG2 / K20Z3)

Date: 2026-04-16 **My build context:** Full bolt-on NA K20Z3 targeting 220-240 WHP on 93 octane. Hondata FlashPro tuned. **Current/planned mods:** Skunk2 Alpha header, Skunk2 Ultra Street intake manifold, bored 66mm DBW throttle body, 3" valved exhaust with QTP cutout, ATI Super Damper, Hondata FlashPro, Acuity fuel rail (planned).

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1. Flex Fuel Basics for the K20Z3

What Is Flex Fuel?

Flex fuel is the ability to run on any blend of gasoline and ethanol -- from pure gasoline (E0) to 85% ethanol (E85) -- with the ECU automatically adjusting fueling and timing in real time based on the actual ethanol content measured by an inline sensor. The system continuously reads the ethanol percentage and interpolates between a gasoline calibration and an E85 calibration, so the engine is always optimally tuned for whatever's in the tank.

What Is E85?

E85 is a fuel blend that's nominally 85% denatured ethanol and 15% gasoline. But here's the thing -- the actual ethanol content varies a lot:

- **Summer blend (warm months):** Typically 70-83% ethanol
- **Winter blend (cold months):** Can drop to 51-63% ethanol (more gasoline added to aid cold starting)
- **Regional variation:** Ethanol content varies by state, supplier, and even individual station
- **The label "E85" is misleading** -- you're rarely getting exactly 85% ethanol. ASTM D5798 allows E85 to range from 51% to 83% ethanol.

This variability is precisely why a flex fuel sensor matters. Without one, you're guessing at the ethanol content and either leaving power on the table (too conservative) or risking engine damage (too aggressive for what's actually in the tank).

E85 vs 93 Octane -- The Chemistry

Property	93 Octane Gasoline	E85 (Nominal)
Octane rating (AKI)	93	~105-108
Energy density (BTU/gal)	~114,000	~76,000-80,000
Stoichiometric AFR	14.7:1	9.8:1
Latent heat of vaporization	Lower	~3x higher than gasoline
Oxygen content	0%	~35% by mass

Key takeaways:

- **Higher octane (105-108 AKI):** Allows significantly more ignition timing advance before knock. That's where the power comes from.
- **Lower energy density (~30% less):** You need roughly 30% more fuel volume for the same energy output. That's why you need larger injectors and more fuel pump capacity.
- **Higher latent heat of vaporization:** E85 cools the intake charge significantly as it evaporates. Denser, cooler charge improves volumetric efficiency and further resists knock.
- **Net effect on an NA engine:** Power gains come from increased timing advance and slightly improved volumetric efficiency from charge cooling. But because NA engines are less knock-limited than turbocharged engines, the gains are more modest.

E85 on NA vs Turbo -- Setting My Expectations

This is the critical distinction I had to wrap my head around. On a turbocharged engine, knock is the primary power limiter. Boost pressure creates enormous cylinder pressures, and the engine is constantly flirting with detonation. E85's massive octane advantage lets the tuner run significantly more boost and timing, often yielding 20-40% more power.

On my naturally aspirated K20Z3:

- The engine isn't as knock-limited. At 11.0:1 compression on 93 octane, there's already reasonable knock margin at most RPM points.
- A tuner can advance timing with E85, but the gains are smaller because the NA engine wasn't leaving as much timing on the table to begin with.
- The charge cooling effect still helps volumetric efficiency, which does add some power.
- **Realistic expectation: 5-15 WHP gain on a full bolt-on NA K20Z3 switching from 93 octane to E85.** This lines up with community dyno data. Some see closer to 5-8 WHP, others report 10-15 WHP depending on the tune, altitude, and how much timing the tuner was willing to add.

Partial Blends (E30, E50, etc.)

Yes, I can run any blend from E0 to E85 with a flex fuel sensor and proper calibration. That's the whole point.

- **E30 (30% ethanol):** A popular "best of both worlds" blend. Roughly 96-98 octane equivalent. Moderate power gain, much less fuel economy penalty than E85. A lot of tuners consider E30-E40 the sweet spot for NA cars.
- **E50 (50% ethanol):** Middle ground. More power than E30, less fuel economy hit than E85.
- **E85:** Maximum octane, maximum power, worst fuel economy.
- **The flex fuel sensor reads in real time** and the ECU interpolates between calibrations continuously. I can fill up with 93, drive until half tank, top off with E85, and the system handles it seamlessly.

2. Hondata FlashPro Flex Fuel Support

Does FlashPro Support Flex Fuel on the 8th Gen SI?

Yes. Hondata FlashPro fully supports flex fuel on the 2006-2011 Civic SI (K20Z3). This was added in firmware/software updates and is a well-established feature. Works on both FA5 (sedan) and FG2 (coupe).

How It Works -- Technical Details

Here's how the FlashPro flex fuel implementation works:

1. **Ethanol content sensor** gets installed inline on the fuel feed line (between the fuel filter and the fuel rail).
2. **The sensor outputs a frequency signal** (50-150 Hz) corresponding to ethanol content (50 Hz = 0% ethanol, 150 Hz = 100% ethanol). It also outputs a fuel temperature signal.
3. **The sensor signal gets wired to an ECU input.** On the 8th gen Civic SI, the standard approach is wiring the sensor signal to the **rear O2 sensor signal wire** (secondary O2 sensor input). The rear O2 is part of the cat efficiency monitor, which isn't needed on a FlashPro-tuned car (the CEL for the rear O2 is already disabled in most FlashPro calibrations). So that ECU input is free.
   - **ECU Pin:** The rear O2 sensor signal wire goes to the secondary HO2S signal input on the ECU. Common references cite connector E pin 4 (SG2) or A33 (ECT2) -- **the exact pin varies by documentation source and I haven't independently verified it. My tuner or Hondata's official wiring guide will confirm the exact pin for the FG2 before wiring.**
4. **FlashPro reads the ethanol content** in real time and displays it on the datalog/dashboard as "Ethanol Content %."

Calibration Strategy -- Interpolation, Not Manual Switching

This was the key question I had, and the answer is good news:

FlashPro interpolates between two calibrations in real time. I do NOT need to manually switch between a "gas tune" and an "E85" tune.

Here's how it works:

- The tuner creates (or I load) a calibration that contains **two fuel and timing tables**: one for the gasoline endpoint (e.g., E0 or E10) and one for the E85 endpoint.
- FlashPro reads the ethanol content from the sensor and **linearly interpolates** between the two tables based on the current ethanol percentage.
- Example: If the sensor reads E40, FlashPro calculates fuel and timing values roughly halfway between the gas table and the E85 table.
- This happens continuously and in real time. No lag, no manual intervention, no risk of running the wrong tune for what's in the tank.

Parameters Adjusted Based on Ethanol Content

FlashPro adjusts the following based on the ethanol content reading:

- **Fuel injection pulse width:** Scaled up as ethanol content increases (to account for E85's lower energy density and richer stoich ratio of 9.8:1 vs 14.7:1).
- **Ignition timing:** Advanced as ethanol content increases (to take advantage of higher octane/knock

resistance).

- **Target AFR (lambda):** Can be set to run slightly richer on E85 for additional charge cooling.
- **VTEC engagement point:** Can be adjusted per calibration, though this is typically set the same for gas and E85 on NA builds.
- **Rev limit, fuel cut, and other parameters** can also differ between the two calibration endpoints if desired.

What FlashPro Displays

On the FlashPro datalogger/dashboard, I'll be able to monitor:

- **Ethanol content (%)** -- real-time reading from the sensor
- **Fuel temperature** -- from the sensor's temperature output
- **Effective timing** -- so I can see the interpolated timing being run
- **Injector duty cycle** -- critical to monitor with larger injectors and E85

This gives me a real-time ethanol content reading that also directly informs the tune. Pretty cool.

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3. The Complete Flex Fuel System -- Everything Needed

3.1 Flex Fuel Ethanol Content Sensor

The industry standard is the **Continental (formerly Siemens/VDO) flex fuel sensor**, commonly called the "GM flex fuel sensor" because GM uses it in their flex fuel vehicles.

- **Part Number:** Continental 13577429 (supersedes older PN 13577394 and 12570260)
- **Also known as:** GM flex fuel composition sensor
- **Price:** $50-80 for a genuine GM/Continental unit (available from GM dealers, Amazon, RockAuto, etc.)
- **Aftermarket kits:** Companies like Hondata, PRL Motorsports, Blox Racing, and others sell "flex fuel kits" that bundle the sensor with the appropriate fittings and sometimes wiring. These typically run $150-250.
- **Sensor specifications:**
  - Reads ethanol content from 0-100%
  - Reads fuel temperature from -40F to 302F (-40C to 150C)
  - Output: 50-150 Hz frequency signal (linear with ethanol content)
  - 3-wire connector: power (12V), ground, signal
  - 3/8" SAE quick-connect fittings (standard GM fuel line size)

Fitment note: The K20Z3 fuel system uses different line sizes than GM vehicles. I'll need adapter fittings (typically -6AN or 5/16" to 3/8" SAE quick-connect adapters) to plumb the sensor inline. Most Honda-specific flex fuel kits include these adapters.

3.2 Sensor Wiring to ECU

- **Signal wire** goes to the rear O2 sensor signal input at the ECU (as described in Section 2).
- **Power wire** goes to a switched 12V source (ignition-on power).
- **Ground wire** goes to a clean chassis/ECU ground.
- **Hondata provides a wiring guide** in the FlashPro software documentation and on their website/forum for the 8th gen Civic SI. The wiring is straightforward -- three wires total.
- **I'll need:** A few feet of automotive-grade wire, soldering supplies or quality butt connectors, and a wiring

diagram for the FG2 rear O2 sensor harness location. The rear O2 connector is accessible behind/under the center console area or by tracing the harness from the cat.

- **Estimated wiring cost:** $10-20 in wire and connectors doing it myself.

3.3 Fuel Lines

E85 is corrosive to certain materials. This is a real concern.

- **Stock Honda fuel lines:** My 8th gen uses a combination of hard metal lines and rubber fuel hoses. The hard metal lines are fine. The rubber sections are the concern.
- **OEM rubber fuel lines on the 8th gen are generally considered E85-compatible** for the short term, as Honda used relatively modern fuel line materials. However, for long-term reliability with sustained E85 use, a lot of owners upgrade to:
  - **PTFE (Teflon) lined stainless braided fuel lines** -- fully E85 compatible and will last indefinitely.
  - **Brands:** Chase Bays, Acuity Instruments, 27WON, or custom -6AN lines.
- **If running occasional E85 or mostly E30-E50 blends,** the stock lines will likely hold up fine for years. If running straight E85 frequently, I should upgrade at least the high-pressure feed line from filter to rail.
- **Estimated cost for upgraded fuel lines:** $80-200 depending on whether I buy a kit or build custom lines.

3.4 Fuel Rail

See Section 4 for my full Acuity deep dive. I'm already planning to buy the Acuity fuel rail.

3.5 Fuel Injectors -- The Critical Upgrade

This is the most important upgrade for E85 capability. My stock K20Z3 injectors are 310cc (at 43.5 PSI / 3 bar, which is stock fuel pressure). E85 requires approximately 30% more fuel volume than gasoline for the same energy output. Here's what that means:

- **Minimum injector size for E85 on a full bolt-on NA K20Z3:** ~410cc
- **Recommended injector size:** 440-550cc

Here's the math:

- Stock 310cc injectors at ~80-85% duty cycle on 93 octane on a full bolt-on K20Z3 making ~220 WHP is already getting tight.
- On E85, I need ~30% more fuel: 310cc x 1.30 = 403cc minimum, and I never want to run injectors above 80-85% duty cycle for reliability.
- So 440cc is the practical minimum. 550cc gives comfortable headroom.

Injector Options for My Build:

Injector	Size	Price (set of 4)	Notes
Injector Dynamics ID1050x	1050cc	$550-650	Massively overkill for NA. Designed for 400-800+ WHP turbo builds. Very low duty cycle on NA can hurt idle quality due to minimum pulse width issues. Not the best choice unless I'm planning turbo.
Injector Dynamics ID1300x	1300cc	$700-750	Even more overkill. Same issues but worse. No.

Injector Dynamics ID725	725cc	$450-500	Good option with headroom for a potential future turbo. Better idle quality than ID1050x on NA. Well-characterized for Hondata.
DeatschWerks DW 440cc	440cc	$280-320	Good entry-level option. Adequate for NA E85. Tight on headroom if I ever add forced induction. DeatschWerks are well-regarded.
DeatschWerks DW 600cc	600cc	$350-400	Better headroom. Still reasonable for NA. Good choice if turbo is in the cards.
Bosch EV14 550cc	550cc	$200-280	Excellent injectors. The Bosch EV14 platform is the gold standard for aftermarket injectors. Great spray pattern, fast response time, excellent idle quality. 550cc is the sweet spot for NA flex fuel. Many tuners swear by these.
Bosch EV14 650cc	650cc	$220-300	Slight additional headroom over 550cc. Still perfectly fine for NA.
Grams Performance 550cc	550cc	$300-350	Good option. Well-characterized for Honda applications.

Decision: Bosch EV14 550cc, NOT ID1050x.

My earlier BOM listed the ID1050x. After cross-checking:

- ID1050x = 1065cc. That's **3.4x** what I need for 250 WHP on E85. At idle and low load, the injector is pulsing so briefly that minimum-pulse-width becomes a drivability problem (stumble, cold start roughness) even with careful tuning.
- EV14 550cc = correct size for my target (max duty cycle ~75% at redline E85, ~60% at redline pump gas) = safe headroom without the low-end drivability penalty.
- EV14 is cheaper (~$200-280 vs ~$546), from the same Bosch OE production line that supplies OEM fuel injectors to half the world's automakers. Better idle quality, faster response time (sub-1ms), well-characterized in Hondata.
- The ONLY reason to pick the ID1050x is if turbo is near-term certain. My current build is NA forever unless explicitly changed -- so the EV14 is the right part.

I'm going with Bosch EV14 550cc. I'll verify the specific variant (0280158040 or similar OE PN) ships with USCAR-to-Honda clip adapters or budget ~$15-25 for them separately. 48mm body, 14mm o-rings top and bottom -- this is the standard K-series top-feed config that both the stock rail and the Acuity rail accept.

3.6 Fuel Pump

My stock fuel pump is a concern with E85.

- The stock pump is rated at approximately 100-120 LPH (liters per hour). On 93 octane with stock injectors, that's adequate.
- With E85 requiring ~30% more fuel volume, the stock pump is being asked to deliver significantly more to support a 220+ WHP NA build on ethanol.
- **At high RPM and high load, the stock pump may not maintain adequate fuel pressure with E85 and larger injectors on a full bolt-on build.** Fuel pressure drop at high RPM is a real risk.

I'm upgrading the fuel pump. Options:

Pump	Flow Rate	Price	Notes
DeatschWerks DW200	255 LPH	$100-130	Direct drop-in for the 8th gen Civic. Very popular. More than enough for NA E85. The standard recommendation.
DeatschWerks DW300C	340 LPH	$120-150	More headroom, direct drop-in. Slightly louder than DW200. Good if turbo is in the future.
Walbro 255 LPH (GSS342)	255 LPH	$60-90	Budget option. Works fine but DeatschWerks units have better Honda-specific fitment kits.
AEM 340 LPH (50-1200)	340 LPH	$110-140	Excellent pump. Comes with universal install kit; may need Honda-specific adapter.

Decision: DeatschWerks DW300C (340 LPH), NOT DW200.

I waffled on this initially. After cross-checking with the tuning-workflow-and-maps.md rule (headroom > margin), the DW200 at 255 LPH is enough for a well-tuned 93-octane build but gets genuinely tight on E85 at my target WHP. Injector duty cycle ceiling is 85% -- the pump needs to deliver at that load without pressure drop, even with the 30% more volume E85 demands. The DW300C at 340 LPH gives me ~30% margin over DW200 for the same money class (~$130-160 vs ~$110-130). That margin is pure reliability insurance under the reliability-first rule.

If I ever go turbo (even modest), the DW300C is still enough. The DW200 is not. Buy once.

3.7 Fuel Filter

- **I'm replacing the stock fuel filter** when doing the flex fuel conversion, just to ensure a clean fuel system.
- The stock OEM Honda fuel filter is compatible with ethanol blends.
- **However, if the stock filter has been in there for years, ethanol can loosen deposits in the fuel system that the filter will catch.** I'll replace the filter at install, and consider replacing it again after the first 1,000 miles on ethanol.
- An OEM Honda fuel filter is fine. Aftermarket from Wix or similar also works.
- **Cost:** $15-30 for the filter.

3.8 Fuel Pressure Regulator

- **My stock fuel pressure regulator is adequate for NA flex fuel use.**
- The stock FPR maintains ~43.5 PSI (3 bar) base fuel pressure. That's fine for 440-550cc injectors on an NA build.
- An aftermarket adjustable FPR isn't required but could be added for fine-tuning. Most tuners adjust fueling through injector pulse width (via FlashPro) rather than changing base fuel pressure.
- **No upgrade needed** for my build.

3.9 Return Line Modifications

- **My 8th gen Civic SI uses a returnless fuel system** (no return line from the fuel rail to the tank). Fuel pressure is regulated at the pump assembly in the tank.
- **No return line modifications needed for flex fuel on an NA build.** The returnless system works fine.
- Converting to a return-style fuel system is possible but unnecessary for NA power levels. More common on high-power turbo builds where precise fuel pressure regulation is critical.

4. Acuity Fuel Rail Deep Dive

Product Identification

- **Part Number:** Acuity 1913-K20 (also listed as Acuity Instruments K-Series Fuel Rail)
- **Application:** K20A, K20A2, K20A3, K20Z1, K20Z3, K24 -- fits all K-series engines with the standard top-feed injector arrangement
- **Price:** $180-250 (varies by retailer and finish)

What Does It Improve Over Stock?

The stock K20Z3 fuel rail is a stamped steel unit. Here's what the Acuity improves:

1. **Fuel volume capacity:** The Acuity rail has a significantly larger internal volume than stock. It acts as a fuel reservoir, reducing pressure drops during high-demand situations (high RPM, aggressive tuning, E85's higher flow demands). Particularly beneficial when running larger injectors and E85.
2. **Fuel distribution:** Larger internal diameter and improved geometry provide more even fuel pressure across all four injectors. The stock rail can have slight pressure variation between cylinders, especially at high RPM.
3. **Injector fitment:** Designed to accept a wide range of aftermarket injectors with standard 14mm o-ring top-feed fitment. Better sealing and more secure injector retention than the stock rail with aftermarket injectors.
4. **Aesthetics:** CNC-machined billet aluminum with anodized finish. Available in multiple colors. Looks way better than the stock stamped steel rail.
5. **AN fitting provision:** Comes with -6AN inlet fitting (or adapter), making it easy to connect aftermarket fuel lines or plumb in the flex fuel sensor.

Material and Construction

- **CNC-machined 6061-T6 billet aluminum**
- **Hard anodized** for corrosion resistance
- **O-ring sealed** injector bores with ethanol-compatible seals
- Available in black, red, blue, and other anodized finishes depending on production run

E85 Compatibility

Yes, the Acuity fuel rail is fully E85 compatible out of the box. The 6061-T6 aluminum and hard anodizing are ethanol-resistant. The o-rings are Viton or equivalent ethanol-compatible material. No modifications or special seals needed for E85. One less thing to worry about.

Fittings and Adapters

- **Ports:** -8 ORB (O-Ring Boss) throughout -- center-feed -8 ORB port and end -8 ORB ports. There are no -6AN fittings on the rail itself; adapters can convert to -6AN if needed.
- **If using stock fuel lines:** An adapter fitting is included or available separately to connect the stock Honda fuel feed line to the rail.
- **If using aftermarket AN fuel lines:** Use -8 ORB to -6AN adapter fittings.
- **FPR/gauge port:** The rail has a port for a fuel pressure gauge or aftermarket FPR connection if desired.

Compatible Injector Sizes

The Acuity rail accepts any injector with:

- Standard 14mm o-ring top seal
- Standard Honda/Denso lower electrical connector pattern (or EV6/USCAR with adapter clips)
- 48mm (short) or 60mm (long) body length (the rail accommodates both with appropriate o-ring positioning)

So it's compatible with essentially all aftermarket injectors sold for K-series applications, including everything I listed in Section 3.5. No restrictions on injector size.

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5. Realistic Power Gains from E85 on NA K20Z3

Community Dyno Data

This is where I had to check my expectations. Real-world dyno results for full bolt-on NA K20Z3 engines switching from 93 octane to E85:

- **Typical gain: 5-15 WHP and 5-10 WTQ**
- Most commonly reported: **8-12 WHP gain**
- The gain comes primarily from **2-4 degrees of additional ignition timing advance** across the power band and slightly improved volumetric efficiency from charge cooling.

Specific data points I've found from the Honda community:

- **Bolt-on K20Z3 (header, intake, exhaust), Hondata FlashPro:** 210 WHP on 93 octane, 222 WHP on E85 -- a gain of ~12 WHP.
- **Similar builds:** Gains of 5-8 WHP are common when the 93 octane tune was already well-optimized and aggressive.
- **Builds where the 93 tune was conservative** (less timing due to marginal fuel quality or tuner caution): Gains can be 12-15 WHP because E85 allows running timing that was being held back.

Where the Power Comes From

1. **Timing advance (primary source):** E85's ~105-108 AKI octane allows 2-5 degrees more timing advance at various RPM points. On a K20Z3 at 11.0:1 compression, this is the main contributor.
2. **Charge cooling (secondary source):** E85's high latent heat of vaporization cools the intake charge by 20-30F, increasing air density slightly. Adds 1-3 WHP.
3. **Combined:** 5-15 WHP total on a well-tuned NA K20Z3.

Do Larger Injectors Offset Gains?

No. Larger injectors don't reduce power. Properly sized and tuned injectors deliver the correct amount of fuel regardless of their maximum capacity. The ECU simply opens them for a shorter duration. There's no "power offset" from larger injectors, provided they are:

- Not so large that minimum pulse width becomes an issue (affects idle quality, not peak power)
- Properly characterized in the FlashPro calibration (injector dead times, slopes, offsets)

The only concern is idle quality and low-RPM smoothness with very oversized injectors (1000cc+ on an NA engine making ~220 WHP). With 440-550cc injectors on my 220+ WHP NA K20Z3, there should be zero drivability compromises.

Is the Main Benefit Power or Knock Resistance?

On my NA K20Z3, it's honestly a combination of both, but the power gains are modest. Here's my honest take:

- The knock resistance is nice but less critical on NA than turbo. My K20Z3 at 11.0:1 compression on 93 octane isn't severely knock-limited in most conditions.
- The power gains are real but modest (5-15 WHP).
- The real-time ethanol reading from the flex fuel sensor is useful for peace of mind and data logging but isn't a game-changer by itself.
- **The biggest practical benefit might be the ability to run E30-E40 blends for a modest power bump with minimal fuel economy penalty,** rather than running straight E85.

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6. Daily Driveability Concerns

Fuel Economy Penalty

- **E85:** I should expect approximately 25-30% worse fuel economy compared to 93 octane gasoline. If I currently get 26 MPG mixed, expect 18-20 MPG on E85.
- **E30:** Roughly 10-12% worse fuel economy. Much more manageable for daily driving.
- **E50:** Roughly 15-20% worse fuel economy.
- **The fuel economy penalty is the single biggest downside of E85 for daily driving.** E85 is often cheaper per gallon than 93 octane (where it's available), but the higher consumption rate means the net cost per mile is usually roughly similar or slightly worse.

Cold Start Issues

- **E85 is harder to ignite in cold weather.** Ethanol has a higher heat of vaporization and lower vapor pressure than gasoline, making cold starts more difficult.
- **Below 40F:** Cold starts on E85 can be noticeably rougher. Extended cranking (2-4 seconds vs. instant start on gasoline) is common.
- **Below 20F:** Cold starts on straight E85 can be very difficult. May require multiple cranking attempts.
- **Mitigation:** Winter-blend E85 (which has more gasoline, often E51-E65) helps significantly. Running E30-E50 in winter is another common strategy.
- **The flex fuel sensor + Hondata system handles the tuning side automatically** -- if I fill up with winter blend E85 (which reads as E55 or whatever it actually is), FlashPro adjusts fuel and timing accordingly. But the physical cold-start difficulty of ethanol remains regardless of the tune.
- **Some tuners program richer cold-start enrichment and longer cranking priming for E85 calibrations** to mitigate this. I'll discuss this with my tuner.

E85 Availability

- **E85 is NOT available at every gas station.** Availability varies enormously by region.
- **Midwest US:** Excellent availability (corn belt states -- Iowa, Illinois, Indiana, Minnesota, etc.)
- **Coasts and Southeast:** Spotty to poor availability. I may need to drive 15-30 minutes to find a station with E85.
- **Use e85prices.com or the Alternative Fuels Station Locator (afdc.energy.gov)** to find E85 stations nearby.

- **This is a key consideration for me.** If the nearest E85 station is inconvenient, I'll be running mostly pump gas anyway, which makes the entire flex fuel investment less impactful (though the system still works perfectly on pump gas).
- **The beauty of flex fuel** is that I'm never stranded -- I can always fill up with regular 93 octane and the car runs fine.

Running Any Blend (E0 to E85)

Yes, with the flex fuel sensor and proper Hondata calibration, I can run any blend from E0 to E85. That's the entire purpose. Practical scenarios:

- Fill up with 93 octane for a road trip where E85 isn't available -- no problem.
- Top off with E85 at a station I pass -- the blend in the tank might read E30 or E40, and the system adjusts automatically.
- Fill up with straight E85 before a dyno session or track day -- maximum power.
- Run E30-E40 for daily driving as a compromise between power and fuel economy.

The system is completely seamless. There's no "wrong" blend and no need to drain the tank or do anything special when switching between fuels.

Fuel System Maintenance with Ethanol

- **Ethanol is hygroscopic** -- it absorbs moisture from the air. Can lead to water accumulation in the fuel tank if the car sits for extended periods.
- **If my car sits for more than 2-3 weeks** with E85 in the tank, I should add a fuel stabilizer designed for ethanol blends, or simply run the tank low and fill with gasoline before letting it sit.
- **Ethanol can loosen old deposits** in the fuel system (tank, lines, filter). After the initial conversion, I need to keep an eye on the fuel filter condition. Replace it after the first 1,000-2,000 miles on ethanol.
- **Injector o-rings and seals:** All o-rings need to be ethanol-compatible (Viton or equivalent). The Acuity rail uses compatible o-rings. If using stock o-rings on new injectors, verify they're ethanol-rated.
- **Fuel tank:** The stock Civic SI fuel tank is polyethylene (plastic) and is E85 compatible.
- **Long-term:** With proper components (E85-safe injector o-rings, quality fuel lines, E85-compatible fuel pump), there's minimal additional maintenance compared to running gasoline.

7. Bill of Materials -- Complete Flex Fuel Kit

Required Components (with Buy Links)

#	Component	Specific Part	Price	Buy Link
1	Flex fuel sensor	Continental / GM 13577429	$50-80	Amazon (ACDelco)
				High Flow Fuel
				Racetronix
2	Sensor fittings/adapters	-6AN or 5/16" to 3/8" SAE QC adapters (pair)	$25-40	Amazon / Summit Racing
3	Sensor wiring	3-conductor automotive wire + connectors	$10-20	Amazon
4	Sensor pigtail connector	GM flex fuel sensor connector pigtail	$10-15	eBay / Amazon
5	Fuel rail	Acuity K-Series 1913-BLK	$180-250	Acuity Direct

				K Series Parts
				Nippon Power
6	Fuel injectors	Bosch EV14 550cc (primary pick)	~$200-280	See decision block below
		alt: Injector Dynamics ID725	~$450	--
		alt: Injector Dynamics ID1050x (only if going turbo later)	~$546	K Series Parts
7	Injector clip adapters (USCAR->Honda)	Required for Bosch EV14 or ID1050x	~$15-25	Included with most injector sets; verify before ordering
8	Fuel pump	DeatschWerks DW300C (9-307-1008) 340 LPH -- primary pick	~$130-160	DeatschWerks Direct
				K Series Parts
				MAPerformance
				HARDmotion
9	Fuel filter	OEM Honda or quality aftermarket	$15-30	AutoZone / Amazon
10	Fuel line (recommended)	PTFE -6AN braided line, filter to rail	$80-150	Amazon / Summit Racing

Optional but Recommended

#	Component	Specific Part	Price (Approx.)	Notes
11	Fuel pressure gauge	Inline -6AN fuel pressure gauge or sensor	$30-60	Monitors fuel pressure, peace of mind
12	Wideband O2 sensor	AEM UEGO or similar	$180-250	Essential for tuning (may already be installed)
13	Professional dyno tune	Hondata FlashPro flex fuel calibration	$300-500	Required. Must include both gas and E85 tables.

Already Owned or Planned

Component	Status
Hondata FlashPro	Already owned
Acuity fuel rail	Planned purchase

Total Cost Estimates

Scenario	Cost Range
Minimum viable (sensor + wiring + injectors + pump + filter + tune)	$685-1,055
Recommended (above + Acuity fuel rail + fuel line + fittings)	$960-1,485
Full build (recommended + fuel pressure gauge + wideband)	$1,170-1,795

Excluding the Acuity fuel rail (already a planned purchase) and FlashPro (already owned), the flex fuel conversion adds approximately $700-1,100 to my build, including a professional dyno tune.

Alternative: Pre-Made Flex Fuel Kit

Some companies sell bundled flex fuel kits for Honda K-series that include the sensor, fittings, wiring harness, and sometimes a mounting bracket. These cost $150-250 for the sensor kit alone (no injectors or pump), but save time on fitment and wiring. Examples:

- **Hondata flex fuel kit** (if available for my model year)
- **PRL Motorsports flex fuel kit**
- **Blox Racing flex fuel sensor kit**

Worth considering for the convenience factor, though buying the sensor and fittings separately is cheaper.

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8. Is Flex Fuel Worth It on an NA K20Z3?

My Honest Assessment

This is the most important section. Here's my unvarnished thinking.

Cost vs Power Gained

- **Cost:** ~$700-1,100 for the conversion (excluding fuel rail and FlashPro I've already planned/own)
- **Power gained on E85:** 5-15 WHP over a well-tuned 93 octane calibration (most likely 8-12 WHP)
- **Cost per WHP gained:** Roughly $60-220 per WHP
- **For comparison:** A set of aftermarket cams (Skunk2 Stage 2, Toda, or similar) costs $400-600 and can add 15-25 WHP on an NA K20Z3 with supporting valvetrain. Cams are a significantly better dollars-per-WHP investment for an NA build.

Complexity of Install and Tune

- **Install complexity:** Moderate. Fuel pump swap requires accessing through the rear seat access panel (the 8th gen has one -- no need to drop the tank). Injector and rail swap is straightforward. Sensor plumbing is moderate. Wiring is simple (3 wires).
- **Total install time:** 4-8 hours for me, including the fuel pump.
- **Tuning complexity:** Requires a tuner experienced with FlashPro flex fuel calibrations. The tune must include both gas and E85 endpoint tables. This isn't a "flash a basemap and go" situation -- a proper flex fuel tune on a full bolt-on car should be done on a dyno with back-to-back pulls on both fuels.
- **Ongoing complexity:** Minimal once properly tuned. Set it and forget it for daily use.

The "Real-Time Octane Reading" Benefit

Let me be direct about what this actually is:

- **It's the flex fuel sensor reading ethanol content (0-100%) and displaying it on the FlashPro datalogger.** That's it.
- **It's NOT an "octane meter."** It doesn't directly measure octane. It measures ethanol percentage, from which effective octane can be inferred (higher ethanol = higher effective octane).
- **The practical value** is knowing exactly what's in my tank at all times and having the ECU automatically optimize for it. Eliminates the guesswork of "is this E85 actually E85 or is it winter blend E60?"
- **But I don't need the full flex fuel system just for this.** If all I want is to know the ethanol content of my fuel, I can buy a standalone ethanol content gauge for $100-150 that just displays the percentage without any ECU integration. The real value of the full flex fuel system is the automatic tune interpolation, not the display.

For My Street Car Targeting 220-240 WHP on 93 Octane

Here's the critical question, answered plainly:

If I'm already targeting 220-240 WHP on 93 octane with my current mod list, adding E85 will bring me to approximately 225-255 WHP (realistically 230-250 WHP on a good E85 tune).

That's a real but modest improvement. My car won't feel dramatically different in daily driving. I'll notice it at the track or on a dyno sheet, but 8-12 WHP isn't a seat-of-the-pants transformation.

The question is whether that 8-12 WHP is worth $700-1,100 plus the ongoing inconvenience and fuel economy penalty of E85.

Would the Money Be Better Spent Elsewhere?

For pure power-per-dollar on an NA K20Z3, yes, there are better investments if I haven't already made them:

1. **Aftermarket cams (Skunk2 Stage 2, Toda A2/A3, Kelford, etc.):** $400-600 for the cams + $200-400 for valvetrain (springs, retainers, seals). Total ~$600-1,000. Expected gain: 15-25 WHP. This is the single best remaining NA power mod. Cams fundamentally change the engine's breathing characteristics and produce larger, more noticeable gains than fuel type changes.
2. **Professional dyno tune optimization on 93 octane:** $300-500. If I haven't already had a thorough dyno tune with my current bolt-on setup, this alone could find 5-10 WHP by optimizing timing, fuel, and VTEC engagement across the entire RPM range.
3. **Port and polish of intake manifold and head:** $500-1,000 (if done by a shop). Meaningful gains on a high-RPM NA engine.

When Flex Fuel DOES Make Sense for My Build

The flex fuel conversion is most justified if:

- **I've already done cams and valvetrain** and I'm looking for the next increment.
- **I've already maximized the 93 octane tune** and my tuner says "we're timing-limited on pump gas."
- **I have easy, convenient access to E85** near my home or daily commute.
- **I want the flexibility** to run any fuel blend without worrying about the tune.
- **I value the data logging** and automatic tune adjustment for peace of mind.
- **I plan to add forced induction later** -- this is honestly the strongest argument (see below).

The Turbo Future-Proofing Argument

This is actually the strongest argument FOR doing the flex fuel conversion:

If I ever plan to add a turbo kit to this engine, the flex fuel system (sensor, pump, injectors, fuel rail, wiring) all carries over directly and becomes essential. On a turbocharged K20Z3, E85 can add 50-100+ WHP over pump gas because the turbo engine is far more knock-limited and benefits enormously from E85's octane advantage.

The $700-1,100 investment in fuel system components now means I don't have to buy them again when I go turbo. If I size the injectors appropriately (ID725 or DW600 instead of 550cc), I'll have a fuel system that supports both NA E85 and moderate-boost turbo applications.

If forced induction is in my future, doing the fuel system now is a smart investment. If my car will remain NA forever, the flex fuel conversion is harder to justify on a cost-per-horsepower basis.

My Bottom Line

For my NA K20Z3 street car already well-modded and targeting 220-240 WHP on 93 octane:

Priority	What I'd Do
If cams are NOT done yet	Do cams + valvetrain first. Larger gains, similar or lower cost, no ongoing inconvenience.
If cams ARE done and 93 tune is maxed	Flex fuel is a reasonable next step, especially with convenient E85 access.
If my primary interest is data/ethanol reading	A standalone ethanol content gauge ($100-150) gives the display without the full conversion. No auto-tune adjustment though.
If turbo is planned eventually	Do the fuel system now (injectors, pump, rail, sensor). Size injectors for turbo (ID725 or DW600). Everything carries over.
If budget is limited	Skip flex fuel. The 8-12 WHP on NA isn't worth $700-1,100 when cams give 15-25 WHP for similar money.

My suggested mod order for maximum value:

1. Finish current bolt-on build and get a quality dyno tune on 93 octane
2. Add cams and valvetrain (if not already planned) -- biggest remaining NA gain
3. Re-tune on 93 octane with cams
4. Then consider flex fuel as a final NA optimization or as turbo preparation
5. If/when going turbo, the fuel system is already done

Quick Reference Summary

Question	Answer
Does FlashPro support flex fuel on 8th gen SI?	Yes, fully supported with real-time interpolation
Real-time interpolation between gas/E85 tunes?	Yes, automatic and seamless based on sensor reading
Power gain from E85 on NA K20Z3?	5-15 WHP (typically 8-12 WHP)
Injector size and part?	Bosch EV14 550cc. ID1050x is overkill and hurts idle.
Fuel pump?	DeatschWerks DW300C (340 LPH). DW200 is too tight on E85 headroom.
Acuity fuel rail E85 compatible?	Yes, fully compatible out of the box
Stock fuel lines OK for E85?	Yes, post-2006 Honda spec -- confirmed E85-safe per 8thcivic FAQ and community reliability reports
Total conversion cost?	$700-1,100 (excluding fuel rail and FlashPro already planned/owned)
Worth it on NA?	Modest gains. Better $/WHP alternatives exist (cams). Best justified as turbo future-proofing.
Can I run any blend E0-E85?	Yes, any blend, any time, seamlessly
E85 fuel economy penalty?	~25-30% worse MPG vs gasoline
Cold start issues?	Yes, below 40F. Worse below 20F. Run lower ethanol blends in winter.

Last updated: 2026-04-16

Complete Fuel System + Flex Fuel Build -- 2009 Civic SI (FG2 / K20Z3)

Date: 2026-04-20 **Theme:** Purple engine bay, Acuity brand consistency, reliability-first **Power targets:** 220-240 WHP on 93 octane, 230-250 WHP on E85 (NA) **Centerpiece:** Acuity Instruments K-Series Fuel Rail -- Satin Purple (1913-PPL) **Companion:** overview.md covers the "is flex fuel on NA worth it" decision and the cost-vs-WHP analysis. This doc is the build spec.

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1. Acuity Fuel Rail

The Purple One Exists from the Factory

I was braced to pay a custom anodizer to strip and re-color a black rail. I don't have to -- **Acuity sells the K-series rail in satin purple straight from the factory as PN 1913-PPL**, same price as any other color. This is the visual anchor of the purple engine bay.

Part Numbers, Colors, Price

SKU	Finish	Price (MSRP)
1913-PPL	Satin Purple	~$199
1913-BLK	Satin Black	~$199
1913-RED	Satin Red	~$199
1913-TEL	Satin Teal	~$199

K Series Parts lists ~$199.86 MSRP with promo pricing occasionally dropping to ~$177. Acuity direct lists $199. Budget $200 + tax/ship. No premium for the purple finish.

Specifications

- **Material:** CNC-machined 6061-T6 billet aluminum, hard-anodized satin purple, laser-etched logo
- **Bore:** 15mm (0.59") ID -- rated "1000+ HP" (wildly over-spec for my 250 WHP NA target, which is fine -- oversize bore stabilizes rail pressure under injector duty)
- **Ports:** -8 ORB center-feed (for tucked/center-fed lines), -8 ORB on both ends (for OEM feed or return-style), 1/8" NPT gauge port on top face
- **Injector bores:** 14mm standard -- accepts Bosch EV14, ID, DW, Grams, KTuned, etc.
- **Included:** Rail, (2) class 10.9 M8 mounting screws, (1) stainless 1/8" NPT port plug

- **NOT included:** Any AN fittings (sold separately; Acuity offers 6 dedicated fittings for this rail)

Fitment with Skunk2 Ultra Street IM + K20Z3

The rail itself is universal across the K-series injector pattern, including the Skunk2 Ultra Street (307-05-0600). The **harder compatibility problem is the IM, not the rail**: Skunk2 Ultra Street uses the early PRB/PRC head + TB pattern, so to run it on my RBC-head K20Z3 I need a **water-bypass adapter** (for the RBC coolant ports the PRB manifold doesn't have) and the **Skunk2 DBW TB adapter (PN 309-05-0120)** to mate my DBW (bored-to-66mm) TB to the PRB-pattern manifold flange. Full intake manifold detail lives in 14-Intake-Manifold/. Once those adapters are in place, **the Acuity rail bolts directly to the manifold with no fitment issue** -- same injector bore pattern as stock.

Install Torque Specs

Fastener	Torque	Notes
Acuity rail M8 mounting screws	10 ft-lb (13.6 Nm)	6mm Allen. Radium's published K-rail guide specifies the same spec -- community reference.
Stock-equivalent rail-to-IM bolt spec	108 in-lb / 9 ft-lb	FSM spec. Acuity uses the same mounting pattern with aluminum IM threads -- be gentle.
Injectors into rail	Hand press + verify seating	Lube O-rings with engine oil, press in, push down to bottom, verify upper O-ring still seated.
-8 ORB fittings into rail	Hand tight + 1/4 to 1/2 turn past O-ring contact	O-ring seal -- never over-torque.
1/8" NPT plug / gauge port	Snug + PTFE tape	Tapered -- seals before fully seated.

Warranty + Reviews

1-year defect warranty from Acuity. Returns accepted within 14 days if unopened. For a billet rail with no moving parts this is more than enough -- the rail either leaks/cracks (warranty) or it lasts the life of the car.

Community take: Acuity K-rail is effectively the default K-series "beyond stock" upgrade alongside Radium 20-0230-02 (black only, ~$150-175, larger 0.69" bore but no purple option) and Hybrid Racing's equivalent. Fit/finish is clean, anodize is even, laser etch is a nice touch. Brand consistency with my Acuity shifter + the purple theme seals it.

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2. Bosch EV14 550cc Injectors

Decision: Bosch EV14 550cc, Not RDX, Not ID1050x

Option	Flow	Verdict
RDX 410cc (Acura OEM)	410 cc/min @ 3 bar	Rejected. No published dead-time data, ~6% set-to-set flow variance. Tight E85 headroom at 240 WHP.
Bosch EV14 550cc (PN 0280158117)	550 cc/min @ 3 bar	Chosen. Bosch OE line, <1% matched, published Hondata dead times, ~$200-280/set, right-sized for NA E85 with margin.
Injector Dynamics ID725	725 cc/min @ 3 bar	Runner-up if turbo becomes real. ~$450-500. Overkill for current NA target.

ID1050x	1065 cc/min @ 3 bar	Rejected (same call as overview.md): 3.4x oversized, minimum pulse-width issues at idle, ~$550.

Bosch EV14 550cc Fitment

- **PN: 0280158117** -- 550 cc/min @ 43.5 PSI / 3 bar
- **Body:** 48mm mid-length (standard short K-series top-feed form)
- **O-rings:** 14mm top + 14mm bottom -- Viton-family, **fully E85-compatible from the factory**. Included with genuine Bosch injectors (verify with the reseller).
- **Impedance:** 12 ohm (high-impedance, same as stock K20Z3 -- plugs into stock injector drivers, no driver box)
- **Max pressure:** 100 PSI (running ~58 PSI, zero concern)
- **Connector:** USCAR/EV6 -- **needs Honda OBD2 adapter clips**

Adapter Clips

Two paths:

1. **One-piece pigtail adapters** (recommended, reversible): $15-25/set of 4. Sources: OBD Innovations, DeatschWerks USCAR-to-Honda, Clean Injection, Injector Nation.
2. **Re-pin the harness**: cleaner, permanent, more labor.

Going with pigtails. Some resellers (Clean Injection, Injector Nation, CP Fuel & Race) bundle the adapters into the injector set price -- **check the listing before buying adapters separately**.

Price per Set of 4

- Bare genuine Bosch 0280158117 (no adapters): **~$200-240** (Polmaxracing, eBay, TRE Performance)
- Flow-matched branded sets with Honda OBD2 adapters included: **~$280-350** (Clean Injection, Injector Nation)
- High Flow Fuel w/ adapter kits: **~$220-280**

Budget: $280 for a set with adapters bundled.

Dead Time Data for Hondata

This matters -- K-series ECU fueling accuracy at idle/low pulse depends on correct dead-time curves.

- **Bosch published spec:** ~0.538 ms @ 14V / 0.846 ms @ 12V (per Ford Racing / EV14 bench data)
- **Hondata's recommendation:** K-series ECUs want scaled dead times closer to **~1.2 ms @ 14V** as a baseline. Bosch's raw spec is from a different bench methodology and causes +25-30% idle trim drift if plugged in directly.
- **Workflow:** Load a Hondata-community-published EV14 550cc dead-time map at install time, then live-tune on the dyno to fine-tune. Don't trust Bosch's raw sheet numbers.

O-rings / Seals

Genuine Bosch ships with 14mm top + 14mm bottom Viton-family O-rings installed. Fully E85-compatible. Budget $5-10 for a spare pack for peace of mind. Lube with engine oil at install.

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3. DW300C Fuel Pump

Part Number + Fitment

- **Kit P/N: DeatschWerks 9-307-1008** -- DW300C 340 LPH pump + complete 8th-gen Civic (2006-2011) install kit
- **Direct drop-in for FG2** -- no hanger modification. Kit includes pump, in-tank strainer, grommet, hose, clamps, Honda-specific electrical connections.
- **Flow:** 340 LPH @ 40 PSI, ~305 LPH @ 60 PSI (my actual rail pressure)
- **E85-compatible by design:** carbon commutators + fully-encapsulated armature
- **PWM-compatible** -- critical because the 8th gen Civic SI pump is PWM-controlled by the ECU, not on/off. A non-PWM-rated pump will have weird current and heat behavior.

Price

~$186-210 across vendors (DeatschWerks direct ~$189, MAPerformance ~$187-199, K Series Parts / HARDmotion / RallySport Direct in the same range). **Budget: $200 including shipping.**

Flow Headroom for 250 WHP E85

At 250 WHP peak on E85, estimated peak fuel volume is ~100-120 LPH at WOT. The DW300C delivers ~305 LPH at 60 PSI -> **~2.5-3x headroom**. Injector duty stays in 70-80% range with 550cc injectors, comfortably under the 85% reliability ceiling. Reliability-first rule satisfied.

Alternatives (Rejected)

Pump	Flow	Why not
Walbro 450 (F90000267)	450 LPH	Overkill for NA. Louder. Designed for high-boost.
AEM 50-1200	340 LPH	Equivalent perf, universal kit -- less clean fit on Honda hanger than DW.
DW65C / DW200	255-265 LPH	Marginal E85 headroom at 240 WHP. DW300C is same price class -- buy once.

Reviews + Warranty

DW300C on the 8th gen Civic is **well-established**: quiet, PWM-compatible, smooth idle, no pressure instability reported on E85 builds up through mid-300 WHP. **3-year no-fault DeatschWerks warranty.** Tilts reliability-first heavily in its favor.

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4. Continental Flex Fuel Sensor

Part Number: Continental 13577429

- Formerly Siemens/VDO, now Continental. Aftermarket PN **SE1004S**. ACDelco sells the same part as 13577429.
- This is the **"small" (shorter) GM flex fuel sensor** -- the one most K-series kits use. Reports 0-100% ethanol + fuel temp.
- Continental's OE Flex Fuel Sensor catalog also lists the larger 13577394 -- different sizes for different GM vehicles. **13577429 is the Honda standard** (smaller, cheaper, plentiful).

Price

- ACDelco 13577429 on Amazon: **~$75-85**
- GMPartsDirect historical: ~$44-55
- High Flow Fuel kit (sensor + -6AN barbs + stainless bracket): **~$95-105** <- easiest path
- Racetronix / Platinum Racing Products: ~$85-115

Budget: $95 (sensor-plus-barbs bundle from High Flow Fuel).

Fittings

- **Sensor ports:** 3/8" SAE EFI quick-connect (male input, female output) -- GM standard
- **On Honda (5/16" feed):** Convert 3/8" QC to -6AN via either bundled barb kit ($95 total path) or bare sensor + 2x 3/8" SAE QC-to--6AN adapters (~$115-145 total) from JBtuned, Summit, or Radium

Mounting

Inline on high-pressure feed, between fuel filter and rail, post-filter. Canonical for three reasons: sensor needs flow to read accurately; upstream-of-rail placement minimizes detection lag on fillups; post-filter keeps the sensor clean.

Physical mount: Bracket on a spare bolt boss on the firewall or battery tray area. **Constraint:** I'll need to delete the stock EVAP canister to make engine-bay room for filter + sensor + fittings (noted by JBtuned + SiriMoto install guides). EVAP delete requires Hondata to suppress the codes -- FlashPro handles cleanly.

Wiring (3 Wires)

Wire	Function	Connects to
12V switched	Sensor power (ignition-on)	Under-hood switched-12V source or add-a-fuse
Ground	Sensor ground	Clean chassis ground near sensor
Signal (50-150 Hz PWM)	Through a converter/gauge module -> 0-5V analog to ECU pin A33 (ECT2)	Innovate MTX-D or similar is the converter

Important: K-series ECU's ECT2 pin reads analog voltage (0-5V), not raw frequency. **Hondata's explicit recommendation:** use a converter module or a flex-fuel gauge with analog output. Do NOT wire the sensor's raw frequency output directly to ECT2.

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5. FlashPro Flex Fuel Setup

ECU Pin: ECT2 at A33

FlashPro's flex fuel implementation on the 8th gen SI uses the **ECT2 input at ECU pin A33** -- the unpopulated secondary coolant-temp sensor pin on the K20Z3, which is a free analog input.

- **Input type:** 0-5V analog, internal 5V pull-up
- **Voltage-to-ethanol mapping:** 0V = 0% ethanol, 5V = 100% ethanol (linear after calibration)
- **Install hook:** Unplug the blue ECT2 connector on the engine (passenger side near lower radiator hose on most K-series) and splice the flex-fuel gauge/converter output there -- signal reaches the ECU without running a new wire through the firewall.

Does FlashPro Read 0-85% Ethanol Linearly? Yes, After 2-Point Calibration

1. **Low-Side Calibration:** Fill tank with E15 or lower (regular pump gas). Run engine, connect FlashPro Live Tuning, confirm flex gauge reads low, adjust **Column 1 of "Ethanol Percentage vs Volts"** table so displayed % matches reality.
2. **High-Side Calibration:** Drain + fill with E65+ (fresh E85). Confirm gauge reads high, adjust **Column 2** of same table.
3. FlashPro linearly interpolates for any intermediate voltage. All blends E0-E85 read correctly once both endpoints are set.

Flex Fuel Tune Workflow -- Pump First, Then E85, Interpolate

Hondata-recommended order:

1. **Tune on pump gas first.** Build the base 93-octane map (or load my existing FlashPro calibration). This is the E0/E10 endpoint.
2. **Tune the ethanol compensation tables** (not separate maps -- compensation overlays):
   - **Ethanol Fuel Compensation:** additional fuel % at each ethanol content step. E85 baseline: ~+30-40%.
   - **Ethanol Timing Compensation:** additional timing advance at each ethanol step. E85 on 11.0:1 K20Z3 with good bolt-ons: +3 to +5 deg.
3. **Dyno-verify E85.** Fresh E85 fill, back-to-back pulls with logs, confirm AFR and timing hit targets.
4. **Live interpolation.** Once both endpoints are calibrated, FlashPro blends automatically based on sensor reading. Fill with any ratio -- ECU handles it seamlessly.

Ethanol % Behavior Across the Range

Ethanol	Timing ?	Fuel ?	WOT AFR	Feel
E0-E10	Baseline	Baseline	12.5-13.0 (0.88 lambda)	Pump-gas tune, ~220-240 WHP
E30-E40	+1 to +2deg	+10-15%	12.2-12.5 (0.85)	Modest bump, minimal MPG hit, ~225-240 WHP
E50	+2 to +3deg	+18-22%	12.0-12.3 (0.82)	Noticeable, moderate MPG hit, ~230-245 WHP
E85 (summer)	+3 to +5deg	+30-40%	11.5-12.0 (0.80)	Peak, ~230-250 WHP, 25-30% MPG penalty

Ethanol Content Gauge In-Car

Because the sensor signal goes through a converter anyway, I get a visible cabin readout. Options:

Gauge	Price	Notes
Innovate MTX-D	~$140	52mm OLED. Ethanol % + fuel temp. Also acts as the 0-5V converter for FlashPro. SiriMoto's default.
Innovate ECB-1	~$300	Ethanol + boost + wideband. Boost channel wasted on NA. Useful if turbo later.
AEM 30-2201 flex kit	~$230	Sensor + gauge + harness bundled

Decision: Innovate MTX-D ($140). Documented SiriMoto path, outputs the voltage FlashPro wants, clean install.

Acuity Fuel Pressure Gauge -- Brand Consistency

Acuity 100 PSI fuel pressure gauge (Marshall Instruments-built, 52mm face), available in **Satin Black** or Polished Stainless -- **not currently offered in purple, but dial face is white/black and coordinates with purple AN fittings and the Acuity rail**. ~$100-130. Screws into the 1/8" NPT port on top of the Acuity rail.

Buying one gives me (1) live rail pressure during install leak check, (2) Acuity brand consistency, (3) early warning for pump failure before fuel trims go haywire. **Budget: $115 (Satin Black).**

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6. Fuel Pressure Regulator

Keep the Stock In-Tank FPR

The 8th gen Civic SI uses a **returnless fuel system** -- regulator is integrated into the in-tank pump hanger, not in the engine bay. Base pressure ~50-58 PSI, PWM-modulated by the ECU.

For 250 WHP NA on E85 with 550cc injectors: stock FPR is adequate. Every flex-fuel 8th gen Civic SI build in the community at or below this power level confirms. Injector duty stays well under 85% at peak, no pressure droop, no benefit from an external adjustable FPR.

Decision: Keep stock. No adjustable FPR.

When Adjustable FPR Would Make Sense

Turbo/SC with rising-rate fuel pressure needs, 1000cc+ injectors needing lower base pressure for idle quality, alcohol-injection or port+DI hybrid setups, or dyno-chasing the last 3-5 WHP at injector duty ceiling. None apply now. If I go turbo later, I'd convert to return-style with a Radium DMR or FUELAB FPR then.

Radium FPR Reference (if ever needed)

- **Radium DMR (Direct Mount Regulator):** 20-1623-00 (black) or 20-1623-01 (green), ~$290-340
- **Radium FPR with Bosch OEM top:** 3 bar = 20-0014-00, 4 bar = 20-0014-01, ~$180-240
- **No purple option from Radium.** Custom anodize $50-100 if ever installed to match theme.

Base Pressure Procedure (If Adjustable FPR Added)

Prime (KOEO x3), read on Acuity gauge, start + warm idle, target **~58 PSI (4 bar) static** -- community standard for 550cc EV14 + Hondata dead-time data. Adjust hex screw 1/4 turn = ~3-5 PSI. Lock with lock nut. Verify at idle, 3500 RPM, and WOT; should hold ±2 PSI.

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7. Fuel Lines

Stock 8th Gen Lines vs. PTFE

Modern Honda fuel lines (post-2006 fluoroelastomer spec) are **generally E85-compatible for moderate-term use** -- documented safe by the 8thcivic FAQ and multiple K-series shops. **Long-term sustained straight E85** is where PTFE starts to matter.

My Plan -- Phased

Phase 1 (initial install): Keep stock feed line tank-to-engine-bay. Add **one short -6AN PTFE section** from filter -> flex sensor -> Acuity rail inlet -- the area getting fittings work anyway. ~$60-90 parts, zero extra labor, upgrades the hardest-working section.

Phase 2 (if needed): If I see stock-line degradation or commit to full-time E85, upgrade full feed line from

tank-hardline to rail. $200-350 parts, $150-300 shop labor if I have them build/crimp.

Options If Upgrading Later

- **Hybrid Racing K-Series Fuel Line Kit:** ~$180-250, PTFE-lined, pre-assembled AN ends
- **Custom -6AN PTFE** (Summit, Fragola, Earl's, Aeroquip components): ~$80-180 for feed only
- **JBtuned 8th Gen Civic SI full fuel system + return kit:** ~$600-900, full replacement + return conversion

Install Difficulty

Proper PTFE field assembly requires a hose vise + crimping tool. For Phase 1 scope (short rail-feed PTFE section), **pre-made -6AN PTFE hoses from Summit/Russell/Fragola** eliminate the crimping requirement entirely.

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8. AN Fittings & Purple Aesthetic

Minimum Fittings (Phase 1)

Location	Fitting	Qty
Acuity rail inlet (center -8 ORB)	-8 ORB to -6AN male adapter OR -8 ORB to Honda QC	1
Acuity rail end ports (unused)	-8 ORB plug	2
Sensor inlet/outlet (3/8" SAE QC)	3/8" SAE QC to -6AN male	2
PTFE line ends	-6AN hose-end fittings (straight or 45deg)	2-4
Fuel filter adapter	5/16" Honda QC to -6AN	1-2

Total ~8-10 pieces. Budget $140-220 for a complete Phase 1 set.

Purple Anodized AN Fittings

- **Fragola Performance Systems** -- top pick. US-made (Connecticut), one-piece design, J-gauge threads. Offers purple anodize across standard -6AN hose ends, NPT adapters, bulkheads.
- **Vibrant Performance** -- purple PTFE hose ends in the 28xxx series, filterable at Summit Racing. $15-30 per fitting.
- **Russell / Edelbrock** -- purple in limited legacy runs, availability thinning.
- **Earl's** -- primarily black/blue/red; occasional purple runs.

Cost Estimate

Fitting	Qty	Est. (purple anodize)
-8 ORB to -6AN male	1-2	$12-18 ea
3/8" SAE QC to -6AN	2	$25-40 ea
-8 ORB plugs	2	$8-12 ea
-6AN straight hose-ends	2-4	$15-30 ea
-6AN 45deg/90deg hose-ends	1-2	$20-40 ea
5/16" Honda QC to -6AN	1	$18-25 ea

Realistic total for full purple fitting set: $140-220. Budget: $180. Black-anodized equivalent saves $40-60. Paying the premium for theme consistency.

9. Fuel Filter

OEM In-Tank + Optional Engine-Bay Inline

On the 8th gen Civic SI, the **OEM fuel filter is in-tank, part of the pump hanger** -- the DW300C 9-307-1008 kit includes a fresh strainer, so it's replaced automatically at the pump install.

Additional engine-bay inline filter is recommended for flex fuel: ethanol can loosen decades-old tank deposits that a single in-tank strainer won't catch, and the Continental sensor downstream benefits from clean fuel.

Options

- **Aeromotive AN-6 Fuel Filter:** ~$90-140, well-regarded, easy sourcing
- **Radium Engineering Fuel Filter Kit:** ~$80-130, replaceable element, -6AN male ends, 10-40 micron
- **Fuelab Prodigy:** ~$140-200, premium

Budget: $90 for an Aeromotive or Radium -6AN 10-micron inline filter mounted engine-bay, upstream of the flex sensor.

No separate OEM disposable fuel filter purchase needed -- the 8th gen's service filter lives in the tank and ships with the pump kit.

10. Purple Summary

Component	Purple Available?	Path
Acuity fuel rail	Yes -- 1913-PPL factory	$199
Bosch EV14 injectors	No	Not visible after install
DW300C pump	No (in-tank, invisible)	N/A
Flex fuel sensor	No (black body)	DIY purple bracket if visible
Radium FPR (if ever)	No	Custom anodize $50-100
AN fittings	Yes -- Fragola, Vibrant	$180 for full set
PTFE hose	Standard stainless/black braid	Purple braid rare/expensive
Fuel pressure gauge	Satin Black or Polished Stainless	$115
Fuel filter	Red/blue/black typical	Not purple
Sensor mount bracket	DIY anodize	$25-50

Visible purple in finished engine bay: Acuity rail (the money shot, sitting on top of the IM) + ~8 purple AN fittings snaking from filter through sensor to rail + DIY bracket if desired. Theme reads clean without paint or re-anodize work.

11. Install Procedure

Combined injectors + rail + pump + sensor + filter + wiring. **Estimated: 6-10 hours over 1-2 weekends solo, not counting dyno.**

Safety Pre-Work

Outdoors or well-ventilated garage. Fire extinguisher in arm's reach. Nitrile gloves + safety glasses. No smoking, no grinding, no pilot lights. Fuel at ~58 PSI -- respect it.

Stage 1 -- Pressure Relief + Battery Disconnect

1. Cold vehicle (overnight rest). Open fuel filler cap for tank vapor relief.
2. Pull **PGM-FI Main Relay 2** (fuel pump relay) from the **auxiliary under-hood fuse/relay box** (top row center per FSM -- verify for my VIN).
3. Start engine, idle until stall (15-60 sec). Crank 2-3 sec after stall to fully bleed rail.
4. Disconnect negative battery terminal (10mm). Tuck cable clear of the post.
5. Reinstall relay (easier for reassembly; pull again if needed during rail work).

Stage 2 -- Remove Stock Rail + Injectors

1. Remove engine/rail cover (4-6x 10mm).
2. Unclip 4 injector connectors.
3. Release Honda QC fitting at stock rail feed -- shop rag ready for residual fuel. Use Honda QC release tool if seized, don't pry.
4. Remove 2x M8 rail-to-IM bolts (108 in-lb / 9 ft-lb). Gently work rail off injector tops.
5. Lift rail + 4 injectors up together.
6. Inspect + replace any damaged lower-injector O-rings in manifold bores.

Stage 3 -- Pre-Assemble Acuity Rail + EV14 Injectors

1. On clean bench: Acuity rail, 4x Bosch EV14 550cc, OBD2 pigtail adapters, spare 14mm O-rings.
2. Verify/install upper + lower O-rings. **Lube with fresh engine oil** (no silicone grease -- degrades rubber).
3. Hand-press each injector upper into the Acuity rail bore. Check O-ring not rolled/pinched.
4. Connect each injector's USCAR plug to Honda OBD2 pigtail. Verify retention clip engagement.
5. Set pre-assembled rail aside on clean rag.

Stage 4 -- Install Rail Assembly onto IM

1. Lube lower-injector O-rings.
2. Line up 4 injectors with IM bores. Press rail straight down -- even pressure both ends, hand pressure only.
3. **Push each injector downward to bottom it; verify upper O-rings still seated in rail.** Re-seat if any backed out.
4. Install 2x M8 mounting screws. **Torque: 108 in-lb (9 ft-lb).**
5. Do not connect fuel feed or injector harness yet -- staged for Stage 6.

Stage 5 -- Tank Drop for DW300C (Actually No Tank Drop)

The 8th gen Civic SI has a rear-seat service panel -- no tank drop required. Saves ~2 hours.

1. Remove rear seat bottom cushion (pull up firmly to release 2 clips, slide rearward out of rear hook).

2. Lift floor carpet, locate fuel pump hanger access cover on passenger side (4-6 Phillips screws or plastic fasteners).
3. Remove cover. Disconnect hanger electrical + fuel line/evap QCs.
4. Remove hanger retaining ring with Honda FSM spanner or large rubber-strap wrench. **Do not use hammer + drift -- cracks the tank.**
5. Lift hanger straight up, be careful of the fuel level float arm.
6. On bench: transfer DW300C pump into the hanger per 9-307-1008 kit instructions -- old pump out, new DW300C in with supplied grommet/hose/clamps, new in-tank strainer, verify electrical pass-through.
7. Reinstall hanger into tank with fresh O-ring/gasket. **Retaining bolts torque: 39 in-lb** (very gentle, plastic tank threads).
8. Reconnect electrical + fuel/evap lines.
9. Pour ~1 gallon of pump gas if tank is empty (need liquid at the strainer for first prime).
10. Reinstall access cover, carpet, seat cushion.

Stage 6 -- Flex Fuel Sensor + Wiring

1. Choose mount location (inline filter-to-rail, post-filter). Fabricate or buy bracket mounted to firewall/battery-tray bolt boss.
2. **EVAP delete** (if going that route): remove canister + hoses. Ensure Hondata tune suppresses EVAP codes before first start.
3. Route line: stock hardline outlet -> inline filter -> Continental 13577429 -> Acuity rail -8 ORB center-feed.
4. Route -6AN PTFE with slack, no sharp bends. Secure with cushioned P-clamps.
5. Torque AN: tube-nut fittings hand tight + 1/4 turn; -6AN to -8 ORB hand tight + 1/4 to 1/2 turn past O-ring contact; QC-to-AN per vendor (usually 15-25 ft-lb).
6. Wire sensor:
   - 12V switched: tap or add-a-fuse in under-hood box (10A)
   - Ground: clean chassis near sensor
   - Signal: into Innovate MTX-D -> gauge's 0-5V output into ECT2 connector at ECU pin A33
7. Verify no wiring chafe near hot exhaust or moving accessories.
8. Pre-key-on visual inspection of every fitting joint.

Stage 7 -- 3-Stage Leak Check

7A -- Cranking only (no start): Reconnect battery. Key on/off/on/off/on -- pump primes 3 cycles fully. **5-minute visual inspection of every joint.** Beading, wetness, smell. Do NOT start engine until 100% leak-free.

7B -- Idle: Start engine (cranking slightly longer than stock -- system refilling from empty). Let warm to ~90degC. Re-inspect joints (vibration + thermal expansion reveal leaks not visible at static pressure). Verify Acuity gauge reads ~50-58 PSI warm idle.

7C -- Post-drive: 20-minute drive cycle, <4000 RPM. Re-inspect on return. Full thermal cycling + real demand variation. Pass = clean. Fail = address before next drive.

Stage 8 -- First Start + Fuel Trims

Live FlashPro monitoring for first 10 min: idle RPM, ECT, STFT, LTFT, rail pressure.

Expected on pump-gas base map: idle ~800 RPM smooth, STFT ±5%, LTFT ±8% first cycle, no CEL. **If trims are**

+10% sustained: likely dead-time mismatch or sensor miscalibration. Don't chase the tune -- calibrate sensor first.

Stage 9 -- Flex Calibration + Dyno Tune

1. **Calibrate Low Side** with known pump gas (Column 1 of "Ethanol % vs Volts" table).
2. Drive 1-2 tanks pump gas, let pump-gas tune stabilize.
3. **Book dyno session** ($500-650, 3 hr expected). Bring fresh E85 for High-Side calibration.
4. At shop: baseline pump-gas pulls, drain, fill E85, calibrate High Side (Column 2), run compensation tune for fuel + timing, back-to-back dyno validation.
5. Archive .fpc file + CSV dyno data for future reference.

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12. E85 Operating Notes

Fuel System Compatibility -- All Green

Every component post-build is E85-safe: polyethylene tank (OEM, E85 OK), DW300C pump (carbon commutators + encapsulated armature), in-tank + inline filters (E85-rated elements), Continental sensor (built for E85), PTFE line section, Acuity rail (6061-T6 aluminum + hard anodize + Viton O-rings), Bosch EV14 injectors (factory E85-compatible Viton seals). **Stock feed hardline is E85-OK medium-term**; replace if degradation appears.

Cold Start

- >40degF: no issue with proper tune
- 20-40degF: cranking 2-4 sec (vs ~1 sec gasoline), slightly rougher first 30 sec -- manageable
- <20degF: straight E85 can fail to start. **Winter-blend E85 (often E55-E65 by law in cold states) helps substantially**. E30-E40 pragmatic in deep winter.
- Tuning mitigation: richer cold-start enrichment + longer cranking priming for high-ethanol end of compensation tables. My tuner dials this.

Fuel Economy

~25-30% worse MPG on E85 vs pump gas (lower energy density). 26 MPG gasoline -> 18-20 MPG E85. E30 ~10-12% worse; E50 ~15-20% worse. E30-E40 is the practical daily sweet spot.

Tank Water Absorption

E85 is hygroscopic but **tolerates ~4% water by volume** before phase separation -- a 13-gallon tank can absorb ~0.5 gal water without phase-separating. In practice a non-issue for a daily driver burning through tanks weekly. For storage >3 months: drain, fill with non-ethanol, or run tank to 1/4 and refill with pump gas. Keep tank full during normal use to minimize condensation air-space. **Burn through, don't store.**

Pump Octane Variance

ASTM D5798 allows "E85" to range **51-83% actual ethanol**. Winter-blend pumps may sell E51-E70 as "E85." Summer: E70-E83. Midwest (corn belt) hits higher end consistently; coasts/Southeast less consistent. **Sensor handles all of this automatically** -- the reason a flash-for-E85 tune without a sensor is dangerous.

Sensor + Gauge Integration

Innovate MTX-D: ethanol % + fuel temp real-time in cabin. FlashPro datalogger captures same data as "Ethanol Content %" in .fpdl logs for post-drive forensics. Both live and forensic -- every fill-up documented.

13. Cost + Phased Plan

Complete Parts Cost (USD, inc. tax/shipping)

#	Component	Part	Cost
1	Fuel rail	Acuity 1913-PPL Satin Purple	$219
2	Rail fitting (inlet adapter + plug)	-8 ORB to -6AN + 1/8 NPT plug	$20
3	Injectors (4)	Bosch EV14 550cc w/ Honda adapters	$280
4	Fuel pump + kit	DW300C 9-307-1008	$200
5	Flex fuel sensor kit	Continental 13577429 + -6AN barbs	$95
6	Flex fuel gauge/converter	Innovate MTX-D	$140
7	Fuel pressure gauge	Acuity 100 PSI Satin Black	$115
8	PTFE -6AN hose	~3 ft braided PTFE	$45
9	Purple AN fittings	~8-10 Fragola/Vibrant	$180
10	Sensor QC-to-AN	3/8" SAE QC to -6AN x2	$50
11	Inline engine-bay filter	Aeromotive -6AN 10-mic	$90
12	Sensor bracket	Custom/aftermarket	$20
13	Wiring	Wire, pigtails, fuse tap	$30
14	Misc hardware	O-rings, P-clamps, ties	$20
Parts subtotal			$1,504
15	Sales tax (~8%)		~$120
16	Flex fuel dyno tune	3 hr pro session	$550
Grand total			~$2,174

Minimum Viable (skip purple premium, gauge, inline filter)

Skip filter (-$90), fuel pressure gauge (-$115), purple premium on fittings (black instead, -$80). **Minimum: ~$1,890 parts + tune.**

Phased Purchase Plan

Phase A -- Hardware (~$1,020 parts + ship): Acuity rail, EV14 injectors + adapters, DW300C, Continental sensor kit, Innovate MTX-D, basic -6AN fittings + short PTFE, engine-bay filter. Complete functional flex-fuel system minus full purple aesthetic + pressure gauge. Bundle-ship from 1-2 retailers. Install in one weekend.

Phase B -- Aesthetic (~$300): Purple AN fittings to replace the black ones from Phase A. Acuity fuel pressure gauge. Pure visual upgrade, swap fittings during a regular oil change.

Phase C -- Tuning ($550-650): Book dyno after ~500 miles. Bring fresh E85 for high-side cal. Walk out with

dialed flex tune.

Phase D -- Future: Full PTFE feed line ($200-300) if stock lines degrade. Return-style + adjustable FPR if/when turbo (not soon).

Dyno Time + Tuning Cost

Base NA tune ~$500 (3 hr). Flex surcharge +$150. Overage ~$150/hr. Deposit ~$150 to book. **Realistic total: $500-650 for complete pump-gas + E85 flex cal.**

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14. Sources

Part Numbers / Pricing

- Acuity K-Series Fuel Rail -- Satin Purple (1913-PPL)
- Acuity K-Series Fuel Rail -- Satin Black
- K Series Parts -- Acuity Satin Purple (1913-PPL)
- Nippon Power -- Acuity Purple Fuel Rail
- Radium K-Series Fuel Rail (20-0230-02)
- Radium K-Rail Install Instructions PDF
- Clean Injection -- K20/K24 550cc Bosch Injectors
- Polmax Racing -- Bosch EV14 550cc Genuine
- CP Fuel & Race -- K20/K24 Bosch EV14 550cc USCAR
- High Flow Fuel -- Bosch EV14 550cc (0280158117)
- Bosch Motorsport EV14 Data Sheet PDF
- DeatschWerks DW300C 9-307-1008
- K Series Parts -- DW300C 9-307-1008
- MAPerformance -- DW300C 9-307-1008
- ACDelco Continental 13577429 on Amazon
- High Flow Fuel -- Continental 13577429 w/ AN Barbs
- Continental OE Flex Fuel Sensors Catalog PDF
- Innovate Motorsports -- Ethanol Content Gauges
- Innovate MTX-D Gauge
- Acuity 100 PSI FP Gauge -- Satin Black
- Radium Engineering FPR

Hondata / Tuning References

- Hondata ECU Connectors Reference
- Hondata Ethanol Parameters (FlashPro flex fuel tuning)
- Hondata Forum -- Flex Fuel Support
- Hondata Forum -- 550cc EV14 Install Info
- Hondata Forum -- Injector Dead Times (550cc)
- Hondata Forum -- ID Dead Times + Calibration
- Hondata Forum -- E85 Sensor + ECT2 Input

Install References

- SiriMoto 8th Gen Civic Si Flex Fuel Kit Install Guide PDF
- Hybrid Racing K-Series Fuel Rail Install Guide
- JBtuned 8th Gen Civic SI Full Fuel System Replacement
- 8thcivic Forum -- K20Z3 Torque Specs
- Nthefastlane -- K-Series Torque Specs

Skunk2 Ultra Street + K20Z3 Compatibility

- Skunk2 Ultra Street Intake Manifold (307-05-0600)
- Hybrid Racing -- Skunk2 Ultra Street
- TF-Works -- Skunk2 PRB-to-RBC TB Adapter

E85 Chemistry / Storage

- PetroClear -- Phase Separation in Ethanol Blends
- Ethanol RFA -- Water Uptake of Ethanol-Gasoline Blends PDF
- NREL Handbook for Handling E85 PDF
- eFlexFuel -- Does E85 Go Bad?

AN Fittings / PTFE

- Summit Racing -- Purple Anodized AN Hose Assemblies
- Fragola Performance Systems
- Vibrant Performance PTFE Hose Ends (Summit Racing)

Tuning Pricing

- XYR Tuned -- Hondata FlashPro Service

Cross-References

- overview.md -- strategic "is flex fuel worth it on NA K20Z3" analysis, cost-vs-WHP breakdown, E85 chemistry background.
- ../10-Hondata-FlashPro/tuning-workflow-and-maps.md -- Daily/Sport two-map reliability-first tune ceilings this calibration plugs into.
- ../14-Intake-Manifold/ -- Skunk2 Ultra Street install, water-bypass adapter, DBW TB adapter spec. Manifold must be finalized before rail install.
- ../11-Wideband-AFR/ -- AEM X-Series 30-0300 wideband. Mandatory before flex tune.
- ../docs/torque-specs.md -- canonical fastener spec table.
- ../docs/maintenance-parts-catalog.md -- add Acuity 1913-PPL, Bosch 0280158117, DW 9-307-1008, Continental 13577429 once finalized.

Last updated: 2026-04-20

Flex Fuel + Fuel System Install Guide -- Acuity Rail + EV14 + DW300C + Continental Sensor

Status: Planning -- Phase 6 (final power adder)

Install companion to acuity-fuel-rail-and-flex-fuel-build.md (parts + wiring deep dive) and overview.md (strategic NA-flex-fuel analysis). Those docs cover WHAT and WHY. This doc is HOW -- the procedure for getting it on the car without starting a fire.

Reality Check Before I Start

This is the most electrically and plumbing-intensive job on the build. The wrenching is moderate, but it touches the fuel system at every level: in-tank pump, feed line, inline sensor, rail, injectors, ECU wiring. Any one of those joints leaking is a fire hazard. Any one wired wrong is either no signal or a damaged sensor.

Time budget, honest:

- **Day 0 (full Saturday):** Phase A depressurize -> Phase B pump -> Phase C sensor plumbing -> Phase D rail + injectors. ~8-10 hours solo.
- **Day 1:** Phase E wiring -> Phase F FlashPro config -> Phase G first-start leak checks. ~4-6 hours.
- **Dyno day (scheduled 2-4 weeks out):** Phase H Tune #3. ~3-4 hours at the shop.

Difficulty: moderate-high. The difficulty lives in fuel risk, not mechanical steps. Rush the leak check, buy a garage fire. Skimp on an O-ring, buy a leak on the highway two weeks later. Every fitting, every O-ring, every clamp -- right the first time.

Bundle opportunity: If doing 14-Intake-Manifold/ around the same time, **combine them.** Rail swap while IM is off the bench saves 2-3 hours of teardown and a round of leak-checking.

Safety First -- Read This Twice

58 PSI of atomized E85 off a loose fitting will find an ignition source I didn't know I had.

1. **No open flames, grinding, pilot lights, or smoking** anywhere in the garage. Water heater pilot is a killer -- shut off gas to the garage or work in the driveway.
2. **Fire extinguisher (Class B or ABC) in arm's reach** -- at the fender, not in the corner.
3. **Depressurize the fuel system before ANY line disconnect.** Phase A. No skipping steps.
4. **No sparks.** Battery disconnected before touching any fuel-carrying component.
5. **Work outdoors or garage door wide open.** Fuel vapor pools low and travels farther than I'd expect.
6. **Safety glasses + nitrile gloves** (not latex -- latex swells with gasoline).
7. **Keep fuel clean.** Every opened fitting is a chance for dirt to enter. Blue shop towels, sealed catch container -- not a coffee can.
8. **If I smell fuel at any point after reassembly, STOP.** Don't crank, don't drive. Find it, fix it, retest.

Prerequisites / Cross-References

Must Be Done FIRST

Prerequisite	Why	Reference
FBO pump gas Tune #2 complete	The E0 endpoint in the flex tune IS the FBO calibration. No point interpolating off a calibration that doesn't exist.	../10-Hondata-FlashPro/tuning-workflow-and-maps.md
Wideband AFR installed and reading	Tuner needs it on the dyno; I need it for Phase G logs.	../11-Wideband-AFR/
FlashPro analog 2 input type VERIFIED	Critical unknown. Frequency-capable or voltage-only? Decides whether MTX-D is inline or standalone. See Decision Tree below.	../10-Hondata-FlashPro/hardware-software-deep-reference.md, acuity-fuel-rail-and-flex-fuel-build.md
Cooling refresh + ignition refresh complete	E85 tune extracts more power = more heat. Weak coil shows up as E85-load misfire.	../09-Cooling-System/, ../06-Ignition-Refresh/
Innovate MTX-D converter ON HAND	Even if analog 2 is frequency-capable, MTX-D is the drop-in fix if Phase F shows no signal. No shipping delay mid-install.	--
Dyno session pre-booked	Flex tune shops book 2-4 weeks out. Book when I order parts, not after install.	../10-Hondata-FlashPro/tuning-workflow-and-maps.md

Bundle Opportunity

- **IM install (14-Intake-Manifold/) is the ideal pairing** -- rail swap trivial with IM off.
- Headers (13-Headers/) separate -- one session per major airflow change.

Tools Required

Critical:

- **Honda fuel line QC release tool set** (5/16" and 3/8" SAE). Prying with a screwdriver wrecks lines and collars.
- **Torque wrench 1/4" drive, 20-200 in-lb** -- rail bolts (108 in-lb) and pump hanger (39 in-lb).
- **Torque wrench 3/8" drive, 10-80 ft-lb** -- AN fittings up to 30 ft-lb.
- **Line wrenches** (5/8", 11/16") -- grip all 6 flats; open-end wrenches round AN hex.
- **Multimeter** -- continuity, 12V verification, sensor signal checks.
- **Scope or frequency-capable DMM (optional)** -- reading Continental's 50-150 Hz direct. If skipped, MTX-D readout is verification.
- **Sockets 10/12/14mm.**
- **Rear seat removal tools** -- Phillips + 10mm for cushion anchors.
- **Fuel pump hanger spanner or rubber-strap wrench** -- for the in-tank retaining ring. **Hammer + drift cracks the plastic tank.**
- **Fire extinguisher, sealed fuel catch container (1-2 gal fuel-rated).**
- **Ratcheting wire crimper, heat shrink, heat gun.** No electrical tape on fuel-system wiring.
- **Nitrile gloves, safety glasses, shop rags/blue towels.**

Also useful: magnetic parts tray, inspection mirror + headlamp, carb cleaner + small brush, zip ties, cushioned P-clamps.

Parts List

Full BOM with prices in acuity-fuel-rail-and-flex-fuel-build.md §13. Install-day checklist:

#	Part	PN	Qty
1	Acuity K-Series Rail, Satin Purple	1913-PPL	1
2	Bosch EV14 550cc injector	0280158117	4
3	USCAR-to-Honda OBD2 injector adapter pigtails	(per vendor)	4
4	DW300C pump kit	9-307-1008	1
5	Continental flex fuel sensor	13577429	1
6	Innovate MTX-D frequency-to-voltage gauge/converter	MTX-D	1
7	PTFE -6 AN braided hose	~3 ft	1
8	-6 AN fittings + adapters (per build doc §8)	--	~8-10
9	Fresh 14mm Viton O-rings (spare)	--	2x set
10	Inline engine-bay filter, -6 AN 10-mic	Aeromotive or Radium	1
11	18 AWG automotive wire (red/black/white)	--	~15 ft each
12	5A add-a-fuse	mini blade	1
13	Heat shrink + crimp terminals + P-clamps	assortment	--

Don't skimp on O-rings. Reusing a stock O-ring on a new EV14 is the #1 cause of post-install leaks. Fresh Viton pack as install-day insurance.

Step-by-Step -- Phased Install

Phase A -- Depressurize the Fuel System

Most-skipped step on the internet. The one that starts fires. Do not skip. Rail sits at ~50-58 PSI at rest.

A1. Level surface, engine off, cold. A2. Open fuel filler cap -- relieves tank vapor pressure. A3. Under-hood **auxiliary fuse/relay box** (driver side). Pull **PGM-FI Main Relay 2** (fuel pump relay). Set aside. A4. Start engine. Runs 15-60 sec on residual rail pressure and stalls. A5. Crank starter 2-3 more seconds after stall. Bleeds last of pressurized fuel through injectors. A6. Key fully off, key out. A7. Disconnect **negative** battery terminal (10mm). Tuck clear of post. A8. **Confirm depressurized** by loosening the rail feed union slowly over 30 seconds with a shop rag wrapped around the joint. Residual drip is normal. **Spray means I didn't depressurize** -- reconnect, repeat A3-A5. A9. Reinstall the PGM-FI relay once confirmed (pull again in Phase G if needed).

Phase B -- DW300C Pump Install (In-Tank)

8th gen has a rear-seat service hatch. No tank drop.

B1. Remove rear seat bottom cushion -- pull firmly at front edge (two plastic clips release), slide rearward to unhook back anchor, lift out. B2. Fold back carpet above the tank. B3. Locate **fuel pump access panel** on passenger side. 4-6 Phillips or plastic fasteners. Remove. B4. **Photograph connections.** Phone camera, 4-5

angles. Reassembly reference. B5. Disconnect pump harness connector (squeeze tab, pull). B6. Disconnect fuel feed line at hanger QC. Shop rag wrapped. Slide collar forward, pull line straight off. B7. Disconnect EVAP line if present at hanger. B8. Unlock **retaining ring** counterclockwise with spanner or rubber-strap wrench. **Never hammer + drift** -- cracks the tank collar. B9. Lift hanger **straight up** -- careful of the fuel level float arm (bends easily). B10. Move to clean bench with small drip pan. B11. Per DW 9-307-1008 kit instructions: remove stock pump from hanger, install DW300C with supplied grommet/hose/clamps, install new in-tank strainer, reconnect internal harness per kit polarity diagram. B12. Inspect hanger-to-tank O-ring/gasket. Replace if crusty (available from DW or dealer). B13. Lower hanger back into tank, mind the keyed orientation and float arm. B14. Hand-thread retaining ring clockwise, then tighten. **Torque: 39 in-lb** per ../docs/torque-specs.md. Plastic threads -- gentle. B15. Reconnect external pump harness, fuel feed QC (tug-test for click engagement), EVAP line. B16. If tank is near-empty, pour **~1 gallon pump gas** now. Pump strainer needs liquid for first prime. B17. Reinstall access panel, carpet, seat cushion. **Leave battery disconnected until end of Phase G.**

Phase C -- Install Continental 13577429 Flex Sensor (Inline)

C1. **Location: between inline engine-bay filter and Acuity rail inlet.** Post-filter (sensor sees clean fuel), close to rail (minimal fillup lag). Firewall or battery tray bolt bosses. C2. **If deleting EVAP canister** (planned per build doc §4): remove canister, cap remaining lines per JBtuned reference. Confirm FlashPro tune suppresses EVAP codes before Phase G start. Keeping EVAP = find space elsewhere (tight). C3. **Mount the sensor bracket** -- simple bent-steel angle from ACE ($5) or purchased from JBtuned/SiriMoto. Mount sensor with **fuel flow arrow pointing toward engine**. Sensor body is marked. C4. **Line routing:** stock hardline -> 5/16" Honda QC-to--6 AN -> inline filter (flow arrow) -> short PTFE -6 AN -> sensor (3/8" SAE QC-to--6 AN each side) -> short PTFE -6 AN -> -8 ORB-to--6 AN adapter -> Acuity rail center-feed. C5. **Cut stock hardline** at filter inlet. Clean square cut, tubing cutter, not a hacksaw. C6. Pre-assembled PTFE -6 AN hoses from Summit/Russell/Fragola save the hose-vise crimping step. Measure twice. C7. **AN torque:**

- -6 AN unions: hand tight + 1/4 turn (O-ring seals at contact, not crush).
- -8 ORB at rail: hand tight + 1/4 to 1/2 turn past O-ring contact.
- QC-to-AN adapters: vendor spec, typically 15-25 ft-lb on AN side.
- **Never over-torque O-ring joints.** Crush damage causes warm-pressurized leaks.

C8. **Secure with cushioned P-clamps.** No chafe points against body, hot exhaust, moving parts. Trace line visually from tank to rail, confirm no hot/vibrating contact.

Phase D -- Acuity 1913-PPL Rail + EV14 Injectors

Much easier with IM off the car. Bundle with ../14-Intake-Manifold/ install. Steps below assume IM on the car.

D1. Remove engine/rail cover (10mm bolts). D2. Unclip 4 injector connectors. Label 1-2-3-4 front-to-rear (masking tape). D3. Disconnect stock feed line at rail union if not already off from Phase A. Shop rag. D4. Remove 2x M8 rail-to-IM bolts (108 in-lb spec). Gently. D5. Lift stock rail straight up -- the 4 injectors come with it (upper O-ring friction). D6. Set stock assembly aside as parts-shelf spare. D7. **Inspect IM injector bores.** Clean rag, small brush + carb cleaner if dirty. No wire brush on sealing surfaces. D8. **Pre-assemble Acuity rail + EV14s on a clean bench:**

- Verify all 8 injector O-rings present (top + bottom x 4). Genuine Bosch ships with them. Replace any

rolled/pinched/discolored.

- **Lube every O-ring with fresh engine oil.** Thin coat. Not silicone grease, not assembly lube.
- **Press each EV14 upper into the Acuity rail.** Straight in, no twisting. Firm hand pressure, no hammer. Verify upper O-ring fully seated.
- Connect USCAR plug to Honda OBD2 pigtail. Clip must click.

D9. Lube lower O-rings just before IM install (no dust pickup). D10. Line up 4 injectors with IM bores. **Lower rail straight down, even pressure both ends.** If one resists, back off -- rolled O-ring. D11. **Push each injector down to fully seat in IM.** Then **verify each upper O-ring is still in the rail bore** -- pressing down sometimes backs an upper out. If so, re-seat. D12. Install 2x M8 bolts. **Torque: 108 in-lb / 9 ft-lb** (per ../docs/torque-specs.md). Alternate bolts, two passes, 50% then final. Aluminum IM threads -- gentle. D13. Connect Phase C fuel feed line to rail center-feed -8 ORB. Hand tight + 1/4 to 1/2 turn past contact. D14. **Do NOT connect injector harness yet.** Staged for Phase G leak check.

Phase E -- Wiring -- Flex Sensor

Read the Decision Tree below BEFORE wiring the signal line.

Continental 13577429 has 3 wires:

Function	Typical Color	Connects To
Sensor power	Red (verify pinout)	Switched 12V
Sensor ground	Black	Clean chassis ground
Frequency signal out	White or Yellow	Per Decision Tree below

Verify colors against Continental 13577429 pinout (ships with High Flow Fuel bundle or Hondata KB).

E1. Route 3-wire harness from sensor to 12V tap, ground, and signal destination. 18 AWG automotive wire, ~6" slack at each end. E2. Cushioned P-clamps every ~12". No chafe against body, hot surfaces, moving linkages. E3. **Power (switched 12V):** add-a-fuse on an accessory circuit in under-hood box, 5A new leg. Or splice into injector harness 12V -- **verify stable during cranking** (some accessory circuits drop during start).

- **Never a constant-12V circuit** -- sensor draw discharges battery during sits.

E4. **Ground:** clean bare-metal chassis point near sensor. Scuff to bare, ring terminal + lock washer + star washer + M6/M8 bolt.

- **Star topology -- no daisy chains.** Flex sensor, MTX-D (if used), and FlashPro analog all go to the same chassis point, not chained. Chained grounds cause voltage drops that shift signal.

E5. **Signal -- follow Decision Tree.**

Flex Sensor Frequency vs Voltage Decision Tree -- MOST IMPORTANT SECTION

FlashPro Original's analog 2 input type is the load-bearing unknown for this install. Two possible answers, two wiring paths. Verify before wiring.

How to verify:

1. Hondata KB -- search "flex fuel K20Z3 analog input" for my specific FlashPro unit.

2. **Ask my tuner.** They've wired more of these than I ever will.
3. Cross-ref ../10-Hondata-FlashPro/hardware-software-deep-reference.md (flagged **[VERIFY]** there).
4. Cross-ref acuity-fuel-rail-and-flex-fuel-build.md §4-§5. Build doc recommends MTX-D path regardless on the conservative assumption analog 2 is voltage-only. That's the safe default.

Path 1 -- FlashPro Analog 2 Reads Frequency Natively

Simpler wiring if verified:

- Continental signal wire -> directly to FlashPro analog 2 (at FlashPro unit or ECU pin A33 / ECT2 depending on unit termination).
- FPM -> Parameters -> Analog Inputs -> Analog 2: **flex fuel frequency preset**, calibrate 50 Hz = 0%, 150 Hz = 100%.
- No MTX-D inline. MTX-D stays as standalone cabin gauge if desired.

Path 2 -- FlashPro Analog 2 Is Voltage-Only (Default Plan)

Default. Build doc §4 recommends this. Documented SiriMoto approach.

- Continental signal wire -> **Innovate MTX-D input.**
- MTX-D reads 50-150 Hz, converts to 0-5V analog.
- MTX-D analog output wire -> **FlashPro analog 2** (or ECU pin A33 / ECT2).
- MTX-D powered: switched 12V + chassis ground + illumination if desired. Gauge in-cabin for live ethanol % readout.
- FPM -> Parameters -> Analog Inputs -> Analog 2: **ethanol % vs voltage** mapping. Per MTX-D docs, typical **0 V = 0% (50 Hz), 5V = 100% (150 Hz)** linear. **Verify against current Innovate MTX-D manual** -- scaling has been updated across revisions.

My default: Path 2. If tuner verifies Path 1 for my specific FlashPro Original, simplify later. Path 2 is the safe baseline.

E6. Complete signal wiring per chosen Path. **Every splice: proper crimp + heat shrink.** No electrical tape. E7. **Continuity check with multimeter before battery reconnect:**

- Power wire to 12V tap: continuous (0 ohms).
- Ground wire to chassis: continuous.
- Signal wire to destination: continuous.
- No shorts to ground on power or signal wires.

Phase F -- FlashPro Configuration

F1. Laptop + FlashPro connected, FPM open. Current calibration = my Tune #2 FBO pump (untouched). F2. FPM -> Parameters -> **Flex Fuel -> Enable**. F3. Set **sensor input source** per Decision Tree path:

- Path 1: analog 2, frequency, 50 Hz = 0% / 150 Hz = 100%, linear.
- Path 2: analog 2, voltage, 0V = 0% / 5V = 100% (or per MTX-D manual), linear.

F4. **Don't touch fuel/timing tables yet.** This flex-ready tune READS the sensor -- the tuner calibrates compensation on the dyno. F5. Save as **FBO_PUMP_FLEX_READY_20260420.fpc**. Keep original FBO pump .fpc as untouched backup. F6. Flash to ECU. Normal flash -- engine off, battery > 12.5V. F7. Verify ethanol sensor appears as monitored parameter in Parameter Monitor. May read 0 or noisy until sensor powered and flowing (next phase).

Phase G -- First-Start Verification (Before Dyno)

Three-stage leak check. Each finds leaks the previous doesn't.

G1. Reconnect battery negative terminal (~96 in-lb / 8 ft-lb). G2. Reconnect injector harness -- all 4 click.

G3. **Stage 1 -- KEY ON, PUMP PRIME, NO START:**

- Key ON (not START), hold 2 sec, pump primes, OFF. Repeat 3 cycles. Builds rail pressure.
- **5-minute visual inspection every joint:** pump hanger to tank, feed QC, every AN fitting from hardline to filter to sensor to rail, -8 ORB rail inlet, every injector upper + lower seat (watch around injector body), unused rail -8 ORB plugs.
- **Any beading/wetness/fuel smell = STOP.** Address leak. Re-prime. Re-inspect. **Do not start.**

G4. **Stage 2 -- IDLE:**

- **Only if Stage 1 is clean.** Start. Longer crank than normal is fine (system refilling after Phase A).
- Idle 5 min, warm to ~90degC. **Re-inspect during warmup** -- vibration + thermal expansion reveal leaks static pressure won't.
- FlashPro datalog. Monitor:
  - **Ethanol reading** (~0% on pump gas). If it stays 0% on a known E85 fill, wiring is wrong (see Common Mistakes).
  - **STFT / LTFT:** ±5% / ±8% first cycle.
  - **Fuel pressure** on Acuity gauge (if installed): 50-58 PSI steady.
  - **No CEL** (EVAP codes suppressed if canister deleted -- verify).
  - **No misfires.**

G5. **Stage 3 -- SHORT DRIVE:**

- **Only if Stage 2 is clean.** 20-minute drive, RPM < 4000, no WOT, mixed city/light highway.
- Return: re-inspect every joint a third time. Thermal + demand reveals slow leaks.
- Post-drive log review: fuel trims within ±10%, no knock counts, ethanol reading stable and sensible.

G6. **Optional but highly recommended sensor verification:**

- Drain tank low or burn it down on a drive.
- Fill a deliberate E30 blend: 30% E85 + 70% pump gas by pump-gallon math.
- Drive 5-10 miles to mix and feed sensor.
- FPM ethanol reading climbs from ~0-10% baseline toward ~30%.
- **No change = sensor dead, wired wrong, or MTX-D not converting.** Diagnose before Phase H.
- **Change but wrong magnitude** = 2-point calibration endpoints need adjustment (build doc §5).

G7. **No WOT on this flex-ready tune.** Compensation tables aren't loaded yet; the ECU uses the E0 fuel/timing map regardless of ethanol reading. Fine for cruise, lean under WOT if E85 in tank. Stay out of WOT until Phase H complete.

Phase H -- Book Dyno Session (Tune #3)

Most complex tune on the build. Budget $400-600, 3-4 hours. See ../10-Hondata-FlashPro/tuning-workflow-and-maps.md for session planning.

H1. **Bring:**

- Laptop + FPM, FlashPro + USB cable.
- .fpc files: STOCK backup, Tune #2 FBO pump, FBO_PUMP_FLEX_READY.
- Phase G datalogs (tuner sees the sensor is reading).
- **Fresh E85** -- 5-gallon jerrycan, verified station. Tank near-empty arriving.

H2. **Pre-session tuner verifies:** wideband idle reading sensible, ethanol sensor tracking real fuel, fuel pressure 58 PSI under load, no pending CELs.

H3. **Session flow (tuner-led):**

- Baseline pump-gas pulls -- confirm E0 endpoint = Tune #2 output.
- Drain tank, fill fresh E85 from jerrycan. Drive/cycle 5-10 miles to flush old fuel and let sensor catch up (reading climbs to 70-83% -- true E85 varies per ASTM D5798).
- **Calibrate high-side** (E85 endpoint) of ethanol % vs voltage using live flex gauge reading.
- **Build Ethanol Fuel Compensation** (~+30-40% at E85) and **Ethanol Timing Compensation** (+3 to +5 deg at E85).
- **Floors (reliability ceilings from tuning-workflow-and-maps):**
  - E85 WOT lambda floor: **0.82 -- never leaner.**
  - Pump gas WOT AFR floor: **12.0 -- never leaner.**
  - Zero knock counts at WOT, both endpoints.
  - Injector duty < 85% at peak, both endpoints.
  - Fuel pressure ±2 PSI at peak WOT.

H4. **Dual-map x E0/E85 = 4 operating corners tuned:**

- Daily + E0 -- primary daily calibration.
- Daily + E85 -- same conservative character, E85-compensated.
- Sport + E0 -- aggressive-within-envelope on pump gas.
- Sport + E85 -- max power corner, ~230-250 WHP target.
- (FlashPro Original switches via laptop flash. Daily/Sport = two .fpc files, each with E0/E85 endpoints. LattePanda install later makes switching a one-tap operation.)

H5. **Archive all calibrations** -- laptop + cloud. Filename convention:

- FLEX_DAILY_E0E85_20260520.fpc
- FLEX_SPORT_E0E85_20260520.fpc
- Dyno sheets + CSV logs filed alongside.

Post-Install Break-In

- **First 200 miles: pump gas only.** Mixed driving, no WOT. Fuel trims stabilize, thermal cycles reveal slow leaks.
- **Then ethanol gradually:** E30 -> E50 -> E85. Each fill, drive 5 miles to mix. Confirm FPM reading tracks.
- **Expect ~25-30% fuel economy drop on E85** -- normal, lower energy density. >30% drop = something wrong (leak, wrong dead times, miscalibrated sensor).
- **Re-check every AN fitting at 100 mi and 500 mi.** Thermal-cycle seep is the #1 post-install failure.
- **Oil change at 500 mi** -- catches any fuel dilution from enrichment early.

First Drive Checklist

- [ ] **No fuel leaks** at any joint (visual + no cabin/bay fuel smell).
- [ ] **Ethanol reading sensible** for current fuel.
- [ ] **Fuel pressure 58 PSI stable** at idle (Acuity gauge if installed).
- [ ] **Injector duty reasonable at cruise** (10-20% at 60 mph typical).
- [ ] **No misfires, no pending CELs, no knock counts.**
- [ ] **Smooth idle** -- rough = dead times probably off.
- [ ] **Smooth cold start** -- if rough, tuner dials in richer cold enrichment on follow-up.

Fueling Protocol

The flex fuel system handles any blend, but don't dump E85 at random:

- **Blend at the pump over multiple fills.** First switch from pump gas to E85: do E50 (half E85 over half pump gas remainder) before straight E85. Not for the car's sake -- for my confidence watching sensor track.
- **Sensor updates within 2-5 miles** of mixed driving after a fresh fill. If it doesn't move after 10 miles, something's wrong.
- **ASTM D5798 allows "E85" to range 51-83% actual.** Sensor handles this automatically. Never assume the pump matches the label.
- **Note consistent stations.** Predictable E-content = predictable power.
- **Don't let E85 sit >3 months.** Hygroscopic. Burn through tanks or drain to empty and fill pump gas for storage.

Common Mistakes

- **Reusing stock injector O-rings on new EV14s.** Looks fine dry, leaks hot and pressurized. Fresh Viton, every time.
- **Installing the flex sensor backwards** (flow arrow toward tank). Sensor may read but lag is much worse; some Continentals won't measure correctly against flow. Check the arrow.
- **Wrong sensor wiring pinout** -- ground not grounded, power on constant-12V, signal grounded instead of routed to analog input. Result: reads 0% or intermittent. Multimeter-verify BEFORE first start.
- **Skipping MTX-D when FlashPro Original analog 2 is voltage-only.** Frequency signal into voltage input = no reading, possible input damage. Default Path 2 unless verified otherwise.
- **WOT on E85 before Tune #3.** Flex-ready tune reads the sensor but has no compensation loaded. Running E85 WOT on pump-gas calibration is 30% lean -- lean knock, hot plugs, bad day. No WOT until post-dyno.
- **Depressurizing the wrong sequence** (battery disconnect before pulling the pump relay and stalling). System stays pressurized; first line disconnect sprays fuel. Phase A in order.
- **Over-torquing AN fittings.** O-rings seal at contact, not crush. Past contact + 1/4 turn damages the O-ring. Warm pressurized leak appears later.
- **Chained grounds.** Sensor ground -> MTX-D ground -> analog ground in a daisy chain causes voltage-drop-induced signal drift. Star topology always.
- **Filling tank before first prime.** Dry strainer can burn the pump on first prime. One gallon of pump gas before

key-on (Phase B16).

- **Teflon tape on AN fittings.** AN seals on flare or O-ring, not threads. Tape shreds into the fuel system.

See Also / Cross-References

- overview.md -- strategic "is flex fuel worth it on NA" analysis, realistic WHP gains, cost-vs-value.
- acuity-fuel-rail-and-flex-fuel-build.md -- **detailed parts + wiring reference.** Source of truth for part numbers, pricing, wiring detail, aesthetics. This install-guide is the procedural companion.
- ../14-Intake-Manifold/ -- bundle fuel rail swap with IM off the car.
- ../10-Hondata-FlashPro/tuning-workflow-and-maps.md -- three-tune plan; this is Tune #3. Daily/Sport rules, reliability-first AFR ceilings.
- ../10-Hondata-FlashPro/hardware-software-deep-reference.md -- FlashPro Original analog 2 input type verification (the load-bearing unknown).
- ../11-Wideband-AFR/ -- AEM X-Series required before any dyno tune.
- ../docs/torque-specs.md -- canonical specs: rail bolts 108 in-lb, pump hanger 39 in-lb, O2 sensor 33 ft-lb, banjo bolts 25 ft-lb.

Last updated: 2026-04-20

Built Motor Bottom End -- Future Phase

Status: Aspirational / Future Planning Only

What This Is (and Isn't)

This folder is **not a current purchase plan**. Nothing on this list is being bought at 170k miles with a 220-250 WHP target. The stock K20Z3 block + rods + crank + cast pistons live forever at that power level with a proper tune.

This is the reference for when I cross these thresholds:

- Power target passes ~270 WHP (stock hypereutectic piston ring lands become the risk)
- I decide to go forced induction (turbo K20 on E85, targeting 400-500 WHP)
- Sustained 9000+ RPM NA race use (not street)
- The bottom end finally announces itself after another 30-50k miles and wants rebuilding

At that point, this file is the build sheet. For now: **don't open this folder for purchasing decisions -- it's a capability plan, not a shopping list.**

Why The Stock Block Is Fine At Current Target

Community consensus (8thcivic.org, k20a.org, HondaTech): stock K20Z3 block is safe to approximately:

- **240-270 WHP NA on pump gas** with a proper tune
- **~350 WHP with moderate boost on pump gas**
- **~400 WHP with moderate boost on E85**

Stock-block failure modes (when they happen):

1. **Cast hypereutectic pistons crack ring lands** under detonation (the #1 failure -- tune-related, not block-related)
2. **Oil pump cavitation** above ~8500 RPM sustained (fixable by upgrading pump, block still fine)
3. **Piston ring gap closure** under sustained boost (ring pack issue, not block)
4. **Rod bolt stretch** under sustained boost >15 psi (rod failure, not block)
5. **Crank flex** above 9000 RPM sustained race use (niche, not street-relevant)

At 220-250 WHP NA, not one of those failure modes is on my radar. The mods that actually protect the stock block at this power level are already planned:

- Hondata FlashPro tune -- prevents detonation (the #1 piston-killer)
- ATI Super Damper -- torsional vibration control
- Cooling system refresh -- thermal stability
- Wideband AFR -- data to verify safe AFR under load

When this folder becomes real, it's because my goals changed.

Target State (If I Go Here Someday)

A bottom end that can reliably handle approximately:

- **400-500 WHP with turbo on E85** (if I ever go forced induction)
- **Sustained 9000+ RPM NA** (if I go race NA)
- **"Forever" reliability** at current or slightly higher NA power levels

Non-negotiable rule (still): reliability over power, always.

Recommended Build Sheet

This is the "if it were done today" recipe, based on current (2025-2026) market research. Prices subject to change.

Pistons -- Wiseco K573M86AP 86mm 11.6:1 ArmorGlide

Spec	Value
Part #	K573M86AP
Price	~$760
Bore	86mm stock
Compression	11.6:1
Material	Forged 2618 aluminum
Coating	ArmorGlide skirt (street durability)
Ring pack	Wiseco standard
Valve reliefs	Flat-top with reliefs

Why this compression ratio: 11.6:1 runs well on pump 93 NA (slight improvement over 11.0:1 stock). If I turbo later on E85, 11.6:1 is manageable with conservative timing (community runs K20 turbos at 10-12 psi, 350-400 WHP on 11-12:1 E85). If I go big boost (15+ psi, 500+ WHP), I'd swap to 9.6:1 pistons then -- but spending money NOW on low-comp pistons I may never use is poor planning.

Alternatives considered:

- CP-Carrillo SC7145 87mm 10.1:1 (~$885) -- future-proof for turbo but soft NA feel; also requires block overbore
- JE FSR 309411 86mm 10.0:1 (~$700) -- turbo-focused, fine alternative to Wiseco
- Manley Platinum 610100-4 11.5:1 (~$640) -- cheaper than Wiseco, quality comparable

Rods -- Brian Crower BC6044 ProH625+ H-Beam

Spec	Value
Part #	BC6044
Price	~$934
Material	4340 forged H-beam
Bolts	ARP Custom Age 625+ (280,000 PSI tensile)
Rated	~900 HP
C-to-C	139.00mm (correct K20Z3 stock length)
Pin	22mm (matches stock)

Weight	~514g

Why this rod:

- 600+ HP safe -- massive headroom above 500 WHP target
- ARP 625+ bolts handle E85 peak cylinder pressures (cylinders pressures, not beams, are usually what fails)
- Overwhelming K-series community track record at 500+ WHP
- Saving ~$560 vs K1 Technologies H-beam ($372) isn't worth it -- the bolt upgrade is cheap insurance

Budget alternative: K1 Technologies 015BW17139L ARP 2000 H-beam at ~$372. Still safe to ~600 WHP crank. Use if budget is the constraint.

Rod Bearings -- ACL Race 4B1946H-STD

Spec	Value
Part #	ACL 4B1946H-STD
Price	~$90
Series	Race Tri-metal (hard alloy)
Clearance	Standard (shoot for ~0.0018-0.0022" target)

Why: proven to 500+ WHP, standard clearance appropriate for reused crank. **King XPG** (~$150) is the "one step up" option with graphite coating -- worth it if budget allows but not necessary.

Main Bearings -- ACL Race 5M1959H-STD

Spec	Value
Part #	ACL 5M1959H-STD
Price	~$140
Target clearance	0.0015-0.0018"

ARP Head Studs -- 208-4701

Spec	Value
Part #	ARP 208-4701
Price	~$185-210
Material	8740 chrome-moly, 190,000 PSI

Same as the cylinder head phase (see ../16-Cylinder-Head/overview.md).

Note on main studs: K20 main caps are well-engineered. Main studs are NOT required at 500 WHP -- skip. Honda OEM main bolts (TTY, one-time-use) are fine.

Oil Pump -- Honda PRB-A01 (RSX Type S)

Spec	Value
Part #	Honda 15100-PRB-A01
Price	~$250-350

Why this part:

- Drop-in OEM Honda quality (no aftermarket-knockoff risk)
- Good to 9500 RPM -- covers future NA high-rev ambitions and all boost scenarios
- The K-series community's "cheap insurance" for high-RPM use -- prevents the classic cavitation-induced bearing death at 8500+ RPM

Skip: Melling high-volume (not K20-specific), Boundary (limited K20 offering), aftermarket knockoffs. 4Piston ported RSX Type S pump ($450-550) is the "better" choice but overkill below 600 WHP.

Crank -- Reuse Stock K20Z3, Prep & Balance

Honda-forged steel crank from the factory. Excellent to 600+ crank HP and 9500 RPM.

Required prep work:

- Magnaflux inspection (mandatory on 170k crank)
- Micropolish journals to standard
- Journal chamfering (teardrop oil holes)
- Full rotating-assembly balance with new pistons/rods/damper: ~$300-500

Total crank prep: ~$400-700 including balance.

Don't buy an Eagle/K1 forged billet crank unless targeting 700+ WHP -- stock is better for this use.

Harmonic Damper -- ATI 918477 Super Damper

Already owned / planned from Phase 15 (Pulleys & Harmonic Balancer). Same part carries over to built motor unchanged. Internally-balanced K20 is compatible with the street 918477 or race 918478.

*Note: P/N 917270 sometimes referenced in online discussions is actually a GM LT1/LT4 damper -- not K20. Use **918477**.*

Head Gasket -- Cometic C4561-030 .030" MLS

Spec	Value
Part #	Cometic C4561-030
Price	~$120
Thickness	.030"
Bore	86mm

.030" thickness matches stock deck height on a stock-bore 11.6:1 CR calc. If block is decked during machine work, step up to C4561-036 (.036") to restore deck clearance.

Misc Consumables

- Timing chain + tensioner + guides (if not done earlier): ~$170
- Water pump (OEM 19200-RBC-013): ~$90
- Valve cover gasket kit: ~$40
- All other cam seals, coolant, oil, Hondabond, Moly 60, assembly lube: ~$200

Machine Shop Menu

Standard prep for a 170k block going back together:

Service	Cost
Hot tank / ultrasonic clean	$75-150
Magnaflux block (crack check)	$75-150
Pressure test block (coolant passages)	$75-150
Deck resurface	$100-200
Bore + hone to 86mm piston spec (plateau finish)	$200-400
Line hone mains	$150-250
Crank polish + chamfer oil holes	$125-250
Balance full rotating assembly (crank + rods + pistons + damper)	$300-500
Total machine work	~$1,100-2,000

Plan on ~$1,200 for a competent shop doing the full menu.

Sleeving -- Only If I Go This Route

Not needed for 500 WHP on stock iron liners. Stock K20Z3 block is safe to that level with iron liners intact. Sleeves become necessary only at:

- Sustained 500+ WHP on boost
- Bore size >89mm (overbore builds)
- Already-cracked stock block

If I ever need sleeves:

- **Darton MID (P/N 400-190-P)** for 86mm-90mm bore range -- $800 parts + $800-1200 install
- **Golden Eagle** -- similar pricing, lifetime warranty on sleeves

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Assembly -- DIY vs Shop

DIY (if I've done an engine build before)

Required tools beyond current kit:

- Plastigauge (multiple clearance ranges)
- Feeler gauges 0.001-0.015"
- Dial bore gauge (ideal -- rental or buy)
- Tapered ring compressor (86mm)
- Piston ring filing tool (or buy pre-gapped rings)
- Clean assembly lube
- ARP Ultra-Torque lube (for stud torques)
- Torque wrench accurate to 100 ft-lb
- Torque angle gauge

Savings: ~$800-1,200 labor. Risk: one mistake = spun bearing / crushed ring / scored bore / ruined build.

Shop Assembly (recommended for first built K20)

Reputable K-series builders:

- **4 Piston Racing** (Indianapolis) -- industry leader, $1,000-1,500 assembly
- **King Motorsports** (Wisconsin) -- K-specialist, premium
- **Drag Cartel** -- race-focused
- **Portflow Design** (Harbor City CA) -- head+short block
- **Bisimoto** -- premium, long lead times
- **Pro Import Tuners**, **1to1 Tuning** -- mid-sized builders

Typical: $1,000-1,500 short block assembly labor, 4-8 week lead time.

Turn-Key Short Blocks (skip parts sourcing entirely)

Builder	Product	Approx Price
4Piston	K-Series Street Short Block (K20)	$4,500-5,500
Drag Cartel	DC N/A K20 Street	$4,800-5,500
Drag Cartel	DC Turbo K20 Street	$5,200-6,000
Golden Eagle	Sleeved K20A Stage 2	$5,500-6,500
SpeedFactory	OEM K20A built 900HP-capable	$6,500-7,500

Customer supplies core. Shipping core one-way ~$150-250.

Total Cost Estimate

Mid-Range DIY Build (my likely target when the time comes)

Item	Cost
Wiseco K573M86AP pistons	$760
Brian Crower BC6044 ProH625+ rods	$934
ACL Race rod + main bearings	$230
ARP head studs 208-4701	$200
Honda PRB-A01 oil pump	$300
Cometic C4561-030 head gasket	$120
Timing chain set + water pump + misc	$400
ATI Super Damper	$0 (already owned)
Crank prep (inspect/polish/balance)	$500
Parts subtotal	~$3,444
Machine work (full menu)	$1,000-1,500
DIY assembly	$0
TOTAL DIY	~$4,444-4,944

Mid-Range Shop Build

Same parts + shop assembly $1,000-1,500. **Total: ~$5,444-6,444**

Turn-Key 4Piston K-Series Street Short Block

~$4,500-5,500 (my core supplied).

High-End Build (if going boost past 500 WHP)

Upgrade	Cost Delta
CP-Carrillo SC7145 87mm + Tec-Coat	+$300
Carrillo CARR rods	+$400
King XPG bearings	+$120
ARP 625+ head studs	+$250
ARP main studs	+$175
Darton sleeves + install	+$1,800
4Piston ported oil pump	+$200
High-end total delta	+~$3,245
Grand total high-end	~$7,700-9,200

Common Mistakes in K-Series Builds

1. **Using cast pistons** -- never acceptable. Always forged 2618.
2. **Mixing rod bolts on OEM rods without resize** -- ARP bolts have different clamp load than OEM; skipping resize = spun bearing.
3. **Skipping full rotating-assembly balance** -- benefit becomes obvious above 7500 RPM.
4. **Wrong bearing clearance** -- too tight = seize, too loose = low oil pressure, knock.
5. **Reusing TTY head bolts instead of ARP studs** -- always upgrade to ARP studs on built motor.
6. **Skipping block pressure test** -- 170k block may have hairline cracks in water jackets that only show under test.
7. **Cheap oil pump or aftermarket knockoff** -- stick to genuine Honda PRB-A01 or verified 4Piston ported.
8. **Wrong compression ratio for fuel** -- 12:1+ on pump gas = detonation = destroyed pistons. Plan fuel before choosing CR.
9. **Skipping the ATI damper** -- stock damper's bonded rubber ring fails on high-RPM use.
10. **No tune or wrong tune after assembly** -- the cheapest build dies fastest without proper tuning.

Parts Interactions With Current Build

Everything I'm already planning carries forward to built motor phase:

- **ATI Super Damper** (Phase 15) -- same part, no change
- **Hondata FlashPro** (Phase 10) -- required for built motor tune, custom tables per build CR

- **Ignition refresh** (Phase 6) -- spark energy scales well; revisit plug gap with boost later
- **Wideband AFR** (Phase 11) -- critical sensor for tuning built motor
- **Cooling system** (Phase 9) -- 500 WHP requires the upgraded cooling
- **Flex fuel** (Phase 19) -- required for E85 on built motor

New purchases needed if/when built motor happens:

- Clutch + flywheel re-evaluation -- the current Exedy Stage 1 / LW flywheel is rated to ~280-300 WHP / ~200-210 lb-ft. At 500 WHP, needs Stage 2 or Competition Clutch equivalent.
- Injectors upgrade -- EV14 550cc (current plan) handles 300 WHP NA E85 max. At 500 WHP on E85, need 1000cc+ (ID1050x or similar).
- Fuel pump upgrade -- DW300C (current plan) adequate to 400 WHP. At 500 WHP on E85, DW400 or DW450 required.
- **Turbo kit** (if forced induction route) -- entirely separate ~$3,000-6,000 purchase
- **Intercooler + piping** (if boost) -- ~$600-1,500

Cross-References

- 16-Cylinder-Head/overview.md -- head side of the puzzle; drop-in head is phase before this
- 05-Timing-Chain-Maintenance/overview.md -- chain + VTC done during any major teardown
- 15-Pulleys-and-Harmonic-Balancer/overview.md -- ATI damper transfers to built motor
- 09-Cooling-System/overview.md -- upgraded cooling is prerequisite for built motor power
- 19-Flex-Fuel-and-Fuel-System/overview.md -- fuel system needs re-evaluation at built-motor power
- docs/torque-specs.md -- ARP torque sequences, bearing clearances
- docs/maintenance-parts-catalog.md -- add built-motor part numbers when the time comes

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Sources

Pistons

- Wiseco K20 86mm Professional Series -- Summit
- Wiseco K573M86AP 86mm 11.6:1 ArmorGlide -- K Series Parts
- CP SC7145 K20 87mm 10.1:1 -- Real Street
- JE 86mm FSR K-Series 10.0:1 -- K Series Parts
- Manley 610100-4 K20 86mm 11.5:1 -- CNC Motorsports

Rods

- Brian Crower BC6044 K20 625+ -- K Series Parts
- BC K20/K20Z rods line
- K1 Technologies K20 139mm H-Beam Kit
- Manley Turbo Tuff 14404-4 K20 -- K Series Parts
- Eagle K20A2/Z H-Beam CRS5470K3D -- K Series Parts

Bearings

- ACL Race Bearing Set K20 -- IPGparts

- ACL 5M1959H-STD Mains -- K Series Parts
- King XP K-series -- King Engine Builders

Oil pump

- 4Piston Ported Type S Oil Pump
- 4Piston Ported RRC Type R Oil Pump

Other

- ARP 208-4701 K-Series Head Studs
- Cometic C4561-030 K20Z3 .030" MLS
- ATI K20 Super Damper 918477 -- Hybrid Racing
- Darton MID Sleeves K20 86-90mm
- Golden Eagle K20A Sleeved Short Block
- 4Piston K-Series Street Short Block (turn-key)

Forum research

- k20a.org -- stock block HP limit thread
- 8thcivic -- K20Z3 handling boost
- 8thcivic -- 9000 RPM oil pump cavitation
- k20a.org -- rod bearing clearance
- k20a.org -- compression for turbo pump vs E85

Last updated: 2026-04-20

Built Motor Bottom End Install Guide -- Future-Phase Reference

Status: FUTURE PHASE -- NOT Current Purchase Plan

Do NOT build at the current 240 WHP target. The stock K20Z3 block is overbuilt for this power level. This document exists as a capability reference for some future version of me who has decided the power ceiling has changed. Nothing in this guide should be purchased, scheduled, or planned for the current build state.

If you're reading this trying to decide whether to do this work: the answer is almost certainly no. Keep reading for the logic.

Reality Check -- Why This Doc Exists

This is a reference in case one of the following happens:

1. **I decide to chase power past 270 WHP NA** -- stock cast hypereutectic piston ring lands become the risk at that level, and forged pistons become a real requirement, not a nice-to-have.
2. **I decide to go forced induction** -- targeting 400+ WHP on E85 boost requires a bottom end designed for peak cylinder pressures the stock rod bolts were not spec'd for.
3. **The bottom end announces itself** -- rod knock, sustained oil-pressure drop, spun bearing, piston slap that won't quiet with warm oil. At 170k+ miles this is a non-zero risk even without power chasing.
4. **Oil consumption goes past 1 quart / 1000 miles** -- but investigate valve seals and PCV first; rings are usually the last suspect, not the first.

At the current 240 WHP pump-gas target none of the above is on the radar. The stock K20Z3 block has a documented community safety window of roughly **240-270 WHP NA pump gas and ~400 WHP boost on E85** with a proper tune and the supporting mods I already have planned (ATI damper, Hondata tune, cooling refresh, wideband AFR). I don't need to open this engine.

What a built motor actually is

Before committing, I need to understand that "build the bottom end" is not an install -- it is an **engine build**. Specifically:

- **The engine comes OUT of the car.** Not the head off. The entire engine on a stand. That's a 1-2 day job by itself.
- **The short block comes apart on a stand.** Another full day of measured, documented disassembly with inspection at every step.
- **Machine shop work is mandatory.** Nobody reassembles a 170k-mile block without a shop honing under a torque plate, checking deck flatness, Magnafluxing the crank, and balancing the new rotating assembly. Mandatory is not an exaggeration. Skipping this step is how people make $4,500 worth of bearings dust inside 1,000 miles.
- **Reassembly with plastigage clearance check on every bearing.** This is the core skill difference between "swapped parts" and "built an engine." Plastigage every rod journal, plastigage every main journal. Every time.
- **Realistic timeline:** 2 weekends of DIY wrenching bookending 1-2 weeks of machine shop turnaround = **3-4**

weeks the car is out of service. Plan transportation for that month before taking anything apart.

Cost ballpark

Path	Cost
DIY with machine shop support (my likely path if I go here)	~$4,444-4,944 parts + ~$800-1,200 machine work
Shop-assembled short block (parts I supply, labor billed)	~$5,444-6,444
Turn-key 4Piston K-Series Street Short Block (drop in, bolt up)	~$4,500-5,500 all in

The turn-key path is genuinely competitive with DIY once I account for the machine shop bill, the specialty tools I don't own (torque plate, rod bolt stretch gauge, dial bore gauge), and the risk of a first-time engine build. Covered in detail under *Turn-Key Alternative* below.

Decision Tree -- WHEN to Build

This is the flowchart that gets checked against reality before any parts are ordered.

Current state	Decision
240 WHP target, block healthy, no knock	DO NOT BUILD. Stock block is overbuilt for this power level.
Power target moves to 270+ WHP NA	Build. Forged pistons and upgraded rod bolts earn their cost here.
Decision made to go forced induction at 400+ WHP on E85	Build. Peak cylinder pressures on boost put the OEM rod bolts out of safe margin.
Rod knock diagnosed (audible from bottom end, oil pressure normal at idle then drops under load)	Build. Or yank and swap short block -- repair labor is the same order of magnitude as a build.
Oil consumption >1 qt / 1000 mi	Investigate first. PCV system, valve stem seals, turbo seals (if applicable). Only build if a leakdown test confirms rings, not some other simpler failure.
Sustained 9000+ RPM NA race use planned	Build. Oil pump cavitation and crank flex become real at that RPM band.

At 240 WHP with a healthy block, this entire folder stays closed. The money belongs to other phases.

Prerequisites and Cross-References

Before this work starts, these things are either already done or on the same schedule:

- **Cylinder head is either already done or bundled into this job.** See ../16-Cylinder-Head/overview.md. It's inefficient to do the head once, then pull the whole engine a second time for the bottom end. Ideally the head is already installed on the fresh short block before the engine goes back in the car. If I already did the head in-car as Phase 16, this built-bottom phase pulls everything back apart.
- **Engine out of the car.** Much bigger job than the head-off-in-car approach documented in the head install guide. The transverse K20 can lift either with or without the transmission attached; I'd take it out with the trans for clearance to drop onto a stand cleanly.
- **Machine shop identified and vetted** -- specifically for K20 work. Not every engine shop does K-series builds correctly. 4 Piston (Indianapolis), Portflow (Harbor City CA), King Motorsports (Wisconsin), Drag Cartel all qualify. A local generic machine shop may or may not -- verify they've done K20s before I hand them a $900 crank.

- **Torque plate available for hone.** This is non-negotiable and the shop should own it. Honing the block without a torque plate simulates unclamped geometry; the bore distorts when the head is bolted back down and the rings don't seal. If a shop says they don't use a torque plate for K20 honing, walk out.
- **Plastigage, telescoping gauge, micrometer, feeler gauges on hand.** These are consumable/inexpensive and live in my toolbox after this.
- **Engine assembly lube** -- Royal Purple Max-Tuff or equivalent. Not standard engine oil. Assembly lube stays in place for first fire; oil will drain off bearings and result in a dry start.
- **Clean assembly area.** Not the garage floor. Not next to the grinder. A cleaned workbench with a lint-free surface, no cardboard shedding, no dust, no debris. Bearing failure at 1,000 miles is almost always a cleanliness failure from assembly, not a parts failure.

Tools Required

This is a materially different tool list from anything else in this build. The engine-out, short-block-apart scope needs specialty hardware I don't own.

Critical specialty tools

Tool	Why	Approximate Cost
Engine crane / cherry picker	To lift the engine out. Rentable at AutoZone / O'Reilly for free with deposit.	Rental $0, purchase $150-250
Engine stand (750+ lb rated)	Block mounts to stand for disassembly and assembly. Own this.	$80-150
Tapered ring compressor, 86mm	Compresses piston rings for insertion into bore. Must be 86mm specific.	$35-60
Torque plate for hone	Shop owns this. If they don't, find a different shop.	N/A (shop tool)
Rod bolt stretch gauge	BC ProH625+ rods are torqued to stretch, not torque value. Stretch gauge measures bolt elongation during torque.	$150-250
Plastigage assortment	Bearing clearance measurement. Multiple ranges.	$15
Telescoping gauge set + 2-3" micrometer	Measure bore and big-end ID. Needed for clearance verification.	$80-120
Feeler gauge set, 0.001"-0.025"	Ring gap measurement.	$10
Engine assembly lube (Royal Purple Max-Tuff or equivalent)	Coats bearings at assembly.	$15
Tap set, chase block threads	Main bolt holes, head bolt holes, oil pan threads -- all get chased before assembly.	$40-80
Piston ring filer	If Wiseco rings need gap adjustment.	$40-90
Dial bore gauge (ideal, rentable)	Precise bore measurement. Shop may do this step anyway.	Rental or $150-300
Bore gauge / snap gauge for big-end	If assembling rods myself.	Included with telescoping set
Dial indicator + magnetic base	Deck height, crank endplay, runout checks.	$40-80

Optional but useful

- **ARP 208-4701 head stud kit** -- if going this far, the head studs are getting upgraded if they aren't already from the head phase.

- **ARP main stud kit** -- **skip for NA or moderate boost.** OEM Honda TTY main bolts are documented safe to ~500 WHP. Main studs only become necessary for sustained 500+ WHP on boost or race use. Not part of this build tier.
- **Good droplight, clean rags by the box, latex gloves.**

Tools I already have that carry over

Everything from the head install -- torque wrenches (in-lb and ft-lb), standard metric sockets, the Lisle 37950 cam alignment tool, in-lb torque wrench for small fasteners. The engine build adds to the collection, not replaces it.

Parts List (From overview.md)

Full detail lives in overview.md. Short-form for this install guide:

Category	Part #	Price	Notes
Pistons	Wiseco K573M86AP 86mm 11.6:1 ArmorGlide	~$760	Stock bore. Forged 2618.
Rods	Brian Crower BC6044 ProH625+ H-beam	~$934	ARP 625+ bolts included, stretch-torqued.
Rod bearings	ACL 4B1946H-STD Race tri-metal	~$90	Standard clearance; verify with plastigage.
Main bearings	ACL 5M1959H-STD Race tri-metal	~$140	Standard clearance; verify with plastigage.
Oil pump	Honda 15100-PRB-A01 RSX Type S pump	~$300	OEM Honda upgrade. Rated to 9500 RPM.
Head gasket	Cometic C4561-030 .030" MLS	~$120	Stock-bore 86mm. Confirm deck height first.
Head studs	ARP 208-4701	~$200	Carried over from head phase if already installed.
Crank	Reuse stock, micropolish + balance + Magnaflux	~$500 shop labor	Do not skip Magnaflux on a 170k crank.
Harmonic damper	ATI 918477 Super Damper	$0 (already owned from Phase 15)	Internally balanced K20 compatible.
Complete gasket kit + seals	Miscellaneous OEM	~$150	Cam seals, VCT seal, valve cover, oil pan, water pump, etc.
Timing chain kit	14401-PNA-004 + guides + tensioner	~$170	Fresh chain on fresh bottom end. Non-negotiable.

Parts subtotal ~$3,364. Machine shop ~$800-1,200. Grand total DIY ~$4,444-4,944.

Alternative: see Turn-Key section at end of this guide.

Alternative: Turn-Key 4Piston Short Block

Before committing to DIY assembly: the 4Piston K-Series Street Short Block ships fully assembled for **$4,500-5,500 delivered** -- Wiseco/CP forged pistons, H-beam rods with upgraded bolts, race bearings, Honda RSX Type S pump, honed under torque plate, balanced assembly, plastigaged and ship-tested.

Factor	DIY	4Piston Turn-Key
Cost	$4,444-4,944	$4,500-5,500
Time	3-4 weeks out of service + my labor	Ships ready; install is engine R&R only
Risk of failure in first 1,000 mi	Non-zero (first-time builder)	4Piston owns warranty
Specialty tools needed	Stretch gauge, bore gauge, assembly tools	None additional
Skill acquired	Engine-building experience	None

Honest recommendation: DIY only if I want the experience and have the time. If the goal is "have a built motor," turn-key is equivalent quality at similar money and dramatically lower risk. DIY for learning, turn-key for pure outcome.

Step-by-Step -- The Phases

This is a big enough job that it's broken into phases. Each phase is a multi-hour or multi-day chunk. Don't try to chain them -- the mental fatigue of 12 straight hours on an engine build is the enemy of bearing clearance precision.

Phase A: Engine Removal (1-2 days)

The biggest single mechanical step and easily outsourced to a shop ($400-800 to R&R an engine). DIY summary:

1. Disconnect battery negative. Drain coolant, engine oil, transmission oil.
2. Remove intake, exhaust downpipe, any mid-pipe blocking engine lift.
3. Remove all accessories: alternator, A/C compressor (refrigerant must be discharged at a shop first -- don't release it yourself), starter, throttle body, intake manifold, valve cover.
4. Disconnect all engine harness connectors and tag them. Photos of everything.
5. Disconnect fuel line at rail, shift cables at trans, clutch hydraulic line at slave.
6. Remove axles from hubs. Remove all engine and transmission mounts.
7. With hoist on factory lift points, lift slowly while rocking the engine/trans out. Drop front subframe if needed for clearance.
8. Split the trans from the engine on the ground -- lighter lifts than pulling the assembly as a unit.
9. Mount bare engine to stand via four-bolt plate.

Realistic time: 1 day if cooperative, 2 days with seized bolts. At 170k miles, assume everything is seized.

Phase B: Short Block Teardown (1 day)

With the engine on the stand (head already off -- see head install guide):

1. Remove timing chain cover and chain (see ../05-Timing-Chain-Maintenance/overview.md). Chain replaced on reassembly -- not reused.
2. Invert block on stand. Remove oil pan, pickup tube, windage tray. Stock oil pump is saved for comparison measurement only; new **Honda 15100-PRB-A01 RSX Type S pump** goes in on reassembly.
3. **Plastigage the OLD rod bearings before removing them** -- baseline data. Was the block in spec before teardown? A journal measuring 0.004" clearance pre-teardown means crank or bearings were already past spec; good data for the machine shop.

4. Remove rod bearing caps one at a time. Bag each cap with its matching rod -- caps are matched to rods from the factory, never swap.
5. Push each piston + rod UP through the top of the block. K20 is designed top-out, not bottom-out; rods won't clear crank counterweights if driven down.
6. Plastigage the OLD main bearings -- same baseline exercise, all 5 mains.
7. Remove main caps (numbered -- photograph anyway). Lift crank out onto clean padded surface, journals contacting nothing.
8. Block is stripped. Bag all hardware, label everything, photograph the block from all sides.
9. **Send to machine shop** with the full service menu: hone to Wiseco 86mm spec under torque plate, deck check/resurface if needed, line hone mains, hot tank clean, Magnaflux block, pressure test coolant passages. Crank: micropolish, chamfer oil holes, full rotating-assembly balance with new rods + pistons + damper, Magnaflux for cracks.

Phase C: Machine Shop Work (1-2 weeks, shop handles)

Shop checklist (pricing in overview.md Machine Shop Menu):

- **Block:** hot tank clean, Magnaflux, pressure test coolant passages, deck flatness check (resurface if warped -- may need thicker gasket to restore deck height), hone to Wiseco spec (typically 400-450 grit plateau cross-hatch), line hone mains.
- **Crank:** polish journals to <=0.0002-0.0005" taper, chamfer oil holes, Magnaflux, full balance with new pistons + rods + ATI damper.
- **Rods (BC):** shop weighs the pair, verifies big-end matches crank journal + bearing combo.
- **Pistons (Wiseco):** inspect for shipping damage, measure ring gaps against actual bore.

If bores are worn beyond hone spec: machine shop quotes an overbore. K573M86AP is stock-bore 86mm -- any overbore needs a different piston P/N (86.5mm or 87mm) and different Cometic gasket bore. Full re-quote event; verify with Wiseco that matched pistons exist at the overbore size before signing off.

Phase D: Pre-Assembly Prep (full day of careful work)

Before a single bolt goes back in, everything gets cleaned and measured.

1. **Thread-chase every bolt hole in the block** -- main, head, oil pan, accessory brackets. Debris in a bolt hole gives false torque reading and bottoms bolts before proper clamp load.
2. **Brush all oil passages** with a long bottle brush. Air-blow clean. Brush again. Grit trapped in a main oil feed comes out at 3,000 RPM and starves a bearing.
3. **Wipe cylinder bores** with ATF on a lint-free cloth until cloth comes out clean white. Twice. Hone residue is invisible and scores piston skirts inside 100 miles.
4. **Crank prep:** journals wiped with assembly lube, verified round to <=0.0002", taper <=0.0005".
5. **Main bearing clearance:** install ACL main halves dry, torque caps to Honda (or ARP) spec, plastigage each journal. **Target: 0.0015-0.0018"** per ACL spec sheet. Out of spec = stop, diagnose with shop. Never "install and see."
6. **Rod bearing clearance:** assemble each BC rod with ACL 4B1946H-STD dry, torque rod bolts to BC stretch spec (~0.005-0.006" -- verify with BC instruction sheet), plastigage each journal. **Target: 0.0015-0.0025"** per ACL spec sheet.
7. **Ring gap verification in the actual honed bore:** push ring square with a piston, measure top ring gap. **Tar**

get 0.018-0.022" for NA pump gas; wider (0.024-0.028") for E85/boost heat expansion. Follow Wiseco sheet, not rules of thumb. File with ring filer 1-2 strokes at a time if under spec -- never over-gap (under-gap is fixable, over-gap = new rings).

8. **Piston-to-wall clearance:** measure with micrometer + dial bore gauge. **Target 0.0025-0.0035" for Wiseco ArmorGlide** per Wiseco spec sheet.
9. **Deck height:** assemble one piston + rod on crank, measure at TDC. Zero or slight positive deck normal; excessive negative deck kills quench.

Document everything on a build sheet. If something comes apart later, the build sheet is the diagnostic baseline.

Phase E: Short Block Assembly (1-2 days careful work)

Clean bench. Latex gloves. Everything lubed. Don't rush.

1. **Main bearing upper halves into block.** Bearing backs DRY (oil between shell and block blocks heat transfer). Assembly lube on the face that contacts the crank.
2. **Lower crank into mains.** Two-person lift -- crank is heavy, bearings don't appreciate being dropped on.
3. **Main bearing lower halves into caps** with assembly lube on the face. Caps numbered and directional -- arrows to front of block.
4. **Torque main bolts per Honda spec** (or ARP if using main studs). Multi-step torque-plus-angle on TTY main bolts. Follow FSM exactly -- see ../docs/torque-specs.md.
5. **Rotate crank by hand** -- smooth with moderate resistance. Any bind = STOP, remove caps, diagnose.
6. **Plastigage one main journal** post-final-torque as sanity check.
7. **Install piston rings** with gap orientations 120deg apart -- never aligned (creates blow-by path).
8. **Rod bearings into big-ends** with assembly lube on face, tang seated in rod notch.
9. **Piston + rod into bore through top of block** with ring compressor. Rod cap OFF. Lower with gentle thumb pressure -- NEVER force. If it doesn't slide, stop: check ring orientation in compressor, check skirt coating isn't catching a bore edge.
10. Guide rod onto crank journal. Install cap + lower bearing half + assembly lube. Factory tang orientation.
11. **Torque rod bolts by STRETCH.** BC ProH625+ specs stretch, not torque. Initial torque to BC's starting value (~40 ft-lb typical), measure bolt length with stretch gauge, torque more, re-measure, until stretch = BC target (~0.005-0.006"). Verify with BC instruction sheet.
12. **Rotate crank after each rod.** Resistance increases slightly per rod, never BINDS. Binding = stop, diagnose -- one tight bearing tang = seized engine at first start.
13. Repeat 7-12 for all 4 pistons.
14. **Oil pump (new Honda 15100-PRB-A01)** torqued to Honda spec. New oil pump chain. New pickup, new gasket.
15. **Windage tray** (reuse stock unless damaged).
16. **Oil pan with fresh gasket.** Small Hondabond bead at case-half corners -- too much clogs the pan.

Short block assembled. Photograph before inverting for head install.

Phase F: Head Install

At this point, follow ../16-Cylinder-Head/install-guide.md (when written) or the equivalent procedure documented in head/overview.md. Whether the head is the 4Piston Pro 156v2 drop-in from Phase 16 or a refreshed stock head, the install procedure on a fresh short block is the same as on an in-car head swap. One difference: cam timing is

set while the engine is still on the stand (easier than in-car), and I can verify cam timing by turning the crank through multiple rotations and confirming no valve-to-piston contact.

Timing chain goes on fresh per Phase 5 procedures.

Phase G: Engine Back In The Car (~1 day)

Reverse of Phase A. Mount engine to transmission (or install alone and mate trans after), lift into bay with crane, drop onto mounts, connect axles, reconnect all harnesses, refill all fluids.

Check oil level TWICE before first crank. Prime the oil system before first start -- see Phase H.

Phase H: First-Start Break-In (CRITICAL)

A fresh engine lives or dies in the first 30 minutes. Do not rush this.

1. **Prime the oil system before first crank.** Pull coil relay so engine can't fire. Crank starter in 10-second bursts (30-second cool breaks) until oil pressure gauge reads positive -- typically 20-30 seconds total cranking. This sends real oil through bearings (assembly lube has been holding them over).
2. **Non-synthetic break-in oil.** Joe Gibbs Driven BR30 or Brad Penn. Synthetic is too slippery for ring-to-bore seat-in. Break-in oil for first 500 miles minimum.
3. **First start.** Plug coils back in. Should fire within 3 seconds. If it cranks past 5 seconds: STOP, re-verify cam timing, fuel, spark.
4. **Hold 2000-2500 RPM for 20 minutes immediately on first fire.** No idle, no rev -- steady RPM. Cam break-in window (same procedure as fresh head in ../16-Cylinder-Head/). Watch coolant rising normal, oil pressure holding, no unusual sounds. Light smoke from assembly-lube burn-off on headers is normal; thick blue-white smoke past 5 minutes is NOT.
5. **Shut off at 20 minutes, cool to room temp, drain break-in oil, refill with fresh break-in oil.** First 20 minutes captures assembly debris and ring-seat metal fines -- throw it away.
6. **Drive 500 miles on break-in oil with VARIED load.** Not steady cruise -- rings need load variation to seat. Accelerate moderately to 4000 RPM, back off, engine brake. No WOT, no sustained high RPM.
7. **Change oil at 500 miles.** Cut filter, inspect. Sparkles normal (bearing break-in); chunks demand investigation.
8. **Switch to full synthetic at 500 miles.** Normal oil cycle resumes.
9. **No WOT until 1000 miles.** Ring seat completion window.
10. **At 1000 miles:** compression + leak-down test all 4 cylinders. Record numbers -- baseline for every future comparison. If rings not seated (compression low, leakdown high), extend break-in 500 more miles and retest.

Post-Build Dyno Retune (MANDATORY)

A fresh engine has materially different operating characteristics than the engine that was on the dyno before teardown. Do not drive on the old calibration past break-in. Book a dyno session immediately after 1000-mile break-in.

What changes:

- Volumetric efficiency curve is different (fresh bore seal, different compression ratio if built CR differs from stock)

- Ignition timing sensitivity -- built motor with forged pistons has more tolerance for advance but also more consequence for detonation
- Fuel requirements shift with any compression change
- New oil pump pressure curve affects VTC response

Budget **$500-800 for a proper dyno retune** -- same as the head-install retune budget in ../10-Hondata-FlashPro/tuning-workflow-and-maps.md. This is the non-negotiable finish line on the build cost.

First Drive Checklist

Before driving beyond the immediate neighborhood after break-in procedures:

- [ ] Oil pressure normal at idle (20+ PSI hot), rises to 45-60 PSI at 3000 RPM
- [ ] Coolant temp stabilizes at thermostat setpoint, no runaway
- [ ] No unusual knocks, taps, or ticks at any RPM (light top-end tick from VTA is normal; bottom-end knock is NOT)
- [ ] No smoke at any throttle position past the first 2 minutes of running
- [ ] No coolant loss over first 30 miles
- [ ] No oil loss over first 100 miles (some consumption during ring seat is normal; leaking is not)
- [ ] No CEL, no pending codes on OBD scan
- [ ] Fuel trims within normal range (±5% LTFT)

Anything that fails this checklist gets diagnosed BEFORE the next drive, not deferred.

Common Mistakes in K-Series Builds

Bearing failure at 1,000 miles is almost always one of these. Every one is preventable.

1. **Assembling without plastigage verification.** "It felt right" is not a measurement. Every bearing gets plastigaged.
2. **Reusing rod bolts.** BC ProH625+ bolts are NEW ONLY. Stretch-torquing yields the bolt; reusing a yielded bolt is how rod caps exit the block.
3. **Wrong ring gap.** Under-gap closes at operating temp and breaks rings; over-gap leaks blow-by. Follow Wiseco sheet, measure in the honed bore, file conservatively.
4. **Skipping torque plate during honing.** Unclamped bore geometry differs from head-torqued geometry -- no torque plate = non-round bores at operating condition = rings don't seal.
5. **WOT before break-in complete.** Rings don't fully seat until 1000 miles of varied load. WOT at 200 miles glazes the bore permanently.
6. **Dirty assembly area.** Dust in an oil passage becomes bearing dust in 1000 miles. Clean bench, lint-free rags, latex gloves, no cardboard.
7. **Skipping machine shop inspection.** A 170k block can have water-jacket cracks, deck warp, bore taper past hone spec. Pressure test, Magnaflux, deck check -- all of it.
8. **Wrong torque spec for hardware.** Hardware spec overrides block spec. ARP follows ARP torque-with-lube; BC rods follow BC stretch. Always the HARDWARE'S sheet.
9. **Pistons in wrong orientation.** Wiseco marks front-of-engine direction. Flipped 180deg = valve reliefs

misaligned = bent valves on first crank.

10. **Skipping oil prime before first start.** Assembly lube covers seconds, not 5+ seconds of firing without pressure. Crank with coils disconnected until gauge reads.
11. **Wrong oil for break-in.** Synthetic too slippery for rings to seat. Break-in oil first 500 miles, no exceptions.
12. **Swapped rod caps or main caps.** Caps are matched from the factory -- bag everything at teardown.

See Also and Cross-References

- overview.md -- the parts I'd pick and the rationale, with full pricing and alternatives
- ../16-Cylinder-Head/overview.md -- head side of the puzzle; ideally done before or bundled with this phase
- ../05-Timing-Chain-Maintenance/overview.md -- timing chain install during reassembly, same procedure on fresh block
- ../15-Pulleys-and-Harmonic-Balancer/overview.md -- ATI Super Damper already owned, transfers unchanged to built motor
- ../10-Hondata-FlashPro/tuning-workflow-and-maps.md -- mandatory post-build dyno retune
- ../11-Wideband-AFR/overview.md -- wideband critical for safe break-in AFR monitoring
- ../09-Cooling-System/overview.md -- upgraded cooling is prerequisite for built motor power level
- ../19-Flex-Fuel-and-Fuel-System/overview.md -- fuel system re-evaluation at built-motor power level
- ../docs/torque-specs.md -- main bolt, rod bolt, ARP, cam cap, head stud torque values
- ../docs/maintenance-parts-catalog.md -- add built-motor part numbers when the phase goes live

Last updated: 2026-04-20

Purple Engine Bay Cosmetics

Status

- **Not yet purchased.** Aesthetic layer -- add in phases alongside mechanical work.
- **Theme anchor:** Acuity purple oil cap + Acuity ESCO-T6 purple shift knob (complements the Acuity 3-way adjustable shifter already owned).

Scope and Rule

Engine bay goes purple. Every part here is **optional** -- nothing affects function. The reliability-first rule still applies: if a purple variant of a part doesn't exist from a reputable brand, I either powder-coat an OEM/existing part or skip it. I'm not buying no-name Amazon "universal purple" caps that might fail seals or adhesive.

Reputable brand list for this project: Acuity, K-Tuned, Hybrid Racing, Mishimoto, Samco Sport, Dress Up Bolts, Skunk2, Password:JDM, NonStopTuning (NST), Koyorad, BC Racing, Radium Engineering.

Summary Purchase Table

Tier 1 -- Critical purple theme anchors (off-the-shelf purple, reputable brands)

Item	Brand / Part #	Price (est. incl. tax+ship)
Oil filler cap	Acuity 1927-PPL (Purple Podium)	~$80
Oil dipstick (K-series)	K-Tuned DP2-K20-PRP	~$72
Dipstick retention spring kit	K-Tuned DP2-SPR-KIT	~$25
Shift knob	Acuity ESCO-T6 Satin Purple 1886-T6P	~$92
Engine-bay titanium bolt kit (purple)	Dress Up Bolts HON-027-TI (FG/FA 06-11)	~$125
Valve cover low-profile hardware (purple)	Skunk2 649-05-0120-PPL family	~$55
Accessory pulley kit (alt + idler -- NOT crank)	NST22015K-Purple (crank piece deleted -- see caveat)	~$180
Fender washer kit (purple)	Password:JDM Purple 6-pack	~$22
Tier 1 Subtotal		~$651

Tier 2 -- Cooling-system purple (covered in 09-Cooling-System/ but lives under this theme)

Item	Brand / Part #	Price
Silicone hose kit (custom purple run)	Samco Sport Civic SI 06-11 purple	~$320
Coolant reservoir	Radium 20-0270-00 w/ custom purple anodize	~$310
T-bolt clamps (stainless -- no purple version from reputable brand)	Hybrid Racing HYB-TBC-00-10	~$45
Radiator cap (Koyorad SK-C13 powder-coated purple)	Koyorad SK-C13 + local powder coat	~$55
Royal Purple Purple Ice additive	01600 (cooling conditioner, mild tint)	~$12
Tier 2 Subtotal (cooling purple)		~$742

Tier 3 -- Custom work (powder-coating / paint on OEM or existing parts)

Item	Approach	Price
Megan Racing front strut bar (already owned, polished)	Powder-coat candy purple before install	$75-120
Stock K20Z3 valve cover	Powder-coat candy/illusion purple (mask mating surfaces correctly)	$150-275
Brake master cylinder cap (46662-S9A-003)	Buy spare OEM, powder-coat purple	$25-40
Power steering reservoir cap (53697-SB3-952)	Buy spare OEM, powder-coat or Plasti-Dip	$15-30
Washer fluid cap	Buy spare OEM, powder-coat or Plasti-Dip	$10-25
Caliper paint (G2 purple epoxy, 1 kit covers all 4)	See 04-Brakes/purple-calipers.md	$55-75
Tier 3 Subtotal (batch at one coater)		~$330-565

Tier 4 -- Nice-to-have accents

Item	Brand / Part #	Price
JDM Sport / AJP low-profile VC washers (purple, secondary set)	AJP K20 Purple	~$25
Purple silicone intake couplers (universal 4-ply, Pegasus/etc.)	Generic 4-ply silicone, 2.75" ID	~$25
Dress Up Bolts fender bolt kit (purple)	DUB Civic 06-11 purple fender kit	~$60
Purple AN fittings (if going -6 AN on fuel/cooling plumbing)	Fragola / Earl's purple anodize	varies
Tier 4 Subtotal		~$110+

Grand Total -- "Full Purple" Package

- Tier 1 only (core theme): ~$651
- Tier 1 + Tier 3 custom (no cooling): ~$981-1,216
- Tier 1 + Tier 2 cooling + Tier 3 custom: ~$1,723-1,958
- Tier 1 + 2 + 3 + 4 (everything): ~$1,833-2,068

Confirmed Purple Parts (detail)

Oil filler cap -- Acuity 1927-PPL Purple Podium

- **Part number:** 1927-PPL
- **Price:** MSRP $76, street $69-80 after shipping
- **Material:** 6061-T6 billet, satin purple anodized
- **Fitment:** Fits all Honda/Acura B, D, F, H, I, J, K, L, R series -- including K20Z3
- **Install:** Hand-tighten only -- o-ring seals on the dovetail groove
- **Source:** acuity-instruments.com/products/purple-podium-oil-cap-for-honda-engines, KSeriesParts, Evasive Motorsports
- **Alternate:** K-Tuned Billet Oil Cap Purple KTD-OLC-PRP -- $45, 7-moon-cut grip, same fitment

Dipstick -- K-Tuned DP2-K20-PRP

- **Part number:** DP2-K20-PRP
- **Price:** $60-72 with shipping

- **Material:** 6061 billet handle (purple anodized), stainless base/wire/level blade, dual o-ring seal
- **Fitment:** K20, K20A, K20A2, K20Z1, K20Z3 -- confirmed for FG2
- **Recommended add-on:** DP2-SPR-KIT spring retention kit ($25) -- prevents pushout after headers install create more crankcase pressure
- **Source:** heeltoeauto.com, kseriesparts.com

Shift knob -- Acuity ESCO-T6 Purple 1886-T6P

- **Part number:** 1886-T6P
- **Price:** MSRP $84, street $75-92 delivered
- **Material:** 6061-T6, 40mm x 70mm barrel, 185g, laser-etched 6-speed pattern
- **Threads:** M10x1.5 (standard Honda, fits Acuity 3-way adjustable shifter)
- **Alternate:** Acuity POCO Low-Profile 1925-PP ($95), or Billetworkz Candy Purple Weighted (500g, $80)
- **Source:** acuityinstruments.com/products/esco-t6-shift-knob-in-satin-purple-anodized-finish

Engine bay bolt kit -- Dress Up Bolts HON-027-TI Purple

- **Part number:** HON-027-TI (Purple finish specifically)
- **Price:** MSRP $120.99, street $108-125 delivered
- **Contents:** 24 pieces 6AL-4V Grade 5 titanium, anodized purple -- covers strut towers, valve cover (supplemental), IM brackets, accessory bolts
- **Fitment:** 2006-2011 Civic FG/FA (confirmed)
- **Source:** evasivemotorsports.com/store/product/dress-up-bolts-titanium-hardware-engine-bay-kit-honda-civic-si-fgfa-06-11-purple/
- **Caveat:** order the full kit at once -- anodize tones shift slightly between batches

Valve cover hardware -- Skunk2 649-05-0120-PPL family

- **Part number:** 649-05-0120 (purple finish available in the family)
- **Price:** ~$50-55
- **Contents:** Stainless steel bolts with purple-anodized low-profile washers, full K-series valve cover set
- **Source:** realstreetperformance.com, titanmotorsports.com

Accessory pulley kit -- NST22015K-Purple ?? IMPORTANT CAVEAT

- **Part number:** NST22015K-Purple (3-piece: crank + alt + idler)
- **Price:** $180-270 depending on configuration
- **Material:** CNC 6061-T6, hard-anodized purple
- **Fitment:** 2006-2011 Civic SI K20
- **?? COMPATIBILITY CONFLICT:** This kit includes a crank pulley. Phase 15 plans already call for an **ATI Super Damper** as the crank pulley. **Don't run both** -- the ATI replaces the crank pulley entirely. Order only the alternator + idler pulleys from NST (contact NST direct -- they'll split the kit), or accept that the NST crank piece goes in the spare bin. Budget ~$180 for alt+idler only.
- **Source:** nonstoptuning.co/store/p49, heeltoeauto.com (NST22015K-Purple)

Password:JDM purple fender washers

- **Price:** ~$20-25 for 5-6 pack

- **Warning:** Amazon is full of fake Password:JDM washers. Buy from Tecnocraft, CorSport, or directly from Password:JDM's official distributors.

Custom Work Required (no purple option exists)

These need powder-coating or paint because no reputable brand sells them in purple:

- **Washer fluid cap** -- OEM 38513-SC4-xxx family. No aftermarket purple billet exists for FG2.
- **Power steering reservoir cap** -- OEM 53697-SB3-952. No aftermarket purple billet.
- **Brake master cylinder cap** -- OEM 46662-S9A-003. Verus/MDR make billet BMC caps but only for newer generations (FK8+, FL5). Werks JDM has an FG2 reservoir tank upgrade ($180) but it's the whole canister, not a cap.
- **Stock K20Z3 valve cover** -- no authentic purple aftermarket cover exists. Spoon is yellow only. Authentic Mugen is red wrinkle only. eBay "purple Mugen" covers are fakes. Skunk2 magnesium cover has cowl-clearance issues on FG2 per Skunk2's own fitment notes.
- **Megan Racing front strut bar** -- already owned in polished finish. Powder-coat before install.

Batching strategy: drop the washer cap + PS cap + BMC cap + strut bar + valve cover off at ONE local powder coater in one trip. Combined cost ~$330-565. Request candy/illusion purple with clear top coat.

Parts with NO Purple Option (don't buy cheap substitutes)

- **BC Racing BR coilover springs** -- only black from BC. Aftermarket 2.5"/65mm springs in purple don't exist from Swift, Eibach, H&R, Tein, or Hyperco (Hyperco will do custom run but $400+). Accept black.
- **Radiator cap (Koyorad)** -- no purple factory option. Powder-coat SK-C13 to purple.
- **Silicone hose clamps (T-bolt)** -- stainless only from Vibrant, HPS, Mishimoto, Hybrid Racing. Hymee-brand purple T-bolts on Amazon have corrosion issues -- don't trust them on engine coolant. Keep stainless.
- **Header / exhaust accents** -- paint fails at exhaust temp. Stay thermally-correct (ceramic coat silver, not purple).
- **Caliper covers** -- MGP doesn't stock purple as a standard color. Custom PPG paint match available but $400+ (see 04-Brakes/purple-calipers.md). G2 paint kit is the recommended approach instead.

Install Strategy

Purple cosmetics are the **last layer** of each mechanical phase:

1. **Phase 09 (Cooling):** order Samco purple hose kit day-1 (4-6 week lead); install with cooling refresh.
2. **Phase 15 (Pulleys):** order NST alt+idler with ATI Super Damper; install together at pulley phase.
3. **Engine bay dress-up (anytime after headers installed):**
   - Install Acuity oil cap (5 min)
   - Install K-Tuned dipstick (5 min, includes spring retention)
   - Install Dress Up Bolts + Skunk2 VC hardware (2-3 hrs -- requires removing intake manifold for VC bolt access if doing the full kit)
   - Install shift knob (2 min)
4. **Batch powder-coat weekend:** drop valve cover + strut bar + fluid caps in one trip.

5. **Fender washers + accents:** install whenever.

Cross-References

- 09-Cooling-System/overview.md -- Samco purple hoses + Radium reservoir live under cooling phase
- 04-Brakes/purple-calipers.md -- G2 purple caliper paint (separate from engine-bay cosmetics)
- 15-Pulleys-and-Harmonic-Balancer/overview.md -- ATI vs. NST crank conflict documented there
- 18-Strut-Bar/overview.md -- Megan bar powder-coat decision tree

Things to Avoid

1. **Amazon "universal purple" caps.** The threads are close but not correct on FG2 necks -- they leak, or worse, pop off under pressure.
2. **eBay "purple Mugen" valve covers.** Fakes. Real Mugen K-series covers are red wrinkle only.
3. **Skunk2 magnesium valve cover.** Not even in purple -- Skunk2 themselves flag cowl interference on 06-15 Civic.
4. **Plasti-Dip on anything that sees heat** (calipers, valve cover, exhaust). Sub-30-day failures are typical.
5. **Mixing Engine Ice coolant with Honda Type 2.** Chemistry is incompatible. See 09-Cooling-System/overview.md.
6. **Cheap purple-anodized T-bolt clamps.** Corrode under coolant exposure. Stainless is the right call.

Last updated: 2026-04-20

Purple Cosmetics Install Guide -- Batched Aesthetic Install

Status: Not started

Reality Check

This is the aesthetic layer. None of it adds a single horsepower. I'm writing this guide because I want the engine bay to look as good as it goes, and because a disorganized purple theme looks worse than no theme -- mismatched shades, off-torque bolts, or three different "purples" next to each other will make me regret the spend.

Roughly 80% of the work is trivial: unscrew the old part, screw in the new purple one, 5-15 minutes each. The other 20% is planning:

- **Samco purple silicone hoses** -- requires cooling-system drain. Bundles with Phase 09. I am not draining coolant twice.
- **Radium 20-0270-00 reservoir, custom purple anodize** -- 4-6 week lead; call Radium direct at 503-244-0990 (custom anodize isn't on their website).
- **Powder-coat batch** -- valve cover + fluid caps + any other cosmetic metal. 1-2 weeks at a local shop, parts-off car.
- **G2 G2165 caliper paint** -- bundles with the 04-Brakes rear install so I'm not pulling calipers twice.

Honest framing: this is a few hundred to a couple thousand dollars spent on how the car looks. That's fine -- I want the bay to look as good as it goes -- but it isn't performance work. Reliability-first still applies: if a purple part creates a mechanical conflict (see NST crank pulley), the purple part loses.

Prerequisites / Cross-References -- READ FIRST

CRITICAL: NST crank pulley CONFLICTS with ATI Super Damper

The NST22015K-Purple kit ships as three pieces (crank + alt + idler). Phase 15 (see 15-Pulleys-and-Harmonic-Balancer/overview.md) puts an **ATI Super Damper 918477** on the crank for proper harmonic damping at 170k miles. Non-negotiable -- the NST billet crank pulley deletes the harmonic damper entirely, and I'm not deleting the damper at this mileage.

What to order: alternator + idler only, purple, direct from NST (nonstoptuning.co). Contact NST -- they'll split the kit on request. Confirm on the order that the crank piece is EXCLUDED. If the seller won't split, buy the full kit and leave the NST crank piece in the spare-parts bin. The ATI 918477 is what goes on the crank, full stop.

Install happens during Phase 15, after the ATI is torqued. See Batch D.

Cooling parts bundle with Phase 09

Samco hoses, Radium reservoir, Hybrid Racing HYB-TBC-00-10 T-bolt clamps, and Royal Purple Purple Ice 01600 all go in during the Phase 09 cooling refresh (09-Cooling-System/overview.md). I am not draining the cooling system twice. Order Samco early enough that it arrives before Phase 09 starts.

G2 caliper paint bundles with the 04-Brakes rear install

Full procedure in 04-Brakes/purple-calipers.md. One G2165 kit covers all four corners. Paint during the rear pad/rotor install when the car is already up on stands and all four wheels are off. The fronts were installed 2026-04-20 (Hawk HPS + R1 Concepts) -- pull them for paint during that same weekend if practical, or accept asymmetric purple until the next brake service.

Powder-coat batch includes strut bar -- but strut bar is DEFERRED

The Megan Racing strut bar is owned in polished finish, but install is deferred until after BC Racing coilovers (Phase 17) -- pillow-ball camber plates change strut-top geometry and the bar may not clear. Don't send the bar for coating until fitment is verified. First powder-coat batch (Batch E) runs without the strut bar.

Valve cover powder coat bundling

The valve cover comes off during Phase 06 (ignition refresh -- plugs, coils, VTC actuator) and again during Phase 16 (head service). Whichever happens first after the coated cover is back from the shop, that's the install moment. Do NOT pull the valve cover just for cosmetics -- pair it with mechanical work that's already opening the area.

Tools Required

- Phillips + flathead screwdrivers
- Metric Allen key set (DUB kit uses 4mm, 5mm, 6mm mostly -- long-reach helps)
- Metric sockets 8mm-17mm
- Torque wrench, inch-pound range (for low-torque valve cover hardware -- a ft-lb wrench is too coarse)
- Torque wrench, 10-50 ft-lb (for pulleys)
- Dielectric grease -- for any electrical connector disturbed during dress-up
- Masking tape + painter's paper -- caliper paint and general masking
- Blue threadlocker (Loctite 243) -- for structural DUB fasteners only, per DUB install notes
- Red Scotchbrite -- caliper paint surface prep
- Brake cleaner -- caliper prep
- Nitrile gloves -- G2 hardener is nasty
- Clean rags and a parts tray

Do NOT threadlock: plastic-covered fasteners, heat-critical coil-on-plug area, the Acuity oil cap (o-ring seal -- hand-tight), the K-Tuned dipstick (o-ring seal), or the shift knob threads (hand-tight only per Acuity).

Parts List

Full pricing and rationale in overview.md. Install-side checklist:

#	Item	Part #	Price
1	Oil filler cap	Acuity 1927-PPL Purple Podium	~$80
2	Dipstick	K-Tuned DP2-K20-PRP	~$72
3	Dipstick retention spring	K-Tuned DP2-SPR-KIT	~$25
4	Shift knob	Acuity ESCO-T6 1886-T6P Satin Purple	~$92

5	Engine-bay titanium bolt kit	Dress Up Bolts HON-027-TI (Purple)	~$125
6	Valve cover hardware	Skunk2 649-05-0120-PPL	~$55
7	Accessory pulleys (alt + idler ONLY)	NST22015K-Purple, crank excluded	~$180
8	Fender washer kit	Password:JDM Purple 6-pack	~$22
9	Silicone hose kit	Samco Sport FG2 purple (custom)	~$280-340
10	Coolant reservoir	Radium 20-0270-00 custom purple anodize	~$310
11	T-bolt hose clamps	Hybrid Racing HYB-TBC-00-10	~$45
12	Radiator cap	Koyorad SK-C13 + local powder coat	~$55
13	Coolant conditioner	Royal Purple Purple Ice 01600	~$12
14	Caliper paint kit	G2 G2165 Purple epoxy	~$55-75
15	Powder-coat batch	Local shop, candy/illusion purple	$330-565

Full program lands around $1,700-2,100.

Do NOT source from Amazon: Password:JDM washers (fakes), "purple anodized" T-bolts (corrode on coolant), "purple Mugen" valve covers (all fake -- real Mugen K-series covers are red wrinkle only).

Step-by-Step -- Batched Install

Five batches. Each attaches to a mechanical job that's already happening, so no assembly opens twice for cosmetic work.

Batch A: Bolt-On Swaps (1-2 hours, anytime)

No dependencies. A clean Saturday afternoon, engine cold, parts tray, shop rag.

A-1: Oil filler cap (Acuity 1927-PPL)

Engine cold. Wipe around the cap, unscrew OEM, verify the Acuity's o-ring is seated in the dovetail groove, thread in clockwise **hand-tight only** -- the o-ring seals on the dovetail; overtightening cracks billet or crushes the o-ring. Keep the OEM cap as spare.

A-2: Dipstick (K-Tuned DP2-K20-PRP)

Engine cold. Pull OEM straight up -- steady pressure, no jerking. Wipe the tube mouth. Verify both o-rings seated on the K-Tuned shaft. Insert until handle bottoms on the tube. If installing the DP2-SPR-KIT retention spring, follow K-Tuned's instructions -- headers (Phase 13) raise crankcase pressure and can push a non-retained dipstick up during spirited driving, so the spring is worth $25 any time after headers.

A-3: Shift knob (Acuity ESCO-T6 1886-T6P)

Acuity's ESCO-T6 threads onto M10x1.5, same as my Acuity 3-way adjustable shifter. Honda's stock shift shaft is reverse-thread, but the aftermarket adjustable and the ESCO-T6 both use standard M10x1.5 right-hand. **Confirm against Acuity's printed instructions** -- their docs are explicit about thread direction per shifter.

Shifter in neutral, parking brake on. Unscrew current knob (CCW on the Acuity shifter). Back off lock ring if present. Clean threads. Thread the ESCO-T6 on by hand CW until it bottoms at the orientation I want. Hand-tight

only -- 6061-T6 billet doesn't need cranked torque. Snug the lock ring against the knob base to lock angular position.

A-4: Dress Up Bolts HON-027-TI (24 pieces)

Slowest sub-step. Rule: **swap one bolt at a time.** Don't pull a whole assembly out.

- Blue threadlocker (Loctite 243) on structural bolts DUB flags -- typically strut tower bolts and intake manifold brackets. No threadlocker on plastic-covered, heat-critical, or anywhere DUB says don't.
- Titanium galls. Thin nickel-based anti-seize on every Ti-into-steel or Ti-into-aluminum thread. **Not copper-based** -- copper accelerates galvanic corrosion on titanium.
- Follow DUB's published torque per-fastener. For anything also in docs/torque-specs.md, use the OEM spec.
- Install the whole kit at once -- anodize tone shifts slightly between batches.

A-5: Skunk2 valve cover hardware (649-05-0120-PPL)

Two options:

- **Valve cover staying on car:** swap the 8 bolts + washers one at a time in star pattern, torqued to OEM valve cover spec (K20Z3 is ~7-10 ft-lb per docs/torque-specs.md -- over-torque warps the cover). Engine cold.
- **Valve cover coming off for powder coat or Phase 06:** skip this step now. Install the Skunk2 hardware when the coated or serviced cover is reinstalled. Saves handling twice.

Batch B: Cooling System Purple (bundled with Phase 09)

Full drain/fill/bleed procedure lives in the Phase 09 install guide. Purple-side additions:

Advance ordering:

- Samco order 6 weeks before Phase 09 start (4-6 week custom color lead).
- Call Radium at **503-244-0990** 6-8 weeks ahead to request custom purple anodize on 20-0270-00. Custom anodize is not on their website -- phone only.
- Hybrid Racing HYB-TBC-00-10 T-bolt kit -- no lead time.
- Royal Purple Purple Ice 01600 -- off the shelf.

At Phase 09 install:

- Replace each OEM hose with the Samco equivalent in the Phase 09 sequence. Shapes are identical -- 8thcivic-confirmed fitment.
- T-bolt clamps on every silicone-to-metal junction (worm-gear fails on silicone). Torque to T-bolt spec (~40-50 in-lb initial), re-check after first heat cycle.
- Radium 20-0270-00 mounts on the passenger strut tower area. Plumb overflow from the radiator cap nipple into the reservoir, pressure-vented return out the top (-6AN fittings, sight tube).
- Fill with Honda Type 2 OL999-9011 per Phase 09 bleed procedure. **Do NOT substitute Engine Ice** for purple tint -- propylene glycol chemistry mixed with Honda Type 2 is a hazard to the water pump and head. Purple look comes from the additive, not a different base coolant.
- After system is bled at operating temp, add Royal Purple Purple Ice 01600 per bottle instructions (~12 oz into the 1.3-gal K20Z3 system). Purple Ice is a Type-2-compatible conditioner.

Batch C: Caliper Paint (bundled with 04-Brakes rear install)

Full procedure in 04-Brakes/purple-calipers.md. Summary:

- One G2 G2165 kit covers all four corners.
- Prep is the critical part: brake clean, wire-brush loose rust, scuff with red Scotchbrite, final brake-clean wipe. No oil, no fingerprints.
- Mask rotor faces, pads, backing plates, brake hose, **slider boots especially** (heat-sensitive rubber -- paint destroys them).
- 2-3 thin coats, 15-20 min tack between coats.
- 2 hr wheel-on cure, wheel back on, torque lugs to **80 ft-lb** (FG2 spec).
- First 50 miles: gentle, no hard stops. Epoxy finishes cross-linking under normal brake heat.
- No power-wash on calipers for 7 days.

Weekend sequence: unbox rear pads + rotors, jack all 4 corners, install rears, **paint all 4 calipers while exposed** , cure overnight, wheels on, Hawk factory bed-in on all four together.

Batch D: NST Pulleys (bundled with Phase 15)

ATI 918477 is already on the crank when this starts.

1. Release the auto tensioner, remove serpentine belt. Replace with fresh OEM or Gates K070663 if showing wear -- new pulleys on a 170k belt invites belt walk I'd misdiagnose as a pulley problem.
2. Swap alternator pulley: hold with correct spanner, back off stock nut, remove stock pulley, install NST purple alt, torque to NST's published spec (~30-35 ft-lb -- **verify against NST's instructions**, spec is model-specific).
3. Swap idler: unbolt single center bolt, install NST purple idler, torque to NST's spec (~20-25 ft-lb -- **verify**).
4. Route belt per underhood diagram. Verify centered in every groove. Rotate engine one full turn by hand (on the crank bolt) and re-verify tracking.
5. Start engine. No belt noise should be present.
6. Re-check tension and tracking at 100 miles.

Batch E: Powder-Coat Batch (1-2 weeks, parts-off)

Long wall-clock time but almost all of it is parts sitting at the coater.

Batch contents (first batch):

- Stock K20Z3 valve cover. Mask mating surfaces (gasket rail, coil-on-plug seats, breather fittings) before coating. Mark masking points with Sharpie before dropoff; show the coater exactly.
- Brake master cylinder cap OEM 46662-S9A-003 -- **buy a spare** so the car stays drivable during the batch.
- Power steering reservoir cap OEM 53697-SB3-952 -- spare.
- Washer fluid cap -- spare.
- Any other purely cosmetic metal decided by batch day.

Batch exclusions:

- **Megan strut bar -- DEFERRED** until Phase 17 coilover fitment verified.
- **Koyorad radiator core -- NEVER.** Powder on fins acts as thermal insulator, reducing cooling efficiency -- directly at odds with Phase 09's E85 heat-headroom goal. Core stays bare aluminum. End-tank-only powder ($75-125) is tolerated but I'm skipping even that.
- **Exhaust-exposed parts** -- paint fails at exhaust temp. Ceramic coat (silver, not purple) for any exhaust part.

Shop selection:

- Local coater with color-match capability.

- Experience with candy/illusion purple over silver base.
- Bring the Acuity oil cap to the shop for in-person color matching against already-installed purple parts.

Sequence at coater: strip -> sandblast -> powder (base + color + clear for candy purple) -> oven cure 375-400degF 15-20 min -> QC inspection.

Reinstall:

- Valve cover goes back on with **fresh gasket** (OEM 12341-RBC-003 or equivalent), fresh half-moon seals, and Skunk2 649-05-0120-PPL hardware. Pair with Phase 06 or Phase 16 -- whichever is first.
- Fluid caps are 30-second swaps whenever.

Don't rush the cure. Hot powder handled with bare hands or recontaminated by oil, fuel, or coolant bubbles at the first real heat cycle. If a coat looks bad at pickup, send it back.

Timing / Batch Planning

Stack the batches so no mechanical job opens twice:

1. **Batch A (bolt-ons) -- anytime.** No dependencies. Do it whenever the kit arrives; delivers visible purple immediately, which makes the slow batches feel less like pure waiting.
2. **Batch E (powder-coat) -- start when the valve cover is expected to come off next.** 1-2 week coater lead; parts are off-car during the window. Pair with Phase 06 ignition refresh (typically first to pull the valve cover). Ship the day it comes off; by reassembly time the coated cover is back.
3. **Batch C (caliper paint) -- during the 04-Brakes rear install weekend.** All four corners in one weekend, bed-in together.
4. **Batch B (cooling purple) -- during Phase 09.** Samco and Radium orders placed 6-8 weeks ahead of Phase 09 start.
5. **Batch D (NST pulleys) -- during Phase 15.** Alt + idler only; ATI on the crank.

Every batch attaches to a mechanical job that's happening anyway. No cosmetic-only teardown. That's the whole design.

First Impression Checklist

Walk-around inspection after all five batches are complete:

- [ ] All purple items are the same shade family under natural daylight -- no Prince-purple anodize next to lavender powder-coat next to Crown-Royal caliper paint.
- [ ] Engine bay reads clean from the passenger side with the hood up (that's where people look first at a show).
- [ ] Caliper paint edges sharp; no overspray on rotor, hub, pad backing plate.
- [ ] Every DUB fastener torqued to spec. Re-check at 1k miles -- titanium bolts settle.
- [ ] No coolant weep at any silicone hose junction after the first heat cycle. Re-torque T-bolts.
- [ ] Shift knob angle correct for hand position; lock ring tight if used.
- [ ] Valve cover gasket not weeping oil, especially at corners.
- [ ] Radium reservoir mounted square, sight tube shows correct level.
- [ ] Samco hose routing does not contact alternator, pulleys, belt, or steering linkage.

Common Mistakes

1. **Ordering the full NST kit without excluding the crank pulley.** Either a paid-for spare in a box, or (worse) the NST billet crank ends up on the engine deleting the harmonic damper I explicitly chose ATI for. Always order NST22015K with crank EXCLUDED.
2. **Powder-coating the radiator core.** Fin powder acts as thermal insulator -- directly undermines the Phase 09 upgrade. Koyorad stays bare aluminum.
3. **Engine Ice instead of Honda Type 2 + Purple Ice.** Propylene glycol with a different inhibitor package. Mixing voids the water pump and head gasket longevity guarantees. Type 2 OL999-9011 is the coolant; Purple Ice 01600 is the conditioner for the tint.
4. **Overlooking Samco lead time.** 4-6 weeks custom. If Phase 09 starts before Samco arrives, I either sit with a drained system (bad for seals) or fill with rubber and drain again later.
5. **Rushing Batch E reinstall.** Slapping the coated valve cover on the engine before full cure bubbles at the first coil-on-plug heat cycle. Let the coat cure fully.
6. **Copper anti-seize on titanium DUB bolts.** Accelerates galvanic corrosion on titanium. Use nickel-based for Ti-into-steel or Ti-into-aluminum.
7. **Amazon Password:JDM fender washers.** Minefield of fakes. Tecnocraft, CorSport, or Password:JDM authorized only.
8. **"Purple anodized" T-bolt clamps on coolant.** Documented corrosion. Stainless only -- Hybrid Racing HYB-TBC-00-10.

Maintenance Notes

- **Park covered if possible.** Direct UV fades purple anodize and purple paint over years. The hood shields most of the bay; cabin UV hits the shift knob, and the calipers see full sun through the wheel.
- **G2 caliper epoxy** -- durable but not forever. Plan on touch-up or repaint every 3-5 years on a regularly driven car. One G2165 kit at each interval.
- **Samco silicone hoses** -- 7-10 year lifespan like any quality silicone. Inspect for surface cracking or age-hardening at every coolant change.
- **Dress Up Bolts titanium** -- anodize on titanium is an oxide-layer interference effect, not dye. Color is indefinitely stable. **Re-torque at 1k miles** (Ti settles) and annually thereafter.
- **Skunk2 valve cover hardware** -- re-torque at 1k miles after install, then at every valve cover gasket replacement.
- **Acuity oil cap** -- billet is stable, o-ring is the wear item. Replace o-ring at oil-change cadence if it shows compression set or cracking. Acuity sells replacements individually.
- **Radium reservoir** -- hard-anodize will outlast the silicone around it. Inspect -6AN fittings for seep at every coolant service.
- **NST pulleys** -- idler bearings are the life-limited item. Listen for whine or rumble at idle; NST sells bearing service kits. The alt pulley has no wear surfaces.

See Also / Cross-References

- overview.md -- full parts list, pricing, tier breakdown, rationale

- ../04-Brakes/purple-calipers.md -- G2 caliper paint full procedure
- ../09-Cooling-System/overview.md -- Samco + Radium + Purple Ice integration
- ../15-Pulleys-and-Harmonic-Balancer/overview.md -- ATI rationale + NST alt/idler install
- ../18-Strut-Bar/ -- Megan bar deferred until post-coilover; powder-coat after Phase 17 fitment check
- ../06-Ignition-Refresh/ -- valve cover pull point; pair with powder-coat install
- ../16-Cylinder-Head/ -- alternate valve cover pull point if Phase 06 already done
- ../docs/torque-specs.md -- OEM torque values for Batches A, B, D
- ../CLAUDE.md -- project-wide context

Last updated: 2026-04-20

Installation References -- Guides, Videos, and Resources

Everything I've collected for each mod -- install guides, video tutorials, community threads, tips from people who've done it before. Organized by mod folder.

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01 -- Clutch & Flywheel (Exedy Stage 1 + Lightweight Flywheel)

Status: Installed

- YouTube Search: Exedy clutch install 8th gen Civic SI K20Z3
- YouTube Search: K20Z3 clutch replacement 8th gen Civic SI flywheel

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02 -- Intake (K&N Typhoon 69-1014TS)

Status: Installed

- YouTube Search: K&N Typhoon 69-1014TS install 8th gen Civic SI

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03 -- Short Shifter + Bushings (Acuity)

Status: Installed

- YouTube Search: Acuity short shifter install 8th gen Civic SI
- YouTube Search: Acuity shifter cable bushings install Civic SI

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04 -- Engine Mounts (Hybrid Racing 70A)

Status: Purchased, Awaiting Install

- YouTube Search: engine mount install 8th gen Civic SI Hybrid Racing Hasport 70A
- YouTube Search: K20Z3 motor mount replacement 8th gen Civic

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05 -- Brakes (Hawk HPS + R1 Concepts Rotors)

Status: Purchased, Awaiting Install

Written Guides I've Found

- Hybrid Racing -- Brake Pad Install (06-11 Civic Si)/26)
- O'Reilly -- How to Replace Brake Pads & Rotors '06-'11 Honda Civic Si
- iFixit -- 2006-2011 Honda Civic Front Brake Pad and Rotor Replacement
- 8thcivic.com -- DIY Removal and Install of Rotors
- 8thcivic.com -- DIY: Changing Rear Brake Pads

Tools I'll Need

- 19mm socket (lug nuts + front caliper bracket bolts)
- 12mm socket (caliper slide pin bolts)
- Brake caliper compressor tool (rear calipers screw in -- I can rent one from AutoZone)
- Torque wrench
- Brake cleaner
- Anti-seize for rotor set screws

Key Notes

- **Bed the Hawk HPS pads properly.** Follow Hawk's bedding procedure: 6-10 moderate stops from 35 mph, then 2-3 harder stops from 45 mph, then drive 5 minutes without heavy braking to cool.
- Front caliper bolts: 19mm. Rear caliper bolts: 12mm.
- Pump brake pedal several times after install until pedal pressure returns.
- Check master cylinder fluid level after compressing pistons.

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06 -- Strut Bar (Megan Racing MR-SB-HC06FU-1P)

Status: Purchased, BLOCKED (Clearance Issue)

- Simple 2-bolt-per-side install once I figure out the clearance issue
- YouTube Search: strut bar install 8th gen Civic SI

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07 -- Hondata FlashPro

Status: Purchased, Awaiting First Flash

Written Guides

- Hondata Official -- FlashPro Civic Si 2006-2011
- 9thgencivic.com -- Hondata FlashPro for Noobs (process is similar for 8th gen)

First Flash Steps

1. Install FlashProManager on a Windows PC
2. Plug in FlashPro via USB, fill out registration, register with Hondata
3. Take FlashPro to car, plug into OBD2, turn ignition ON (don't start) for 5 seconds to record VIN
4. Back at PC: select "Online, Lock" and choose a calibration (start with stock + intake base map)
5. Upload calibration to FlashPro
6. At car: plug in FlashPro, ignition ON, press Program button for 1+ second
7. Wait ~5-6 minutes for flashing to complete (light stops flashing)
8. Start car, verify no CELs
9. **Save stock calibration as backup FIRST before making any changes**

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08 -- Valved Exhaust (QTP QTEC30 3")

Status: Planned

- YouTube Search: QTP exhaust cutout install DIY
- YouTube Search: electric exhaust cutout install wiring
- I'll be taking this to a muffler shop for the welding/fabrication work

09 -- Headers (Skunk2 Alpha 412-05-1930 + Berk HFC)

Status: Planned

Written Guides

- Hybrid Racing -- Skunk2 Alpha Header Install (06-11 Civic Si)/32)
- 8thcivic.com -- Recent Skunk2 Alpha Header Install Tips and Q&A
- 8thcivic.com -- Official Skunk2 Alpha Header Chat/Review Thread
- 8thcivic.com -- Skunk2 Alpha Header Install

Key Install Tips (from the community)

- **PB Blaster the manifold studs 3-5 days before.** Heat cycle + spray daily.
- Skunk2 does NOT include hardware -- only the "donut gasket." I'll need to buy: spring bolts, new exhaust flange bolts, new exhaust manifold gasket.
- Buy the Skunk2 O2 sensor extender -- I will need it.
- Remove the "Axle Shaft" cover (3 bolts) for maximum working space.
- To remove the old manifold, flip it and pull out from the cat converter area. Same process in reverse for the new header.
- EP3 OEM donut gasket part number: 18229-S6D-T31
- Budget 6-10 hours DIY. I'm leaning toward having a shop do this one ($300-500 labor) given the seized stud risk at 170k miles.

10 -- Intake Manifold (Skunk2 Ultra Street 307-05-0600)

Status: Planned

- YouTube Search: Skunk2 Ultra Street intake manifold install K20 K-series
- YouTube Search: K20Z3 intake manifold swap 8th gen Civic SI
- **Take photos of ALL vacuum line routing before removal** -- I know I'll forget where something goes otherwise
- **Block off coolant passages and run a bypass hose**

11 -- Pulleys (ATI Super Damper 918477 + NST Kit)

Status: Planned

- YouTube Search: ATI Super Damper pulley install K-series K20
- YouTube Search: crank pulley replacement K20Z3
- **Use a harmonic balancer puller -- never pry** (damages crank seal)
- I can rent the puller free from AutoZone

12 -- Flex Fuel & Fuel System

Status: Planned

- YouTube Search: 8th gen Civic SI flex fuel install Hondata
- YouTube Search: fuel injector replacement K20Z3
- YouTube Search: fuel pump replacement 8th gen Civic SI DeatschWerks
- Fuel pump is accessed through the rear seat access panel (no need to drop the tank)

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13 -- Clutch Hydraulics (Hybrid Racing HYB-CMC-01-20)

Status: Planned (Priority Purchase)

Written Guides

- Hybrid Racing Official Install Guide -- CMC Upgrade -- **Step-by-step with photos, this is the main one**
- importnut.net -- Hybrid Racing 8th Gen Civic CMC Upgrade -- Detailed walkthrough with photos
- 8thcivic.com -- 8th Gen CMC Upgrade Thread
- Scribd -- 06+ Civic Si CMC Swap Guide (PDF)

Install Overview

Difficulty: Moderate. 3-5 hours. Access is tight.

What the Kit Includes (HYB-CMC-01-20)

- Clutch master cylinder (CMC)
- Concentric slave cylinder (CSC)
- Stainless steel braided clutch line
- Firewall seal
- PTFE grease
- Hardware and clips

Step-by-Step Summary

1. **Access:** Remove lower dash panel to get to the pedal assembly
2. **Clamp reservoir line:** Use clamping pliers to pinch the reservoir fluid line that feeds the clutch master cylinder -- keeps fluid from draining everywhere
3. **Disconnect clutch line at CSC:** Using a metric line wrench, remove the factory clutch line from the slave cylinder. Hold the CSC fitting with a second wrench.
4. **Remove old CMC:** Remove the two mounting nuts from the firewall side. The CMC is buried under the cowl and brake master cylinder -- **I'll be doing this almost entirely by feel**
5. **Remove clevis pin:** Disconnect the pushrod from the clutch pedal
6. **Swap pushrod/clevis:** Transfer the pushrod and clevis from the old CMC to the new Hybrid Racing CMC (straightforward)
7. **Install new CMC:** Mount in reverse, torque mounting nuts to spec
8. **Install new CSC:** Remove transmission inspection plate to access the slave cylinder. Swap old CSC for new one.
9. **Install braided line:** Route the new stainless braided clutch line from CMC to CSC

10. **Fill and bleed:** Fill reservoir with **Motul RBF 600** brake fluid. Bleed the system thoroughly -- gravity bleed or pressure bleed. Air bubbles = spongy pedal.

11. **Test:** Pump clutch pedal multiple times. Should feel firm with consistent engagement point. Road test through all gears.

Tips from the Community

- **The CMC is the hardest part** -- it's completely buried. Work by feel.
- **Use a turkey baster** to remove old fluid from the reservoir before disconnecting anything
- **Bleed thoroughly.** Any air in the system will make the pedal spongy. I'm considering a Motive pressure bleeder (~$50) so I can do it solo.
- **Motul RBF 600** is the recommended fluid (594F dry boiling point). DOT 4 is minimum spec.
- **Have paper towels and a drip pan ready** -- brake fluid will drip on everything

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14 -- Suspension (BC Racing BR + Full Overhaul)

Status: Planned

Written Guides -- Coilover Install

- 8thcivic.com -- DIY: Installing Coilovers on 8th Gen Civics
- 8thcivic.com -- Full Coilover Install Tips/Tricks
- 8thcivic.com -- MaXpeedingRods Coilover Installation Guide (process applies to any coilover)
- vtec.academy -- 8th Gen Civic Coilover Install
- civicforums.com -- How to Install Coilovers on the 8th Gen Honda Civic

Written Guides -- Bushings

- 8thcivic.com -- Energy Suspension Bushing Kit, Front Bushing on Rear Trailing Arm Install
- 8thcivic.com -- Suspension Bushings Replacement Options

YouTube Searches

- BC Racing BR coilover install 8th gen Civic SI FG2
- coilover install 8th gen Civic SI front rear
- rear camber arm install 8th gen Civic SI Skunk2 Godspeed
- sway bar install 8th gen Civic SI Progress
- polyurethane bushing install 8th gen Civic
- rear trailing arm bushing replacement Honda Civic
- ball joint replacement 8th gen Honda Civic
- tie rod end replacement Honda Civic
- roll center adjuster install lowered Civic

Key Install Notes

- **No spring compressor needed** -- coilovers come pre-assembled
- **Bushing removal requires a press** -- a shop press or bushing removal tool set
- **Do ALL the work in one session** -- the car is already apart, no point doing this piecemeal
- **Anti-seize on all bolts** during reassembly

- **Drive carefully to the alignment shop immediately after**
- **Re-torque all fasteners after 100 miles**

General Resources

8th Gen Civic SI Community

- 8thcivic.com -- The primary 8th gen Civic forum. Honestly the single best resource out there for this platform.
- r/CivicSi -- Reddit community
- Honda-Tech -- General Honda forum

Hybrid Racing Install Guides Hub

- guides.hybrid-racing.com -- Step-by-step guides for a lot of 8th gen Civic SI parts. Really well done.

General DIY Channels (YouTube)

- **ChrisFix** -- Excellent brake, suspension, and general maintenance tutorials
- **BrenTuning** -- Hondata/tuning content
- **SpeedFactory Racing** -- K-series builds
- **BriansMobile1** -- Thorough Civic SI tutorials

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